



# Emergency Ambulance Service

## Clinical Practice Guidelines

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**Hato Hone  
St John**

Ambulance personnel cannot legally supply or administer prescription medicines to patients unless they have authority to practise (ATP), or they are a registered health practitioner with the ability to independently supply or administer prescription medicines described within their scope of practice. In addition, services restrict the use of some items of clinical equipment and the performance of some clinical procedures to personnel at specified practice levels.

ATP is the authorisation of individual personnel to use these CPGs by the ambulance service Medical Director/Clinical Director. Personnel may not use these CPGs without ATP. ATP is granted at a specified practice level and the practice levels are listed in the table. Each practice level has an associated list of authorised medicines and interventions, that defines the medicines that may be administered and the interventions that may be provided by personnel when treating patients. Personnel may be individually credentialed within their service to administer additional medicines or to perform additional interventions that are not within the list. Conversely, personnel may be individually credentialed within their service to only administer some of the medicines, or to only provide some of the interventions from within the list. Other interventions that are not described within the authorised medicines and interventions (for example automated defibrillation) may be provided by all personnel.

The authorised medicines and interventions for those with an Extended Care Paramedic (ECP) ATP encompasses all authorised medicines and interventions described for the Paramedic ATP in the following table. All ECP-specific skills are outlined within the ECP CPGs.

| Skill                                    | EMT | Paramedic | ICP | CCP |
|------------------------------------------|-----|-----------|-----|-----|
| Adrenaline IN, IM, nebulised and topical | ✓   | ✓         | ✓   | ✓   |
| Aspirin PO                               | ✓   | ✓         | ✓   | ✓   |
| Glucagon IM                              | ✓   | ✓         | ✓   | ✓   |
| Glyceryl trinitrate SL                   | ✓   | ✓         | ✓   | ✓   |
| Ibuprofen PO                             | ✓   | ✓         | ✓   | ✓   |
| Ipratropium nebulised                    | ✓   | ✓         | ✓   | ✓   |
| Laryngoscopy (airway obstruction)        | ✓   | ✓         | ✓   | ✓   |
| Loratadine PO                            | ✓   | ✓         | ✓   | ✓   |
| Methoxyflurane inhaled                   | ✓   | ✓         | ✓   | ✓   |
| Ondansetron IM                           | ✓   | ✓         | ✓   | ✓   |
| Paracetamol PO                           | ✓   | ✓         | ✓   | ✓   |
| Prednisolone PO                          | ✓   | ✓         | ✓   | ✓   |
| Prednisone PO                            | ✓   | ✓         | ✓   | ✓   |

|                                                                                               |   |   |   |   |
|-----------------------------------------------------------------------------------------------|---|---|---|---|
| Positive end expiratory pressure (PEEP)                                                       | ✓ | ✓ | ✓ | ✓ |
| Salbutamol MDI and nebulised                                                                  | ✓ | ✓ | ✓ | ✓ |
| Supraglottic airway (SGA)                                                                     | ✓ | ✓ | ✓ | ✓ |
| Tramadol PO                                                                                   | ✓ | ✓ | ✓ | ✓ |
| Adrenaline IV (cardiac arrest only)                                                           |   | ✓ | ✓ | ✓ |
| Amiodarone IV (cardiac arrest only)                                                           |   | ✓ | ✓ | ✓ |
| Cefazolin IV                                                                                  |   | ✓ | ✓ | ✓ |
| Ceftriaxone IM and IV                                                                         |   | ✓ | ✓ | ✓ |
| Clopidogrel PO                                                                                |   | ✓ | ✓ | ✓ |
| Continuous positive airway pressure (CPAP)                                                    |   | ✓ | ✓ | ✓ |
| Droperidol IM, IV and SC                                                                      |   | ✓ | ✓ | ✓ |
| Enoxaparin SC                                                                                 |   | ✓ | ✓ | ✓ |
| Fentanyl IM, IN, IV and SC                                                                    |   | ✓ | ✓ | ✓ |
| Glucose IV                                                                                    |   | ✓ | ✓ | ✓ |
| Heparin IV                                                                                    |   | ✓ | ✓ | ✓ |
| Hydrocortisone IM and IV                                                                      |   | ✓ | ✓ | ✓ |
| IO access                                                                                     |   | ✓ | ✓ | ✓ |
| IV cannulation                                                                                |   | ✓ | ✓ | ✓ |
| Ketamine IM, IV and PO (analgesia only)                                                       |   | ✓ | ✓ | ✓ |
| Levetiracetam IV                                                                              |   | ✓ | ✓ | ✓ |
| 1% lignocaine IO and SC                                                                       |   | ✓ | ✓ | ✓ |
| Manual defibrillation                                                                         |   | ✓ | ✓ | ✓ |
| Metoprolol PO                                                                                 |   | ✓ | ✓ | ✓ |
| Midazolam IM (seizures, acute behavioural disturbance and care during last days of life only) |   | ✓ | ✓ | ✓ |
| Midazolam IV (seizures and care during last days of life only)                                |   | ✓ | ✓ | ✓ |

|                                      |  |   |   |   |
|--------------------------------------|--|---|---|---|
| Morphine IM, IV and SC               |  | ✓ | ✓ | ✓ |
| Naloxone IM and IV                   |  | ✓ | ✓ | ✓ |
| Olanzapine PO                        |  | ✓ | ✓ | ✓ |
| Ondansetron IV                       |  | ✓ | ✓ | ✓ |
| Oxytocin IM and IV                   |  | ✓ | ✓ | ✓ |
| 0.9% sodium chloride IV              |  | ✓ | ✓ | ✓ |
| Synchronised cardioversion           |  | ✓ | ✓ | ✓ |
| Tenecteplase IV                      |  | ✓ | ✓ | ✓ |
| Ticagrelor PO                        |  | ✓ | ✓ | ✓ |
| Tranexamic acid IV, IN and nebulised |  | ✓ | ✓ | ✓ |
| Adenosine IV                         |  |   | ✓ | ✓ |
| Adrenaline (all routes)              |  |   | ✓ | ✓ |
| Amiodarone IV                        |  |   | ✓ | ✓ |
| Atropine IV                          |  |   | ✓ | ✓ |
| Calcium chloride IV                  |  |   | ✓ | ✓ |
| Chest decompression (needle)         |  |   | ✓ | ✓ |
| Cricothyroidotomy                    |  |   | ✓ | ✓ |
| Endotracheal intubation              |  |   | ✓ | ✓ |
| Fascia iliaca block                  |  |   | ✓ | ✓ |
| Finger thoracostomy                  |  |   | ✓ | ✓ |
| Ketamine (all routes)                |  |   | ✓ | ✓ |
| Labetalol IV                         |  |   | ✓ | ✓ |
| Magnesium IV                         |  |   | ✓ | ✓ |
| Metaraminol IV                       |  |   | ✓ | ✓ |
| Midazolam IM and IV                  |  |   | ✓ | ✓ |



|                                 |  |  |   |   |
|---------------------------------|--|--|---|---|
| Pacing                          |  |  | ✓ | ✓ |
| Rocuronium IV                   |  |  | ✓ | ✓ |
| Ropivacaine SC                  |  |  | ✓ | ✓ |
| 8.4 % sodium bicarbonate IV     |  |  | ✓ | ✓ |
| Glyceryl trinitrate IV          |  |  |   | ✓ |
| Mechanical ventilation          |  |  |   | ✓ |
| Rapid sequence intubation (RSI) |  |  |   | ✓ |
| Suxamethonium IV                |  |  |   | ✓ |

## More Information

### Version history

**28 October 2024**

1.0.5

**Update:** Added nebulised route to tranexamic acid.**11 September 2023**

1.0.4

**Update:** Addition of individual credentialing and guidance. LMA changed to SGA. Addition and removal of medicines.**22 November 2022**

1.0.1

**Update:** Added ECP references and CCP skills.

# General Principles

## Standing orders

- These CPGs incorporate standing orders and personnel are required to adhere to them.
- Use of these standing orders does not require the Medical Director/Clinical Director to countersign each use. However, use will be subject to the audit process within each service.
- Not all services carry all the medicines and equipment described within these CPGs and some services carry additional medicines and equipment, for which additional standing orders are issued as required.
- The words “must” and “should” appear throughout the CPGs. The word “must” means that personnel are always required to follow this instruction. The word “should” means that personnel are required to follow the instruction unless there is a good clinical reason not to.
- Some personnel have an expanded delegated scope of practice, for example Extended Care Paramedics (ECPs). Such personnel are provided with additional CPGs which supersede these CPGs in specific circumstances.

## General principles of treatment

- Although not listed in each guideline, all patients require a primary and secondary survey with appropriate intervention as required.
- Unless otherwise specified, all medicine doses and fluid volumes in these CPGs are for adults, and children whose weight has been rounded to 50 kg or more. See the ['paediatric drug dose'](#) guideline for children whose weight has been rounded to less than or equal to 40 kg.
- For the purposes of these CPGs, a patient is an adult if they are aged 16 years or over. Note: in the Auckland area, children should only be taken to Starship Hospital if aged less than 15 years of age.

## Clinical trials

- The services are committed to improving clinical knowledge and patient outcomes by taking part in clinical trials. Such involvement in clinical trials improves the overall care that patients receive.
- Personnel are required to adhere to protocols and enter all eligible patients into clinical trials undertaken by the services.

## Seeking clinical advice from personnel who are not at the scene

- All requests for clinical advice from personnel who are not at the scene must be sought from one of the following:

# General Principles

- » Personnel designated to provide clinical advice on behalf of the service:
    - » For personnel who have ATP with HEMS this is via the Medical Director or their delegate.
    - » For personnel who have ATP with HHStJ this is via the Clinical Desk.
    - » For personnel who have ATP with Northern Rescue (NRHL), this is via the Medical Director or their delegate (Senior Clinical Person).
    - » For personnel who have ATP with SRSL this is via the Medical Director or their delegate.
    - » For personnel who have ATP with WFA this is via the Clinical Desk.
  - » The person described within these CPGs, for example a designated medical specialist as part of a destination policy or pathway.
  - » A registered health professional who is directly involved in the patient's care.
- The advice and the name of the person providing it must be recorded in the appropriate section of the patient report form (PRF).
  - When seeking clinical advice, use **ISBAR**:
    - I Identify yourself.** State your name, practice level, vehicle call sign and where you are calling from.
    - S Situation.** State a succinct reason for calling, for example, "I am calling for permission to administer adrenaline IM" or "I am calling for advice".
    - B Background.** Briefly describe the background of the incident.
    - A Assessment.** Describe your assessment of the patient. Ensure that any information that is likely to be required is available, for example, vital signs or a 12 lead ECG.
    - R Recommend and review.** State what you think is required and listen carefully to instructions. Review and confirm the plan before ending the call.

## Safe restraint during transport

- During transport the patient and personnel should be safely restrained.
- If being transported on a stretcher, the patient should be restrained using:
  - » The supplied stretcher belts if the patient is an adult or large child, or
  - » Specific weight/size appropriate neonatal/paediatric devices supplied by the service, if the patient is a small child.
- Children must not be transported in the arms of an adult.
- See the '[normal birth](#)' guideline for additional guidance on the restraint of a newborn baby.

# General Principles

- Rarely, it may not be possible to restrain the patient on a stretcher. For example, if the patient is morbidly obese. In this setting the risk to the patient must be balanced against the risk of not being transported.
- There are some circumstances when it is not possible to provide vital treatment to a patient during transport, with personnel restrained. In this setting, the risk to personnel must be balanced against the risk to the patient of not receiving the required treatment.
- The following principles must be applied if personnel or the patient is unrestrained:
  - There must be an explicit decision that the balance of risk is such that it is appropriate to transport without restraint, and
  - The vehicle must travel at normal road speed (even if under lights), and
  - The driver must ensure that the nature of their driving is modified to keep the patient and personnel as safe as possible during transport.

## More Information

### Version history

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**16 June 2026**

1.1.0.1

**Update:** Removed PPE content and added advice on safe restraint during transport.

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**22 November 2022**

1.0.1

**Update:** Added clinical advice, PPE, providing treatment during transport. Removed alternative care pathways.

Personnel may only administer medicines and provide interventions from within the authorised list associated with their practice level, as per [CPG EAS 1.1](#), unless further described below.

## When the treatment is a medicine that is not described within these CPGs

- For example, the administration of insulin.
- This is only permissible when personnel are instructed to do so by the on call doctor or their delegate, for the service.

## When the treatment is a medicine that is not an authorised medicine for the person administering it

- For example, an EMT is administering midazolam, or a Paramedic is administering ketamine for dissociation.
- This is only permissible in the following circumstances:
  - 1 When instructed to do so as a result of seeking clinical advice, or
  - 2 When instructed to do so by a registered medical or nursing practitioner (see later in this guideline), or
  - 3 When the treatment is provided by a student under the direct supervision of ambulance personnel (see later in this guideline).

## When the treatment is an authorised medicine, but the administration differs from that described in these CPGs

- For example, administering higher doses than described or using alternative indications.
- This is only permissible in the following circumstances:
  - 1 An ICP or CCP may do so when it is in the best interest of the patient, or
  - 2 When instructed to do so as a result of seeking clinical advice, or
  - 3 When instructed to do so by a registered medical or nursing practitioner (see later in this guideline).

## When the treatment is a medicine being administered or a procedure being performed, by personnel with a scope of practice\* in addition to that of paramedicine

- For example, a Paramedic who is also a registered nurse is placing a urinary catheter, or a Nurse Practitioner is administering medicines from within their Nurse Practitioner scope of practice.



- A registered health practitioner with a scope of practice in addition to paramedicine, may choose to administer a medicine or perform a procedure that is not within their list of authorised interventions, provided all the following criteria are met:
  - 1 The treatment is within their additional scope of practice\* as defined by their registering authority, and
  - 2 The treatment is in the best interest of the patient, and
  - 3 The treatment is consistent with the principles contained within these CPGs.

\*A treatment is within a practitioner's scope of practice if it can be provided by them:

- Independently making the decision without anyone else instructing them to do so, and
- Without anyone else needing to chart it or prescribe it, and
- Without anyone else needing to countersign afterwards.

## When the treatment is being provided following instructions from a registered medical or nursing practitioner who is external to the service

- For example, an EMT is administering adrenaline following the instructions of a PRIME responder on scene, or a Paramedic is administering medicines following the instructions of hospice personnel or following a telehealth consult.
- This is only permissible in the following circumstances:
  - 1 When a patient has pre-supplied medicines for administration in the event of an emergency, for example a patient with a last days of life care plan. All personnel should administer such medicines, including providing treatments outside their authorised medicines and interventions, provided:
    - a The medicine is indicated, and
    - b There are clear written instructions, and
    - c Personnel have sufficient knowledge and skill to provide the treatment.
  - 2 When a patient has written instructions for a specific care plan. All personnel should follow such instructions provided:
    - a The instructions are clear and apply to the current circumstances, and
    - b The instructions are consistent with good practice, and
    - c Personnel have sufficient knowledge and skill to provide the treatment.
  - 3 When instructed to do so by a registered medical or nursing practitioner involved in the patient's care, including via a telehealth consultation. All personnel should follow such instructions, including providing treatments outside their authorised medicines and interventions, provided:
    - a The instructions are consistent with good practice, and
    - b The instructions are clear, and

- c Personnel have sufficient knowledge and skill to provide the treatment.

## When the problem is immediately life-threatening and it is not possible to seek clinical advice

- Rarely, a patient may have an immediately life-threatening problem and personnel may be unable to seek timely clinical advice to gain permission to provide treatment that is immediately life-saving, that is outside their authorised medicines or interventions.
- Personnel may administer treatment in this circumstance, provided they have made all reasonable attempts to obtain clinical advice and they have sufficient knowledge and skill to provide the treatment. In addition, as soon as practical they must:
  - Complete a comprehensive PRF and refer the PRF for clinical audit, and
  - Speak to the on-call doctor (or delegate) for that service during daylight hours to discuss the incident.
- **Examples of immediately life-saving treatments:**

There are very few treatments that are considered immediately life-saving, but examples include:

  - Antibiotic administration for meningococcal septicaemia.
  - Chest decompression for tension pneumothorax.
- **Examples of treatments that are not considered immediately life-saving:**
  - Endotracheal intubation.
  - IV medicines during cardiac arrest.

## When the treatment is being provided under supervision

### The following groups of personnel may provide treatments under supervision

- Personnel enrolled in the New Zealand Diploma in Ambulance Practice may provide treatments from within the EMT list (see [CPG EAS 1.1](#)), under the supervision of personnel with ATP as an EMT, Paramedic, ECP, ICP or CCP.
- Personnel enrolled in a paramedicine degree programme in New Zealand or Australia may provide treatments from within the Paramedic list (see [CPG EAS 1.1](#)), under the supervision of personnel with ATP as a Paramedic, ECP, ICP or CCP.
- Personnel who are registered as a Paramedic with a current annual practicing certificate, who do not have ATP as a Paramedic, may provide treatments from within the Paramedic list (see [CPG EAS 1.1](#)), under the supervision of personnel with ATP as a Paramedic, ECP, ICP or CCP.
- Personnel employed in a service specific programme<sup>1</sup> to gain ATP as an ECP may provide treatments from within the ECP list (see [CPG EAS 1.1](#)), under the supervision of personnel with ATP as an ECP.

- Personnel employed in a service specific programme<sup>2</sup> to gain ATP as a CCP may provide treatments from within the CCP list (see [CPG EAS 1.1](#)) under the supervision of personnel with ATP as a CCP.
- Service specific programmes:
  - <sup>1</sup> The ECP internship programme in Hato Hone St John.
  - <sup>1</sup> ECP training in Wellington Free Ambulance.
  - <sup>2</sup> The CCP internship programme in Hato Hone St John, HEMS Clinical Limited, Northern Rescue Helicopter Limited and Search and Rescue Services Limited.
  - <sup>2</sup> CCP training in Wellington Free Ambulance.

## Additional information

- Providing treatment under supervision does not change a practitioner's scope of practice nor change any conditions set by Te Kaunihera Manapou Paramedic Council.
- Student doctors, student nurses and New Zealand Defence Force medics undertaking clinical placements and/or electives are required to have a document with them describing what treatments they may provide under supervision.
- Personnel with specialist endorsement as an ICP, who do not have ATP as an ICP, may not provide treatments from the ICP list (see [CPG EAS 1.1](#)) under the supervision of personnel with an ATP as an ICP or CCP, unless there is a separate written agreement with their employer to do so.
- Personnel enrolled in the PGDipHSc (ECP), and personnel with specialist endorsement as an ECP, who do not have ATP as an ECP, may not provide treatments from the ECP list (see [CPG EAS 1.1](#)) under the supervision of personnel with ATP as an ECP, unless they have a written agreement with their employer to do so.
- Personnel enrolled in the PGDipHSc (CCP), and personnel with specialist endorsement as a CCP, who do not have ATP as a CCP, may not provide treatments from the CCP list (see [CPG EAS 1.1](#)) under the supervision of a personnel with ATP as a CCP, unless they have a written agreement with their employer to do so.

## When treatment is being provided under supervision

- The person must have been taught how to provide the treatment, and
- The supervising person must have the treatment within their own authorised medicines and interventions, and
- The supervising person has overall responsibility for safe clinical oversight of the treatment, and the person providing the treatment is responsible for practising within their scope of practice, within their competence, and within any conditions set by Te Kaunihera Manapou Paramedic Council, and

- The supervising person must provide direct supervision in such a way that they can immediately intervene if required, and
- The patient (if they have capacity) should be asked to consent to treatment being provided by a student.

## Audit

- The PRF must be sent for audit whenever treatment is provided that is not described within these CPGs, except when treatment is provided under supervision.
- The person providing the treatment is responsible for ensuring the PRF is sent for audit.

## More Information

### Version history

16 June 2026

1.1.0.1

**Update:** Guidance on treatment under supervision updated.

22 November 2022

1.0.1

**Update:** Multiple changes.

# Patient Capacity

This guideline contains information on patient capacity. It should be read in conjunction with the guidelines on [‘treatment and/or transport without informed consent’](#), [‘treatment and referral decisions’](#), and [‘informed consent’](#).

## General information

- The term capacity is used to describe the ability of the patient to make an informed choice and give informed consent regarding treatment and/or transport. The term competency can also be used and in general these terms are interchangeable. Use of the term capacity aligns language with police and mental health personnel.
- An assessment of capacity should occur if the patient is declining treatment and/or transport, particularly when this is associated with a significant risk of serious harm.
- When assessing capacity:
  - The patient is presumed to have capacity unless there are reasonable grounds for believing they do not.
  - The implications of the patient's decision cannot be used to determine capacity. For example, making a decision that most reasonable people would consider unwise does not, by itself, indicate a lack of capacity.
  - Being intoxicated may impair memory and/or understanding, but by itself does not prevent the patient from having capacity.
  - Suicidal intent is not taken into account when assessing capacity.
- Have a low threshold for seeking clinical advice if the situation is complex, difficult to resolve, or there is a significant risk of serious harm to the patient without treatment and/or transport.

## To assess capacity

- Explain the following points clearly:
  - Their clinical condition using language they can understand, and
  - The recommendations for treatment and/or transport, and
  - The benefits of accepting the recommendations, and
  - The risks involved in declining the recommendations.
- Ask if they have any questions.
- After 2-3 minutes, ask them to explain back to you their clinical condition, their decision, and confirm that they understand the risks involved in making that decision.

# Patient Capacity

- The patient has capacity if they demonstrate they can:
  - Understand information and the decision they are making, and
  - Understand and consider the consequences of their decision, and
  - Remember information to the extent needed to make a decision, and
  - Communicate their decision in any way.
- The patient does not have capacity if they clearly do not meet all of the above criteria.
- The patient has diminished capacity if they understand some but not all of the information provided, or some but not all of the consequences of declining recommendations. For example, a patient who can make simple decisions about medicines but struggles to understand and consider treatment options for a life-threatening condition.
- The patient should be presumed to have diminished capacity if they have severely impaired vital signs, for example severe acute hypotension.
- The patient has fluctuating capacity if their ability to understand or remember information changes over time, with periods of time when they have capacity and periods of time when they do not. For example, a patient who has severe acute hypoxia and does not have capacity, but improves with oxygen administration and then has capacity.

## If the patient has capacity

- Treatment and/or transport cannot be provided without their informed consent, except when:
  - They have attempted suicide, and treatment and/or transport is indicated for a life-threatening condition, or
  - Treatment and/or transport is indicated to prevent suicide, or
  - They have been sectioned under the Mental Health Act and treatment and/or transport is occurring under the instructions of a Duly Authorised Officer.
- In all other circumstances, the patient may decline recommendations for treatment and/or transport, even if doing so risks serious harm, including death.
- In situations required by law, notification to an appropriate agency must occur and the patient cannot decline this. See the [‘vulnerable persons’](#) guideline.

## If the patient does not have capacity

- If a person legally able to provide informed consent on behalf of the patient is contactable, this person may make decisions on behalf of the patient.

# Patient Capacity

- In all other circumstances, treatment and/or transport may occur if personnel believe this is clearly in the best interest of the patient. When making these decisions take into account:
  - » The known previously expressed views of the patient, and
  - » Any information within an advance directive, and
  - » The risks and benefits of providing treatment and/or transport versus not doing so, and
  - » The views, if known, of family/whānau, caregivers and healthcare providers.
- When providing treatment and/or transport without informed consent:
  - » Encourage the patient to accept recommendations, and
  - » Involve the patient's family/whānau, caregivers, and/or healthcare providers when appropriate.

## If the patient has diminished or fluctuating capacity

- Allow the patient to provide informed consent and make decisions within the level of their capacity. For example, a patient with diminished capacity may decline non-essential medications such as paracetamol, but cannot decline treatment for a potentially life-threatening condition, unless this is in their best interest.
- Balance the risks and benefits of treatment and/or transport versus not doing so. Take into account:
  - » The degree of impairment of capacity, and
  - » The threat to life from the underlying condition, and
  - » The known previously expressed views of the patient, and
  - » Any information within an advance directive, and
  - » The views, if known, of family/whānau, caregivers and healthcare providers.
- If a person legally able to provide consent on behalf of the patient is contactable, make treatment and/or transport decisions based on the above considerations, in conjunction with this person.
- When providing treatment and/or transport without informed consent:
  - » Encourage the patient to accept recommendations, and
  - » Involve the patient's family/whānau, caregivers and/or healthcare providers when appropriate.

## If the patient is a child

- If the child is aged 16 years or older they are treated as an adult.
- If the child is aged under 16 years:



# Patient Capacity

- » They should be treated as having diminished capacity, considering their level of maturity. For example, a child aged nine years may decline analgesia but cannot decline treatment for a serious condition.
- » If they are old enough to be involved in the discussion, ascertain and consider their views, considering their level of maturity and their understanding of the issues.
- » Involve a parent or guardian when possible and appropriate.

## Documentation of capacity

- Document the patient's ability to demonstrate they can:
  - » Understand the information provided and the decision they need to make.
  - » Understand and consider the consequences of their decisions.
  - » Remember the information provided.
  - » Communicate their decisions.
- For example:
  - » We explained to the patient that they had xx condition.
  - » We explained that the risks of not being transported to hospital were xx.
  - » When asked to describe their understanding, the patient was able to describe the information we had provided and the risk of xx if they were not transported.
  - » They were able to remember information, and they communicated with us clearly about their decisions.

## More Information

### Version history

|                                                           |       |
|-----------------------------------------------------------|-------|
| <b>26 November 2025</b>                                   | 1.0.7 |
| <b>New:</b> New guideline replacing 'Patient Competency'. |       |
| <b>22 November 2022</b>                                   | 1.0.1 |
| <b>Update:</b> Multiple changes.                          |       |

# Informed Consent

## Introduction

- Informed consent is an interactive process involving communication between personnel and a patient, during which the patient gains an understanding of their condition and makes an informed choice regarding their treatment and/or transport options.
- Personnel have a statutory obligation to abide by the Code of Health and Disability Services Consumers' Rights. One of the rights is to make an informed choice and to give informed consent, noting that a patient who has capacity to make informed healthcare decisions has the right to refuse or withdraw consent at any time.
- When a patient does not have capacity to make informed healthcare decisions, it may be appropriate to provide treatment without informed consent, when this is in the best interest of the patient. Additional information is contained within the relevant guidelines titled '[patient capacity](#)' and '[treatment and referral decisions](#)'.
- Assessing capacity and making decisions regarding treatment, referral and/or transport can be complex. Seek clinical advice if the situation is difficult to resolve.

## Obtaining informed consent

- Personnel must obtain informed consent whenever it is feasible to do so.
- When obtaining informed consent, personnel must:
  - » Fully inform the patient by providing an explanation (using non-clinical language) including:
    - » The nature of their condition.
    - » The recommendations being made.
    - » The reasons for the recommendations.
    - » The benefits and risks of the proposed treatments and/or transport, including the benefits and risks of any alternative courses of action.
    - » Introducing the people who will be providing treatment. The patient must be explicitly informed if a student is present or will perform any interventions and be given the opportunity to decline the student's involvement.
- Consider the patient's capacity to make informed decisions.
- Allow the patient and their family/whānau to ask questions.
- Fully answer the patient's questions, and family/whānau questions as appropriate.

# Informed Consent

## Providing information

- Good communication is the key factor in obtaining informed consent. It is important to take the time required to ensure the patient understands the issues as much as possible.
- Utilise a translator if appropriate.
- Clinical judgement is required when providing information to a patient:
  - It is not feasible to provide all of the information relating to each condition.
  - The patient should receive the information that another patient in the same setting would reasonably expect to receive.
  - The clinical setting must be considered. For example, detailed discussion is inappropriate if a patient has an immediately life-threatening problem or is in severe pain. In such settings it is appropriate to initiate treatment while explaining the treatment that is being provided.

## Documentation

- Personnel are not required to routinely document that informed consent was obtained. However, personnel must document when a patient declines to accept information or to give consent.

## More Information

### Version history

|                                                                                               |       |
|-----------------------------------------------------------------------------------------------|-------|
| <b>26 November 2025</b>                                                                       | 1.0.7 |
| <b>Update:</b> Minor changes to reflect the change from patient 'competency' to 'capacity'.   |       |
| <b>22 November 2022</b>                                                                       | 1.0.1 |
| <b>Update:</b> Introduction and added reference to family/whānau being able to ask questions. |       |

# Treatment and Referral Decisions

- Whenever personnel are assessing a patient, the following decisions must be made:
  - Is treatment required?
  - Is referral to a medical facility required?
  - If referral to a medical facility is required, what type of medical facility is most appropriate?
  - If referral to a medical facility is required, what mode of transport is most appropriate?

## Obligations of personnel

- Personnel must convey these decisions to the patient as clear recommendations. When making decisions and conveying recommendations personnel must:
  - Fully assess the patient including taking a history, performing a primary and secondary survey, measuring appropriate vital signs, and consider the early warning score (EWS) when appropriate.
  - Consider all available information, including from family/whānau or caregivers, and non-clinical information such as social factors.
  - Fully assess the patient's capacity to provide informed consent. See the '[patient capacity](#)' guideline.
  - Obtain informed consent whenever it is feasible to do so, by fully informing the patient regarding their condition, the recommendations being made to them, the reasons for the recommendations and the benefits and risks of the proposed treatments and any alternative courses of action.
  - Act in the patient's best interest if the patient does not have capacity.
  - Insist on treatment and/or transport if it is clearly in the best interest of a patient who does not have capacity. See the '[treatment and/or transport without informed consent](#)' guideline.
  - Fully document the assessment, interventions and recommendations.
  - Seek clinical advice if the situation is difficult to resolve.

## Deciding if the patient requires referral to a medical facility

- Not all patients assessed by ambulance personnel require referral to a medical facility. It is appropriate for a patient with minor illness or injury to be managed in the community, provided:
  - The obligations previously outlined have all been followed, and
  - The patient receives appropriate verbal and written advice on what to do if they do not improve, including when to seek further help or advice, and
  - The non-transport pause and checklist is used, and
  - Appropriate documentation is completed.

# Treatment and Referral Decisions

## Non-transport pause and checklist

- If a patient is being given a recommendation by ambulance personnel that transport to a medical facility by ambulance is not required, the crew must pause briefly to go through the non-transport checklist (below) and agree that non-transport is the right decision. If consensus is unable to be easily achieved, personnel should have a low threshold for recommending the patient is transported.
- The following non-transport checklist must be completed prior to leaving the scene:
  - The patient has been fully assessed including a set of vital signs and appropriate investigations, and
  - There are no unexplained abnormal vital signs, and
  - The EWS has been considered if EWS greater than or equal to five, or there is a vital sign in the red or blue zone, and
  - Serious illness or injury has been reasonably excluded, and
  - No red flags requiring transport to an ED are present, and
  - The patient is seen to mobilise safely (when able to normally do so), noting that if the patient is unable to mobilise there must be a minor or long-standing condition preventing this, and
  - The patient and/or caregivers have been given a verbal and written explanation of when to seek further help or advice.

## Deciding where the patient should be referred

- If a patient is being referred to a medical facility, referral should be to the most appropriate medical facility considering:
  - The patient's expected healthcare requirements, including investigation and treatment, and
  - The most effective and efficient way of meeting those requirements, and
  - Specific referral and transport guidance contained within the relevant guideline.
- The patient may not require referral to an ED. It is preferable to refer the patient to a primary care facility, provided that:
  - The patient's healthcare requirements can be reasonably met at that facility, and
  - It is reasonable and practical to refer the patient to that facility.

## The mode of transport

- Not all patients requiring transport to a medical facility require transport in an ambulance. It is appropriate to recommend alternative transport provided all of the following criteria are met:
  - The patient is very unlikely to require treatment or intervention during transport, and
  - The referral and transport guidance within each guideline are followed, and

# Treatment and Referral Decisions

- » A reasonable and appropriate alternative form of transport is available, and
- » Personnel are reasonably assured the patient and family/whānau will comply with transport arrangements, and
- » Advice is provided to the patient's family/whānau on what to do if the patient deteriorates en route.

## When the patient or family/whānau/caregivers insist on transport by ambulance

- Family/whānau and caregivers have a role in providing information on the known views of the patient. However, they cannot insist on treatment and/or transport that is not clinically indicated.
- If the insistence appears to be based upon genuine concern and no other reasonable transport option is available, then the patient should be transported by ambulance. If the insistence appears to be based on maliciousness, convenience or petty concerns, then personnel may decline to transport the patient provided they:
  - » Explain the reasons for not providing transport, and
  - » Fully document the circumstances.
- Seek clinical advice if the situation is difficult to resolve.

## When a registered health professional insists on transport by ambulance

- A registered health professional might insist on transport by ambulance that personnel do not assess as being clinically indicated.
- Personnel should try to resolve this by achieving consensus via collegial discussion, taking into account that the registered health professional may know the patient well. However, if consensus cannot be achieved, personnel should follow the principles contained under the previous heading.

## The deterioration early warning score (DEWS)

- The DEWS is a standardised scoring system that is being progressively introduced into aged residential care facilities, to help identify clinical severity and deterioration, using a color-coded system (yellow, orange, red and blue) for risk levels.
- The DEWS uses an aggregate score based on vital signs and other patient factors such as changes in cognition, oral intake and mobility.
- A DEWS of:
  - » 1-9 indicates the patient may have a serious condition.
  - » 10-15 indicates the patient is likely to have a serious condition.

# Treatment and Referral Decisions

- » 16-20 or any factor in the red zone, indicates the patient may have a life threatening condition.
- » Greater than 20 or any item in the blue zone, indicates the patient is likely to have a life threatening condition.
- There is no threshold at which a specific DEWS mandates treatment or transport. The DEWS should be considered, along with all other information, including any information within an advance directive, when making treatment and transport decisions.

## More Information

### Version history

|                                                                                        |         |
|----------------------------------------------------------------------------------------|---------|
| <b>16 June 2026</b>                                                                    | 1.1.0.1 |
| <b>Update:</b> Non-transport decision-making and Early Warning Score guidance updated. |         |
| <b>26 November 2025</b>                                                                | 1.0.7   |
| <b>Update:</b> Multiple changes                                                        |         |
| <b>22 November 2022</b>                                                                | 1.0.1   |
| <b>Update:</b> Removed duplicate content and general updates.                          |         |

This guideline is for the treatment and/or transport of patients without their informed consent. It should be read in conjunction with the guidelines on [‘patient capacity’](#), [‘informed consent’](#), and [‘treatment and referral decisions’](#).

## If the patient has capacity

- Treatment and/or transport cannot be provided without their informed consent, except when:
  - They have attempted suicide, and treatment and/or transport is indicated for a life-threatening condition, or
  - Treatment and/or transport is indicated to prevent suicide, or
  - They have been sectioned under the Mental Health Act and treatment and/or transport is occurring under the instructions of a Duly Authorised Officer.
- In all other circumstances, the patient may decline recommendations for treatment and/or transport, even if doing so risks serious harm, including death.
- In situations required by law, notification to an appropriate agency must occur and the patient cannot decline this. See the [‘vulnerable persons’](#) guideline.

## If the patient does not have capacity

- If a person legally able to provide informed consent on behalf of the patient is contactable, this person may make decisions on behalf of the patient.
- In all other circumstances, treatment and/or transport may occur if personnel believe this is clearly in the best interest of the patient. When making these decisions take into account:
  - The known previously expressed views of the patient, and
  - Any information within an advance directive, and
  - The risks and benefits of providing treatment and/or transport versus not doing so, and
  - The views, if known, of family/whānau, caregivers and healthcare providers.
- When providing treatment and/or transport without informed consent:
  - Encourage the patient to accept recommendations, and
  - Involve the patient’s family/whānau, caregivers, and/or healthcare providers when appropriate.

## If the patient has diminished or fluctuating capacity



- Allow the patient to provide informed consent and make decisions within the level of their capacity. For example, a patient with diminished capacity may decline non-essential medications such as paracetamol, but cannot decline treatment for a potentially life-threatening condition, unless this is in their best interest.
- Balance the risks and benefits of treatment and/or transport versus not doing so, taking into account:
  - The degree of impairment of capacity, and
  - The threat to life from the underlying condition, and
  - The known previously expressed views of the patient, and
  - Any information within an advance directive, and
  - The views, if known, of family/whānau, caregivers and healthcare providers.
- If a person legally able to provide consent on behalf of the patient is contactable, make treatment and/or transport decisions based on the above considerations, in conjunction with this person.
- When providing treatment and/or transport without informed consent:
  - Encourage the patient to accept recommendations, and
  - Involve the patient's family/whānau, caregivers and/or healthcare providers when appropriate.

## The role of family/whānau, caregivers, and healthcare providers

- Family/whānau, caregivers, and healthcare providers have a role in providing information on the known views of the patient. However, they cannot:
  - Provide informed consent on behalf of the patient, unless they have the legal ability to do so.
  - Insist on treatment and/or transport that is not clinically indicated.

## The role of people legally able to provide informed consent on behalf of the patient

- A person may legally provide informed consent on behalf of a patient, if that person:
  - Is the parent or guardian of a child aged under 16 years, or
  - Holds an active enduring power of attorney (EPOA) for personal care and welfare, or
  - Is legally appointed as a welfare guardian.
- A person legally able to provide informed consent on behalf of the patient may make treatment and/or transport decisions on behalf of the patient.
  - This includes declining the recommendations of personnel.
  - This does not include the ability to insist on treatment and/or transport that is not clinically indicated.

- » Seek clinical advice if the person appears to be making a decision that is not in the patient's best interest.
- For an EPOA to be active, a registered medical practitioner must determine the patient does not have capacity, and a lawyer must activate the EPOA.
- It is common for a person to be nominated to hold an EPOA, but for the EPOA to not yet be active. In this setting the person nominated is not legally able to provide informed consent on behalf of the patient, but their views (if known) should be taken into account.

## If the patient is a child

- If the child is aged 16 years or older they are treated as an adult.
- If the child is aged under 16 years:
  - » They should be treated as having diminished capacity, considering their level of maturity. For example, a child aged nine years may decline analgesia but cannot decline treatment for a serious condition.
  - » If they are old enough to be involved in the discussion, ascertain and consider their views, considering their level of maturity and their understanding of the issues.
  - » Involve a parent or guardian when possible and appropriate.
- If a parent or guardian is not present:
  - » Attempt to contact a parent or guardian to discuss informed consent.
  - » If Oranga Tamariki have guardianship, the duty social worker is the point of contact on 0508 326459.
  - » If immediate treatment and/or transport is required for a serious clinical condition, this should be provided and an attempt made to contact a parent or guardian as soon as it is feasible to do so.
- Parents and/or guardians may provide informed consent on behalf of a child.
  - » This includes declining the recommendations of personnel.
  - » Seek clinical advice if the parent or guardian appears to be making a decision that is not in the child's best interest.

## If the patient requires treatment and/or transport and is physically resisting

- Ensure scene safety.
- Assess and manage the risks using the TENR tool.
- Administer sedation only if required to ensure the safety of the patient or personnel. See the '[acute behavioural disturbance](#)' guideline.

- Provide restraint only if required to ensure the safety of the patient or personnel. See the [‘acute behavioural disturbance’](#) guideline.
- Call for assistance from police if there is a significant risk of physical harm to personnel, the patient, or bystanders.

## The TENR tool

- T** Assess the **threat** of physical harm. For example, consider:
  - » Intent.
  - » Capability.
  - » Opportunity.
  - » The physical environment.
- E** Assess and mitigate the **exposure** to physical harm. For example, consider:
  - » Safety and a safety plan.
  - » What resources are required.
  - » The consequences of the safety plan failing.
- N** Assess the **necessity** to act. For example:
  - » Consider the urgency of the need for treatment and/or transport.
  - » Throughout the course of the incident, continually reassess the necessity to act.
- R** Develop a **response** plan.
  - » The response must be proportionate and appropriate for the level of risk.
  - » The response must be in accordance with these CPGs and with the law.
  - » Throughout the course of the incident continually reassess the response plan.

## The role of police

- The primary role of police is to ensure the safety of ambulance personnel, the patient and bystanders.
- Police should be requested when:
  - There is a clear requirement to provide treatment and/or transport for a life-threatening condition, and
  - The patient is physically resisting treatment and/or transport, and
  - There is a significant risk of serious physical harm to ambulance personnel, the patient or bystanders.
- Police only have limited legal powers to restrain the patient if a crime has not been committed, or there is no immediate risk of serious physical harm, especially when the patient is on their own property.
- It is not usually the role of police to assist in physically restraining the patient, but this may be required until the patient has been safely sedated and/or restrained.
  - Once the patient is safely sedated and/or restrained, the TENR assessment should be repeated, and police may not be required to be present during transport.
- If police have been requested but are not attending:
  - Escalate the situation to an on duty ambulance manager.
  - Do not provide treatment and/or transport if it is unsafe to do so.
- If police are not attending or have left the scene, and it is unsafe to provide treatment and/or transport without police, it may be necessary as a last resort to leave the scene. In this situation:
  - Ensure comprehensive documentation, and
  - Implement a safety plan for the patient whenever it is feasible to do so.

## Seeking clinical advice

- Treatment and/or transport without consent is a complex and nuanced area of clinical care. The time taken to discuss and share decision-making in high-risk clinical situations is rarely time wasted.
- Have a low threshold for seeking clinical advice if the situation is complex, difficult to resolve, or there is a significant risk of serious harm to the patient without treatment and/or transport.

## Documentation

- Comprehensive documentation is always required, but is particularly important if:
  - The patient requires sedation and/or restraint, or

- » Treatment and/or transport is not occurring, and the patient is at significant risk of serious harm or death.
- » Ensure that the following points are covered:
  - » The assessment of the patient's capacity, and
  - » The assessment of the balance of risks and benefits when determining the need for treatment and/or transport without informed consent, and
  - » The threat to the patient's life, and
  - » The threat to the safety of ambulance personnel, and
  - » Any sedation and/or restraint provided, and
  - » Any clinical advice received, and
  - » Any requests for assistance, for example from police.

## More Information

### Version history

26 November 2025

1.0.7

**New:** Guideline released.

# Documentation

## General principles

- Documentation must be accurate and complete.
- Comprehensive documentation is particularly important when a patient is not transported to a medical facility.
- Documentation must be objective and subjective statements avoided. Take particular care to avoid subjective statements about the behaviour or demeanour of the patient, their family/whānau or other healthcare professionals.
- As a general rule, a third party (for example, the Health and Disability Commissioner) will assume that something is not recorded if it did not occur.
- A third party should be able to read the documentation and understand what happened and why.
- One person will usually lead the patient assessment and treatment and this person should usually complete the PRF. However, all clinical personnel in the crew are responsible for ensuring the documentation is accurate and complete.
- A separate PRF is required for each patient assessed. This includes for example, a separate PRF for mother and her newborn baby.
- The use of abbreviations should be minimised.
- Terminology specific to ambulance personnel (for example R codes) must not be used.

## Information documented in the PRF

- The PRF must include all of the following:
  - Patient details (including NHI if possible).
  - History of the presenting complaint.
  - Past medical and surgical history (if applicable).
  - Family and social history (if applicable).
  - Relevant prescribed medicines and known allergies.
  - Key positive and negative findings from examination.
  - At least one set of **'vital signs'**, including an early warning score (if applicable).

# Human Factors and Teamwork

## General principles

- Human factors, also known as crew resource management, encompass a set of principles focusing on the skills of teamwork, team leadership and communication.
- Good teamwork reduces the possibility that human factors and/or human error will result in harm, while helping ensure the patient receives timely and effective treatment.
- It is the responsibility of all personnel to utilise the principles of good teamwork irrespective of clinical hierarchy, organisational position, experience, or interpersonal challenges.
- Human factors and teamwork are most important during a complex or emergency incident, but the principles are important during all incidents.

## Initial actions and the pre-primary survey

Utilise the **STEP-UP** acronym:

- S** Self check.
- T** Team check and team brief.
- E** Environmental scan.
- P** Primary survey.
- U** Update the team.
- P** Plan and priorities.

## Principles of good teamwork

- Establish a team leader.
- Communicate effectively.
- Utilise resources effectively.
- Utilise pause points, reassess the situation frequently and keep the team updated.

# Human Factors and Teamwork

## Additional information

### The initial actions and the pre-primary survey

- The pre-primary survey, also known as the zero-point survey, is a structured approach to the initial actions when approaching the scene/patient and it overlaps with the primary survey.
- The self check helps ensure individuals are appropriately prepared. For example, are:
  - Wearing appropriate PPE.
  - Mentally ready. If the scene is likely to be stressful, performance may be improved with simple techniques such as controlled breathing, reviewing available resources (for example, the CPG app), and visualising and rehearsing tasks mentally.
- The team check and team brief helps to ensure the team is appropriately prepared. For example:
  - A team leader is identified.
  - Roles are allocated.
  - Team members' PPE is checked.
  - The team is briefed on what is known and what the initial priorities are likely to be.
- The environmental scan helps ensure the environment is optimised. For example:
  - Safety is ensured.
  - Space and light are optimised.
- Updating the team with the plan and the priorities optimises good teamwork.

### Principles of good teamwork

- Establish a team leader, who:
  - Is the most appropriate person and is clearly identified. This may not necessarily be the team member with the highest practice level.
  - Identifies the plan and the priorities.
  - Directs the actions of the team, including tasking specific people to specific tasks, utilising pause points, and ensuring the team is regularly updated on the plan and priorities. This helps ensure the team have the same shared mental model.
  - Avoids performing physical tasks unless absolutely necessary, as this risks the team leader becoming task-focused and losing situational awareness.
  - Seeks input from other team members as required.
  - Utilises resources effectively, for example:
    - Utilises all members of the team.



# Human Factors and Teamwork

- » Ensures team members perform tasks simultaneously whenever feasible.
- » Utilises bystanders and other healthcare providers as appropriate.
- All team members communicate effectively, for example:
  - » Utilise clear and concise language.
  - » Utilise closed loop communication. For example, if tasked to gain IV access, state when this is complete.
  - » Ensure important information is communicated directly to the team leader, and unnecessary communication is minimised.
  - » Utilise a flattened hierarchy that enables all team members to feel comfortable raising a concern, without fear that they will be criticised.
  - » Utilise a graded approach to escalation of concerns.

## Graded escalation of concerns

- Even in the presence of a flattened hierarchy, it is important to take a graded approach to escalation of concerns. This ensures the best outcome for the patient and that the team continues to function effectively.
- Use the 'PACE' acronym (probe, alert, challenge, emergency) to escalate a concern. Move to the next step if your concern is not resolved and move straight to the last step if there is an immediate risk of severe harm.
- Below is an example of PACE used during an RSI:
  - P** Probe: "Are you sure you are happy with the saturations of the patient?"
  - A** Alert: "The patient's saturations are 90% and decreasing"
  - C** Challenge: "We should abandon this attempt and reoxygenate"
  - E** Emergency: "STOP the intubation attempt, we need to reoxygenate the patient".
- Seek clinical advice if the situation is difficult to resolve.

## Interaction with other healthcare personnel

- Other healthcare personnel are often at the scene and may be able to provide assistance.
- Personnel should utilise such assistance if appropriate, including allowing access to medicines and/or equipment, provided other healthcare personnel:

# Human Factors and Teamwork

- » Identify themselves (verbally is acceptable and documentation is not routinely required), and
  - » Appear to be following good clinical practice.
- The name of other healthcare personnel treating the patient must be recorded on the PRF.
  - If other healthcare personnel appear to be following poor clinical practice, ambulance personnel must politely, but firmly, decline their assistance and document this on the PRF.

## More Information

### Version history

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**16 June 2026**

1.1.0.1

**Update:** Introduced PACE communication framework.

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**22 November 2022**

1.0.1

**New:** Guideline released.

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# Handover

Use the **IMIST AMBO** acronym when handing over a patient to another healthcare provider:

- I** Identification of the patient.
- M** Mechanism of injury or the medical complaint.
- I** Injuries identified or information related to the medical complaint.
- S** Signs and symptoms.
- T** Treatment provided and trends.
- A** Allergies.
- M** Medicines.
- B** Background, including previous medical history.
- O** Other, including information on the patient's social situation.

## Additional information

- Handover is a very important part of the patient's care. Loss of important information during handover is a common contributor to adverse incidents and patient harm.
- Prior to handover, review the details so that they can be delivered in a proficient and succinct manner. It is important to determine what information to include because irrelevant information risks the possibility of important information not being noted.
- When handing over to a team, inform the team leader if there is a problem with the primary survey and ask if they want the handover before or after moving the patient from the stretcher. The preference is to provide a handover before moving the patient from the stretcher, as this helps focus the team members on listening to the handover.
- Aim to deliver the handover in 30-60 seconds. Pause at the end of the handover and ask if there are any questions.
- Complete the PRF and ask if there are any further questions before leaving.

## More Information

# Handover

## Version history

|                                                                     |       |
|---------------------------------------------------------------------|-------|
| <b>22 November 2022</b>                                             | 1.0.1 |
| <b>Update:</b> ○ in IMIST AMBO and additional information expanded. |       |

# Status Codes

- Status codes are:
  - » A numerical estimate of the severity of a patient's clinical condition.
  - » Qualitative and require clinical judgement.
  - » Allocated to a patient after considering the nature of the illness or injuries, the vital signs and the potential threat to life.
  - » Not directly altered by the mechanism of injury, the physical environment (for example, trapped or not trapped) or the patient's age.
- Status codes are not ideal for pregnant women as the code cannot always be used to describe the potential threat to life for the unborn baby and the mother. Consideration should be given to adjusting the status code of the mother to take into account the threat to life for the unborn baby if the baby is at risk.

| Status | Triage priority | Triage tag colour |
|--------|-----------------|-------------------|
| Zero   | Dead            | Black/white       |
| One    | Immediate       | Red               |
| Two    | Urgent          | Orange/yellow     |
| Three  | Delayed         | Green             |
| Four   | Delayed         | Green             |

## Examples

The examples below are indicative only and are not a complete list.

# Status Codes

## Status one

- Obstructed airway or airway needing intervention to prevent obstruction.
- Severe stridor.
- Severe respiratory distress.
- Severe shock.
- Multi-system major trauma.
- Cardiac arrest or post cardiac arrest.
- Cardiogenic shock.
- ST elevation myocardial infarction.
- Dysrhythmia causing severe cardiovascular compromise.
- GCS of less than or equal to 9.

## Status two

- Moderate stridor.
- Moderate respiratory distress.
- Moderate shock.
- Two or more fractures (including closed fractures) involving the shaft of the femur, the tibia or the humerus.
- Fractures or dislocations with clear signs of limb ischaemia. Note: there may be abnormal sensation or movement distal to the injury but there must be signs of limb ischaemia for the patient to be status two.
- Isolated spinal cord injury.
- Dysrhythmia causing moderate cardiovascular compromise.
- Myocardial ischaemia with clinically significant symptoms, or signs on 12 lead ECG, which persist following treatment with nitrates. Note: the patient is status two if opiates are administered for the pain of myocardial ischaemia.
- GCS of 10-13.

# Status Codes

## Status three

- Mild stridor.
- Mild respiratory distress.
- Dysrhythmia causing mild cardiovascular compromise.
- Myocardial ischaemia with symptoms, or signs on 12 lead ECG, that have resolved following treatment with nitrates. Note: the patient is status two if the signs or symptoms return and further administration of nitrates is required.
- Isolated fracture of one long bone. This includes the shaft of the femur and compound fractures, provided there are no signs of limb ischaemia.
- Dislocations of joints without clear signs of distal limb ischaemia.
- Spinal pain without signs or symptoms of spinal cord injury.
- Loss of consciousness with normal or near normal (GCS 14 or 15) recovery.
- Transient ischaemic attack.

## Status four

- Isolated minor fractures.
- Strains and sprains.
- Lacerations where bleeding has been controlled.
- Fever without systemic signs of sepsis.
- Headache with normal neurological findings.

## More Information

### Version history

22 November 2022

1.0.1

**Update:** Status code examples and triage 'Urgency' to 'Priority'.

Task cards and other information is available [here](#)

- This is a summary of the initial management of a major incident. Further details are contained within ambulance service operational plans and procedures.
- Initial priorities include the following:
  - Declare a major incident.
  - Establish a clear command structure utilising a central incident control point.
  - Establish clear communication with the dispatcher through one single point.
  - Triage the patients.
  - Prioritise the patients for treatment.
  - Prioritise the patients for transport.

## Initial actions for the first ambulance crew at scene

- Each vehicle has a major incident folder within it.
  - Decide who will be performing the Operations Manager and Triage Officer roles.
  - Don an appropriate major incident vest.
  - Follow the steps in the appropriate task card.
- Place an initial sitrep using **METHANE** as soon as possible:
  - M** Major incident declaration.
  - E** Exact location of incident.
  - T** Type of incident.
  - H** Hazards (significant) identified.
  - A** Access and egress.
  - N** Number (estimated) of patients.
  - E** Emergency services already present and extra resources required.
- Perform primary (or initial) triage using status codes.
  - Begin by asking all patients that can walk to move to a specified safe area.
  - Apply the appropriate triage wristband.



- » During primary triage, interventions should usually be limited to opening airways (using positioning) and compressing life-threatening external bleeding. Whenever possible, other emergency service personnel and/or bystanders should be utilised to provide these.

- Provide an updated sitrep.

| Status | Triage priority | Triage tag colour |
|--------|-----------------|-------------------|
| Zero   | Dead            | Black/white       |
| One    | Immediate       | Red               |
| Two    | Urgent          | Orange/yellow     |
| Three  | Delayed         | Green             |
| Four   | Delayed         | Green             |

## Secondary triage

- Perform secondary triage, at the casualty collection point if one is set up, utilising a primary and secondary survey on all patients in order of their priority as determined by primary triage.
- Whenever feasible, allocate each patient a number using a patient information tag.
- Using the casualty tracking form, create a central record of all patients that includes the number allocated to each patient, their approximate age, a summary of their injuries and their status.
- If a patient's status changes:
  - » Replace the triage wristband with a new wristband, and
  - » Update the casualty tracking form, and
  - » Update the patient information tag.

## Treatment

- Initiate treatment on patients in order of priority as determined by their triage priority. Appropriate prioritisation of treatment optimises outcomes. The general principles are:
  - » Treatment decisions should be made by experienced personnel who task others to provide those treatments.
  - » Patients within each triage priority who have the greatest chance of survival, with the least expenditure of time, equipment and personnel, should be treated first.

- » Treatments that are highly unlikely to be successful (for example, CPR) should not usually be performed unless there are sufficient personnel and a good clinical reason to do so.
- » Treatments that take significant time (for example, rapid sequence intubation) should not usually be performed unless there are sufficient personnel and a good clinical reason to do so.
- » The greater the number of patients, the greater the importance of restricting the treatments to those that are immediately life-saving.

## Transport

- Task experienced personnel to coordinate transport.
- Distribute patients across hospitals/facilities in proportion to the capacity of those facilities, provided it is feasible to do so.
- Do not commence transport until informed of the destination and do not change the destination en route without seeking permission from the transport coordinator.
- Document all transport using the casualty tracking form, ensuring that the facility destination and transporting vehicle is recorded for each patient.
- Patients should be transported to hospitals/facilities capable of meeting their immediate resuscitation and treatment/intervention needs, provided it is feasible to do so.
- Family members who are also patients should be transported to the same hospital/facility, provided it is feasible to do so.
- Consideration should be given to concurrently transporting low priority patients in seated positions if this can be done safely.

## Patients considered almost certain to die

- It is impossible to predict which patients will die no matter what treatment they receive. However, in the setting of a major incident some patients have injuries that are so severe that death is highly likely. Examples include patients with:
  - » Respiratory arrest.
  - » Severe shock with a falling heart rate.
  - » A GCS of 3 with bilateral dilated and unreactive pupils.
- Patients considered almost certain to die may initially be triaged as red. However, if the number of severely injured patients is high, it is appropriate for patients considered almost certain to die to have treatment and transport initially withheld, allowing other immediate and urgent patients to be treated

and transported first. Patients considered almost certain to die should then be reassessed and further decisions made regarding treatment and transport.

## Additional Information

### HAZMAT incidents

- HAZMAT incidents are rare but should be considered in the following situations:
  - Multiple casualties with similar non-traumatic symptoms.
  - Unexplained symptoms in responding personnel such as chest pain, respiratory distress or headache. This is especially important if multiple patients have similar symptoms.
  - Road traffic crashes involving vehicles carrying hazardous substances.
- The STEP 123 system can be used to guide the initial approach:
  - 1 One casualty:** Approach using normal procedures.
  - 2 Two casualties:** Approach with caution, consider all options.
  - 3 Three or more casualties:** Do not approach.
- If a HAZMAT incident is suspected:
  - Stay clear of the area, upwind if possible.
  - Formalise a safe forward point.
  - Contact the dispatcher advising them of a possible HAZMAT situation, and request backup from FENZ.

## More Information

### Version history

16 June 2026

1.1.0.1

**Update:** Major incident management guidance updated.

22 November 2022

1.0.1

**Update:** Added primary triage and HAZMAT incidents. Removed triage flowchart.

# Requesting a Helicopter

## Making the request

- Phone the Air Desk whenever feasible.
- If a phone call is not feasible, make a radio call and include all the following information using the **RAS** acronym:
  - R** **Request:** the main reason a helicopter is being requested.
  - A** **Assessment:** a 1-2 sentence summary of the patient's clinical condition.
  - S** **Specific requirements:** describe any likely specific requirements. For example, RSI, winching, or the destination hospital is not the usual main hospital for the area.

## Request criteria

- A** **Access:** access is difficult, and a helicopter is the most appropriate means of extrication.
- N** **Number:** the number of patients at the scene exceeds the capacity of road resources. Personnel should inform the dispatcher of the number of patients requiring transport, leaving the dispatcher to arrange the most appropriate resources, which may include a helicopter/s.
- T** **Time:** the patient has a clinical condition with a time sensitive need for transport, and a helicopter will result in a clinically significant time saving in the patient arriving at the most appropriate hospital, and the time saving is expected to improve the patient's outcome.
- S** **Skill:** the patient has a clinical condition with a time sensitive need for specific skills, and a helicopter will result in a clinically significant time saving in appropriately skilled personnel reaching the patient, and the time saving is expected to improve the patient's outcome. Personnel should request the skills required, leaving the dispatcher to arrange for skilled personnel to provide backup, which may be by helicopter.

## Preparing for helicopter arrival

- Designate personnel to secure an appropriate landing site.
- Update the helicopter crew prior to their arrival with:
  - A description of the landing site and any obvious hazards, particularly wires.
  - An update on the patient's condition.

## Additional information

# Requesting a Helicopter

## The advantages and disadvantages of using a helicopter

- Deciding to use a helicopter requires clinical judgement that balances the advantages and disadvantages.
- Helicopters can reduce the time for skilled personnel to get to the patient and reduce the time for the patient to get to the most appropriate hospital.
- Transporting a patient by helicopter can compromise the ability to provide treatment to the patient, in comparison to the same level of personnel treating the patient in a road ambulance. This is because helicopters:
  - Have less space than a road ambulance.
  - Are noisy and this restricts communication, even with modern headsets.
  - Have reduced light at night in order to maintain night vision for the helicopter crew.

## Dispatch of a helicopter

- When a helicopter is requested, Air Desk personnel are required to determine if dispatch is appropriate, and if so to dispatch the most appropriate helicopter, which may not be the closest helicopter to the scene. The following factors are considered when dispatching a helicopter:
  - Availability of helicopters.
  - Composition and availability of clinical crew.
  - Clinical interventions required by the patient.
  - Specific helicopter requirements. For example, winching.
  - The hospital destination.
- To make the most appropriate dispatch decision, Air Desk personnel require as much information as possible from the scene. This is why a phone call is preferable to a radio call because it enables clinical discussion that cannot be achieved over the radio for example, local road conditions that may cause additional delays.
- When placing a radio call, it is important to include all the required information, as failure to do so will result in a delay while this is obtained.

## Time saving

- Clinical judgement is required when determining that time saving is clinically significant.
- Guidance on the required time saving for specific clinical conditions is described within the additional information section of relevant guidelines.

# Requesting a Helicopter

## Landing sites

- In many areas of New Zealand there are commonly used landing sites and these should be used whenever feasible.
- If it is not feasible to use a commonly used landing site, select a landing site that is:
  - Sufficiently removed from the scene to minimise the effects of downwash.
  - As large as possible. A minimum of 20 metres by 20 metres is preferred.
  - As flat and firm as possible. Less than five degrees of slope is preferred.
  - Free of sand or loose soil if possible.
  - Free of surrounding wires, poles and trees if possible.
  - Free of loose objects. Remove loose objects that may move with the helicopter downwash.
  - Free of long grass. Less than 30 cm is preferred.
  - Free of livestock.
- It is not necessary to routinely utilise FENZ personnel to secure a landing site, but they should be requested if they are required to help ensure the landing site is safe. For example, the landing site is on a public road.
- It is not necessary to mark the outer edges of the landing site. Do not use objects such as cones that may move in the helicopter downwash. Low intensity lights may be used to mark the landing site at night, but this is not routinely required.
- At night, do not use headlights to mark the landing site or shine lights toward the helicopter, both of which may impair the night vision of the helicopter crew.
- Turn vehicle beacons on as this helps the helicopter crew locate the site but note that a request may be made to have these turned off as the helicopter crew approach.
- Confirm whether a liaison radio channel will be used as the helicopter approaches the scene. When on final approach the pilot may ask for a 'sterile cockpit' (a term used to describe restricted communication within the helicopter) and this may prevent the helicopter crew from talking to ground personnel during this time.
- Be prepared for the possibility that the helicopter crew may choose an alternative landing site.

## More Information

### Version history

16 June 2026

1.1.0.1

# Requesting a Helicopter

**Update:** Helicopter dispatch guidance updated.

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**22 November 2022**

1.0.1

**Update:** Multiple changes.

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There are various types of plans or documents personnel must be aware of. These include:

- Advance directives.
- Advance care plans.
- Shared goals of care forms.
- Do not resuscitate or do not attempt resuscitation orders/requests.
- Allow natural death orders/requests.

## Advance directives

An advance directive is a written or oral instruction by which a patient indicates their choices regarding possible future treatments. An advance directive comes into effect when a patient no longer has capacity to make informed healthcare decisions or is unable to communicate their decisions. Most commonly, an advance directive is used to indicate that a patient does not want specific treatments, for example a patient may not want resuscitation in the event of cardiac arrest.

A patient cannot legally demand or refuse treatments in an advance directive which they cannot legally demand or refuse when they have capacity. For example, a patient cannot demand that a treatment is provided if the treatment is not clinically indicated, and a patient cannot refuse treatment for attempted suicide if the attempt is life-threatening. This is a complex area of clinical care. Seek clinical advice if a patient with a valid advance directive has attempted suicide.

## Advance care plans

Advance care plans are a means by which a patient indicates their wishes and choices regarding their dying process. Advance care plans often contain more detailed information than an advance directive regarding the patient's wishes.

## Shared goals of care

- Patients in hospital level care or aged residential care may have a 'shared goals of care' form, which documents a conversation between patient, family/whānau, and clinicians regarding shared goals of care and treatment. The shared goals of care form establishes treatment priorities into four categories:
  - Curative and/or restorative.
    - Treatment aims to prolong life.
    - All indicated treatments are appropriate.
  - Curative and/or restorative with emphasis on quality of life.



- » Treatment aims to prolong life and enhance its quality.
- » Resuscitation in the event of cardiac arrest may not be indicated.
- » May be appropriate for non-invasive CCP interventions.
- » Improving quality of life.
  - » Treatment aims to control symptoms and enhance wellbeing.
  - » Treatment in the community is preferred.
  - » Resuscitation in the event of cardiac arrest is not indicated.
  - » Limited interventions such as IV fluids or antibiotics may be appropriate.
- » Care during the last days of life.
  - » Treatment is limited to that which provides comfort and alleviates suffering.
- When this information is in an advance care form or clinical record, it can be used to guide the appropriateness of interventions and the decision to provide treatment.
- When this information is on a discharge summary or other non-official document, it should be considered in the context of the whole clinical situation and discussions that occur with family/whānau.

## Do not resuscitate orders/requests

Do not resuscitate (DNR) and do not attempt resuscitation (DNAR) orders are terms usually used to describe a medical decision that a patient should not be resuscitated in the event of cardiac arrest. Such decisions are commonly made in hospitals and aged residential care facilities if the patient is coming to the end of their natural life.

DNR and DNAR are imprecise terms when used in isolation. For example, there are many patients for whom some form of resuscitation (for example, treatment for choking or anaphylaxis) would be appropriate, but resuscitation in the event of cardiac arrest would be inappropriate. When treating a patient described as having a DNR or DNAR order, always clarify what the order means in terms of the treatments that are appropriate for that patient.

## Allow natural death orders/requests

Allow natural death (AND) is a term that is commonly promoted as a more useful and precise term than DNR or DNAR. An allow natural death request is usually made for a patient who is coming to the end of their natural life and in whom life-saving treatments should be withheld in the event of a life-threatening illness or injury. Treatments for relief of pain and suffering are never withheld and a patient with an allow natural death request should:

- Receive usual treatments (for example, IV pain relief for a fractured neck of femur) in the event of an illness or injury that is not immediately life-threatening.

- Not receive resuscitative treatments (for example, ventilation, CPR or 0.9% sodium chloride IV for shock) in the event of an illness or injury that is immediately life-threatening.

## Complying with directives and plans

When making decisions, ambulance personnel must consider all available information, including the information in an advance directive or an advance care plan. These are usually written, but a clearly described verbal advance directive or advance care plan must also be considered. Personnel must comply with the requests in an advance directive or advance care plan provided that:

- They are available, preferably in written form, and
- They apply to the current situation, and
- They are clear.

If there is uncertainty, resuscitation and/or treatment should commence when personnel believe it is in the best interest of the patient. At the same time personnel should urgently seek additional information from the family/whānau and if available, advice from a registered health practitioner who is directly involved in the patient's care.

## Medical information adjuncts

- Some patients have medical information adjuncts, such as a bracelet or tattoo, indicating that they do not wish to be resuscitated.
- Medical information adjuncts usually have no legal standing as a legitimate advance directive and usually contain insufficient detail to allow appropriate decisions to be made.
- Decisions regarding resuscitation must be based on all the available information, including the information contained within a medical information adjunct, noting that the presence of such information is likely to indicate that the patient has a firm view on what treatments they do not wish to receive.

## The End of Life Choice Act 2019

- The End of Life Choice Act 2019 gives patients who have a terminal illness and meet specific eligibility criteria, the legal right to request medicines for assisted dying.
- It is illegal for a registered healthcare professional to discuss the option of assisted dying with a patient unless the patient asks about it first. If a patient asks ambulance personnel about the option of assisted dying, personnel should recommend the patient discusses this with their usual healthcare provider.

- A person requesting assessment for eligibility to receive assisted dying must be assessed by two independent registered medical practitioners. Ambulance personnel have no role in this assessment process.
- Assisted dying must be directly overseen by a registered medical practitioner. Ambulance personnel have no role in this oversight process.
- Ambulance personnel may be called to a patient at the time of the assisted dying processes. If this occurs, the issues will be very complex and ambulance personnel must seek clinical advice.

## More Information

### Version history

|                                                                               |         |
|-------------------------------------------------------------------------------|---------|
| <b>16 June 2026</b>                                                           | 1.1.0.1 |
| <b>Update:</b> Added shared goals of care guidance.                           |         |
| <b>22 November 2022</b>                                                       | 1.0.1   |
| <b>Update:</b> Added End of Life Choice Act and medical information adjuncts. |         |

# Care During the Last Days of Life

This guideline is for management of symptoms in patients during the last days of their life.

- Locate and follow the patient's advance care plan if this is available.
- Contact healthcare personnel coordinating the patient's care and seek their advice if possible. Consider seeking support from an ECP if the patient's usual healthcare provider is unavailable.
- Offer the opportunity for a prayer/karakia (or equivalent) if appropriate.
- Administer medication to relieve symptoms related to their condition, noting that patients that are already taking opiates or other palliative medications may need higher doses than anticipated to achieve symptom control. There is no total maximum dose and the goal is to provide adequate symptom relief without intentionally hastening the patient's death.
- For pain:
  - Administer an opiate IV/IM/SC. Repeat as required.
  - Administer droperidol 1.25 mg IV or 2.5 mg IM/SC for pain that does not respond to an opiate and administer additional doses as required.
- For shortness of breath:
  - Administer midazolam 1-2 mg IV or 2.5-5 mg IM/SC and administer additional doses as required.
  - Administer an opiate IV/IM/SC for shortness of breath that does not respond to midazolam and administer additional doses as required.
- For agitation or anxiety:
  - Administer midazolam 1-2 mg IV or 2.5-5 mg IM/SC and administer additional doses as required.
  - Administer droperidol 1.25 mg IV or 2.5 mg IM/SC for agitation that does not respond to midazolam and administer additional doses as required.
- For nausea:
  - Administer droperidol 1.25 mg IV or 2.5 mg IM/SC and administer additional doses as required.
- Seek clinical advice if there will be a delay until primary care, palliative care or hospice staff are able to attend the patient and provide ongoing symptom management.

## Referral and transport

- Ambulance personnel may administer medicines and agree on a plan with the patient and/or family/whānau that transport to a medical facility is not required, ensuring consideration is given to the requirements for ongoing symptom control.

# Care During the Last Days of Life

- Whenever possible, follow the wishes of the patient regarding transport to a medical facility, considering the views of the family/whānau and the patient's healthcare providers. If transport is required, the preferred destination is a hospice provided this is arranged prior to arrival.

## Additional information

### General information

- Care during the last days of life is for patients who are recognised as dying and where death is likely to occur in the next days. This is not the same as palliative care, which can often be provided for many months before a patient dies.
- Seek clinical advice if there is any doubt whether care during the last days of life is appropriate.
- The most important aspects of care during the last days of life are to ensure that:
  - The patient is provided with adequate comfort and relief of symptoms, and
  - The patient and their family/whānau feel safe and supported, and
  - There is good communication with the patient, their family/whānau and the patient's healthcare providers.
- It is usually inappropriate to commence treatment for sepsis or hypotension.
- Most hospice and palliative care staff utilise the SC route for administration of medicines:
  - Administration of medicines via the SC route is usually effective and is the preferred option.
  - When administering medicines via the SC route, the doses administered should be the same as an IM dose.
  - An SC access line may be in place and personnel should follow administration of a medicine SC with a flush of 0.5-1 ml.
- Personnel may choose to administer medicines via the IV route provided IV access is easy to obtain.
- If IV access has been gained and the patient is not being transported, discuss the possibility of leaving the IV access in place with the patient and/or family/whānau, particularly if it is likely that additional IV medicines will be administered within 24 hours.

### Changes that indicate a person is in their last days of life

- Recognising the last days of life can be difficult because:
  - Signs and symptoms may be subtle and vary in intensity.
  - Life-threatening but reversible palliative care emergencies can mimic the last days of life.
  - There is usually a degree of uncertainty over the timing of a person's death.

# Care During the Last Days of Life

- » Sometimes the patient's condition unexpectedly improves.

## Signs and symptoms (not all need to be present)

|                                      |                                                                                                                        |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Speed of deterioration               | Becomes faster (hours/days) as death becomes more imminent.                                                            |
| Weakness                             | Reduced mobility, bed bound, profound weakness.                                                                        |
| Social interaction and communication | Withdrawn, reduced ability or unable to communicate. If able to communicate, may express a realisation they are dying. |
| Consciousness                        | Reduced level of consciousness, confused, delirious, may be agitated.                                                  |
| Appetite and swallowing              | Loss of appetite, not eating or drinking, unable to swallow, no longer taking oral medications.                        |
| Skin                                 | Pale, mottled or cyanosed lips, nose, ears, fingers. Cold extremities.                                                 |
| Breathing                            | Irregular breathing, Cheyne-Stokes, irregular, or agonal, periods of apnoea (awake or asleep).                         |
| Cardiovascular                       | Tachycardia, bradycardia, hypotension.                                                                                 |
| Secretions                           | Persistent noisy mouth, throat or bronchial secretions.                                                                |
| Toileting                            | New incontinence or diminished bowel/bladder function.                                                                 |

## Palliative care emergencies

- Unexpected deterioration due to a potentially reversible palliative care emergency can mimic the last days of life.
- Referral to an ED is appropriate if active treatment fits the patient's goals of care.
- Superior vena cava obstruction:
  - » Superior vena cava obstruction may occur due to compression or invasion by a tumour.
  - » Obstruction occurs over days to weeks and symptoms are usually worse when bending forward or lying flat. Symptoms may include:
    - » Upper body swelling/oedema, and/or
    - » Distension of upper body veins, and/or
    - » Cough, dyspnoea, or stridor.

# Care During the Last Days of Life

- Sudden onset of limb weakness. This may be a sign of metastatic spinal cord compression, and it may be appropriate for these patients to be transported to an ED.

## More Information

### Version history

16 June 2026

1.1.0.1

**Update:** Guideline redeveloped.

22 November 2022

1.0.1

**Update:** Treatment options for symptom relief, referral and transport.

# Verification of Death

Death may be verified when:

- ✓ There are clear and obvious signs of death such as decomposition, rigor mortis or decapitation, or
- ✓ All the clinical criteria below are met.

## Clinical criteria

- To verify death using clinical criteria there must be:
  - No signs of breathing for one minute, and
  - No palpable central pulse, and
  - No audible heart sounds, and
  - Pupils that are dilated and unreactive to light.
- After ten minutes, all the above examinations must be repeated and at this time the cardiac rhythm must be asystole. Document this as the time death has been verified.
- A patient may be dead but may not be in asystole at the second examination after ten minutes. For example:
  - There may be slow broad complexes. If this is the case, wait until asystole is present before verifying death.
  - A patient with a pacemaker may have electrical activity generated by the pacemaker for many hours after death. In this setting it is appropriate to verify death despite electrical activity on the ECG, provided all the other clinical criteria are met.

## Additional information

### General principles

- The process described in this guideline is for verifying that death has occurred. It is not to be used to determine whether resuscitation attempts are futile as these two decisions are very different.
- Death must be clear and unequivocal if the condition of the body is used to verify death.
- A clinical assessment confirming the absence of signs of life must occur if death is not clear and unequivocal.
- The entire chest and abdomen must be exposed when examining the patient for signs of breathing and this examination must occur over an uninterrupted period of one minute.
- The carotid or femoral site may be used when examining for a palpable central pulse in a patient aged over one year. In a patient aged up to one year, palpation of the brachial pulse is recommended. There



# Verification of Death

must be certainty that the pulse is not palpable and palpation for a minimum of ten seconds is required.

- Auscultation for heart sounds should occur over the expected site of the apex beat of the heart. In most patients this is over the 5<sup>th</sup> intercostal space in the midclavicular line. There must be certainty that heart sounds cannot be heard and listening for a minimum of ten seconds is required.
- The pupils must be dilated but no exact pupil size is specified. The pupils must be unreactive to light and this requires the use of a focal light source.
- The clinical assessment must be performed twice, with a minimum of ten minutes between the two assessments. The reason for this is that the patient may be in asystole for 5-10 minutes and then spontaneously develop return of a beating heart. This is sometimes called auto-resuscitation or the Lazarus phenomenon.

## Deaths that must be reported to the Coroner

- Deaths meeting any of the following criteria must be reported to the Coroner via police:
  - Due to suicide.
  - Due to trauma or violence.
  - Due to an unnatural cause. For example, drowning, poisoning or asphyxiation.
  - During or as a result of a complication of a surgical, dental or medical procedure.
  - During birth or as a result of a complication of pregnancy/birth.
  - In custody. For example, in prison or a police cell.
  - In compulsory care. For example, detained under the Mental Health Act.
  - In all other circumstances where a registered health practitioner confirms they are unable to complete a Medical Certificate of Cause of Death.

## After death has been verified

- Inform those present that the patient has died and provide support as required.
- Offer the opportunity for a prayer/karakia (or equivalent) if appropriate.
- Follow tikanga principles for Māori patients and whānau, whenever appropriate:
  - Allow whānau to accompany the tūpāpaku (deceased person).
  - Move the tūpāpaku feet first whenever feasible.
  - Avoid transporting the tūpāpaku via public, food or waste areas whenever feasible.
  - Avoid removing taonga (a treasured item, for example, a pounamu necklace), but if this is required seek consent before doing so and allow whānau the option of removing taonga.

# Verification of Death

- Request police if the death needs to be reported to the Coroner or if death has occurred in a public place.
- If death has occurred in a private place and does not need to be reported to the Coroner, contact a registered medical or nurse practitioner directly involved in the patient's care and ask if they are able to complete a Medical Certificate of Cause of Death:
  - If the medical or nurse practitioner confirms they can complete a Medical Certificate of Cause of Death, inform the family/whānau they can proceed with funeral arrangements by contacting a funeral director.
  - If the medical or nurse practitioner confirms they are unable to complete a Medical Certificate of Cause of Death, inform the family/whānau that the death will need to be reported to the Coroner and request police.
  - If the medical or nurse practitioner cannot be contacted and the death was not unexpected, inform the family that they will need to contact a funeral director who will liaise with the medical or nurse practitioner as soon as possible. In this setting it is acceptable for the body/tūpāpaku to remain in the house and a written summary of the clinical events (on the ACS) must be left with the family/whānau.
- Clinical equipment (for example, endotracheal tubes and IV lines) should usually be removed unless:
  - Doing so may disturb evidence at a crime scene, or
  - The clinical equipment could have contributed to death, or
  - Police request that clinical equipment is left in place.

## Documentation

- A verification of death form must be completed for every patient whose death is verified by ambulance personnel.
- If a verification of death form is completed, a PRF must also be completed. In addition, a photo of the completed form must be taken in the ePRF.
- If a patient is unable to be identified, a verification of death form can still be completed. In these instances, document descriptors of the patient whose death is being verified, for example their clothing, their sex, and approximate age.

## More Information

### Version history

16 June 2026

1.1.0.1

**Update:** Documentation guidance updated.

22 November 2022

1.0.1

# Verification of Death

**Update:** Tikanga information and general updates.

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# Oxygen Administration

- Few guidelines contain specific instructions on oxygen administration and clinical judgement is required.
- Oxygen should usually only be administered if the patient has an SpO<sub>2</sub> less than 92% on air, airway obstruction, or a specific indication as described within these CPGs.
- Use the simplest device and lowest flow rate required to achieve an SpO<sub>2</sub> of 92-96% (or 88-92% if at risk of hypercarbia) unless specific devices and/or flow rates are described in individual guidelines.
- If pulse oximetry is unreliable or unavailable, administer oxygen as appropriate for the patient's clinical condition.
- The oxygen flow rates usually used are:
  - Nasal prongs 1-4 litres/minute.
  - Simple mask 6 litres/minute.
  - Nebuliser mask 8 litres/minute.
  - Reservoir mask 10 litres/minute.
  - Manual ventilation bag 10 litres/minute.
- If a manual ventilation bag is being used to provide ventilation, use the following PEEP settings:
  - Adult or child:
    - Do not attach PEEP during cardiac arrest.
    - Apply PEEP at 5 cmH<sub>2</sub>O for all other conditions. CCPs may choose to increase PEEP if clinically indicated.
    - Refer to the '[cardiogenic pulmonary oedema](#)' guideline if the patient is an adult and has cardiogenic pulmonary oedema.
  - Neonate:
    - Apply PEEP at 5 cmH<sub>2</sub>O, including during cardiac arrest.

## Additional information

### Oxygen administration

- Oxygen is a treatment for hypoxia and not a general treatment for patients that are ill or injured.
- Oxygen administration is restricted to those patients that have an indication to receive it, because when oxygen levels within blood are higher than normal:
  - Blood vessels (particularly small arteries) vasoconstrict. This has the potential to lower blood flow to tissues and organs.
  - Inflammation is increased and this may worsen inflammatory states.

# Oxygen Administration

- Routine oxygen administration during neonatal resuscitation appears to make outcomes worse. This is why oxygen is reserved for deterioration despite initial ventilation.

## Delivery devices

- For most patients, nasal prongs or a simple mask is sufficient.
- Reservoir masks should be reserved for patients requiring a high concentration of inspired oxygen. For most patients 10 litres/minute is sufficient oxygen flow through a reservoir mask. The oxygen flow is sufficient if the reservoir bag is not completely deflating.
- Manual ventilation bags should be reserved for patients requiring assistance with their breathing or requiring PEEP. For most patients 10 litres/minute is sufficient oxygen flow through a manual ventilation bag. The oxygen flow is sufficient if the reservoir bag is not completely deflating.
- Ensure a high degree of vigilance if a manual ventilation bag and mask is being held against the face of a small child who is spontaneously breathing, as a small child with respiratory exhaustion may not generate sufficient pressure to open the valves within the manual ventilation bag, and this may result in asphyxiation.

## Pulse oximetry

- A pulse oximetry reading does not indicate how well a patient is breathing. How well a patient is breathing is determined by clinical examination and end tidal carbon dioxide (ETCO<sub>2</sub>) measurement if this is available.
- Pulse oximetry can be unreliable if the patient is very vasoconstricted, shaking, moving, has very dirty fingers, or has been exposed to toxins such as carbon monoxide (CO) or nitrites.
- Do not spend long periods of time trying to obtain a pulse oximetry reading, noting that a common cause of failure to get a reading is severe vasoconstriction.
- Low pulse oximetry readings may be invalid if the SpO<sub>2</sub> waveform on the monitor is damped, flat or irregular. A wide and regular waveform is a sign that the pulse oximetry reading is likely to be valid.

## Cyanosis

- A patient may be significantly hypoxic without being cyanosed because cyanosis is usually only detectable with an SpO<sub>2</sub> less than 80%.
- Cyanosis is much more difficult to detect in a patient who is anaemic, has a darker skin tone, or has been exposed to CO.

# Oxygen Administration

- Central cyanosis (for example, of the mouth and lips) is usually due to severe hypoxia. Peripheral cyanosis (for example, of the extremities) in the absence of central cyanosis is usually due to vasoconstriction.

## Oxygen administration and patients at risk of developing hypercarbia

- Some patients have carbon dioxide clearance that is altered by oxygen administration and excess oxygen administration in these patients may cause hypercarbia.
- Patients at high risk of hypercarbia include patients with COPD, morbid obesity, those on home oxygen and those on home CPAP or BiPAP.
- Oxygen flow rates should be titrated to the patient's known normal SpO<sub>2</sub> if this is known. If this is not known, titrate the oxygen flow rate to an SpO<sub>2</sub> of 88-92%.
- Bronchodilators should be nebulised using air if possible:
  - If using air to nebulise bronchodilators, administer bronchodilators using a flow rate of 8 litres/minute and administer oxygen via nasal prongs if required, adjusting the oxygen flow rate to achieve an SpO<sub>2</sub> of 88-92%.
  - If using oxygen to nebulise bronchodilators, alternating five minutes with the mask on and five minutes with the mask off should only occur if the SpO<sub>2</sub> climbs to above 92% during nebuliser delivery. This is done to limit oxygen exposure whilst delivering most of the nebulised bronchodilator.
- The signs of a rising carbon dioxide level are usually confusion, drowsiness, agitation and a falling level of consciousness. If a patient is suspected of developing hypercarbia, oxygen administration should not be discontinued immediately. Instead, oxygen administration should be reduced to a lower flow rate (targeting an SpO<sub>2</sub> of 88-92%) and the patient reassessed.

## Oxygen administration and smoke inhalation

- Oxygen administration is not routinely required following smoke inhalation and should be reserved for patients that have suspected CO poisoning.
- See the '[poisoning from gases](#)' guideline for more information.

## Oxygen administration following bleomycin treatment

- Oxygen administration can cause severe lung inflammation in patients previously treated with bleomycin (a chemotherapy drug) and should usually only be administered to maintain an SpO<sub>2</sub> of 88-92%.
- This sensitivity to oxygen is lifelong following exposure to bleomycin.

# Oxygen Administration

## Oxygen administration in confined spaces

- Use caution when administering oxygen in a confined space, for example, in a tank, pipe or silo.
- Oxygen administration in a confined space may lead to an increased oxygen concentration within the ambient air. An increase in the oxygen concentration within ambient air to as little as 24% may significantly increase the risk of fire and/or explosion.
- Only administer oxygen within a confined space if:
  - The clinical indication is very strong, and
  - FENZ personnel are present and are monitoring the oxygen concentration within the space.

## PEEP (positive end expiratory pressure)

- PEEP improves oxygenation and reduces work of breathing via the following physiological effects:
  - The expiratory pressure reduces lung collapse.
  - The positive pressure reduces venous return to the heart and blood flow through lung vessels, reducing the amount of fluid entering the lungs.
  - The expiratory pressure increases the functional residual capacity. From this more expanded resting position, less work is required for inspiration because of the non-linear compliance of the lungs.
- In cardiac arrest:
  - PEEP is not applied to adults and children during CPR because it reduces the blood flow achieved during CPR. If ROSC is achieved it is appropriate to apply PEEP, but this is not an immediate priority.
  - PEEP is applied to neonates during CPR because the cause of cardiac arrest is usually respiratory failure and the balance of risk is in favour of improving ventilation.
- PEEP reduces cardiac output and should be used with caution in patients showing signs of shock.

## More Information

### Version history

16 June 2026

1.1.0.1

**Update:** Added PEEP guidance.

22 November 2022

1.0.1

**Update:** Indications for oxygen, oxygen administration threshold and guideline simplified.

# Vital Signs

## Recording vital signs

### When vital signs must be recorded

- Vital signs must be recorded as part of patient assessment:
  - A set of vital signs should include as a minimum: respiratory rate, heart rate, blood pressure, capillary refill time, SpO<sub>2</sub>, GCS and temperature.
  - Personnel should have a lowered threshold for recording two sets of vital signs if any of the first set were abnormal, or at the outer limits of normal.
- Vital signs must be recorded when they are a prerequisite to providing treatment. For example, a blood pressure must be recorded before administering GTN.
- Vital signs must be recorded following treatment that has been initiated in response to abnormal vital signs.

### When vital signs do not have to be recorded

- Not all vital signs have to be recorded if the patient has a time critical problem and the results will not change the treatment that is provided. For example, a full set of vital signs is not required prior to adrenaline administration if a patient has anaphylaxis that is clearly immediately life-threatening.
- Vital signs do not need to be routinely recorded if the patient has a minor problem (for example, a minor digit laceration). However, if a patient appears to have more than minor peripheral injuries, a set of vital signs must be recorded before making a recommendation that transport to a medical facility is not required.
- Clinical judgement is required if the patient is receiving care during the last days of life. In this setting, vital signs are not a prerequisite for providing treatment and it is usually not appropriate to take recordings or perform examinations that will cause additional unnecessary discomfort.
- When vital signs are not recorded (or are unable to be recorded), the reason for this must be documented on the PRF.

### The frequency of vital sign recordings

- Clinical judgement is required when determining how frequently to record vital signs:
  - Vital signs should usually be recorded every 10-15 minutes for a patient that is status one or status two, noting that vital signs are not required if the patient has a time critical problem, and the result will not change the treatment that is provided.
  - Vital signs should usually be recorded every 20-30 minutes for a patient that is status three.



# Vital Signs

- Some vital signs are monitored continually (for example, heart rate via ECG leads) but are recorded at intervals. It is appropriate to record these if a significant change occurs, or other vital signs (such as blood pressure) are being recorded simultaneously.
- Consider recording a set of vital signs just prior to, or on arrival, at ED.

## Specific vital signs

- **Respiratory rate.** Tachypnoea is a subtle but important sign that a patient is unwell or injured. The respiratory rate must be counted and not estimated. The trend of the respiratory rate is more important than a single recording.
- **Heart rate.** Unexplained tachycardia is a subtle but important sign that a patient is unwell or injured. The trend of the heart rate is more important than a single recording. Baseline heart rate is usually reduced in young and/or fit adults, and tachycardia may not be present if the patient is taking a beta blocker.
- **Blood pressure.** Blood pressure alone is a poor indicator of the adequacy of cardiac output. Take note of the pulse pressure and the trend, noting that a narrowed pulse pressure is a sign of vasoconstriction, usually in response to reduced cardiac output. A standing and lying/sitting blood pressure should be measured if postural hypotension may have contributed to the patient's clinical condition.
- **Capillary refill time.** In the absence of hypothermia or significant peripheral vascular disease, a prolonged capillary refill time is a sign of vasoconstriction, usually in response to reduced cardiac output. The trend of the capillary refill time is more important than a single recording.
- **GCS.** Carefully determine each component noting that the motor score is the most important component of the GCS.
- **SpO<sub>2</sub>.** This measures how well a patient is oxygenated, but does not measure how well a patient is breathing (ventilating). How well a patient is breathing is determined by clinical examination and use of ETCO<sub>2</sub>. Failure of the pulse oximeter probe to record an SpO<sub>2</sub> is often an indication that the patient is vasoconstricted and poorly perfused.
- **Blood glucose.** The blood glucose concentration does not need to be routinely recorded in all patients. It should be recorded to rule out hypoglycaemia or hyperglycaemia, considering the overall clinical picture. For example, a blood glucose must be recorded in all:
  - Patients with diabetes.
  - Patients with an altered level of consciousness.
  - Patients who are unwell without an obvious cause.
  - Patients with signs of infection. In particular, children under five years of age are particularly prone to hypoglycaemia if they have severe infection.
  - Patients with poisoning where hypoglycaemic medicines may have been taken.
  - Patients with signs or symptoms of a stroke.

# Vital Signs

» Patients who have had a seizure.

➤ **Temperature.** Hyperthermia is most commonly due to infection, but normothermia does not rule out infection. There is no specific temperature that correlates well with severity of illness, however:

- » A temperature must be recorded if the patient has a clinical condition that does not involve trauma, and a recommendation is being made that transport to a medical facility is not required.
- » A temperature greater than 40°C is potentially life-threatening and should result in a patient being given a recommendation to be transported to a medical facility.
- » Hypothermia is an important clinical sign as it is often a sign of low cardiac output. A temperature below 35°C should usually result in a patient being given a recommendation to be transported to a medical facility.
- » The measured temperature can differ between sites at the same time, for example the temperature at the tympanic site may differ from the temperature at the axillary site.
- » Tympanic thermometers may be unreliable in small children if the probe will not adequately fit their external auditory meatus. An axillary measurement is preferred if a child is under six months of age.

➤ **ETCO<sub>2</sub>.**

- » Must be measured continually via capnography and regularly recorded if the patient has been intubated with an endotracheal tube (ETT).
- » Should be measured when a supraglottic airway (SGA) is in situ, noting that the trend is more important than individual recordings, as these are affected by any leak.
- » May be measured in a spontaneously breathing patient using nasal prongs if these are available, noting that clinical examination must be used in conjunction with the waveform and the ETCO<sub>2</sub>.
- » May be measured if the patient is being ventilated via a manual ventilation bag and mask, noting that the trend is more important than individual recordings, as these are affected by any leak around the mask.

## Other clinical signs

➤ Other clinical signs are elicited by examining the patient. They are just as important to document as the vital signs and include:

- » Features of general concern such as pallor or sweating.
- » Airway noise such as stridor or grunting.
- » Lung sounds such as wheeze or crackles.
- » Signs of increased work of breathing such as indrawing and nasal flaring.
- » Interaction and activity, particularly in small children.
- » The ability to mobilise normally without assistance.

# Vital Signs

## More Information

### Version history

|                                                          |         |
|----------------------------------------------------------|---------|
| <b>16 June 2026</b>                                      | 1.1.0.1 |
| <b>Update:</b> Added temperature to routine vital signs. |         |
| <b>22 November 2022</b>                                  | 1.0.1   |
| <b>Update:</b> Multiple changes.                         |         |

# Analgesia for Acute Pain

See the '[acute exacerbation of chronic pain](#)' guideline, the '[care during the last days of life](#)' guideline, or the relevant additional guideline (for example '[headache](#)'), if this is applicable to the patient's presentation.

## Medicines for mild pain

- Administer paracetamol.
- Consider administering ibuprofen.

## Medicines for moderate pain

- Administer an opiate (IV, IN or IM) and titrate further doses to effect.
- Administer paracetamol.
- Consider administering ibuprofen and/or tramadol.
- Consider administering methoxyflurane only if:
  - An opiate cannot be administered because of a contraindication or caution, or
  - Opiate administration will be significantly delayed, or
  - There will be severe worsening of pain during a procedure or extrication.

## Medicines for severe pain

- Administer an opiate, preferably IV, and titrate further doses to effect.
- Consider administering paracetamol and/or ibuprofen once initial improvement is achieved.
- Consider administering methoxyflurane only if:
  - An opiate cannot be administered because of a contraindication or caution, or
  - Opiate administration will be significantly delayed, or
  - There will be severe worsening of pain during a procedure or extrication.
- Consider administering medicine adjuncts for analgesia.
- If pain remains difficult to control:
  - Consider administering ketamine.
  - Consider commencing a fentanyl infusion.

# Analgesia for Acute Pain

## Medicine dosing and additional guidance

### Paracetamol

- Dosage:
  - 1.5 g PO for an adult weighing greater than 80 kg.
  - 1 g PO for an adult weighing 80 kg or less.
  - See the paediatric drug dose tables for a child.
- Paracetamol is not an initial priority for moderate to severe pain, but will usually reduce the need for subsequent analgesia and improve the quality of pain relief.
- The described doses are higher than those usually recommended in many references. However, the described doses are safe, as an initial loading dose, provided they are not administered repeatedly.

### Ibuprofen

- Dosage:
  - 600 mg PO for an adult weighing greater than 80 kg.
  - 400 mg PO for an adult weighing 80 kg or less.
  - See the paediatric drug dose tables for a child.
- Ibuprofen is particularly useful in the context of soft tissue, musculoskeletal or female genitourinary pain.
- Ibuprofen is not an initial priority for moderate to severe pain but will usually reduce the need for subsequent analgesia and improve the quality of pain relief.

### Tramadol

- Dosage: 50 mg PO.
- Do not routinely administer tramadol if an opiate or ketamine is administered because tramadol does not usually provide significant additional pain relief and may worsen side effects. However, tramadol may be administered if the patient has taken tramadol before and found it effective.

### Methoxyflurane

- Dosage:
  - 3 ml inhaled.

# Analgesia for Acute Pain

- » A second dose may be administered if the patient is aged greater than or equal to 12 years, and no cautions are present.
- » If a caution is present limit the dose to 3 ml.
- » In the event of repeated ambulance presentation, the maximum dose within 7 days is 15 ml.
- Methoxyflurane should be reserved for when:
  - » An opiate cannot be administered because of a contraindication or caution, or
  - » Opiate administration will be significantly delayed (for example greater than 20 minutes), or
  - » There will be severe worsening of pain during a procedure (for example during relocation of a joint but not during IV insertion), or during extrication.

## Fentanyl

- IV bolus dosage:
  - » 10-50 mcg IV for an adult.
  - » See the paediatric drug dose tables for a child.
  - » Repeat as required every five minutes.
- IN dosage:
  - » 100 mcg for an adult.
  - » See the paediatric drug dose tables for a child.
  - » Halve the dose if the patient is frail or has signs of shock.
  - » Repeat as required every 20 minutes.
- IM and SC dosage:
  - » 50-100 mcg IM/SC for an adult.
  - » See the paediatric drug dose tables for a child.
  - » Halve the dose if the patient is frail or has signs of shock.
  - » Repeat as required, every 20 minutes, up to a maximum of three doses.
- IV infusion dosage:
  - » Start at 1 mcg/kg/hour, up to a maximum of 150 mcg/hour.
  - » Administer additional boluses as required, every five minutes.
  - » Adjust the infusion rate as required, every 15-20 minutes.
- The decision to choose between fentanyl and morphine requires clinical judgement. Fentanyl is usually preferred if the patient:

# Analgesia for Acute Pain

- » Requires analgesia for a short period of time only, or
- » Has signs of shock.
- The decision on the route of administration requires clinical judgement:
  - » The IV route is preferred for severe pain.
  - » The IN route is preferred if IV access may not be required, or will be delayed.
  - » The IM/SC route should be reserved for patients in whom the IV or IN route is clinically inappropriate.

## Morphine

- IV bolus dosage:
  - » 1-5 mg IV for an adult.
  - » See the paediatric drug dose tables for a child.
  - » Repeat as required every five minutes.
- IM and SC dosage:
  - » 5-10 mg IM/SC for an adult.
  - » See the paediatric drug dose tables for a child.
  - » Halve the dose if the patient is frail or has signs of shock.
  - » Repeat as required, every 20 minutes, up to a maximum of three doses.
- The decision to choose between morphine and fentanyl requires clinical judgement. Morphine is usually preferred if the patient:
  - » Requires analgesia for a longer period of time, or
  - » Is receiving care during the last days of life.
- The decision on the route of administration requires clinical judgement:
  - » The IV route is preferred for severe pain.
  - » The IM/SC route should be reserved for patients in whom the IV route is clinically inappropriate.

## Ketamine

- IV dosage:
  - » 0.2 mg/kg IV for an adult, up to a maximum of 20 mg.
  - » See the paediatric drug dose tables for a child.
  - » Administer over approximately 15 minutes.
  - » Repeat as required.

# Analgesia for Acute Pain

- IM and PO dosage:
  - 0.5 mg/kg IM or PO for an adult, up to a maximum of 50 mg.
  - See the paediatric drug dose tables for a child.
  - Repeat as required, every 20 minutes, up to a maximum of three doses.
- Ketamine is particularly useful in the context of severe pain associated with traumatic injuries or burns.
- Prior to the administration of ketamine for analgesia (as opposed to dissociation), a sufficient dose of an opiate should be administered, so that further doses are not providing additional analgesia. This will usually require 150-200 mcg of fentanyl, or 15-20 mg of morphine, for an adult.
- When administering ketamine IV for pain, administering the dose over approximately 15 minutes may result in a reduction in adverse effects.

## Intraosseous lignocaine

- Lignocaine may be administered IO for significant bone pain associated with intraosseous administration.
- Dosage:
  - 50 mg (5 ml of 1% lignocaine) over two minutes for an adult.
  - See the paediatric drug dose tables for a child.
  - Wait one further minute before administering intraosseous fluid.
  - May be repeated once after 15 minutes.
- Intraosseous administration will result in lignocaine uptake into the circulation, and this may be harmful, particularly if the patient has shock. Uptake is minimised by waiting a minimum of one minute between lignocaine administration and fluid administration.

## Digital ring blocks using 1% lignocaine

- 1% lignocaine may be administered for moderate to severe pain associated with isolated injuries to digits.
- Dosage:
  - 1-2 ml per injection site. The maximum for multiple digits is 20 ml for an adult.
  - See the paediatric drug dose tables for the maximum dose for a child.
  - May be repeated once after 30 minutes if the block was effective and pain returns.
- Examine and record the presence or absence of sensation in the digits before administration.



# Analgesia for Acute Pain

## Fascia iliaca block using 0.375% ropivacaine

- A fascia iliaca block using 0.375% ropivacaine may be administered for pain associated with a fractured neck of the femur, or a fractured proximal shaft of femur, where pain has not been adequately controlled with an opiate, and transport time (including extrication time) to hospital is greater than 30 minutes.
- Dosage:
  - 40 ml of 0.375% ropivacaine for an adult weighing greater than 60 kg.
  - 30 ml of 0.375% ropivacaine for an adult weighing 60 kg or less.
  - Administer the block once only.
- Clinical judgement is required when requesting backup to perform the block, taking into account:
  - The severity of the patient's pain after administration of an opiate, and
  - The time it will take for backup to arrive and perform the block, and
  - The transport time to hospital.
- The goal of a fascia iliaca block is to provide additional analgesia and reduce the need for opiates and/or ketamine, particularly if the patient is elderly and/or frail.
- The block may be performed if the patient is taking an anticoagulant and/or antiplatelet medicines, but have a low threshold for stopping the procedure if it is difficult, and do not attempt multiple passes with the needle.
- Following administration monitor heart rate, blood pressure and GCS every 15 minutes.

## Medicine adjuncts for analgesia

- Medicine adjuncts for analgesia do not have analgesic properties, however, may have an adjunctive role.
- Prior to administration of a medicine adjunct a sufficient dose of an opiate should be administered so that further doses are not providing additional analgesia. This will usually require 150-200 mcg of fentanyl or 15-20 mg of morphine for an adult.

## Midazolam

- Consider administering midazolam as an adjunct for severe pain, if
  - The patient is aged greater than or equal to 12 years, and
  - Pain is associated with muscle spasm or anxiety, and
  - Adequate analgesia is not being achieved with an opiate.

# Analgesia for Acute Pain

- Dosage:
  - 0.5 - 1 mg IV.
  - Repeat as required every 10 minutes.
  - There is no maximum total dose, but the patient must be rousable and able to obey commands.
- Midazolam does not have analgesic properties, but it may have an adjunctive role in reducing severe pain associated with muscle spasm or anxiety.
- Prior to administering midazolam, a sufficient dose of an opiate should be administered, so that further doses are not providing additional analgesia. This will usually require 150-200 mcg of fentanyl or 15-20 mg of morphine for an adult.
- Continually monitor the patient's SpO<sub>2</sub>, breathing and level of consciousness, until the effects of midazolam have clearly worn off.

## Droperidol

- Consider administering droperidol, usually in conjunction with an opiate, as an adjunct for severe pain, associated with one of the following:
  - Chronic use of opiates.
  - Severe psychological distress.
  - Nausea and/or vomiting.
- Dosage: 0.625 mg IV/IM, once only.

## Additional information

### General principles

- Analgesia is usually best achieved by taking a multimodal approach including:
  - Physical interventions, for example positioning, splinting and cold therapy.
  - Psychological interventions, for example demonstrating empathy and utilising distraction techniques.
  - The administration of medicines.
- Administering a combination of medicines with different mechanisms of action usually provides better analgesia than one medicine.
- The type, dose and route of medicines administered requires clinical judgement, and should be selected and escalated in proportion to the level of the patient's pain.

# Analgesia for Acute Pain

- As a minimum, an assessment of pain severity, preferably using a patient generated pain score/rating, should be recorded:
  - Prior to the administration of analgesia, and
  - On arrival at a medical facility if transported by ambulance, or prior to leaving the scene if the patient is not transported by ambulance.

## Referral and transport

- A patient administered paracetamol and/or ibuprofen and/or tramadol for pain associated with a minor condition may be given a recommendation that transport to a medical facility via ambulance is not required.
- A patient may be administered one dose of IN/IM fentanyl, and/or one dose of methoxyflurane, and/or one dose of droperidol, may be given a recommendation that transport to a medical facility via ambulance is not required, provided:
  - Their condition does not require transport to an ED by ambulance, and
  - They have a GCS of 15 and can mobilise safely, and
  - Adequate pain control is achieved (for example a joint is relocated), and
  - They are advised to be seen in primary care for a review, if required, and
  - They are in the care of a competent adult for the next six hours, and
  - They are advised not to drive a vehicle or operate machinery for at least 24 hours.
- A patient receiving treatment for severe pain, especially if administered an opiate, ketamine, or midazolam, should be given a recommendation that transport to an ED via ambulance is required. Any deviation from this requires a clearly documented reason, and the safeguarding aspects in the bullets above covered.

## Accompanying the patient during transport

- Personnel with an appropriate practice level should accompany the patient during transport. For example, a paramedic should usually accompany the patient if an opiate, ketamine, or midazolam has been administered.
- However, clinical judgement is required, and it may occasionally be appropriate for personnel who have administered an opiate, ketamine, or midazolam, to hand over to an EMT to transport the patient, providing:
  - The patient has adequate control of their pain, and
  - The patient is rousable and can obey commands, and

# Analgesia for Acute Pain

- » The patient is not expected to require additional opiate, ketamine, or midazolam during transport, and
- » Medicines are not handed over to the EMT for administration during transport.
- If a fascia iliaca block has been performed, the ICP/CCP/ECP should remain with the patient for 10 minutes. If after 10 minutes, no adverse effects are observed, it is appropriate to hand care of the patient to a Paramedic or EMT crew to transport the patient.

## Response and/or transport by helicopter

- Backup for analgesia should usually occur by road and is not time sensitive.
- If a patient is in acute severe pain, which is unable to be managed by personnel on scene (including via clinical advice), response by helicopter may be indicated if it can locate at least at 60 minutes faster than backup by road.

## More Information

### Version history

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**16 June 2026**

1.1.0.1

**Update:** Guideline redeveloped.

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**22 November 2022**

1.0.1

**Update:** Multiple changes.

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- Assess the patient and consider the possibility of a new clinical problem unrelated to their chronic condition.
- Locate and follow the patient's care plan/pain management plan if they have one.
- Seek clinical advice from a clinician directly involved in the patient's usual care if feasible.
- Use non-pharmacological pain reduction strategies if feasible:
  - Encourage movement even during an acute episode.
  - Consider heat or cold for localised pain.
  - Encourage meditation or other relaxation techniques.
- Utilise the patient's own prescribed oral analgesia, in preference to administration of medicines from ambulance personnel.
- If administration of analgesia is required, use the following principles to help guide treatment choices:
  - Administer paracetamol and/or ibuprofen PO as per the '[analgesia](#)' guideline.
  - Consider early use of droperidol as per the '[analgesia](#)' guideline.
  - Avoid the administration of parenteral opiates, ketamine, and/or midazolam, if feasible.
  - Do not administer methoxyflurane.
  - Consider seeking support from an ECP.

## Referral and transport

- The goal is to avoid transport to ED by ambulance, provided a reduction in the patient's pain can be achieved.
- If transport is required, transport the patient to a clinician directly involved in their usual care if feasible.
- A patient may be administered one dose of droperidol and/or one dose of an opiate, and be given a recommendation that transport to a medical facility via ambulance is not required, provided:
  - Their condition does not require transport to an ED by ambulance, and
  - They have a GCS of 15 and can mobilise safely, and
  - An adequate reduction in their pain is achieved, and
  - They are advised to be seen in primary care for a review, if required, and
  - They are in the care of a competent adult for the next six hours, and
  - They are advised not to drive a vehicle or operate machinery for at least 24 hours.

- A patient administered ketamine, midazolam, or multiple doses of an opiate, should be given a recommendation that transport to an ED via ambulance is required. Any deviation from this requires a clearly documented reason, and the safeguarding aspects in the bullets above covered.

## Additional information

- Chronic pain is defined as an ongoing unpleasant sensory experience that is not correlated with the severity of pathology.
- About 20% of the population live with a degree of chronic or persistent pain, and patients may experience serious acute pathology unrelated to this. For example, a patient may develop acute appendicitis on a background of chronic lower abdominal pain.
- Acute exacerbations of chronic pain are physiologically very different to other forms of acute pain and administration of standard analgesic medicines is often not the most effective treatment.
  - Medications are usually only moderately effective in changing the patient's perception of their pain.
  - The goal is to modify their experience of pain to the extent they are better able to function, rather than a reduction in pain to a certain level.
- Although the following principles are important for all patients, they are particularly important in the context of chronic pain:
  - Demonstrate empathy and provide reassurance.
  - Discuss what is usually effective with the patient and their family/whānau and involve them in shared decision making.
  - Manage patient and family/whānau expectations. Improvement in function or a modest reduction in pain is a more appropriate goal than complete relief of pain.
- If a patient is calling frequently and does not have a care/management plan, forward their details to the appropriate manager.

## More Information

### Version history

16 June 2026

1.1.0.1

**Update:** Analgesia and transport guidance updated.

22 November 2022

1.0.1

**New:** Guideline released.

# Vulnerable Persons

This guideline is for vulnerable persons who are at risk of harm.

- Determine the level of risk.
- Develop a safeguarding plan:
  - Take actions to make the person safe. This may include removing the person from the threat or arranging to have the source of the threat removed from the person.
  - Involve appropriate agencies or support services as required.
- Seek clinical advice if the situation becomes difficult to resolve.

## Low risk of harm

- Explain referral options and obtain consent to refer the patient to an appropriate support service.
- If consent is given, make the referral.
- Ensure the person's response is documented, even if consent for a referral is not given.

## Medium to high risk of harm

- Involve police if required.
- Explain referral options and obtain consent to refer the patient to an appropriate support service.
- Consider taking actions without the patient's consent if:
  - There is a serious threat to the patient's life or health, or
  - The best interests of the patient require the situation to be reported to obtain support or protection.
- Refer the patient to an appropriate service or make a report of concern as appropriate.

# Vulnerable Persons

## Immediate risk of harm

- Request assistance from police early, if required.
- Take steps to either:
  - Transport the patient to a place of safety (for example, an appropriate medical facility or police station), or
  - Arrange for the patient to be transported to a place of safety by a trusted person, or
  - Arrange for the source of the threat to be removed from the environment.
- Report concerns to an appropriate agency (for example, to Oranga Tamariki if the patient is aged less than 18 years, an elder abuse response service if the patient is aged 65 years or over, or police) as soon as possible, even if the patient is transported to a medical facility.

## Additional information

### General principles

- Ambulance personnel are in the unique position of seeing people in their own environment on an unplanned basis. As such, an assessment of whether the person is vulnerable should routinely occur as part of patient assessment.
- Personnel must legally take reasonable steps to protect people from harm who are vulnerable or at risk. The principles which inform this guideline are taken from the following legislation:
  - The Crimes Act 1961.
  - Children's Act 2014.
  - Oranga Tamariki Act 1989.
- To be classified as vulnerable, there must be a genuine risk of harm to the patient. People with 'general' social issues who are not at risk of harm are not usually categorised as 'vulnerable' in the context of these CPGs. However, general social issues can be risk factors for future vulnerability/harm and personnel should attempt to minimise these where feasible.
- If there are general social concerns about a patient, personnel should involve the patient's family/whānau, and make referrals to and/or provide contact information for social services, as appropriate. Examples of general social issues that do not usually meet the threshold of being vulnerable include (but are not limited to):
  - Those who require home help.
  - Those whose primary carer is fatigued and requires additional support.
  - Those who require assistance managing their medications.
  - Those requiring support interacting with government agencies such as Work and Income.



# Vulnerable Persons

## Determining the level of risk

- Determining the level of risk can be challenging and clinical judgement is required.
- Personnel should consider the entire situation and utilise all available information when determining the level of risk including health, social and environmental factors.
- If there is uncertainty as to the level of risk, personnel should seek input from an appropriate agency (for example, Oranga Tamariki or an elder abuse response service) or seek clinical advice.

## Developing a safeguarding plan

- A safeguarding plan is a set of actions designed to reduce the patient's risk of harm. A safeguarding plan must be developed for all vulnerable people and should include actions needed to make them safe, depending on the level of risk. Examples of actions that form part of a safeguarding plan are:
  - Transporting the person to a place of safety or arranging for them to be transported to a place of safety by a trusted person.
  - Arranging for the source of the threat to be removed from the environment.
  - Involving police.
  - Referring the patient to a support service.
  - Reporting concerns to an appropriate agency.
- Referring the patient to a support service:
  - Personnel should be familiar with support services available in their local area and refer the patient to an appropriate service as part of the safeguarding plan.
  - Examples of support services include:
    - The patient's usual primary healthcare provider, for example, their GP.
    - Local elder abuse response services, for adults aged 65 years or over.
    - Oranga Tamariki, for children aged less than 18 years (including for an unborn child).
    - Hospital-based social workers.
    - Women's Refuge.
    - Telehealth support services (for example, the Elder Abuse Response Line).
    - Local Māori health providers.
  - Referrals should usually be made by phone, however making a referral using other methods (for example, via ePRF as part of a pathway) is appropriate if the patient has otherwise been appropriately safeguarded and the referral is non-urgent.
  - If the patient is at low risk of harm, consent is required prior to making a referral. However, if the patient is at medium to high risk of harm and does not provide consent, personnel should consider

# Vulnerable Persons

whether taking actions (including referring the patient to a support service) without their consent is in the best interest of the patient.

- Reporting concerns:
  - If the patient is at immediate risk of harm due to abuse and/or neglect (including self-neglect), personnel must report their concerns to a statutory agency such as Oranga Tamariki (for children aged less than 18 years including unborn children), police, or a designated elder abuse response service (for adults aged 65 years or over).
  - If the patient is at medium to high risk of harm, personnel should consider reporting their concerns, particularly if the patient is aged less than 18 years.
  - Concerns must be reported by phone, unless the agency's personnel (for example, police) attend the scene.
  - Consent is not required for personnel to report their concerns, however the patient and their family/whānau should be informed if it is feasible and safe to do so.
  - Concerns must be reported irrespective of whether the patient is transported to a medical facility. This is because information loss is common in healthcare, and it cannot be assumed that another organisation will make a report of concern and pass on the concerns and observations of ambulance personnel.

## Abuse and neglect

- The most common cause of vulnerability is neglect, which is defined as someone who is responsible for another person persistently failing to meet the basic needs of the dependent person. Examples include:
  - Physical neglect: not providing necessities like food, shelter and clothing.
  - Neglectful supervision: leaving the dependent person without someone appropriate looking after them.
  - Neglect of medical care: failing to seek medical attention for the dependent person when required, or withholding medication.
- Self-neglect is when the behaviour of a person threatens their own safety. For example, an older adult not providing themselves with adequate food, water, clothing, shelter, or medication. Self-neglect should be managed in the same way as neglect, for example, referring the patient to a support service or reporting concerns to an appropriate agency.
- Neglect can be difficult to identify because personnel and families/whānau may have different expectations for hygiene, nutrition and parenting.
- Abuse is defined as any non-accidental act resulting in harm to another person. Examples include:
  - Physical abuse: hitting, shaking, burning, poisoning, strangling, inappropriate use of restraints, or anything else that could cause physical injury.

# Vulnerable Persons

- » Psychological and emotional abuse: ridicule, threats, harassment, humiliation and preventing the person from making decisions.
- » Sexual abuse: inappropriate physical contact, exposing the dependent person to sexually explicit material, or encouraging sexually inappropriate behaviour.
- When assessing a patient's vulnerability, it is important to objectively consider the 'whole picture' and the overall risk of harm to the patient, rather than only focusing on the event that has just occurred. This is because abuse and neglect are rarely an isolated event, and the most recent event may be part of a wider episode of harm.

## Documentation

- Personnel must objectively document their concerns and actions taken in the PRF, describing:
  - » Exactly what the patient and other people at the scene said.
  - » The type of wellbeing and/or safety concern, including the social situation and environment.
  - » The actions personnel have taken and who has been informed.
- Personnel using ePRF must record the above information in the Vulnerable Person tab under 'Clinical Tools'. This is because recording concerns and actions taken in this tool enables a review group to assess safeguarding plans for vulnerable people.

## More Information

### Version history

22 November 2022

1.0.1

**New:** Guideline released.

# Psychological Wellness

This guideline is for personnel to help them pro-actively support their psychological wellness and that of their colleagues. It is based on the MANERS psychological first aid tool developed by the Victorian Ambulance Service and is designed to be used as an adjunct to other forms of psychological support.

## The MANERS psychological first aid tool

- M** Minimise exposure.
- A** Acknowledge the event.
- N** Normalise reactions.
- E** Educate as required.
- R** Review, restore or refer.
- S** Self-care.

### Minimise exposure

- Minimise exposure to distressing situations. For example, if a person is psychologically distressed speak to them and their manager about the possibility of taking time away from work.
- Move the person to a place of calm and safety if they are distressed. This may not always be immediately possible during an incident, but if a person is distressed remove them from the situation as soon as possible and have someone be with them.

### Acknowledge the event

- Talk openly if an incident was potentially distressing.
- Ask the person if they are OK.
- Allow the person to talk about how they have been affected, but respect people who do not want to talk about it.
- Ensure conversations are sensitive, respectful and confidential.
- Be prepared to listen without trying to provide solutions.

# Psychological Wellness

## Normalise reactions

- Reassure the person it is normal to be affected.
- Recognise that individual people are affected differently and that this is normal.
- Acknowledge the person's feelings without being judgemental.

## Educate as required

- Talk about the MANERS psychological first aid tool.
- Talk about what helps you.
- Encourage techniques that promote psychological wellness.
- Ensure the person knows the support services available through their service and externally.

## Review, restore or refer

- Encourage the person to return to normal duties when ready.
- Arrange a follow-up phone call or visit to check on how they are doing.
- Refer the person to support services if required, for example peer support or professional assistance.

## Self-care

- Maintain a healthy lifestyle ensuring adequate sleep, exercise, nutrition and leisure activity.
- Encourage a culture of openly talking about how we are feeling.
- Identify your own barriers to psychological wellness and develop a proactive approach to ensuring you are well.
- Develop skills that improve communication and help deal with conflict.
- Access peer support and/or professional support as required.

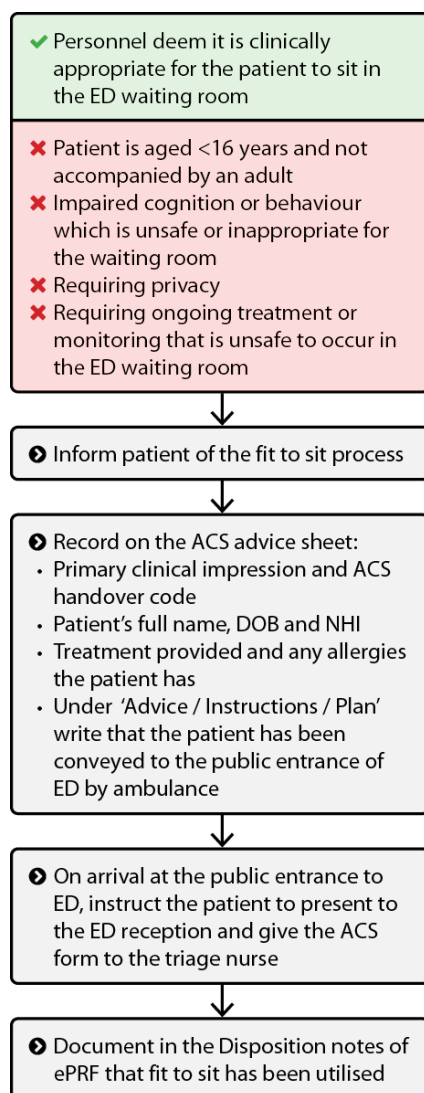
# Psychological Wellness

## At the end of each shift

- Pause briefly at the end of each shift (preferably with your crew) and think about or discuss:
    - What was the best thing we did for a patient or family today?
    - What is something we did today that we could do better next time?
    - What are we going to do when we get home to help maintain our psychological wellness?
-

# Fit to Sit

Use this guideline for patients who are being transported to an ED, to determine whether they are 'fit to sit' in the ED waiting room.



## Additional information

### General principles

- Fit to sit aims to minimise the time that ambulance personnel spend at ED when transporting patients with low acuity problems who are safe and appropriate to sit in the ED waiting room.
- As part of this process, personnel convey the patient to the public entrance of ED, provide the patient with an ambulance care summary (ACS) advice sheet and instruct them to present to the triage nurse at the ED reception. Because the patient is provided with an ACS advice sheet with a handover code, ED staff can access the patient's ACS and a clinical handover is not required.

# Fit to Sit

## Explaining the fit to sit process

- Use the following bullet points to explain the fit to sit process:
  - You will be provided with a piece of paper containing information which enables hospital personnel to access the paperwork we have written about your condition.
  - Once we have directed you to the public entrance of ED, you should present to reception and hand over your form (ACS advice sheet) to the triage nurse.
  - While you are waiting, you will have access to reception staff who will be able to address any concerns you have accordingly. Please advise staff if there is any deterioration in your condition.

## More Information

### Version history

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9 June 2025

1.0.5.3

**New:** Guideline released.

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# Radio Notification to Hospital

## Format for radio notification (R40) to hospital

**[Hospital]** ED this is **[call sign]** how do you read?

Patient is a status **[#]** , **[##]** year/month old male/female

NHI is \_\_\_\_\_ (if known)

Chief complaint is \_\_\_\_\_

Relevant information is (include time of onset if applicable): \_\_\_\_\_

Abnormal vital signs are:

GCS \_\_\_\_\_ (motor) \_\_\_\_\_

HR \_\_\_\_\_ BP \_\_\_\_\_ RR \_\_\_\_\_

SpO<sub>2</sub> \_\_\_\_\_ on \_\_\_\_\_ L/min O<sub>2</sub>

Temp \_\_\_\_\_ BGL \_\_\_\_\_

Significant treatment currently underway: \_\_\_\_\_

ETA: **[##]** minutes.

Copy, see you in **[receiving location in dept]**

**[Call sign]** out.

## Additional information

- The goal of the radio transmission to hospital is to provide a succinct summary of the relevant information required for ED staff to allocate a bedspace, prepare a resuscitation team, and/or initiate in-hospital protocols such as a trauma call. It should usually take approximately 30 seconds.
- If detailed information needs to be conveyed or a detailed conversation is anticipated, the notification is best conveyed by phone.
- Pause after asking "How do you read?" This allows ED staff to answer and confirm the radio signal is clear. If the signal is not confirmed as clear, consider waiting a few minutes and trying again.
- Use the phonetic alphabet when stating the patient's NHI.
- Clarify the patient's age with individual numbers. For example, an "eighteen, one – eight, year old male, or a "ninety, nine-zero" year old female. Provide an estimate if the age is not known.

# Radio Notification to Hospital

- The chief complaint should be a few words. For example, stroke, STEMI or post cardiac arrest. If the likely diagnosis is unclear, it is appropriate to describe the most concerning signs or symptoms. For example, severe shock, chest pain or shortness of breath.
- The relevant information should be brief and only contain the information required for ED staff to determine where the patient is going to go and what personnel they are going to notify. For example, if the ECG has evolving signs of myocardial ischaemia, or the patient is anticoagulated and has signs of shock. A description of additional detailed relevant history should wait for the verbal handover in hospital.
- Only abnormal vital signs should be described. If there are no abnormal vital signs, it is appropriate to state that the vital signs are normal.
  - If the GCS is abnormal, describe the total GCS and the motor score. For example, "GCS is 10 with a motor score of 4".
  - If the SpO<sub>2</sub> is abnormal, describe what oxygen flow rate is being delivered and via what device.
- Only describe significant treatments that are currently being provided. For example, CPAP, an adrenaline infusion, ventilation, or CPR is in progress. A description of all other 'standard' treatments (for example, gaining IV access) should wait for the verbal handover in hospital.
- At the end of the radio transmission, acknowledge where you have been asked to go to within the department (if applicable) and end the transmission with the vehicle call sign and "out", to indicate that the radio channel is now free for other users.

## More Information

### Version history

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**16 June 2026**

1.1.0.1

**Update:** Hospital notification guidance updated.

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**9 June 2025**

1.0.5.3

**New:** Guideline released.

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# Hot Debrief (STOPP 10)

STOPP 10 should be used as a hot debriefing checklist following any critical or complex case.

## **S** Summarise the case

- » Summarise the entire case.
- » Everyone in the team is encouraged to contribute.

## **T** Things that went well

- » Acknowledge everything that went well.
- » This can be from an individual, clinical, operational, team, or wellbeing perspective.

## **O** Opportunities to improve

- » Consider what can be improved.
- » This can be from a clinical, operational, team, or wellbeing perspective.

## **P** Psychological welfare

- » Discuss the feelings and emotions you may have, and what you may expect over the coming days.
- » Options of additional support, such as employee counselling services, peer support, or managerial support, should be discussed.

## **P** Points to action

- » Decide on the next steps forward. Is a cold debrief required? Were there any adverse clinical events, equipment issues, or operational issues that require reporting? Do support services need activating?

## Additional information

- A hot debrief following critical or complex cases should usually occur.
- Utilise the [STOPP 10 hot debriefing checklist](#), which should take less than 10 minutes to complete and can be run by any member of the clinical team.
- Ensure a respectful, supportive space where contribution is welcomed but not compulsory.

# Hot Debrief (STOPP 10)

- Hot debriefs should reflect on team performance, covering technical skills, decision-making, communication, leadership, and use of time and resources.
- The process should highlight strengths, identify areas for improvement, and raise awareness of available clinical and psychological support.
- Benefits of hot debriefing include improved team cohesion, morale and psychological wellbeing, and it promotes a culture of learning, patient safety, and quality improvement.
- If any action points are identified, for example escalation to a manager, ensure this is tasked to an appropriate person.

## More Information

### Version history

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**16 June 2026**

1.1.0.1

**New:** Guideline released.

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# Early Warning Score

This guideline describes the use of the New Zealand Early Warning Score (EWS).

## When an EWS should be recorded

- An EWS should be recorded for patients who are status two or status three, and are having vital signs recorded.
- A minimum of two EWS should be recorded if the patient is being transported, one with the initial set of vital signs and one just before, or on arrival at, the medical facility.
- A minimum of one EWS must be recorded prior to leaving the scene, if a patient is not being transported by ambulance and has had vital signs recorded.
- If there is concern that the patient is deteriorating, consider recording the EWS regularly, for example every 30 minutes.

## When an EWS does not need to be recorded

- An EWS does not need to be recorded if the patient:
  - Has a time sensitive condition, and the EWS will not alter treatment or transport decisions, or
  - Has a minor condition and a set of vital signs has not been recorded.

## Calculating and recording the EWS

- Use the tables below or the calculator within the CPG app to calculate the EWS.
- Record the EWS on the PRF, preferably in the vital signs section.
- The EWS, and the trend of any changes, should be included in the handover to other healthcare personnel.

## Additional information

### General information

- The EWS is a standardised scoring system used across hospitals and healthcare facilities, to help identify clinical severity and deterioration, using a colour-coded system (yellow, orange, red and blue) for risk levels.
- The EWS has versions for non-pregnant adults, children, and pregnant women.

# Early Warning Score

- The EWS uses an aggregate score based on vital signs and clinical observations.
- Using the EWS helps standardise language, especially during handover, helps identify subtle deterioration, and supports clinical judgment.

## Incorporating the EWS into clinical decisions

- The EWS should be considered as an additional vital sign.
- The EWS will not always correlate with clinical acuity. For example, a patient:
  - With an immediate threat to life from STEMI may have a low EWS.
  - Without an immediate threat to life from SVT may have a high EWS.
- The trend of the EWS is more important than a single calculation, and the trend is an important part of the handover to other healthcare personnel.
- The EWS should be considered along with all other clinical information when making treatment and/or referral decisions.
- A slow deterioration of the EWS may help detect a deterioration in the patient's condition, which otherwise may go undetected because the change is subtle and slow.
- A deterioration in the EWS should prompt a review of the:
  - Patient's status code, and
  - Patient's treatment, and
  - Need for backup, and
  - Need to seek clinical advice.
- Clinical judgment is required on when to calculate and record an EWS. An EWS is not routinely required if it will not change the decision to treat or transport the patient. Examples include patients:
  - With major trauma.
  - Receiving care during the last days of life.
  - Being transferred between facilities with a time sensitive condition, for example stroke for clot retrieval.
- The EWS must be considered when a patient is being given a recommendation that transport by ambulance to a medical facility is not required, and vital signs have been recorded. An EWS greater than or equal to five, or any vital sign in the red or blue zone, must trigger a review of the non-transport decision, and clear documentation on the clinical rationale is required if non-transport remains the outcome.

# Early Warning Score

## EWS tables

| Adult EWS                             |              |                                      |       |                                            |           |                                                                                      |         |                                                                                                |      |
|---------------------------------------|--------------|--------------------------------------|-------|--------------------------------------------|-----------|--------------------------------------------------------------------------------------|---------|------------------------------------------------------------------------------------------------|------|
|                                       | 10+          | 3                                    | 2     | 1                                          | 0         | 1                                                                                    | 2       | 3                                                                                              | 10+  |
| Respiratory rate                      | ≤4           | 5-8                                  |       | 9-11                                       | 12-20     |                                                                                      | 21-24   | 25-35                                                                                          | ≥36  |
| SpO <sub>2</sub> %                    |              | ≤91                                  | 92-93 | 94-95                                      | ≥96       |                                                                                      |         |                                                                                                |      |
| Supplemental O <sub>2</sub> required  |              |                                      |       |                                            | No        |                                                                                      | Yes     |                                                                                                |      |
| Heart rate                            | ≤39          |                                      | 40-49 |                                            | 50-89     | 90-109                                                                               | 110-129 | 130-139                                                                                        | ≥140 |
| Systolic BP                           | ≤69          | 70-89                                | 90-99 | 100-109                                    | 110-219   |                                                                                      |         | ≥220                                                                                           |      |
| Temperature (°C)                      |              |                                      | ≤34.9 | 35.0-35.9                                  | 36.0-37.9 | 38.0-38.9                                                                            | ≥39.0   |                                                                                                |      |
| Level of Consciousness                | Unresponsive | Voice or Pain                        |       |                                            | Alert     |                                                                                      |         |                                                                                                |      |
| Total EWS score zone from table above |              | EWS 1-5 May have a serious condition |       | EWS 6-7 Likely to have a serious condition |           | EWS 8-9 or any vital sign in the red zone<br>Very likely to have a serious condition |         | EWS 10+ or any vital sign in the blue zone<br>Very likely to have a life-threatening condition |      |

| Maternity EWS                        |     |     |       |   |       |   |       |       |     |
|--------------------------------------|-----|-----|-------|---|-------|---|-------|-------|-----|
|                                      | 10+ | 3   | 2     | 1 | 0     | 1 | 2     | 3     | 10+ |
| Respiratory rate                     | ≤5  | 6-9 |       |   | 10-20 |   | 21-25 | 26-30 | ≥31 |
| SpO <sub>2</sub> %                   |     | ≤91 | 92-94 |   | ≥95   |   |       |       |     |
| Supplemental O <sub>2</sub> required |     |     |       |   | No    |   | Yes   |       |     |

# Early Warning Score

|                                       |              |                                             |       |                                                   |         |                                                                                             |         |                                                                                                       |      |
|---------------------------------------|--------------|---------------------------------------------|-------|---------------------------------------------------|---------|---------------------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------|------|
| Heart rate                            | ≤39          | 40-49                                       |       | 50-59                                             | 60-99   | 100-119                                                                                     | 120-129 | 130-139                                                                                               | ≥140 |
| Systolic BP                           | ≤69          | 70-79                                       | 80-89 | 90-99                                             | 100-139 |                                                                                             | 140-159 | 160-199                                                                                               | ≥200 |
| Diastolic BP                          |              |                                             |       |                                                   | 40-89   |                                                                                             | 90-109  | ≥110                                                                                                  |      |
| Temperature (°C)                      |              | ≤34.9                                       |       | 35.0-35.9                                         | 36-37.9 | 38-38.9                                                                                     |         | ≥39.0                                                                                                 |      |
| Level of Consciousness                | Unresponsive | Voice or Pain                               |       |                                                   | Alert   |                                                                                             |         |                                                                                                       |      |
| Total EWS score zone from table above |              | <b>EWS 1-5</b> May have a serious condition |       | <b>EWS 6-7</b> Likely to have a serious condition |         | <b>EWS 8-9 or any vital sign in the red zone</b><br>Very likely to have a serious condition |         | <b>EWS 10+ or any vital sign in the blue zone</b><br>Very likely to have a life-threatening condition |      |

## Paediatric EWS 0-11 months

|                                      | 10+ | 3     | 2     | 1      | 0       | 1       | 2        | 3        | 10+ |
|--------------------------------------|-----|-------|-------|--------|---------|---------|----------|----------|-----|
| Respiratory rate                     | ≤9  | 10-19 | 20-25 | 26-29  | 30-49   | 50-54   | 55-69    | ≥70      |     |
| Respiratory distress                 |     |       |       |        | None    | Mild    | Moderate | Severe   |     |
| SpO <sub>2</sub> %                   |     |       | ≤90   | 91-94  | ≥95     |         |          |          |     |
| Supplemental O <sub>2</sub> required |     |       |       |        | None    |         | <4 L/min | ≥4 L/min |     |
| Heart rate                           | ≤59 | 60-79 | 80-89 | 90-109 | 110-159 | 160-169 | 170-179  | ≥180     |     |
| Capillary refill (central)           |     |       |       |        | <3 secs |         |          | ≥3 secs  |     |
| Systolic BP                          | ≤49 | 50-55 | 56-64 | 65-74  | 75-99   | 100-119 | 120-149  | ≥150     |     |



# Early Warning Score

|                                              |              |                                             |                                                   |  |                                                                                             |  |                                                                                                       |  |  |
|----------------------------------------------|--------------|---------------------------------------------|---------------------------------------------------|--|---------------------------------------------------------------------------------------------|--|-------------------------------------------------------------------------------------------------------|--|--|
| <b>Level of Consciousness</b>                | Unresponsive |                                             |                                                   |  | Alert,<br>Voice,<br>Pain                                                                    |  |                                                                                                       |  |  |
| <b>Total EWS score</b> zone from table above |              | <b>EWS 1-5</b> May have a serious condition | <b>EWS 6-7</b> Likely to have a serious condition |  | <b>EWS 8-9 or any vital sign in the red zone</b><br>Very likely to have a serious condition |  | <b>EWS 10+ or any vital sign in the blue zone</b><br>Very likely to have a life-threatening condition |  |  |

| Paediatric EWS 1-4 years                   |              |       |       |       |                          |         |          |          |     |
|--------------------------------------------|--------------|-------|-------|-------|--------------------------|---------|----------|----------|-----|
|                                            | 10+          | 3     | 2     | 1     | 0                        | 1       | 2        | 3        | 10+ |
| <b>Respiratory rate</b>                    | ≤4           | 5-14  | 15-17 | 18-19 | 20-39                    | 40-44   | 45-54    | ≥55      |     |
| <b>Respiratory distress</b>                |              |       |       |       | None                     | Mild    | Moderate | Severe   |     |
| <b>SpO<sub>2</sub>%</b>                    |              |       | ≤90   | 91-94 | ≥95                      |         |          |          |     |
| <b>Supplemental O<sub>2</sub> required</b> |              |       |       |       | None                     |         | <4 L/min | ≥4 L/min |     |
| <b>Heart rate</b>                          | ≤59          | 60-69 | 70-79 | 80-89 | 90-139                   | 140-149 | 150-159  | ≥160     |     |
| <b>Capillary refill (central)</b>          |              |       |       |       | <3 secs                  |         |          | ≥3 secs  |     |
| <b>Systolic BP</b>                         | ≤55          | 56-59 | 60-75 | 76-89 | 90-109                   | 110-124 | 125-159  | ≥160     |     |
| <b>Level of Consciousness</b>              | Unresponsive |       |       |       | Alert,<br>Voice,<br>Pain |         |          |          |     |

# Early Warning Score

|                                              |                                             |                                                   |                                                                                             |                                                                                                       |
|----------------------------------------------|---------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Total EWS score</b> zone from table above | <b>EWS 1-5</b> May have a serious condition | <b>EWS 6-7</b> Likely to have a serious condition | <b>EWS 8-9 or any vital sign in the red zone</b><br>Very likely to have a serious condition | <b>EWS 10+ or any vital sign in the blue zone</b><br>Very likely to have a life-threatening condition |
|----------------------------------------------|---------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|

| Paediatric EWS 5-11 years             |              |                                      |       |                                            |                    |                                                                                      |          |                                                                                                |     |
|---------------------------------------|--------------|--------------------------------------|-------|--------------------------------------------|--------------------|--------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------|-----|
|                                       | 10+          | 3                                    | 2     | 1                                          | 0                  | 1                                                                                    | 2        | 3                                                                                              | 10+ |
| Respiratory rate                      | ≤4           | 5-11                                 | 12-14 | 15-19                                      | 20-29              | 30-34                                                                                | 35-44    | ≥45                                                                                            |     |
| Respiratory distress                  |              |                                      |       |                                            | None               | Mild                                                                                 | Moderate | Severe                                                                                         |     |
| SpO <sub>2</sub> %                    |              |                                      | ≤90   | 91-94                                      | ≥95                |                                                                                      |          |                                                                                                |     |
| Supplemental O <sub>2</sub> required  |              |                                      |       |                                            | None               |                                                                                      | <4 L/min | ≥4 L/min                                                                                       |     |
| Heart rate                            | ≤49          | 50-59                                | 60-69 | 70-79                                      | 80-129             | 130-139                                                                              | 140-155  | ≥156                                                                                           |     |
| Capillary refill (central)            |              |                                      |       |                                            | <3 secs            |                                                                                      |          | ≥3 secs                                                                                        |     |
| Systolic BP                           | ≤59          | 60-69                                | 70-79 | 80-89                                      | 90-119             | 120-139                                                                              | 140-169  | ≥170                                                                                           |     |
| Level of Consciousness                | Unresponsive |                                      |       |                                            | Alert, Voice, Pain |                                                                                      |          |                                                                                                |     |
| Total EWS score zone from table above |              | EWS 1-5 May have a serious condition |       | EWS 6-7 Likely to have a serious condition |                    | EWS 8-9 or any vital sign in the red zone<br>Very likely to have a serious condition |          | EWS 10+ or any vital sign in the blue zone<br>Very likely to have a life-threatening condition |     |

## Paediatric EWS 12-16 years

# Early Warning Score

|                                              | 10+          | 3                                           | 2     | 1                                                 | 0                  | 1                                                                                           | 2        | 3                                                                                                     | 10+  |
|----------------------------------------------|--------------|---------------------------------------------|-------|---------------------------------------------------|--------------------|---------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------|------|
| <b>Respiratory rate</b>                      | ≤4           | 5-9                                         | 10-11 | 12-14                                             | 15-24              | 25-29                                                                                       | 30-34    | ≥35                                                                                                   |      |
| <b>Respiratory distress</b>                  |              |                                             |       |                                                   | None               | Mild                                                                                        | Moderate | Severe                                                                                                |      |
| <b>SpO<sub>2</sub>%</b>                      |              |                                             | ≤90   | 91-94                                             | ≥95                |                                                                                             |          |                                                                                                       |      |
| <b>Supplemental O<sub>2</sub> required</b>   |              |                                             |       |                                                   | None               |                                                                                             | <4 L/min | ≥4 L/min                                                                                              |      |
| <b>Heart rate</b>                            | ≤39          | 40-49                                       | 50-59 | 60-65                                             | 66-109             | 110-119                                                                                     | 120-135  | 136-139                                                                                               | ≥140 |
| <b>Capillary refill (central)</b>            |              |                                             |       |                                                   | <3 secs            |                                                                                             |          | ≥3 secs                                                                                               |      |
| <b>Systolic BP</b>                           | ≤75          | 76-79                                       | 80-89 | 90-99                                             | 100-135            | 136-149                                                                                     | 150-189  | ≥190                                                                                                  |      |
| <b>Level of Consciousness</b>                | Unresponsive |                                             |       |                                                   | Alert, Voice, Pain |                                                                                             |          |                                                                                                       |      |
| <b>Total EWS score zone from table above</b> |              | <b>EWS 1-5</b> May have a serious condition |       | <b>EWS 6-7</b> Likely to have a serious condition |                    | <b>EWS 8-9 or any vital sign in the red zone</b><br>Very likely to have a serious condition |          | <b>EWS 10+ or any vital sign in the blue zone</b><br>Very likely to have a life-threatening condition |      |

## More Information

### Version history

16 June 2026

1.1.0.1

**New:** Guideline released.

# Procedural Dissociation

This guideline is for all patients greater than or equal to five years requiring dissociation for a painful procedure, for example limb alignment or cardioversion. Seek clinical advice if the patient is aged less than five years. See the '[acute behavioural disturbance](#)' guideline if the patient requires sedation for control of agitation.

## Preparation for dissociation checklist

- Informed consent gained if appropriate.
- Plan verbalised to team.
- Oxygen being administered.
- One person monitoring airway, breathing and vital signs:
  - SpO<sub>2</sub>.
  - 3-lead ECG.
  - BP.
  - Nasal capnography, if available.
- Equipment prepared:
  - Manual ventilation bag and mask.
  - Airway adjuncts.
  - Suction.
- Visual and audible stimuli reduced if appropriate.

## Dissociation procedure

- Administer ketamine:
  - 1 mg/kg IV up to a maximum of 150 mg. This may be repeated once.
  - 5 mg/kg IM up to a maximum of 400 mg, once only.
  - Halve all doses if a caution is present.
- Ensure the patient is adequately dissociated before performing the procedure.
- Continue to monitor airway, breathing and vital signs until the effects of ketamine have clearly worn off.

# Procedural Dissociation

## More Information

### Version history

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**16 June 2026**

1.1.0.1

**New:** Guideline released.

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# Exacerbation of Asthma

This guideline is for patients aged greater than or equal to two years with a provisional diagnosis of an exacerbation of asthma.

## Mild to moderate asthma

- Follow the patient's asthma action plan if they have one.
- Administer bronchodilators using one of the following:
  - Administer a beta two agonist (for example a combination steroid and beta two agonist or salbutamol) via an inhaler. Consider using a spacer and administer up to six doses. Or,
  - Administer 5 mg of nebulised salbutamol in combination with 0.5 mg of nebulised ipratropium using oxygen as the driving gas.
- Administer an oral steroid:
  - 40 mg of prednisone PO for an adult.
  - See the paediatric drug dose tables for a child.
- Administer further doses of nebulised salbutamol as required.
- Consider the likelihood that transport may not be required if the patient rapidly improves with bronchodilators via an inhaler or following one dose of nebulised bronchodilators.

## Severe asthma

- Administer 5 mg of nebulised salbutamol in combination with 0.5 mg of nebulised ipratropium using oxygen as the driving gas.
- Administer continuous nebulised salbutamol until improvement occurs.
- Administer adrenaline IM if the patient is not improving:
  - Administer 0.5 mg IM for an adult.
  - See the paediatric drug dose tables for a child.
- Begin transport without delay, providing most treatments en route.
- Administer magnesium IV:
  - Administer 10 mmol (2.5 g) IV over approximately 15 minutes for an adult.
  - See the paediatric drug dose tables for a child and administer over approximately 15 minutes.
  - Administer a second dose if the patient is not improving.

# Exacerbation of Asthma

- Repeat adrenaline IM every 10 minutes if the patient is deteriorating and adrenaline IV is not being administered.
- Consider application of NIV if the patient is not improving. See the '[NIV](#)' guideline.
- Administer a steroid:
  - An oral steroid if the patient is able to swallow, using the doses described in mild to moderate asthma, or
  - Hydrocortisone IV:
    - 100 mg IV for an adult.
    - See the paediatric drug dose tables for a child.

## Immediately life-threatening asthma

- All treatment for severe asthma.
- Administer adrenaline IV:
 

**Adrenaline for an adult:**

  - Administer an adrenaline infusion via an infusion pump. Start at a rate of 0.5 mg/hour and adjust the rate as required, or
  - Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
    - Administer as an infusion. Start at 2 drops/second and adjust the rate as required, or
    - Administer a bolus of 0.01 mg adrenaline (10 ml) IV every 1-2 minutes as required.

**Adrenaline for a child aged 5-14 years:**

  - Administer an adrenaline infusion via an infusion pump using service guidelines, or
  - Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
    - Administer as an infusion. Start at 1 drop/second and adjust the rate as required, or
    - Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.

**Adrenaline for a child aged less than five years:**

  - Administer an adrenaline infusion via an infusion pump using service guidelines, or
  - Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.
- An ICP or CCP may administer ketamine if an adult patient is severely agitated and this is preventing the ability to safely provide treatment or transport:
  - Administer 0.5 mg/kg of ketamine IV, up to a maximum of 50 mg. This may be repeated every 10 minutes as required.
  - Consider commencing an infusion using service guidelines.

# Exacerbation of Asthma

## Backup

- Backup from an ICP/CCP should be requested if the patient has severe or immediately life-threatening asthma.

## Referral and transport

- All patients with severe to immediately life-threatening asthma must receive a clear recommendation to be transported to an ED by ambulance.
- Patients with mild to moderate asthma who have clearly improved may receive a clear recommendation for treatment without transport providing the following criteria are met:
  - No red flags are present.
  - The patient is observed for a minimum of 20 minutes following completion of the last bronchodilator administration.
  - Prednisone is provided (if indicated and available), and an information sheet with the contents clearly explained.
  - The patient is advised to see an appropriate healthcare professional within two days.
  - If the patient has signs of a chest infection (for example fever or purulent sputum), the patient should be seen by an appropriate healthcare professional within 12 hours. This should usually be in a primary care facility.

## Red flags (should be seen in ED, usually via ambulance)

- Severe or immediately life-threatening asthma.
- More than one dose of nebulised bronchodilators has been administered (not including MDI).
- Significantly abnormal vital signs.
- SpO<sub>2</sub> less than 92% when breathing air.
- Unable to talk in full sentences.
- Unable to mobilise normally.
- Precipitous or 'brittle' asthma.



# Exacerbation of Asthma

## Additional information

### General principles

- Asthma is characterised by reversible bronchospasm. It is caused by an inflammatory state within the lungs resulting in recurrent attacks of breathlessness and wheezing. It is often associated with mucus plugging of airways.
- Brittle or precipitous asthma is defined as rapid progression from a well to a serious condition within 1-2 hours.

### Determining the severity of asthma

#### Summary table (not all clinical features need to be present)

| Clinical feature                     | Mild to moderate asthma    | Severe asthma                                | Immediately life-threatening asthma |
|--------------------------------------|----------------------------|----------------------------------------------|-------------------------------------|
| Words per breath                     | Able to speak in sentences | Able to speak a few words per breath         | Unable to speak                     |
| Wheeze                               | Usually have a loud wheeze | May only have a quiet wheeze                 | May not have a wheeze               |
| Chest/neck indrawing and positioning | No significant indrawing   | Significant indrawing and tripod positioning | Marked indrawing, unless exhausted  |
| SpO <sub>2</sub>                     | Normal                     | Usually > 90%                                | Rapidly falling                     |
| Agitation                            | Usually not present        | May have agitation                           | Severe agitation                    |
| Level of consciousness               | Normal                     | May be confused                              | Falling LOC                         |

### Inhalers and spacers

- If the patient has a mild or moderate exacerbation of their asthma, it is preferable to commence with administration of a beta two agonist via an inhaler, either a pressurised metered dose inhaler (pMDI) or a dry powder inhaler.
- If the patient has their own inhaler this should be used in preference to one from the ambulance.
- Many patients have a combination inhaler containing a steroid and a long acting beta two agonist, for example budesonide and formoterol. This may be referred to as single maintenance and reliever therapy

# Exacerbation of Asthma

(SMART) or anti-inflammatory and reliever (AIR) therapy. Use of a SMART/AIR inhaler during an exacerbation is as effective as use of an inhaler containing a short acting beta two agonist, such as salbutamol.

- Use of a spacer is not always required. The most important factor is to use the technique the patient is most familiar with.
- Inhalers and spacers from the ambulance should only be used for a single patient. The inhaler and the spacer from the ambulance may be left with the patient following use.
- Dry powder inhalers must not be used with a spacer.

## Children

- Gaining IV access in young children is a balance of risk. It may cause distress and worsen their work of breathing, but will be required if their exacerbation is severe or immediately life-threatening.
- Steroid administration does not usually have a role in children aged less than five years because it does not generally alter the course of their asthma exacerbation. However, an oral steroid is indicated if the child has a clear history of asthma and has previously received oral steroids.

## Oral steroid administration

- Prednisone tablets are preferred if the patient is able to swallow tablets, but they may be administered the same dose of prednisolone syrup.
- Providing the patient with prednisone:
  - » If the patient is aged greater than or equal to 12 years and is not transported, additional prednisone should be provided to them by ambulance personnel, unless the patient already has an asthma action plan and their own supply of steroid.
  - » Provide the patient with an additional four days (four x 40 mg) of prednisone.
  - » Advise the patient and any carers, including in writing, that the patient:
    - » Takes 40 mg (two 20 mg tablets) at approximately the same time each day, for the next four days until finished, in addition to their other usual medicines.
    - » Should be reviewed in primary care for a review of their treatment within two days.
- Provide the patient and any carers with an asthma and a prednisone information sheet, if available, ensuring the information is explained to them.

# Exacerbation of Asthma

## Adrenaline administration

- Adrenaline IM is reserved for severe asthma that is not improving with nebulised bronchodilators.
- A dose of 0.5 mg adrenaline IM is appropriate for the majority of adults. Paramedics, ICPs and CCPs may make a decision to reduce the dose, particularly if the patient is small, frail or has ischaemic heart disease.
- Adrenaline IV is reserved for immediately life-threatening asthma.
- When administering adrenaline IV to patients aged greater than or equal to five years, an infusion is preferred over IV boluses because this reduces the adverse effects of surges of adrenaline.

## Ketamine administration

- Ketamine administration requires clinical judgement, and should only occur if the patient's agitation is severe enough to impair the ability to safely provide treatment and/or transport. Although ketamine may have some bronchodilator activity, in this setting the focus is on treating severe agitation. Personnel should seek clinical advice if uncertain.
- Patients with severe agitation are likely to require RSI and backup must be requested early.

## The differential diagnosis for asthma

- COPD. Patients usually:
  - Have an established COPD diagnosis.
  - Have a history of smoking.
  - Are not usually symptom free between exacerbations.
- Cardiogenic pulmonary oedema. Patients usually:
  - Have been supine, for example in bed, prior to onset of symptoms.
  - Are hypertensive, clammy and peripherally vasoconstricted.
  - Have crackles and wheeze which is loudest in the lower zones.
- Pneumonia. Patients usually have:
  - Signs of sepsis.
  - A productive cough.
  - Wheeze and/or crackles that is unilateral or limited to one lobe/area.
- Anaphylaxis. Patients usually have:

# Exacerbation of Asthma

- » Rapid onset over minutes.
- » Systemic signs of shock.
- » A rash.
- Vocal cord dysfunction. Patients usually have:
  - » Stridor rather than wheeze.

## Dynamic hyperinflation

- Dynamic hyperinflation (gas trapping) occurs when the amount of gas within the lungs increases in the presence of severe bronchoconstriction.
- This occurs because the resistance to gas leaving the lungs during expiration is higher than the resistance to gas entering the lungs during inspiration.
- Dynamic hyperinflation rarely causes pneumothorax, but it commonly causes reduced venous return to the heart by increasing intrathoracic pressure.
- Dynamic hyperinflation occurs in spontaneously breathing patients when their asthma is severe, but those most at risk of life-threatening dynamic hyperinflation are those receiving positive pressure ventilation or NIV. For this reason, the ventilation rate must be kept at six breaths/minute in patients receiving positive pressure ventilation for immediately life-threatening asthma.

## Tension pneumothorax

- Tension pneumothorax due to asthma is very rare unless the patient is receiving positive pressure ventilation or NIV. Even in this setting, if the patient is deteriorating with signs of shock the cause is usually dynamic hyperinflation rather than tension pneumothorax.
- Needle chest decompression in the setting of immediately life-threatening asthma carries a significant risk of causing iatrogenic pneumothorax. Chest decompression should only be undertaken if there are very convincing clinical signs of a tension pneumothorax and the preferred technique is finger thoracostomy, provided the patient is ventilated (see '[tension pneumothorax](#)' guideline).
- Diagnosing tension pneumothorax can be very difficult in the presence of immediately life-threatening asthma because:
  - » Breath sounds are already reduced because the patient is moving very little air.
  - » The jugular veins are already distended because of raised intrathoracic pressure.
  - » Cardiac output is already reduced because of dynamic hyperinflation.
  - » The percussion note is already hyperresonant because of dynamic hyperinflation.

# Exacerbation of Asthma

- In the setting of immediately life-threatening asthma, the convincing signs of tension pneumothorax are most likely to be:
  - A very clear difference in breath sounds and percussion note between the two sides, and
  - Signs of a progressively falling cardiac output, or reduction in  $\text{ETCO}_2$ , in the absence of signs of dynamic hyperinflation.
- Consider utilising ultrasound, if available, to assist with the diagnosis of pneumothorax.

## Other causes of bronchospasm

- Bronchodilators do not have a significant role in the treatment of bronchospasm resulting from smoke inhalation, toxic gas inhalation or chest infection. However, bronchodilators may be administered if bronchospasm is prominent (usually withholding adrenaline and steroids unless the underlying condition is suspected to be complicated by asthma).

## Response and/or transport by helicopter

- Asthma is time sensitive to treatment, but rarely time dependent to hospital providing suitable skill is with the patient.
- Response by helicopter may be indicated if appropriate clinical skill can be delivered 15 minutes faster than by road for severe respiratory compromise, or 30 minutes faster than by road for moderate respiratory compromise.
- Transport by helicopter may be indicated if the patient has severe respiratory compromise not improving with treatment and it will save more than 30 minutes compared to road transport.
- For helicopter transport criteria to be met, the road transport time will usually need to be greater than 100 minutes.

## More Information

### Version history

16 June 2026

1.1.0.1

**Update:** Assessment, treatment and referral guidance updated.

22 November 2022

1.0.1

**Update:** Multiple changes.

This guideline is for adults with a provisional diagnosis of an exacerbation of COPD.

## Mild to moderate COPD

- Follow the patient's COPD action plan if they have one.
- Administer bronchodilators using one of the following:
  - Administer a beta two agonist (for example a combination steroid and beta two agonist or salbutamol) via an inhaler. Consider using a spacer and administer up to six doses. Or,
  - Administer 5 mg of nebulised salbutamol in combination with 0.5 mg of nebulised ipratropium using air as the driving gas.
- Titrate oxygen administration, usually via nasal prongs, to an SpO<sub>2</sub> of 88-92% or to the patient's usual SpO<sub>2</sub> if this is known.
- Administer 40 mg of prednisone PO.
- Administer further doses of nebulised salbutamol as required.
- Consider the likelihood that transport may not be required if the patient rapidly improves with bronchodilators via an inhaler or following one dose of nebulised bronchodilators.

## Severe COPD

- Administer 5 mg of nebulised salbutamol in combination with 0.5 mg of nebulised ipratropium usually using air as the driving gas.
- Titrate oxygen administration, usually via nasal prongs, to an SpO<sub>2</sub> of 88-92% or to the patient's usual SpO<sub>2</sub> if this is known.
- Administer continuous nebulised salbutamol until improvement occurs.
- Administer magnesium IV:
  - Administer 10 mmol (2.5 g) IV over approximately 15 minutes.
  - Administer a second dose if the patient is not improving.
- Consider application of NIV if the patient is not improving. See the '[NIV](#)' guideline.
- Consider 0.5 mg of midazolam IV for severe anxiety. Administer sparingly and repeat only if required, every five minutes, provided the patient is able to obey commands at all times.
- Administer a steroid:

- » 40 mg of prednisone PO if the patient is able to swallow, or
- » 100 mg of hydrocortisone IV.

## Imminent respiratory arrest

- Administer adrenaline IV in addition to the treatments for severe COPD:
  - » Administer an adrenaline infusion via an infusion pump. Start at a rate of 0.5 mg/hour and adjust the rate to the patient's condition, or
  - » Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000):
    - » Administer as an infusion. Start at 2 drops/second and adjust the rate to the patient's condition, or
    - » Administer a bolus of 0.01 mg adrenaline (10 ml) IV every 1-2 minutes as required.
- Do not administer midazolam.

## Backup

- Backup from an ICP/CCP should be requested if the patient has severe anxiety or imminent respiratory arrest.

## Referral and transport

- All patients with severe (or worse) COPD must receive a clear recommendation to be transported to an ED by ambulance.
- Patients with mild to moderate COPD who have clearly improved may receive a clear recommendation for treatment without transport providing the following criteria are met:
  - » No red flags are present.
  - » The patient is observed for a minimum of 20 minutes following completion of the last bronchodilator administration.
  - » Prednisone is provided (if indicated and available), and an information sheet with the contents clearly explained.
  - » The patient is advised to see an appropriate healthcare professional within two days.
  - » If the patient has signs of a chest infection (for example fever or purulent sputum), the patient should be seen by an appropriate healthcare professional within 12 hours. This should usually be in a primary care facility.

# Exacerbation of Chronic Obstructive Pulmonary Disease (COPD)

CPG EAS 2.2 St John New Zealand

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## Red flags (should be seen in ED, usually via ambulance)

- Severe COPD exacerbation.
- More than one dose of nebulised bronchodilators has been administered (not including MDI).
- Significantly abnormal vital signs.
- New onset of altered mental status.
- SpO<sub>2</sub> less than 88% (or > 3% less than the patient's normal baseline).
- Has not improved to their usual respiratory state.
- Acute functional deterioration (for example, unable to mobilise normally).

## Additional information

### General principles

- COPD is a term used to encompass chronic inflammatory and destructive diseases within the lung, including chronic bronchitis and emphysema. The bronchoconstriction present in COPD is not completely reversible.
- COPD should be suspected in patients with chronic respiratory illness, particularly if they have risk factors such as age over 50 years, long-term exposure to cigarette smoke (including second hand exposure), or long-term exposure to environmental or industrial pollutants.

## Determining the severity of COPD

### Summary table (not all clinical features need to be present)

| Clinical feature                     | Mild to moderate COPD       | Severe COPD                                          | Imminent respiratory arrest |
|--------------------------------------|-----------------------------|------------------------------------------------------|-----------------------------|
| Words per breath                     | Able to speak in sentences  | Only able to speak a few words per breath            | Unable to speak             |
| Wheeze                               | Usually have a wheeze       | May not have a wheeze                                | May not have a wheeze       |
| Chest/neck indrawing and positioning | Usually have some indrawing | Usually have severe indrawing and tripod positioning | May not have indrawing      |



|                               |                         |                                        |                  |
|-------------------------------|-------------------------|----------------------------------------|------------------|
| <b>SpO<sub>2</sub></b>        | Near their normal level | Significantly below their normal level | Rapidly falling  |
| <b>Agitation</b>              | Usually not present     | May have agitation                     | Severe agitation |
| <b>Level of consciousness</b> | Normal                  | May be confused                        | Falling LOC      |

## Inhalers and spacers

- If the patient has a mild or moderate exacerbation of their COPD, it is preferable to commence with administration of a beta two agonist via an inhaler, either a pressurised metered dose inhaler (pMDI) or a dry powder inhaler.
- If the patient has their own inhaler this should be used in preference to one from the ambulance.
- Many patients have a combination inhaler containing a steroid and a long acting beta two agonist, for example budesonide and formoterol. This may be referred to as single maintenance and reliever therapy (SMART) or anti-inflammatory and reliever (AIR) therapy. Use of a SMART/AIR inhaler during an exacerbation is as effective as use of an inhaler containing a short acting beta two agonist, such as salbutamol.
- Use of a spacer is not always required. The most important factor is to use the technique the patient is most familiar with.
- Inhalers and spacers from the ambulance should only be used for a single patient. The inhaler and the spacer from the ambulance may be left with the patient following use.
- Dry powder inhalers must not be used with a spacer.

## General principles of oxygen and nebulised bronchodilator administration

- Only administer oxygen if the patient has an SpO<sub>2</sub> less than 88%. Titrate the oxygen flow to maintain an SpO<sub>2</sub> of 88-92%.
- Bronchodilators should be nebulised using air if possible:
  - If using air to nebulise bronchodilators, administer bronchodilators using a flow rate of 8 litres/minute and administer oxygen via nasal prongs if required, adjusting the oxygen flow rate to achieve an SpO<sub>2</sub> of 88-92%.
  - If using oxygen to nebulise bronchodilators, administer bronchodilators using a flow rate of 8 litres/minute. If the SpO<sub>2</sub> climbs to above 92% during nebuliser delivery, alternate five minutes with the mask on and five minutes with the mask off. This is done to limit oxygen exposure whilst delivering most of the nebulised bronchodilator.

## Oxygen administration and hypercarbia

- Some patients have carbon dioxide clearance that is altered by oxygen administration and excess oxygen administration in these patients may cause hypercarbia.
- Patients at high risk of hypercarbia include patients with COPD, morbid obesity, those on home oxygen and those on home CPAP or BiPAP.
- The signs of a rising carbon dioxide level are usually confusion, drowsiness, agitation and a falling level of consciousness. If a patient is suspected of developing hypercarbia, oxygen administration should not be discontinued immediately. Instead, oxygen administration should be reduced to a lower flow rate (targeting an SpO<sub>2</sub> of 88-92%) and the patient reassessed.
- Consider assisting the patient's ventilation early using a manual ventilation bag and mask if:
  - SpO<sub>2</sub> continues to fall below 80% despite treatments, or
  - The patient is becoming exhausted, or
  - The patient is suspected of developing hypercarbic respiratory failure despite lowering the oxygen flow.
- A T-piece may be used if available to administer nebulised medications if the patient's ventilation is being assisted with a manual ventilation bag.

## Differential diagnosis for COPD

- Asthma. Patients usually:
  - Have an established asthma diagnosis.
  - Are symptom free between exacerbations.
- Cardiogenic pulmonary oedema. Patients usually:
  - Have been supine, for example in bed, prior to onset of symptoms.
  - Are hypertensive, clammy and peripherally vasoconstricted.
  - Have crackles and wheeze which is loudest in the lower zones.
- Pneumonia. Patients usually have:
  - Signs of sepsis.
  - A productive cough.
  - Wheeze and/or crackles that is unilateral or limited to one lobe/area.
- Anaphylaxis. Patients usually have:
  - Rapid onset over minutes.

- » Systemic signs of shock.
- » A rash.
- Vocal cord dysfunction. Patients usually have:
  - » Stridor rather than wheeze.

## Oral steroid administration

- Prednisone tablets are preferred for patients with COPD, but they may be administered the same dose of prednisolone syrup.
- Providing the patient with prednisone.
  - » If the patient is not transported, additional prednisone should be provided to them by ambulance personnel, unless the patient already has a COPD action plan and their own supply of steroid.
  - » Provide the patient with an additional four days (four x 40 mg) of prednisone.
  - » Advise the patient and any carers, including in writing, that the patient:
    - » Takes 40 mg (two 20 mg tablets) at approximately the same time each day, for the next four days until finished, in addition to their other usual medicines.
    - » Should be reviewed in primary care for a review of their treatment within two days.
- Provide the patient and any carers with a COPD and a prednisone information sheet, if available, ensuring the information is explained to them.

## Magnesium administration

- The evidence for magnesium administration in patients with COPD is weaker than the evidence in patients with asthma. However, some patients with COPD have a significant reversible component to their bronchospasm and magnesium may be helpful in this cohort.

## Adrenaline administration

- Adrenaline administration is reserved for imminent respiratory arrest.
- The decision to administer adrenaline must include weighing up the potential benefits against the potential risks. Adrenaline may result in bronchodilation, but it may also cause tachydysrhythmias, severe hypertension, and myocardial ischaemia. Patients with COPD are at very high risk of the adverse effects of adrenaline and this is why it is reserved for imminent respiratory arrest.
- Adrenaline administration (particularly IV bolus administration) can make the patient tachypnoeic and look or feel awful. It is important to differentiate this from a worsening of their COPD and not to

automatically respond by administering more adrenaline.

- An IV infusion is preferred over IV boluses because this reduces the adverse effects of surges of adrenaline.

## Midazolam administration

- Midazolam administration in patients with an exacerbation of COPD is not commonly required but may be administered for relief of severe anxiety.
- It is important to differentiate between severe anxiety and severe respiratory distress, noting that midazolam has a role in treating anxiety but not in treating respiratory distress.
- The decision to administer midazolam requires clinical judgement that balances the possible benefit from a reduction in anxiety against the possible risk of worsening the patient's respiratory effort.
- Midazolam must be administered sparingly and the patient must be able to obey commands at all times.

## Response and/or transport by helicopter

- COPD is time sensitive to treatment, but rarely time dependent to hospital providing suitable skill is with the patient.
- Response by helicopter may be indicated if appropriate clinical skill can be delivered 15 minutes faster than by road for severe respiratory compromise, or 30 minutes faster than by road for moderate respiratory compromise.
- Transport by helicopter may be indicated if the patient has severe respiratory compromise not improving with treatment and it will save more than 30 minutes compared to road transport.
- For helicopter transport criteria to be met, the road transport time will usually need to be greater than 100 minutes.

## More Information

### Version history

16 June 2026

1.1.0.1

**Update:** Assessment, treatment and referral guidance updated.

22 November 2022

1.0.1

**Update:** Multiple changes.



# Foreign Body Airway Obstruction

## Consciousness and ventilation

### If the patient is conscious and ventilation is adequate

- Immediate intervention is not required.
- Be prepared to intervene if deterioration occurs.

### If the patient is conscious and ventilation is inadequate

- Perform up to five back blows.
- Perform up to five chest thrusts.
- Alternate between cycles of back blows and chest thrusts until the obstruction is cleared, or the patient becomes unconscious.

### If the patient is unconscious and ventilation is adequate

- Place the patient on their side.
- Be prepared to intervene if deterioration occurs.

### If the patient is unconscious and ventilation is inadequate

- Attempt to remove the foreign body under direct vision with a finger sweep and move sequentially through the steps below if the obstruction is not cleared:
  - Attempt to remove the foreign body using a laryngoscope and Magill forceps.
  - Perform five chest compressions and recheck the airway using a laryngoscope and Magill forceps.
  - Commence CPR and try to ventilate using a manual ventilation bag and mask.
  - Intubate with an endotracheal tube, inserting the tube as far as possible and then withdrawing the tube to the usual position.
  - Perform a cricothyroidotomy.

## Backup

- Backup from an ICP/CCP should be requested for patients with signs of persisting foreign body airway obstruction following initial intervention.

# Foreign Body Airway Obstruction

## Referral and transport

- The patient must be transported to an ED by ambulance if the obstruction is unable to be removed or they have been unconscious.
- The patient should be transported to an ED by ambulance if the obstruction is removed and the patient has persistent symptoms, for example, shortness of breath, coughing, chest pain, stridor or the sensation of a foreign body.
- A clear recommendation may be made that transport is not required if the obstruction is removed and the patient has no persisting symptoms.

## Additional information

### General

- Foreign body airway obstruction occurs most commonly in young children, the elderly, the intoxicated and the intellectually impaired.
- A patient that has adequate ventilation and/or is coughing does not require immediate intervention.
- Check the patient for signs of relief of the obstruction between each intervention.
- When performing back blows, position the patient with their head below their shoulders, provided this is feasible.
- Abdominal thrusts (also known as the Heimlich manoeuvre) are no longer recommended because they are associated with a risk of intra-abdominal injury and do not appear to be associated with a higher chance of success than chest thrusts.
- During intubation, the reason the endotracheal tube is inserted as far as possible and then withdrawn is to try to push the foreign body down one bronchus, allowing ventilation to occur through the other bronchus.

## Oesophageal obstruction and foreign body ingestion

- Commonly, the foreign body is lodged at the top of the oesophagus and the patient does not have airway obstruction.
- The patient is often in significant distress and/or is unable to swallow saliva.
- Providing the patient has a normal airway and normal breathing, encouraging the patient to take small sips of fluid (which may be carbonated) may cause the obstruction to dislodge.

# Foreign Body Airway Obstruction

- If oesophageal obstruction persists, the patient requires transport to a hospital with surgical facilities as anaesthesia and endoscopy is usually required to remove the foreign body.
- Although glucagon may reduce oesophageal contraction, there is no role for glucagon administration in patients with oesophageal obstruction.
- If the patient has ingested a foreign body, refer to the '[poisoning from miscellaneous causes](#)' guideline.

## More Information

### Version history

22 November 2022

1.0.1

**Update:** Added referral and transport. Removed foreign body ingestion instructions.



# Non-Invasive Ventilation (NIV)

This guideline is for adult patients who have a condition requiring application of NIV. Seek clinical advice if the patient is a child.

CPAP is indicated if the patient has moderate to severe respiratory distress with the following conditions:

- Cardiogenic pulmonary oedema.
- Pulmonary oedema due to drowning.
- Pulmonary oedema due to asphyxiation (negative pressure pulmonary oedema).
- Asthma or COPD (and BiPAP is not available).
- Undifferentiated respiratory problem that is not improving with treatment.
- An SpO<sub>2</sub> of less than 92% due to a respiratory problem despite treatment (less than 88% if COPD or known chronic hypoxia).

BiPAP is indicated if the patient has moderate to severe respiratory distress with the following conditions:

- Asthma or COPD.

NIV is contraindicated if the patient has:

- Ineffective breathing or inadequate ventilatory drive, or
- Upper airway obstruction, or
- Active vomiting.

Use NIV with caution if the patient has:

- An altered level of consciousness, or
- Signs of shock, or
- Clinical suspicion of pneumothorax.

## Application of CPAP

- Commence at 5 cmH<sub>2</sub>O.
- Increase to 10 cmH<sub>2</sub>O if it is being tolerated.
- Consider increasing to 15 cmH<sub>2</sub>O if the patient is not improving.

# Non-Invasive Ventilation (NIV)

- Titrate  $\text{FiO}_2$  as required to maintain target  $\text{SpO}_2$ .

## Application of BiPAP

- Commence at:
  - 10  $\text{cmH}_2\text{O}$  inspiratory positive airway pressure (IPAP).
  - 5  $\text{cmH}_2\text{O}$  expiratory positive airway pressure (EPAP).
- Consider increasing the IPAP up to 15  $\text{cmH}_2\text{O}$  if the patient is not improving.
- Titrate  $\text{FiO}_2$  as required to maintain target  $\text{SpO}_2$ .

## Additional information

### General principles

- Non-invasive ventilation (NIV) is a form of oxygenation and ventilation support via a sealed face mask in acute respiratory failure.
- Continuous positive airway pressure (CPAP) and bi-level positive airway pressure (BiPAP) are forms of NIV which deliver positive pressure breathing support. In general terms, CPAP improves oxygenation, while BiPAP improves both oxygenation and ventilation.
- Effective NIV is dependent on achieving an adequate mask seal. Ensure:
  - The correct mask size is selected, and
  - Consideration of the contour of the face (especially the bridge of the nose and cheeks) and adjust the mask positioning as required, and
  - Dentures (if present) are kept in situ, and
  - The patient is allowed to adopt a position most comfortable for them.
- Some patients may not easily tolerate the mask and may need continuous coaching and reassurance.
- Medicines may be administered for severe respiratory distress and/or agitation as described in the relevant guidelines but must not be administered specifically to facilitate NIV without seeking clinical advice.

### The physiological effects of NIV

- The physiological effects of CPAP are similar to those of PEEP but provide positive pressure during inspiration as well as expiration. BiPAP provides different pressures during inspiration compared to expiration.

# Non-Invasive Ventilation (NIV)

- During application of NIV, the positive pressure assists small and medium sized airways to remain open during expiration, reducing lung collapse. This may also aid expiration in patients with bronchospasm by reducing dynamic hyperinflation.
- The positive pressure during inspiration means that less work is required, reducing the work of breathing.
- The positive pressure in the thoracic cavity reduces the preload (filling) of the right ventricle by reducing venous return to the heart.
- The positive pressure in the thoracic cavity increases the afterload of the right ventricle. This reduces blood flow through lung vessels, reducing the amount of fluid entering the lungs.
- The expiratory pressure increases the amount of air remaining in the lungs at the end of expiration (functional residual capacity), and this causes the lungs to be more expanded. From this more expanded resting position, less work is required for inspiration because of the non-linear compliance of the lungs, particularly when the lungs are wet.

## Haemodynamic effects

- NIV can reduce cardiac output and should be used with caution in patients showing signs of shock. The reduction in cardiac output may be significant in patients with:
  - A clinical condition reducing right ventricular filling, such as hypovolaemia or dehydration.
  - A clinical condition increasing right ventricular afterload, such as pulmonary embolism.

## NIV in asthma

- In severe asthma, the airflow limitation is most pronounced in the expiratory phase, resulting in increased end-expiratory lung volumes, also known as 'breath stacking' or 'dynamic hyperinflation'.
- NIV may overcome the patient's intrinsic PEEP secondary to hyperinflation by providing extrinsic PEEP, therefore reducing the pressure gradient in which the patient has to breathe against. This may reduce muscle fatigue and improve alveolar ventilation and gas exchange.
- Have a low threshold for seeking clinical advice in this setting and prepare for the possibility that RSI may be required.

## More Information

### Version history

16 June 2026

**Update:** Guideline redeveloped.

1.1.0.1

# Non-Invasive Ventilation (NIV)

22 November 2022

1.0.1

**New:** Guideline released.

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# Stridor

This guideline is for any form of upper airway obstruction secondary to infection or swelling other than croup. If stridor from [‘croup’](#) or [‘anaphylaxis’](#) is present, see the relevant guideline.

- Administer 5 mg of nebulised adrenaline if stridor is causing moderate or severe respiratory distress.
- Repeat nebulised adrenaline as required every ten minutes.

## Backup

- Backup from an ICP/CCP should be requested if the patient has severe respiratory distress or is deteriorating despite nebulised adrenaline.

## Referral and transport

- All patients with stridor must be given a clear recommendation to be transported to an ED by ambulance.

## Additional information

### General principles

- Stridor is an abnormal high pitched noise created when air is moving through a narrowed airway. It is a clinical sign and not a diagnosis. Consider the cause of the stridor using the differential diagnosis section in this guideline.
- Stridor is predominantly inspiratory, but may have an expiratory component (biphasic stridor).
- Stridor must be distinguished from wheeze.
- It is important to keep children as calm as possible, because stridor will usually get worse if they become upset or cry.
- The dose of nebulised adrenaline is the same for children and adults.
- Angioedema may respond to nebulised adrenaline but do not administer adrenaline IM or IV. This is because angioedema does not improve with parenteral adrenaline and the risks from the adverse effects of adrenaline outweigh any possible benefits.

### Differential diagnosis for stridor

- Croup. Patients usually:
  - Are aged six months to two years.

# Stridor

- » Have an onset of illness over several days.
- » Have a barking cough that is worse at night.
- » Have a fever but appear systemically well.
- Epiglottitis/supraglottitis. Patients usually:
  - » Are aged older than five years.
  - » Have an onset of illness over a day or two.
  - » Have an intensely sore throat.
  - » Have difficulty swallowing and may drool.
  - » Have a fever and appear systemically unwell.
- Tracheitis. Patients usually:
  - » Are aged 5-15 years.
  - » Have an onset of illness over a day or two.
  - » Do not have difficulty swallowing
  - » Have a fever and appear systemically very unwell.
- Foreign body airway obstruction. Patients usually:
  - » Are young children, elderly, frail, intoxicated or intellectually impaired.
  - » Have a history of choking preceding the onset of stridor.
- Anaphylaxis. Patients usually have:
  - » Rapid onset over minutes.
  - » Systemic signs of shock.
  - » A rash.
  - » Bronchospasm rather than stridor.
- Angioedema. Patients usually have:
  - » Onset over 30-60 minutes.
  - » Oedema that is isolated to the face, tongue or mouth.
  - » No systemic signs of shock.
  - » No bronchospasm.

## More Information

Version history

|                                                         |         |
|---------------------------------------------------------|---------|
| <b>16 June 2026</b>                                     | 1.1.0.1 |
| <b>Update:</b> Differential diagnosis guidance updated. |         |
| <b>22 November 2022</b>                                 | 1.0.1   |
| <b>Update:</b> Backup instructions.                     |         |

# Croup

This guideline is for children with a provisional diagnosis of croup. See the '[stridor](#)' guideline if the cause is unclear.

- Take a history and perform an examination, including:
  - A focused respiratory assessment.
  - Features in the croup severity table.
  - Features in the transport/referral criteria.
- Focus on excluding sinister causes of stridor other than croup.
- Consider a diagnosis other than croup if the child has any of the following:
  - History of worsening sore throat for several days.
  - Fever  $>40^{\circ}\text{C}$ .
  - Petechiae or mottled skin.
  - Signs of shock.
  - Difficulty swallowing or drooling.
  - Trismus, or difficulty opening mouth.

## Mild respiratory distress

- Administer prednisolone syrup PO. See the paediatric drug dose tables for dosages.
- Consider the likelihood that transport may not be required.

## Moderate to severe respiratory distress

- Administer 5 mg of nebulised adrenaline.
- Repeat nebulised adrenaline as required every ten minutes.
- Administer prednisolone syrup PO if the patient can swallow. See the paediatric drug dose tables for dosages.

## Backup

- Backup from an ICP/CCP should be requested if the patient has severe respiratory distress or is deteriorating despite nebulised adrenaline.



# Croup

## Referral and transport

- The parents/caregivers must be given a clear recommendation that the child is transported to a medical facility by ambulance if adrenaline has been administered. Transport should usually be to an ED but may be to a primary care or urgent care facility if the child is rapidly improving and only one dose of adrenaline has been administered.
- The parents/caregivers of a child with mild respiratory distress may receive a recommendation for treatment without transport (see 'red flags' below). The parents/caregivers must be advised that the patient needs to be seen by an appropriate healthcare professional within 24 hours because a further course of oral steroids is required.

## Red flags (should be seen in ED, usually via ambulance)

- Two or more features of severe croup.
- SpO<sub>2</sub> less than 94% when breathing air.
- Drooling or difficulty swallowing.
- Difficulty opening mouth.
- Signs of shock.
- Lethargy.
- Nebulised adrenaline has been administered.

## Worsening advice

- Provide parents/caregivers with verbal and written advice if the child is not transported to a medical facility and consider photographing this advice using their phone. For example:
  - Administer paracetamol using package instructions for fever or discomfort.
  - Encourage regular feeding/drinking.
  - Comfort them upright on your lap or in your arms if they are not settling.
  - Call Healthline on 0800 611116 if:
    - Their breathing is getting faster or noisier, or
    - They are not feeding or drinking normally, or
    - You are worried.
  - Call 111 for an ambulance if:

# Croup

- » Their lips or face goes blue, or
- » They cannot swallow, or
- » They are floppy or hard to wake, or
- » They are struggling to breathe, or
- » They have pauses in their breathing.

## Additional information

### General principles

- Croup is a viral infection of the upper airway. It is the most common cause of stridor in children, especially children aged six months to two years.
- The child usually has an onset of illness over the preceding days, a barking cough that is worse at night and a fever.
- It is important to keep the child as calm as possible, because stridor will usually get worse if they become upset or cry. Although a young child is more likely to remain calm if they are kept in the arms of a parent, they must be transported in an approved restraint and not in the arms of a parent.

### Croup severity table (not all features need to be present)

| Sign/symptom          | Mild            | Moderate               | Severe                                          |
|-----------------------|-----------------|------------------------|-------------------------------------------------|
| Barking cough         | Occasional      | Frequent               | Constant, or minimal due to exhaustion          |
| Stridor               | Minimal at rest | Easily audible at rest | Prominent inspiratory and occasional expiratory |
| Retractions           | None to mild    | Visible at rest        | Marked or severe                                |
| Agitation or lethargy | None            | None or limited        | Marked lethargy or agitation                    |
| Cyanosis              | None            | None                   | Dusky or cyanotic                               |

## More Information

### Version history

16 June 2026

1.1.0.1

**Update:** Assessment, red flags and referral guidance updated.

22 November 2022

1.0.1

**Update:** Backup, referrals, transport criteria and treatment instructions.

## If the airway is patent:

- Administer oxygen and/or ventilate via the stoma using a neonatal face mask if possible. If this proves difficult, consider intubation via the stoma if a tracheostomy tube is absent (or removed). This is the default route for oxygenation and ventilation as it is appropriate for both laryngectomy and tracheostomy patients.
- When ventilating via the stoma, assess for a significant air leak through the mouth. If this is present, cover the stoma with an occlusive dressing and ventilate through the upper airway in the normal way.
- If it is difficult to establish whether the patient has a laryngectomy or tracheostomy, provide oxygen via both the stoma and the mouth. Adding a mask to the patient's face may achieve additional oxygenation via a partially patent upper airway in the tracheostomy patient.
- Consider other causes for respiratory distress if the stoma is patent.

## If the airway is not patent:

- Inspect the stoma. Only remove potentially blocked devices if necessary and try to leave the outer tracheostomy tube and any internal prosthetic devices in place.
- Apply suction using a suction catheter.
- If normal breathing is restored following removal of a blocked inner tracheostomy tube, the tube can be cleaned and re-inserted.
- If a suction catheter cannot be passed:
  - The stoma or tube is likely to be completely blocked.
  - If the tube has a cuff, deflate it but do not remove the tube and re-attempt suction as material may have been dislodged.
  - If this approach fails, remove the tube completely.
- Ventilate the patient after the tube has been removed, using one of the following techniques:
  - A neonatal mask over the stoma.
  - A size 6 ETT via the stoma.
  - Cover the stoma and use a facemask or ETT via the mouth if air leaks from the mouth.

## Backup

- Backup from an ICP/CCP should be requested if the airway patency is not rapidly resolved or significant respiratory distress persists.

## Referral and transport

- Unless the problem is simple and easily resolved and the patient returns to their normal state, they should receive a clear recommendation to be transported to an ED by ambulance.

## Additional information

### General principles

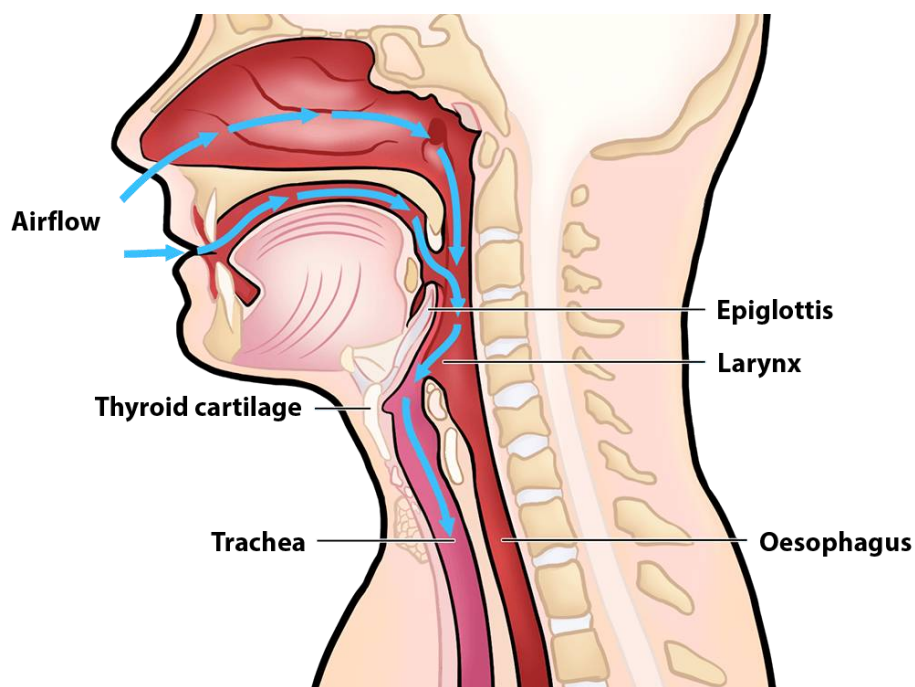
- Patients and carers will frequently be familiar with the management of tracheostomy / laryngectomy emergencies. Consider their advice and follow any action plans.
- Speaking aids should be transported with the patient whenever feasible.
- Features of concern associated with a stoma include:
  - Inflamed or swollen mucosa.
  - Exudate (discharge).
  - Growth of new tissue (granulation).
  - Pain around the site.
  - Bleeding around the site.
- Signs of laryngeal tube or stoma occlusion:
  - Respiratory distress or arrest.
  - Noisy breathing.
  - Blood, tissue or secretions around the stoma or tracheotomy tube if present.
  - Increased work of breathing.
  - Agitation, tachypnoea, diaphoresis, pallor or hypoxia.

### Tracheostomy

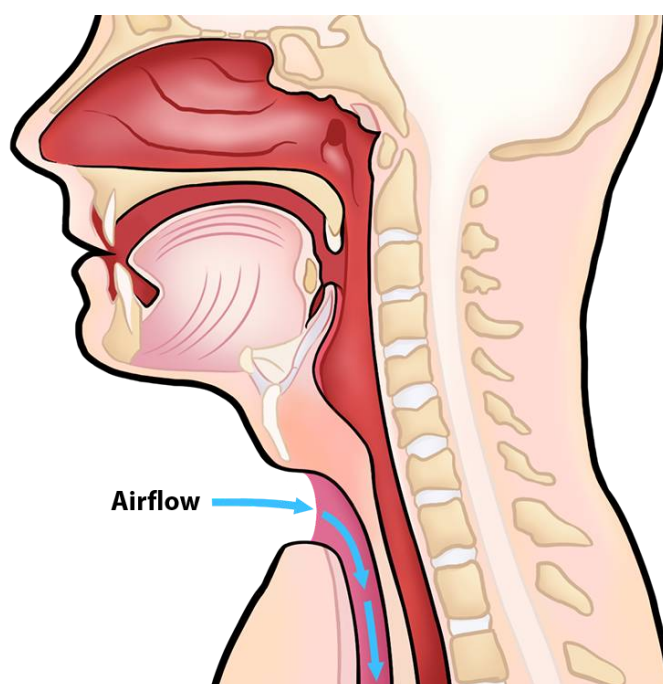
- A tracheostomy stoma is created from the skin into the trachea to form an airway below the larynx. The larynx remains intact.
- Tracheostomy patients can breathe through their stoma and potentially their mouth/nose to some degree.
- A tracheostomy tube with or without an inner tube may be present.

### Laryngectomy

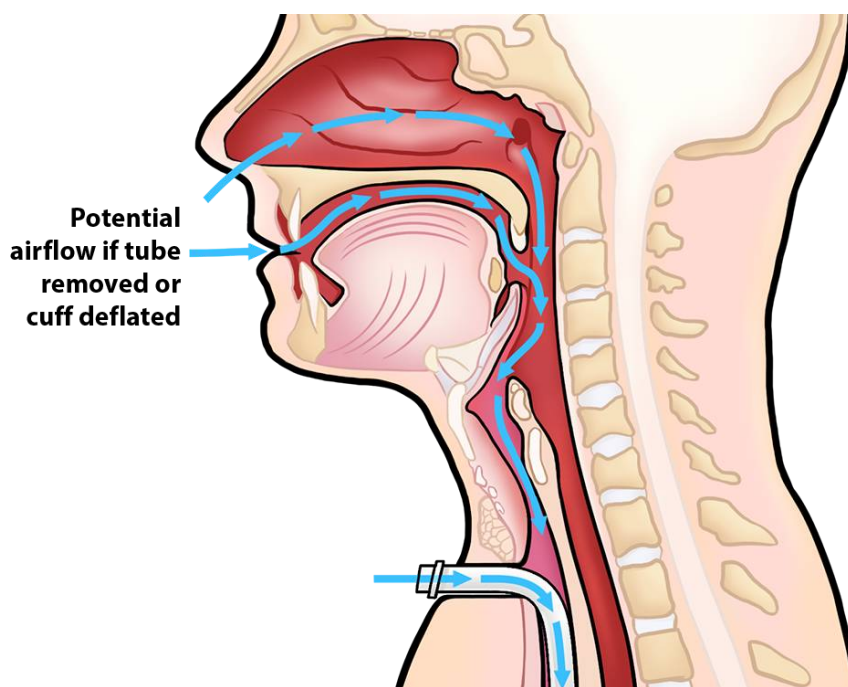
- In laryngectomy, the larynx is completely removed and a stoma is created from skin to trachea to form an airway. The trachea is only connected to the stoma; so these patients cannot be oxygenated, ventilated or intubated via the mouth.
- Usually, a tube will not be present in laryngectomy patients. Other devices such as laryngeal buttons, heat-moisture exchangers (HME) or tracheo-oesophageal puncture ('TOP') speaking valves may be seen.



Normal airway



Laryngectomy

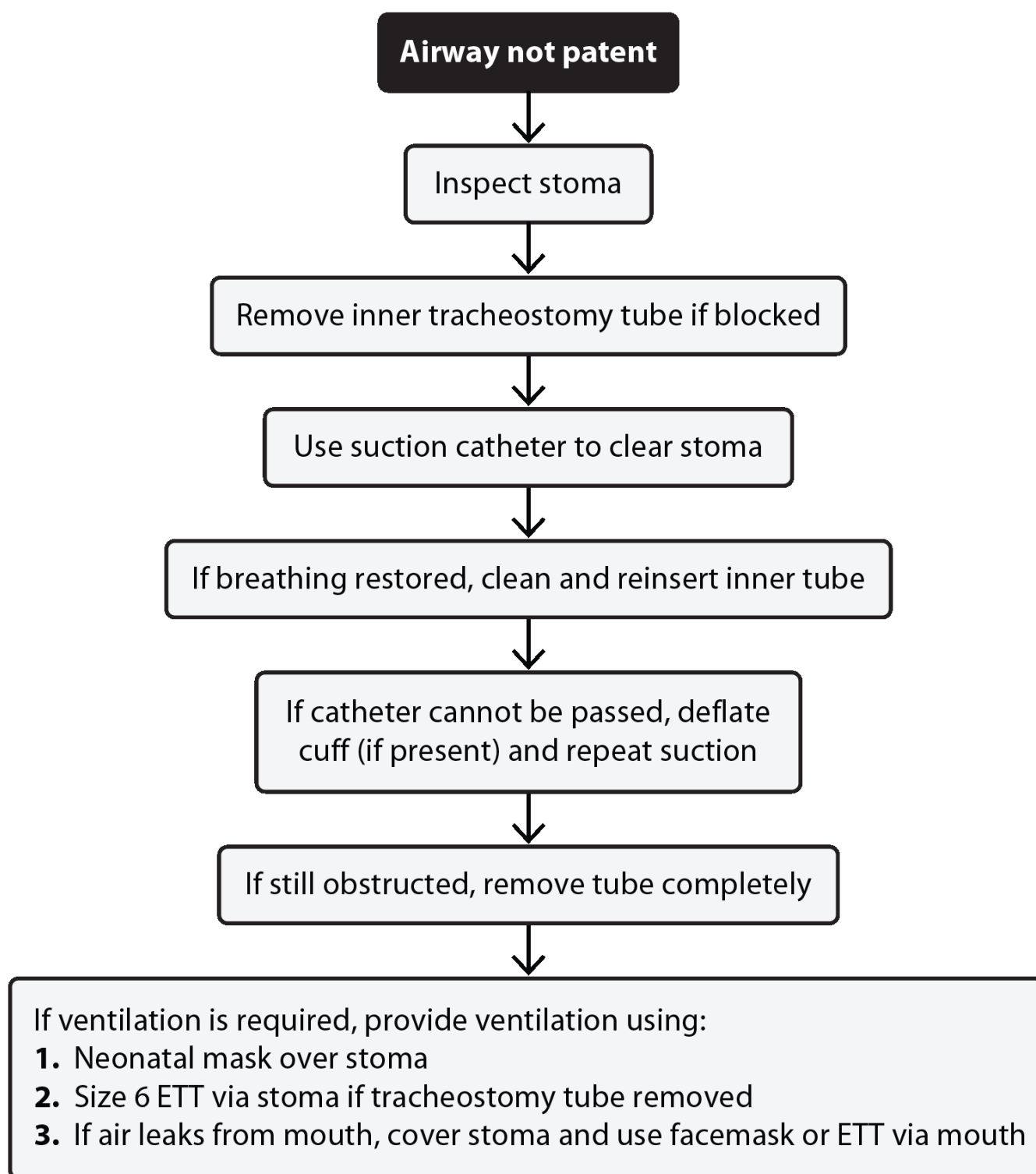


Tracheostomy

## Tracheostomy tubes

- There are many different brands and tubes available usually held in situ by tape around the neck.
- The tube is usually synthetic but some people may still have silver tubes:
  - Single lumen: An outer tube only and holds the tracheostomy open.
  - Double lumen: Within the outer tube is a disposable/reusable inner cannula which can be easily removed for cleaning.
  - Cuffed tubes are typically used for patients who are mechanically ventilated or cannot swallow. Cuffed tubes prevent air from escaping from around the tube, and prevent aspiration. A cuff needs to be deflated whilst speaking valves are in situ.





## More Information

## Version history

|                                 |       |
|---------------------------------|-------|
| 11 September 2023               | 1.0.4 |
| <b>New:</b> Guideline released. |       |

# Bronchiolitis

This guideline is for patients aged 2-23 months who have a provisional diagnosis of bronchiolitis. Refer to the [‘exacerbation of asthma’](#) guideline if the patient is aged greater than or equal to two years.

- Don airborne level PPE.

## Mild to moderate bronchiolitis

- Consider administering 0.9% sodium chloride nasal drops if nasal congestion is present.
- Consider administration of paracetamol if temperature is  $>39^{\circ}\text{C}$  and causing discomfort.
- Encourage small and frequent feeds.

## Severe bronchiolitis

- Administer oxygen if required to maintain an  $\text{SpO}_2$  of  $>92\%$ . This will usually only require 0.5-1 litres/minute via nasal prongs.
- Administer salbutamol if the patient is aged greater than 12 months:
  - Administer one puff of salbutamol via a metered dose inhaler (MDI) and spacer, followed by six breaths. Repeat this six times. Or,
  - Administer 5 mg of nebulised salbutamol using oxygen as the driving gas.
  - Repeat as required if there are clear signs of improvement.
- If respiratory arrest appears imminent, consider assisting ventilation with a manual ventilation bag and mask with PEEP.

## Backup

- Backup from a CCP should be requested if the patient has severe respiratory distress or is likely to require ventilatory support.

## Referral and transport

- If one or more red flags are present, the patient must be given a clear recommendation to be transported to an ED, usually by ambulance.
- If no red flags are present, the patient is usually suitable to be given a clear recommendation to remain in the community, with worsening advice, and be seen in primary care within 24 hours.

# Bronchiolitis

## Red flags (should be seen in ED, usually via ambulance)

- Features of moderate to severe bronchiolitis.
- Any history of apnoea during this illness.
- Lethargy or reduced alertness.
- Aged less than 12 weeks (corrected for prematurity).
- Dehydration.
- Temperature >39°C and aged less than one year.
- Chronic respiratory, cardiac, or neuromuscular disease.
- Suspected pneumonia.

## Worsening advice

- Provide parents/caregivers with verbal and written advice if the child is not immediately transported to a medical facility and consider photographing this advice using their phone. For example:
  - Ensure your child is seen in primary care within 24 hours.
  - Phone Healthline on 0800 611 116 if:
    - They have had no wet nappies for 12 hours, or
    - Their fluid intake has been less than half their usual amount for 12 hours.
  - Call 111 for an ambulance if:
    - Their lips or face goes blue, or
    - They cannot swallow, or
    - They are floppy or hard to wake, or
    - They are struggling to breathe, or
    - They have pauses in their breathing.

## Additional information

### General principles

- Bronchiolitis is a viral lower respiratory tract infection most often caused by respiratory syncytial virus (RSV), which is common in those aged less than one year.

# Bronchiolitis

- The illness usually peaks on day 3-5 and resolves over 7-10 days. The cough may persist for up to four weeks.
- Bronchiolitis is the most common cause of hospitalisation in children aged less than one year.
- Most cases of bronchiolitis are mild and self-limiting, but some develop severe illness which is associated with increased respiratory effort, cyanosis, difficulty feeding and dehydration.
- Thick secretions are typical, and younger infants may struggle to clear these secretions.
- The management of bronchiolitis is focused on supporting oxygenation and feeding until the patient recovers. There is no role for bronchodilators, steroids or antibiotics in patients aged less than 12 months, unless there is clinical concern that another condition is present (e.g., pneumonia or asthma). In older infants (greater than 12 months), a trial of salbutamol may be administered, but benefit should be assessed before repeat doses are provided.
- To correct the patient's age for prematurity, this is done by establishing the age they would be if born on their due date. For example, if the baby is 16 weeks old, but was born eight weeks premature, their corrected age is eight weeks.

## Nasal drops

- Nasal drops may clear thick secretions and improve work of breathing and feeding.
- To administer nasal drops:
  - Place the infant in a slight recline.
  - Instil 2-3 drops of 0.9% sodium chloride into each nostril and allow 20-30 seconds for loosening of secretions.
  - Gently clear the nasal passages with a syringe or soft suction catheter.
- Avoid deep or over-frequent suctioning, which may worsen mucosal swelling.

## Determining the severity of bronchiolitis table (not all features need to be present)

| Sign/symptom         | Mild                | Moderate          | Severe                     |
|----------------------|---------------------|-------------------|----------------------------|
| Behaviour            | Normal              | Some irritability | Irritable and/or lethargic |
| Respiratory rate     | Less than 60/minute | 60-69/minute      | ≥70/minute                 |
| Accessory muscle use | None to mild        | Moderate          | Marked                     |

# Bronchiolitis

|                     |                             |                                                           |                                                      |
|---------------------|-----------------------------|-----------------------------------------------------------|------------------------------------------------------|
| Wheeze              | None or end-expiratory only | Entire expiratory phase                                   | Inspiratory and expiratory                           |
| Oxygen saturations  | >92% (room air)             | Initial 90-92% (room air), corrected with low flow oxygen | Initial < 90% and not corrected with low flow oxygen |
| Apnoea              | None                        | Brief, not requiring stimulus to resolve                  | Increasingly frequent or prolonged                   |
| Heart rate          | <160/minute                 | 160-169/minute                                            | >170/minute                                          |
| Feeding             | Normal                      | Reduced                                                   | Reluctant or unable                                  |
| Wet nappies         | Normal                      | Less than half of usual                                   | None for 12 hours                                    |
| Early Warning Score | White zone                  | Yellow/orange zone                                        | Red/blue zone                                        |

## Risk factors for severe bronchiolitis

- There are a number of factors which increase the risk of developing severe bronchiolitis, which include:
  - Premature birth less than 37 weeks gestation.
  - Aged less than ten weeks.
  - Prenatal or postnatal smoke exposure.
  - Māori or Pacific People.
  - Low socioeconomic status.
  - Breastfed for less than two months.
  - Failure to thrive.
  - Chronic lung disease.
  - Congenital heart disease.
  - Chronic neurological disease.
- Parents/caregivers of patients with mild bronchiolitis and any of the above risk factors should be made aware of them, but alone they do not mean a referral to ED is required (rather close monitoring for deterioration is needed).

## Differentiating bronchiolitis from other respiratory illness

- Other illnesses may appear initially like bronchiolitis and should be considered in patients with mild illness.

# Bronchiolitis

- Pneumonia will often present with a fever greater than 39°C, or crackles in a single area of the chest on auscultation of the lungs. Infants with pneumonia will also often appear unwell.
- Asthma presents with wheeze, which can be brought on by a viral illness, however this is usually in older children. In general, asthma is not diagnosed in very young children.
- Inhaled foreign body will often have a history of a choking/gagging episode and/or focal findings in the chest. Patients with concern for an inhaled foreign body should be transported to an ED for further assessment.

## More Information

### Version history

16 June 2026

1.1.0.1

**New:** Guideline released.

# Myocardial Ischaemia

This guideline is for adults with suspected myocardial ischaemia. If STEMI is present, treat using the relevant STEMI guideline, which may be '[STEMI: primary PCI pathway](#)', or '[STEMI: fibrinolytic pathway](#)'.

- Acquire a 12 lead ECG.
  - Acquire an additional 12 lead ECG using a V4R lead position if an inferior STEMI is suspected.
  - Acquire an additional 12 lead ECG using V7-9 posterior lead positions if a posterior STEMI is suspected.
- Administer 300 mg of aspirin PO.
- Administer 0.4 mg of GTN SL.
- Repeat GTN SL every five minutes provided it relieves symptoms. Increase the dosing interval to 10 minutes if a caution is present.
- Administer an opiate IV if there is significant pain despite GTN.
- Acquire additional 12 lead ECGs every 15-30 minutes looking for evolving changes.

## Backup

- EMTs should request backup if the patient has significant pain despite GTN, noting that a Paramedic is appropriate for most patients.
- Backup from an ICP/CCP should be requested if the patient has:
  - Bradydysrhythmia, or
  - Tachydysrhythmia, or
  - Signs of shock.

## Referral and transport

- All patients with suspected myocardial ischaemia should receive a clear recommendation to be transported to a medical facility by ambulance.
- Transport should usually be to an ED but may be to a primary care facility if this is specifically equipped to investigate patients with suspected myocardial ischaemia.

## Additional information



# Myocardial Ischaemia

## General

- Chest pain/discomfort or shortness of breath without an obvious noncardiac cause in adult patients must be considered to be possibly myocardial ischaemia until proven otherwise. The distribution of the autonomic nerve supply to the intrathoracic and upper abdominal organs is such that the pain/discomfort from myocardial ischaemia may mimic the pain from many other causes in terms of location, sensation and radiation.
- Even if myocardial ischaemia is considered unlikely, a patient with chest pain/discomfort will usually require assessment in an ED because of the possibility of a potentially life-threatening cause, for example, pulmonary embolism, myocarditis, oesophageal tear/rupture, aortic dissection and pneumothorax.
- Personnel must have a very low threshold for clearly recommending that a patient with chest pain/discomfort or shortness of breath without an obvious cause is transported to an ED by ambulance.
- A 12 lead ECG should be obtained in all patients with either typical or atypical symptoms, noting that a normal 12 lead ECG does not rule out myocardial ischaemia. Up to 50% of patients having an acute myocardial infarction have a 12 lead ECG that is initially normal. In the absence of a clear diagnosis, a 12 lead ECG should be repeated every 15-30 minutes looking for evolving ECG changes.
- It is possible for young patients to have myocardial ischaemia. The cause is not usually atherosclerosis of the coronary arteries, but may be from other conditions such as coronary artery spasm, coronary artery dissection or aortic dissection. If a young patient has symptoms suggesting myocardial ischaemia the patient must be given a clear recommendation to be transported to a medical facility by ambulance.

## Atypical symptoms

- Some patients have atypical pain or discomfort, including any combination of face, jaw, neck, arm or upper abdominal discomfort.
- Treat as myocardial ischaemia if you suspect atypical symptoms are due to this.

## Silent myocardial ischaemia

- Some patients have silent myocardial ischaemia without pain or discomfort.
- Patients who are elderly or have diabetes are particularly at risk of this because they may have autonomic neuropathy.
- Symptoms may include shortness of breath, fatigue, weakness, non-specific malaise or feeling light-headed.

# Myocardial Ischaemia

- To make a provisional diagnosis of silent myocardial ischemia there must be signs of myocardial ischaemia on the 12 lead ECG. These signs include ST depression, T wave inversion and T wave flattening.
- The predominant indication for GTN is the presence of chest pain or discomfort, but GTN may be administered if the patient has silent myocardial ischaemia. Only administer repeat doses of GTN if it is clearly associated with improvement of the signs of ischaemia on the 12 lead ECG.

## Response and/or transport by helicopter

- Patients with myocardial ischaemia and not STEMI, should usually be transported by road.
- Transport by helicopter may be indicated if the patient has ongoing significant myocardial ischaemia despite treatment and it will save more than 60 minutes compared to road transport.
- For helicopter transport criteria to be met, the road transport time will usually need to be more than 120 minutes.

## More Information

### Version history

11 September 2023

1.0.4

**Update:** Multiple changes.

23 February 2023

1.0.2.1

**Update:** GTN cautions updated to align with GTN guideline.

# The Diagnosis of STEMI

- To have a clear diagnosis of STEMI the patient must have:
  - A clinical presentation consistent with STEMI, and
  - A 12 lead ECG meeting diagnostic criteria for STEMI, and
  - An absence of clear 12 lead ECG mimics of STEMI.
- The 12 lead ECG diagnostic criteria for STEMI in New Zealand have been defined by the National Cardiac Network as:
  - More than or equal to 2 mm (200  $\mu$ V) of ST elevation in two or more leads V1-3, or
  - More than or equal to 1 mm (100  $\mu$ V) of ST elevation in two or more contiguous leads in any other area.

## High risk ECG patterns without STEMI

- Consider transporting the patient to a hospital with the facilities for PCI, without activating the PCI pathway, if the patient has symptoms of myocardial ischaemia that are not clearly resolving with treatment, and
  - Isolated ST elevation in aVR, or
  - Wellens pattern, or
  - de Winter T waves.
- Seek clinical advice if transport will bypass a regional or base hospital, or if transport time will exceed 90 minutes.

## Additional information

### General

- While ST segment elevation is required for STEMI diagnosis, other concurrent ECG changes can support the diagnosis, including T wave changes, evolving ST elevation and reciprocal changes.
- Most monitors provide an automated analysis of the 12 lead ECG, but the analytic software has limitations which may indicate STEMI is present when it is not. Personnel must always independently evaluate the ECG and consider the clinical scenario before making a diagnosis.
- If STEMI is present, the ST elevation will usually be evolving and this is often most evident when two or more ECGs are acquired approximately 15 minutes apart.
- New left bundle branch block is a rare finding in STEMI and should not be considered diagnostic of STEMI without seeking clinical advice.

# The Diagnosis of STEMI

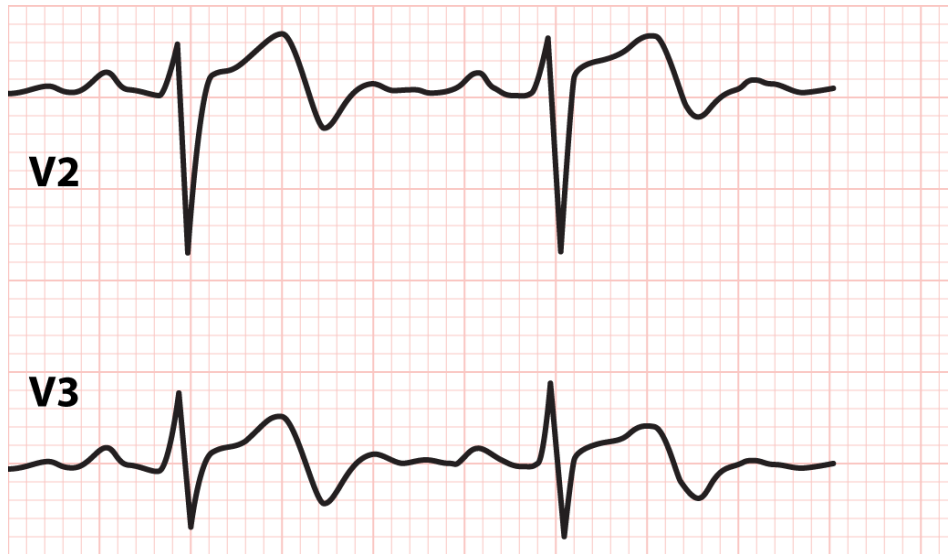
- Cardiac arrest may cause transient ST elevation to be present for 20-30 minutes after return of spontaneous circulation, particularly when adrenaline has been administered. Unless the STEMI was diagnosed prior to the cardiac arrest, ST elevation post cardiac arrest should not be immediately treated as STEMI. Consider delaying or repeating the 12 lead ECG 10 minutes following ROSC, obtain serial ECGs 15 minutes apart, and have a low threshold for seeking clinical advice.

## High risk ECG patterns

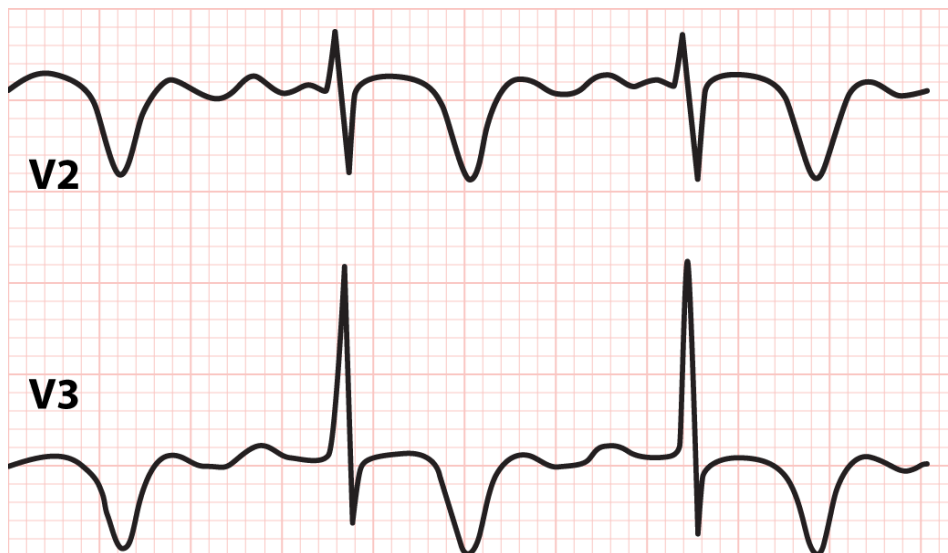
- Some high risk ECG patterns are associated with proximal occlusion of coronary arteries, but are not diagnostic of STEMI. Examples include:
  - Isolated ST elevation in aVR.
  - Wellens pattern (deeply inverted or biphasic T waves in leads V2-V3).
  - de Winter T waves (up sloping ST depression with peaked T waves in leads V2-V3).
- If the patient has a high risk ECG pattern they may need PCI, but they do not usually need immediate PCI.
- If the patient has a high risk ECG pattern, and symptoms of myocardial ischaemia that are not clearly resolving with treatment, they should be transported to a hospital with cardiac catheter facilities, without activating the PCI pathway, provided this is feasible and safe. Note that in this setting, the facilities for immediate PCI are not required.
- If transport will be prolonged, seek clinical advice, which may include administration of dual antiplatelet therapy and/or anticoagulation.

## Wellens pattern

# The Diagnosis of STEMI



**TYPE A: Biphasic T waves**



**TYPE B: Deeply and symmetrically inverted T waves**

## de Winter T waves



# The Diagnosis of STEMI

## STEMI mimics

### Common causes of ST elevation

- Acute pericarditis or myocarditis
- Benign early repolarisation
- Left ventricular aneurysm
- Left bundle branch block
- Left ventricular hypertrophy
- Ventricular paced rhythm
- Hyperkalaemia
- Non-ACS myocardial injury
- Subarachnoid haemorrhage
- Brugada syndrome
- Post cardiac arrest

### Common causes of ST depression

- Bundle branch block.
- Left ventricular hypertrophy
- Ventricular paced rhythm
- Digoxin toxicity
- Tachycardia/rate-related
- Non-ACS myocardial injury

## More Information

### Version history

16 June 2026

1.1.0.1

**Update:** ECG interpretation guidance updated.

11 September 2023

1.0.4

**New:** Derived from previous 'Assessment for myocardial ischemia' guideline. Left bundle branch known to be new removed.

# STEMI: Primary PCI Pathway

This guideline is for adults with ST elevation myocardial infarction (STEMI) where transport to a hospital with facilities for immediate PCI can clearly occur within 90 minutes of STEMI diagnosis.

- Transmit the 12 lead ECG to the destination hospital.
- Administer 300 mg of aspirin PO.
- Confirm whether any of the following cautions to PCI are present:
  - Time of symptom onset was greater than 12 hours ago.
  - Requires assistance with their activities of daily living.
  - Severe comorbidities that significantly limit their life expectancy.
- Activate the primary PCI pathway, ensuring the following aspects are discussed if this is occurring by phone:
  - The details of any cautions to PCI that are present.
  - Confirm if ticagrelor is to be administered.
- Gain IV access, preferably in the left arm.
- Administer 180 mg of ticagrelor PO if asked to do so by the receiving clinician.
- Administer an opiate IV as required for moderate to severe pain.
- Administer 0.4 mg of GTN SL with caution and withhold GTN SL if signs of poor perfusion are present.
- If cardiac arrest occurs:
  - Commence resuscitation using the '[cardiac arrest](#)' guideline.
  - Administer tenecteplase IV (from the tables in the '[fibrinolysis](#)' guideline) if ROSC does not occur within five minutes of commencing resuscitation.
- On arrival at hospital transfer the patient directly to the PCI lab, whenever this is feasible.

## Activating the primary PCI pathway

- North Shore Hospital:
  - Available if the patient will arrive in ED between 0730hrs and 1530hrs, Monday to Friday (excluding public holidays).
  - Place an early radio call to ED.
  - HHStJ personnel: Lifepak 15 transmission site codes:
    - Monday-Friday 0730-1530 NR03.

# STEMI: Primary PCI Pathway

- Auckland City Hospital:
  - Available 24/7.
  - Auckland area: place an early radio call to ED if the patient will arrive in ED between 0730hrs and 1530hrs, Monday to Friday. Outside of these hours phone the interventional cardiologist on 0800 366 666.
  - Northland area: phone the interventional cardiologist on 0800 366 666.
  - HHStJ personnel: Lifepak 15 transmission site code NR01.
- Middlemore Hospital:
  - Available if the patient will arrive in ED between 0730hrs and 1530hrs, Monday to Friday (excluding public holidays).
  - Place an early radio call to ED.
  - HHStJ personnel: Lifepak 15 transmission site codes:
    - Monday-Friday 0730-1530 NR02.
- Waikato Hospital:
  - Available 24/7.
  - Phone the on call cardiologist on 07 839 8676.
  - HHStJ personnel: Lifepak 15 transmission site code CR01 or NR07.
- Tauranga Hospital:
  - Available if the patient will arrive in ED between 0730hrs and 1530hrs Monday to Friday (excluding public holidays).
  - Phone the cardiac catheter lab coordinator on 027 7047712.
  - HHStJ personnel: Lifepak 15 transmission site code CR04.
- Wellington Hospital
  - Available 24/7.
  - WFA personnel: phone the Cardiology Nurse Coordinator on 021 504746.
  - All other personnel: call the on call doctor for HHStJ via the Clinical Desk.
  - HHStJ personnel: Lifepak 15 transmission site code CLINICAL DESK.
- Nelson Hospital:
  - Available if the patient will arrive in ED between 0800hrs and 1600hrs, Monday to Friday (excluding public holidays).
  - Call the on call cardiologist on 0800 728 4929, extension 2.
  - HHStJ personnel: Lifepak 15 transmission site code SI02.



# STEMI: Primary PCI Pathway

- Christchurch Hospital:
  - Available 24/7.
  - Call the on call cardiology registrar on 0800 728 4929, extension 1.
  - HHStJ personnel: Lifepak 15 transmission site code SI01.
- Dunedin Hospital:
  - Available 24/7.
  - Call the on call cardiologist on 0800 728 4929, extension 3.
  - HHStJ personnel: Lifepak 15 transmission site code SI03.
- Seek clinical advice if contact is not able to be made with the clinician listed above.

## Backup

- If a Paramedic is present, backup from an ICP/CCP is not routinely required.
- EMTs must request backup from a Paramedic or ICP/CCP.
- Backup from an ICP/CCP should be requested if the patient has:
  - Bradydysrhythmia, or
  - Tachydysrhythmia, or
  - Signs of shock.

## Additional information

### General principles

- Minimise scene time and avoid unnecessary delays as outcomes are primarily dependent on how quickly the occluded coronary artery is opened.
- Primary PCI is the preferred reperfusion pathway provided the patient can clearly be transported to a hospital with the facilities for immediate PCI within 90 minutes of the diagnosis of STEMI. However, the balance of risk is in favour of administering fibrinolysis if there will be any delays preventing transport being achieved within 90 minutes.
- Be prepared to treat cardiac arrest but do not routinely attach defibrillation pads unless the patient is being transported by helicopter. Cardiac arrest will occur in approximately 5% of patients, the initial rhythm will usually be ventricular fibrillation, and most patients will achieve return of spontaneous circulation after 1-2 shocks.

# STEMI: Primary PCI Pathway

- IV access is preferably gained in the left arm because the interventional cardiologist will usually gain access via the right radial artery.
- Have a low threshold for administering antiemetics.
- Severe comorbidities that significantly limit life expectancy include dementia, severe COPD, severe heart failure, a requirement for aged residential care and metastatic cancer with weight loss.

## Response and/or transport by helicopter

- Response by helicopter may be indicated if a CCP is required for the management of a dysrhythmia or cardiogenic shock, and helicopter arrival at scene is at least 15 minutes faster than a road response.
- Transport by helicopter may be indicated if the patient is following the primary PCI pathway, and it will save more than 15 minutes compared to road transport.
- For helicopter transport criteria to be met, the road transport time will usually need to be more than 75 minutes.

## More Information

### Version history

11 September 2023

1.0.4

**New:** Guideline released.

# STEMI: Fibrinolysis Pathway

This guideline is for adults with ST elevation myocardial infarction (STEMI) where transport to a hospital with facilities for immediate PCI cannot clearly occur within 90 minutes of STEMI diagnosis.

- Transmit the 12 lead ECG to the destination hospital.
- Administer 300 mg of aspirin PO.
- Complete the **fibrinolytic therapy checklist** (below). Seek urgent clinical advice prior to fibrinolysis if any cautions are present.
- Ensure hypertension (a systolic BP greater than 180 mmHg or a diastolic BP greater than 110 mmHg) is controlled before fibrinolysis. See **controlling hypertension** (below) if hypertension is present.
- Administer fibrinolysis (see **dosing tables** below).
- If cardiac arrest occurs prior to fibrinolysis:
  - Commence resuscitation using the '[cardiac arrest](#)' guideline.
  - Administer IV tenecteplase (see **dosing tables** below) if ROSC does not occur within five minutes of commencing resuscitation.
- Administer an opiate IV as required for moderate to severe pain.
- Administer 0.4 mg GTN SL with caution and withhold GTN SL if signs of poor perfusion are present.
- Transport the patient to a hospital with the facilities for immediate PCI and contact the appropriate clinician (see **hospital facilities** below).
- Enroute to hospital:
  - Measure vital signs every ten minutes and continue to ensure hypertension (if present) is controlled.
  - Monitor the patient closely for signs of complications, particularly bleeding.
  - Repeat a 12 lead ECG every 30 minutes but do not routinely transmit these.

## Fibrinolytic therapy checklist

- Seek clinical advice if any of the following are present:
  - Suspected aortic dissection.
  - Major surgery, severe trauma, or severe brain injury within the last six weeks.
  - Intracranial surgery within the last six months.
  - Ischaemic stroke within the last six months.
  - Previous intracerebral haemorrhage.

# STEMI: Fibrinolysis Pathway

- » Known cerebral aneurysm, arteriovenous malformation, or tumour.
- » More than ten minutes of CPR.
- » Non-compressible vascular puncture within the last 24 hours.
- » Internal bleeding within the last six weeks.
- » Lumbar puncture or epidural insertion within the last six weeks.
- » TIA within the last three months.
- » Known bleeding disorder.
- » Taking an anticoagulant.
- » A systolic BP greater than 180 mmHg or a diastolic BP greater than 110 mmHg despite treatment.
- » Known to be pregnant or less than two weeks postpartum.
- » The time of symptom onset was greater than 12 hours ago.
- » Requires assistance with their activities of daily living.
- » Severe comorbidities that significantly limit their life expectancy.

## Controlling hypertension

- Ensure an opiate has been administered IV, and be generous with dosing.
- If the systolic BP is greater than 180 mmHg or the diastolic BP is greater than 110 mmHg, commence the following treatment to maintain a systolic BP of 140-160 mmHg and a diastolic BP below 110 mmHg:
  - » Administer 0.4 - 0.8 mg of GTN SL every five minutes, as required. Move to the next step if hypertension is not controlled within approximately 10 minutes of commencing GTN.
  - » Continue to administer GTN SL and administer labetalol IV:
    - » Administer a 20 mg bolus IV every five minutes, as required.
    - » If more than three doses are required, commence an IV infusion at 1 mg/kg/hour, up to a maximum of 150 mg/hour.
    - » Adjust the rate as required by 20 mg/hour every 10 minutes, titrating to effect.
    - » Administer additional boluses of 20 mg IV every five minutes as required.
  - » If labetalol is contraindicated, or the target BP is not met within 30 minutes of commencing labetalol, seek clinical advice to discuss further options.

## Fibrinolysis if aged less than 75 years

- Administer tenecteplase IV as per the dosing table below.
- Administer enoxaparin SC as per the dosing table below.

# STEMI: Fibrinolysis Pathway

- Administer 300 mg of clopidogrel PO.
- Administer 30 mg of enoxaparin IV:
  - Dilute 100 mg to a total of 10 ml using 0.9% sodium chloride.
  - Administer 30 mg (3 ml) IV.

## Tenecteplase IV

## Enoxaparin SC

|          | Tenecteplase IV | Enoxaparin SC  |
|----------|-----------------|----------------|
| < 60 kg  | 30 mg (6 ml)    | 60 mg (0.6 ml) |
| 60-69 kg | 35 mg (7 ml)    | 70 mg (0.7 ml) |
| 70-79 kg | 40 mg (8 ml)    | 80 mg (0.8 ml) |
| 80-89 kg | 45 mg (9 ml)    | 90 mg (0.9 ml) |
| ≥ 90 kg  | 50 mg (10 ml)   | 100 mg (1 ml)  |

## Fibrinolysis if aged greater than or equal to 75 years

- Administer tenecteplase IV as per the dosing table below.
- Administer enoxaparin SC as per the dosing table below.
- Administer 75 mg of clopidogrel PO.
- Do not administer enoxaparin IV.

## Tenecteplase IV

## Enoxaparin SC

|          | Tenecteplase IV  | Enoxaparin SC   |
|----------|------------------|-----------------|
| < 60 kg  | 15 mg (3 ml)     | 45 mg (0.45 ml) |
| 60-69 kg | 17.5 mg (3.5 ml) | 50 mg (0.5 ml)  |
| 70-79 kg | 20 mg (4 ml)     | 60 mg (0.6 ml)  |
| 80-89 kg | 22.5 mg (4.5 ml) | 70 mg (0.7 ml)  |
| ≥ 90 kg  | 25 mg (5 ml)     | 75 mg (0.75 ml) |

## Hospitals with the facilities for immediate PCI

- Auckland City Hospital:

# STEMI: Fibrinolysis Pathway

- » Available 24/7.
- » Call the cardiology registrar on 021 906624.
- » HHStJ personnel: Lifepak 15 transmission site code NR01.

## ➤ Waikato Hospital:

- » Available 24/7.
- » Call the cardiology registrar on 07 839 8676.
- » HHStJ personnel: Lifepak 15 transmission site code CR01 or NR07.

## ➤ Tauranga Hospital:

- » Available if the patient will arrive in ED between 0730hrs and 1530hrs Monday to Friday (excluding public holidays).
- » Phone the cardiac catheter lab coordinator on 027 7047712.
- » HHStJ personnel: Lifepak 15 transmission site code CR04.

## ➤ Wellington Hospital:

- » Available 24/7.
- » WFA personnel: call the Cardiology Nurse Coordinator on 021 504746.
- » All other personnel: call the on call doctor for HHStJ via the Clinical Desk.
- » HHStJ personnel: Lifepak 15 transmission site code CLINICAL DESK.

## ➤ Nelson Hospital:

- » Availability is variable.
- » Call the on call cardiologist on 0800 728 4929, extension 2, to confirm availability prior to transport.
- » HHStJ personnel: Lifepak 15 transmission site code SI02.

## ➤ Christchurch Hospital:

- » Available 24/7.
- » Call the cardiology registrar on 0800 728 4929, extension 1.
- » HHStJ personnel: Lifepak 15 transmission site code SI01.

## ➤ Dunedin Hospital:

- » Available 24/7.
- » Call the on call cardiologist on 0800 728 4929, extension 3.
- » HHStJ personnel: Lifepak 15 transmission site code SI03.

# STEMI: Fibrinolysis Pathway

## Backup

- Backup from an ICP/CCP should be requested if the patient has:
  - Bradydysrhythmia, or
  - Tachydysrhythmia, or
  - Signs of shock.

## Additional information

### General principles

- Fibrinolytic therapy must be initiated as soon as possible because outcomes are predominantly determined by the time to reperfusion of the coronary artery. For example, fibrinolytic therapy has a higher priority than administering pain relief or commencing extrication.
- If the patient is being transported to a medical facility for fibrinolytic therapy personnel should initiate staging via the Air Desk as soon as is feasible.
- Fibrinolytic therapy may be administered via IO access if IV access cannot be obtained, but IO access must be secure.
- Be prepared to treat cardiac arrest but do not routinely attach defibrillation pads unless the patient is being transported by helicopter. Cardiac arrest will occur in approximately 5% of patients, the initial rhythm will usually be ventricular fibrillation, and most patients will achieve return of spontaneous circulation after 1-2 shocks.
- Following fibrinolytic therapy, the patient should be transported to a hospital with facilities for immediate PCI, whenever this is feasible and safe, because approximately 30% of patients receiving fibrinolytic therapy will not achieve reperfusion and will require rescue PCI.
- Some patients with severe comorbidities may not be eligible for rescue PCI and this will be determined when seeking clinical advice following the checklist. In this setting patients should be transported to the most appropriate hospital, in discussion with the advising clinician.
- Have a low threshold for administering antiemetics.

### The fibrinolytic therapy checklist

- The conditions within the checklist place the patient at increased risk of bleeding, and/or alter the risk/benefit balance, associated with fibrinolytic therapy.

# STEMI: Fibrinolysis Pathway

- Aortic dissection may involve the coronary arteries and cause STEMI. Aortic dissection usually presents with a sudden onset of severe pain that is maximal at the time of onset and often described as sharp, tearing or stabbing and may radiate to the back.
- Major surgery is any form of surgery where if subsequent bleeding developed, it could be life-threatening. Examples include surgery within a body cavity, surgery involving the spine and any form of joint replacement.
- Severe trauma is trauma severe enough to cause an injury where if subsequent bleeding developed, it could be life-threatening. Examples include rib fractures, intra-abdominal injury, and pelvic fractures.
- Severe brain injury is an injury that was severe enough to result in an injury visible on a CT scan.
- Intracranial surgery includes any operation or procedure involving the brain.
- Ischaemic stroke within the last six months includes thrombotic and embolic strokes, even if the patient was successfully treated.
- Previous intracerebral haemorrhage includes all forms of previous bleeding within the brain.
- A cerebral aneurysm is a known weakness in the wall of a cerebral artery causing dilation of the vessel. If the patient has not heard the term 'cerebral or brain aneurysm' it is highly unlikely the patient has one.
- An arteriovenous (AV) malformation is an abnormal connection between an artery and a vein. The vein is exposed to arterial pressure because there is no capillary bed to reduce the pressure so the vein dilates and may rupture. If the patient has not heard of the term 'AV malformation' it is highly unlikely the patient has one.
- A cerebral tumour is any form of brain tumour, primary or metastatic.
- Non-compressible vascular puncture includes any invasive procedure involving a non-compressible site, for example an organ biopsy.
- Internal bleeding includes gastrointestinal bleeding, bleeding from the urinary tract and bleeding from the lung. 'Coffee ground' vomit is very non-specific and is not a reliable clinical sign of gastrointestinal bleeding. Normal menstrual bleeding is not considered to be internal bleeding.
- A lumbar puncture is a procedure in which a needle is inserted between two lumbar vertebrae, most commonly to collect cerebrospinal fluid for diagnostic testing. An epidural insertion is a procedure in which a needle is inserted between two vertebrae, most commonly for the placement of a local anaesthetic agent.
- Anticoagulants include medicines such as warfarin, dabigatran, rivaroxaban and apixaban, but not antiplatelet medicines such as aspirin, clopidogrel and ticagrelor.



# STEMI: Fibrinolysis Pathway

- Hypertension must be controlled before and after fibrinolysis because uncontrolled hypertension increases the risk of intracranial haemorrhage following fibrinolysis. Paramedics should seek clinical advice regarding labetalol administration if an ICP/CCP is not immediately available and GTN administration has not successfully decreased the patient's blood pressure.
- Severe comorbidities that significantly limit their life expectancy include dementia, severe COPD, severe heart failure, a requirement for aged residential care and metastatic cancer with weight loss.

## Informed consent

- Written consent is not required for fibrinolytic therapy, but the patient needs to provide informed consent. For example, inform the patient that:
  - They are having a 'heart attack' and that this is life-threatening.
  - The 'heart attack' is due to a clot blocking one of the blood vessels in their heart.
  - A medicine will be administered to help break down the clot and this reduces the chance they will die or be left with permanent heart damage.
  - The earlier the medicine is administered, the more likely it is to work.
  - Rarely the medicine can cause bleeding, for example inside the body. Very rarely this bleeding can cause death but the risk of this is very low.
  - The medicine is more likely to help them than to harm them.

## Complications following fibrinolytic therapy

- Treat the patient using the appropriate guideline if complications from STEMI, for example dysrhythmia, occur.
- Seek clinical advice if complications from fibrinolytic therapy occur, for example bleeding.

## Response and/or transport by helicopter

- Time to a PCI-capable hospital is generally less critical following fibrinolysis, however rescue PCI may be required in some cases.
- Response by helicopter may be indicated if a CCP is required for the management of a dysrhythmia or cardiogenic shock, and helicopter arrival at scene is at least 15 minutes faster than a road response.
- Transport by helicopter may be indicated if it will save more than 30 minutes compared to road transport.
- For helicopter transport criteria to be met, the road transport time will usually need to be more than 90 minutes.

# STEMI: Fibrinolysis Pathway

## More Information

### Version history

|                                                                                                             |         |
|-------------------------------------------------------------------------------------------------------------|---------|
| <b>3 August 2026</b><br><b>Update:</b> Anticoagulant treatment for fibrinolysis.                            | 1.1.0.6 |
| <b>16 June 2026</b><br><b>Update:</b> Treatment guidance updated.                                           | 1.1.0.1 |
| <b>11 September 2023</b><br><b>New:</b> Guideline redeveloped.                                              | 1.0.4   |
| <b>23 February 2023</b><br><b>Update:</b> Fixed drug volume error in tenecteplase volume for >90kg patient. | 1.0.2.1 |

- Receive a handover from the treating staff including:
  - Where in the receiving hospital the patient is expected.
  - Who the accepting clinician is.
  - Whether or not the accepting clinician requires an update en route.
- Commence transport without delay.
- Monitor and record vital signs every 10 minutes.
- Seek clinical advice for any significant changes in patient condition that are not easily managed.

## Treatment of hypertension if fibrinolysis has been administered

- If the systolic BP is greater than 180 mmHg or the diastolic BP is greater than 110 mmHg, commence the following treatment to maintain a systolic BP of 140-160 mmHg and a diastolic BP below 110 mmHg:
  - Ensure an opiate has been administered IV and be generous with dosing.
  - Administer 0.4 - 0.8 mg of GTN SL every five minutes, as required. Move to the next step if hypertension is not controlled within approximately 10 minutes of commencing GTN.
  - Continue to administer GTN SL and administer labetalol IV:
    - Administer a 20 mg bolus IV every five minutes, as required, while setting up an infusion.
    - Commence an IV infusion at 1 mg/kg/hour, up to a maximum of 150 mg/hour.
    - Adjust the rate as required by 20 mg/hour every 10 minutes, titrating to effect.
    - Administer additional boluses of 20 mg IV every five minutes as required.
- If labetalol is contraindicated, or target BP is not met within 30 minutes of commencing labetalol, seek clinical advice to discuss further options.

## Additional information

### General principles

- Transfer may be occurring because the patient:
  - Is being transferred for primary PCI, or
  - Has received fibrinolytic therapy and is being transferred so that they are in the right hospital for immediate coronary intervention (rescue PCI) in the event of failure of reperfusion.
- Be prepared to treat cardiac arrest but do not routinely attach defibrillation pads unless the patient is being transported by helicopter.
- Have a low threshold for administering antiemetics.

## More Information

### Version history

|                                                                                               |         |
|-----------------------------------------------------------------------------------------------|---------|
| <b>16 June 2026</b><br><b>Update:</b> Inter-hospital transfer and treatment guidance updated. | 1.1.0.1 |
| <b>11 September 2023</b><br><b>Update:</b> Multiple changes.                                  | 1.0.4   |

# Cardiogenic Pulmonary Oedema

This guideline is for adults with a provisional diagnosis of cardiogenic pulmonary oedema.

- Acquire a 12 lead ECG.
- Administer 0.8 mg of GTN SL.
- Continue to administer 0.8 mg GTN SL every five minutes as above if the patient is not improving. Reduce the dose to 0.4 mg SL and consider increasing the dosing interval to 10 minutes if caution is present.
- Apply CPAP if the patient has moderate to severe respiratory distress that is not rapidly improving (see the '[NIV](#)' guideline). If backup for CPAP is unavailable or delayed, apply PEEP:
  - Apply PEEP at 10 cmH<sub>2</sub>O. Focus on ensuring a tight seal with the mask and do not assist the patient's breathing unless it is ineffective.
  - Consider increasing PEEP to 15 cmH<sub>2</sub>O if the patient is not improving.
  - Use PEEP with caution if the patient has signs of shock.
- Commence a GTN infusion IV if the patient is not rapidly improving:
  - Commence the infusion at 2 mg/hr.
  - Adjust the rate as required by 0.5 mg/hr every 5-10 minutes, titrating to effect.
- Consider administering 10-20 mcg of fentanyl IV or 1-2 mg of morphine IV every five minutes as required, for severe anxiety.

## Backup

- Backup from an ICP/CCP should be requested if the patient has:
  - Moderate to severe respiratory distress which is not improving, or
  - STEMI, or
  - Bradydysrhythmia, or
  - Tachydysrhythmia, or
  - Signs of shock.

## Referral and transport

- All patients treated for cardiogenic pulmonary oedema must be given a clear recommendation to be transported to a medical facility by ambulance.

# Cardiogenic Pulmonary Oedema

- The medical facility should usually be an ED, unless the patient is rapidly improving and can be transported to an appropriate primary care facility.

## Additional information

### General

- Cardiogenic pulmonary oedema may present as a new onset of symptoms (commonly caused by myocardial ischaemia involving the left ventricle), or as an acute decompensation in a patient with heart failure.
- Patients with a history of chronic heart failure may have crackles on auscultation but no increased shortness of breath. These patients do not require treatment under this guideline.
- Opiates are not a treatment for cardiogenic pulmonary oedema but may be used for symptom relief of severe anxiety.
- Allow the patient to adopt the most comfortable position, placing their legs in a dependent position if this is feasible.

## The differential diagnosis

- Cardiogenic pulmonary oedema. Patients usually:
  - Have been supine, for example in bed, prior to the onset of symptoms. They may have an audible wheeze. If so, the wheeze is usually worse bilaterally in the lower zones.
  - Have crackles, worse in the lower zones.
  - Are hypertensive, clammy and peripherally vasoconstricted.
- COPD. Patients usually:
  - Have an established COPD diagnosis.
  - Have a history of smoking.
  - Are not symptom free between exacerbations.
- Asthma. Patients usually:
  - Have an established asthma diagnosis.
  - Are symptom free between exacerbations.
- Pneumonia. Patients usually have:
  - Signs of sepsis.
  - A productive cough.

# Cardiogenic Pulmonary Oedema

- » Wheeze or crackles that is unilateral or limited to one lobe/area.
- Negative pressure pulmonary oedema. Patients usually have:
  - » A history of an obstructed airway.
  - » Crackles which are widespread and uniform throughout the lung fields.
  - » Signs of hypovolaemia.

## Negative pressure pulmonary oedema

- During any form of asphyxiation, negative pressure can develop within the thoracic cavity as a result of inspiratory efforts occurring against a closed airway.
- This negative pressure can result in the rapid development of pulmonary oedema because the pressure gradient generated between lung capillaries and alveoli can result in fluid moving from capillaries into alveoli.
- The patient will have signs and symptoms of pulmonary oedema and will also usually have signs of hypovolaemia because the fluid that has entered the lungs has come from the circulation.
- Do not administer GTN but it is appropriate to administer CPAP or PEEP. Refer to the relevant guideline.

## More Information

### Version history

|                                                                                                   |         |
|---------------------------------------------------------------------------------------------------|---------|
| <b>16 June 2026</b>                                                                               | 1.1.0.1 |
| <b>Update:</b> Treatment guidance updated.                                                        |         |
| <b>11 September 2023</b>                                                                          | 1.0.4   |
| <b>Update:</b> Multiple changes.                                                                  |         |
| <b>22 November 2022</b>                                                                           | 1.0.1   |
| <b>Update:</b> CPAP and PEEP. Added phosphodiesterase inhibitors, removed cautions for GTN spray. |         |

- The treatment and the urgency of treatment provided to a patient with a dysrhythmia is determined by the level of their cardiovascular compromise.
- There is a continuum of compromise and determining the level of compromise requires clinical judgement that divides this continuum into four categories.

## Not compromised

- The patient is not compromised if their vital signs are normal (aside from heart rate which is abnormal by definition) and they have no symptoms of myocardial ischaemia.
- The assessment of the level of cardiovascular compromise can be further validated by a visual assessment of the status code. If the patient 'looks status four' the patient is usually not compromised.

## Mildly compromised

- The patient is mildly compromised if their vital signs are normal (ignoring the heart rate which is abnormal by definition), and they have mild symptoms of myocardial ischaemia. For example, when the patient has:
  - Normal blood pressure and peripheral capillary refill time, and
  - Normal level of consciousness, and
  - Mild shortness of breath, or
  - Mild chest pain/discomfort.
- The assessment of the level of cardiovascular compromise can be further validated by a visual assessment of the status code. If the patient 'looks status three' the patient is usually mildly compromised.

## Moderately compromised

- The patient is moderately compromised if their vital signs are near normal (ignoring the heart rate which is abnormal by definition), and they have moderate symptoms of myocardial ischaemia. For example, when the patient has:
  - Normal blood pressure with a prolonged capillary refill time, and
  - Normal level of consciousness, and
  - Moderate shortness of breath, or
  - Moderate chest pain/discomfort.
- The assessment of the level of cardiovascular compromise can be further validated by a visual assessment of the status code. If the patient 'looks status two' the patient is usually moderately compromised.



## Severely compromised

- The patient is severely compromised if their vital signs are markedly abnormal (ignoring heart rate which is abnormal by definition) or they have severe symptoms of myocardial ischaemia. For example, when the patient has:
  - » Hypotension, or
  - » An altered level of consciousness, or
  - » Severe shortness of breath, or
  - » Severe chest pain/discomfort.
- The assessment of the level of cardiovascular compromise can be further validated by a visual assessment of the status code. If the patient 'looks status one' the patient is usually severely compromised.

## Summary table

### Not compromised

- Normal vital signs.
- No symptoms of myocardial ischaemia.
- 'Looks status four'.

### Mildly compromised

- Normal vital signs.
- Mild shortness of breath.
- Mild symptoms of myocardial ischaemia.
- 'Looks status three'.

## Moderately compromised

- Normal BP but prolonged CRT.
- Moderate shortness of breath.
- Moderate symptoms of myocardial ischaemia.
- 'Looks status two'.

## Severely compromised

- Hypotension.
- Altered level of consciousness.
- Severe shortness of breath.
- Severe symptoms of myocardial ischaemia.
- 'Looks status one'.

## More Information

### Version history

16 June 2026

1.1.0.1

**Update:** Cardiovascular compromise assessment updated.

11 September 2023

1.0.4

**Update:** Guideline renamed. Summary table simplified.

# Ventricular Tachycardia

This guideline is for adults with ventricular tachycardia (VT), or an undifferentiated broad complex tachycardia with a ventricular rate greater than or equal to 150/minute, provided the patient is not in cardiac arrest. Seek clinical advice if the patient is a child.

- Attach defibrillation pads in the anterior/posterior position when feasible.
- Acquire a 12 lead ECG.
- Do not administer GTN, even if the patient has cardiac chest pain.
- Determine the level of 'cardiovascular compromise'.
- If VT is secondary to cyclic antidepressant poisoning, treat as per the 'poisoning from medicines' guideline.

## If the patient is not compromised or is mildly compromised

- Administer 300 mg of amiodarone IV over approximately 30 minutes.
- Administer a further 150 mg of amiodarone IV over approximately 30 minutes if the patient remains in VT.

## If the patient is moderately compromised or is severely compromised

- Administer a 'dissociative dose' of ketamine unless unconscious, and
- Cardiovert ('action list') using maximum joules in synchronised mode.
- Cardiovert up to a total of three times if the rhythm fails to revert.
- Seek clinical advice if the rhythm does not revert.

## If torsades de pointes is present or has occurred

- Cardiovert as above.
- Administer 10 mmol (2.5 g) of magnesium IV over approximately 15 minutes and withhold amiodarone. Repeat once if episodes of torsades de pointes continue.

## Backup

- Backup from an ICP/CCP should be requested for all patients in VT, even if it has spontaneously reverted.
- Seek urgent clinical advice if an ICP/CCP is not immediately available.

# Ventricular Tachycardia

## Referral and transport

- All patients that have been in VT, even if it has spontaneously reverted, must be given a clear recommendation to be transported to an ED by ambulance.

## Additional information

### General principles

- The rhythm is broad complex if the QRS duration is greater than 0.12 seconds.
- In VT the ventricular rate is usually greater than 150/minute and usually associated with cardiovascular compromise.
- A broad complex tachycardia in a compromised patient should be assumed to be VT and treated accordingly.

### Differentiating VT from SVT with abnormal conduction

- Supraventricular tachycardia (SVT) with abnormal conduction (for example bundle branch block) can mimic VT.
- Differentiating VT from SVT with abnormal conduction can be difficult. Treat the rhythm as VT if you are uncertain.
- VT is most likely if the patient is older, has ischaemic heart disease or is compromised.
- SVT with abnormal conduction is most likely if the patient is younger, does not have ischaemic heart disease or is not compromised.
- The following ECG characteristics may help determine whether a rhythm is most likely to be VT or SVT with abnormal conduction:
  - Precordial concordance is more likely with VT.
  - The longer the QRS duration, the more likely it is to be VT.
  - Regular frequency of QRS complexes is more likely with VT.
  - Right axis deviation is more likely with VT.
  - Capture beats or fusion beats indicate atrioventricular dissociation and are more likely with VT.

# Ventricular Tachycardia

## Amiodarone and VT

- If amiodarone is commenced the full dose should be administered, even if the rhythm reverts to sinus rhythm, unless severe hypotension or bradycardia occurs.
- Amiodarone should be administered if there is recurrent VT, even if the runs of VT spontaneously revert.

## Dissociation before cardioversion

- Dissociation is required prior to cardioversion unless the patient is unconscious.
- See the [‘preparation for dissociation’](#) checklist.
- Paramedics should seek clinical advice if dissociation is required for cardioversion and an ICP/CCP is not immediately available.

## VT in special circumstances

- VT secondary to poisoning is most commonly associated with cyclic antidepressant overdose. Amiodarone should not be administered in this setting because it can be associated with severe worsening of shock without resolution of the rhythm. See the [‘poisoning from medicines’](#) guideline.
- VT in children is usually due to underlying structural heart disease:
  - If moderate to severe cardiovascular compromise is present, cardioversion is usually the safest option.
  - Amiodarone should only be administered following clinical advice.
- Torsades de pointes, also called polymorphic VT, can occur when the QT interval is prolonged. This will often revert spontaneously but may recur frequently.
- Patients with a previous history of VT may have an implanted cardioverter defibrillator (ICD). If the VT rate is below 180/minute, the ICD may not activate. In these cases, synchronised cardioversion may still be required.

## Response and/or transport by helicopter

- Most patients in VT will respond to cardioversion and a helicopter will not usually be required.
- Transport or response by helicopter may be indicated if the patient:
  - Has VT and the helicopter will arrive at scene at least 15 minutes sooner than a road-based CCP, or

# Ventricular Tachycardia

- » Has VT with severe cardiovascular compromise not improving with treatment and it will save more than 15 minutes compared with road transport.
- » Has VT regardless of cardiovascular compromise and it will save more than 30 minutes compared with road transport.
- For helicopter transport criteria to be met, the road transport time will usually need to be more than 75 minutes for severe cardiovascular compromise, or 90 minutes for other presentations.

## More Information

### Version history

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**16 June 2026**

1.1.0.1

**Update:** Maximum cardioversion attempts increased.

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**11 September 2023**

1.0.4

**Update:** Multiple changes.

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# Supraventricular Tachycardia

This guideline is for patients aged greater than or equal to 12 years with supraventricular tachycardia (SVT) and a ventricular rate greater than or equal to 150/minute. Seek clinical advice if the patient is aged less than 12 years.

- Acquire a 12 lead ECG.
- Exclude the possibility that the rhythm is sinus tachycardia secondary to another clinical condition. Only use this guideline if the dysrhythmia appears to be the primary problem.
- Determine the level of '[cardiovascular compromise](#)'.

## If the patient is not severely compromised

- Attempt up to two Valsalva manoeuvres.
- If the rhythm fails to revert:
  - Administer 6 mg adenosine IV, and
  - Administer 12 mg adenosine IV if the rhythm fails to revert.

## If the patient is severely compromised

- Reconsider the diagnosis as it is rare for SVT to cause severe compromise.
- Do not administer adenosine as the risk of precipitating cardiac arrest is very high.
- Administer a '[dissociative dose](#)' of ketamine unless unconscious, and
- Cardiovert ('[action list](#)') using maximum joules in synchronised mode.
- Cardiovert up to a total of three times if the rhythm fails to revert.

## Backup

- Backup from an ICP/CCP should be requested if the rhythm fails to revert following a Valsalva manoeuvre.

## Referral and transport

- A patient with SVT that does not resolve must be given a clear recommendation to be transported to ED by ambulance.
- A patient who does not have a known history of SVT and whose rhythm reverts to sinus rhythm following treatment must be given a clear recommendation to be seen in an ED or medical facility.

# Supraventricular Tachycardia

- A patient who has a previous history of SVT and whose rhythm reverts to sinus rhythm following treatment should be given a clear recommendation to be seen in primary care for a review of their condition provided that:
  - There are no ongoing signs or symptoms of myocardial ischaemia, and
  - The patient is given a copy of their 12 lead ECG.

## Additional information

### SVT and cardiovascular compromise

- SVT is normally well tolerated provided there is no significant underlying heart disease.
- It is very important to differentiate SVT causing cardiovascular compromise from sinus tachycardia secondary to another clinical condition, such as cardiogenic shock, hypovolaemia, or sepsis.
- If the primary problem is SVT causing cardiovascular compromise, the patient will usually have been well before suddenly developing palpitations. If not, the diagnosis should be reconsidered.

### Differentiating SVT from atrial fibrillation

- When atrial fibrillation is very fast (ventricular rates of 160-200/minute) the rhythm can appear regular, like SVT.
- When the rhythm is SVT the heart rate recorded on the monitor does not usually vary by more than one or two beats/minute. If the rhythm is very fast atrial fibrillation the heart rate recorded by the monitor will usually vary.

### Differentiating SVT from atrial flutter

- When atrial flutter has 2:1 block, the rhythm may appear to be SVT with a rate of 150/minute and it may be very difficult to tell between the two.
- Printing the rhythm strip at 50 mm/second (if this is possible within the monitor configuration) may allow the atrial flutter waves to be seen.
- If there is uncertainty, treat the rhythm as atrial flutter and not SVT.

### The Valsalva manoeuvre

- A Valsalva manoeuvre creates a sustained positive pressure in the thorax. This causes a reduction in venous return which causes sympathetic nervous system stimulation. When spontaneous breathing



# Supraventricular Tachycardia

recommences, a sudden increase in arterial pressure results in vagal stimulation that may terminate SVT.

- Patients require coaching to produce a good Valsalva manoeuvre:
  - Place the patient in a sitting position.
  - Ask the patient to blow as hard as possible into a 20 ml (or larger) syringe to try and move the plunger.
  - Continue the manoeuvre for a minimum of 15-20 seconds.
  - When the patient stops blowing, simultaneously lay the patient flat and raise their legs.
- If cardioversion occurs, it usually occurs after the end of the Valsalva manoeuvre.

## Dissociation before cardioversion

- Dissociation is required prior to cardioversion unless the patient is unconscious.
- See the '[preparation for dissociation](#)' checklist.
- Paramedics should seek clinical advice if dissociation is required for cardioversion and an ICP/CCP is not immediately available.

## Response and/or transport by helicopter

- Most patients in SVT will respond to treatment and a helicopter will not usually be required.
- Transport or response by helicopter may be indicated if the patient has SVT and severe cardiovascular compromise not improving with treatment and:
  - The helicopter will arrive at scene at least 15 minutes sooner than a road-based CCP, or
  - It will save more than 15 minutes compared with road transport.
- For helicopter transport criteria to be met, the road transport time will usually need to be more than 75 minutes.

## More Information

### Version history

16 June 2026

1.1.0.1

**Update:** Maximum cardioversion attempts increased.

11 September 2023

1.0.4

**Update:** Multiple changes.

# Supraventricular Tachycardia

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# Atrial Fibrillation

This guideline is for adults with a primary clinical problem of atrial fibrillation (AF).

- Use this guideline if the patient has signs of both AF and atrial flutter on their ECG.
- Treat in conjunction with the '[myocardial ischaemia](#)' and/or '[cardiogenic pulmonary oedema](#)' guidelines if these are present.
- Seek clinical advice if the patient is a child.
- Assess for the possibility that AF is associated with a non-cardiac clinical condition, for example sepsis or dehydration. If this is considered possible, do not administer medicines for rate control.
- Acquire a 12 lead ECG.
- Determine the level of '[cardiovascular compromise](#)'.

## If the patient is not compromised

- Do not administer medicines for rate control.

## If the patient is mildly compromised

- Do not administer medicines for rate control.
- Seek clinical advice if the ventricular rate is predominantly greater than 130/minute, and transport time will exceed 90 minutes. Advice may include the administration of metoprolol or amiodarone.

## If the patient is moderately compromised

- Do not administer medicines for rate control if the ventricular rate is predominantly less than 130/minute.
- If the ventricular rate is predominantly greater than 130/minute:
  - Administer 25 mg metoprolol PO once, or
  - If metoprolol is contraindicated, administer 300 mg amiodarone IV over 30 minutes.

# Atrial Fibrillation

## If the patient is severely compromised

- Reconsider the diagnosis because it is very rare for atrial fibrillation to cause severe cardiovascular compromise.
- Cardiovert if the ventricular rate is predominantly greater than 130/minute and the patient is not known to have chronic AF:
  - Administer a '**dissociative dose**' of ketamine unless unconscious, and
  - Cardiovert ('**action list**') using maximum joules in synchronised mode.
  - Cardiovert up to a total of three times if the rhythm fails to revert.
- Seek clinical advice if the patient is known to have chronic AF. Advice may include the administration of amiodarone and/or metaraminol.

## Backup

- Backup from an ICP/CCP is not usually required, unless amiodarone or cardioversion is indicated.
- Seek clinical advice if amiodarone is indicated and backup from an ICP/CCP will be delayed. Advice may include instructions to administer amiodarone.
- Seek clinical advice if cardioversion is indicated and backup from an ICP/CCP is not immediately available. Advice may include instructions to perform cardioversion.

## Referral and transport

- If the patient is not compromised, they should receive a recommendation to be assessed in primary/urgent care within 24 hours.
- If the patient is mildly, moderately or severely compromised, they should receive a recommendation to be assessed in an ED.

## Additional information

### AF and cardiovascular compromise

- The most common cause of cardiovascular compromise due to AF is a fast ventricular rate causing symptomatic myocardial ischaemia or cardiogenic pulmonary oedema. In this setting the goal is to control the ventricular rate and to improve cardiac function.

# Atrial Fibrillation

- It is rare for AF to cause severe cardiovascular compromise and other conditions, particularly septic shock, must be ruled out. If severe cardiovascular compromise is present, this usually requires a ventricular rate that is consistently greater than 150/minute.
- Cardioversion is likely to be ineffective if the patient has chronic AF.

## AF and other non-cardiac clinical problems

- It is common for non-cardiac clinical conditions, such as sepsis, fever, or dehydration, to be associated with a new onset of AF, or with a rapid ventricular rate associated with chronic or paroxysmal AF.
- If AF is associated with a non-cardiac clinical condition, it is crucial that the underlying condition is treated in hospital before administering metoprolol or amiodarone, as these may cause a significant fall in cardiac output and/or blood pressure.

## The differential diagnosis

- Atrial flutter:
  - Atrial flutter is usually associated with an atrial rate of 300/minute, and this may appear as a 'sawtooth like' P wave pattern on the ECG.
  - If the atrio-ventricular conduction ratio is varying, this will produce an irregular ventricular rate, which may resemble AF.
  - If there is uncertainty, vagal stimulation, for example from a Valsalva manoeuvre, may produce a transient increase in the atrio-ventricular conduction ratio, which may make it easier to confirm the rhythm.
  - Patients may have atrial flutter interspersed with AF. In this setting treat using the AF guideline.
- SVT:
  - If the ventricular rate associated with AF is very fast, for example 160-200/minute, the rhythm may appear regular, like SVT.
  - If the rhythm is AF, the heart rate recorded by the monitor will usually vary by 10-20 beats per minute, or more, over time. Conversely, if the rhythm is SVT, the heart rate recorded by the monitor will usually not vary by more than one to two beats per minute.

## Response and/or transport by helicopter

- Most patients in atrial fibrillation will respond to treatment and a helicopter will not usually be required.
- Transport or response by helicopter may be indicated if the patient has rapid atrial fibrillation and severe cardiovascular compromise not improving with treatment and:

# Atrial Fibrillation

- » The helicopter will arrive at scene at least 15 minutes sooner than a road-based CCP, or
  - » It will save more than 15 minutes compared with road transport.
- For helicopter transport criteria to be met, the road transport time will usually need to be more than 75 minutes.

## More Information

### Version history

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**16 June 2026**

1.1.0.1

**New:** Guideline released.

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**11 September 2023**

1.0.4

**Update:** Multiple changes.

# Atrial Flutter

This guideline is for adults with a primary clinical problem of atrial flutter.

- Use the '[atrial fibrillation](#)' guideline if the patient has signs of both atrial fibrillation and atrial flutter on their ECG.
- Treat in conjunction with the '[myocardial ischaemia](#)' and/or '[cardiogenic pulmonary oedema](#)' guidelines if these are present.
- Seek clinical advice if the patient is a child.
- Acquire a 12 lead ECG.
- Determine the level of '[cardiovascular compromise](#)'.

## If the patient is not compromised

- Do not administer medicines for rate control.

## If the patient is mildly compromised

- Do not administer medicines for rate control.
- Seek clinical advice if the ventricular rate is predominately greater than 130/minute, and transport time will exceed 90 minutes. Advice may include the administration of metoprolol or amiodarone.

## If the patient is moderately compromised

- Do not administer medicines for rate control if the ventricular rate is predominantly less than 130/minute.
- Administer 300 mg amiodarone IV over 30 minutes if the ventricular rate is predominantly greater than 130/minute.

## If the patient is severely compromised

- Cardiovert if the ventricular rate is predominantly greater than 130/minute:
  - Administer a '[dissociative dose](#)' of ketamine unless unconscious, and
  - Cardiovert ('[action list](#)') using maximum joules in synchronised mode.
  - Cardiovert up to a total of three times if the rhythm fails to revert.

# Atrial Flutter

## Backup

- Backup from an ICP/CCP is not usually required, unless amiodarone or cardioversion is indicated.
- Seek clinical advice if amiodarone is indicated and backup from an ICP/CCP will be delayed. Advice may include instructions to administer amiodarone.
- Seek clinical advice if cardioversion is indicated and backup from an ICP/CCP is not immediately available. Advice may include instructions to perform cardioversion.

## Referral and transport

- If the patient is not compromised, they should receive a recommendation to be assessed in primary/urgent care within 24 hours.
- If the patient is mildly, moderately or severely compromised, they should receive a recommendation to be assessed in an ED.

## Additional information

### Atrial flutter and cardiovascular compromise

- The most common cause of cardiovascular compromise due to atrial flutter, is a fast ventricular rate causing symptomatic myocardial ischaemia or cardiogenic pulmonary oedema. In this setting the goal is to control the ventricular rate and to improve cardiac function.
- It is uncommon for atrial flutter to cause severe compromise unless the ventricular rate is greater than 150/minute.

## The differential diagnosis

- Atrial fibrillation (AF):
  - AF usually has an irregularly irregular ventricular rate, without visible P waves.
  - If the rhythm is atrial flutter and the atrio-ventricular conduction ratio is varying, this will produce an irregular ventricular rate, which may resemble AF.
  - Patients with paroxysmal AF may have runs of atrial flutter, interspersed with AF.
- SVT:
  - If the SVT is at a rate of 150/minute, the rhythm may resemble atrial flutter with a 2:1 atrio-ventricular conduction ratio.



# Atrial Flutter

- » Atrial flutter will not revert with vagal simulation, for example from a Valsalva manoeuvre, but this may produce a transient increase in the conduction ratio which may make it easier to confirm the rhythm.

## Response and/or transport by helicopter

- Most patients in atrial flutter will respond to treatment and a helicopter will not usually be required.
- Transport or response by helicopter may be indicated if the patient has rapid atrial flutter and severe cardiovascular compromise not improving with treatment and:
  - » The helicopter will arrive at scene at least 15 minutes sooner than a road-based CCP, or
  - » It will save more than 15 minutes compared with road transport.
- For helicopter transport criteria to be met, the road transport time will usually need to be more than 75 minutes.

# Cardioversion Action List

- ✓ Ensure the ECG electrodes are in place.
  - ✓ Place pads in the anterior/posterior position when feasible.
  - ✓ Ensure the defibrillator is in manual mode.
  - ✓ Select synchronised mode.
  - ✓ Confirm there are detection symbols with most QRS complexes.
  - ✓ Ensure the patient is dissociated if indicated.
  - ✓ Select maximum joules and charge the defibrillator.
  - ✓ Press and hold the shock button until the shock is delivered.
  - ✓ Determine the rhythm and reassess vital signs.
  - ✓ If administering a second or third cardioversion, confirm the defibrillator is still in synchronised mode and the patient is adequately dissociated.
-

# Bradycardia

This guideline is for adults with bradycardia. Refer to the relevant guideline if the bradycardia is suspected to be secondary to another clinical condition (for example, hyperkalaemia). Bradycardia in children is usually due to hypoxia or hypovolaemia and treating the underlying cause takes priority over drug therapy.

- Acquire a 12 lead ECG and determine the rhythm.
- Initiate treatment if the heart rate is less than 40/minute and there is moderate or severe cardiovascular compromise.
- Consider administering 0.6 mg of atropine IV, particularly if the rhythm is a narrow complex bradycardia. Administer further doses as required if the bradycardia is responsive to atropine.
- Administer adrenaline IV if the bradycardia is unresponsive to atropine, or atropine is not administered:
  - Administer an adrenaline infusion via an infusion device. Start at a rate of 0.5 mg/hour and adjust the rate as required, or
  - Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
    - Administer as an infusion. Start at 2 drops/second and adjust the rate as required, or
    - Administer a bolus of 0.01 mg adrenaline (10 ml) IV every 1-2 minutes as required.
- Stop adrenaline administration and initiate pacing if the bradycardia is unresponsive to adrenaline IV.

## Pacing action list

- ✓ Place the pads in either the anterior/posterior (recommended) or apex/sternum position in addition to ECG electrodes.
- ✓ Select pacing.
- ✓ Confirm pacing is in demand mode and that there are detection symbols with most QRS complexes.
- ✓ Set the pacing rate to 70/minute.
- ✓ Select current and increase this until pacing capture occurs. Confirm there is a pacing spike before each QRS complex.
- ✓ Increase the current 10 mA above the capture threshold.
- ✓ Administer an opiate IV and add ketamine IV if required.
- ✓ Confirm there is mechanical capture with a palpable pulse or other signs of increased cardiac output.
- ✓ Increase the pacing rate to 80/minute if there is electrical capture, but no signs of increased cardiac output.

# Bradycardia

- ✓ Change to fixed or non-demand mode if pacing is ineffective due to artefact.

## Backup

- Backup from an ICP/CCP should be requested if the patient has moderate to severe cardiovascular compromise.

## Referral and transport

- All patients with symptomatic bradycardia should be given a clear recommendation to be transported to an ED by ambulance.

## Additional information

### General principles

- Bradycardia in adults is most commonly caused by:
  - Myocardial ischaemia, particularly when the SA node or AV node is ischaemic. Inferior myocardial ischaemia is much more likely to involve these nodes than anterior or anterolateral myocardial ischaemia.
  - Structural heart disease involving the sinoatrial node, atrioventricular node or the conduction system. This is most common in elderly patients.
- Bradycardia caused by a problem high in the conduction system, for example at the level of the sinoatrial node or atrioventricular node, is most likely to be responsive to atropine or adrenaline.
- Bradycardia caused by a problem low in the conduction system, for example below the atrioventricular node, is most likely to be responsive to pacing.

### The pharmacological treatment of bradycardia

- Medicines that increase the heart rate will also increase myocardial oxygen demand. The decision to administer such medicines must take into account the potential advantage of an improvement in cardiac output versus the potential disadvantage of an increase in myocardial oxygen demand.
- When administering IV adrenaline, an IV infusion is preferred over IV boluses because this reduces the adverse effects of surges of adrenaline.

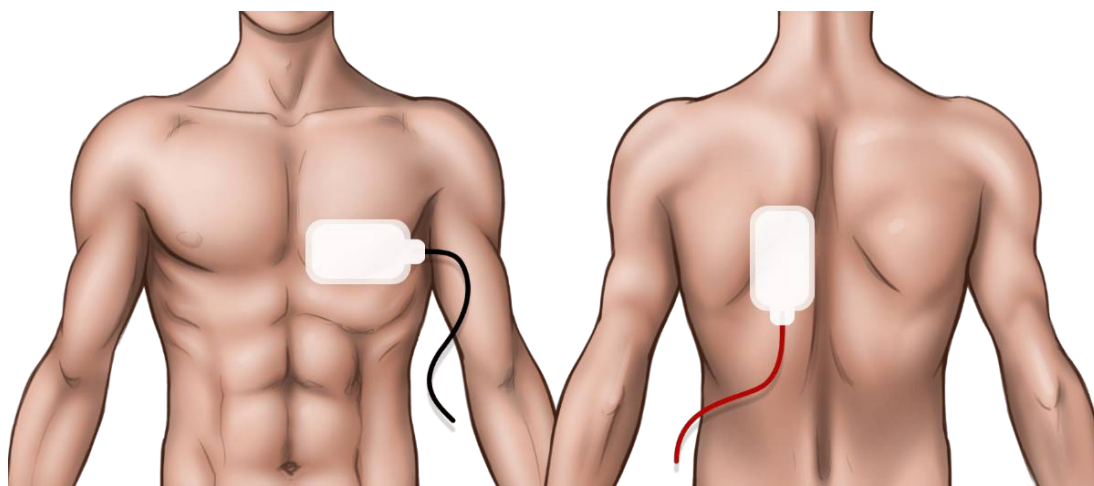
# Bradycardia

## Bradycardia secondary to poisoning with beta blockers

- Bradycardia may be prominent in a patient who has taken a large dose of a beta blocker, particularly if taken in combination with a calcium channel blocker.
- A patient who has taken large doses of these medicines may require high dose adrenaline by infusion. Seek clinical advice if the bradycardia or hypotension does not rapidly respond to adrenaline IV.
- Glucagon has no role in the out-of-hospital treatment of poisoning from beta-blockers because it rarely provides a sustained heart rate rise in addition to high dose adrenaline and requires much higher doses than carried by ambulance personnel.

## Pacing

- Most patients will require a current setting of 50-90 mA to achieve electrical capture. Patients that are obese or have COPD are likely to require a higher energy setting to achieve capture.
- Clinical judgement must be used when determining the maximum current level that is delivered, noting that it should be rare to exceed 150 mA.
- False capture may occur when electrical artefact resembles electrical capture. This electrical artefact is created by the current passing between the pacing pads and the most common cause is poor pad placement combined with insufficient mA.



Anterior/posterior pad placement

## Response and/or transport by helicopter

- Most patients with bradycardia will respond to treatment and a helicopter will not usually be required.

# Bradycardia

- Transport or response by helicopter may be indicated if the patient has bradycardia and severe cardiovascular compromise not improving with treatment and:
  - The helicopter will arrive at scene at least 15 minutes sooner than a road-based CCP, or
  - It will save more than 15 minutes compared with road transport.
- For helicopter transport criteria to be met, the road transport time will usually need to be more than 75 minutes.

## More Information

### Version history

11 September 2023

1.0.4

**Update:** Multiple changes.

# Cardiogenic Shock

This guideline is for adults with a provisional diagnosis of cardiogenic shock. Treat in conjunction with the appropriate guideline if myocardial ischaemia, STEMI or cardiogenic pulmonary oedema is also present. Seek clinical advice if the patient is a child.

- Acquire a 12 lead ECG and determine the rhythm.
- Move to the appropriate guideline if the primary cause is dysrhythmia.
- Administer 0.9% sodium chloride IV, provided the patient has no signs or symptoms of pulmonary oedema:
  - Administer 500 ml IV. This may be repeated once, if required.
  - Stop fluid administration if the patient develops signs or symptoms of pulmonary oedema.
- Administer metaraminol IV if the systolic blood pressure is less than 100 mmHg:
  - Administer via an infusion device:
    - Administer an initial bolus of 0.5-1 mg IV.
    - Start the infusion at 2 mg/hour and adjust the rate as required.
  - Alternatively, administer a bolus of 0.5-1 mg IV every 5-10 minutes as required.
- Consider administering adrenaline IV if the blood pressure is unresponsive to metaraminol or the patient is hypotensive and bradycardic:
  - Administer an adrenaline infusion via an infusion device. Start at a rate of 0.5 mg/hour and adjust the rate as required, or
  - Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
    - Administer as an infusion. Start at 2 drops/second and adjust the rate as required, or
    - Administer a bolus of 0.01 mg adrenaline (10 ml) IV every 1-2 minutes as required.

## Backup

- Backup from an ICP/CCP should be requested.

## Referral and transport

- Transport the patient to a hospital with a cardiac catheter lab whenever it is feasible and safe to do so.

## Additional information

# Cardiogenic Shock

## General

- Cardiogenic shock is caused by the heart being unable to pump adequately. Common causes include:
  - Acute myocardial infarction.
  - Acute valve rupture.
  - Dysrhythmia (particularly VT).
  - Myocarditis.
  - A progressive reduction in cardiac function in end-stage or chronic heart failure.
- Other conditions that can present like cardiogenic shock include pulmonary embolism, cardiac tamponade and tension pneumothorax.
- Cardiogenic shock secondary to poor left ventricular function is commonly associated with the development of pulmonary oedema. This is unlikely to respond to 0.9% sodium chloride IV, which must be administered with caution as it may make pulmonary oedema worse.
- Cardiogenic shock secondary to poor right ventricular function is commonly caused by acute inferior myocardial infarction which is more likely to respond to 0.9% sodium chloride IV.

## Vasopressor administration

- Metaraminol is the preferred vasopressor because it does not increase myocardial oxygen consumption and is less likely to cause tachycardia and/or tachydysrhythmia than adrenaline.
- The decision to administer adrenaline must weigh up the potential benefit of improving cardiac output against the potential risks of raising myocardial oxygen consumption and causing tachycardia and/or tachydysrhythmia. In general, it is appropriate to commence an adrenaline infusion when there has not been a response to metaraminol and/or shock is associated with bradycardia.

## More Information

### Version history

11 September 2023

1.0.4

**Update:** Multiple changes.



# Cardiac Arrest

This guideline is for adults and children in non-traumatic cardiac arrest. See the '[cardiac arrest secondary to trauma](#)' guideline if the cardiac arrest is secondary to trauma.

## Initial actions (the first 1-4 minutes)

- Perform continuous chest compressions while the defibrillator is being attached and charged.
- Defibrillate immediately if the cardiac rhythm is VF or VT using a single shock and immediately recommence chest compressions:
  - Maximum joule setting for an adult.
  - See the paediatric drug dose tables for joule settings for a child.
- Place an SGA and commence ventilation, preferably using a timing indicator.

## Subsequent actions

- Perform two-minute cycles of high performance CPR between rhythm checks.
- Administer adrenaline IV every four minutes:
  - 1 mg IV for an adult.
  - See the paediatric drug dose tables for a child.
- Consider placing an ETT.
- Establish a team leader and utilise the [cardiac arrest checklist](#).

## If the rhythm is VF or VT

- Defibrillate using single shocks every two minutes.
- Immediately recommence chest compressions after each shock.
- If in VF or VT after three shocks, including shocks delivered prior to ambulance arrival:
  - Administer amiodarone:
    - 300 mg IV for an adult.
    - See the paediatric drug dose tables for a child.
  - Consider changing to anterior/posterior pad placement or performing double sequential defibrillation.
- If in VF or VT after six shocks, including shocks delivered prior to ambulance arrival, administer a second dose of amiodarone:
  - 150 mg IV for an adult.
  - See the paediatric drug dose tables for a child.

# Cardiac Arrest

- Seek clinical advice if VF or VT persists.

## If torsades de pointes is present or has occurred

- Withhold amiodarone.
- Administer magnesium IV:
  - 10 mmol (2.5 g) over 1-2 minutes for an adult.
  - See the paediatric drug dose tables for a child.
  - Repeat once if torsades de pointes recurs or persists.

## If the patient is in PEA

- Correct immediately reversible causes.
- Administer 0.9% sodium chloride IV:
  - 1-2 litres IV for an adult.
  - 20-40 ml/kg IV for a child.

## If patient movement is interfering with resuscitation

- Administer 1 mg/kg of ketamine IV, up to a maximum of 150 mg.
- This may be repeated once if required.
- If significant movement continues, consider administering 2 mg/kg of rocuronium IV up to a maximum of 200 mg, provided an ETT has been placed, and the position confirmed using ET<sub>CO</sub><sub>2</sub>.

## Backup

- Backup from an ICP/CCP should be requested.

## Cardiac arrest checklist

- ✓ Team leader identified.
- ✓ Roles allocated and confirmed.
- ✓ CPR feedback device and metronome on if available.
- ✓ Defibrillator screen visible to the team leader.
- ✓ Airway adjunct or ETT in place and ventilation adequate.

# Cardiac Arrest

- ✓ ETCO<sub>2</sub> attached and waveform visible.
- ✓ Oxygen flow and supply adequate.
- ✓ IV/IO access secure, two sites if possible.
- ✓ Consider underlying and reversible causes.
- ✓ Appropriate medicines administered.
- ✓ Initiate documentation.
- ✓ Communication with family/bystanders.

## Additional information

### Deciding to start resuscitation

- Resuscitation should start unless there is a clear reason not to. Clear reasons for not starting resuscitation include:
  - Signs of rigor mortis or post-mortem lividity.
  - A clear advance directive not to receive resuscitation for cardiac arrest.
  - Scenarios where resuscitation is futile or clearly not in the best interest of the patient. Examples include unwitnessed cardiac arrest with asystole as the initial rhythm, patients who are dying from cancer and patients with severe life-limiting disease.
- Ambulance personnel must consider all available information, including the patient's known wishes (for example, their advance directive) and act in what they believe is the best interest of the patient.
- If there is doubt regarding the appropriateness of a resuscitation attempt, resuscitation should begin while further information is gained.
- Nobody has the right to demand resuscitation on behalf of a patient, including those with an Enduring Power of Attorney and family/whānau, but the known wishes of the patient must be considered.
- Family/whānau do not have the right to decline resuscitation on behalf of a patient unless they have an Enduring Power of Attorney. See the [‘treatment and/or transport without informed consent’](#) guideline for more information.
- There must be clear documentation regarding decisions made.

# Cardiac Arrest

## Deciding to stop resuscitation

- There is no absolute time at which it is possible to say that further resuscitation is futile. Stopping resuscitation requires clinical judgement on the likelihood of survival, considering the following:
  - » The cause of the cardiac arrest.
  - » Whether or not the cardiac arrest was witnessed.
  - » Whether or not there was bystander CPR.
  - » The response time.
  - » The initial rhythm.
  - » The total estimated time in cardiac arrest.
  - » The end tidal CO<sub>2</sub>.
  - » Whether ROSC has occurred at any time.
  - » The patient's comorbidities.

## Deciding to stop resuscitation: EMTs

- Resuscitation must continue until a Paramedic, ICP or CCP arrives and makes a decision to stop, or advice is received via the Clinical Desk that it is appropriate to stop. If backup is not available, and the Clinical Desk cannot be contacted:
  - » If the arrest was unwitnessed and no shock is advised, the prognosis is very poor and it is appropriate to stop resuscitation if there are no signs of ROSC 20 minutes after the onset of resuscitation by ambulance personnel.
  - » For other circumstances it is appropriate to stop resuscitation if there are no signs of ROSC 40 minutes after the onset of resuscitation by ambulance personnel.

## Deciding to stop resuscitation: Paramedics, ICPs and CCPs

- It is usually appropriate to stop resuscitation:
  - » 20 minutes after the onset of resuscitation by ambulance personnel in poor prognosis scenarios.
  - » 40 minutes after the onset of resuscitation by ambulance personnel in good prognosis scenarios.
  - » Earlier than described above, if it becomes clear that it was inappropriate to have commenced resuscitation, or the rhythm has deteriorated into asystole for more than a few minutes despite resuscitation.

# Cardiac Arrest

## Movement during cardiac arrest

- The patient may move during cardiac arrest if cerebral blood flow is being maintained by very good CPR.
- This can cause personnel to believe that the patient is not in cardiac arrest and may lead to delayed defibrillation and/or inappropriate pauses in CPR.
- It is appropriate to briefly pause CPR to confirm the patient is in cardiac arrest.
- If rocuronium is administered, neuromuscular blockade will continue if sustained ROSC is achieved and the patient will then require an adequate level of sedation.

## The prognosis of cardiac arrest

There is no one factor that can be used to determine the prognosis of an individual patient in cardiac arrest. Multiple factors must be considered, noting that in most patients there will be a mixture of prognostic factors.

# Cardiac Arrest

## Poor prognostic factors

- Secondary cardiac arrest
- Unwitnessed
- No bystander CPR
- Response time > 8 minutes
- Initial rhythm asystole or PEA
- Time in cardiac arrest > 30 minutes
- Severe comorbidities
- Living in aged residential care
- Age > 80 years
- $\text{ETCO}_2 < 15 \text{ mmHg}$  or falling despite CPR

## Good prognostic factors

- Primary cardiac arrest
- Witnessed
- Bystander CPR
- Response time < 8 minutes
- Initial rhythm VT or VF
- Time in cardiac arrest < 30 minutes
- No severe comorbidities
- Living independently
- Age  $\leq 80$  years
- $\text{ETCO}_2 > 25 \text{ mmHg}$  with CPR

- Severe comorbidities are chronic diseases that significantly limit a patient's life expectancy. Examples include dementia, severe COPD, severe heart failure, a requirement for aged residential care and metastatic cancer with weight loss.
- The reactivity of the patient's pupils during and immediately following cardiac arrest is not prognostic and there is no role for including pupil reactivity in decision making.

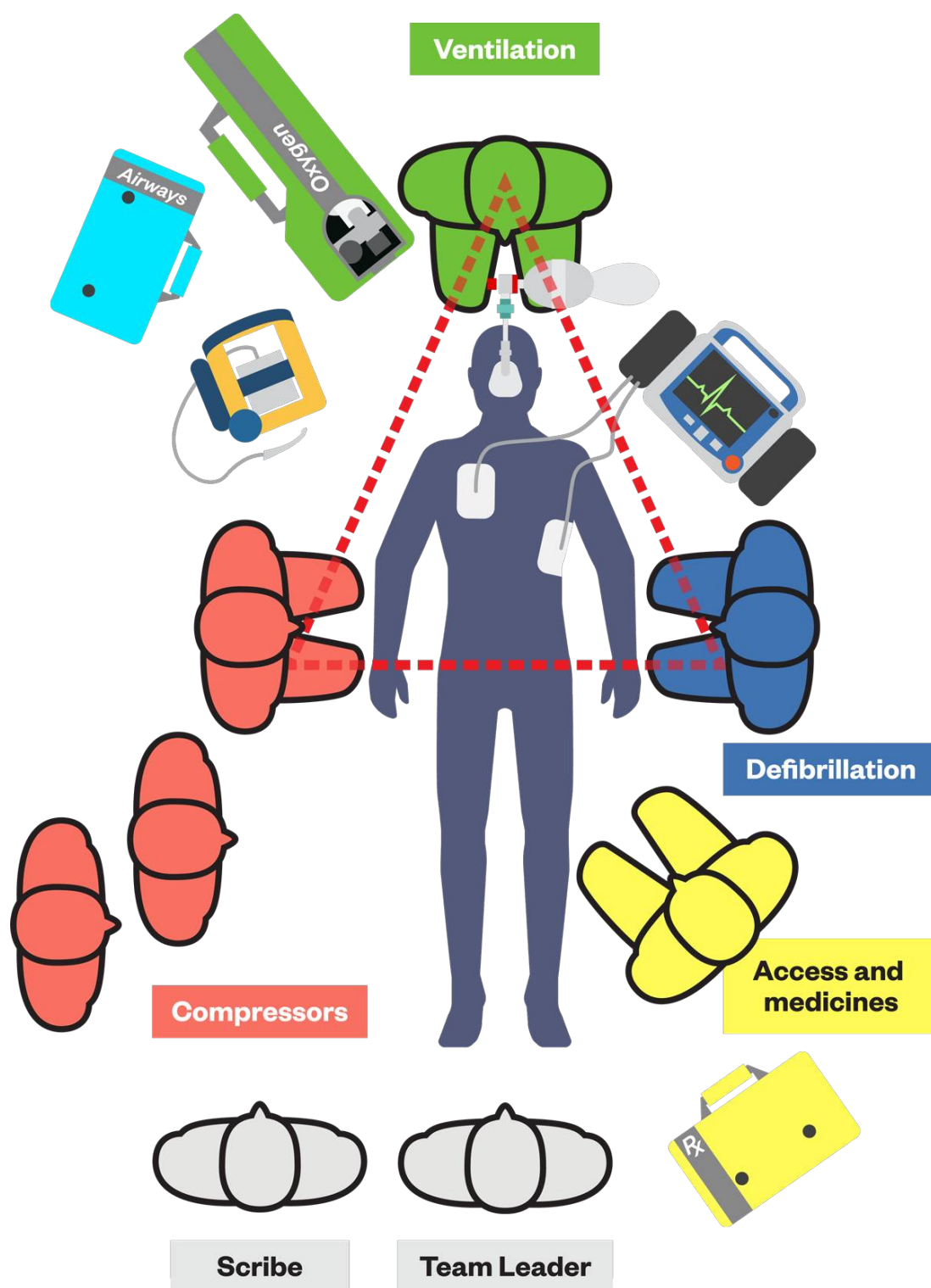
## High performance CPR

- High performance CPR encompasses the concepts of:
  - Optimising chest compression depth, rate, and recoil.

# Cardiac Arrest

- » Minimising interruptions to chest compressions.
- » Avoiding over-ventilation.
- Perform chest compressions at a rate of 100-120/minute, ensuring adequate depth and complete recoil.
- CPR should be performed for two minute cycles between rhythm checks. The person performing chest compressions should ideally change every two minutes, or earlier if tired.
- A CPR feedback device and/or metronome must be used if available.
- Pauses in CPR for rhythm checks, defibrillation or ventilation should be minimised, and usually last no longer than three seconds. A pause of up to 10 seconds is permitted for a pulse check in PEA.
- The person performing chest compressions should hover approximately 5 cm above the patient's chest during the pause.

# Cardiac Arrest



## Ventilation in high performance CPR

- Ventilation has a low priority during the first few minutes of resuscitation if the patient has had a primary cardiac arrest.



# Cardiac Arrest

- Once ventilation has commenced, the ventilation rate should be no more than 10/minute, as rates higher than this increase intrathoracic pressure and therefore reduce venous return during CPR.
- When ventilation is provided via a bag and mask:
  - For an adult the CPR compression to ventilation ratio is 30:2. This ratio prioritises chest compressions on the basis that an adult is most likely to have had a primary cardiac arrest. If an adult has had a cardiac arrest secondary to asphyxiation or respiratory failure, alter the ratio to 15:2.
  - For a child the CPR compression to ventilation ratio is 15:2 (exception – the ratio is 3:1 for newborns). The 15:2 ratio reduces the priority of chest compressions on the basis that a child is most likely to have had a cardiac arrest secondary to respiratory failure. If a child has had a primary cardiac arrest, alter the ratio to 30:2.
- When ventilation is being provided via an SGA or ETT:
  - Perform continuous chest compressions and ventilate adults and children at a rate of 10/minute, preferably using a ventilation timing light.
  - If continuous chest compressions clearly impair ventilation via an SGA, interrupt the chest compressions to provide ventilations using the ratios described above.
- Intubation is not a high priority during the initial management of cardiac arrest. If ROSC has not been achieved within 10 minutes, intubation with an ETT may occur provided this does not interrupt chest compressions.

## IV access and medicines

- Gaining IV access and administering medicines has a lower priority than providing high performance CPR.
- IV access is preferred over IO access. Clinical judgement should be used and IO access should usually be obtained if IV access cannot be gained, and resuscitation is ongoing.
- All doses should be flushed with a running line of 0.9% sodium chloride or a minimum of 20 ml 0.9% sodium chloride.

## End tidal CO<sub>2</sub>

- End tidal CO<sub>2</sub> (ETCO<sub>2</sub>) is a marker of the blood flow being achieved during CPR. With good CPR an ETCO<sub>2</sub> of greater than 20 mmHg should usually be achieved, unless there is an obstruction to blood flow, for example from a pulmonary embolus or tension pneumothorax.
- ETCO<sub>2</sub> must be used to confirm ETT placement and must be continuously measured if the patient has been intubated with an ETT.

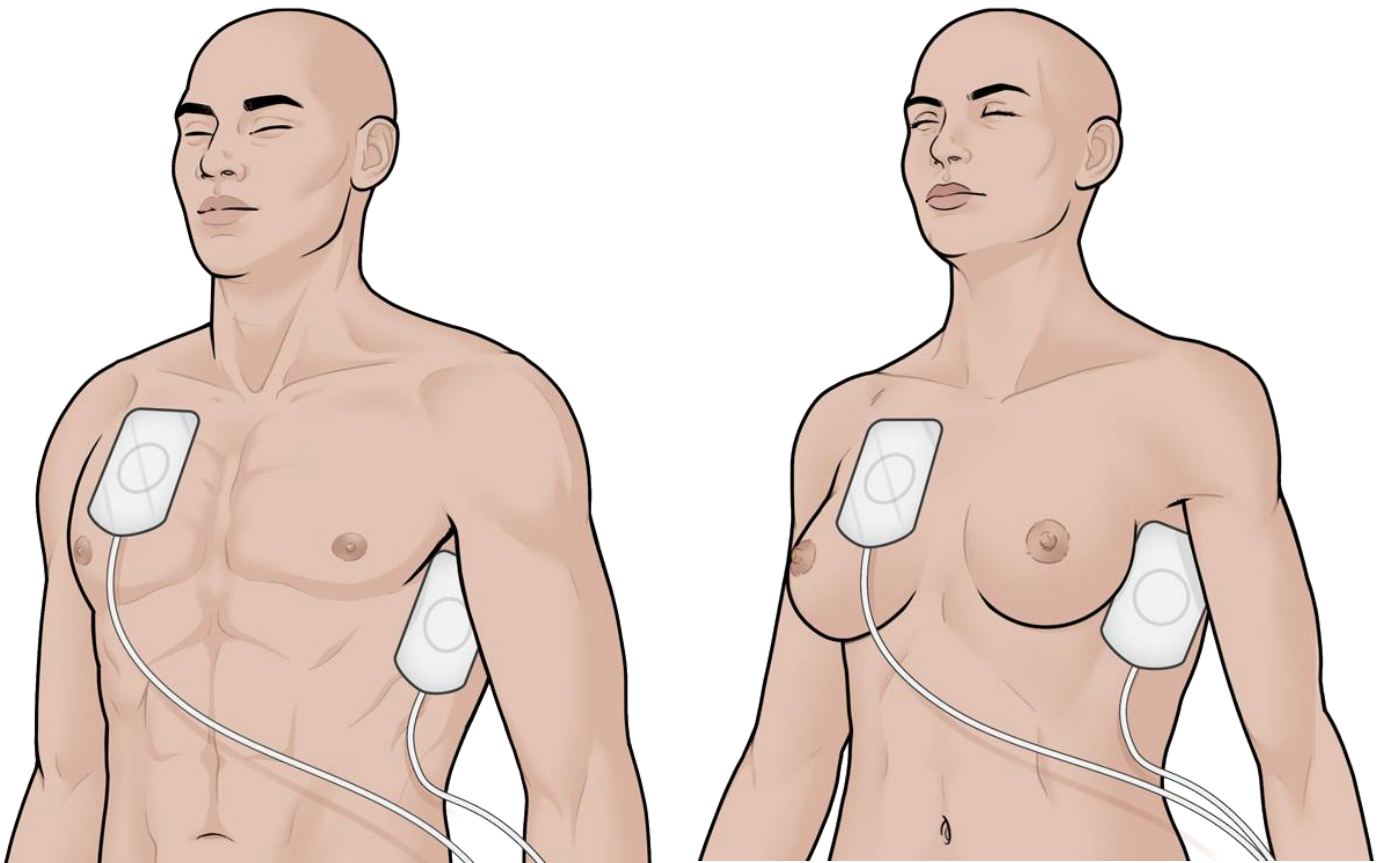
# Cardiac Arrest

- ETCO<sub>2</sub> should be continuously measured if the patient is being ventilated via an SGA, noting that the ETCO<sub>2</sub> will be reduced if there is significant leak around the cuff. In this setting the trend of the ETCO<sub>2</sub> is more important than individual recordings.
- ETCO<sub>2</sub> may be measured if the patient is being ventilated via a face mask, noting that the ETCO<sub>2</sub> will be reduced if there is significant leak around the mask. In this setting the trend of the ETCO<sub>2</sub> is more important than individual recordings.
- A sudden increase in ETCO<sub>2</sub> is usually an indicator that ROSC has occurred.
- A falling ETCO<sub>2</sub> despite CPR is a poor prognostic factor and must prompt a focus on ensuring high performance CPR is being provided and the ventilation rate is at or below 10/minute.
- An ETCO<sub>2</sub> of less than 15 mmHg despite CPR is a very poor prognostic factor.

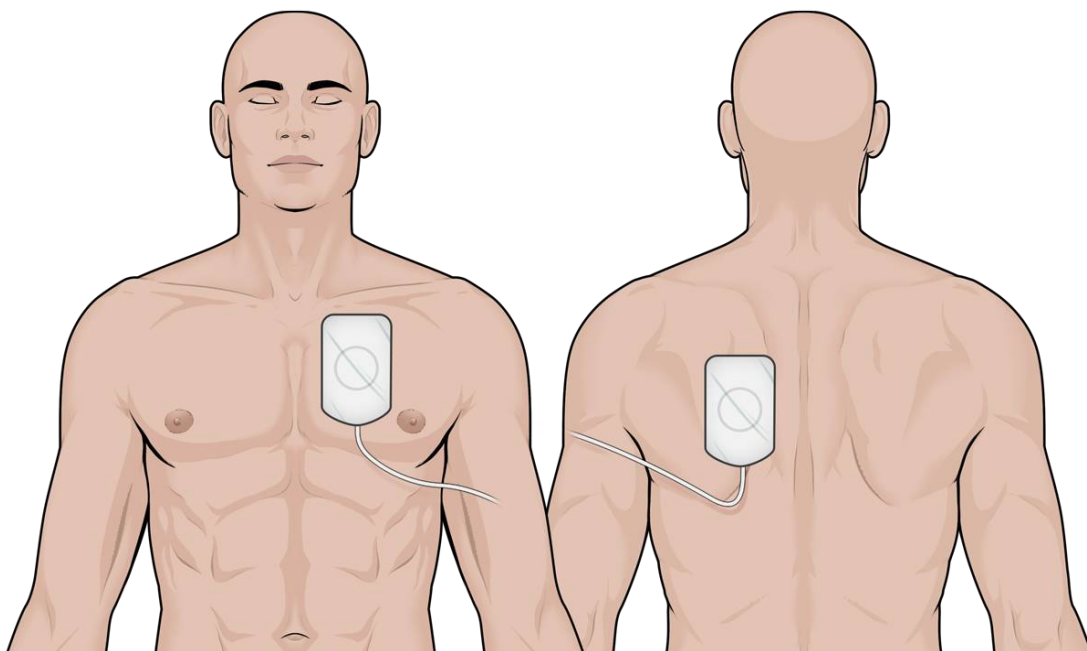
## Defibrillation

- Pad placement and adherence is important. It is worth taking the extra time to ensure the pads are placed in the ideal location and are well adhered.
- Routinely place pads in the anterior/posterior position for small children and morbidly obese patients.
- If VF or VT persists or recurs after three shocks, consider whether it is recurrent, or refractory. Because CPR is initiated immediately after each shock, it is not always possible to differentiate:
  - VF or VT that is successfully reverted with each shock and then recurs, and thus is recurrent, versus
  - VF or VT that is not successfully reverted with each shock, and thus is refractory.
- If VF or VT is clearly recurrent, changing the defibrillation strategy is not required.
- If VF or VT is clearly or possibly refractory, the defibrillation strategy should be changed.
- The two options when changing the defibrillation strategy are:
  - Anterior/posterior pad placement, or
  - Double sequential defibrillation (DSD).
- Anterior/posterior pad placement is the preferred approach when one manual defibrillator is present, or if the team are not fully trained in DSD.
- DSD is the preferred approach when two manual defibrillators are present, and the team is fully trained in DSD.

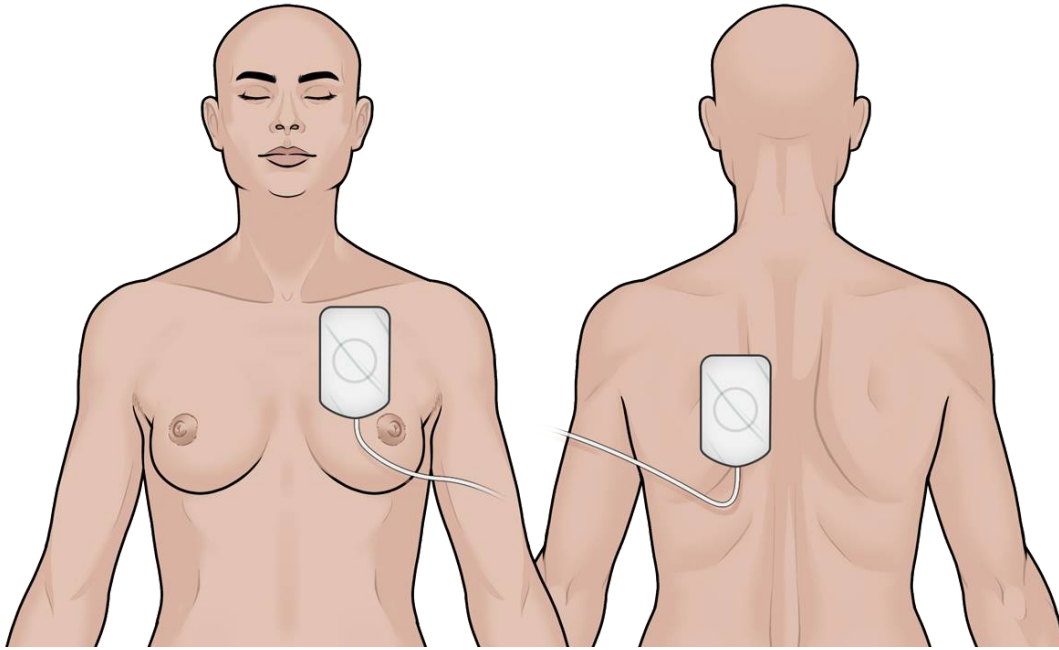
# Cardiac Arrest



**Antero-lateral pad placement for defibrillation**



# Cardiac Arrest

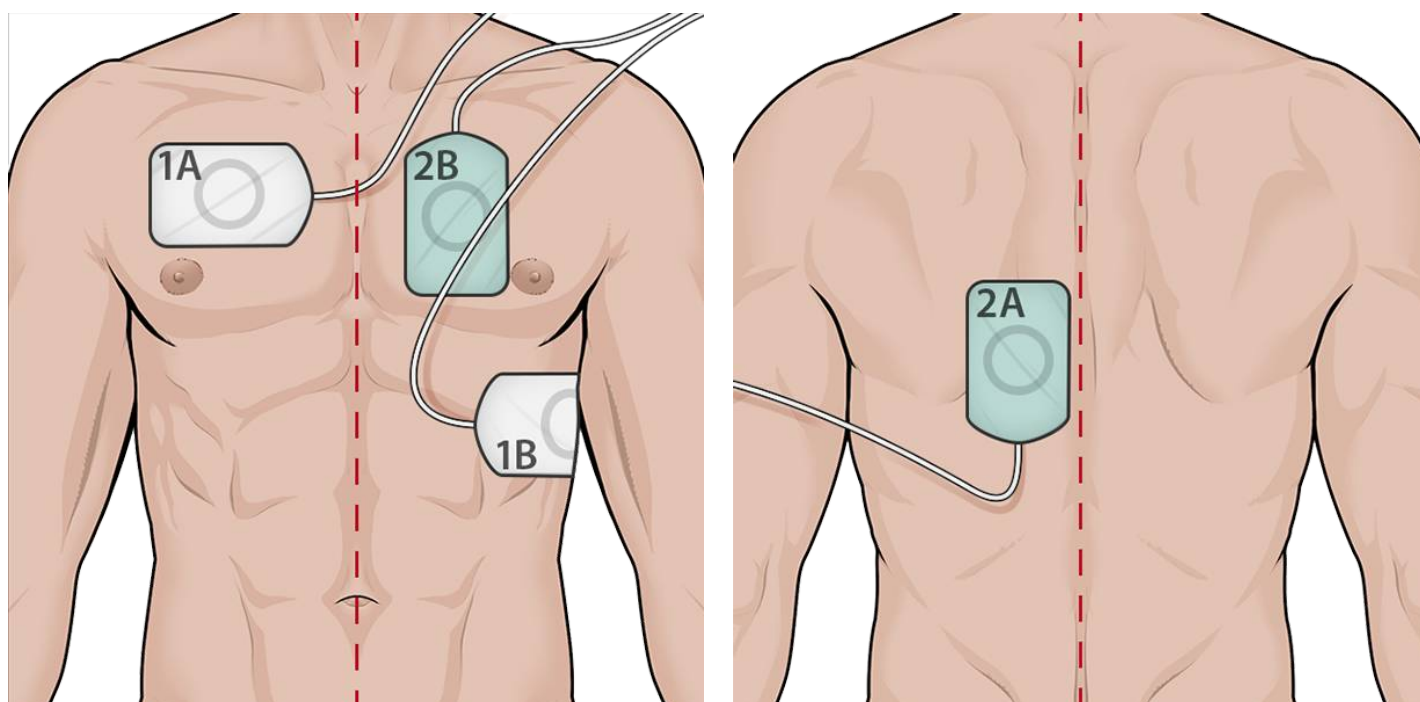


**Antero-posterior pad placement for defibrillation**

## Double sequential defibrillation

- Double sequential defibrillation (DSD) is the provision of two sequential (approximately one second apart) shocks, using two manual defibrillators, with the pads attached in two different vectors, one in the antero-lateral vector and the other in the antero-posterior vector.
- When providing DSD:
  - » Only those personnel specifically trained in the technique may provide DSD.
  - » Use only manual defibrillators.
  - » Ensure none of the pads are touching.
  - » One person must discharge both defibrillators sequentially, to avoid simultaneous defibrillation.

# Cardiac Arrest



**Pad placement for double sequential defibrillation**

## Pulseless electrical activity (PEA)

- PEA is present when a patient in cardiac arrest has a rhythm that should be associated with cardiac output but is not. PEA is a clinical condition and not an abnormal rhythm.
- PEA is often secondary to a non-cardiac problem. The history immediately prior to cardiac arrest is very important in helping to determine what the cause may be.
- Potentially reversible causes of PEA include:
  - Hypoxia.
  - Hypothermia.
  - Hypovolaemia (including anaphylaxis).
  - Hyper/hypokalaemia (and other metabolic abnormalities).
  - Tension pneumothorax.
  - Tamponade (cardiac).
  - Toxins (poisoning).
  - Thrombosis (pulmonary and coronary).

# Cardiac Arrest

- Some references include hydrogen ions (acidosis) and hypoglycaemia in the list of causes of PEA. However, neither of these cause cardiac arrest in the absence of another clinical problem, even when severe. Hypoglycaemia should be ruled out following ROSC.
- Consideration should be given to transporting the patient to hospital with CPR en route if a potentially reversible cause is identified that cannot be treated at the scene. To be successful, such a decision must be made very early in the resuscitation attempt.
- It is common for PEA to degenerate into asystole with time. During this process slow (less than 30/minute) broad complexes may be present. This is treated as asystole and not PEA.
- Survival is rare if the rhythm deteriorates from PEA into asystole despite resuscitation and prolonged resuscitation attempts in this setting are usually inappropriate.
- In general, patients that survive cardiac arrest with an initial rhythm of PEA do so because the underlying problem is immediately identified and corrected.
- If the patient remains in PEA despite resuscitation attempts, consider the possibility that there is cardiac output that cannot be detected clinically. In this setting stop chest compressions and observe the ETCO<sub>2</sub> for one minute while ventilating at a rate of 10/minute:
  - If the ETCO<sub>2</sub> rapidly falls to below 10 mmHg the patient is in cardiac arrest.
  - If the ETCO<sub>2</sub> is maintained at or above 10 mmHg, the patient has low cardiac output and should be treated accordingly.
- If available, ultrasound can be useful to assess for cardiac activity and reversible causes of PEA arrest.

## Asystole

- Confirm the rhythm is asystole by checking:
  - The cables.
  - The leads.
  - Which lead is selected on the screen.
  - The amplitude.
- Exclude bradycardia which can look like asystole at a glance, particularly in children.
- Survival from cardiac arrest with an initial rhythm of asystole is rare and prolonged resuscitation attempts in this setting are usually inappropriate, particularly if the cardiac arrest was unwitnessed.

## Single responder resuscitation

# Cardiac Arrest

- Attach and use a defibrillator immediately. It is acceptable to use the defibrillator in automatic mode until additional personnel arrive.
- Perform CPR with a focus on chest compressions, utilising bystanders to help.
- Interventions such as IV access or intubation are not a priority and should only occur after additional trained personnel arrive.

## Transport to hospital with CPR en route

- Transport to hospital with CPR en route should be rare because:
  - There is no convincing evidence it improves patient outcomes.
  - The quality of the resuscitation attempt (particularly the quality of CPR) is usually compromised, unless mechanical CPR is available.
  - Unrestrained personnel are at risk in the back of a moving ambulance.
- However, a small number of patients may benefit from transport to hospital with CPR en route, particularly if there is a potentially reversible cause that ambulance personnel are unable to treat (for example, hypothermia and poisoning). To be successful, such a decision must be made very early in the resuscitation attempt.
- Provide as much pre-hospital warning as possible.

## Response and/or transport by helicopter

- It is uncommon for a helicopter to locate a patient within a timeframe that provides meaningful clinical benefit.
- Response by helicopter may be indicated if the patient is suspected to be in cardiac arrest and the helicopter will arrive on scene:
  - Prior to a road ambulance resource and a resuscitation attempt is expected, or
  - To deliver a CCP skill that will likely improve the patient's outcome (for example, cardiac arrest secondary to drowning).

## Mechanical CPR

- Mechanical CPR does not usually have a role early in cardiac arrest but should be considered when:
  - It is not safe to perform manual chest compressions, for example in a moving ambulance.
  - There is insufficient room or personnel to perform chest compressions.
  - A prolonged resuscitation attempt is anticipated.



# Cardiac Arrest

- One trained person must be present to supervise the mechanical CPR device during use.

## Defibrillator failure action list

- Use this action list if a defibrillator fails and there is not another defibrillator (including an AED) immediately available. At each defibrillator intervention, pause briefly to determine if the problem has been fixed.
  - Task specific personnel to focus on resuscitating the patient.
  - Task specific personnel to focus on troubleshooting the defibrillator.
  - Ensure the pads are attached and connected.
  - Ensure all components of the defibrillator are physically connected.
  - Ensure the ECG leads are attached.
  - Change the lead shown on the screen so that the rhythm is visible.
  - Attach and connect a new set of pads.
  - Switch to automatic mode if in manual mode.
  - Turn the defibrillator off for 30 seconds and turn it back on again.
  - Seek urgent clinical advice if the situation is not resolved.

## Cardiac arrest in special situations

### Cardiac arrest secondary to drowning

- Prioritise the ventilation aspect of CPR and use a CPR ratio of 15:2 unless an ETT is in place.
- Place an ETT if ROSC is not achieved in the first few minutes.
- IV medicines have a very low priority.
- See the '[drowning](#)' guideline for additional information.

### Cardiac arrest secondary to hanging

- Prioritise the ventilation aspect of CPR and use a CPR ratio of 15:2 unless an ETT is in place.
- Cervical spine immobilisation is not routinely required. This is because clinically significant cervical spine injury following hanging is extremely rare unless the patient has fallen the height of their body.
- The survival rate is low, but survivors usually come from the group that are in PEA and get ROSC within 5-10 minutes with good CPR alone.



# Cardiac Arrest

- Prolonged resuscitation in the presence of asystole is inappropriate.
- IV medicines have a very low priority.

## Cardiac arrest secondary to asthma

- Focus on using a ventilation rate of only 6/minute to avoid dynamic hyperinflation (gas trapping).
- IV adrenaline has a high priority.
- Exclude tension pneumothorax, noting this is rare. Needle chest decompression carries a significant risk of causing pneumothorax and finger thoracostomy is the preferred technique if chest decompression is required.

## Cardiac arrest secondary to anaphylaxis

- IV adrenaline has a high priority.
- If the patient is in PEA and not immediately responding to resuscitation, escalate the adrenaline doses:
  - For an adult escalate the second dose to 3 mg, the third dose to 5 mg and then revert to standard 1 mg dosing.
  - For a child escalate the doses following the same principle.
- 0.9% sodium chloride IV has a high priority:
  - For an adult administer 2-3 litres of 0.9% sodium chloride IV.
  - For a child administer 40-60 ml/kg of 0.9% sodium chloride IV.

## Cardiac arrest secondary to cyclic antidepressant poisoning

- The cardiac toxicity of cyclic antidepressants is partly caused by blockade of sodium channels within the heart and may be reduced by a large bolus of sodium ions.
- Administer 8.4% sodium bicarbonate IV:
  - 100 ml for an adult.
  - 2 ml/kg for a child.
- Administer 0.9% sodium chloride IV:
  - 1-2 litres for an adult.
  - 20-40 ml/kg for a child.

# Cardiac Arrest

- Do not administer amiodarone because it can be associated with severe worsening of shock without resolution of the rhythm.

## Cardiac arrest during pregnancy greater than 20 weeks

- In pregnancy where gestation is greater than 20 weeks, the uterus may impede venous return through the inferior vena cava in the supine position. Manually displace the uterus to the left to alleviate this.
- Consider transporting the patient with CPR en route (focusing on good chest compressions) if ROSC is not immediately achieved and time to a hospital is less than 10 minutes. Provide as much pre-hospital warning as possible. In this setting the primary reason for emergency caesarean section is to improve the chance of survival for the mother.

## Cardiac arrest secondary to hypothermia

- See the '[hypothermia](#)' guideline.

## Cardiac arrest secondary to hyperkalaemia

- See the '[hyperkalaemia](#)' guideline.

## Cardiac arrest secondary to opiate poisoning

- The best treatment is CPR that includes a focus on ventilation.
- There is no role for naloxone because cardiac arrest is secondary to respiratory arrest and once cardiac arrest has occurred naloxone has no useful effect. If ROSC occurs, naloxone should still not be administered because it may be associated with seizures, hypertension, pulmonary oedema or severe agitation.

## Cardiac arrest and implanted defibrillators/pacemakers

- Implanted defibrillators and pacemakers are usually situated in the soft tissue under the left clavicle.
- Place defibrillation pads at least 8 cm from the implanted device if possible and consider utilising the anterior/posterior position.

# Cardiac Arrest

## Cardiac arrest occurring in infants during sleep

- Prioritise the ventilation aspect of CPR using a CPR ratio of 15:2.
- Beware of misdiagnosing severe bradycardia as asystole.
- Survival is very low, but survivors tend to come from the group of patients that get ROSC within 5-10 minutes with good CPR alone.
- Prolonged resuscitation in the presence of asystole is inappropriate.
- IV medicines have a very low priority.
- Transport to hospital with CPR en route is inappropriate.

## More Information

### Version history

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**16 June 2026**

1.1.0.1

**Update:** Resuscitation guidance updated.

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**11 September 2023**

1.0.4

**Update:** Multiple changes.

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# Treatment following ROSC

- Ensure an adequate airway and adequate breathing:
  - Ventilate to an  $\text{ETCO}_2$  of 35-45 mmHg if an ETT or SGA has been placed.
  - If the patient is ventilated via an ETT, administer sedation and neuromuscular blockade if required, using the '[post intubation](#)' guideline.
- Ensure an adequate blood pressure, targeting a systolic BP of greater than or equal to 100 mmHg in an adult, and greater than or equal to the normal predicted systolic BP in a child.
- Acquire a 12 lead ECG. Treat as per the appropriate guideline if STEMI or dysrhythmia is present.
- Utilise the post cardiac arrest checklist prior to commencing transport.

## If the systolic BP is lower than the target

### For an adult:

- Consider administering 500 ml of 0.9% sodium chloride IV. Repeat once if required.
- Administer metaraminol IV:
  - Administer via an infusion device:
    - Administer an initial bolus of 0.5-1 mg metaraminol IV.
    - Start the infusion at 2 mg/hour and adjust the rate as required.
  - Alternatively, administer a bolus of 0.5-1 mg metaraminol IV every 5-10 minutes as required.
- Consider administering adrenaline if the blood pressure is unresponsive to metaraminol, or the patient is hypotensive and bradycardic:
  - Administer an adrenaline infusion via an infusion device. Start at a rate of 0.5 mg/hour and adjust the rate as required, or
  - Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
    - Administer as an infusion. Start at 2 drops/second and adjust the rate as required, or
    - Administer a bolus of 0.01 mg adrenaline (10 ml) IV every 1-2 minutes as required.

### For a child aged 5-14 years:

- Consider administering 10 ml/kg of 0.9% sodium chloride IV. Repeat once if required.
- Administer a bolus of metaraminol IV (see the paediatric drug dose tables) every 5-10 minutes as required.
- Consider administering adrenaline if the blood pressure is unresponsive to metaraminol or the patient is hypotensive and bradycardic:

# Treatment following ROSC

- » Administer an adrenaline infusion via an infusion pump using service guidelines, or
- » Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
  - » Administer as an infusion. Start at 1 drop/second and adjust the rate as required, or
  - » Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.

## For a child aged less than five years:

- Consider administering 10 ml/kg of 0.9% sodium chloride IV. Repeat once if required.
- Administer a bolus of metaraminol IV (see the paediatric drug dose tables) every 5-10 minutes as required.
- Consider administering adrenaline IV if the blood pressure is unresponsive to metaraminol or the patient is hypotensive and bradycardic.
  - » Administer an adrenaline infusion via an infusion pump using service guidelines, or
  - » Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.

## Backup

- Backup from an ICP/CCP should be requested.

## Referral and transport

- Transport to a hospital with PCI facilities if the patient has STEMI provided this is feasible and safe.
- Transport to a hospital with intensive care facilities if the patient is not obeying commands, provided this is feasible and safe.

## Post cardiac arrest checklist

- ✓ Airway and ventilation are adequate/secure.
- ✓ Vital signs, including blood glucose level.
- ✓ Backup for RSI called for if indicated.
- ✓ IV access secure.
- ✓ ETCO<sub>2</sub> 35-45 mmHg if ventilated.
- ✓ Sedation and neuromuscular blockade if indicated.

# Treatment following ROSC

- ✓ BP above target.
- ✓ 12 lead ECG.
- ✓ Confirm appropriate transport destination.

## Additional information

### General principles

- Return of spontaneous circulation (ROSC) is the presence of a palpable pulse, or clear signs of spontaneous circulation (such as non-agonal breathing, normal  $\text{ETCO}_2$  or active movement) in the absence of CPR.
- The most important aspects of post cardiac arrest care are to maintain an adequate airway, breathing and circulation, avoiding hypoxia, hyperventilation and hypotension.
- In the first few minutes following ROSC, the patient's physiological state is often quite unstable and it is important to monitor the patient closely and intervene as required. It is appropriate to prepare for transport during this time and to complete the post cardiac arrest checklist, but do not immediately commence transport unless the patient has an unresolved time critical problem.

### Targeted temperature management

- Targeted temperature management (TTM) and the avoidance of hyperthermia may improve outcomes following cardiac arrest in patients who remain unconscious.
- If core temperature monitoring is available, target a temperature of 35-36°C. Following ROSC, most patients covered with a single blanket will achieve a temperature of approximately 36°C. If core temperature monitoring is not available, cover the patient with a blanket.
- It is useful to document the patient's tympanic temperature post ROSC, but this is not a priority.
- There is no role for active warming unless the patient has had a cardiac arrest secondary to hypothermia.
- There is no role for active cooling unless the patient has hyperthermia.

### Blood pressure

- There is reduced blood flow to the brain post cardiac arrest, even if blood pressure is normal.
- Any fall in blood pressure will further reduce blood flow to the brain and this is why a systolic blood pressure greater than 100 mmHg is targeted for an adult, and a blood pressure above the normal

# Treatment following ROSC

predicted systolic blood pressure is targeted for a child.

## Response and/or transport by helicopter

- Response by helicopter may be indicated if the patient has sustained ROSC and the helicopter will arrive on scene at least 15 minutes sooner than a road-based CCP.
- Transport by helicopter may be indicated if the patient has sustained ROSC and it will save more than 30 minutes compared with road transport.
- For helicopter transport criteria to be met, the road transport time will usually need to be more than 90 minutes.

## More Information

### Version history

16 June 2026

1.1.0.1

**Update:** Post-resuscitation treatment guidance updated.

11 September 2023

1.0.4

**Update:** Multiple changes.

This guideline is for ICPs/CCPs providing inter-hospital transfer of adult patients with acute aortic dissection. Seek clinical advice if the patient is a child.

- Receive a handover from hospital staff including:
  - Where in the receiving hospital the patient is expected.
  - Who the accepting clinician is.
  - Whether or not the accepting clinician requires an update en route.
- Continue or commence a labetalol infusion to maintain the target systolic blood pressure and heart rate (see below). Seek clinical advice if an infusion has been commenced that is not labetalol.
- Commence transport without delay.
- Monitor and record vital signs every five to ten minutes. Utilise arterial line monitoring if available, but this should not delay transfer.
- Continue blood administration initiated by a medical practitioner following their verbal instructions, but this should not delay transfer.
- Seek clinical advice for any changes in patient condition that are not easily managed.

## Managing blood pressure and heart rate

- If the systolic BP is greater than 120 mmHg or the heart rate is greater than 80/minute, commence treatment with labetalol to maintain a target systolic BP of 100-120 mmHg and a target heart rate of 50-80/minute:
  - Administer a 20 mg bolus IV every five minutes, as required, while setting up an infusion.
  - Commence an IV infusion at 1 mg/kg/hour, up to a maximum of 150 mg/hour.
  - Adjust the rate as required by 20 mg/hour every 10 minutes, titrating to effect.
  - Administer additional boluses of 20 mg IV every five minutes as required.
- If labetalol is contraindicated, or the target systolic BP or target heart rate are not met within 30 minutes of commencing labetalol, seek clinical advice to discuss further options.

## Additional information

### General principles

- Acute aortic syndrome is a spectrum of life-threatening aortic pathologies including acute aortic dissection, intramural haematomas, and penetrating atherosclerotic ulcers.



- Acute aortic dissection is caused by a tear in the intimal layer of the aorta, allowing blood to dissect into the medial layer and form a false lumen. This can compromise blood flow and may lead to aortic rupture or other life-threatening complications.
- Most patients develop an aortic dissection due to increased aortic wall stress (for example, hypertension), or due to structural aortic valve or ascending aorta abnormalities. Younger patients may have an inherited disease, for example a connective tissue disorder.
- Acute aortic dissection is most commonly classified as:
  - Type A: involving the ascending aorta.
  - Type B: involving the aorta beyond the left subclavian artery only.
- The role of the ICP/CCP is to ensure rapid and safe transfer, to control blood pressure and heart rate, and be prepared to treat any complications that may occur.
- Have a low threshold for administering antiemetics.

## Complications

- Suspect complications if the patient becomes hypotensive.
- Complications from acute aortic dissection can include:
  - Extension of the dissection flap.
  - Aortic rupture.
  - Aortic vessel branch occlusion.
  - Aortic valve incompetency (regurgitation).
  - Coronary dissection (usually with clinical signs of acute MI).
  - Pericardial effusion with tamponade.

## Managing blood pressure and heart rate

- Management of blood pressure and heart rate should be initiated as soon as possible to prevent the dissected aorta from expanding further, with the risk of rupture.
- The target blood pressure and heart rate should ideally be reached within 30 minutes. It is acceptable for the heart rate to be less than 50/minute if this is required to achieve a systolic blood pressure of less than 120 mmHg.

## The administration of blood products

- Transfer should not be delayed for the provision of blood products.
- Blood transfusion may continue if this has been commenced by a medical practitioner, following their verbal instructions.
- Have a low threshold for seeking clinical advice.

## More Information

### Version history

|                          |         |
|--------------------------|---------|
| 16 June 2026             | 1.1.0.1 |
| New: Guideline released. |         |

This guideline is for patients who are in cardiac arrest secondary to trauma.

- Do not initially perform chest compressions.
- Perform the following tasks, focusing on the most likely reversible causes:
  - Control external bleeding.
  - Ventilate at 10 breaths/minute via an ETT or SGA and attach ETCO<sub>2</sub>.
  - Perform bilateral finger thoracostomies if chest injury is possible.
  - Gain IV or IO access and administer 0.9% sodium chloride:
    - 2-3 litres for an adult.
    - 40-60 ml/kg for a child.
  - Arrange for blood to be administered if available.
  - Splint the pelvis if pelvic injury is possible.
  - Align long bone fractures that are significantly displaced.
  - Administer TXA IV:
    - 2 g for an adult.
    - See the paediatric drug dose tables for a child.
- Begin transport as soon as possible.
- Consider commencing chest compressions once the above tasks are complete.
- Stop resuscitation if the rhythm deteriorates into asystole for more than a few minutes.

## Referral and transport

- Transport the patient to the most appropriate major trauma hospital, which will usually be the closest major trauma hospital.

## Additional information

### General

- A small number of patients have a primary cardiac arrest directly preceding their trauma. If this is suspected treat as a primary cardiac arrest.
- Cardiac arrest secondary to trauma has a poor prognosis and in most cases is caused by severe hypovolaemia. However, cardiac arrest in this setting is not futile and survival is possible provided the resuscitation attempt is focused on immediately correcting reversible causes.

- Resuscitation should stop if the rhythm deteriorates into asystole for more than a few minutes.
- Resuscitation should occur en route to hospital whenever feasible and safe.
- Cardiac output may be difficult to detect clinically. Attach ETCO<sub>2</sub> and use this as a measure of blood flow.
- Consider the possibility that the patient has cardiac output that cannot be detected clinically if the rhythm is PEA.
- In the absence of an immediately reversible cause, it is usually inappropriate to commence resuscitation on a patient who is in cardiac arrest and trapped, unless the patient can be extricated in the next 1-2 minutes.

## Priorities in cardiac arrest secondary to trauma

- Treatments and tasks should be performed rapidly and simultaneously if possible.
- Chest compressions are not initially performed in the setting of cardiac arrest secondary to trauma because:
  - The priority is to treat potentially reversible causes, including tension pneumothorax, hypovolaemia from blood loss and airway obstruction and/or hypoxia.
  - The person performing chest compressions may impair access to the patient and may impair the resuscitation attempt.
  - The person performing chest compressions is best tasked with a higher priority intervention.
- Chest compressions may be performed once the initial priorities have been treated and resources allow.
- Only call for blood if there is an established protocol in the area for blood to be delivered to the scene.

## More Information

### Version history

28 October 2024

1.0.5

**Update:** Multiple changes.

11 September 2023

1.0.4

**New:** Guideline released.

This guideline is for determining which patients meet criteria for major trauma in the out-of-hospital setting. Patients with major trauma should be transported to the most appropriate major trauma hospital, utilising the principles within this guideline.

## Life-threatening problems requiring immediate intervention

- Transport the patient to the closest appropriate medical facility if the patient has a life-threatening problem requiring immediate intervention that cannot be provided by personnel on scene.
- Activate staging via Comms if the medical facility is not a major trauma hospital.
- Notify staff in the staging facility as soon as possible that staging is being activated.

## Severe traumatic brain injury (TBI)

- Transport the patient to a major trauma hospital with neurosurgical facilities provided this is feasible and safe, if the patient has TBI and any of the following are present:
  - a RSI is indicated for severe TBI.
  - b Lateralising neurological signs.
  - c Clinically obvious penetrating brain injury.

## Multi-system trauma

- Transport the patient to a tertiary major trauma hospital if the patient has multi-system trauma, provided this is feasible and safe.

## Abnormal primary survey and injury patterns

- Transport to the most appropriate major trauma hospital if the patient has any of the following clinical features:
  - a Manageable airway obstruction.
  - b Respiratory distress.
  - c Shock.
  - d A motor score less than or equal to five.
  - e Penetrating injury to the neck or torso.
  - f Flail chest.
  - g Clinically obvious pelvic fracture.

- h** More than one long bone fracture.
- i** Severe limb injury with amputation, severe bleeding or severe tissue disruption\*.
- j** Severe facial injury\*\*.

\* Transport to a major trauma hospital with facilities to manage patients with severe limb injuries, provided this is feasible and safe.

\*\* Transport to a major trauma hospital with facilities to manage patients with severe facial injury, provided this is feasible and safe.

## Burns

- Refer to the '[burns](#)' guideline.

## Spinal cord injury

- Refer to the '[spinal cord injury](#)' guideline.

## Additional risk factors

- Consider transporting the patient to a major trauma hospital even if the patient does not clearly have major trauma if there are significant additional risk factors.
- Examples include:
  - a** Additional signs or symptoms of concern, such as severe pain.
  - b** High risk mechanisms of concern, such as ejection from a vehicle.
  - c** Additional patient risk factors of concern, such as pregnancy.

## Children

- If transport is occurring by helicopter, have a low threshold for transporting children with severe injuries to a tertiary hospital, even if major trauma has not been specifically identified.

## Staging

- Consider staging the patient at another medical facility en route to a major trauma hospital if:
  - a** The patient meets criteria to be transported to a major trauma hospital, and
  - b** Transport by road directly to the major trauma hospital is not feasible because of transport time, and

- c The patient has a life-threatening problem requiring immediate intervention that cannot be provided by personnel at the scene, and
- d The patient can be transported to the staging medical facility significantly faster than suitable backup can locate at the scene.

## Major Trauma Hospitals

The below table lists the major trauma hospitals in New Zealand.

| Facilities                          |        |            |          |             |               |
|-------------------------------------|--------|------------|----------|-------------|---------------|
| Northland and Auckland              | Trauma | Severe TBI | Tertiary | Severe Limb | Severe Facial |
| Whangārei Hospital                  | ✓      |            |          |             |               |
| Auckland City Hospital              | ✓      | ✓          | ✓        |             |               |
| Starship Children's Hospital        | ✓      | ✓          | ✓        | ✓           | ✓             |
| Middlemore Hospital <sup>1</sup>    | ✓      |            | ✓        | ✓           | ✓             |
| <b>Te Manawa Taki (Midland)</b>     |        |            |          |             |               |
| Waikato Hospital                    | ✓      | ✓          | ✓        | ✓           | ✓             |
| Tauranga Hospital                   | ✓      |            |          |             |               |
| Gisborne Hospital                   | ✓      |            |          |             |               |
| Rotorua Hospital                    | ✓      |            |          |             |               |
| Taranaki Base Hospital              | ✓      |            |          |             |               |
| <b>Lower North Island</b>           |        |            |          |             |               |
| Hawke's Bay Regional Hospital       | ✓      |            |          | ✓           |               |
| Whanganui Hospital                  | ✓      |            |          |             |               |
| Palmerston North Hospital           | ✓      |            |          | ✓           |               |
| Wellington Regional Hospital        | ✓      | ✓          | ✓        | ✓           | ✓             |
| <b>South Island</b>                 |        |            |          |             |               |
| Nelson Hospital                     | ✓      |            |          |             |               |
| Te Nikau Grey Hospital <sup>2</sup> | ✓      |            |          |             |               |

|                                 |   |   |   |   |   |
|---------------------------------|---|---|---|---|---|
| Christchurch Hospital           | ✓ | ✓ | ✓ | ✓ | ✓ |
| Timaru Hospital <sup>2</sup>    | ✓ |   |   |   |   |
| Dunedin Hospital <sup>3</sup>   | ✓ | ✓ | ✓ | ✓ | ✓ |
| Southland Hospital <sup>2</sup> | ✓ |   |   |   |   |

## Key:

- Trauma: designated major trauma hospital.
- Severe TBI: major trauma hospital with neurosurgical facilities.
- Tertiary: tertiary major trauma hospital.
- Limb: major trauma hospital with personnel and facilities to manage patients with severe limb injuries.
- Facial: major trauma hospital with personnel and facilities to manage patients with severe facial injury.

1. All children aged less than 15 years of age in the Auckland area who are status one or status two, even if major trauma has not been specifically identified, should be transported to Starship Hospital unless an unresolved immediately life threatening problem requires transport to Middlemore Hospital. Whenever feasible transport injured family members to the same hospital site.

2. Te Nikau Grey Hospital, Timaru Hospital, and Southland Hospital are only designated to receive patients with major trauma by road. Patients in the catchment area of these hospitals being transported by helicopter will be transported to Nelson Hospital, Christchurch Hospital, or Dunedin Hospital whenever feasible and safe.

3. Dunedin Hospital has limited neurosurgical cover. Prior to a patient with severe TBI being transported to Dunedin Hospital by air, helicopter personnel must confirm neurosurgical availability at Dunedin Hospital. If Dunedin Hospital does not have neurosurgical cover, the patient will be transported to Christchurch Hospital whenever feasible and safe.

## Additional information

### Life-threatening problems requiring immediate intervention

- It should be rare for a patient with a life-threatening problem to be transported to a medical facility that is not a major trauma hospital, because delays to definitive care worsen outcomes.
- However, some patients have a life-threatening problem requiring immediate intervention that cannot be provided by personnel at the scene, such that there is a high risk of death before reaching a major trauma



hospital and the problem may be able to be rectified at a closer medical facility. Examples include but are not limited to:

- » Severe airway obstruction despite manual techniques and airway adjuncts.
- » Inadequate breathing.
- » Severe external bleeding that is not controlled.

- The closest appropriate medical facility will usually be a hospital, but sometimes it will be a medical centre, particularly in remote areas.
- The decision to transport a patient with a life-threatening problem to a medical facility that is not a major trauma hospital requires clinical judgement and personnel must have a low threshold for seeking clinical advice. The decision should consider the nature of the patient's injuries, the rate of deterioration, the relative proximity of the medical facilities and the personnel available at the medical facility.
- Personnel in the receiving medical facility must be notified as soon as possible, preferably before leaving the scene.
- Staging must be activated via Comms, preferably before leaving the scene, if the medical facility is not a major trauma hospital.

## Severe TBI

- Patients with any of the following clinical features have a high probability of requiring urgent neurosurgery and/or neuro-intensive care and should be transported to a major trauma hospital with neurosurgical facilities, provided this is feasible and safe:
  - » RSI is indicated for severe TBI. These patients usually require neuro-intensive care and may require urgent neurosurgery.
  - » Lateralising neurological signs, for example unilateral pupil dilatation or unilateral weakness. These patients usually require urgent neurosurgery for extradural or subdural bleeding.
  - » Clinically obvious penetrating brain injury. These patients usually require neurosurgery.
- In addition to the above features, transport to a major trauma hospital with neurosurgical facilities should be strongly considered in patients with a progressively falling GCS that is not due to sedation.
- Personnel should have a low threshold for seeking clinical advice if transport to a major trauma hospital with neurosurgical facilities will be prolonged, particularly if the patient is not intubated and ventilated.

## Multi-system trauma

- Multi-system trauma is defined as the patient having major injuries to more than one body region. Examples include, but are not limited to:

- » A patient with more than one long bone fracture and a flail chest, or
- » A patient with severe facial injury and a clinically obvious pelvic fracture.
- Patients with multi-system trauma will usually benefit from transport to a tertiary major trauma hospital, provided this is feasible and safe. This is because tertiary major trauma hospitals have additional personnel and facilities to manage patients with multi-system trauma.
- Personnel should seek clinical advice prior to commencing transport, if transport to a tertiary major trauma hospital will be prolonged.

## Abnormal primary survey

- Airway obstruction:
  - » Clinical judgement is required when determining that the patient has manageable airway obstruction rather than life-threatening airway obstruction requiring immediate intervention.
  - » For the majority of patients their airway obstruction is manageable and they can be adequately oxygenated using airway adjuncts and supplemental oxygen. If this is the case the patient should be transported to a major trauma hospital.
- Respiratory distress:
  - » A patient with chest wall bruising or isolated rib fractures will often have pain when taking a deep breath, but to have respiratory distress the patient must have clinical signs of difficulty breathing or severe pain with normal breathing.
- Shock:
  - » Shock is a clinical diagnosis and cannot be tightly defined using specified vital signs.
  - » Clinical signs of shock include tachycardia (unless the patient is beta-blocked or has 'end-stage' shock when the heart rate is falling), a narrowed pulse pressure, vasoconstriction, and an altered level of consciousness (this usually occurs late in shock particularly in children and young adults and usually manifests as agitation with preservation of the ability to obey commands).
  - » Blood pressure is a poor guide to the severity of shock and must be considered as part of the overall clinical picture. Blood pressure may only begin to fall when shock is severe and blood pressure varies with age, sex, degree of fitness, and medications.
  - » If IV fluid is clinically indicated, the patient has shock.
- Motor score of the GCS:
  - » A motor score of less than or equal to five is a more useful predictor of clinically important TBI than the GCS.
  - » Consider transporting patients with a falling GCS or severe agitation to a major trauma hospital, even if they are obeying commands.

- » A patient with alcohol or drug intoxication who has a motor score of less than or equal to five following a mechanism of injury consistent with TBI, should be presumed to have severe TBI until proven otherwise, even if it is suspected that alcohol or drug intoxication is contributing to the altered level of consciousness. Factors that point towards a higher risk of TBI include a high risk mechanism of injury, clear evidence of physical trauma to the head or face, or a progressively falling level of consciousness.

## Injury patterns

- Penetrating injury to the neck or torso:
  - » To meet the definition of penetrating injury to the neck or torso, there must be a strong clinical impression that the injury has penetrated:
    - » The deep tissues when the injured region is the neck.
    - » The thoracic cavity when the injured region is the chest.
    - » The abdominal cavity when the injured region is the abdomen or pelvis.
  - » Clinical judgement may be used if the patient has a penetrating injury that appears to only involve skin or subcutaneous tissue and the patient's vital signs are normal, however a high index of suspicion is required for these injuries, and transport should usually occur to a major trauma hospital.
- Severe facial injury:
  - » To meet the criteria for severe facial injury the patient should have either obvious facial deformity or severe tissue disruption.
  - » This will usually be due to a high force mechanism of injury.
- Flail chest:
  - » Flail chest is a clinical diagnosis. There must be clinical signs of paradoxical chest wall movement with breathing.
  - » The patient usually has severe pain, but pain alone is not a diagnostic sign of flail chest.
- Clinically obvious pelvic fracture:
  - » A clinically obvious pelvic fracture is defined as an obvious major deformity or clear evidence of a pelvic fracture visible through a wound.
  - » The most common symptom of a pelvic fracture is the presence of pelvic pain, but the presence of pelvic pain alone is not sufficient to diagnose a clinically obvious pelvic fracture.
  - » There is no role for springing/stressing the pelvis for signs of instability or crepitus, because the pelvis may be severely unstable without these signs being present and the force required to elicit signs may cause harm.
- More than one long bone fracture:

- » For the purposes of meeting criteria for major trauma, a fractured long bone requires the patient to have a clinically obvious fracture of the shaft of the femur, tibia or humerus.
- » A fracture that is clinically isolated to the neck of femur or to the ankle is not considered a long bone fracture.
- » No distinction is made between closed and compound fractures for the purpose of meeting criteria for major trauma.
- Severe limb injury with amputation, severe bleeding or severe tissue disruption:
  - » To meet criteria for major trauma, amputation or severe limb injury must occur proximal to the wrist or the ankle.
  - » If direct pressure is not able to control severe bleeding or a tourniquet remains in place the patient should be transported to a major trauma hospital with vascular surgical facilities provided this is feasible and safe. If the bleeding has been controlled without a tourniquet and the limb has normal perfusion transport may occur to a hospital that is not a major trauma hospital, provided it has surgical facilities.
  - » Severe tissue disruption cannot be tightly defined, however it will usually meet one or both of the following criteria:
    - » Severe disruption of bones and/or tissue, or
    - » Poor or absent distal perfusion.

## Additional risk factors

- Consider transport to a major trauma hospital if the patient has injuries and additional risk factors, but does not meet the criteria for having major trauma.
- Signs or symptoms of concern. Examples include, but are not limited to:
  - » Severe abdominal pain.
  - » Severe spinal pain.
  - » Signs of limb ischaemia or severely impaired limb perfusion.
- High risk mechanisms of concern. Examples include, but are not limited to:
  - » Ejection from a vehicle.
  - » Person hit by a vehicle at >30 km/hr.
  - » Fall greater than three metres.
- Patient risk factors of concern. Examples include, but are not limited to:
  - » Elderly.
  - » Pregnancy.
  - » Taking an oral anticoagulant or dual antiplatelet therapy.

- » Known bleeding disorder.

- Even in the presence of additional risk factors, if the patient has apparently minor injuries and normal vital signs, clinical judgement should be used and transport should usually occur to the most appropriate hospital, rather than to a major trauma hospital. This is particularly the case if a major trauma hospital is significantly further away than the alternative hospital.

## Determining the most appropriate major trauma hospital

- Not all major trauma hospitals in New Zealand have the facilities required to treat all the injuries a patient with major trauma may have. Clinical judgement must be used when determining which major trauma hospital the patient is transported to, taking into account:
  - » The information within this guideline, and
  - » The patient's expected treatment requirements, and
  - » The transport time to the relevant hospitals.
- In most cases, the most appropriate major trauma hospital will be the closest major trauma hospital. However, in some cases there will be a choice of major trauma hospitals that the patient could be transported to and in this setting the patient should be transported to the major trauma hospital with the most appropriate facilities to meet the expected treatment needs of the patient, provided this is feasible and safe.
- Personnel should seek clinical advice if they are uncertain.

## Activation of staging by ambulance personnel

- Patients with major trauma should be transported directly to a major trauma hospital, provided this is feasible and safe. However, it may be appropriate for the patient to be transported via another medical facility while a helicopter or road ambulance is simultaneously dispatched to immediately transport the patient onward to a major trauma hospital. This is termed staging and should only occur when all the following apply:
  - » The patient meets criteria to be transported to a major trauma hospital, and
  - » Transport by road directly to the major trauma hospital is not feasible because of transport time, and
  - » The patient has a life-threatening problem requiring immediate intervention that cannot be provided by personnel at the scene, and
  - » The patient can be transported to the staging medical facility significantly faster than suitable backup can locate at the scene.
- When a medical facility is being used as a staging point by ambulance personnel:
  - » The aim of treatment at the staging medical facility is to provide immediate resuscitation/treatment and prepare the patient for immediate onward transport.

- » Personnel must notify Comms that the medical facility is being used as a staging point, prior to arrival at the medical facility and preferably before leaving the scene.
  - » Personnel in the staging medical facility must be notified as soon as possible by personnel that the medical facility is being used as a staging facility.
  - » Personnel should hand the patient over to medical facility personnel and remain to assist if required.
  - » An appropriate helicopter and crew will be dispatched as soon as possible and preferably before the patient leaves the scene. If it is not feasible or safe to use a helicopter, a road ambulance will be dispatched provided this is feasible and safe.
  - » Air Desk personnel are responsible for arranging onward transport from the staging facility to a major trauma hospital and updating staging facility personnel on the ETA of the helicopter/road ambulance.
- When a helicopter is being dispatched to a medical facility that is being used as a staging facility by ambulance personnel:
- » The helicopter mission will be dispatched as an out-of-hospital job and not as an inter-hospital transfer.
  - » The clinical care of the patient during transfer will be provided by the helicopter crew.

## Hanging

- While hanging is a traumatic injury, and many of these patients will require management in a hospital with ICU facilities, these patients usually do not require urgent surgery or advanced procedures. For this reason, patients following hanging should be transported to the most appropriate hospital which may not be a major trauma hospital.

## Determining the most appropriate medical facility

- If the patient does not meet criteria to be transported directly to a major trauma hospital, they should be transported to the most appropriate medical facility, considering:
- » The location of the scene.
  - » The anticipated healthcare needs of the patient.
  - » Where the patient lives.
- The patient should be transported to a medical facility capable of meeting their anticipated healthcare needs provided this is feasible and safe. For example, a patient with a compound fracture should be transported to a hospital with orthopaedic surgical facilities and a patient with minor injuries should be transported to an appropriate primary care facility.

## Patients that rapidly improve without treatment

- A patient may initially meet criteria for major trauma but then rapidly improve without specific treatment.
- For example, a patient may have lost consciousness and then rapidly recovered, or had respiratory distress from an emotional cause that has rapidly improved.
- Provided the patient is very clearly improving and meets no other criteria for major trauma, clinical judgement should be used and transport should occur to the most appropriate medical facility.

## Response and/or transport by helicopter

- A response by helicopter may be indicated if the helicopter will locate at least 15 minutes sooner than a road-based CCP.
- Transport by helicopter may be indicated if the patient has an immediately life-threatening problem (for example, severe shock) and it will save more than 15 minutes compared with road transport. For criteria to be met the road transport time will usually need to be more than 75 minutes.
- Transport by helicopter may be indicated if the patient does not have an immediately life-threatening problem (for example, controlled arterial bleeding without severe shock) and it will save more than 30 minutes compared with road transport. For criteria to be met the road transport time will usually need to be more than 90 minutes.

## More Information

### Version history

28 October 2024

1.0.5

**Update:** Multiple changes.

# Burns

- Administer oxygen if the patient has suspected carbon monoxide poisoning as a result of clinically significant smoke inhalation. See the '[poisoning from gases](#)' guideline for more information.
- Cool the burn for at least 20 minutes:
  - This should be at the scene unless there is an immediately life-threatening problem in the primary survey.
  - Remove all clothing and decontaminate the patient if the burns are due to chemical exposure.
- Irrigate chemical burns to the eye for at least 30 minutes.
- Estimate burn depth and size after cooling.
- Cover burns with cling film after cooling.
- Administer 0.9% sodium chloride IV if the patient has signs of hypovolaemia, or the burn area is greater than 20%:
  - 1 litre for an adult.
  - 20 ml/kg for a child.
  - Administer further 0.9% sodium chloride if transport time is greater than one hour, or if required for hypovolaemia.
- Administer nebulised bronchodilators (using the '[exacerbation of asthma](#)' guideline) if bronchospasm is prominent.
- Administer nebulised adrenaline (using the '[stridor](#)' guideline) if stridor is present.
- Consider RSI if there are immediate concerns for airway compromise, or if the burn extensively involves the face/head and transport time will be prolonged. Have a low threshold for seeking clinical advice.

## Backup

- Backup from a CCP or should be requested if the patient has airway burns or is unable to obey commands.

## Referral and transport

- Transport the patient to a regional burn centre if the burn size is greater than 10% of TBSA in an adult or greater than 5% of TBSA in a child, whenever it is feasible and safe to do so.
- The regional burn centres are Middlemore Hospital, Waikato Hospital, Hutt Hospital and Christchurch Hospital. Note that Hutt Hospital is not a major trauma hospital and if there are signs or symptoms of



# Burns

major trauma in addition to the burn injury the patient must be transported to a major trauma hospital.

- If it is not feasible or safe to transport the patient to a regional burn centre, the patient should be transported to a major trauma hospital.
- Patients with burns involving the airway should usually be transported to a regional burns centre. However, if there are signs of airway compromise, the patient should be transported to the closest hospital with personnel able to perform RSI.
- Patients with burns involving the face, hands or genitals may require treatment in a regional burn centre. However, provided the burn size is less than 10% of TBSA in an adult or less than 5% of TBSA in a child, treatment is not usually time sensitive and the patient should usually be transported to the most appropriate hospital, and subsequently transferred if required.
- Patients with a burn secondary to an electrical injury should be transported to a major trauma hospital, even if the burn area is small.

## Additional information

### Cooling, irrigating and dressing burns

- Burns are preferably cooled using cool (not ice cold) running water:
  - There is benefit in cooling the burn up to three hours post injury.
  - Beware of hypothermia during cooling, particularly if the patient is a small child or the burn is large.
  - It is acceptable to shorten the duration of cooling for burns greater than 40% if it risks causing clinically significant hypothermia.
  - Cool the burn but keep the patient warm.
- Chemical burns of the eye are potentially vision-threatening and irrigation must continue for at least 30 minutes. Chemical burns of the eye are not time critical in terms of transport to hospital and irrigation should occur at the scene whenever feasible, because it is irrigation that makes a difference to the outcome. The method of irrigation must ensure cross contamination of an uninjured eye is avoided.
- Dressing burns is not a priority but may make burns less painful. Cling film should be applied after cooling and may provide some analgesia.
- Burn gels provide analgesia but are not a substitute for 20 minutes of cool running water. If burn gels are used these should be applied after cooling is complete.

### Airway burns and inhalational injury

- Airway burns occur when there is direct thermal injury to the airway and this requires the patient to be enclosed in a small space (for example a car) or exposed for a significant period (for example trapped in a

# Burns

house fire).

- It is common to see patients with superficial 'flash' burns to the face, for example from lighting a barbeque or lighting a fire with an accelerant. These burns often appear clinically concerning but they are not usually associated with airway burns, and patients rarely require intubation for airway protection in this setting.
- The signs of airway burns include:
  - Burn involving the inside of the mouth or nose.
  - Loss of nasal hair.
  - Stridor.
  - Voice change.
- Inhalational injury occurs when lung parenchyma is affected by chemicals within smoke. Inhalational injury does not usually require specific treatment in the out-of-hospital setting, other than general supportive treatment.
- The signs of inhalation injury include:
  - Hypoxia.
  - Respiratory distress.
  - Bronchospasm.
  - Black sputum.
- Bronchospasm following inhalational injury is not usually responsive to bronchodilators, however these may be administered using the '[exacerbation of asthma](#)' guideline, but do not administer magnesium or steroids.

## Decontamination

- The most appropriate form of decontamination for a patient with chemical burns is a shower of approximately three minutes' duration, including a domestic shower. If heat has not been involved, the burn does not require cooling and in this setting the water should preferably be warm.
- Decontamination using a high pressure shower must be avoided if the patient has burns, as this may worsen the burn injury.
- Decontamination using a fire service hose (even on low pressure) should be avoided unless it is the only option, as this is likely to result in hypothermia.
- Decontamination using a fire service low pressure decontamination unit is acceptable, provided one is immediately available. However, decontamination should not be delayed waiting for one to arrive.

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- The method of decontamination must ensure cross (or self) contamination is avoided.

## Estimating burn depth and size

- Estimate burn depth only after cooling is complete:
  - Superficial burns do not have blisters and are red and painful like sunburn.
  - Partial thickness burns have blisters, weep fluid and are painful.
  - Full thickness burns are charred, white, leathery and usually painless.
- Estimate burn size only after cooling is complete:
  - Do not include superficial burns in the estimate of burn size.
  - The preferred method of estimating burn size is to use a piece of paper the same size as the patient's hand (including their fingers). This represents 1% of TBSA.
  - It is very easy to overestimate burn size.

## Electrical injury

- Patients with burns following electrical injury may have significant muscle and nerve damage along the pathway of the current, even if the external burn area is small.
- Continuous ECG monitoring is required because dysrhythmia may occur.

## Fluid loss from burns

- Shock as a result of fluid loss following burns takes several hours to develop.
- If a patient with burns has shock, look for an alternative cause.

## Response and/or transport by helicopter

- Transport by helicopter may be indicated if the patient has burns that meet major trauma pathway to a regional burn facility (>5% in paediatric patients, >10% in adults), and it will save more than 60 minutes compared with road transport.
  - For helicopter transport criteria to be met, the road transport time will usually need to be more than 120 minutes.
- Transport or response by helicopter may be indicated if the patient has TBSA >30% or airway burns, and:
  - The helicopter will arrive at scene at least 15 minutes sooner than a road-based CCP, or

# Burns

- » It will save more than 30 minutes compared with road transport.
- » For helicopter transport criteria to be met, the road transport time will usually need to be more than 90 minutes.

## More Information

### Version history

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**28 October 2024**

1.0.5

**Update:** Multiple changes.

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# Crush Injury

Use this guideline for adults with an entire lower limb (or more) trapped under a weight for more than 60 minutes. Seek clinical advice if the patient is a child.

- If there is significant risk of asphyxiation, the weight must be released as soon as possible. In other circumstances, it is appropriate to delay release of the weight for up to 20 minutes while preparing for release syndrome.
- Consider seeking clinical advice early.
- Apply a tourniquet on the limb(s) if possible.
- Gain IV access (preferably in two sites) and administer a minimum of two litres of 0.9% sodium chloride. Administer further fluid as required.
- Monitor the cardiac rhythm continuously.
- Administer continuous nebulised salbutamol approximately ten minutes prior to release of the weight.
- As the weight is being released:
  - Administer 100 ml of 8.4% sodium bicarbonate IV over one minute, and
  - Administer a large flush of 0.9% sodium chloride IV via a running line, and
  - Administer 6.8 mmol (1 g) of calcium chloride IV over one minute.
- Administer further doses of 8.4% sodium bicarbonate and calcium chloride if signs of hyperkalaemia persist or recur.
- Administer 2 g of tranexamic acid IV.

## Referral and transport

- Transport the patient direct to a major trauma hospital whenever this is feasible and safe.

## Additional information

### Crush injury

- Crush injury occurs when tissue is crushed underneath a heavy object with subsequent muscle damage.
- The degree of damage and the subsequent complications are directly related to the amount of tissue that is crushed, the weight of the object and the length of time that the weight is in place before it is released.
- Crush injury is rarely seen in isolation due to the mechanism of injury. Always look for other traumatic injuries and clinical problems (such as hypothermia) that are likely to coexist.

# Crush Injury

- Patients with severe crush injury are an important subgroup of patients with major trauma, because they may have a cardiac arrest in association with release syndrome soon after extrication.

## Release syndrome

- Release syndrome is a combination of severe shock, acidosis, hyperkalaemia and dysrhythmia that occurs immediately following release of the weight, when severe crush injury has occurred.
- For release syndrome to occur a significant amount of tissue must be crushed (equivalent to at least an entire leg) and the weight must be in place for at least one hour.

## Pathophysiology of crush injury and release syndrome

- Crush causes direct injury to muscles. Prolonged crush causes further damage by causing ischaemia. As muscle cells die, acid, cellular proteins (in particular myoglobin) and potassium leak out of cells.
- Acid, myoglobin and potassium may be contained within the limb(s). On release of the pressure, reperfusion occurs and may result in:
  - Many litres of fluid rapidly moving into the crushed area, reducing circulating volume and causing hypovolaemia. This will be exacerbated if there is also uncontrolled bleeding.
  - A rapid release of acid, myoglobin and potassium into the circulation.
- Acid interferes with normal cellular function, particularly in the heart.
- Potassium interferes with normal cardiac conduction and may cause severe dysrhythmia, including cardiac arrest.
- Inflammatory mediators cause an inflammatory response within the lungs that may cause severe pulmonary oedema and impaired oxygenation. If this occurs it usually develops over several hours following release.
- Myoglobin blocks the kidney's tubules and may cause renal failure.

## Treatment

- Tourniquets:
  - Applying tourniquets is a balance of risk. Tourniquets help contain the toxic products within the crushed area and also help control bleeding, but in the absence of severe bleeding only have a role if applied prior to release of the weight.
  - If tourniquets have been applied, they should usually remain in place until the patient is in a clinical environment where life-threatening hyperkalaemia and/or severe bleeding can be treated. However,

# Crush Injury

re-evaluate the need for tourniquets following release of the weight and consider releasing the tourniquets one at a time if the limb(s) do not appear badly injured and transport time to hospital is longer than 30 minutes.

- A minimum of two litres of 0.9% sodium chloride should be administered, even if the patient does not appear to be hypovolaemic. This is termed pre-loading. Pre-loading offers protection in three ways:
  - Increased intravascular volume helps dilute the released products.
  - Sodium ions help protect the cardiac cell membranes from the effects of potassium.
  - Increased urine flow through the kidneys helps prevent myoglobin blocking the tubules.
- Salbutamol stimulates beta-2 receptors and causes potassium to move into cells, lowering the potassium concentration in blood.
- Calcium provides protection to cardiac cell membranes from potassium.
- Sodium bicarbonate provides protection in three ways:
  - Sodium ions help protect cardiac cell membranes from the effects of potassium.
  - The bicarbonate raises blood pH, causing potassium to move into cells which lowers the potassium concentration in blood.
  - The bicarbonate raises urinary pH which reduces myoglobin deposition in kidney tubules.
- Always ensure an adequate flush between administering calcium and sodium bicarbonate because precipitation will occur if they are combined.
- Amputation is only rarely required. If amputation is being considered outside the setting of a major incident, seek clinical advice whenever feasible.

## More Information

### Version history

28 October 2024

1.0.5

**Update:** Multiple changes.

11 September 2023

1.0.4

**Update:** IV glucose and reference to PATCH trial removed.

# Bleeding following Trauma

This guideline is for patients with suspected or confirmed internal bleeding, or clinically significant external bleeding, following blunt or penetrating trauma.

- Immediately control life-threatening external bleeding:
  - Apply manual external pressure.
  - Apply an arterial tourniquet if life-threatening bleeding from a limb is not easily controlled by manual external pressure.
  - Consider packing the wound if arterial tourniquet application is not indicated, using a haemostatic dressing if one is available.
- Apply a chest seal device to penetrating chest injuries or a standard dressing if a chest seal device is not available.
- Apply a pelvic splint if the mechanism of injury could have caused a fractured pelvis.
- Administer 0.9% sodium chloride IV (using a fluid warming device if available), if the patient has signs of hypovolaemia:
  - 500 ml IV for an adult.
  - 10 ml/kg IV for a child.
  - Repeat as required, minimising the total volume.
- Administer tranexamic acid IV:
  - 2 g IV for an adult.
  - See the paediatric drug dose tables for a child.
- If shock is severe:
  - Expedite transport.
  - Call for blood, provided there is an established protocol for this in the area, particularly if extrication will be delayed and/or transport will be prolonged.
  - Consider seeking clinical advice and commencing a metaraminol infusion in an adult, if transport time is prolonged and the systolic BP is less than 80 mmHg systolic. Titrate the infusion rate to maintain a systolic BP of 80-90 mmHg.
- Provide active external warming whenever feasible.
- Splint long bone fractures.
- Notify receiving hospital personnel early. Activate a critical haemorrhage code if indicated, provided there is an established protocol for activation by ambulance personnel in the area.



# Bleeding following Trauma

## Backup

- Backup from a Paramedic must be requested if the patient has signs of hypovolaemia.
- Backup from an ICP/CCP must be requested if the patient has severe shock.
- Backup must not delay rapid transport to an appropriate hospital if shock is severe.

## Referral and transport

- Expedite transport, particularly when there are signs of severe shock. This is particularly important with penetrating truncal trauma when the transport time to an appropriate hospital is short.
- Transport the patient direct to a major trauma hospital whenever this is feasible and safe.
- Transport the patient direct to a major trauma hospital with vascular surgical facilities if there are clear signs of arterial bleeding from a large artery, for example the brachial, popliteal or femoral artery. Clinical judgement must be used if a major trauma hospital with vascular surgical facilities is significantly further away than the closest major trauma hospital, and personnel should have a low threshold for seeking clinical advice in this setting.

## Additional information

### Arterial tourniquet application

- Applying a tourniquet is a balance of risk.
  - Tourniquets help control severe bleeding but can cause significant tissue ischaemia and damage (for example to nerves) and should only be used when external pressure is insufficient to control life-threatening bleeding.
  - Applying a tourniquet will cause ischaemia of the limb distal to the tourniquet and this may compromise the success of surgical reattachment.
- When applying a tourniquet:
  - Remove clothing from the limb if possible.
  - Apply as distally as possible.
  - Do not apply over a joint.
  - Tighten until bleeding clearly stops. The tourniquet must be tight enough to stop arterial flow.
  - Leave the wound exposed so that it can be observed for bleeding.
  - Record the time of application.

# Bleeding following Trauma

- Reassess the tourniquet frequently following treatment. It may need to be further tightened if blood pressure improves.
- If the tourniquet is applied to a forearm or lower leg, the presence of two bones may limit the pressure that can be applied to vessels. If bleeding continues despite the tourniquet being tightened maximally, place the tourniquet on the upper arm or thigh.
- The tourniquet may need to be very tight if applied over a large thigh. In this setting a second tourniquet may be required proximal to the first tourniquet.
- Re-evaluate the need for a tourniquet after bleeding is controlled, IV access has been obtained and appropriate fluid resuscitation has been commenced:
  - If the source of the bleeding is a single lacerated artery, for example a lacerated brachial or radial artery, it is preferable that bleeding is controlled with direct pressure and then the tourniquet is released, particularly when transport time to hospital is over 30 minutes.
  - If bleeding cannot be controlled with direct pressure the tourniquet should be reapplied.
- A tourniquet may be used to provide pressure over a focal dressing, for example directly over a lacerated brachial artery. In this setting the tourniquet needs to be tight enough to control bleeding and not necessarily tight enough to stop arterial flow.

## Pelvic splinting

- It is often difficult to determine accurately that a pelvis is fractured by clinical examination. Assume that the pelvis is fractured if the patient has pain in the pelvic area or is unable to report pain and has a suitable mechanism of injury.
- Remove clothing prior to placing a splint whenever feasible.
- Firmly splint the pelvis using a service supplied dedicated pelvic splint.
- Do not use a sheet to splint the pelvis because a sheet cannot be applied firmly enough.

## Penetrating chest wounds and chest seal devices

- Use a service supplied dedicated chest seal provided one is available.
- Do not seal the wound with other devices (such as a defibrillator pad) as this risks the development of tension pneumothorax.
- A chest seal device is not required if the patient has been intubated and ventilated, as positive pressure in the chest from ventilation will prevent air entry into the chest via the wound.

# Bleeding following Trauma

- Do not attempt to create a sealed dressing on three sides to form a valve, as this is rarely effective.

## Administration of 0.9% sodium chloride

- 0.9% sodium chloride IV should be titrated to clinical signs of intravascular volume and perfusion.
- Large volumes of crystalloid may worsen bleeding by contributing to coagulopathy, particularly in the context of severe and uncontrolled bleeding:
  - This is why the total volume of 0.9% sodium chloride should be minimised.
  - Clinical judgment is required when determining the volume administered, but it should be uncommon to exceed one litre in an adult or 20 ml/kg in a child, and it should be rare to exceed two litres in an adult or 40 ml/kg in a child.
  - Have a low threshold for seeking clinical advice if shock is severe and transport time to the most appropriate hospital is prolonged.
- Ambient temperature IV fluid can cause hypothermia which worsens bleeding by contributing to coagulopathy. IV fluid should be warmed using a dedicated warming device if one is available.
- Do not warm IV fluid in microwaves because this can result in unreliable fluid temperatures and/or may damage the bag.

## Tranexamic acid (TXA)

- TXA reduces bleeding by reducing clot break down via fibrinolysis, and this is associated with reduced mortality rates in bleeding secondary to trauma.
- TXA is indicated if there are signs of clinically significant bleeding or signs of hypovolaemia.
- TXA administration is not a priority and it should usually be administered with the first dose of 0.9% sodium chloride.

## Blood

- Requests for blood to be delivered to the scene must not compromise rapid transport to the most appropriate hospital.
- Blood should only be requested if:
  - Shock is severe, and
  - There is an established protocol in the area for blood to be delivered to the scene, and

# Bleeding following Trauma

- » Blood can be administered 15 or more minutes before the patient can arrive in the most appropriate hospital.
- Blood may be available via staging at a small hospital/medical facility en route to the most appropriate hospital. In this setting personnel must have a low threshold for seeking advice from the on-call doctor for the service.
- ICPs/CCPs may continue blood administration initiated by a medical practitioner following their verbal instructions but should have a low threshold for seeking clinical advice.
- Requests for blood to be delivered to the scene outside of an established protocol must be made via the on-call doctor for the service. In this setting:
  - » It will usually take approximately 15 minutes to arrange for blood to be prepared, plus transport time to the scene. This means that for such a request to be appropriate, the patient should usually have severe shock and a prolonged transport time.
  - » Personnel cannot administer blood without specific instructions from the on-call doctor for the service.

## Patient temperature

- It is important to keep the patient warm because hypothermia worsens bleeding by contributing to coagulopathy.
- Passive warming (for example with external blankets) is rarely effective, and the patient should also be actively warmed.
- It is important to document the patient's temperature.

## Bleeding from wounds

- The best method of controlling severe external bleeding is usually by providing firm and sustained manual pressure using a dressing directly over the site of the bleeding, before applying a focal dressing and a pressure bandage.
- A haemostatic dressing (if available) will reduce bleeding, but do not delay manual external pressure while obtaining a haemostatic dressing.
- Topical adrenaline is not useful in the setting of severe external bleeding. Topical adrenaline may be considered in the setting of non-severe bleeding if a haemostatic dressing is unavailable.
  - » Dilute adrenaline to 1:10,000 using 0.9% of sodium chloride. For example, dilute 2 mg of adrenaline to a total of 20 ml.
  - » Soak a dressing in 1:10,000 adrenaline. Place the dressing into the wound or over the bleeding site, and continue to provide pressure over the wound.

# Bleeding following Trauma

## Critical haemorrhage codes and the assessment of blood consumption score

- Critical haemorrhage codes such as Code Crimson and Code Red are used in many major trauma hospitals to accelerate in-hospital clinical interventions to stop life-threatening bleeding.
- Notify ED personnel as soon as possible if shock is severe, to enable early activation of a critical haemorrhage code. Where an agreed protocol is in place this may be activated by transporting personnel.
- The criteria for critical haemorrhage code activation are:
  - An assessment of blood consumption (ABC) score greater than or equal to 2, or
  - The patient has received blood products.
- The ABC score is composed of (score one for each):
  - Heart rate greater than or equal to 120/min.
  - Systolic BP less than or equal to 90 mmHg.
  - Penetrating truncal trauma.
  - A positive ultrasound scan for free fluid in the abdomen, pelvis, or chest.

## Penetrating objects

- Penetrating objects should usually be removed from a limb, as this will allow direct pressure to be more reliably applied. Clinical judgement should be used if the object is large (for example a knife rather than a nail) and appears to be near an artery.
- Do not remove penetrating objects from the head or truncal area because severe bleeding may occur that cannot be compressed.
- Provide manual pressure directly around the object if severe bleeding is present and the object is not removed.

## Metaraminol

- Metaraminol is not a priority in patients with haemorrhage. However, as shock progresses vasodilation will often develop, and it is important to minimise the total volume of IV fluid administered.
- Metaraminol should only be administered to adult patients who continue to have severe hypotension following IV fluid administration.
- If a metaraminol infusion has been commenced, this should be accompanied by further fluid boluses if indicated.

# Bleeding following Trauma

- The aim of administering metaraminol is not to restore normal perfusion, but to maintain sufficient perfusion of vital organs without increasing blood loss.

## More Information

### Version history

28 October 2024

1.0.5

**New:** Guideline released.

Use this guideline for assessing patients who can obey commands and have a mechanism of injury consistent with a strike or blow to the head.

- In addition to assessing for physical injuries, assess the patient for signs or symptoms of traumatic brain injury and/or concussion.
  - Assess the patient's GCS.
  - Assess the patient's mental status.
  - Assess for symptoms of concussion such as headache and nausea.
  - Assess for signs of concussion such as vomiting, disorientation or reduced attention.
  - Assess coordination and balance by observing the patient walk and performing the finger-nose test.
- The patient should be presumed to have traumatic brain injury and/or concussion if any abnormal signs or symptoms are present, or if there is any abnormality detected in memory, coordination or balance.

## Red flags

**Should be seen in ED, usually via ambulance**

- Loss of consciousness at any time.
- Abnormal GCS.
- Abnormal mentation.
- Seizure following the injury.
- Taking an anticoagulant or dual antiplatelet therapy, or has a known bleeding disorder.
- Severe signs or symptoms of concussion.
- Focal neurological deficit.
- More than one episode of vomiting.

## Orange flags

**Should be seen in primary care within 48 hours**

- Mild or moderate signs or symptoms of concussion.
- Recent concussion episode.

## Referral and transport

- If one or more red flags are present:
  - The patient should be given a clear recommendation to be transported to an ED, noting that transport by ambulance may not be required.
  - Additional clinical judgement should be used if the patient is in hospital-level care or has dementia. In this setting, it may be appropriate for the patient to remain in the community, however personnel must liaise with an appropriate registered healthcare professional linked to the facility and provide appropriate worsening advice.
- If any orange flags are present (and no red flags):
  - Clearly recommend the patient immediately stops activity/sport that might result in further head impact/injury.
  - Clearly recommend the patient is assessed in primary care within 48 hours.
  - Provide the patient, carers, and any accompanying adults, with an appropriate information sheet (if available), along with an explanation of the advice within it.
- The patient may be given a recommendation to continue with normal activity/sport if there are clearly no signs or symptoms of concussion. Clinical judgement is required that considers the nature of the impact/injury, because the onset of signs or symptoms of concussion may be delayed. If there is any doubt, the patient should be given a clear recommendation to immediately stop activity/sport that might result in further head impact/injury until the patient can be assessed in primary care.

## Additional information

### General principles

- If the patient has had loss of consciousness, or has an abnormal GCS, or has abnormal mentation, they should receive a clear recommendation to be transported to an ED by ambulance, because further assessment is required in case subsequent intracranial bleeding develops.
- Clinical judgement is required when determining the patient has had loss of consciousness from the history given by bystanders. It is common for the patient to appear to be stunned for several seconds after a head strike and this is often reported by bystanders as being knocked out. This is not considered loss of consciousness unless there is a clear history of the patient being unconscious for a period of time that is in excess of a few seconds.
- The concussion assessment may be used for a child provided they are old enough to cooperate with having a history taken and being examined.
- The threshold for recommending transport to an ED must be lowered if the patient has alcohol and/or drug intoxication.



## Concussion

- Concussion is a form of brain injury where no detectable injury is present on CT brain imaging, but the patient has signs or symptoms of altered brain function.
- Concussion can occur without loss of consciousness.
- The associated symptoms are often significant and may impair brain function for many months.
- Concussion is most often thought of as being associated with contact sports, but also commonly occurs with assaults, falls and minor road traffic crashes.
- Repeated concussion can have long-term effects and this is why medical assessment is required prior to resuming activities that risk further concussion.

## Patients taking an anticoagulant or taking dual antiplatelet therapy or with a bleeding disorder

- There is an increased risk of intracranial bleeding following a head strike in these patient cohorts, even if there has been no loss of consciousness, and many will require a CT scan of their brain.
- Clinical judgment is required and the patient may not need to be seen in an ED if:
  - They have had a clearly trivial head strike (for example bumping their head on an open window), and
  - They have no external signs of head trauma (for example bruising, wounds or tenderness), and
  - They have no signs or symptoms of concussion, and
  - This is their only red flag.
- If transport to ED is recommended, it does not necessarily need to be by ambulance, provided the patient has a GCS of 15, is asymptomatic, is accompanied by a competent adult and suitable private transport is available within an hour or two.
- If transport to ED is not recommended, the patient should receive worsening advice, along with a recommendation to be seen in primary care within 48 hours, and to remain with a competent adult until seen.
- Dual antiplatelet therapy consists of more than one antiplatelet medicine, such as aspirin and ticagrelor.

## The elderly

- The elderly are at increased risk of intracranial bleeding, particularly subdural bleeding, following minor head injury.

- Cerebral atrophy expands the subdural space and exposes cerebral bridging veins to an increased risk of rupture, even with minor injury such as a fall where the patient appears to be uninjured.
- Subdural bleeding may occur slowly over many days, and the patient may present with subtle signs and/or symptoms such as a slight reduction in cognitive function.
- The elderly may not recognise the signs and/or symptoms of a brain injury, may not seek clinical help in a timely manner, and are at increased risk of falling following a minor brain injury.
- There is no specific age at which a patient can be considered to be 'elderly' and clinical judgement is required. The risk of a subdural haemorrhage increases over the age of 65 years, and the risk is increased further if the patient is frail.

## The finger-nose test

- Ask the patient to put the tip of their index finger on their nose.
- Hold your finger approximately 30 cm away and ask the patient to touch your finger.
- Slowly move your finger and ask them to alternately touch their nose, then your finger, then their nose etc.
- A patient with normal coordination will successfully do this. A patient with abnormal coordination will miss or overshoot.

## Clearance following a minor head strike at sporting events

- It is a common request for clearance to be provided at sporting events, to enable the injured person to continue with the sporting activity.
- Repeated minor brain injuries can predispose patients to very serious consequences including permanent cognitive and neurological deficits, and so clearance to resume activity should be carefully considered by personnel.
- A number of sporting codes have their own clearance and documentation processes. Provided these are formally structured, these may be followed. If there is a contradiction between the sporting code clearance processes and the guidance within this CPG, ambulance personnel must follow the CPG and seek clinical advice if this is difficult to resolve.
- The minimum safe period of time between minor brain injuries is unclear. The specific stand down period for a patient who is not cleared to return to the activity/sport should be provided by a healthcare professional in primary care, and not ambulance personnel.

More Information

Version history

|                                                                      |         |
|----------------------------------------------------------------------|---------|
| 16 June 2026                                                         | 1.1.0.1 |
| <b>Update:</b> Assessment, red flags and transport guidance updated. |         |

# Severe Traumatic Brain Injury

Use this guideline for patients who cannot obey commands and have a mechanism of injury consistent with traumatic brain injury (TBI).

- Routinely administer oxygen.
- Ensure an open airway and adequate ventilation.
- Measure the blood glucose concentration and treat accordingly.
- Administer 0.9% sodium chloride IV if there are signs of hypovolaemia, or the systolic blood pressure is less than 120 mmHg in an adult, or less than the normal predicted systolic blood pressure in a child:
  - 500 ml for an adult.
  - 10 ml/kg for a child.
  - Repeat as required, minimising the total volume.
- Administer metaraminol IV in addition to 0.9% sodium chloride, if the systolic blood pressure is less than 120 mmHg in an adult, or less than the normal predicted systolic blood pressure in a child.

## Metaraminol for an adult:

- Administer via an infusion device:
  - Administer an initial bolus of 0.5-1 mg metaraminol IV.
  - Start the infusion at 2 mg/hour and adjust the rate as required.
- Alternatively, administer a bolus of 0.5-1 mg metaraminol IV every 5-10 minutes as required.

## Metaraminol for a child:

- Administer a bolus of metaraminol IV (see the paediatric drug dose tables) every 5-10 minutes as required.
- If the patient has seizure activity:
  - Administer midazolam using dosing from the '[seizures](#)' guideline, and
  - Administer levetiracetam, even if seizure activity has ceased with midazolam:
    - 2 g IV for an adult.
    - See the paediatric drug dose tables for a child.
- Intubate using RSI if indicated. See the '[rapid sequence intubation](#)' guideline.
- If the patient has been intubated and has clinical signs of raised intracranial pressure:
  - Administer 8.4% sodium bicarbonate IV over approximately one minute via a running line, and anticipate the need to increase ventilation over the next 5-15 minutes:
    - 100 ml for an adult.

# Severe Traumatic Brain Injury

- » 2 ml/kg for a child.
- » Alternatively, administer hypertonic sodium chloride as per service guidelines.
- If the patient has a compound skull fracture, administer cefazolin IV:
  - » 1 g IV for an adult.
  - » See the paediatric drug dose tables for a child.

## Backup

- Backup from a CCP should be requested if the patient has a GCS less than or equal to 10.

## Referral and transport

- Transport the patient to a major trauma hospital with neurosurgical facilities provided this is feasible and safe, if the patient has TBI and any of the following are present:
  - » RSI is indicated for severe TBI.
  - » Lateralising neurological signs.
  - » Clinically obvious penetrating brain injury.
- In addition to the above features, transport to a major trauma hospital with neurosurgical facilities should be strongly considered in patients with a progressively falling GCS that is not due to sedation.
- Personnel should have a low threshold for seeking clinical advice if transport to a major trauma hospital with neurosurgical facilities will be prolonged, particularly if the patient is not intubated and ventilated.
- The hospitals in New Zealand with neurosurgical facilities are:
  - » Starship Hospital.
  - » Auckland City Hospital.
  - » Waikato Hospital.
  - » Wellington Hospital.
  - » Christchurch Hospital.
  - » Dunedin Hospital. Note that Dunedin Hospital has limited neurosurgical cover. Prior to a patient with severe TBI being transported to Dunedin Hospital by air, personnel must confirm neurosurgical availability at Dunedin Hospital. If neurosurgical cover is not available, the patient should be transported to Christchurch Hospital whenever feasible and safe.
- Transport all other patients to a major trauma hospital whenever feasible and safe.

# Severe Traumatic Brain Injury

## Additional information

### General principles

- The goal of treating a patient with severe TBI is to:
  - Recognise severe TBI, and
  - Minimise or prevent secondary brain injury, and
  - Treat other life-threatening injuries if present, and
  - Transport the patient to the most appropriate major trauma hospital whenever this is feasible and safe.
- Secondary brain injury occurs when a further physiological insult occurs to the brain after the primary (initial) injury. Secondary brain injury increases mortality and worsens neurological recovery in survivors. Common causes of secondary brain injury include:
  - Hypoxia.
  - Hypoventilation.
  - Hypotension.
  - Hyperventilation
- A patient with alcohol or drug intoxication who cannot obey commands following trauma should be presumed to have severe TBI until proven otherwise, even if it is highly suspected that alcohol or drug intoxication is the cause of the patient's altered level of consciousness.
- Hypoglycaemia can mimic severe TBI. A blood glucose concentration should always be measured.
- Sedation increases the risk of secondary brain injury and must be avoided unless the patient's level of agitation is such that treatment and/or transport cannot be provided safely. If sedation is required, the preferred medicine is ketamine.

### Airway, oxygenation and ventilation

- Oxygen is routinely administered to a patient with severe TBI to help prevent hypoxia, even if their SpO<sub>2</sub> is greater than or equal to 92%. The choice of administration device and flow rate is a pragmatic one, but for most patients a simple mask at 6 litres/minute is appropriate.
- Although there is a risk the patient may have a basal skull fracture, placement of a nasopharyngeal airway is not contraindicated if this is required to maintain an open airway, for example if there is trismus.
- Capnography is compulsory for all intubated patients. The target ET<sub>CO</sub><sub>2</sub> for patients ventilated via an endotracheal tube is 30-35 mmHg. This is intended to ensure that arterial CO<sub>2</sub> levels are at the lower end of normal. See the '[intubation and ventilation](#)' section.

# Severe Traumatic Brain Injury

## Cerebral blood flow and blood pressure

- Cerebral blood flow is dependent on the cerebral perfusion pressure (CPP).
- CPP is determined by the mean arterial pressure (MAP) and the intracranial pressure (ICP) as per the formula:  $CPP = MAP - ICP$ .
- A reduction in CPP leads to cerebral ischaemia which worsens outcome in patients with severe TBI.
- Most patients who have been intubated and ventilated will need a combination of intravascular fluid administration and metaraminol (usually via an infusion device) to maintain an adequate CPP.
- Additional clinical judgement is required if the patient has hypovolaemic shock and severe TBI.
  - In this setting it may not always be possible to achieve a normal systolic blood pressure and a clinical treatment plan that targets the most life-threatening injuries is appropriate. For example, if shock is very severe a treatment plan that tolerates a blood pressure below normal may be appropriate.
  - Personnel should have a low threshold for seeking clinical advice, particularly if the transport time is prolonged.

## Raised intracranial pressure

- Raised ICP following severe TBI is associated with poor neurological outcomes and can lead to brain compression, herniation, and reduced cerebral blood flow.
- Clinical signs of raised ICP include:
  - Lateralising neurological signs, for example, unilateral pupil dilation.
  - Posturing.
  - Severe hypertension with bradycardia or tachycardia.
  - Irregular respirations.
- 8.4% sodium bicarbonate and hypertonic saline may reduce ICP through an osmotic effect. Raising the serum sodium concentration raises the osmolality of extracellular fluid, causing water to move from the intracellular space to the extracellular space, which may reduce cerebral oedema and reduce ICP.
- 8.4% sodium bicarbonate will cause a brief rise in  $CO_2$  levels which will usually cause a brief rise in end tidal  $CO_2$ . Anticipate the possibility that an increase in the ventilation rate by approximately 20% may be needed, for approximately 5-15 minutes, after sodium bicarbonate administration.

## Skull fractures

# Severe Traumatic Brain Injury

- The presence or absence of a skull fracture does not correlate with the severity of brain injury and significant time should not be spent assessing for the presence of a skull fracture.
- A skull fracture may be present without brain injury and severe brain injury may be present without a skull fracture.
- Most skull fractures cannot be diagnosed clinically and require imaging, for example a CT scan.
- Cerebrospinal fluid (CSF) may leak from the nose or ear following a basal skull fracture. The presence of a suspected CSF leak is not an indication for transport to a hospital with neurosurgical facilities as neurosurgery (if indicated) is usually scheduled months after the injury.

## Response and/or transport by helicopter

- Response by helicopter may be indicated for patients with severe TBI requiring RSI if it will arrive at scene at least 15 minutes sooner than a road-based CCP.
- Transport by helicopter may be indicated if it will save more than 30 minutes compared with road transport. For helicopter transport criteria to be met, the road transport time will usually need to be more than 90 minutes.

## More Information

### Version history

28 October 2024

1.0.5

**Update:** Multiple changes.



# Limb and/or Soft Tissue Injuries

- Fractures with significant displacement should be realigned as soon as possible, particularly if there is impaired perfusion or sensation distal to the injury, and this should occur out-of-hospital whenever feasible and safe.
- Administer cefazolin IV if a compound fracture is present:
  - 1 g IV for an adult.
  - See the paediatric drug dose tables for a child.
- Consider relocating a dislocated joint. See the relevant guideline.

## Backup

- Backup from a Paramedic should be requested if the patient has significant pain or a compound fracture is present.
- Backup from an ICP/CCP should be requested if the patient is likely to require dissociation.

## Referral and transport

- Transport the patient to a hospital with orthopaedic surgical facilities whenever this is feasible and the patient has any of the following:
  - A clearly obvious fracture of the shaft of the femur or tibia, or
  - A clearly obvious fracture of the neck of femur, or
  - A clearly obvious compound fracture, except those isolated to the tip of a digit, or
  - A fracture with significant displacement, even if this has been realigned.
- All patients on the West Coast of the South Island with orthopaedic injuries, including compound fractures, should usually be transported to Te Nikau Grey Hospital unless major trauma is present. If transport is occurring by helicopter, use clinical judgement and consider transporting to a hospital with orthopaedic surgical facilities.

## Additional information

### General principles

# Patella Dislocation

- An attempt to relocate a dislocated patella should usually occur.
- Administer appropriate analgesia.
- Grasp the patella and push it medially (inwards) while simultaneously straightening the knee.

## Referral and transport

- The patient should be transported for clinical review and an x-ray post relocation, if it is the first time that the patella has dislocated. This could be at a primary care facility and transport by ambulance may not be required.
- Transport for clinical review is not required if:
  - The patella has dislocated before, and
  - The patella relocates, and
  - There is relief of pain, and
  - The patient can actively bend their knee.
- A patient administered fentanyl, methoxyflurane or midazolam may be given a recommendation not to be transported to an ED by ambulance provided the patient:
  - Has a GCS of 15, and
  - Is in the care of a competent adult, and
  - Is instructed not to drive a vehicle or operate machinery for at least 24 hours.
- Provide advice on taking analgesia post relocation. Regular paracetamol and/or an anti-inflammatory (following the package instructions) are usually appropriate.
- If the patient is not transported, provide advice to avoid unnecessary movement of the knee and be seen in primary care (which may include a physiotherapist) within 48 hours.
- The patient must be given a clear recommendation to be seen in a medical facility if the patella does not relocate.

## Additional information

### General principles

- Patella dislocation occurs most commonly in adolescents or young adults, following twisting on a bent knee.

# Patella Dislocation

- The patella dislocates to the lateral (outwards) aspect of the knee and this needs to be confirmed by palpating it. The patient may describe swelling on the inside of the knee, but this results from prominence of the underlying femur which is no longer covered by the patella.
- Relocation of the patella will usually be indicated by:
  - Relief of pain, and
  - Return of a normal knee shape, and
  - Return of an improved range of motion of the knee joint.
- Complications are mainly associated with first dislocations. A piece of bone may be torn off the patella during dislocation or relocation. An x-ray is required if it is the first time the patella has dislocated, or pain is not immediately relieved by relocation.

## More Information

### Version history

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**28 October 2024**

1.0.5

**Update:** Multiple changes.

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# Shoulder Dislocation

- An attempt to relocate a dislocated shoulder may occur provided:
  - The patient has had a previous dislocation of the same joint, and
  - The shoulder is dislocated anteriorly, and
  - There is no clear evidence of acromioclavicular joint dislocation, and
  - There is no clear evidence of a fracture involving the humerus, and
  - The dislocation is a result of malpositioning and/or a relatively minor force.
- To attempt relocation of a dislocated shoulder:
  - Administer appropriate analgesia.
  - Administer low dose midazolam IV (as per the '[analgesia](#)' guideline) if required, provided the patient can obey commands at all times. Relocate the shoulder joint using an established technique, for example the Stimson or modified Kocher's technique.
  - A maximum of two attempts may be made.
  - Place the arm in a sling post relocation.
- Place the arm in the most comfortable position if relocation does not occur.

## Referral and transport

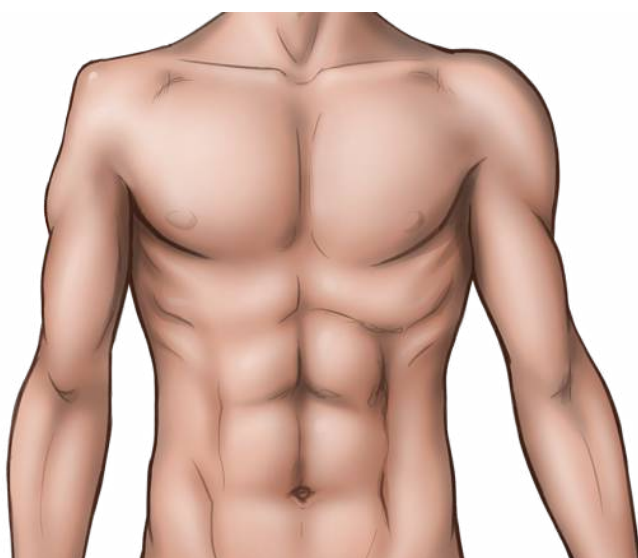
- The patient must receive a clear recommendation to be seen in an appropriate medical facility if the shoulder does not relocate. Many primary care providers will not attempt to relocate a dislocated shoulder, and transport to primary care should only occur if staff at the facility have been contacted prior to commencing transport, and have agreed to see the patient.
- If the shoulder relocates:
  - Place the arm in a sling.
  - Recommend the patient keeps their arm in a sling, avoids using the arm unnecessarily, and is seen in primary care (which may include a physiotherapist) within 48 hours.
  - Provide advice on taking analgesia. Regular paracetamol and/or an anti-inflammatory medicine (following advice on the packet instructions) is usually appropriate.
- A patient administered methoxyflurane, fentanyl or midazolam may be given a recommendation not to be transported to ED provided the patient:
  - Has a GCS of 15, and
  - Is in the care of a competent adult, and
  - Is instructed not to drive a vehicle or operate machinery for at least 24 hours.

# Shoulder Dislocation

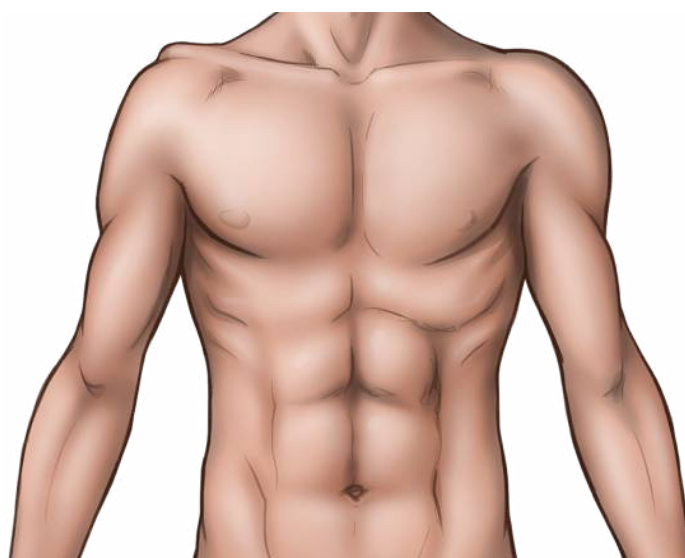
## Additional information

### General principles

- Approximately 95% of dislocations are anterior. This typically produces a squared shoulder appearance.
- The shoulder must be examined to ensure that the deformity relates to the shoulder joint and not the acromioclavicular joint situated at the outer end of the clavicle.
- Shoulder relocation is usually indicated by:
  - A palpable or audible clunk, and
  - Relief of pain, and
  - Return of a normal shoulder shape, and
  - Return of normal (or near normal) motion of the shoulder joint.



Squared shoulder appearance



Acromioclavicular joint dislocation

- Complications are mainly associated with relocation of first dislocations, and dislocations that are associated with fractures or significant trauma.
  - The complications include displacing a fracture sustained at the time of injury or causing a fracture during relocation. This is the reason for only attempting relocation if the patient has an anterior dislocation of the shoulder in the setting of previous dislocation.
- Relaxation of muscles is one of the most important aspects of relocation. This requires reassurance, analgesia, sustained traction and patience.
- The shoulder should relocate with sustained traction and no force should be used during the manoeuvres described.

# Shoulder Dislocation

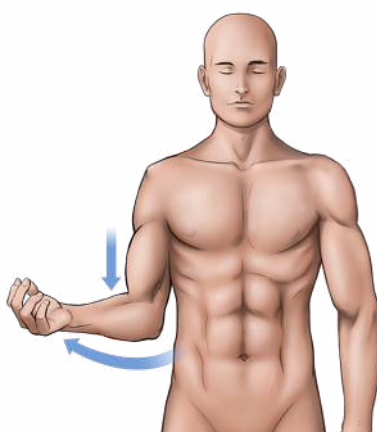
- Multiple techniques have been described for relocating a dislocated shoulder.
  - Most of them rely on adequate muscle relaxation and sustained traction.
  - If the patient has previously experienced successful relocation using a specific technique, it is acceptable to try this technique even if it is not one of the techniques described here.

## The modified Kocher's technique

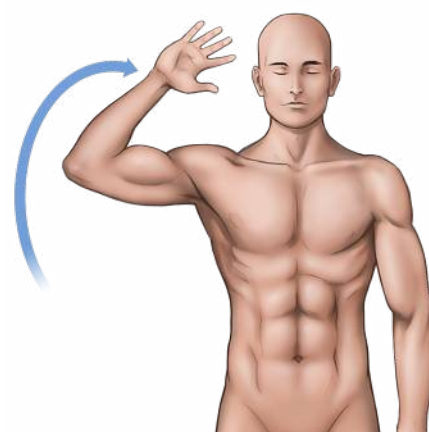
- Position the patient supine or sitting, with the arm by their side.
  - Bend the elbow to 90°.
  - Apply traction to the humerus and slowly externally rotate the arm until resistance is felt (usually approximately 45°).
  - Slowly abduct the arm, as if to scratch the back of the head with the patient's hand.
- Massage the head of the humerus if the shoulder does not relocate.



A



B



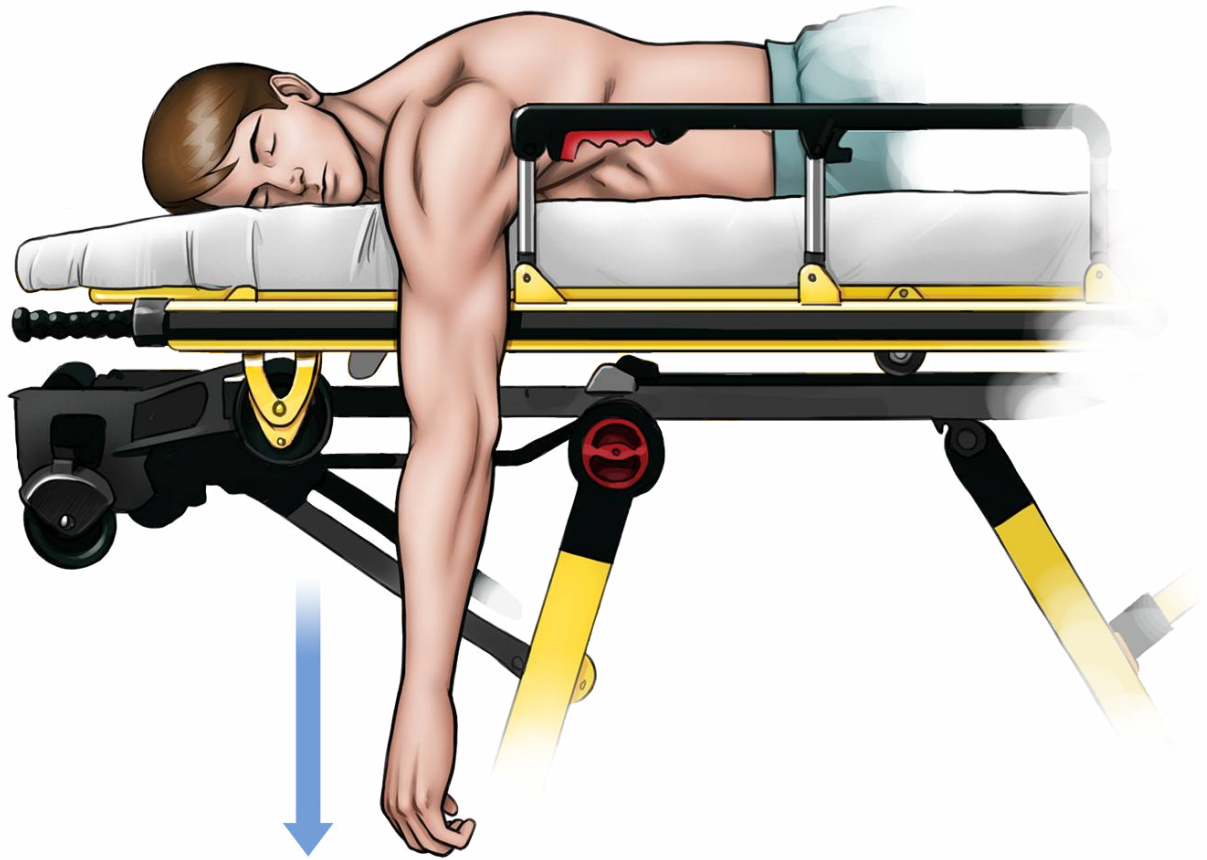
C

## The Stimson technique

- Place the patient prone on the stretcher with their affected arm hanging down, ensuring the stretcher height is such that their arm does not touch the ground.
- Apply continuous downward traction on the hand or wrist for several minutes.
- Maintain traction and gently rotate (supinate) the hand and wrist outwards if the shoulder does not relocate after several minutes. Maintain this position for several minutes.

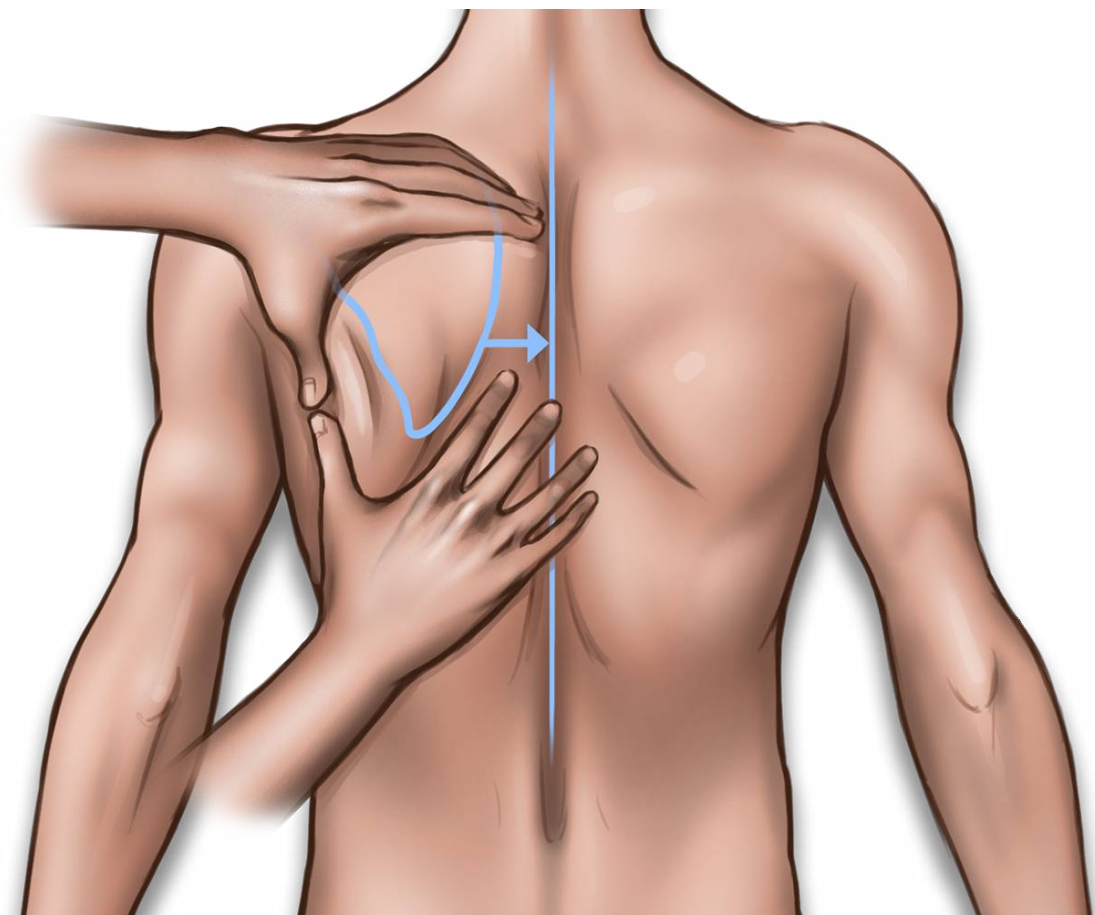
# Shoulder Dislocation

- Apply scapular rotation if the shoulder does not relocate after several minutes. Push the lower pole of the scapula (shoulder blade) towards the spine, whilst maintaining downward traction on the arm.



**Stimson technique**

# Shoulder Dislocation



Scapular rotation

## Providing sedation

- If midazolam is being administered:
  - » Routinely administer oxygen and task one person to continually monitor the patient's SpO<sub>2</sub>, breathing and level of consciousness.
  - » Have a manual ventilation bag and mask immediately available.
  - » During midazolam administration the patient must be able to obey commands at all times.

## More Information

### Version history

28 October 2024

**Update:** Multiple changes.

1.0.5



# Digit Dislocation

- An attempt to relocate a dislocated digit (finger or toe) joint should usually occur.
- Examine and record the presence of sensation.
- Administer a digital ring block.
- Administer additional analgesia if required.
- Apply longitudinal traction and hyper-extend the digit until relocation occurs.
- Splint the digit post relocation.

## Referral and transport

- If the digit does not relocate, the patient must receive a clear recommendation to be seen in an ED or in primary care, as soon as possible. Transport by ambulance is not usually required.
- If the digit relocates:
  - Splint the digit to the adjacent digits.
  - An x-ray is always required as there is a high association with fracture. This is non-urgent, and may occur within 24 hours provided there are no signs of a compound fracture, and the digit is aligned with normal distal perfusion.

## More Information

### Version history

28 October 2024

1.0.5

**Update:** Multiple changes.

# Other Dislocations

This guideline is for dislocations or severe deformities of the wrist, elbow, knee, ankle or hip joint.

- An attempt to relocate/realign a dislocated or severely deformed wrist, knee or ankle joint should usually occur, particularly when there is impaired sensation or perfusion distal to the injury, unless time to hospital is less than 15 minutes.
  - » Administer 1 mg/kg of ketamine IV (up to a maximum of 150 mg) to achieve dissociation. Use the 'preparation for dissociation' checklist.
  - » Provide sustained longitudinal traction of the limb with an assistant providing counter-traction above the injury site.
  - » Splint the joint post relocation/realignment.
  - » Place the limb in the most comfortable position if relocation/realignment does not occur.
- Do not attempt to relocate a dislocated hip or elbow without seeking clinical advice. See 'additional information'.

## Referral and transport

- Transport the patient to a hospital with orthopaedic surgical facilities whenever this is feasible and safe, even if the joint has been relocated/realigned, because surgery is usually required.

## Additional information

### General principles

- A dislocated or severely deformed joint should be relocated/realigned as soon as possible because this will relieve pain, and the longer the joint is dislocated or severely deformed, the higher is the risk that damage will be done to nerves, blood vessels and ligaments.
- If there is compromised sensation or perfusion distal to a dislocated/severely deformed joint (including compromised perfusion of skin overlying the dislocation), relocation/realignment is urgent and should occur out-of-hospital whenever this is feasible and safe. Time to relocation/realignment is more important than time to hospital, because once in hospital the patient will require to be handed over and then seen by medical staff before relocation/realignment can occur.
- Dislocations associated with significant force have an increased likelihood of being associated with fractures. It is not possible to define 'significant' force and clinical judgement is required. However, the presence of associated fractures (including compound fractures) does not change the need to relocate/realign the joint if this is clinically indicated.
- Even in the absence of compromised perfusion or sensation, a dislocation or severe deformity of the wrist, ankle or knee should be relocated/realigned as soon as possible because of the high risk of

# Other Dislocations

damage to nerves and blood vessels. This should usually occur out-of-hospital unless time to hospital is less than 15 minutes.

- If relocation is indicated because of distal ischaemia but is not achieved, applying longitudinal traction is the best method for restoring circulation, if this is feasible.
- The affected limb must always be assessed for perfusion, sensation, and movement distal to the injury. This assessment must be repeated after any attempt to relocate the joint.

## Wrist dislocation

- Wrist dislocation is frequently accompanied by fractures. The presence or absence of fractures, including compound fractures, does not alter the need to relocate/realign the wrist if clinically indicated, and the technique is the same.
- The wrist should be relocated/realigned urgently because it may be associated with damage to arteries or nerves.
- If relocation/realignment is indicated because of distal ischaemia but is not achieved, applying longitudinal traction is the best method for restoring circulation, if this is feasible.

## Elbow dislocation

- Deformity of the elbow may be due to either dislocation or fracture, and is an emergency because it may be associated with damage to the brachial artery or associated nerves.
- Relocation or realignment may be required if there is a prolonged transport time or there is impaired sensation or perfusion distal to the injury. There is a small risk of worsening the nerve or blood vessel damage during the relocation/realignment procedure.
- For these reasons, personnel must seek clinical advice prior to attempting to relocate or realign an elbow joint.

## Ankle dislocation

- Dislocation may involve the subtalar joint or the ankle joint itself. The clinical appearance is similar for both types.
- Ankle dislocation is frequently accompanied by fractures. The presence or absence of fractures, including compound fractures, does not alter the need to relocate/realign the ankle if clinically indicated, and the technique is the same.

# Other Dislocations

- Relocation will often correct with a 'clunk'. Dislocations with fractures may produce a grating sensation with no clear end point. If this occurs continue applying traction until normal alignment is achieved.

## Knee dislocation

- A dislocated knee joint should be relocated/realigned urgently.
- A dislocated knee is an emergency because it may be associated with damage to the popliteal artery or peroneal nerve. This is in contrast to a dislocated patella, which is not associated with arterial or nerve damage.

## Hip dislocation

- A dislocated hip most commonly occurs in the setting of a prosthetic joint.
- A dislocated native (non-prosthetic) hip joint is time sensitive because there is a risk of ischaemia of the femoral head.
- Relocation usually requires deep sedation or anaesthesia and for this reason should usually only occur in hospital.
- Occasionally a patient may suffer recurrent dislocations of a prosthetic hip joint and know that their hip is relatively easy to relocate. In this setting only attempt relocation following clinical advice.

## More Information

### Version history

28 October 2024

1.0.5

**Update:** Multiple changes.

# Spinal Cord Injury

This guideline is for patients with clinically significant signs and/or symptoms of acute spinal cord injury following trauma.

- Assess for the presence of other major injuries.
- Ensure the patient is appropriately immobilised, positioned, protected from pressure injury and kept warm.
- Administer 0.9% sodium chloride IV if there are signs of hypovolaemia, or the systolic blood pressure is less than 120 mmHg in an adult, or less than the normal predicted systolic blood pressure in a child:
  - 500 ml for an adult.
  - 10 ml/kg for a child.
  - Repeat as required, minimising the total volume.
- Administer metaraminol IV in addition to 0.9% sodium chloride, if the systolic blood pressure is less than 120 mmHg in an adult, or less than the normal predicted systolic blood pressure in a child.

## Metaraminol for an adult:

- Administer via an infusion device:
  - Administer an initial bolus of 0.5-1 mg metaraminol IV.
  - Start the infusion at 2 mg/hour and adjust the rate as required.
- Alternatively, administer a bolus of 0.5-1 mg metaraminol IV every 5-10 minutes as required.

## Metaraminol for a child:

- Administer a bolus of metaraminol IV (see the paediatric drug dose tables) every 5-10 minutes as required.
- Consider administering adrenaline IV if the blood pressure is unresponsive to metaraminol or the patient is hypotensive and bradycardic.

## Adrenaline for an adult:

- Administer an adrenaline infusion via an infusion pump. Start at a rate of 0.5 mg/hour and adjust the rate as required, or
- Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
  - Administer as an infusion. Start at 2 drops/second and adjust the rate as required, or
  - Administer a bolus of 0.01 mg adrenaline (10 ml) IV every 1-2 minutes as required.

## Adrenaline for a child aged 5-14 years:

- Administer an adrenaline infusion via an infusion pump using service guidelines, or

# Spinal Cord Injury

- » Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
  - » Administer as an infusion. Start at 1 drop/second and adjust the rate as required, or
  - » Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.

## Adrenaline for a child aged less than five years:

- » Administer an adrenaline infusion via an infusion pump using service guidelines, or
- » Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.

## Referral and transport

- Transport the patient direct to a spinal cord impairment (SCI) centre whenever this is feasible and safe.
- The SCI centres have the following catchment areas:
  - » Middlemore Hospital: adults in the upper two-thirds of the North Island.
  - » Christchurch Hospital: adults in the lower third of the North Island and in the South Island.
  - » Starship Hospital: children in the North Island.
  - » Christchurch Hospital: children in the South Island.
- The patient should be transported to an SCI centre if there are other signs of major trauma in addition to that of spinal cord injury, provided this is feasible and safe, as the SCI centres are also major trauma hospitals.
- If it is not feasible or safe to transport to an SCI centre, for example the patient has other major injuries and is deteriorating, the patient should be transported to the most appropriate major trauma hospital.
- Personnel should consider seeking clinical advice prior to commencing transport if this will involve a prolonged flight.
- Urinary catheterisation is not usually required, but may occur if a transport time in excess of two hours is expected, personnel are formally trained and authorised to perform urinary catheterisation and this will not cause a significant delay in commencing transport.

## Additional information

### General principles

- A patient with acute spinal cord injury following trauma should be treated in a designated SCI centre as soon as possible after injury. This is because outcomes are optimised when surgery to decompress the spinal cord is performed urgently and this is only possible 24 hours a day, seven days a week in the SCI centres.

# Spinal Cord Injury

- The patient should be transported directly to an SCI centre unless there is a specific clinical reason to transport the patient to a closer major trauma hospital (for example, an immediately life-threatening problem). This is because secondary transfers incur delays that impair outcomes.
- Patients with non-traumatic spinal cord impairment are not covered by this guideline. They should be transported to the most appropriate hospital and then transferred to an SCI centre.

## Defining spinal cord injury in the out-of-hospital setting

- Clinically significant signs and/or symptoms of spinal cord injury require the patient to have one of the following:
  - Paraplegia.
  - Quadriplegia.
  - Clinically significant limb weakness.
  - Clinically significant loss of sensation.
- Clinical judgement must be used when determining a patient has clinically significant limb weakness or clinically significant loss of sensation, particularly if the SCI centre is a significant distance from the scene and the clinical signs/symptoms are not clear. A delay in transfer to an SCI centre may result in harm if the patient has spinal cord injury, but conversely it is important to avoid transporting patients that do not require it over a long distance to an SCI centre. Personnel must seek clinical advice if they are uncertain.

## The adequacy of breathing

- Inadequate breathing is uncommon following spinal cord injury and usually only occurs with high cervical cord injury.
- Patients with diaphragmatic breathing should be transported to an SCI centre provided this is feasible and safe, the patient has adequate oxygenation with supplemental oxygen and their breathing is not deteriorating.

## Shock

- Loss of sympathetic outflow from the spinal cord following spinal cord injury can cause neurogenic shock, and in this setting the patient is usually vasodilated below the site of injury.
- The sympathetic nervous supply to the heart leaves the spinal cord in the mid-thoracic area. Spinal cord injury above the mid-thoracic area may impair sympathetic nervous supply to the heart, resulting in an absence of tachycardia despite shock.

# Spinal Cord Injury

- It is preferable to transport the patient direct to an SCI centre. However, if the patient is deteriorating and it is possible that hypovolaemic shock is present, the patient should be transported to the closest major trauma hospital, particularly if transport to an SCI centre will be significantly prolonged. Personnel must have a low threshold for seeking clinical advice.

## Spinal cord blood flow and blood pressure

- Spinal cord blood flow is dependent on the spinal cord perfusion pressure following spinal cord injury, in the same way that cerebral blood flow is dependent on cerebral perfusion pressure following TBI.
- A reduction in spinal cord perfusion pressure leads to spinal cord ischaemia which worsens outcomes.
- The pressure around the spinal cord is commonly raised in patients with spinal cord injury and this is why maintaining a normal systolic blood pressure is important.
- Metaraminol is the preferred vasopressor unless the patient's blood pressure is unresponsive to metaraminol or the patient is hypotensive and bradycardic.

## Cervical spinal cord neuropraxia

- Also sometimes called spinal shock and cervical cord concussion, this is temporary loss of motor and/or sensory function followed by recovery over a few minutes to a few hours.
- It is due to bruising and/or stretching of the cervical spinal cord and is often associated with hyperflexion or hyperextension of the neck.
- Most commonly the patient experiences immediate symptoms with any combination of the following: burning pain, numbness, tingling, weakness or paralysis. All four limbs are usually involved, but the patient may experience symptoms in only some limbs.
- Commonly the patient does not have a cervical fracture and may be completely symptom free following recovery from their symptoms.
- There is a high association between cervical cord neuropraxia and pre-existing cervical stenosis (narrowing of the cervical canal through which the spinal cord runs). If cervical stenosis is present this often requires urgent surgery.
- The symptoms of cervical neuropraxia may have resolved by the time ambulance personnel reach the scene:
  - The history must be recorded and passed on to medical staff, as this is likely to change the subsequent investigation of the patient.
  - The patient's cervical spine should not be cleared clinically, even though the patient may not have any symptoms.



# Spinal Cord Injury

- » The patient should be transported by ambulance to a hospital with CT scanning facilities.
- The patient does not need to be transported to an SCI centre, provided the signs and symptoms of cervical spinal cord neuropraxia are minor or have completely resolved.

## Response and/or transport by helicopter

- Transport by helicopter may be indicated for spinal cord injury if it will save more than 60 minutes compared with road transport.
- For criteria to be met, the road transport time will usually need to be more than 120 minutes.

## More Information

### Version history

28 October 2024

1.0.5

**Update:** Multiple changes.

# Cervical Spine Immobilisation

- If all the following criteria are met, the patient's cervical spine can be cleared clinically:
  - A normal level of alertness, and
  - No complaint of pain in the midline of the cervical spine, and
  - No tenderness to palpation at the posterior midline of the cervical spine, and
  - No signs or symptoms of spinal cord injury, and
  - No pain or other factors that might distract the patient from the pain of a cervical spine injury.
- If the patient's cervical spine cannot be cleared clinically:
  - Place a firm cervical collar if the patient has:
    - Significant posterior midline tenderness, or
    - Signs or symptoms of spinal cord injury.
  - Place a lanyard around the patient's neck if a firm cervical collar has not been placed and consider using head blocks, rolled towels or manual stabilisation of the neck.

## Referral and transport

- Transport the patient to an ED by ambulance if their cervical spine has not been cleared clinically.
- Most patients with a possible cervical spine injury will require radiological imaging and many will require a CT scan.
- The ED should preferably be within a hospital that has CT scanning facilities whenever this is feasible and safe, particularly when the patient has had a firm cervical collar placed.

## Additional information

### Clearing the cervical spine clinically

- The criteria described above may be used in a child, provided the child is old enough to cooperate with having a history taken and being examined.
- Examine the patient:
  - Feel for midline tenderness by palpating the posterior cervical spine from the skull to the prominence of the first thoracic vertebrae. Lateral muscle tenderness is not a sign of cervical spine injury.
  - Assess for normal sensation to light touch in the limbs.
  - Assess for normal movement of the limbs.
  - Assess for signs of decreased alertness.

# Cervical Spine Immobilisation

- Signs and symptoms of possible spinal cord injury are present if there is altered sensation or altered motor power (strength) in the limbs.
- Deciding if the patient has pain that might cause distraction from the pain of a cervical spine injury requires clinical judgement. To be considered distracting, the pain must be significant enough to prevent the patient from noticing that their neck is sore.
- A firm cervical collar or lanyard is not required if the cervical spine is cleared clinically.

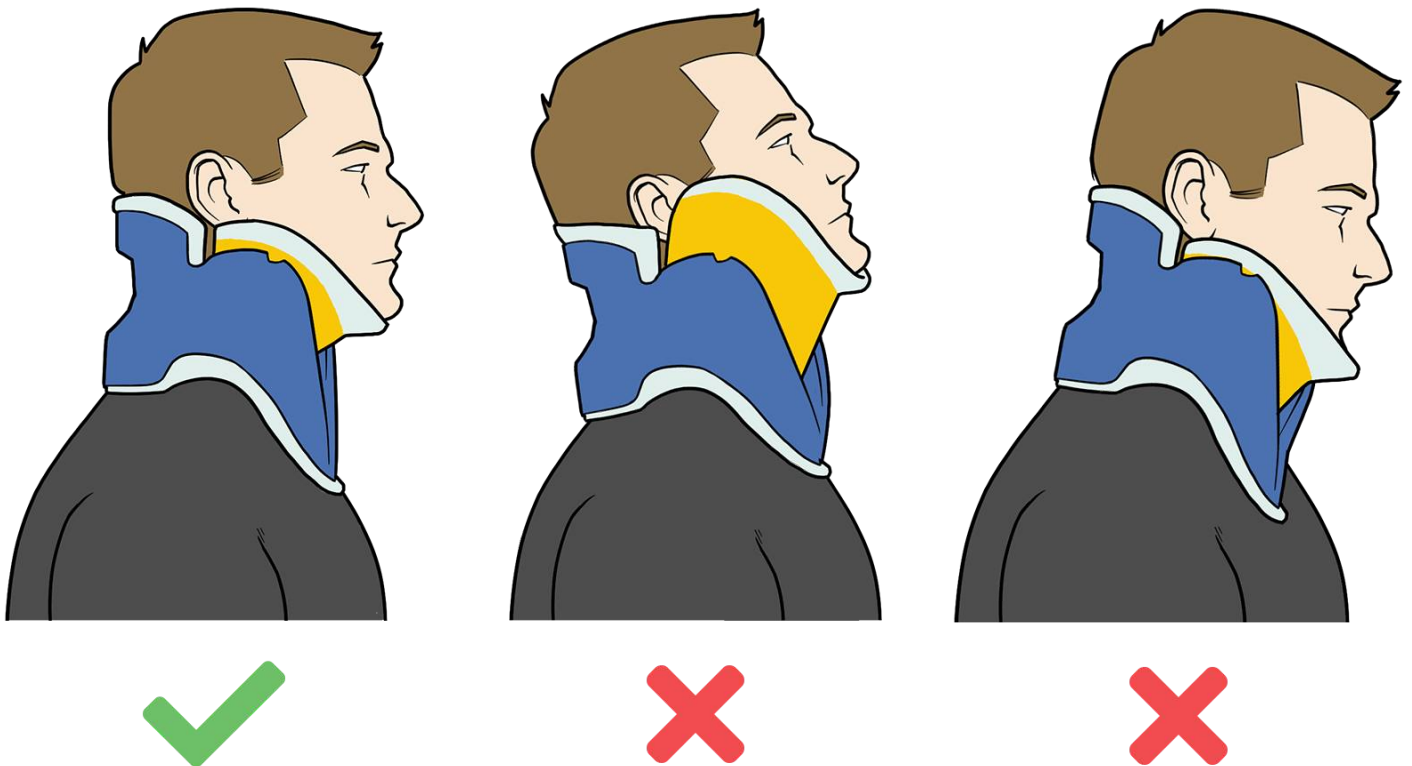
## Factors that increase risk

- The following factors increase the risk of cervical spine injury:
  - Road crash involving rollover or ejection.
  - Fall from a significant height. For example, more than one metre or more than five stairs in an adult, particularly if headfirst.
  - Injury involving significant extension or flexion. For example, a collapsed rugby scrum.
  - Injury involving axial loading of the spine. For example, diving headfirst into shallow water.
  - Pre-existing cervical spine abnormalities. For example, rheumatoid arthritis or ankylosing spondylitis.
  - The elderly or frail.

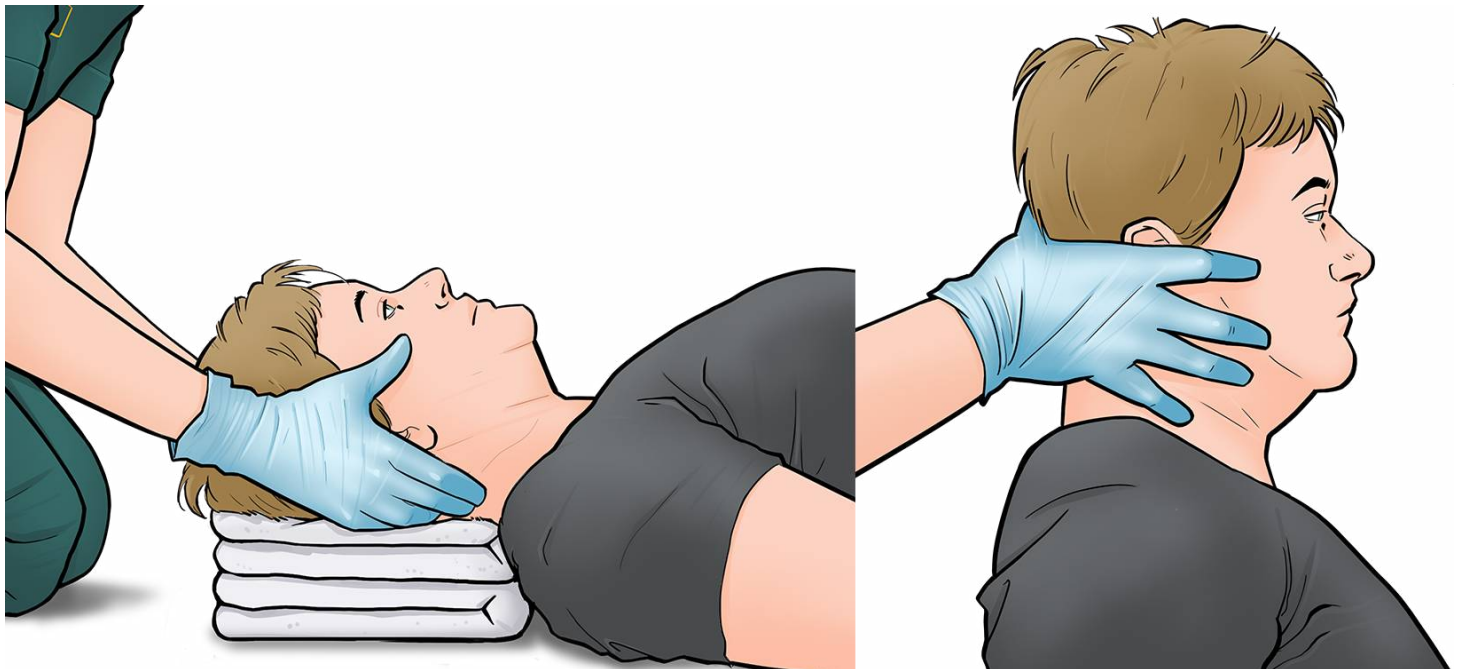
## General principles of immobilisation and firm cervical collars

- The patient should always be positioned with their spine in neutral alignment. If the patient's spine is not aligned, for example there is significant angulation, the spine must be aligned immediately.
- For most adults in the supine position, achieving neutral alignment will require 3-4 cm of flat pillow or 1-2 folded towels behind the head, noting that if pre-existing kyphosis is present the patient may require more than this. Conversely, small children may require padding under the thoracic spine to avoid neck flexion from their relatively large head.
- A firm cervical collar should not be routinely placed if the patient's cervical spine cannot be cleared clinically. Deciding to place a firm cervical collar requires clinical judgement that balances the benefits and risks:
  - A firm cervical collar will limit movement of the cervical spine but there is no good evidence that this reduces the risk of secondary spinal cord injury.
  - A firm cervical collar may worsen neck pain, promote the development of pressure areas, make airway management more difficult and raise intracranial pressure.

# Cervical Spine Immobilisation

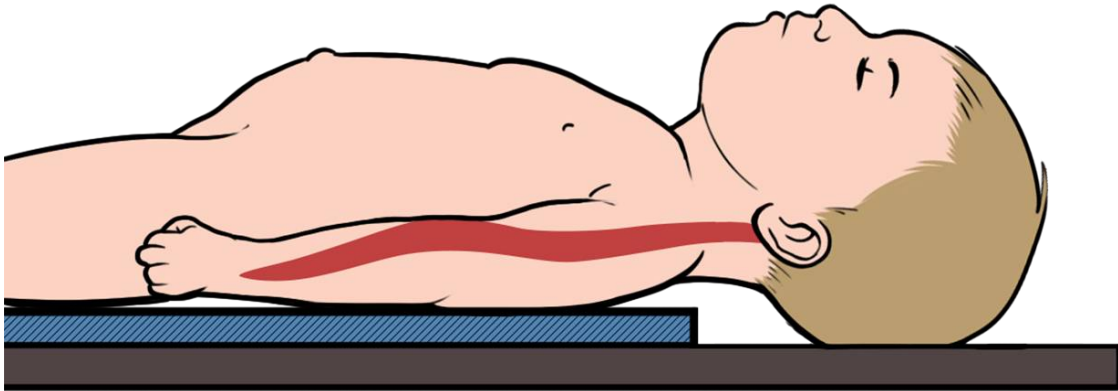


Neutral alignment with a firm cervical collar



Manual stabilisation

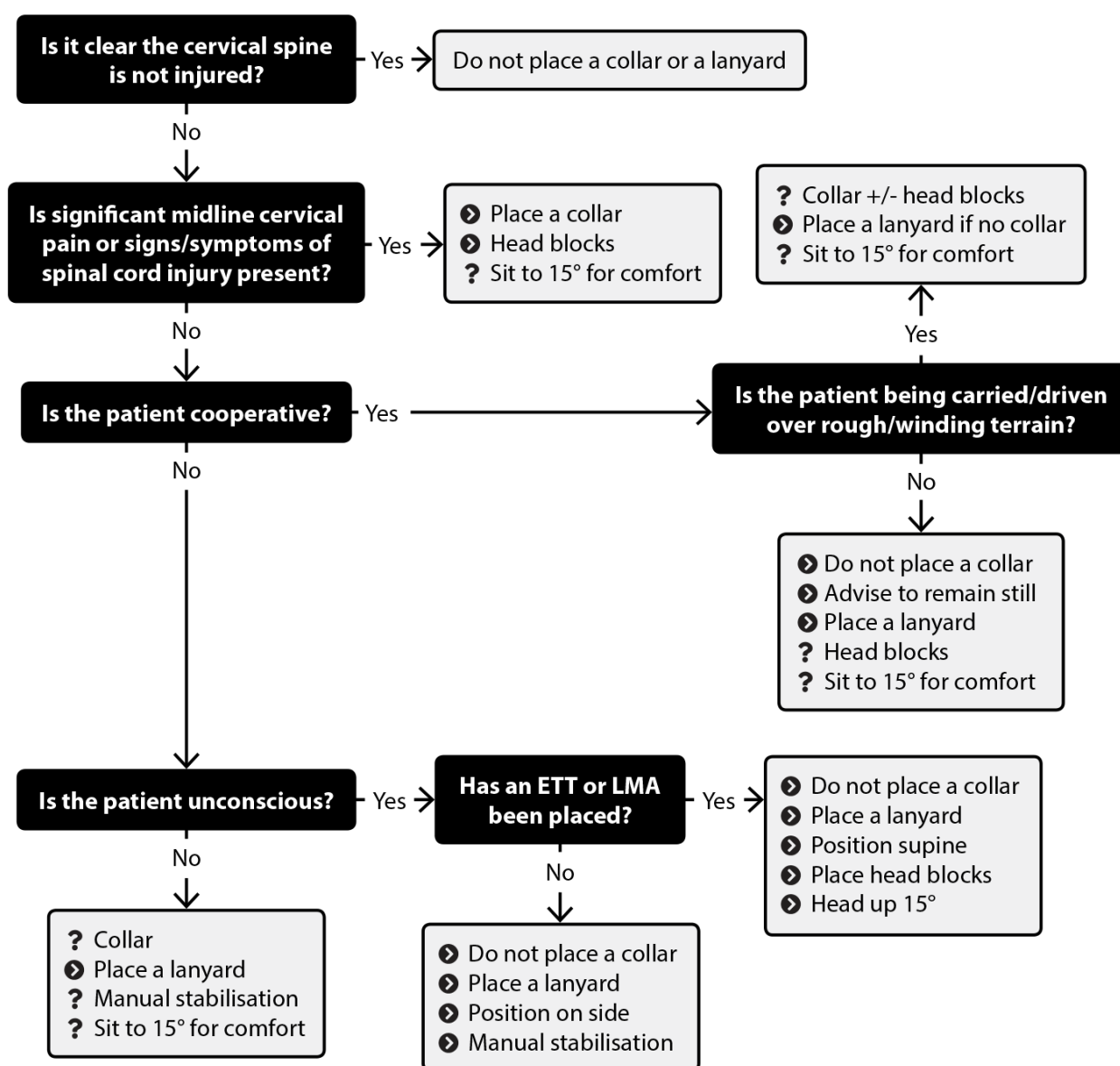
# Cervical Spine Immobilisation



Thoracic padding in a small child

## Cervical spine immobilisation summary

# Cervical Spine Immobilisation

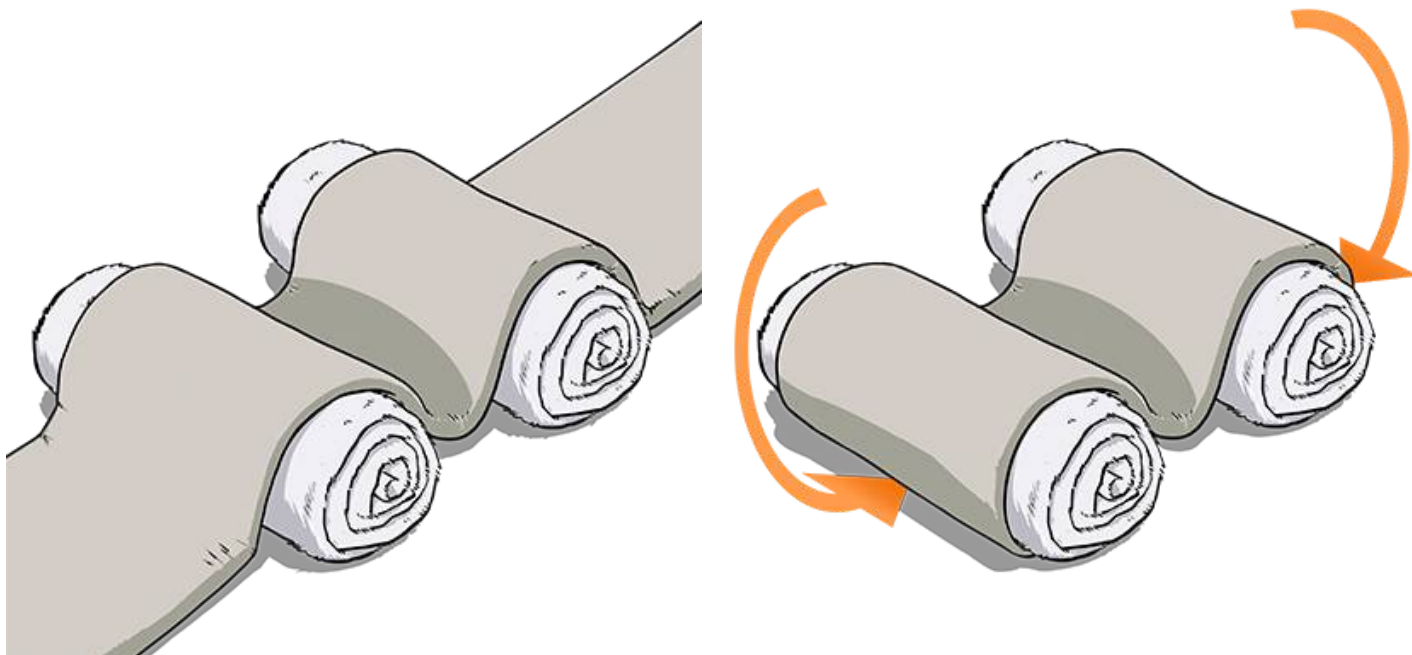


## Additional notes on immobilisation

- > Do not place tape across the patient's head and/or chin when the patient is on an ambulance stretcher because this has no useful role in immobilisation and risks creating a fulcrum effect that may worsen injury. It is acceptable to use tape for a brief period during extrication on a device such as a scoop stretcher or combi-carrier.

# Cervical Spine Immobilisation

- Consider providing manual stabilisation during extrication from a vehicle, but this is not required if the patient is cooperative, is able to extricate themselves and is instructed to keep their head and neck still.
- Cervical spine immobilisation is not required if there is a penetrating injury to the neck.
- Cervical spine immobilisation is not routinely required following hanging. Clinically significant cervical spine injury following hanging is extremely rare and should only be considered a possibility if the patient has fallen a height that is greater than or equal to the height of their body.
- If it is not possible to place a firm cervical collar, for example if the patient has severe obesity, alternative options for stabilisation (including manual stabilisation) should be considered.
- If the patient has a pre-existing abnormality of the anatomy of their cervical spine (for example ankylosing spondylitis or rheumatoid arthritis), placing a firm cervical collar may be unsafe because this may cause the patient's normal anatomical position to be altered. Maintain the normal anatomical position of the patient's spine, noting this may require the patient to sit and be provided with additional pillows.



Using rolled towels and a sheet as head blocks

## Rigid boards, scoop stretchers and combi-carriers

- Rigid boards, scoop stretchers and combi-carriers do not have a role in providing immobilisation of the spine.

# Cervical Spine Immobilisation

- All these devices carry the risk of creating pressure injury if a patient is on one for longer than 30 minutes. If this is expected, the patient should be removed from the device prior to beginning transport whenever this is feasible and safe.
- If the patient is transported on such a device, it must be removed as soon as possible after arrival at hospital.

## Allowing the patient to walk/mobilise

- If the cervical spine has not been cleared clinically and the patient is conscious:
  - The patient should be placed supine on a stretcher with their spine in neutral alignment, as soon as this is feasible and safe.
  - It is acceptable to ask a cooperative patient to walk a few steps to a stretcher, provided they are instructed to keep their head and neck as still as possible.
  - Once on a stretcher the patient should not be asked to walk or to sit in a chair.
  - When transferring the patient from a stretcher to a bed, the preferred technique is to keep the patient supine and to use a sliding or lifting device, particularly when a firm cervical collar has been placed. However, it is acceptable to ask a cooperative patient to move themselves from a stretcher to an immediately adjacent bed, provided they are instructed to keep their head and neck as still as possible.

## Cervical collars placed by other personnel

- Sometimes cervical collars (including soft cervical collars) have been placed by other healthcare providers prior to ambulance arrival.
- If a collar has been placed prior to ambulance arrival:
  - The collar should be removed if the patient's cervical spine can be cleared clinically.
  - The cervical spine immobilisation guidelines should be followed if the patient's cervical spine cannot be cleared clinically and this may involve removing or replacing the collar.
  - Removing or replacing the collar must be explained in a collegial manner to the other healthcare providers as a requirement under these CPGs.
  - The patient must receive an explanation that this is not a criticism of the clinical care they have already received.
  - If conflict occurs with the other healthcare providers, personnel should not persist with removing the collar and should begin transport from the scene. En route the patient should be reassessed and consideration given to the collar being removed or replaced.



# Cervical Spine Immobilisation

## Prophylaxis of nausea and vomiting

- Prophylactic administration of antiemetics is not routinely required for a patient with an immobilised cervical spine.
- Consider administering an antiemetic if:
  - The patient has nausea, or
  - The nature of the patient's injuries or their position is such that vomiting would be particularly problematic, or
  - Transport by road or via helicopter is prolonged.

## More Information

### Version history

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28 October 2024

1.0.5

**Update:** Multiple changes.

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# Tension Pneumothorax

## If the patient is spontaneously breathing and is not peri-arrest/in cardiac arrest

- Decompress the affected side(s) using a specific chest decompression device.
  - The preferred site is the 5<sup>th</sup> intercostal space in the anterior axillary line.
  - Decompress in the 2<sup>nd</sup> intercostal space in the midclavicular line if the preferred site is not feasible.
- Decompress the affected side(s) using finger thoracostomy in the 5<sup>th</sup> intercostal space in the anterior axillary line, if a specific chest decompression device is not available or decompression with a device was not successful. Consider administering a [dissociative dose](#) of ketamine prior to the procedure.

## If the patient is ventilated or is peri-arrest/in cardiac arrest

- Decompress the affected side(s) using finger thoracostomy in the 5<sup>th</sup> intercostal space in the anterior axillary line.

## Following chest decompression

- Administer 0.9% sodium chloride IV if the patient has signs of hypovolaemia:
  - 500 ml IV for an adult.
  - 10 ml/kg IV for a child.
  - Repeat as required, minimising the total volume.
- Administer 1 g cefazolin IV. See the paediatric drug dose tables for a child.
- Dress thoracostomy wounds using a chest seal device if available. If not available, use a standard dressing.
- Reopen the thoracostomy wounds if signs of tension pneumothorax recur.
- Consider placing an endotracheal tube (ETT) in the thoracostomy wound to maintain patency if:
  - The wound has been reopened because signs of tension pneumothorax have recurred, or
  - Reopening the wound will be difficult during transport due to the wound location.

## Backup

- Backup from an ICP/CCP must be requested.

## Referral and transport

- Transport the patient to a major trauma hospital if the tension pneumothorax is secondary to trauma, whenever this is feasible and safe.

# Tension Pneumothorax

## Additional information

### Pneumothorax

- Pneumothorax can occur under three circumstances:
  - Traumatic pneumothorax most commonly occurs following blunt trauma, noting that even in the setting of penetrating trauma a pneumothorax is not always present.
  - Primary spontaneous (idiopathic) pneumothorax occurs when the lung tissue is normal. The classic presentation is a tall young male with sudden onset of dyspnoea and pleuritic chest pain, often at the shoulder tip. It often occurs secondary to rupture of a congenital weakness (bleb) of the pleura.
  - Secondary spontaneous pneumothorax occurs secondary to abnormal lung tissue. The pleural rupture occurs as a consequence of lung disease, for example COPD, asthma or autoimmune lung disease.
- Pneumothorax without tension is common following trauma and does not usually require intervention in the out-of-hospital setting. Even in the setting of positive pressure ventilation, intervention is not usually required unless there are signs of significantly impaired oxygenation or ventilation, or tension pneumothorax is present.

### Tension pneumothorax

- Tension pneumothorax is a cause of obstructive shock that occurs when a pneumothorax is under positive pressure that is high enough to impair venous return to the right heart.
- Tension pneumothorax occurs when the volume of air within the pleural space progressively increases with each inspiration, but is unable to escape during expiration because the pleural tear acts as a one-way valve. With each breath the pressure within the pleural space increases until it is higher than the venous pressure within the superior and inferior vena cava, resulting in a progressive fall in venous return to the right heart and cardiac arrest if not treated.
- Tension pneumothorax is uncommon and only occurs in a small subset of patients with pneumothorax, usually those that are receiving positive pressure ventilation.
- The signs and symptoms of tension pneumothorax include:
  - Progressively worsening shock. This is an important defining feature. If the patient does not have progressively worsening shock they do not have tension pneumothorax.
  - Progressively worsening dyspnoea.
  - Distended jugular veins. This occurs as a result of impaired venous return to the right heart. Very rarely, distended jugular veins may not be present if the patient has both tension pneumothorax and severe hypovolaemic shock.
  - Tracheal deviation away from the affected side. This is an extremely late sign as it requires very high pressure within the thorax and is rarely seen.

# Tension Pneumothorax

## Treatment

- Unnecessary chest decompression carries risks. However, untreated tension pneumothorax will cause death and the balance of risk is in favour of chest decompression if the diagnosis is clear, or the patient has signs consistent with tension pneumothorax and is progressively deteriorating with shock despite supportive treatment.
- IV fluid is usually helpful because most patients have coexisting bleeding, and raising venous pressure within the superior and inferior vena cava will produce a temporary improvement in venous return to the right heart.
- If a specific chest decompression device is available (for example a Turkel needle), this must be used in preference to an IV cannula because it is safer and more effective.
- Decompression in the 5<sup>th</sup> intercostal space in the anterior axillary line is safer and easier than decompression in the 2<sup>nd</sup> intercostal space in the midclavicular line because there is usually less distance to the pleura.
- Needle decompression is associated with the risks of:
  - Creation of a pneumothorax.
  - Haemorrhage (from vessel injury and of particular concern in the 2<sup>nd</sup> intercostal space).
  - Air embolism.
  - Failure to decompress the chest. Failure to reach the pleura is common with a needle technique, particularly if the chest wall is thick.
- Finger thoracostomy is usually more effective and safer than needle decompression, but is associated with an increased risk of infection.
- With either technique it is vital to ensure that the pleura has been reached and it is worth taking extra time to be confident this has occurred.
- The most important sign of success following decompression is a sudden improvement in the severity of shock.
- It is possible for the tissues to close around a finger thoracostomy wound, particularly if the chest wall is thick, and thus a tension pneumothorax may recur following finger thoracostomy. Reopen the thoracostomy wound and widen the aperture if signs of tension pneumothorax recur.
- When placing an ETT in the thoracostomy wound:
  - Select the same ETT size that would be suitable for the patient's airway.
  - Insert the ETT approximately 10 cm through the wound.
  - Inflate the balloon and secure the ETT to ensure it remains in the chest.

# Tension Pneumothorax

- » Connect the ETT to a one-way valve if available.

## Response and/or transport by helicopter

- Response by helicopter may be indicated if it will arrive at scene at least 15 minutes sooner than a road-based CCP.

## More Information

### Version history

28 October 2024

1.0.5

**Update:** Multiple changes.

# Amputation

This guideline is for patients with complete or near-complete amputations of limbs, extremities or digits.

- Compress significant external bleeding and apply a tourniquet if required.
- Recover the amputated part(s) provided it is feasible and safe to do so.
- Take a photo of the amputated part(s) if possible.
- Protect the amputated part(s):
  - Wash the amputated part(s) using clean running water or 0.9% sodium chloride if severely contaminated, but this is not a priority and significant time should not be taken to accomplish this.
  - Wrap in a clean damp towel or dressing.
  - Place into a clean (preferably sealed) plastic bag and place this into a second plastic bag.
  - Place a mixture of water and ice (icy slush) in the second (outer) plastic bag. It is acceptable to use cold packs and water if ice is not available.
  - Do not allow the part(s) to come into direct contact with ice, or into prolonged direct contact with fluid.
- Administer cefazolin IV:
  - 1 g IV for an adult.
  - See the paediatric drug dose tables for a child.

## Backup

- Backup from an ICP/CCP should be requested if the patient has severe shock.

## Referral and transport

- Patients with isolated amputated limbs, extremities or digits should be transported directly to a hospital with acute plastic surgical facilities whenever it is feasible and safe, provided:
  - Amputation is proximal to the distal inter-phalangeal joint, and
  - The amputated part(s) are available to be transported with the patient, and
  - The amputated part(s) are not severely crushed or mangled, and
  - The amputated part(s) have been cut and not avulsed (pulled) from the body, and
  - The patient can be transported to a hospital with acute plastic surgical facilities within four hours of injury.
- If there are signs or symptoms of major trauma in addition to the amputation, the patient must be transported to a major trauma hospital whenever feasible and safe.

# Amputation

- The following hospitals have acute plastic surgical facilities:
  - Middlemore Hospital.
  - Waikato Hospital.
  - Hutt Hospital.
  - Christchurch Hospital.
  - Dunedin Hospital.
- Text or email a photo of the amputated part(s) to the receiving surgical team via the Clinical Desk, if possible.
- Have a low threshold for seeking clinical advice, particularly if transport time to hospital will be prolonged or if it is unclear whether the patient should be transported to a hospital with plastic surgical facilities.

## Additional information

### General principles

- Surgical reattachment of an amputation is usually only possible within four hours of the injury.
- Transporting the patient to a hospital that does not have acute plastic surgical facilities incurs a clinically significant delay that will reduce the chance of successful reattachment.
- Surgical reattachment of an amputation is usually only feasible if the amputated part(s) have been cut from the body and are intact. However, parts of crushed, mangled or avulsed extremities may still be used by the surgical team as part of the surgical repair, and should be transported with the patient whenever feasible.
- Have a low threshold for seeking clinical advice if the patient has any of the following features, because they are unlikely to have their amputation successfully reattached:
  - Has peripheral vascular disease, or
  - Has diabetes, or
  - Is aged greater than 75 years.

### Tourniquet application

- Applying a tourniquet is a balance of risk.
  - Tourniquets help control severe bleeding but can cause significant tissue ischaemia and damage (for example to muscles and nerves) and should only be used when direct pressure is insufficient to control severe bleeding.
  - Applying a tourniquet will cause ischaemia of the limb distal to the tourniquet and this may compromise the success of surgical reattachment.

# Amputation

- When applying a tourniquet:
  - Remove clothing from the limb if possible.
  - Apply as distally as possible.
  - Do not apply over a joint.
  - Tighten until bleeding clearly stops. The tourniquet must be tight enough to stop arterial flow.
  - Leave the wound exposed so that it can be observed for bleeding.
  - Record the time of application.
- Reassess the tourniquet following treatment. It may need to be further tightened if blood pressure improves.
- If the tourniquet is applied to a forearm or lower leg, the presence of two bones may limit the pressure that can be applied to vessels. If bleeding continues despite the tourniquet being tightened maximally, place the tourniquet on the upper arm or thigh.
- The tourniquet needs to be very tight if applied over a thigh, particularly if the thigh is large. Occasionally a second tourniquet may be required proximal to the first tourniquet.
- Re-evaluate the need for a tourniquet after bleeding is controlled, IV access has been obtained and appropriate fluid resuscitation has been commenced.
  - If the source of bleeding is a single lacerated artery, for example a lacerated brachial or radial artery, it is preferable that bleeding is controlled with direct pressure and then the tourniquet is released, particularly when transport time to hospital is over 30 minutes.
  - If bleeding cannot be controlled with direct pressure the tourniquet should be reapplied.
- A tourniquet may be used to provide direct pressure over a dressing, for example directly over a lacerated brachial artery. In this setting the tourniquet needs to be tight enough to control bleeding and not necessarily tight enough to stop arterial flow.

## Response and/or transport by helicopter

- Transport by helicopter may be indicated if it will save more than 30 minutes compared with road transport and re-attachment is considered possible.
- For criteria to be met, the road transport time will usually need to be more than 90 minutes.

## More Information



# Amputation

## Version history

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**28 October 2024**

1.0.5

**Update:** Multiple changes.

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# Eye Injuries

This guideline is for patients with blunt or penetrating injuries that are isolated to the eye and/or the immediate tissues around the eye.

- Position the patient in a head up position.
- Irrigate chemical burns for at least 30 minutes.
- Apply pressure to external bleeding, but do not apply pressure to the globe if there is a possibility of a penetrating injury.
- Attempt to flush out a 'simple floating' foreign body using 0.9% sodium chloride.
- Examine and record the following:
  - The appearance of the eye and surrounding tissues.
  - Pupil size, shape and equality.
  - Pupil reactivity to light, including the consensual light reflex.
  - Eye movement, vertically and horizontally.
  - An approximate assessment of the vision in each eye.
- Administer an antiemetic if the patient has an injury that appears to have penetrated the globe. See the ['nausea and/or vomiting'](#) guideline.

## Backup

- Backup is rarely required, but backup from an ECP may assist with diagnosis and treatment.

## Referral and transport

- The patient must be given a clear recommendation to be assessed in an ED if any of the following are present:
  - Injuries that appear to penetrate the globe or a mechanism of injury that suggests penetration is likely.
  - Lacerations involving the eyelid.
  - Deep and/or severe pain.
  - Hyphaema (bleeding into the anterior compartment).
  - Suspected retrobulbar haematoma.
  - Chemical injury.
  - New reduction in vision.

# Eye Injuries

- » Abnormal pupillary light reflex.
  - » Reduced ability to move the eye.
  - » Swelling of the eyelids preventing full examination of the eye.
- The patient must be given a clear recommendation to be assessed in a primary care facility or in an ED if any of the following are present:
    - » Suspected foreign body, provided it does not appear to have penetrated the globe.
    - » The sensation of a foreign body persisting after removal.
    - » Suspected corneal abrasion.
  - The patient should be given a clear recommendation to remain in the community and seek follow up in primary care if they are concerned, if any of the following are present:
    - » Subconjunctival haemorrhage with otherwise normal findings on examination.
    - » Bruising of the soft tissues around the eye ('black eye'), with or without subconjunctival haemorrhage, with otherwise normal findings on examination.
  - Have a lowered threshold for recommending the patient is assessed in an ED if you are uncertain of the severity of injury, or the patient is taking an anticoagulant.
  - Avoid transporting the patient higher than 300 metres above sea level provided this is feasible and safe, if it is possible the globe has been penetrated.
  - It is not necessary to routinely transport the patient to a hospital with facilities for eye surgery and the patient should usually be transported to the most appropriate ED or medical facility, and referred to an eye surgeon if required. However, transport to a hospital with facilities for eye surgery should occur if it is clear the globe has been penetrated, provided it is feasible and safe to do so, and the hospital has an ED.

## Additional information

### General principles

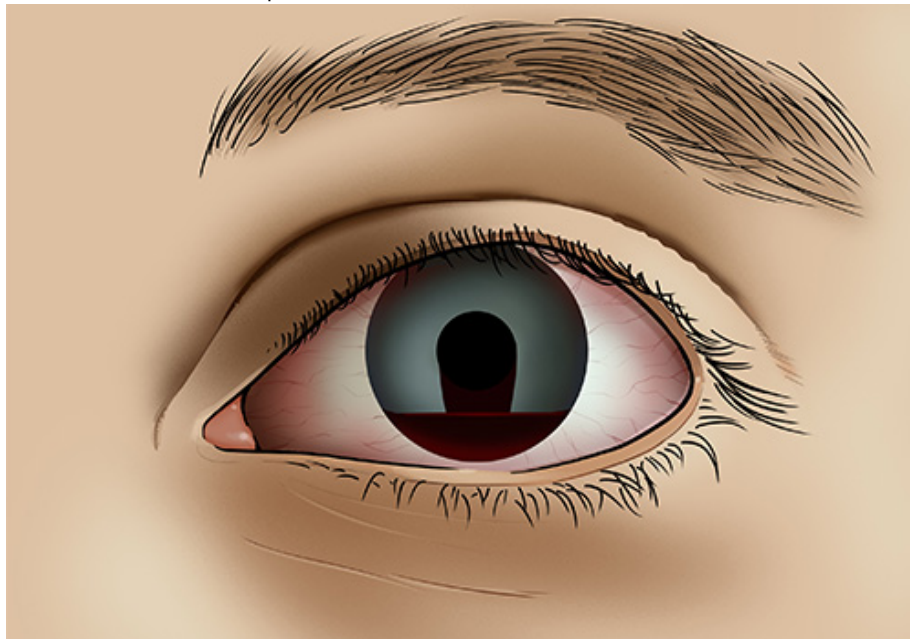
- Contact lenses should be removed whenever feasible.
- Chemical burns of the eye are potentially vision-threatening and irrigation must continue for at least 30 minutes. Chemical burns of the eye are not time critical in terms of transport to hospital and irrigation should occur at the scene whenever feasible, because it is irrigation that makes a difference to the outcome. The method of irrigation must ensure cross contamination of an uninjured eye is avoided.
- If a specific buffering agent is available (such as diphoteryne), this should be used because it is more effective than irrigation with saline or water.
- Raised intraocular pressure may worsen outcomes, particularly if the injury has penetrated the globe:

# Eye Injuries

- » Placing the patient in a semi-sitting position lowers intraocular pressure.
  - » Transporting the patient at altitude lowers the external pressure and increases the risk that intraocular contents may leak from the globe if there has been a penetrating injury.
  - » Vomiting raises intraocular pressure and this is why an antiemetic is routinely administered if the injury appears to have penetrated the globe.
- Do not apply lignocaine or ropivacaine topically to the eye because the solutions contain a preservative that may cause harm.

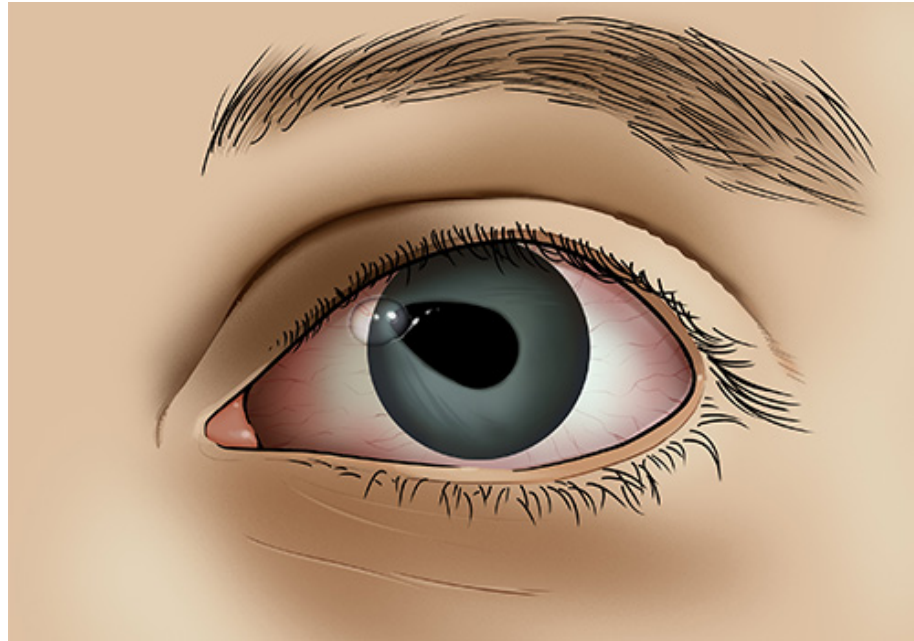
## Common injury patterns

- **Corneal abrasion** commonly occurs with a glancing blow to the eye. Corneal abrasions are not always clinically obvious and the diagnosis often requires the use of fluorescein (a dye) and a blue light to make it fluoresce. Assume the patient has a corneal abrasion if they have a mechanism of injury consistent with a glancing blow and pain during blinking.
- **Hyphaema** is bleeding within the anterior chamber and usually occurs after a direct blow to the eye. A layer of blood may be visible and the patient will usually have blurred vision which may be red-stained.

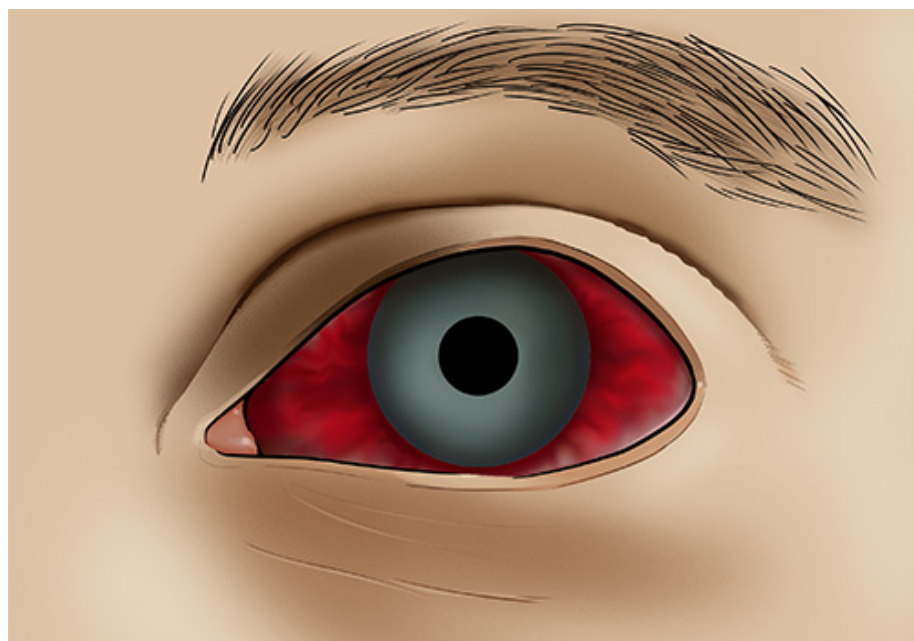


- **Foreign bodies** usually require removal under local anaesthesia in a medical facility. Foreign bodies are easily missed and personnel should assume a foreign body is present if the patient has persistent pain with blinking.
- **Penetrating injury** is often obvious but may be subtle. If the pupil has developed a new tear drop shape, the tear drop often points in the direction of the injury.

# Eye Injuries



- **Retrobulbar haematoma** occurs when there is bleeding behind the globe. This rapidly raises intraocular pressure and is immediately vision-threatening. The patient will usually have deep pain, proptosis (bulging) of the eye, reduced vision and a 'blood blister' like appearance of the conjunctiva. Urgent lateral canthotomy (a simple procedure that can be performed in ED) may be vision saving.
- **Retinal detachment** occurs when the retina detaches from the layer underneath. It is often associated with flashes of light and a 'curtain like' reduction in vision.
- **Subconjunctival haemorrhage** may appear spectacular, but does not extend over the cornea so vision is not affected. It resolves over several days and requires no active treatment or follow up provided the rest of the eye assessment is normal and the patient is not taking an anticoagulant. Subconjunctival haemorrhage often occurs spontaneously and may occur after rubbing the eye or following trauma to the head.



# Eye Injuries

- **'Black eyes'** (for example from a punch) are infrequently associated with eye trauma because of the shielding effect of the facial bones. The bruising resolves over days and requires no active treatment or follow up provided the rest of the eye assessment is normal.
- **Capsicum spray** causes intense discomfort but does not cause harm. Flush the eyes (prolonged irrigation is not required) and advise the patient that with time the pain will settle.
- **Photokeratitis** occurs when the eyes are unprotected from exposure to high levels of ultraviolet light, for example as a result of light reflected from snow or ice, or light from an arc welder. Photokeratitis is also often referred to by a number of other terms including snow blindness, arc eye and welder's flash. Symptoms often arise several hours after exposure and commonly include pain, photophobia, lacrimation and blepharospasm (tight closure of the eyelids). Patients with photokeratitis do not usually require referral to a doctor and should be advised to protect their eyes from light, not wear contact lenses and take oral pain relief. Symptoms usually settle over 1-2 days and they should seek further clinical assessment if their symptoms are not improving.

## Dressings

- Dressing an eye injury is not always required, but ensure exposed eye surfaces are kept moist, for example by intermittently applying saline drops.
- All that is usually required is to tape the eyelid closed.
- If a covering is required (for example to avoid others being upset by seeing the wound), tape the eyelid closed and place a dressing loosely over the face, ensuring there is no pressure on the eye.

## Response and/or transport by helicopter

- Transport by helicopter may be indicated if it will save more than 60 minutes compared with road transport, and
  - » There is a clinically obvious penetrating eye injury, or
  - » There is unilateral total vision loss.
- For criteria to be met, the road transport time will usually need to be more than 120 minutes.

## More Information

### Version history

28 October 2024

**Update:** Multiple changes.

1.0.5

# Eye Injuries

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# Wounds

This guideline is for patients with minor wounds including lacerations, abrasions and skin tears. See the '[limb and/or soft tissue injuries](#)' guideline if the wound is more than minor, or is associated with a compound fracture. See the '[bleeding following trauma](#)' guideline if the bleeding is severe.

- Compress external bleeding.
- Assess sensation, movement and blood supply in the area of (and distal to) the wound.
- Thoroughly irrigate the wound if it is visibly contaminated or was caused by a dirty object, ensuring all visible foreign material is removed.
- Administer 1 g cefazolin IV for an adult if the wound is large and contaminated. See the paediatric drug dose tables for a child.
- Close the wound or arrange for wound closure if this is required.
- Apply an appropriate dressing.

## Backup

- Consider requesting backup from an ECP if one is available, for wounds requiring closure:
  - Ambulance personnel at the scene should speak directly to the ECP to determine if it is appropriate for the ECP to close the wound.
  - Taking a photo and sending it to the ECP (with the patient's permission) may aid decision making.
  - Ambulance personnel do not need to remain on scene until the ECP arrives, provided the ECP accepts the referral and bleeding is adequately controlled.

## Referral and transport

- Provide a clear recommendation for the patient to be assessed in an ED if any of the following are present:
  - Penetrating injuries of unknown depth.
  - Significant tissue loss, tissue damage, tissue ischaemia or crush injury.
  - Signs of distal ischaemia.
  - Loss of distal sensory or motor function.
  - Significant wound contamination.
  - Signs of infection with systemic involvement.
  - Wound of cosmetic concern, for example involving the face or hands.
  - Aged less than two years and requiring closure.



# Wounds

- If ECP backup or referral is not appropriate, provide a clear recommendation for the patient to be assessed in primary care if any of the following are present:
  - Requiring closure that cannot be achieved by ambulance personnel.
  - Secondary to a bite.
  - Signs of infection without systemic involvement.
  - Tetanus prophylaxis is required.
- Have a low threshold for recommending the patient is seen in a medical facility if they have significant comorbidities or are immunosuppressed.

## Additional information

### General principles

- Wounds requiring closure (for example using sutures, staples or glue) should usually be closed within six hours of injury.
- Wounds to the face, hands, feet, genitals, or over joints may require specialist treatment. Seek clinical advice if uncertain which hospital or primary care facility the patient should be assessed in.
- Wounds caused by a dirty object or that are visibly contaminated, require thorough decontamination via irrigation:
  - Several minutes of irrigation is usually required.
  - The volume of irrigating fluid is more important than the nature of the fluid, provided it is clean.
  - Clean running water or 0.9% sodium chloride may be used.
  - Have a low threshold for recommending the patient is seen in a medical facility if the wound cannot be adequately decontaminated.

### Tetanus-prone wounds and tetanus prophylaxis

- Tetanus is a life-threatening disease caused by exotoxin released from the bacterium *Clostridium tetani*, which is commonly found in soil.
- Wounds are tetanus-prone if they are:
  - Contaminated, or
  - Caused by a dirty or rusty object.
- Clearly recommend that patients with a tetanus-prone wound are seen in a medical facility for tetanus prophylaxis within 24 hours if their immunisation status is unknown, or their last tetanus immunisation was more than five years ago.

# Wounds

## Skin tears

- Skin tears are wounds caused by a shearing force resulting in a separation of skin layers.
- Skin tears can be partial thickness (separation of the epidermis from the dermis) or full thickness (separation of both the epidermis and dermis from underlying structures).
- Skin tears are an important cause of morbidity in the elderly.
- As soon as possible attempt to realign the skin edges using a clean technique.
- The use of wound tape/strips should usually be avoided, but may be considered if it is difficult to retain the skin edges in place. Instruct the patient not to attempt to remove wound tape/strips if they are used.
- Provide a clear recommendation for the patient to be seen in a medical facility (preferably within primary care) as soon as possible, if the skin edges cannot be aligned.
- Apply an appropriate dressing:
  - Ensure the dressing immediately over the involved skin is 'non-stick'.
  - Apply an additional dressing that holds the non-stick dressing and underlying skin gently (but not tightly) in place. For example, use a bandage over a non-stick dressing.
  - If a dressing is stuck around the skin tear, consider drawing on the dressing (with an arrow) the direction in which it should be removed.
  - Provide advice to the patient and/or caregivers to keep the area dry.
  - Provide advice to be followed up in primary care.

## Dressing minor wounds

- There is no evidence that any form of dressing is better than another, but the following principles are useful:
  - Covered wounds heal faster than non-covered wounds.
  - Wounds that are moist (but not wet) heal faster than wounds that are dry.
  - The wound should be clean before being dressed.
  - The skin edges should be dry and aligned before being dressed.
  - Whenever possible, the part of the dressing directly over the wound should be 'non-stick'. If a 'non-stick' dressing is not available and gauze is being used, this must be moistened prior to application.
- Large grazes can be difficult to dress. Advise the patient to keep the graze moist (for example, using petroleum jelly) if possible during healing.

# Wounds

## More Information

### Version history

|                                  |       |
|----------------------------------|-------|
| 28 October 2024                  | 1.0.5 |
| <b>Update:</b> Multiple changes. |       |

# Hyperglycaemia

This guideline is for patients with hyperglycaemia and suspected diabetic ketoacidosis (DKA) or suspected hyperosmolar hyperglycaemic non-ketotic state (HHNS).

- Measure the patient's blood ketone levels if possible.
- Administer 0.9% sodium chloride IV if the patient has signs of hypovolaemia or poor perfusion:
  - 1 litre for an adult over one hour.
  - 20 ml/kg for a child over one hour.
  - Repeat as required.

## Backup

- Backup from a Paramedic should be requested if the patient has signs of hypovolaemia or poor perfusion.
- Backup from an ICP/CCP should be requested if the patient has severe shock.

## Referral and transport

- The patient must be given a clear recommendation to be transported to an ED by ambulance if DKA or HHNS is suspected.

## Additional information

### Diabetic ketoacidosis

- DKA develops in patients with type one diabetes who receive insufficient insulin, leading to clinically significant hyperglycaemia.
- Patients with DKA have:
  - Hyperglycaemia with a blood glucose concentration that is usually greater than 20 mmol/litre.
  - Hypovolaemia from a combination of osmotic diuresis secondary to hyperglycaemia, reduced oral intake and vomiting.
  - Acidosis from metabolism of fatty acids to ketones. The most common sign of this is tachypnoea. The patient's breath may have a fruity smell from ketones.
- The patient may have non-specific abdominal pain or vomiting.
- There is no role for out-of-hospital administration of insulin because rapid falls in glucose risk causing cerebral oedema.

# Hyperglycaemia

- Hyperventilation is a normal physiological response to metabolic acidosis and the patient should not be coached to lower their respiratory rate.
- Patients on sodium-glucose cotransporter-2 (SGLT2) inhibitors (for example, empagliflozin) may present with euglycaemic DKA, where DKA occurs without the usual increase in blood glucose level. If this is suspected, measuring blood ketone levels should occur if possible and the patient treated as per this guideline.

## Hyperglycaemia without acidosis

- Patients with type two diabetes can develop clinically significant hyperglycaemia without acidosis, because there is sufficient insulin present to prevent cells shifting to predominantly metabolising fatty acids. This may be referred to as hyperosmolar hyperglycaemic non-ketotic state (HHNS), hyperosmolar non-ketosis (HONK) or hyperosmolar hyperglycaemic state (HHS)..
- The patient may be very hypovolaemic from osmotic diuresis but will not usually have significant acidosis.
- The principles of out-of-hospital treatment are the same as for a patient with DKA.

## Patients with diabetes who are unwell

- Patients with diabetes often have significant comorbidities including ischaemic heart disease, peripheral vascular disease and renal impairment.
- They are at increased risk of developing infection, silent myocardial ischaemia and metabolic or electrolyte disorders. Have a low threshold for referring the patient to a doctor, even if there are no signs of clinically significant hyperglycaemia.

## IV fluid and hyperglycaemia

- Rapid boluses of IV fluid are avoided if the cause of hypovolaemia is hyperglycaemia, unless the patient has severe shock. This is because rapid boluses of IV fluid may contribute to cerebral oedema:
  - Rapid boluses of fluid may cause a rapid fall in glucose (by dilution) and this causes a rapid fall in osmolality. A rapid fall in osmolality causes water to shift into the brain and this may cause cerebral oedema.
  - Children and young adults are most at risk of the adverse effects from cerebral oedema because they do not have cerebral atrophy.

# Hyperglycaemia

## More Information

### Version history

|                                  |       |
|----------------------------------|-------|
| 11 September 2023                | 1.0.4 |
| <b>Update:</b> Multiple changes. |       |

# Hypoglycaemia

This guideline is for patients with a blood glucose concentration less than 3.5 mmol/litre. See the '[neonatal resuscitation](#)' guideline if the patient is a newborn baby.

- Administer 10-20 g of glucose PO provided the patient is sufficiently conscious to swallow safely.
- Administer glucose IV if the patient has a significantly altered level of consciousness or cannot swallow safely:
  - 100 ml of 10% glucose IV for an adult.
  - 2 ml/kg of 10% glucose IV for a child.
- Administer glucagon if unable to gain IV access:
  - 1 mg of glucagon IM for a patient aged five years and over.
  - 0.5 mg of glucagon IM for a child aged less than five years.
- Repeat the glucose measurement every ten minutes until the glucose concentration is consistently greater than 3.5 mmol/litre. Administer further doses of glucose if required, but do not repeat glucagon IM.

## Referral and transport

- A patient may receive treatment for hypoglycaemia and be given a clear recommendation for treatment without transport providing the following criteria are met:
  - No red flags are present.
  - A competent adult can stay with the patient for the next four hours.
  - The patient eats a meal (preferably with complex carbohydrates).
  - The patient is given the hypoglycaemia information sheet, which is explained to them and the accompanying adult.
  - The patient is given a clear recommendation to have their treatment reviewed (for example by an appropriate healthcare professional or diabetes service personnel). If the patient is aged less than or equal to 18 years or has been recently diagnosed with diabetes, this review should occur within 24 hours if feasible.

## Red flags (Should be seen in ED, usually via ambulance)

# Hypoglycaemia

- Overdose (including accidental) of insulin or oral hypoglycaemics.
- No known history of diabetes.
- Non-isolated episode.
- Taking an oral hypoglycaemic medicine and has hypoglycaemia without an obvious cause.
- Does not recover fully.
- Unable to mobilise normally.
- Unable to achieve a blood glucose concentration greater than 3.5 mmol/litre, ten minutes (or more) after glucagon or the last glucose administration.

## Additional information

### Causes of hypoglycaemia

- Hypoglycaemia usually occurs in a patient taking insulin or an oral hypoglycaemic medicine.
- Less common causes of hypoglycaemia include severe sepsis (particularly in young children), poisoning with medicines that lower blood glucose and liver failure.
- Some oral hypoglycaemic medicines are excreted primarily by the kidneys. If a patient taking an oral hypoglycaemic medicine develops hypoglycaemia without an obvious cause, unrecognised deterioration in kidney function may be a contributor.

### Measuring blood glucose concentration

- Do not measure blood glucose concentration using the patient's glucose meter because the result may be inaccurate.
- Use capillary blood and not blood from other sources because glucose meters are calibrated to use capillary blood.
- Check the history of the patient's glucose meter whenever possible. In particular rule out recurrent hypoglycaemia.

### Treating hypoglycaemia with oral glucose

- Oral glucose is best administered as a simple carbohydrate that is rapidly absorbed.



# Hypoglycaemia

- » Examples include 10% glucose (100 ml contains 10 g of glucose), glucose tablets or gel, sugar dissolved in water, 'non-diet' jam or similar glucose containing spreads and 'non-diet' soft drinks.
  - » 10-20 g of glucose is approximately half a cup of 'non-diet' fruit juice or soft drink, one tablespoon of sugar, one tablespoon of honey or six squares of chocolate.
  - » Most oral glucose gels contain 10-20 g of glucose per dose and can be administered to all patients, regardless of age.
  - » Document oral glucose as the approximate number of grams of oral glucose administered, or a description of the food ingested.
- Glucose gel may be spread on the gums, tongue and inside of the cheeks of a baby or small child.
  - Hypoglycaemia may occur several hours later as glucose is metabolised. This is why the instructions within the information sheet are important and why a meal must be eaten. The meal should preferably contain a complex carbohydrate such as brown rice, whole grain bread, porridge or beans.

## Treating hypoglycaemia with glucagon

- Glucagon will be ineffective if the patient has depleted glycogen stores. Examples include if the patient does not have diabetes, is a young child, has sepsis, or has not eaten food for more than 12 hours. In this setting glucagon will not be harmful, but the focus must be on administration of glucose.

## Hypoglycaemia in children aged less than five years

- Hypoglycaemia in children aged less than five years is most commonly due to:
  - » An illness in combination with no food intake and depleted glycogen stores, or
  - » Severe sepsis, or
  - » Poisoning with oral hypoglycaemic medicines.
- Children not known to have diabetes are usually malnourished or glycogen depleted for another reason, and glucagon is unlikely to be effective.
- Although metabolic disorders can cause hypoglycaemia in young children, this is rare. Do not delay treatment of hypoglycaemia in young children pending diagnostic tests, other than measuring a capillary glucose concentration. Although some references advocate delaying treatment until a blood sample is taken for analysis of possible metabolic disorders, treatment always takes priority over blood tests in this setting.

## Response and/or transport by helicopter

# Hypoglycaemia

- Response by helicopter may be indicated if it will arrive at scene at least 30 minutes sooner than a road-based Paramedic, and the patient is unconscious.

## More Information

### Version history

11 September 2023

1.0.4

**Update:** Multiple changes.

# Poisoning from Gases

This guideline is for poisoning following exposure to gases, solvents, smoke and/or fumes.

- Ensure scene safety.
- Assess for symptoms of carbon monoxide (CO) poisoning.
- Measure the carboxy-haemoglobin level if suitable equipment is available.
- Administer oxygen via a reservoir mask at 10 litres/minute if:
  - CO poisoning is suspected and the patient is symptomatic, or
  - The carboxy-haemoglobin level is greater than 10%.
- Administer nebulised bronchodilators if clinical signs of bronchospasm are prominent.

## Backup

- Backup from a Paramedic should be requested if the patient has significantly abnormal vital signs.
- Backup from a CCP should be requested if the patient has:
  - A GCS less than 10 with poor airway and/or poor breathing, or
  - Agitation causing a severe to immediately life-threatening risk to safety, that is not easily controlled.

## Referral and transport

- The patient must receive a clear recommendation to be transported to an ED by ambulance if CO poisoning is suspected, their vital signs are significantly abnormal, or bronchospasm requires more than one dose of nebulised bronchodilators.
- The patient should usually receive a clear recommendation that transport to an ED following exposure to solvents and/or fumes is not required, even if they are symptomatic, provided there are no signs of abnormal physiology.
- The patient requires a mental health assessment if they have attempted suicide. See the [‘attempted suicide’](#) and [‘non-suicidal self-harm’](#) guidelines for more information.

## Additional information

### Carbon monoxide (CO) poisoning

- CO poisoning occurs when there is significant exposure to CO, resulting in CO binding to the haemoglobin molecule and displacing oxygen.

# Poisoning from Gases

- CO poisoning may occur as a suicide attempt or following accidental exposure in an enclosed space, for example exposure to vehicle exhaust fumes, gas heaters/BBQs, burning charcoal and/or smoke.
- CO poisoning usually causes (in order of worsening severity of poisoning): headache, nausea, vomiting, chest tightness, shortness of breath, tachycardia, confusion, falling level of consciousness, dysrhythmias and cardiac arrest. Although some references describe the patient developing a cherry red skin colour, this is rarely seen.
- Measure the carboxy-haemoglobin level using a CO-oximeter if one is available.
- The patient must receive a clear recommendation to be assessed in an ED if CO poisoning is suspected and:
  - The patient is symptomatic, or
  - The carboxy-haemoglobin level is greater than 10%.
- Measurement of SpO<sub>2</sub> is unreliable if the patient has CO poisoning.

## Solvents and/or fumes

- Exposure to solvents and/or fumes commonly causes a combination of the following: headache, nausea, vomiting, dizziness, shortness of breath and irritation of the eyes, nose, mouth or throat.
- The symptoms will usually resolve over 1-2 hours provided the patient is removed from the source.
- The presence of symptoms does not necessarily indicate exposure to a toxic level and is not an indication to routinely recommend transport to an ED.
- The possibility of CO poisoning should be reasonably ruled out and the patient should receive a recommendation that transport to ED is not required unless there are objective signs of abnormal physiology. Examples of abnormal physiology include an SpO<sub>2</sub> less than 94% breathing air, persisting clinical signs of bronchospasm despite one dose of nebulised bronchodilators and clinical signs of pulmonary oedema.

## Chlorine gas

- Chlorine is commonly used throughout New Zealand in industry and home swimming/spa pools.
- When chlorine comes into contact with mucous membranes, hydrochloric acid is formed resulting in local irritation/inflammation.
- Patients commonly experience a burning sensation in the eyes, nose, mouth or throat, coughing, chest tightness, shortness of breath and nausea. If the exposure is severe the patient may develop signs of pulmonary oedema.

# Poisoning from Gases

- Mucous membranes should be liberally flushed using water, for example from a shower.
- The symptoms will usually resolve over 1-2 hours provided the patient is removed from the source.
- The presence of symptoms does not indicate exposure to a toxic level and is not an indication to routinely recommend transport to an ED, unless there are signs of abnormal physiology. Examples of abnormal physiology include an SpO<sub>2</sub> less than 94% breathing air, persisting clinical signs of bronchospasm despite one dose of nebulised bronchodilators and clinical signs of pulmonary oedema.

## Hydrogen sulphide (H<sub>2</sub>S) poisoning

- H<sub>2</sub>S impairs oxygen utilisation at a mitochondrial level. The oxygen levels in the blood and tissues may be normal, but oxygen cannot be utilised by cells.
- H<sub>2</sub>S is produced when sulphur combines with organic material, and/or from some industrial processes.
- A subset of H<sub>2</sub>S poisoning colloquially known as 'detergent suicide' involves the mixing of chemicals (usually an acid and a source of sulphur) within a confined space.
- H<sub>2</sub>S is very toxic and there is danger to personnel. Indicators may include:
  - A sulphurous odour.
  - Notes/signs warning of the danger.
  - Taped doors and/or windows.
  - A container with liquid/chemicals within the space.
  - A scene that presents like a carbon monoxide poisoning but with no pipe leading from the exhaust to the vehicle.
- In low concentrations H<sub>2</sub>S can cause mucous membrane irritation, cough, nausea and vomiting. At higher concentrations, it can cause shortness of breath, altered consciousness, seizures, respiratory arrest, and cardiac arrest. The evolution of symptoms may be rapid and the progress from exposure to death can occur within minutes.
- If H<sub>2</sub>S poisoning is suspected:
  - Do not enter the enclosed space, and request urgent assistance from FENZ and/or HAZMAT personnel.
  - If the patient is conscious and able to move, call on them to exit the confined space and move to fresh air. Ask them to remove their clothing down to underwear.
  - If the patient is unconscious, wait for FENZ or HAZMAT personnel with breathing apparatus and keep everyone away from the confined space (do not break windows or attempt rescue).
- Once the patient is removed from the immediate area of danger:

# Poisoning from Gases

- » Decontamination of the patient is not required unless liquid or crystallised chemicals are on the skin or clothes, but clothes should be removed as soon as possible.
- » No specific PPE for personnel is required, other than standard contact precautions unless there is a risk of fluid or crystal contamination. In this instance, personnel should wear gloves, a gown/apron, eye protection and a mask.
- » Decontamination of personnel is not required unless there is direct contact with chemicals or crystals.
- » Treatment is supportive, including administration of high flow oxygen.
- » The administration of cyanide antidotes may have a role, and if a cyanide antidote kit is immediately available, the contents should be administered.
- » In the setting of cardiac arrest, resuscitation should usually commence unless there is a clear reason not to. However, cardiac arrest following H<sub>2</sub>S poisoning is almost certainly fatal and prolonged resuscitation is inappropriate.

## Decontamination

- Decontamination prior to transport is only required if there are chemicals on the patient's skin or clothing:
  - » Clothing should be removed, leaving underwear on, and the patient decontaminated using water.
  - » The most appropriate form of decontamination is a shower, including a domestic shower, of approximately three minutes' duration.
  - » Decontamination using a fire service decontamination unit is acceptable, provided one is immediately available. However, decontamination should not be delayed while waiting for one to arrive.
- Off gassing (where the patient exhales dangerous levels of a gas) has never been documented to occur. All cases of apparent off gassing were caused by chemicals on the patient's clothes and this is why clothing contaminated with chemicals must be removed prior to transport.

## More Information

### Version history

11 September 2023

1.0.4

**Update:** BGL removed, SpO<sub>2</sub> changed to less than 92%, new hydrogen sulphide guideline added.

# Poisoning from Medicines

This guideline is for poisoning from medicines, or when the cause of poisoning is unknown.

- Administer naloxone if opiate poisoning is suspected and the patient has a significantly impaired level of consciousness, or significantly impaired breathing:
  - 0.1-0.4 mg of naloxone IV every five minutes for an adult, or
  - 0.8 mg of naloxone IM every ten minutes for an adult.
  - See the paediatric drug dose tables for a child.
- Administer 0.9% sodium chloride IV if the patient has signs of hypovolaemia or poor perfusion:
  - 1 litre IV for an adult.
  - 20 ml/kg for a child.
  - Repeat as required.
- If cyclic antidepressant poisoning is suspected and the patient has QRS prolongation:
  - Administer 8.4% sodium bicarbonate IV:
    - 100 ml IV for an adult.
    - 2 ml/kg IV for a child.
  - Administer 0.9% sodium chloride IV:
    - 1-2 litres for an adult.
    - 20-40 ml/kg for a child.

## Backup

- Backup from a Paramedic should be requested if the patient has abnormal vital signs.
- Backup from a CCP should be requested if the patient has:
  - A GCS less than 10 with poor airway and/or poor breathing, or
  - Acute behavioural disturbance causing a severe to immediately life-threatening risk to safety, that is not easily controlled.

## Referral and transport

- The patient must receive a clear recommendation to be transported to an ED if they have abnormal vital signs or the medicine/dose taken is potentially harmful. See the '[patient capacity](#)' guideline for more information if the poisoning is a suicide attempt and the patient declines the recommendation.
- The patient requires a mental health assessment if they have attempted suicide. This does not routinely require transport to an ED and assessment should occur in the community, provided:

# Poisoning from Medicines

- » The poisoning is clearly not harmful, and
- » An appointment can be made directly with a mental health team, and
- » The patient can be observed by a competent adult until assessment occurs.
- » See the '[attempted suicide](#)' and '[non-suicidal self-harm](#)' guidelines for more information.

## Additional information

### General principles

- The treatment of poisoning is rarely medicine specific and is focused on:
  - » Supporting airway, breathing and circulation.
  - » Treating agitation if it is impairing the ability to safely provide treatment and/or transport. See the '[acute behavioural disturbance](#)' guideline for more information.
  - » Ensuring the patient receives appropriate additional assessment and/or follow up.
- An altered level of consciousness following poisoning from medicines is usually caused by benzodiazepines, antidepressants, antipsychotics, opiates, sedatives or a combination of these.

### Naloxone

- Naloxone is not indicated in the treatment of poisoning associated with an altered level of consciousness unless opiate poisoning is strongly suspected. Even when opiate poisoning is suspected, naloxone only has a role when there is clinically significant impairment of consciousness or breathing.
- Naloxone administration may be associated with seizures, hypertension, pulmonary oedema or severe agitation. The goal is to administer the minimum amount of naloxone required to produce an adequate improvement in the patient's level of consciousness and/or breathing.

### Police

- Assistance from police is not routinely required but should be requested if there is:
  - » Significant risk of injury to the patient, personnel or bystanders, or
  - » The patient has agitation causing a severe to immediately life-threatening risk to safety, or
  - » More than minimal restraint is required.
- Police do not have the legal authority to restrain a patient on their own property, unless there is immediate danger to the patient, personnel or bystanders, including as a result of being unable to treat the patient.



# Poisoning from Medicines

## Paracetamol poisoning

- Significant paracetamol poisoning may cause acute liver failure, but this is treatable provided treatment occurs early.
- A patient with significant paracetamol poisoning is commonly asymptomatic in the first 6-12 hours and then usually develops nausea, vomiting and non-specific abdominal pain.
- A patient with suspected paracetamol poisoning requires transport to an ED for serum paracetamol measurement, even if asymptomatic.

## Cyclic antidepressant poisoning

- The new generation cyclic antidepressants are less toxic than many of the older generation tricyclic antidepressants, but the patient may develop an altered level of consciousness, seizures, tachycardia, tachydysrhythmias and shock.
- Part of the toxicity from cyclic antidepressants comes from the drug binding to sodium channels within the heart and this may be reduced by the administration of a large dose of sodium ions.

## Atypical antipsychotic poisoning

- Examples of atypical antipsychotics include quetiapine, risperidone and olanzapine.
- Signs and symptoms of poisoning can include:
  - Sedation.
  - Tachycardia and/or hypotension.
  - Respiratory depression.
  - Anticholinergic effects including delirium and agitation.
  - Dystonic reactions.
  - Seizures and dysrhythmia in rare cases.
- The onset of toxicity usually occurs within 1-2 hours and significant symptoms should be evident within four hours.
- Management should be supportive, focusing on maintaining the airway, breathing and circulation.

## Serotonin syndrome

# Poisoning from Medicines

- Also known as serotonin toxicity, serotonin syndrome occurs as a result of raised serotonin levels within the brain.
- Although most commonly thought of as occurring following poisoning with selective serotonin reuptake inhibitors (SSRIs), serotonin syndrome may also occur when medicines or substances that increase serotonin levels are taken in combination. Examples include SSRIs, MDMA, amphetamines, antidepressants and tramadol.
- Signs and symptoms include tachycardia, tachypnoea, hypertension, sweating, hyperthermia, tremor, rigidity, confusion, agitation and seizures. If severe, the patient will be unconscious with severe shock.
- Treatment is supportive. Uncover the patient and measure the temperature regularly. Commence cooling and administer 0.9% sodium chloride IV if the temperature is greater than 39°C. Treat agitation using low dose midazolam and avoid high dose droperidol as this may worsen serotonin syndrome. Seek clinical advice if agitation is not rapidly resolved.
- Nausea and/or vomiting must not be treated with ondansetron as this may worsen serotonin syndrome.

## Beta-blocker and/or calcium channel blocker poisoning

- Bradycardia and/or hypotension may be prominent with beta blocker and/or calcium channel blocker poisoning, particularly if taken in combination.
- If haemodynamic support is required, an adrenaline infusion is preferred over metaraminol.
- Glucagon has no role in the out-of-hospital treatment of poisoning from beta-blockers and/or calcium channel blockers because it rarely provides a sustained heart rate rise in addition to high dose adrenaline and requires much higher doses than carried by ambulance personnel.

## Colchicine poisoning

- Colchicine is a medicine that is sometimes used in the treatment of gout.
- It is extremely poisonous with a high mortality rate. There are no effective treatments once it is absorbed.
- Patients with possible colchicine poisoning must be transported to an ED without delay (even if asymptomatic) because specific gut decontamination techniques may be required urgently.

## Iron poisoning

- Iron is found in many multivitamins and is potentially lethal in poisoning. There are early gastrointestinal effects including abdominal pain, vomiting and diarrhoea, which may be severe or cause bleeding. This is followed by a delayed systemic response with shock, acidosis, liver failure and kidney failure.

# Poisoning from Medicines

- Patients may improve between these two phases, and must be transported to an ED if iron poisoning is suspected even if they do not have significant symptoms.
- The hospital treatment of iron overdose involves whole bowel irrigation and antidote therapy (desferrioxamine) if indicated.
- Out-of-hospital management is supportive, and intravenous fluid should be administered if indicated for hypovolaemia or poor perfusion.

## More Information

### Version history

11 September 2023

1.0.4

**Update:** Multiple changes.

This guideline is for poisoning from recreational drugs including alcohol.

- Administer naloxone if opiate poisoning is suspected and the patient has a significantly impaired level of consciousness, or significantly impaired breathing:
  - 0.1-0.4 mg of naloxone IV every five minutes as required for an adult, or
  - 0.8 mg of naloxone IM every ten minutes as required for an adult.
  - See the paediatric drug dose tables for a child.
- Administer 0.9% sodium chloride IV if the patient has signs of hypovolaemia or poor perfusion:
  - 1 litre IV for an adult.
  - 20 ml/kg IV for a child.
  - Repeat as required.

## Backup

- Backup from a Paramedic should be requested if the patient has significantly abnormal vital signs..
- Backup from a CCP should be requested if the patient has:
  - A GCS less than 10 with poor airway and/or poor breathing, or
  - Agitation causing a severe to immediately life-threatening risk to safety, that is not easily controlled.

## Referral and transport

- The patient must be given a clear recommendation to be transported to an ED if they have significantly abnormal vital signs.
- Clinical judgement is required taking into account the social circumstances and the nature of the drug, but most patients should usually receive a recommendation that transport to an ED is not required, provided all the following are met:
  - They are able to obey commands, and
  - They do not have significantly abnormal vital signs, and
  - There is no history, signs or symptoms of acute traumatic brain injury, and
  - Hypoglycaemia has been excluded, and
  - They can mobilise safely, and
  - They are in a safe situation, for example they are with a competent adult.
- Clinical judgement is required when determining an intoxicated patient is safe to be taken into the care of police. For this to occur the patient must meet all the criteria listed above.

- Clinical judgement is required if the patient has had a seizure, particularly following ingestion of chemicals commonly referred to as synthetic cannabinoids. It is usually appropriate to manage the patient without transport to an ED if the patient has stopped seizing and has recovered to be able to mobilise safely. Advice on providing supervision until the patient has fully recovered should be provided to family and/or friends whenever feasible.
- The patient does not require a mental health assessment unless they have attempted suicide.

## Additional information

### General principles

- The treatment of poisoning from recreational drugs is rarely drug specific and is focused on supportive care.
- Often recreational drugs are adulterated, so the stated dose and composition may be unreliable.
- 0.9% sodium chloride IV does not speed recovery from alcohol poisoning and is not indicated unless there are signs of hypovolaemia, poor perfusion, or dehydration.
- Do not transport drugs or vomit to hospital to help identify the drug.

### Naloxone

- Naloxone is not indicated in the treatment of poisoning associated with an altered level of consciousness unless opiate poisoning is strongly suspected. Even when opiate poisoning is suspected, naloxone only has a role when there is clinically significant impairment of consciousness or breathing.
- Naloxone administration may be associated with seizures, hypertension, pulmonary oedema or severe agitation. The goal is to administer the minimum amount of naloxone required to produce an adequate improvement in the patient's level of consciousness and/or breathing.

### Treatment of alcohol and/or recreational drug poisoning at events

- At large events, specific facilities and staff may be organised to manage and treat patients with alcohol and/or recreational drug poisoning. In the absence of significant complications, such as severe airway obstruction, these patients are usually suitable for treatment at such facilities.
- Patients with an altered level of consciousness may be adequately managed provided their airway, breathing and circulation are adequate and they are closely monitored.
- Clinical judgement is required when deciding if a patient should be transported to an ED. In general, transport to an ED should occur if the patient:

- » Has signs or symptoms consistent with TBI, or
  - » Tolerates an airway adjunct, or
  - » Has hypoglycaemia that does not respond to treatment, or
  - » Has very abnormal physiology (for example poor airway, poor breathing, signs of shock, or moderate to severe hypothermia).
- Some patients with gamma hydroxybutyrate (GHB) poisoning will tolerate an airway adjunct for 20-30 minutes and then rapidly improve. In the absence of other reasons for transport, such patients are suitable for treatment at the facility provided a doctor or ICP/CCP is present.
  - Patients must be observed regularly. In general, half hourly observations of respiratory rate, SpO<sub>2</sub>, heart rate and GCS are appropriate.
  - Most patients will improve with time. There are no specific treatments for alcohol poisoning, but some treatments may be provided without transporting the patient to an ED:
    - » 0.9% sodium chloride IV for signs of hypovolaemia, poor perfusion or dehydration.
    - » Antiemetics for nausea and/or vomiting.
    - » Glucose IV for hypoglycaemia. Alcohol impairs gluconeogenesis and may occasionally cause hypoglycaemia, particularly in children and adolescents. A patient may be treated for hypoglycaemia and not transported to ED provided only one dose of 10% glucose is required and the non-transport criteria within the '[hypoglycaemia](#)' guideline are followed.
  - The patient may be discharged once they are obeying commands and able to mobilise safely. Prior to discharge they should receive advice (preferably written) on safe alcohol and/or recreational drug consumption.

## Specific recreational drugs

- Recreational drugs are often taken in combination (particularly with alcohol) producing uncertain and compounding effects.
- Common recreational drugs include:
  - » **Gamma hydroxybutyrate (GHB)** may cause the patient to be deeply unconscious with a poor airway, poor breathing and intermittent apnoea. Commonly the patient requires assisted ventilation and improves rapidly after 20-30 minutes. The patient may take longer to improve if another sedative (for example alcohol) has also been ingested.
  - » **MDMA** may cause an altered level of consciousness, seizures and hyperthermia.
  - » **Ketamine** may cause hallucinations or an altered level of consciousness.
  - » **Amphetamines and methamphetamine** may cause severe hypertension, tachycardia and disturbed behaviour. The latter may be severe and may be associated with violence or attempted suicide.

- » **Cathinones** (for example mephedrone) are amphetamine-like. They may cause hypertension, tachycardia, hallucinations, paranoia, panic attacks and disturbed behaviour. The latter may be severe.
- » **Cannabis and cannabinoids** may cause mental dissociation, anxiety, tachycardia, palpitations, chest pain, nausea and vomiting.
- » **'Synthetics'** is a broad term used to describe a mixture of synthetic chemicals that are usually added to dried plant material enabling them to be smoked. These drugs may cause altered level of consciousness, seizures, agitation and cardiac arrest.
- » **Cocaine** may cause severe hypertension, tachycardia, intracranial haemorrhage, coronary artery spasm and myocardial ischaemia.

## Seizures following recreational drug use

- Seizures following recreational drug use are common, particularly with synthetics.
- The seizures are usually self-limiting and commonly the patient will refuse assessment and/or transport.
- Clinical judgement must be used as forcibly initiating assessment and/or transport often requires a combination of significant restraint and/or sedation, which may be more life-threatening than the effects of the recreational drug.
- It is usually appropriate to manage the patient without transport to an ED if the patient has stopped seizing and has recovered to be able to mobilise safely. Advice on providing supervision until the patient has fully recovered should be provided to family and/or friends whenever feasible.

## Police

- Assistance from police is not routinely required but should be requested if there is:
  - » Significant risk of injury to the patient, personnel or bystanders, or
  - » The patient has agitation causing a severe to immediately life-threatening risk to safety, or
  - » More than minimal restraint is required.
- Police do not have the legal authority to restrain a patient on their own property, unless there is immediate danger to the patient, personnel or bystanders, including as a result of being unable to treat the patient.

## More Information

## Version history

|                                  |       |
|----------------------------------|-------|
| 11 September 2023                | 1.0.4 |
| <b>Update:</b> Multiple changes. |       |



This guideline is for patients with poisoning from miscellaneous causes including swallowed objects, that do not fit into another guideline.

- Administer 0.9% sodium chloride IV if the patient has signs of hypovolaemia or poor perfusion:
  - 1 litre IV for an adult.
  - 20 ml/kg IV for a child.
  - Repeat as required.

## Backup

- Backup from a Paramedic should be requested if the patient has abnormal vital signs.
- Backup from a CCP should be requested if the patient has:
  - A GCS less than 10 with poor airway and/or poor breathing, or
  - Agitation causing a severe to immediately life-threatening risk to safety, that is not easily controlled.

## Referral and transport

- The patient must receive a clear recommendation to be transported to an ED if they have abnormal vital signs or the poison is potentially harmful. See the [‘patient capacity’](#) guideline for more information if the poisoning is a suicide attempt and the patient declines the recommendation.
- A clear recommendation must be made to be assessed in an ED if the patient has ingested a foreign body and the swallowed object is potentially dangerous, such as a battery, a magnet (especially if there is more than one) or a sharp object such as a pin.
- A clear recommendation may be made to remain in the community if the patient is asymptomatic and there is clear evidence that the swallowed object is not dangerous, for example a button or marble. In this setting, the patient or parent/caregiver should be advised that the object will most likely pass through the bowel and that medical advice should be sought if symptoms develop.
- The patient requires a mental health assessment if they have attempted suicide. This does not routinely require transport to an ED and assessment should occur in the community, provided:
  - The poisoning is clearly not harmful, and
  - An appointment can be made directly with a mental health team, and
  - The patient can be observed by a competent adult until assessment occurs.
  - See the [‘attempted suicide’](#) and [‘non-suicidal self-harm’](#) guidelines for more information.

## Additional information

### General principles

- The treatment of poisoning is rarely poison specific and is focused on:
  - Supporting airway, breathing and circulation.
  - Treating agitation if it is impairing the ability to safely provide treatment and/or transport. See the [‘acute behavioural disturbance’](#) guideline for more information.
  - Ensuring the patient receives appropriate additional assessment and/or follow up.
- Do not induce vomiting as this has limited therapeutic benefit and may cause oesophageal injury.
- There is no role for the transport of chemicals or vomit to hospital to help identify the poison.

### Battery ingestion

- Batteries (especially button or disc batteries) can cause severe injury to the oesophagus or bowel and may need to be surgically removed. This is a time sensitive condition that should be assessed in hospital without significant delay, noting that transport by ambulance may not be required. If a battery of the same size is available it should also be taken to ED.
- Button batteries can be mistaken for coins. If the patient is thought to have swallowed a coin, but it is possible that it is a button battery, the patient should be treated as if a battery has been swallowed.

### Acid or alkali ingestion

- Ingestion of strong acid or alkali may cause significant injury to the oesophagus. This is particularly the case with alkaline products such as dishwashing powders and bleach.
- Some inflammation of the oropharynx is common but significant injury and/or swelling of the oropharynx is uncommon.
- Encourage sips of water provided the patient's airway and swallow are normal and this does not induce vomiting.
- Have a low threshold for administering an antiemetic.

### Organophosphate poisoning

- Organophosphate (OP) poisoning occurs most commonly with deliberate ingestion of insecticides containing an OP. It is possible for OP poisoning to occur following skin contact with chemicals containing an OP, but this requires very significant exposure.

- Organophosphates inhibit the activity of the enzyme cholinesterase. This causes acetylcholine (ACh) to build up at neural and neuromuscular junctions. The build-up of ACh causes:
  - Salivation.
  - Lacrimation.
  - Defaecation and vomiting.
  - Urination.
  - Bradycardia.
  - Bronchoconstriction and bronchial secretions.
  - Muscle twitching and muscle weakness.
- Treatment is support of the patient's airway, breathing and circulation, in addition to treatment with atropine. Atropine will reverse most of the effects of the ACh:
  - Administer 1.2 mg atropine IV. Repeat every five minutes until there are signs of adequate atropinisation.
  - Repeated and escalating doses may be required.
  - It is likely that additional atropine will need to be delivered to the treating crew.
  - Seek clinical advice if the patient is a child.
- Adequate atropinisation will be indicated by:
  - Resolution of bradycardia.
  - Drying of secretions.
  - Resolution of wheeze.
- Adequate PPE is achieved by wearing gloves and a gown/overalls, unless the patient is in a confined space with an aerosolised OP.
- Decontamination prior to transport is not required if the patient has ingested an OP. The patient's vomit will contain OP and contact PPE should be worn to prevent contact with vomit.
- Decontamination prior to transport is required if there are OP chemicals on the patient's skin or clothing:
  - Clothing should be removed, leaving underwear on, and the patient decontaminated using water.
  - The most appropriate form of decontamination is a shower, including a domestic shower, of approximately three minutes' duration.
  - Decontamination using a fire service decontamination unit is acceptable, provided one is immediately available. However, decontamination should not be delayed while waiting for one to arrive.
- Off gassing (where the patient exhales dangerous levels of OP) has never been documented to occur. All cases of apparent off gassing were caused by OP on the patient's clothes and this is why clothing must be removed prior to transport.

## Nitrite poisoning

- Nitrites can be found in food preservatives, room deodorisers, liquid incense and coolant.
- Nitrite poisoning causes methaemoglobinemia which reduces the amount of oxygen which can bind to haemoglobin and causes tissue hypoxia, cyanosis, hypotension and can cause rapid deterioration into cardiac arrest.
- Patients may present with severe hypoxia which is often unresponsive to supplemental oxygen administration, extreme tachypnoea, extreme tachycardia and agitation. If IV access is gained, blood may also be noted to be a dark brown colour.
- Initial management is supportive care and rapid transport to definitive care. The antidote to nitrite poisoning is methylene blue, which will usually be available in ED. Early notification is essential to allow the receiving hospital staff to prepare.
- If the patient deteriorates into cardiac arrest and transport time to ED is less than 10 minutes, transport with resuscitation en route for antidote therapy.

## Cyanide poisoning

- Cyanide impairs oxygen utilisation at a mitochondrial level. The oxygen levels within blood and tissues are normal, but oxygen cannot be used by cells.
- Cyanide is used in industry (particularly in mining, electroplating and plastics) and by hunters (pesticides).
- Cyanide may be present in high concentrations within smoke from house fires that involve synthetic furnishings.
- Most cyanide poisoning occurs when it is ingested, usually deliberately. Cyanide is poorly absorbed through skin, but poisoning can occur from skin contact with high concentrations.
- Patients with cyanide poisoning have non-specific signs and symptoms:
  - Anxiety, nausea and headache.
  - Tachycardia.
  - Tachypnoea.
  - Falling level of consciousness (when severe).
  - Cardiac arrest (when very severe).
- PPE and decontamination:
  - No specific PPE (other than normal body fluid precautions) or decontamination, is required if cyanide has been ingested.

- » In the unlikely event that cyanide is present on the patient's skin or clothing, decontamination should occur in the same way as described for organophosphates.

## ➤ Treatment:

- » The initial treatment is supportive.
- » Specific cyanide antidotes may be available at sites where cyanide is being used. Cyanide antidotes include hydroxocobalamin, amyl nitrite and sodium thiosulfate.
- » Personnel may administer the contents of cyanide antidote kits, following the instructions within them, if there is a history consistent with cyanide poisoning and the patient is symptomatic. Personnel should seek clinical advice if uncertain, but this should not delay treatment if the patient is showing signs of poisoning.

## More Information

### Version history

11 September 2023

1.0.4

**Update:** Multiple changes.

22 November 2022

1.0.1

**Update:** Referrals for foreign body parts and oxygen administration threshold (paraquat poisoning).

# Hyperkalaemia

This guideline is for patients aged greater than or equal to 12 years, with suspected hyperkalaemia or a known blood potassium concentration of greater than 6 mmol/L. Move to the '[crush injury](#)' guideline if the patient has crush injury. Seek clinical advice if the patient is aged less than 12 years.

- Monitor the cardiac rhythm continuously.
- Acquire a 12 lead ECG.
- Determine the severity of ECG changes.

## No ECG changes

- Provide general supportive treatment but do not administer medicines for hyperkalaemia.

## Mild to moderate ECG changes

- Administer continuous nebulised salbutamol.

## Severe ECG changes

- Administer continuous nebulised salbutamol.
- Administer 6.8 mmol (1 g) of calcium chloride IV, over approximately one minute via a running line.
- Administer a large flush of 0.9% sodium chloride IV via a running line.
- Administer 100 ml of 8.4% sodium bicarbonate IV, over approximately one minute via a running line.
- Administer 500 ml of 0.9% sodium chloride IV if there are signs of hypovolaemia. Repeat as required, using caution if the patient has end stage renal failure.
- Repeat the calcium chloride and sodium bicarbonate after 20 minutes if severe ECG changes persist or recur.
- Administer an adrenaline infusion IV if bradycardia persists despite the treatment above:
  - Administer an adrenaline infusion via an infusion device. Start at a rate of 0.5 mg/hour and adjust the rate as required, or
  - Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
    - Administer as an infusion. Start at 2 drops/second and adjust the rate as required, or
    - Administer a bolus of 0.01 mg adrenaline (10 ml) IV every 1-2 minutes as required.

## Cardiac arrest

- Provide the following treatments in addition to usual resuscitation:
  - Administer 6.8 mmol (1g) of calcium chloride IV, as a bolus via a running line.

# Hyperkalaemia

- » Administer a large flush of 0.9% sodium chloride IV via a running line.
- » Administer 100 ml of 8.4% sodium bicarbonate IV, as a bolus via a running line.
- Repeat the calcium chloride and sodium bicarbonate if ROSC is not achieved within 10 minutes of the first doses.
- If ROSC is achieved, provide additional treatment as per severe ECG changes (as above).

## Backup

- Backup from an ICP/CCP should be requested if the patient has severe ECG changes.

## Referral and transport

- The patient should be given a clear recommendation for transport to an ED by ambulance if hyperkalaemia is suspected or known to be present.

## Additional information

### General principles

- Potassium is a cation that is vital to the process of generating action potentials within cells, particularly cardiac and skeletal muscle cells.
- The intracellular concentration of potassium is much higher than the extracellular concentration. As the extracellular concentration rises this reduces the gradient across the cell membrane, impairing the generation of action potentials. Impaired action potentials most commonly present as an abnormal ECG.
- Hyperkalaemia is defined as a serum (extracellular) potassium concentration greater than 5.5 mmol/L, but adverse cardiovascular effects from hyperkalaemia do not usually occur until the serum concentration is greater than 6 mmol/L.
- Hyperkalaemia is usually asymptomatic. Any symptoms associated with hyperkalaemia are usually those associated with reduced cardiac output due to dysrhythmia.
- The level at which hyperkalaemia is associated with abnormal cardiac conduction is dependent on the patient's usual potassium concentration, and the rate of potassium concentration increase. For example, if a patient has a very slow rise in their potassium concentration over many days, they may not experience adverse effects until the concentration exceeds 7-8 mmol/L. Conversely, a patient whose potassium concentration rapidly rises over a few hours, may experience adverse effects at a concentration of 6 mmol/L.

# Hyperkalaemia

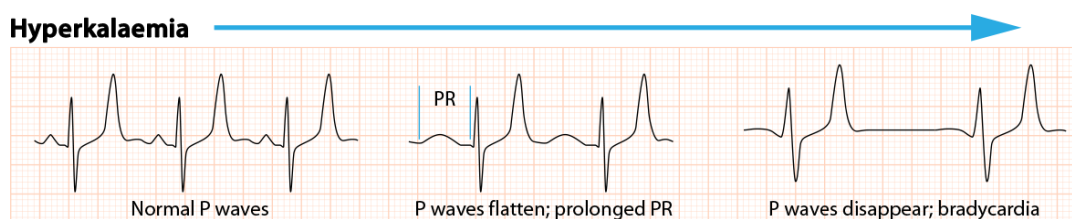
- Blood test results may indicate hyperkalaemia is present if blood has been haemolysed during phlebotomy or testing. This does not require specific treatment if the patient's ECG is not indicative of hyperkalaemia.

## Causes of hyperkalaemia

- The most common cause is end stage renal failure, particularly if the patient is on dialysis and has had their usual dialysis schedule disrupted.
- Other causes can include:
  - Rhabdomyolysis associated with prolonged immobility.
  - Metabolic acidosis associated with severe sepsis.
  - Very severe diabetic ketoacidosis.
- The patient is more likely to develop hyperkalaemia if they are taking an angiotensin converting enzyme inhibitor, an angiotensin receptor blocker, an NSAID and/or a potassium sparing diuretic. However, these medicines are usually only associated with hyperkalaemia in the context of acute impairment of kidney function.

## ECG features of hyperkalaemia

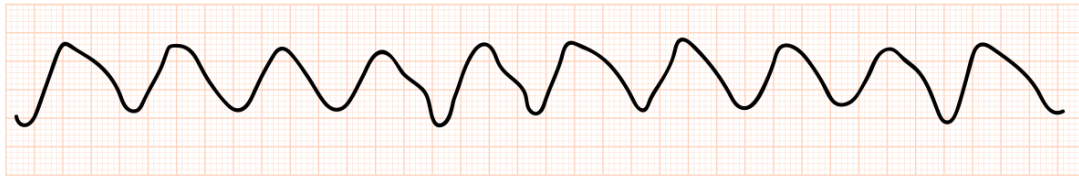
- The ECG features are dependent on the potassium concentration, the rate of rise of the concentration and the patient's usual potassium concentration.
- Typically, if the rate of rise is slow, the following ECG features will usually be seen sequentially as the potassium concentration increases:
  - Peaked T waves.
  - Flat P waves or loss of P waves.
  - Mild to moderate broadening of the QRS complex (0.12-0.16 seconds).
  - Severe broadening of the QRS complex (greater than 0.16 seconds).
  - Bradycardia.
  - Fusion of the QRS complex with the T wave forming sine waves.





# Hyperkalaemia

**Sine wave pattern**



- Mild to moderate ECG changes include peaked T waves, and/or flattening or loss of P waves, without QRS broadening and without bradycardia.
- Severe ECG changes include broadening of the QRS complex or bradycardia.
- If the potassium concentration rises very quickly the above ECG features may not occur and instead the patient may develop VT and/or VF.

## Treatment of hyperkalaemia

- The flush of 0.9% sodium chloride between calcium administration and sodium bicarbonate administration is required, as precipitation will occur if these electrolytes mix.
- Calcium ions provide electrical stabilisation of cardiac cells, but do not reduce the potassium concentration.
- Sodium bicarbonate causes hydrogen ions to leave the intracellular space, and these are replaced by potassium ions. This will usually reduce the potassium concentration by 0.5-1 mmol/L. The reduction in concentration is temporary (usually an hour or two) but helps allow time for the underlying cause to be identified and treated.
- Continuous nebulised salbutamol stimulates beta 2 receptors, causing potassium to move into the intracellular space. This will usually reduce the potassium concentration by 0.5-1 mmol/L. The reduction in concentration continues provided salbutamol administration continues, but as soon as salbutamol administration ceases the concentration will rise again.
- If the patient is hypotensive or has signs of poor perfusion in the absence of bradycardia, this is likely to be due to hypovolaemia and/or sepsis, rather than hyperkalaemia and the patient should be treated accordingly.
- If the patient has rhabdomyolysis associated with prolonged immobility, they are likely to have concurrent hypovolaemia which should be treated accordingly. They may also need additional treatment for the underlying cause of the immobility, for example poisoning.
- If bradycardia persists despite pharmacological treatment, the heart rate is usually more likely to respond to an adrenaline infusion than to pacing. Adrenaline will also further lower the potassium concentration via stimulation of beta 2 receptors.

# Hyperkalaemia

## Response and/or transport by helicopter

- Response by helicopter may be indicated if the patient has severe hyperkalaemia and it will arrive at scene at least 30 minutes sooner than a road-based CCP.
- Transport by helicopter may be indicated if the patient has severe hyperkalaemia not responding to treatment and it will save more than 30 minutes compared to road transport. For criteria to be met, the road transport time will usually need to be more than 90 minutes.

## More Information

### Version history

|                                                             |         |
|-------------------------------------------------------------|---------|
| <b>16 June 2026</b><br><b>New:</b> Guideline released.      | 1.1.0.1 |
| <b>11 September 2023</b><br><b>New:</b> Guideline released. | 1.0.4   |

# Bacterial Sepsis

This guideline is for patients with a provisional diagnosis of bacterial sepsis. See the '[meningococcal septicaemia](#)' guideline if this is suspected.

- Assess whether antibiotics are indicated (all the following must be present):
  - One or more clinical features indicating antibiotics (see the sepsis features table), and
  - The patient is aged greater than or equal to 12 years, and
  - Time to hospital is greater than 30 minutes.
- Obtain blood cultures prior to administration of antibiotics if there is a local protocol in place.
- Administer an antibiotic if indicated:
  - 1 g cefazolin IV if the source of sepsis appears to be the soft tissues or a joint.
  - 2 g ceftriaxone IV for all other sources of sepsis.
  - Seek clinical advice regarding antibiotic administration if the patient is aged less than 12 years and is very unwell.
- Administer 0.9% sodium chloride IV if there are signs of hypovolaemia or poor perfusion:
  - 1-2 litres for an adult.
  - 20-40 ml/kg for a child.
  - Repeat once if required.
- Assess for features of concern (see the sepsis features table).

## Treatment of hypotension

- Administer metaraminol IV if the systolic blood pressure is less than 100 mmHg in an adult, or significantly less than the normal predicted systolic blood pressure in a child.

### Metaraminol for an adult:

- Administer via an infusion device:
  - Administer an initial bolus of 0.5-1 mg metaraminol IV.
  - Start the infusion at 2 mg/hour and adjust the rate as required.
- Alternatively, administer a bolus of 0.5-1 mg metaraminol IV every 5-10 minutes as required.

### Metaraminol for a child:

- Administer a bolus of metaraminol IV (see the paediatric drug dose tables) every 5-10 minutes as required.

# Bacterial Sepsis

- Consider administering adrenaline IV if the blood pressure is unresponsive to metaraminol, or the patient is hypotensive and bradycardic.

## Adrenaline for an adult:

- » Administer an adrenaline infusion via an infusion pump. Start at a rate of 0.5 mg/hour and adjust the rate as required, or
- » Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
  - » Administer as an infusion. Start at 2 drops/second and adjust the rate as required, or
  - » Administer a bolus of 0.01 mg adrenaline (10 ml) IV every 1-2 minutes as required.

## Adrenaline for a child aged 5-14 years:

- » Administer an adrenaline infusion via an infusion pump using service guidelines, or
- » Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
  - » Administer as an infusion. Start at 1 drop/second and adjust the rate as required, or
  - » Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.

## Adrenaline for a child aged less than five years:

- » Administer an adrenaline infusion via an infusion pump using service guidelines, or
- » Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.

## Sepsis features table

# Bacterial Sepsis

## Clinical features indicating antibiotics

- History or CNS examination
  - Objective evidence of new onset of an altered mental status.
  - Known neutropenia.
- Respiratory
  - New need for oxygen via reservoir mask to maintain  $\text{SpO}_2 > 92\%$  (or  $> 88\%$  in known COPD).
- Blood pressure
  - Systolic blood pressure  $\leq 90$  mmHg.
  - Systolic blood pressure  $> 40$  mmHg below known normal.
- Circulation and hydration
  - Heart rate  $> 130$ /minute.
- Skin
  - Mottled or ashen appearance.
  - Petechiae or purpura.

## Features of concern

- History or CNS examination
  - History from family/caregivers of new onset of an altered mental status.
  - History of acute deterioration of functional ability.
  - Impaired immune system (illness or drugs, including oral steroids).
  - Trauma, surgery or invasive procedures in the last six weeks.
- Respiratory
  - Tachypnoea.
- Blood pressure
  - Systolic blood pressure 91-100 mmHg.
- Circulation and hydration
  - Heart rate 91-130/minute
  - New onset dysrhythmia.
  - Reduced urinary output.
- Temperature
  - Tympanic temperature  $< 36^\circ\text{C}$ .
- Skin

# Bacterial Sepsis

- Signs of potential infection, including redness, swelling or discharge at surgical site or breakdown of wound.

## Backup

- Backup from an ICP/CCP should be requested if the patient has hypotension that is not rapidly responding to 0.9% sodium chloride IV.

## Referral and transport

- A patient with one or more clinical features indicating antibiotics (see the sepsis features table) must be given a clear recommendation to be transported to an ED by ambulance.
- A patient with one or more features of concern (see the sepsis features table) must be given a clear recommendation to be assessed by a doctor within two hours, and should usually be given a clear recommendation to be transported to ED by ambulance.
- Clinical judgement is required when determining whether a patient with one or more features of concern is given a recommendation to be transported to ED by ambulance:
  - Most patients should receive a recommendation to be transported to an ED by ambulance, particularly if the patient is living independently.
  - Being seen in primary care may be the best option if the patient is in an aged residential care facility, is very frail, or has dementia.
- Have a lowered threshold for recommending transport if the trajectory of illness is one of deterioration, and/or the patient has factors that increase the susceptibility to developing sepsis.

## Additional information

### General principles

- Sepsis is present when a microorganism invades a normally sterile part of the body, resulting in an immune response producing systemic signs and symptoms.
- The infecting microorganism is usually a bacterium, but may be a virus, a fungus or a parasite.
- Sepsis is usually isolated to a specific organ, for example the skin, urinary tract or lungs, but the immune response to sepsis affects all organs in the body.
- Sepsis has a significant mortality rate and early recognition with treatment improves outcomes.

# Bacterial Sepsis

- There are no criteria that can be used to tightly define the presence or absence of sepsis. Always consider sepsis a possibility if a patient is unwell without an obvious cause or has unexplained abnormal vital signs. The patient may have non-specific or very subtle signs or symptoms for example, new onset of altered mental status or decreased frequency of urination.

## Factors that increase susceptibility to developing sepsis

- Comorbidities that increase the risk of developing sepsis can include:
  - Chronic lung, heart, liver or kidney disease.
  - Cancer.
  - Immunocompromise.
  - Diabetes.
  - Morbid obesity.
- Factors that increase the risk of developing sepsis can include:
  - Age  $\geq 65$  years or  $\geq 45$  years for Māori, or  $> 40$  years for Pacific People.
  - Challenges with health literacy.
  - Frailty or disability.
  - Drug or alcohol addiction.
  - Complex social circumstances.
  - An indwelling venous line or urinary catheter.
  - Surgery or an invasive procedure within the last four weeks.
  - A breach of skin integrity, for example wounds or blisters.
  - Birth, termination of pregnancy or miscarriage in the last four weeks.

## Interpreting vital signs

- Although fever is commonly associated with sepsis, the presence or absence of fever is not a diagnostic sign. A fall in temperature is a very important sign that is often associated with sepsis in the elderly. Rigors indicate that bacteria are likely to be present in the blood.
- New onset of tachydysrhythmia (particularly atrial fibrillation) is common in the elderly with sepsis.
- A patient with pre-existing hypertension may have a significant fall in blood pressure without developing hypotension.
- Mental status:

# Bacterial Sepsis

- » Changes in mental status may be subtle and a history of an acute change from family and/or carers is important.
- » A change in mental status may present as irritability, altered behaviour or agitation, particularly in young children and those with pre-existing intellectual impairment.

## Withdrawing blood for culture

- If a local protocol is in place for obtaining blood cultures in the setting of sepsis, it should be followed.
- Not all ambulances will carry blood culture bottles as they have a short shelf life. If antibiotics are indicated and no blood culture bottles are available, they must be administered without any delay.

## Antibiotics in the setting of palliative care and care during the last days of life

- Sepsis with clinical features indicating antibiotics is common in patients who are receiving palliative care, and/or at the end of their life.
- It is usually of benefit to administer antibiotics when the patient has a life expectancy of months to years.
- Antibiotics should not routinely be administered to patients who are receiving care during the last days of life, as they can prolong the dying process.
- Wherever possible ascertain the patient's wishes and/or consult with their usual care provider. If an advance care plan is available this will also help to guide this decision.

## Response and/or transport by helicopter

- Response by helicopter may be indicated if the patient:
  - » Requires vasopressor support, and it will arrive at scene at least 15 minutes sooner than a road-based CCP, or
  - » Requires 0.9% sodium chloride and/or antibiotics, and it will arrive at least 60 minutes sooner than a road-based paramedic.
- Transport by helicopter may be indicated if the patient is not responding to vasopressor support and it will save more than 30 minutes compared to road transport. For criteria to be met, the road transport time will usually need to be more than 90 minutes.

## More Information



# Bacterial Sepsis

## Version history

|                                                                                                             |         |
|-------------------------------------------------------------------------------------------------------------|---------|
| <b>16 June 2026</b><br><b>Update:</b> Ketamine dosage updated.                                              | 1.1.0.1 |
| <b>11 September 2023</b><br><b>Update:</b> Guideline renamed to specify bacterial sepsis. Multiple changes. | 1.0.4   |

# Meningococcal Septicaemia

This guideline is for patients with a clinical diagnosis of meningococcal septicaemia.

- Don droplet level PPE.
- Administer ceftriaxone:
  - 2 g IV for an adult.
  - See the paediatric drug dose tables for a child.
  - Administer the dose IM if IV access cannot be immediately obtained.
- Administer 0.9% sodium chloride IV if there are signs of hypovolaemia or poor perfusion:
  - 1 litre for an adult.
  - 20 ml/kg for a child.
  - Repeat as required.
- Measure the blood glucose concentration and treat accordingly.
- Gain IV access in a second site if possible.

## Treatment of hypotension

- Administer metaraminol IV if the systolic blood pressure is less than 100 mmHg in an adult, or significantly less than the normal predicted systolic blood pressure in a child.

### Metaraminol for an adult:

- Administer via an infusion device:
  - Administer an initial bolus of 0.5-1 mg metaraminol IV.
  - Start the infusion at 2 mg/hour and adjust the rate as required.
- Alternatively, administer a bolus of 0.5-1 mg metaraminol IV every 5-10 minutes as required.

### Metaraminol for a child:

- Administer a bolus of metaraminol IV (see the paediatric drug dose tables) every 5-10 minutes as required.

- Consider administering adrenaline IV if the blood pressure is unresponsive to metaraminol or the patient is hypotensive and bradycardic.

### Adrenaline for an adult:

- Administer an adrenaline infusion via an infusion pump. Start at a rate of 0.5 mg/hour and adjust the rate as required, or

# Meningococcal Septicaemia

- » Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
  - » Administer as an infusion. Start at 2 drops/second and adjust the rate as required, or
  - » Administer a bolus of 0.01 mg adrenaline (10 ml) IV every 1-2 minutes as required.

## Adrenaline for a child aged 5-14 years:

- » Administer an adrenaline infusion via an infusion pump using service guidelines, or
- » Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
  - » Administer as an infusion. Start at 1 drop/second and adjust the rate as required, or
  - » Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.

## Adrenaline for a child aged less than five years:

- » Administer an adrenaline infusion via an infusion pump using service guidelines, or
- » Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.

## Backup

- Backup must be requested from an ICP/CCP.

## Referral and transport

- The patient must be transported to an ED by ambulance.
- Transport the patient to a hospital with intensive care facilities whenever this is feasible and safe.

## Additional information

### General principles

- Meningococcal septicaemia is uncommon and has a high mortality rate. It is a systemic infection from the bacterium *Neisseria meningitidis*.
- Meningococcal septicaemia is time critical in terms of commencing treatment and antibiotics should be administered out-of-hospital whenever possible.
- Patients with early meningococcal septicaemia commonly have non-specific influenza-like symptoms. Antibiotics are not indicated unless there are clear clinical signs of meningococcal septicaemia such as petechiae or purpura (see below).
- Meningococcal septicaemia commonly triggers disseminated intravascular coagulation (DIC).

# Meningococcal Septicaemia

- » DIC causes a combination of bleeding and clotting within small blood vessels throughout the body.
  - » DIC occurs in all organs but can be seen in the skin as small spots. The spots are not like a rash and will not blanch if they are pressed on.
  - » Petechiae are small spots about the size of the tip of a pen. They are due to bleeding from small capillaries in the skin. They can be seen anywhere and the patient should be examined fully to exclude them. They usually develop over time and if the diagnosis is being considered, the patient's skin should be re-examined every 10-15 minutes looking for new petechiae.
  - » Purpura are larger spots that look like small bruises. They result from a combination of bleeding and ischaemia of the skin and are usually very obvious.
- As meningococcal bacteria die they release endotoxin. The body's immune response to endotoxin can cause profound worsening of shock following antibiotic administration. Be prepared to treat this with further 0.9% sodium chloride IV, metaraminol IV, and adrenaline IV if required. It is rare for significant amounts of endotoxin to be released from other bacteria.
  - Hypoglycaemia is common in patients with meningococcal septicaemia. Have a low threshold for re-measuring the blood glucose concentration if transport time is prolonged or the patient has a falling level of consciousness.

## Meningitis

- Meningitis is inflammation of the meninges of the brain and is a completely different disease to meningococcal septicaemia. It is more common and has a low mortality rate.
- The infection is most commonly viral, but may be bacterial. *Neisseria meningitidis* is only one of many bacteria that can cause bacterial meningitis.
- The patient will usually present with headache and signs of infection. They may have nausea, a stiff neck and photophobia.
- Meningitis is not usually time critical in terms of commencing antibiotic treatment and administration of antibiotics IV should only occur if they meet criteria as outlined in the '[bacterial sepsis](#)' guideline.

## Advice to personnel following exposure to a patient with suspected meningococcal infection

- Meningococcal bacteria are present in the oro/nasopharynx in approximately 10% of the asymptomatic population and are spread between people via oro/nasopharyngeal droplets.
- The risk of infection in personnel following treatment and/or transport of a patient with meningococcal disease is low and no different from the rest of the population, unless personnel have had close contact with the patient.

# Meningococcal Septicaemia

- Close contact with the patient is defined as contact resulting in transmission of oro/nasopharyngeal droplets between the patient and personnel. For example, mouth to mouth ventilation or the patient coughing or sneezing directly into the face of personnel not wearing a mask.
- Personnel who have not had close contact with the patient do not require antibiotic prophylaxis.
- Personnel who have had close contact with the patient:
  - Are at slightly increased risk of subsequent meningococcal infection than the rest of the population.
  - Should notify their service via the usual reporting process.
  - Should notify public health, for example via the local 0800 number, including the name and NHI of the patient. Public health staff will contact personnel to discuss antibiotic prophylaxis.
- Antibiotic prophylaxis is not time critical and can be commenced several days following exposure. The decision to take antibiotic prophylaxis requires discussion on the balance of risk.
- Personnel should seek clinical advice if they require urgent advice.

## Response and/or transport by helicopter

- Response by helicopter may be indicated if the patient has suspected meningococcal septicaemia and it will arrive at scene at least 15 minutes sooner than a road-based CCP.
- For criteria to be met, the road transport time will usually need to be more than 75 minutes.

## More Information

### Version history

11 September 2023

1.0.4

**Update:** Multiple changes.

# Cellulitis

This guideline is for patients aged greater than or equal to 12 years. Seek clinical advice if the patient is aged less than 12 years.

- If the patient has suspected bacterial sepsis, move to the ['bacterial sepsis'](#) guideline.
- If the patient has one or more red flags, provide a clear recommendation to be seen in an ED.
- If no red flags are present, provide a clear recommendation to be seen in primary care within 12 hours or refer to an ECP.
- Administer 1 g cefazolin IV if the patient is not being directly referred to a medical facility or to an ECP. Wait a minimum of 20 minutes before leaving, to ensure there are no signs or symptoms of severe allergy.
- Have a lowered threshold for recommending transport if the trajectory of illness is one of deterioration, and/or the patient has factors that increase the susceptibility to developing sepsis.

## Red flags

### Should be seen in ED, usually via ambulance

- Severe pain, blistering or skin necrosis.
- Inability to safely mobilise, when normally able to do so.
- Neutropenia or chemotherapy within the last four weeks.
- Temperature > 40°C.
- An associated abscess.
- Involves > 10% of body surface area.
- Rapidly spreading.
- Involves the face or genitals.
- Significant lymphangitis.
- A hot or painful joint.
- Associated with a bite wound.

# Cellulitis

## Additional information

### General principles

- Severe pain, blistering or skin necrosis may indicate a deeper infection (for example, necrotising fasciitis).
- Chemotherapy includes any form of IV or oral chemotherapy for cancer treatment.
- 'Rapidly' spreading cannot be tightly defined and requires clinical judgement.
- Significant lymphangitis is present if there is red tracking or severe pain in the area of the lymph drainage/lymph nodes of the limb.
- A hot or painful joint indicates the possibility of septic arthritis (an infected joint) and this requires assessment in a hospital.
- If the patient is not being referred directly or transported to a medical facility, and cefazolin IV has been administered, remove the IV line prior to leaving the scene.

### Referral and transport

- Provide advice if the patient is not being transported to a medical facility:
  - Keep the limb elevated if possible.
  - Take paracetamol and/or ibuprofen for analgesia.
  - Call an ambulance if signs of severe systemic illness (for example feeling faint, severe nausea, severe vomiting or rigors) develop or the patient becomes unable to mobilise.

### More Information

#### Version history

11 September 2023

1.0.4

**Update:** Referral and transport instructions changed. Antibiotic changed. Red flag and additional information simplified.

# Viral Respiratory Illness

This guideline is for patients aged greater than or equal to 5 years with signs or symptoms of a viral respiratory illness, irrespective of the specific diagnosis.

- Don appropriate PPE.
- Have the patient wear a medical mask if this is feasible.
- If the patient has one or more red flags, provide a clear recommendation to be seen in an ED.
- If the patient does not have red flags but has one or more orange flags, provide a clear recommendation to be seen in primary care within 24 hours or refer to an ECP.
- Have a lowered threshold for recommending transport if the trajectory of illness is one of deterioration, and/or the patient has factors that increase the susceptibility to developing sepsis.

## Red flags

### Should be seen in ED, usually via ambulance

- New onset of an altered mental status.
- SpO<sub>2</sub> < 94% on air (unless normal for the patient).
- Signs of shock.
- New onset dysrhythmia.
- Mottled or ashen appearance.
- Inability to safely mobilise, when normally able to do so.
- Severe pleuritic chest pain.
- Severe headache.
- Neutropenia or chemotherapy within the last four weeks.
- Temperature > 40°C.

## Orange flags

### Should be seen in primary care within 24 hours



# Viral Respiratory Illness

- Tachypnoea.
- Pregnant.
- Immunocompromise.
- Ischaemic heart disease.
- COPD.
- Diabetes.
- Morbid obesity.
- Purulent sputum.
- Age  $\geq 65$  years or  $\geq 45$  years for Māori, or  $> 40$  years for Pacific People.
- Challenges with health literacy.
- Frailty or disability.
- Complex social circumstances.

## Additional information

- The term 'viral respiratory illness' encompasses a range of illnesses such as influenza, respiratory syncytial virus (RSV) and coronaviruses (including COVID-19).
- The patient will most commonly present with:
  - Non-specific symptoms (for example muscle aches, fatigue, malaise and headache), and
  - Respiratory symptoms (for example cough, sore throat and chest discomfort), and fever.
- Chest pain is common with viral respiratory illness:
  - It may be dull in nature despite having a respiratory or musculoskeletal source. It is important to fully assess the patient, including acquisition of a 12 lead ECG.
  - There is an increased risk of myocardial ischaemia and myocarditis. The patient should be treated as possibly having myocardial ischaemia or myocarditis if:
    - The pain is not clearly associated with the respiratory cycle or coughing, or
    - The nature of the pain seems to be cardiac in nature, or
    - The 12 lead ECG shows signs of myocardial ischaemia or myocarditis.

# Viral Respiratory Illness

- Viral respiratory illnesses are generally seasonal (usually winter and spring months in New Zealand) and the prevalence of a specific illness in the community at any given time needs to be considered before making a provisional diagnosis of a specific viral respiratory illness.
- Include an assessment for signs of meningococcal septicaemia, which in the very early phase may be indistinguishable from viral respiratory illnesses, particularly influenza.
- If more than two orange flags are present, have a lowered threshold for recommending the patient is assessed in an ED.
- If the patient is not transported by ambulance, the patient should be provided a viral respiratory illness information sheet and advised to:
  - Administer paracetamol and/or ibuprofen if indicated.
  - Keep well hydrated.
  - Call their usual healthcare provider or Healthline if there is no improvement.
  - Call for an ambulance if there is sudden deterioration in their condition.

## More Information

### Version history

11 September 2023

1.0.4

**New:** This guideline replaces the previous 'Influenza' guideline.

# Urinary Tract Infection

This guideline is for patients aged greater than or equal to 12 years. Seek clinical advice if the patient is aged less than 12 years.

- If the patient has suspected bacterial sepsis, move to the '[bacterial sepsis](#)' guideline.
- If the patient has one or more red flags, provide a clear recommendation to be seen in an ED.
- If the patient does not have red flags but has one or more orange flags, provide a clear recommendation to be seen in primary care within 24 hours or refer to an ECP.
- Have a lowered threshold for recommending transport if the trajectory of illness is one of deterioration, and/or the patient has factors that increase the susceptibility to developing sepsis.

## Red flags

**Should be seen in ED, usually via ambulance**

- Flank/loin pain.
- Severe pain.
- Significant haematuria.
- Urinary retention.
- Inability to mobilise normally.
- Neutropenia or chemotherapy within the last four weeks.
- Temperature > 40°C.

## Orange flags

**Should be seen in primary care within 24 hours**

# Urinary Tract Infection

- Dysuria.
- Temperature 38-40°C.
- Moderate pain.
- Male.
- Aged < 15 years.
- Aged ≥ 65 years.
- Pregnancy.
- Immunocompromise.

## Additional information

- The classic presentation is usually a triad of:
  - Dysuria (pain on urination).
  - Frequency (frequent urination).
  - Lower abdominal discomfort.
- Urinary tract infections are a common cause of an altered mental status in the elderly.
- If more than two orange flags are present, have a lowered threshold for recommending the patient is assessed in an ED.
- In women, lower urinary tract infections are usually self-limiting. Antibiotics may shorten the duration of the illness and reduce the incidence of recurrence, but are not required in all cases. If the patient is being given a recommendation for self-care, provide the following advice:
  - Take paracetamol and/or ibuprofen for analgesia.
  - Ensure adequate hydration.
  - See a pharmacist for advice.
  - Be seen in primary care if not improving.
- Lower urinary tract infections are uncommon in males and the patient should be given a clear recommendation to be seen in primary care.
- Flank/loin pain indicates that pyelonephritis is possible.
- Significant haematuria requires blood to be clearly visible in the urine. Pink urine does not indicate significant haematuria.

# Urinary Tract Infection

- Chemotherapy includes any form of IV or oral chemotherapy for cancer treatment.

## More Information

### Version history

11 September 2023

1.0.4

**Update:** Now accommodates all patients aged 12 and above. Referral and transport instructions changed. Flag table and additional information simplified.

# Sore Throat

- If the patient has suspected bacterial sepsis, move to the '[bacterial sepsis](#)' guideline.
- If the patient has one or more red flags, provide a clear recommendation to be seen in an ED.
- If the patient does not have red flags but has one or more orange flags, provide a clear recommendation to be seen in primary care within 24 hours or refer to an ECP.
- Have a lowered threshold for recommending transport if the trajectory of illness is one of deterioration, and/or the patient has factors that increase the susceptibility to developing sepsis.

## Red flags

### Should be seen in ED, usually via ambulance

- Signs of airway compromise.
- Severe pain.
- Drooling or severe difficulty swallowing.
- Abnormal speech.
- Neutropenia or chemotherapy within the last four weeks.
- Temperature > 40°C.

## Orange flags

### Should be seen in primary care within 24 hours

- Temperature 38-40°C.
- Moderate pain.
- Aged < 15 years.
- Aged 16-45 years with any of the following features:
  - Māori or Pacific People, or
  - Live in a low socioeconomic area of the North Island, or
  - Have a past history or family history of rheumatic fever.

# Sore Throat

## Additional information

- Sore throats are commonly viral in origin and not usually a serious clinical problem. However, the immune response to untreated throat infections due to Group A Streptococcus may cause rheumatic fever, with subsequent heart and kidney disease. Rheumatic fever is particularly prominent within Māori and Pacific Island children and is a significant cause of preventable morbidity and mortality in New Zealand.
- Epiglottitis is a bacterial infection of the upper airway. It is now relatively rare as a result of immunisation. Historically it was most common in children aged 2-7 years but it is now more common in adults. The patient usually has an onset of illness over a day or two, a very sore throat, difficulty swallowing (which may cause drooling) and a high grade fever. Epiglottitis is an emergency because the risk of airway occlusion is relatively high.
- Tracheitis is a bacterial infection of the trachea. It is relatively uncommon and mainly affects children. It is most commonly due to secondary bacterial infection following a viral infection.
- Pharyngeal abscess formation is usually associated with an onset of illness over days, a very sore throat, difficulty swallowing and a high grade fever. It is usually a complication of:
  - Bacterial pharyngitis in young children, or
  - Tonsillitis in young adults, or
  - Trauma from a foreign body.
- Chemotherapy includes any form of IV or oral chemotherapy for cancer treatment.

## More Information

### Version history

11 September 2023

1.0.4

**Update:** Referral and transport instructions changed. Red flag and additional information simplified.

# Fever

Use this guideline to help determine which patients with fever require referral to a medical facility and if so, to which type of medical facility. If the patient has presumed bacterial sepsis, refer to the '[bacterial sepsis](#)' guideline. If the patient has a fever and a suspected source of infection, refer to the relevant guideline.

- Assess paediatric patients using the paediatric assessment triangle, including an assessment of the flags in the relevant table.
- Assess adult patients, including an assessment of the flags in the relevant table.
- Administer paracetamol if the patient has a temperature greater than 39 degrees and the fever is causing discomfort:
  - 1.5 g PO for an adult weighing greater than 80 kg, or
  - 1 g for an adult weighing 80 kg or less.
  - See the paediatric dosage tables for a child.
- Encourage intake of oral fluids as tolerated.

## Flag table for patients aged less than five years

### Red flags

**Should be seen in ED, usually via ambulance**



# Fever

- **Appearance:**
  - Pale, mottled or cyanosed.
  - Petechiae or purpura.
  - Abnormal or weak cry.
  - No response to social cues.
  - Difficult to rouse or does not stay awake when roused.
- **Respiratory:**
  - Grunting.
  - Respiratory rate >50/minute.
  - Moderate or severe increased work of breathing.
  - SpO<sub>2</sub> <94% on air.
- **Circulation and hydration:**
  - Severe tachycardia or bradycardia.
  - Peripheral capillary refill time >3 seconds.
  - Moderate to severe dehydration.
  - Has not passed urine or had a wet nappy in the past 18 hours.
- **Other:**
  - Known neutropenia or chemotherapy within the last four weeks.
  - Pain in a single joint or a single muscle area.
  - Rigors.
  - Neck stiffness.
  - Focal neurological signs.
  - Concern regarding neglect or non-accidental injury.

## Orange flags

**Should be seen in primary care within 6 hours**

# Fever

- **Appearance:**
  - Pallor reported by caregiver but not seen by personnel.
  - Not responding to social cues normally.
  - Decreased activity.
  - Poor feeding.
- **Respiratory:**
  - Mild increased work of breathing.
  - Respiratory rate 40- 50/minute.
  - Crackles audible on auscultation.
- **Circulation and hydration:**
  - Mild dehydration
  - Peripheral capillary refill time 2-3 seconds.
  - Reduced urinary output or frequency.
- **Other:**
  - Illness for longer than five days.
  - Non-weight bearing or not mobilising appropriately.
  - Immunocompromise.
  - Help from a healthcare provider has been sought more than once within 24 hours.

Flag table for patients aged greater than or equal to five years

Red flags

Should be seen in ED, usually via ambulance

# Fever

- Objective evidence of new onset of an altered mental status.
- Significantly abnormal vital signs.
- Rigors.
- Known neutropenia or chemotherapy within four weeks.
- Moderate or severe dehydration.
- Severe pleuritic chest pain.
- Severe or worsening headache.
- Neck stiffness.
- Petechiae or purpura.
- Pain or tenderness in the flank or back.
- Abdominal pain with tenderness on palpation.
- Pain in a single joint or muscle area.

## Orange flags

### Should be seen in primary care within 12 hours

- History from family/caregivers of new onset of an altered mental status.
- Immunocompromise.
- Help from a healthcare provider has been sought more than once within 24 hours.
- Purulent sputum.
- Mild dehydration.

## Referral and transport

- Provide a clear recommendation to be transported to ED by ambulance if the patient is aged less than 12 months or has one or more red flags.
- If no red flags are present, but one or more orange flags are present:

# Fever

- » Provide a clear recommendation that the patient is seen in primary care or an ED within six hours if the patient is aged less than 5 years, or
- » Provide a clear recommendation that the patient is seen in primary care or an ED within 12 hours if the patient is aged greater than or equal to 5 years, or
- » Refer the patient to an ECP for further review.
- If the patient is not transported by ambulance, the patient should be advised to:
  - » Administer paracetamol and/or ibuprofen if indicated.
  - » Keep well hydrated.
  - » Call their usual healthcare provider or Healthline if there is no improvement.
  - » Call for an ambulance if there is sudden deterioration in their condition.
- Have a lowered threshold for recommending transport if the trajectory of illness is one of deterioration, and/or the patient has factors that increase the susceptibility to developing sepsis.

## Additional information

### General principles

- Fever is common and has a number of causes. The most common are viral or bacterial infection, but environmental stress, pulmonary embolism, and inflammatory conditions may all cause fever in the absence of clinical infection.
- Patients who appear dehydrated should be encouraged to drink fluids where this is appropriate.
- Paracetamol has a role where patients have a high fever and this is causing discomfort. If the patient is well appearing and comfortable, paracetamol should generally not be administered.

### Fever in children

- The most common cause for fever in children is a viral infection but it is important that bacterial infection is considered.
- Children can be challenging to assess and treat. Parents are often very concerned, even when the patient appears well. Their concerns should be considered when making recommendations around referral and/or transport.
- Fever usually causes an increase in the heart rate and respiratory rate. Clinical judgement must be used when determining that the patient's vital signs are significantly abnormal.
- Tympanic thermometers may be unreliable in small children. An axillary measurement is preferred if a child is under six months of age.

# Fever

- Parents may wish for their child to be transported by ambulance despite a recommendation to the contrary. In this setting transport by ambulance should occur if no other reasonable transport option is available.
- An apparent improvement following the administration of antipyretics (such as paracetamol and/or ibuprofen) may be due to masking of symptoms and does not rule out a serious infection.
- The threshold for transport to an ED must be lowered in children with coexisting chronic diseases such as respiratory disease.
- The threshold for transport to an ED must be lowered if help from a healthcare provider has been sought more than once within 24 hours.

## Red flags

- Rigors indicate that bacteria may be present in the blood.
- Neutropenia is a low neutrophil count. Neutropenia usually occurs post chemotherapy and the patient and their family/whānau will usually know if they may be neutropenic.
- Chemotherapy includes any form of IV or oral chemotherapy for cancer treatment.
- Meningitis may present with drowsiness, headache and/or neck stiffness. Neck stiffness does not usually occur below the age of one year.
- Petechiae or purpura commonly occur with meningococcal septicaemia.
- Pyelonephritis may present with pain or tenderness in the flank or back.
- A patient with fever, abdominal pain and tenderness on palpation often requires a surgical review.
- Septic arthritis may present with pain in a single joint.
- Myositis and fasciitis may present with tenderness in a single muscle area.

## More Information

### Version history

11 September 2023

1.0.4

**New:** Amalgamating previous guidelines 'Fever in patients under five years' and 'Fever in patients five years and over'.

# Infectious Disease Precautions

| Infectious disease                  | Level of PPE required | Level of vehicle cleaning required |
|-------------------------------------|-----------------------|------------------------------------|
| Chicken pox                         | Airborne              | Additional                         |
| Clostridium difficile diarrhoea     | Contact               | Standard                           |
| ESBL                                | Contact               | Additional                         |
| Gastroenteritis, type not specified | Contact               | Standard                           |
| Hepatitis A                         | Contact               | Standard                           |
| Hepatitis B                         | Standard              | Standard                           |
| Hepatitis C                         | Standard              | Standard                           |
| HIV                                 | Standard              | Standard                           |
| Viral respiratory illness           | Airborne              | Standard                           |
| Measles                             | Airborne              | Standard                           |
| Meningitis, type not specified      | Standard              | Standard                           |
| Meningococcal disease               | Droplet               | Standard                           |
| MRSA                                | Contact               | Additional                         |
| MRO, type not specified             | Contact               | Additional                         |
| Mumps                               | Droplet               | Standard                           |
| Norovirus with vomiting             | Airborne              | Additional                         |
| Norovirus without vomiting          | Contact               | Additional                         |
| Pneumonia, type not specified       | Standard              | Standard                           |
| Rotavirus                           | Contact               | Standard                           |
| Rubella                             | Airborne              | Standard                           |
| Tuberculosis                        | Airborne              | Standard                           |
| VRE                                 | Contact               | Additional                         |
| Whooping cough                      | Droplet               | Standard                           |

# Infectious Disease Precautions

ESBL – extended spectrum beta-lactamase producing organism.

MRSA – methicillin resistant staphylococcus aureus.

MRO – multi-resistant organism.

VRE – vancomycin resistant enterococci.

## PPE levels

|          | Gloves | Medical mask | Eye protection | Gown/overalls/apron | N95 mask           |
|----------|--------|--------------|----------------|---------------------|--------------------|
| Standard | ✓      | -            | ?              | -                   | -                  |
| Contact  | ✓      | -            | ?              | ✓                   | -                  |
| Droplet  | ✓      | ✓            | ?              | ?                   | Aerosol procedures |
| Airborne | ✓      | -            | ✓              | ?                   | ✓                  |

## Vehicle cleaning and disinfection

### Standard level of cleaning and disinfection

- Open all vehicle doors for ten minutes with nobody in the vehicle, if the infectious disease was airborne.
- Wear gloves.
- Decontaminate and disinfect surfaces contaminated with body fluid:
  - Decontaminate using a cleaning solution, removing all visible soiling.
  - Wipe with a disinfectant and allow to dry.
- Remove used linen.
- Wipe down the stretcher and all surfaces touched by the patient with disinfectant and allow to dry.
- Wipe down all surfaces in the back of the vehicle touched by personnel (such as the monitor) with an approved disinfectant wipe and allow to dry.
- Clean the floor if visibly dirty.
- Replace linen.
- Wash and dry hands.
- Once disinfected surfaces are dry, the vehicle may be used for other patients.

# Infectious Disease Precautions

## Additional level of cleaning and disinfection

- Standard level cleaning and disinfection plus:
    - Wear gloves and overalls/gown and/or an apron.
    - Wipe down all interior surfaces (including in the front of the vehicle) that the patient or personnel may have touched, with disinfectant and allow to dry.
    - Clean the floor.
  - Once disinfected surfaces are dry, the vehicle may be used for other patients.
-



# Seizures

- Measure the blood glucose concentration and treat accordingly.
- Administer midazolam if the seizure is generalised and continues for longer than five minutes, or seizures are recurrent:
  - Administer midazolam IV:
    - 5 mg of midazolam IV for an adult. Consider reducing the dose to 3 mg if a caution is present.
    - See the paediatric drug dose tables for a child.
    - This may be repeated once after five minutes.
  - Administer midazolam IM if IV access has not already been obtained:
    - 10 mg of midazolam IM for an adult. Consider reducing the dose to 5 mg if a caution is present.
    - See the paediatric drug dose tables for a child.
    - This may be repeated once after ten minutes.
- Administer levetiracetam IV if the seizure continues or recurs after two parenteral doses of midazolam:
  - 2 g of levetiracetam for an adult.
  - See the paediatric drug dose tables for a child.
  - Repeat once after ten minutes for an adult, if the seizure continues or recurs ten minutes after the first dose.
- Consider performing RSI if the seizure continues.

## Backup

- Backup from a CCP must be requested if there is severe airway obstruction, persistent severe hypoxia or levetiracetam is indicated.

## Referral and transport

- The patient should receive a clear recommendation to be transported to an ED by ambulance if this is the first time they have had a seizure, unless the cause is clearly recreational drug use and the patient has recovered to be able to mobilise safely.
- The patient may receive a clear recommendation not to be transported to a medical facility by ambulance, even if midazolam has been administered, providing the following criteria are met:
  - No red flags are present.
  - The patient is left in the care of a competent adult.

# Seizures

- » The patient is given a clear recommendation to be seen in primary care within 72 hours for a review of their treatment.
- If transport is required, this should usually be to an ED, but could be to primary care if the patient is rapidly improving and is well known to staff at that facility.

## Red flags (should be seen in ED, usually via ambulance)

- No known history of epilepsy.
- Significant change in their usual pattern of seizures.
- More than one dose of parenteral midazolam administered by ambulance personnel.
- Does not recover to a safe postictal state.
- Injury requiring transport.

## Additional information

### General principles

- Most seizures will terminate spontaneously after 2-3 minutes.
- There is no immediate urgency to treat seizures with medication as long as the patient and their airway are protected by positioning.
- Some patients have pre-prescribed medicines to be administered via the rectal, nasal or buccal route. All personnel may administer such medicines, even if not within their delegated scope of practice, provided they have been prescribed for that patient and the seizure continues for longer than five minutes, or seizures are recurrent.

### Treating seizures with midazolam

- The preferred route is IV, but do not delay administration IM while IV access is obtained because the evidence is that midazolam by the IM route is as effective as IV route.
- Midazolam IV may take 2-3 minutes to stop the seizure.
- A brief period of apnoea is common following successful treatment with midazolam IV.
- The maximum number of doses of midazolam administered by ambulance personnel is two by any route if the patient does not stop seizing. For example, two IV doses, one IM dose and one IV dose, or two IM

# Seizures

doses. This maximum of two doses may be administered in addition to any benzodiazepine that has already been administered prior to ambulance arrival.

- There is no maximum number of midazolam doses if the seizures stop with midazolam administration and the patient has normal/purposeful movement between seizures. Examples of normal/purposeful movement include localising to pain, sitting up, rolling over and pulling on an oxygen mask.
- Wait ten minutes before administering midazolam IV after IM administration. This is to allow adequate time for the IM dose to be absorbed.

## Prolonged seizures

- Prolonged seizures (for example, those lasting greater than 30 minutes) alter brain receptor activity, making pharmacological control of seizures more difficult. The longer the seizure persists before being controlled, the more difficult control becomes.
- Prolonged seizures are life-threatening because they are commonly associated with hypoxia, hypercarbia, metabolic acidosis (lactate is produced during vigorous skeletal muscle activity), hyperthermia (as a result of prolonged muscle activity) and aspiration.

## Following the seizure

- Position the patient on their side.
- Maintain airway and breathing.
- Monitor pulse oximetry and administer oxygen if required.

## The postictal state

- Following a seizure it is common for the patient to have an altered level of consciousness with drowsiness, confusion, agitation or amnesia. This is called the postictal state and usually lasts for 5-60 minutes.
- Sometimes the patient may appear to be in a postictal state, but is actually continuing to have seizures. Suspect this if the patient has rhythmic eye movements, dilated pupils, persistent tachycardia or failure to improve.
- During the postictal state the patient does not usually have capacity to make decisions and it is common for the patient to refuse assessment and/or transport. Clinical judgement is required taking into account the balance of risks, but it may not be in the best interest of the patient to forcibly initiate assessment and/or transport. Seek clinical advice if the situation is difficult to resolve.

# Seizures

## Types of seizures

- Seizures may be classified as generalised (grand mal) or partial (focal or localised). Partial seizures may be simple (where the patient retains awareness of their surroundings) or complex (where the patient loses awareness of their surroundings).
- A patient having partial seizures may present without obvious convulsions or motor activity and may be able to obey commands and interact during seizure activity. The patient may present with any combination of the following:
  - Habitual repetitive movements (automatisms).
  - Sensory symptoms including visual or auditory hallucinations.
  - Emotional outbursts or unusual feelings, such as feeling like they are outside their body.
  - Blank gaze.
- The most common cause of partial seizures is temporal lobe epilepsy.
- Partial seizures may respond to midazolam and/or levetiracetam. If the seizure lasts longer than five minutes and is causing distress, it is appropriate to treat the seizures with anticonvulsants.

## Non-epileptic seizures (pseudo-seizures or psychogenic seizures)

- Non-epileptic seizures occur when there is motor activity that looks clinically like seizures, but there is no electroencephalography (EEG) evidence of seizure activity in the brain.
- In general, healthcare personnel cannot reliably distinguish between epileptic seizures (EEG positive) and non-epileptic seizures (EEG negative).
- It is very common for partial seizures from temporal lobe epilepsy to be initially misdiagnosed as non-epileptic seizures.
- The presence of tongue biting, or incontinence of urine or faeces lowers the likelihood of non-epileptic seizures.
- A patient with non-epileptic seizures may not have conscious control over their motor activity. In the absence of a previously confirmed diagnosis of non-epileptic seizures, treat the patient using the 'seizures' guideline.
- The majority of patients with non-epileptic seizures will subsequently be diagnosed with a medical problem and a proportion have true epilepsy. Only a minority will be diagnosed as having a mental health problem such as conversion disorder or a personality disorder.

# Seizures

## Seizures following recreational drug use

- Seizures following recreational drug use are common, particularly with the chemicals commonly referred to as synthetics.
- The seizures are usually self-limiting and commonly the patient will refuse assessment and/or transport.
- Clinical judgement must be used as forcibly initiating assessment and/or transport often requires a combination of significant restraint and/or sedation, which may be more life-threatening than the effects of the recreational drug.
- It is usually appropriate to manage the patient without transport to an ED if the patient has stopped seizing and has recovered to be able to mobilise safely. Advice on providing supervision until the patient has fully recovered should be provided to family and/or friends whenever feasible.

## Febrile seizures in children

- Febrile seizures are associated with a rapid temperature rise, rather than any specific temperature and usually occur in children aged less than six years.
- The most common cause of febrile seizures is a viral illness.
- Fever associated with infection usually confers some benefit to the patient and does not cause harm provided it is less than 40°C. For this reason rapid and/or aggressive cooling is not indicated unless the temperature is higher than 40°C. Unless this is the case, cool slowly by uncovering the child.

## Seizures during pregnancy

- Eclampsia occurs when a pregnant patient with pre-eclampsia has one or more generalised seizures.
- If the patient is more than 20 weeks pregnant and eclampsia is suspected, refer to the '[eclampsia](#)' guideline.

## Response and/or transport by helicopter

- Response by helicopter may be indicated if the patient has prolonged seizures and it will arrive at scene at least 15 minutes sooner than a road-based CCP.
- Transport by helicopter may be indicated if the patient has prolonged seizures not resolving with treatment and it will save more than 30 minutes compared to road transport. For criteria to be met, the road transport time will usually need to be more than 90 minutes.

# Seizures

## More Information

### Version history

|                                                         |         |
|---------------------------------------------------------|---------|
| <b>16 June 2026</b>                                     | 1.1.0.1 |
| <b>Update:</b> Referral and transport guidance updated. |         |
| <b>11 September 2023</b>                                | 1.0.4   |
| <b>Update:</b> Multiple changes.                        |         |

# Acute Behavioural Disturbance

This guideline is for patients aged greater than or equal to 12 years with acute behavioural disturbance, exhibiting signs of agitation and significantly abnormal behaviour which is posing a clear risk to themselves, personnel or bystanders. Seek clinical advice if the patient is aged less than 12 years of age.

- Determine the level of risk to safety of the patient, personnel and bystanders.
- Identify any underlying clinical condition causing the behaviour and provide appropriate treatment using the relevant CPG. Examples of conditions that may present with acute behavioural disturbance include:
  - Acute alcohol and drug intoxication, including drug overdose and serotonin syndrome.
  - Drug or alcohol withdrawal.
  - Acute psychosis.
  - Dementia.
  - Traumatic brain injury.
  - Hypoglycaemia or hyperglycaemia.
  - Sepsis or another generalised inflammatory process.
  - Epilepsy.
  - Neurodevelopmental disorders.
- The patient should be treated with the minimum amount of sedation and restraint to achieve psychomotor control and minimise the risk of harm.

## Mild to moderate risk to safety

- Attempt verbal de-escalation and move sequentially through the following steps if the level of agitation continues to pose a risk to safety.
- Involve the patient's family/whānau where appropriate.
- Consider calling for police assistance.
- Provide safe restraint if required.
- Administer olanzapine provided the patient will take an oral medicine:
  - 10 mg of olanzapine PO. Consider reducing the dose to 5 mg if a caution is present.
  - The dose may be repeated once after 20 minutes.
- Administer droperidol if olanzapine is not administered or is ineffective:
  - 10 mg of droperidol IM/IV. Consider reducing the dose to 5 mg if a caution is present.
  - The dose may be repeated once after 20 minutes.

# Acute Behavioural Disturbance

- Administer midazolam if droperidol is ineffective:
  - 2-3 mg of midazolam IV every five minutes as required. Consider reducing the dose to 1-2 mg if a caution is present.
  - 10 mg of midazolam IM. Consider reducing the dose to 5 mg if a caution is present.
  - The IM dose may be repeated once after 20 minutes.
- If the patient has an altered level of consciousness:
  - Position the patient on their side and provide safe restraint.
  - Administer oxygen.
  - Continually monitor the patient's airway, breathing and vital signs.
  - Gain IV access if not already achieved.

## Severe to immediately life-threatening risk to safety

- Call for urgent police assistance and consider leaving the scene.
- Attempt verbal de-escalation and/or provide restraint only if it is safe to do so.
- Administer ketamine:
  - 1 mg/kg of ketamine IV (up to a maximum of 150 mg) every five minutes as required, or
  - 5 mg/kg of ketamine IM, rounded to the nearest 100 mg (up to a maximum of 400 mg).
  - The IM dose may be repeated once after 20 minutes.
- Seek clinical advice if the situation is not easily controlled.
- Once control has been obtained:
  - Position the patient on their side and provide safe restraint.
  - Administer oxygen.
  - Continually monitor the patient's airway, breathing and vital signs.
  - Gain IV access if not already achieved.
  - Maintain a sedated state as required using 2-3 mg of midazolam IV every five minutes.
  - Administer additional ketamine IV if a severe or immediately life-threatening risk to safety persists or recurs.
  - Request police escort during transport.



# Acute Behavioural Disturbance

## Backup

- Backup from a Paramedic should be requested if there is a mild to moderate risk to safety and verbal de-escalation is unsuccessful.
- Backup from an ICP/CCP must be requested if there is a severe to immediately life-threatening risk to safety.

## Referral and transport

- The patient must be transported to an ED by ambulance if droperidol, midazolam or ketamine has been administered.
- The patient must be transported to an ED by ambulance if olanzapine has been administered, unless:
  - The patient has a known mental health problem or intellectual disability, and
  - The patient's agitation improves and they are safe, and
  - The patient is with a competent adult who can remain with them, and
  - Personnel have arranged follow up with mental health personnel, or other appropriate healthcare professional.

## Additional information

### General principles

- Acute behavioural disturbance is an inclusive term covering other diagnoses such as agitated delirium, excited delirium or hyperactive delirium.
- To have agitation, the patient must have an abnormal increase in motor activity, for example trying to climb off the stretcher or actively resisting assessment, treatment and/or transport. Anxiety and/or signs of mental distress alone are not enough to define the patient as being agitated.
- Distasteful, annoying or unusual behaviours are not grounds for physical restraint or the administration of sedative medicines.

### Verbal de-escalation

- Verbal de-escalation is often under-utilised and must occur prior to providing sedation and/or restraint unless there is a severe or immediately life-threatening risk to safety.
- A gentle and calm approach should be used, as a confrontational approach from personnel may escalate the patient's behaviour. Avoid procedures that may antagonise the situation (for example, obtaining a

# Acute Behavioural Disturbance

blood pressure), noting that clinical judgement is required.

- Friends and/or family/whānau may be helpful, however personnel should ask them to desist if they are making the situation worse.
- The key aspects to successful verbal de-escalation are:
  - Allow sufficient time. 15-20 minutes may be required.
  - Maintain a safe distance between you and the patient.
  - Have only one person verbally engage with the patient.
  - Introduce yourself, state you are there to help, and provide reassurance.
  - Use a calm voice and adopt a non-threatening stance. Sit if it is feasible and safe to do so.
  - Use short sentences and keep messages simple. Repetition is essential as the patient will usually have a short attention span.
  - Minimise the number of people in the immediate vicinity.
  - Limit unnecessary noise and distractions. For example, limit radio noise.
  - Actively listen to the patient and try to gain an understanding of their concerns.
  - Try to establish rapport and offer choices if appropriate. For example, offer a drink or a cigarette.
  - Avoid provocative statements. For example, do not say “if you don’t calm down, we will have to sedate you”.
  - Once rapport has been established, offer medicine if appropriate. For example, “I see you are very uncomfortable, would you like some medication to help?”

## Determining the level of risk to safety

- Determining the level of risk to safety requires clinical judgement that considers the risk to the patient and to personnel.

# Acute Behavioural Disturbance

## Mild to moderate risk to safety

Examples include, but are not limited to:

- Verbally aggressive
- Actions not involving immediate risk of serious harm to personnel, for example pushing or grabbing
- Pulling at equipment
- Trying to climb off the stretcher
- Agitation preventing control of moderate external bleeding

## Severe to immediately life-threatening risk to safety

Examples include, but are not limited to:

- Dangerous physical aggression
- Wielding a weapon
- Actions involving immediate risk of serious harm to personnel, for example punching or kicking
- Destruction of physical surroundings
- Trying to get out of a moving ambulance
- Agitation preventing control of severe or life-threatening external bleeding

## Providing restraint

- Restraint must only be provided when necessary as it often increases the level of agitation.
- Restraint carries risks that must be balanced against the risks of not appropriately assessing, treating or transporting the patient.
- Use the minimum amount of restraint required to ensure safety, using a tiered approach that matches the level of restraint to the risk to safety.
- For example:
  - For agitation causing a mild to moderate risk to safety, the patient may only require guidance with a hand on their arm/shoulder/back or may require someone to hold their hands.
  - For agitation causing a severe to immediately life-threatening risk to safety, the patient is likely to require their limbs to be restrained.
- All forms of physical restraint must be recorded on the PRF.

# Acute Behavioural Disturbance

- Never restrain the patient face down and never restrain the patient with weight on their neck, chest, back or abdomen, as these risk causing respiratory arrest (positional asphyxia).
- There is a high risk of injury to the patient and/or personnel if the patient has severe agitation. In this setting it is vital that there is a planned and coordinated team approach to simultaneous provision of safe sedation and restraint.

## Involving police

- Assistance from police is not routinely required but should be requested if there is:
  - Significant risk of injury to the patient, personnel or bystanders, or
  - The patient has agitation causing a severe to immediately life-threatening risk to safety, or
  - More than minimal restraint required.
- Police do not have the legal authority to restrain a patient on their own property, unless there is immediate danger to people, including immediate danger as a result of being unable to treat the patient.
- Police do not have the legal authority to restrain a patient in a public space, unless there is immediate danger to people (as above), or the patient is considered to be 'disturbing the peace'.
- In rare circumstances where an immediate threat to life exists, it may be appropriate for a taser to be used by police to gain initial control, so that sedation and restraint can be safely provided. In this setting there must be an explicit discussion with police regarding the risks of using a taser, noting that these predominantly relate to the risk of injury associated with the fall following deployment of the taser.

## Providing sedation

- Sedation carries risks that must be balanced against the risks of not appropriately assessing, treating or transporting the patient.
- Use the minimum amount of sedation required to ensure safety, using a tiered approach that matches the choice and dose of sedative medicine to the level of risk posed to safety.
- The safest approach is usually to provide restraint and sedation, because sedation minimises the amount of restraint required and vice versa.
- When administering droperidol:
  - The IM route is as effective as the IV route.
  - IV administration may occur if IV access is already in place, but IM administration should not be delayed while IV access is obtained.
- When administering midazolam:

# Acute Behavioural Disturbance

- » The preferred route is IV, but this route is often not feasible initially and IM administration should not be delayed while IV access is obtained.
- » No maximum IV dose has been described because the balance of risk is in favour of continuing to administer midazolam if the patient remains agitated.
- When administering ketamine:
  - » Ketamine should not be administered to patients with dementia unless all other options have been exhausted.
  - » The goal is to produce a state of dissociation so that the patient can be safely restrained, and IV access obtained.
  - » IV access should be obtained whenever possible and IV midazolam or droperidol administered as the ketamine effect wears off, with the aim of transporting the patient in a sedated state.
  - » If IV access cannot be obtained, IM midazolam or droperidol should be administered using the doses described under mild to moderate risk to safety, in anticipation of the ketamine effect wearing off.
  - » Uncontrolled severe agitation carries a risk of precipitating cardiac arrest. For this reason, sedation must be administered if the patient is severely agitated, even if they have been successfully physically restrained.

## More Information

### Version history

16 June 2026

1.1.0.1

**Update:** Ketamine dosage updated.

11 September 2023

1.0.4

**Update:** Guideline renamed. Multiple changes.

# Stroke

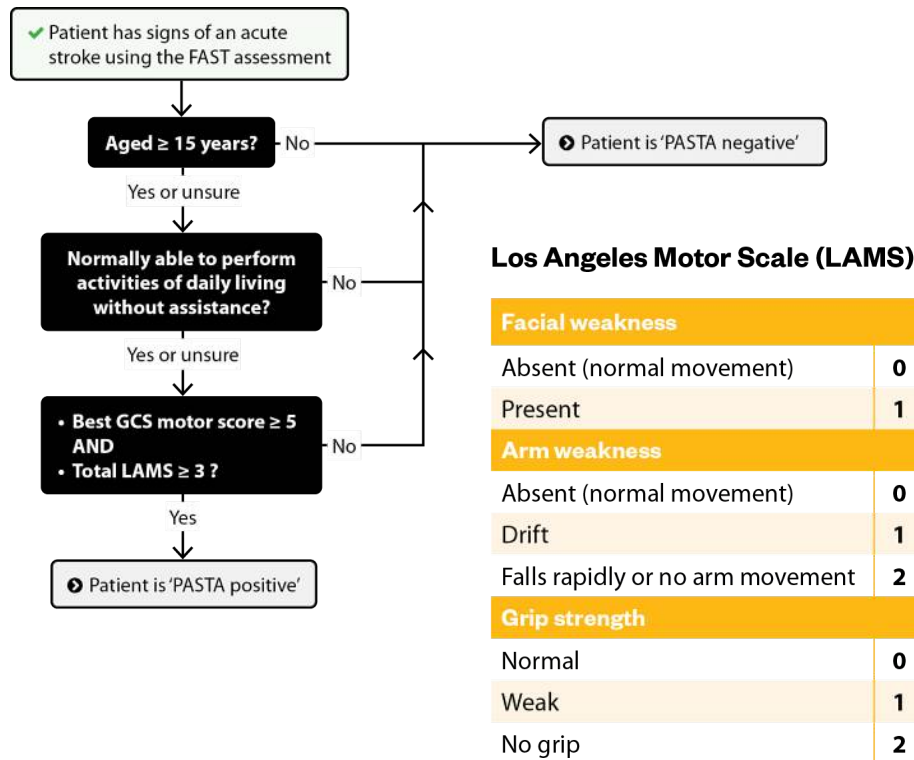
This guideline is for adults with signs of an acute stroke. Seek clinical advice if the patient is a child. Do not use this guideline if:

- ✗ The FAST assessment is normal (negative), or
  - ✗ The patient is hypoglycaemic or has been treated for hypoglycaemia, or
  - ✗ The patient has had a seizure.
- 

- Assess the patient using the FAST assessment to confirm signs of an acute stroke.
- Perform additional assessment using the PASTA tool.
- Transport to a designated stroke hospital without delay if:
  - The patient will arrive in hospital within eight hours of the last known well time, or
  - The patient has woken with signs of a stroke ('wake up stroke'), was symptom free prior to going to sleep, and will arrive in hospital within four hours of waking.
- Gain IV access, preferably an 18G in the antecubital fossa, provided this does not delay extrication and/or transport.

# Stroke

## Pre-hospital acute stroke triage in Aotearoa (PASTA)



## Backup

- Backup is rarely required.
- If an EMT is present, there is no role for backup to be requested solely for IV access.

## Referral and transport

- Provide a clear recommendation for the patient to be transported to an ED by ambulance.
- Designate the patient as status two and transport to a designated stroke hospital, if they can be transported within the time frames outlined above.
- Designate the patient as status three and transport to the most appropriate hospital (preferably a designated stroke hospital, if feasible and safe), if the patient cannot be transported within the time frames outlined above, or if the last known well time cannot be determined.
- Transport a relative with the patient whenever this is feasible and safe as the relative is likely to be required to help provide the history and/or be involved in treatment discussions.

# Stroke

- Transport under lights is not routinely required but should be considered if doing so will result in a clinically significant time saving.
- Alert hospital staff early if the patient is status two and include the following information:
  - FAST results, and
  - PASTA result (positive or negative), and
  - Last known well time.
- Provide a handover while the patient is on the ambulance stretcher and convey the patient direct to the CT scanner if asked to do so, provided this is not associated with significant delay.

## Region specific information/pathways

### Northland

- The designated stroke hospital is Whangārei Hospital.

### Auckland

- The designated stroke hospitals are:
  - North Shore Hospital.
  - Waitakere Hospital.
  - Auckland City Hospital.
  - Middlemore Hospital.
- Phone the Auckland City Hospital on call neurologist on 0800 1 PASTA (0800 172 782) to discuss hospital destination, if the patient:
  - Is PASTA positive, and
  - Is outside the Auckland City Hospital catchment area, and
  - Will arrive in the catchment ED outside 0800-1600 Monday to Friday.
- Do not phone the on call neurologist unless all of the above criteria are met, and do not divert patients from outside the Auckland City Hospital catchment area to Auckland City Hospital, unless told to do so by the on call neurologist.

### Midland

- The designated stroke hospitals are:
  - Thames Hospital.
  - Tauranga Hospital.
  - Waikato Hospital.



# Stroke

- » Whakatane Hospital.
- » Rotorua Hospital.
- » Gisborne Hospital.
- » Taranaki Base Hospital.

## Central/Lower North Island

- The designated stroke hospitals are:
  - » Hawke's Bay Regional Hospital.
  - » Whanganui Hospital.
  - » Palmerston North Hospital.
  - » Wairarapa Hospital.
  - » Hutt Hospital.
  - » Wellington Regional Hospital.
- If the patient is in the Ōtaki area, is PASTA positive, and is approximately equidistant between Palmerston North and Wellington, the patient should be transported to Wellington Regional Hospital.

## South Island

- The designated stroke hospitals are:
  - » Nelson Hospital.
  - » Wairau Hospital.
  - » Greymouth Hospital.
  - » Christchurch Hospital.
  - » Ashburton Hospital.
  - » Timaru Hospital.
  - » Oamaru Hospital.
  - » Dunstan Hospital.
  - » Lakes District Hospital.
  - » Dunedin Hospital.
  - » Southland Hospital.
- If the patient is PASTA positive and approximately equidistant between Ashburton and Christchurch, the patient should be transported to Christchurch Hospital.

## Additional information

## General principles

- A patient is having a stroke until proven otherwise if there are new abnormalities as detected by the FAST assessment.
- Hypoglycaemia can cause signs and symptoms that mimic a stroke, and these may persist for many hours following treatment. Treat the patient using the '[hypoglycaemia](#)' guideline and not this guideline if the patient is hypoglycaemic or has acutely received treatment for hypoglycaemia, even if there are signs of a stroke.
- Seizures can cause signs and symptoms that mimic a stroke during the postictal phase, and these may persist for many hours following the seizure. Treat the patient using the '[seizures](#)' guideline and not this guideline if the patient has had a seizure, even if there are signs of a stroke.
- A patient with a stroke will have signs and symptoms that relate to the part of the brain that has lost blood supply. Most commonly these include any combination of:
  - Unilateral face weakness.
  - Unilateral arm weakness.
  - Unilateral leg weakness.
  - Speech disturbance.
  - Visual disturbance.
- The FAST assessment detects approximately 85% of patients with an ischaemic stroke.
- The FAST assessment can miss posterior circulation strokes, which may present with abrupt onset of double vision, visual field loss, abnormal coordination and/or abnormal balance that is out of proportion to any associated vertigo. If an acute posterior circulation stroke is confidently suspected the patient should be transported to a designated stroke hospital, provided it is feasible and safe to do so.
  - Personnel should seek clinical advice if transport to a designated stroke hospital will bypass another hospital.
  - ED personnel should receive notification prior to arrival, but a stroke physician should not be contacted directly.
- IV access should be obtained if this is easy to achieve. An 18G in the antecubital fossa is preferred because this optimises IV contrast administration during CT imaging. Multiple attempts to gain IV access should not occur because of the subsequent risk of bleeding if fibrinolytic treatment is administered.

## The FAST test

-  **Face:** Look for new onset of unilateral facial weakness. Ask the patient to smile and show their

# Stroke

teeth/gums. If there is new asymmetry the FAST test is positive.

- A Arm:** Look for new onset of unilateral arm weakness. Ask the patient to raise their arms to 90° from the body, with their palms facing upward. Ask the patient to close their eyes and keep their arms raised. Look for inability to raise one arm or for one arm that drifts downward. If there is new weakness in either arm the FAST test is positive.
- S Speech:** Look for new onset of abnormal speech. Ask the patient to repeat a sentence and listen for slurring of words. Ask the patient to name several common objects shown to them and observe any difficulty or inability to name them. If there is new onset of abnormal speech the FAST test is positive.
- T Time:** If the FAST test is positive, determine and record the last known well time. The last known well time is the time that the patient was last seen or known to be symptom free. If the patient has woken up with symptoms, record the last known well time (this will usually be the time they went to sleep) in addition to the time of waking.

## The PASTA tool

- The PASTA tool is designed to help select patients that are most likely to receive reperfusion therapy.

## Helicopter criteria

- Transport by helicopter is indicated if the patient:
  - Is PASTA positive, and
  - Will reach a designated stroke hospital within eight hours of the last known well time, and
  - Helicopter transport will save more than 30 minutes compared with road transport.
- Depending on helicopter availability and assuming a 20 minute scene time, for helicopter criteria to be met the road transport time will usually need to be more than 100 minutes.

## Reperfusion therapy

- The earlier reperfusion therapy is provided for an ischaemic stroke, the more likely the patient is to make a good recovery.
- Delays to reperfusion therapy worsen outcomes and ambulance personnel have an important role in identifying patients with stroke, minimising on scene and transport times, transporting patients to the most appropriate hospital and ensuring timely notification prior to arrival.
- Two forms of reperfusion therapy are available in New Zealand:
  - Fibrinolytic therapy is available at all designated stroke hospitals.

# Stroke

- » Stroke clot retrieval is available at Auckland City Hospital, Wellington Regional Hospital and Christchurch Hospital.
- Patients who are outside the timeframes described in this guideline may still be a candidate for reperfusion therapy. For this reason, unnecessary delays in extrication and/or transport should be minimised for all patients, regardless of their status.
- Stroke clot retrieval is a procedure where the occluded cerebral artery is accessed and the clot removed, thus re-establishing blood flow. It is potentially a very effective treatment for selected patients with an ischaemic stroke. However, do not bypass a designated stroke hospital to transport a patient direct to a hospital with stroke clot retrieval facilities, unless there is an established pathway for this in the area.

## Haemorrhagic stroke

- A patient with a haemorrhagic stroke may present very similarly to a patient with an ischaemic stroke and it is not possible to confidently distinguish between an ischaemic stroke and a haemorrhagic stroke without a CT scan.
- Follow this guideline even if haemorrhagic stroke is suspected.

## Hypertension

- Controlling hypertension, even if severe, may be associated with worsening of outcome. For this reason, treatment should not be provided to control hypertension without seeking clinical advice or being advised to do so by a stroke physician.

## Response and/or transport by helicopter

- Transport by helicopter is rarely required for patients with stroke.
- Transport by helicopter may be indicated if the patient is:
  - » PASTA positive, and
  - » It will save more than 30 minutes compared to road transport
  - » The patient will clearly arrive in a stroke hospital within four hours from onset.
- For criteria to be met, the road transport time will usually need to be more than 90 minutes.

## More Information

# Stroke

## Version history

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**9 June 2025**

1.0.5.3

**Update:** Guideline redeveloped.

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# TIA (Transient Ischaemic Attack)

- Assess the patient using the FAST test.
- Treat using the '[stroke](#)' guideline if the patient has any signs or symptoms of stroke that persist.
- Treat as a transient ischaemic attack (TIA) only if all signs and symptoms have completely resolved.

## Referral and transport

- The patient must be seen by a doctor and this should usually be in an ED. Transport by ambulance may not be required if the patient has access to private transport, providing the patient will not be driving.
- The patient may be seen in primary care (preferably by their own GP) provided primary care staff are spoken to by ambulance personnel and an appointment is confirmed for the same day.

## Additional information

- To have a TIA the patient must have signs or symptoms of a stroke that have completely resolved.
- A patient who has had a TIA is at increased risk of subsequently developing a stroke and this is why the patient must be assessed by a doctor on the same day.
- A patient with a TIA needs a series of investigations that usually includes radiology and this is why most patients should be assessed in an ED.

## More Information

### Version history

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9 June 2025

1.0.5.3

**Update:** Multiple changes.

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9 June 2025

1.0.5.3

**Update:** Multiple changes.

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This guideline is for ICP/CCPs providing inter-hospital transfer of adult patients for stroke clot retrieval (SCR). Seek clinical advice if the patient is a child.

- Receive a handover from hospital staff and commence transport without delay.
- Ensure you are supplied with an ampoule of hydrocortisone by hospital staff, unless this is already available to ambulance personnel.
- If applicable, ensure the alteplase infusion is still running and continue this until it is complete.
- Monitor and record vital signs including GCS every ten minutes.
- Examine the patient regularly for signs of angioedema.
- Phone the receiving clinician shortly after departure and confirm the ETA. Clarify if a second update call is required en route.
- Seek clinical advice for any significant changes in patient condition that are not easily managed.

## Treatment of hypertension for patients who have received fibrinolysis

- If the systolic BP is greater than 180 mmHg or the diastolic BP is greater than 110 mmHg commence treatment with labetalol to maintain a systolic BP of 140-160 mmHg and a diastolic BP below 110 mmHg:
  - Administer a 20 mg bolus IV every five minutes as required, while setting up an infusion.
  - Commence an IV infusion at 1 mg/kg/hour, up to a maximum of 150 mg/hour.
  - Adjust the rate as required by 20 mg/hour every 10 minutes, titrating to effect.
  - Administer additional boluses of 20 mg IV every five minutes as required.
  - If labetalol is contraindicated, or target BP is not met within 30 minutes of commencing labetalol, seek clinical advice to discuss further options.

## Stopping the alteplase infusion

- Stop the alteplase infusion and seek clinical advice if:
  - The systolic BP is greater than 200 mmHg, or
  - The diastolic BP is greater than 120 mmHg, or
  - There are signs of intracranial haemorrhage (for example sudden onset of headache, fall in GCS by more than two points, or sudden worsening of stroke signs), or
  - There are signs of severe bleeding, or
  - The patient develops angioedema.

## Treatment of angioedema

- Stop the alteplase infusion.
- Administer 100 mg of hydrocortisone IV.
- Administer 5 mg of adrenaline nebulised if there is stridor or concern about airway compromise.

## Additional information

### General principles

- Stroke clot retrieval (SCR) is a procedure where the occluded cerebral artery is accessed and the clot removed, thus re-establishing blood flow.
- SCR is also known as percutaneous stroke intervention (PSI), endovascular clot retrieval (ECR) and thrombectomy.
- SCR is a very effective treatment for stroke, particularly if a large clot is blocking a proximal artery, because fibrinolytic therapy in this setting often fails to achieve reperfusion.
- The earlier SCR occurs the better is the patient's outcome. The potential 'treatment window' for clot retrieval from time of symptom onset is much longer than with fibrinolytic therapy, and is up to 24 hours in selected patients.
- For a patient to be selected for SCR, a CT scan is required which shows a proximal occlusion and fibrinolytic therapy will usually be commenced while arrangements are being made for SCR to occur.
- The role of the ICP/CCP in inter-hospital transfer is to ensure rapid and safe transfer, while being prepared to treat any complications that may occur.

### Patient handover at the referring hospital

- Receive a handover from personnel at the referring hospital, including:
  - The history and any relevant additional information, for example allergies.
  - Current therapy including whether an alteplase infusion is running and if so, the total dose to be administered, the rate of infusion and whether additional bag/syringe changes are required en route.
  - The patient's vital signs.
  - The name and phone number of the receiving clinician.
  - Any relevant medical records.



- Minimise delays during handover, patient preparation and leaving the hospital. Personnel should aim to leave the hospital within 15 minutes of arrival.
- Do not insert additional IV lines unless absolutely necessary.

## Fibrinolytic therapy

- If alteplase is being administered this requires an IV infusion.
  - Ideally the infusion should be set up prior to leaving the referring hospital so that no additional changing of bags or syringes is required en route.
  - Ensure the rate of infusion is known in case the infusion has to be stopped and restarted during transfer.

## Hypertension

- Hypertension increases the risk of intracranial haemorrhage, particularly when fibrinolytic therapy has been administered.
- Do not use GTN SL in this setting because it may be associated with a rapid fall in BP which may worsen brain ischaemia.
- The treatment of hypertension for patients who have not received fibrinolysis should be patient-specific, by seeking clinical advice.

## Angioedema

- Angioedema is an uncommon complication of fibrinolytic therapy, but may occur in approximately 1% of patients, particularly those treated with alteplase.
- Angioedema may occur during or up to two hours after the infusion, and may be hemilingual.
- Do not administer adrenaline IM because it has no useful role in treating angioedema and risks haemorrhage following fibrinolytic therapy.
- Do not administer adrenaline IV unless there are convincing signs of anaphylaxis, because adrenaline via the IV route has no role in treating angioedema and risks causing hypertension which increases the risk of intracranial haemorrhage following fibrinolytic therapy.
- It is rare for angioedema to cause life-threatening airway obstruction, but if this occurs it should be treated using the same approach as in other patients, noting that:
  - Intubation with RSI must be approached with extreme caution, as intubation is likely to be difficult.

» A surgical approach to the airway is likely to be associated with significant bleeding.

## More Information

### Version history

|                                                                                                          |         |
|----------------------------------------------------------------------------------------------------------|---------|
| <b>16 June 2026</b>                                                                                      | 1.1.0.1 |
| <b>Update:</b> Hypertension management updated.                                                          |         |
| <b>11 September 2023</b>                                                                                 | 1.0.4   |
| <b>Update:</b> Removal of GTN patches, promethazine and metoprolol IV. Dosing for labetalol had changed. |         |

# Headache

This guideline is for adults with primary headache. Do not use this guideline if the headache is thought to be secondary to another underlying cause such as meningitis or concussion.

- Provide a quiet, cool, dark environment if possible.
- Assess the patient, including an assessment of the features contained within the flag table below.
- Administer droperidol 1.25 mg IV/IM if headache is severe. Consider reducing the dose to 0.625 mg if a caution is present.
- Administer ondansetron as per the '[nausea and vomiting](#)' guideline if the patient has significant nausea and/or vomiting.
- Administer paracetamol and/or ibuprofen PO as per the '[analgesia](#)' guideline.
- Do not administer an opiate or methoxyflurane.
- Consider administering 1 litre of 0.9% sodium chloride IV over 20 minutes if the patient has significant nausea and/or vomiting or has signs of dehydration.
- Administer 15 litres/minute of oxygen via reservoir mask for 15-20 minutes if the patient has a provisional diagnosis of cluster headache.

## Red flags

**Should be seen in ED, usually via ambulance**

# Headache

- Headache or neck pain following recent neck manipulation.
- Neck pain or neck stiffness.
- Sudden onset of severe headache.
- Temperature > 38°C (in the absence of influenza symptoms).
- Persistent vomiting following administration of antiemetics.
- Focal neurological signs.
- New onset of altered level of consciousness, including a history of altered level of consciousness with the onset of the headache.
- Worsening headache following recent trauma to the head.
- Taking an anticoagulant, dual antiplatelet therapy or has a known bleeding disorder.
- Signs of temporal arteritis.
- Suspected pre-eclampsia.
- Previous history of intracranial bleeding.
- Family history of cerebral vascular abnormalities.
- Onset during sexual activity or exercise.
- Headache associated with seizure.
- Suspected mastoiditis.
- Inability to mobilise in a manner that is normal for the patient.
- Lack of significant improvement following treatment.

## Backup

- Backup is rarely required, however ECP backup should be considered to enable the provision of additional treatments.

## Referral and transport

# Headache

- If one or more red flags are present the patient must be given a clear recommendation to be assessed by a doctor within two hours and should usually be given a clear recommendation to be transported to ED, usually by ambulance.
- If the patient has one or more red flags and is being referred to primary care:
  - The patient's anticipated clinical needs must be able to be safely met in primary care, and
  - A clinician within primary care must be contacted directly by ambulance personnel who agrees to accept the patient, and
  - Safe transport (if required) must be available.
- If no red flags are present, the patient is usually suitable to be given a clear recommendation to remain in the community.
- If no red flags are present and they are clearly improving, the patient may be administered any combination of 0.9% sodium chloride IV, ibuprofen, paracetamol, ondansetron or droperidol without being transported to a medical facility.

## Worsening advice

- All patients (and their family/whānau/carers) who are not immediately referred or transported to a medical facility, must be provided with the following worsening advice, preferably in writing:
  - Be seen in primary care or phone Healthline on 0800611 116 if their symptoms are not improving within 12 hours.
  - Remain in the care of an adult and do not drive a vehicle or operate machinery for 12 hours, if droperidol has been administered.
  - Call 111 for an ambulance if they develop:
    - Persistent vomiting.
    - A severe worsening of the headache.
    - Confusion or inability to be easily woken.
    - Fever.
    - Seizure.
    - Inability to mobilise if usually able to do so.

## Additional information

### Primary headache

- Examples of primary headache include tension headache, migraine, and cluster headache.

# Headache

- The management goal is not to completely resolve the headache, but to reduce pain, settle nausea and/or vomiting, rehydrate, and enable the patient to sleep/rest.
- There is no good evidence that administration of 0.9% sodium chloride IV is an effective treatment, unless the patient has nausea and/or vomiting or has signs of dehydration.
- For patients with a provisional diagnosis of cluster headache, a short period of inhaling high concentration oxygen may resolve the headache by causing cerebral vasoconstriction. This is why 15 litres/minute of oxygen should be administered via a reservoir mask for 15-20 minutes.

## Red flags

- Vertebral artery dissection may present with sudden onset of neck pain and/or headache which may follow injury or neck manipulation.
- Meningitis may present with headache, fever, neck stiffness, photophobia, nausea, and vomiting. The symptoms of meningitis are similar to those of migraine, except that the pain from migraine:
  - Usually comes on over an hour or two, whereas the pain from meningitis usually comes on more slowly.
  - Is usually throbbing, whereas the pain from meningitis is usually more constant.
  - Usually lasts only for a few hours, whereas the pain from meningitis persists.
- Subarachnoid haemorrhage may present with sudden onset of severe headache (thunderclap headache) and/or headache associated with neck stiffness.
- A temperature of >38 degrees in the absence of a viral respiratory illness may be due to meningitis or an intracerebral infection such as encephalitis.
- Persistent vomiting that continues following the administration of anti-emetics may be indicative of intracranial pathology.
- Intracerebral haemorrhage usually presents with sudden onset of severe headache and focal neurological signs. The patient will also have a falling level of consciousness if the intracerebral haemorrhage is severe.
- Anticoagulants include warfarin, dabigatran, and rivaroxaban. Dual antiplatelet therapy consists of the patient taking more than one antiplatelet medicine, such as aspirin and ticagrelor.
- Temporal arteritis (also known as giant cell arteritis) is an inflammatory condition affecting the blood vessels supplying the temporal area of the head. It is an emergency because untreated it can lead to blindness. It most commonly occurs in patients over the age of 60 years and may present with any combination of:
  - Headache
  - Fever.

# Headache

- » Jaw pain (which may get worse with chewing).
  - » Altered vision.
  - » Scalp sensitivity.
  - » Stiff aching joints.
- Headache during pregnancy is a potential sign of pre-eclampsia. If this is suspected, refer to the ['eclampsia'](#) guideline.
  - Onset of headache during sexual activity or exercise may be associated with subarachnoid haemorrhage.
  - Onset of severe headache during exercise (particularly strenuous exercise or heavy lifting) may be associated with subarachnoid haemorrhage or vertebral artery dissection.
  - Signs of suspected mastoiditis include erythema and tenderness over the mastoid bone.

## Analgesia

- Opiates should not be administered for primary headache but may be administered if the headache has a secondary cause such as presumed subarachnoid haemorrhage or intracerebral haemorrhage.
- Methoxyflurane does not have a role in treating primary headache syndromes.
- Personnel should recommend the patient takes other medicines such as triptans (for example, sumatriptan or rizatriptan) that have been prescribed to the patient for headaches if the patient has not taken these.

## More Information

### Version history

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9 June 2025

1.0.5.3

**Update:** Guideline redeveloped.

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# Vertigo

This guideline is for adult patients with vertigo.

- Assess the patient, including an assessment of the features contained within the red flags.
- Administer droperidol 1.25 mg IV/IM. Consider reducing the dose to 0.625 mg if a caution is present.
- Administer ondansetron as per the ['nausea and vomiting'](#) guideline if the patient has significant nausea and/or vomiting.
- Consider administering 1 litre of 0.9% sodium chloride IV over 20 minutes if the patient has significant nausea and/or vomiting, or has signs of dehydration.
- Perform an Epley manoeuvre if the patient has known or suspected BPPV.

## Red flags

### Should be seen in ED, usually via ambulance

- Signs of stroke.
- Headache.
- Unable to walk unaided.
- Neck pain.
- Visual disturbance.
- Abnormal coordination during the finger-nose test.
- Horizontal nystagmus that persists for more than 10 seconds with the head still.
- Rotational or vertical nystagmus.
- Altered level of consciousness.
- Abnormal vital signs.
- History of recent trauma, especially head or neck injury.
- Symptoms that do not improve when the head is still.
- Acute loss of hearing.



# Vertigo

## Backup

- Backup is rarely required, however ECP backup should be considered to enable the provision of additional treatments.

## Referral and transport

- If one or more red flags are present the patient must be given a clear recommendation to be assessed by a doctor within two hours, and should usually be given a clear recommendation to be transported to ED, usually by ambulance.
- If the patient has one or more red flags and is being referred to primary care:
  - The patient's anticipated clinical needs must be able to be safely met in primary care, and
  - A clinician within primary care must be contacted directly by ambulance personnel who agrees to accept the patient, and
  - Safe transport (if required) must be available.
- If no red flags are present, the patient is usually suitable to be given a clear recommendation to remain in the community.
- If no red flags are present and they are clearly improving, the patient may be administered any combination of 0.9% sodium chloride IV, droperidol or ondansetron, without being transported to a medical facility.

## Worsening advice

- All patients (and their family/whānau/carers) who are not immediately referred or transported to a medical facility, must be provided with the following worsening advice, preferably in writing:
  - Be seen in primary care or phone Healthline on 0800611 116 if their symptoms are not improving within 24 hours.
  - Remain in the care of an adult and do not drive a vehicle or operate machinery for 12 hours, if droperidol has been administered.
  - Call 111 for an ambulance if they develop:
    - Headache or neck pain.
    - Limb weakness.
    - Confusion or inability to be easily woken.
    - Loss of normal hearing or normal sight.
    - Inability to mobilise unaided.

# Vertigo

## Additional information

### General principles

- Vertigo is the false sensation that the body or its surroundings are moving or spinning and is usually accompanied by nausea and loss of balance. It is important to differentiate this from a feeling of light headedness.
- The three main causes of vertigo are cerebellar stroke, benign paroxysmal positional vertigo (BPPV) and vestibular neuritis. Cerebellar stroke is less common but much more serious and can easily be missed. BPPV and vestibular neuritis are both common and benign.

### Cerebellar stroke

- Cerebellar stroke is usually ischaemic and occurs most commonly in patients aged over 50 years.
- The symptoms of cerebellar stroke usually come on suddenly (over a few minutes) and are associated with:
  - Vertigo that is not altered by head position.
  - Nausea and vomiting.
  - Loss of coordination with an abnormal finger-nose test.
  - Loss of balance with an inability to walk unaided.
- Hearing is not affected and tinnitus is not a feature.
- Cerebellar stroke may be associated with vertebral artery dissection and this may be associated with neck pain and/or recent head or neck injury.

### Benign paroxysmal positional vertigo (BPPV)

- BPPV is caused by calcium particles (called otoliths) which become dislodged and float to abnormal positions in the inner ear.
- The symptoms of BPPV usually come on slowly over 12-24 hours and are associated with:
  - Brief but very intense periods of vertigo that occur with changes in head position, for example sitting up or rolling over. Vertigo usually resolves over 30-60 seconds when the head is kept still.
  - Nausea is common but vomiting is uncommon.
  - Symptoms may be worse in the morning and may be worse when the head is tilted to a particular side.
- Hearing is not affected and tinnitus is not a feature.

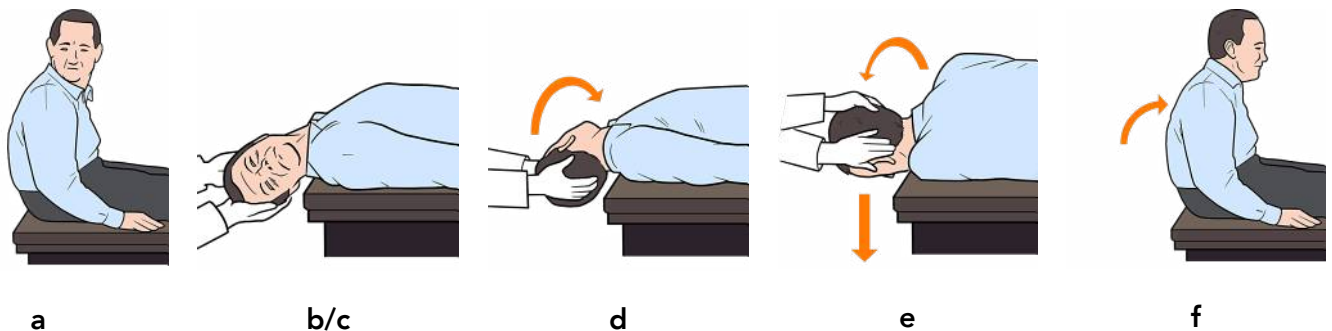
# Vertigo

- Mildly abnormal balance may be present, but the finger-nose test is normal.

## The Epley manoeuvre

- The Epley manoeuvre aims to reposition the calcium particles in the inner ear, and resolve symptoms of vertigo in patients with BPPV.
- During the manoeuvre the patient is likely to require coaching and reassurance. Vomiting is common, and preparation should be made for this. Movements between each position are rapid and the patient is likely to experience an initial worsening of their vertigo and associated symptoms.
- There are several variations, but the following is recommended:
  - Sit the patient upright on a bed or stretcher and rotate their head to face one side, preferably toward the side that makes symptoms worse.
  - Assist the patient to lie back, keeping their head turned. Recline their head, using a pillow under their shoulders or with their head hanging off the end of the bed/stretcher, and their ear parallel with the floor.
  - Hold the patient's head in this position for 60 seconds.
  - Quickly turn the patient's head to face the opposite side and hold this position for 60 seconds.
  - Turn the head further, so that the patient is facing downward (this may require the patient to move their body to accommodate this). Hold this position for 60 seconds.
  - Assist the patient into a sitting position and rotate their head forward.
- Repeat the Epley manoeuvre once if the symptoms do not resolve.

## The Epley manoeuvre



## Vestibular neuritis

- Vestibular neuritis is caused by an inflammation of the vestibular nerve.

# Vertigo

- The most common cause of vestibular neuritis is a viral infection.
- The symptoms of vestibular neuritis usually come on over 12-24 hours and are associated with:
  - Vertigo that is altered by position, but not as dramatically as BPPV.
  - Nausea and vomiting.
  - Reduced hearing (especially on one side) and/or tinnitus (buzzing noise).
- Mildly abnormal balance may be present when walking, but the finger-nose test is normal.

## Nystagmus

- Nystagmus is involuntary, rapid and repeated small movements of the eyes.
- To look for nystagmus, ask the patient to keep their head still and watch your finger while you move it from one side of their field of vision to the other. Pause your finger for ten seconds and observe the patient's eyes.
- Nystagmus that stops within ten seconds of the patient focusing on your finger is usually due to a peripheral benign cause such as BPPV or vestibular neuritis.
- Nystagmus that persists for longer than ten seconds when the patient is focusing on your finger is usually due to a central cause such as cerebellar stroke.

## The finger-nose test

- Ask the patient to put the tip of their index finger on their nose.
- Hold your finger approximately 30 cm away and ask the patient to touch your finger.
- Slowly move your finger and ask them to alternately touch their nose, then your finger, then their nose etc.
- Test both sides.
- A patient with normal coordination will successfully do this. A patient with abnormal coordination will miss or overshoot.

## More Information

## Version history

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**9 June 2025**

1.0.5.3

**Update:** Guideline redeveloped.

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## Initial assessment

- A child's general appearance is an important consideration when determining how severe their illness or injury is, the need for treatment and the response to therapy.
- The paediatric assessment triangle (PAT) is a form of assessment that can be used to help determine the severity of illness or injury. The PAT involves an assessment of activity, breathing and circulation and is performed at the same time as the primary survey.

### Activity

Movement, interaction, tone

**Abnormal:** Inactive, lethargic, abnormal or absent cry or speech, failure to interact with people or objects, floppy.

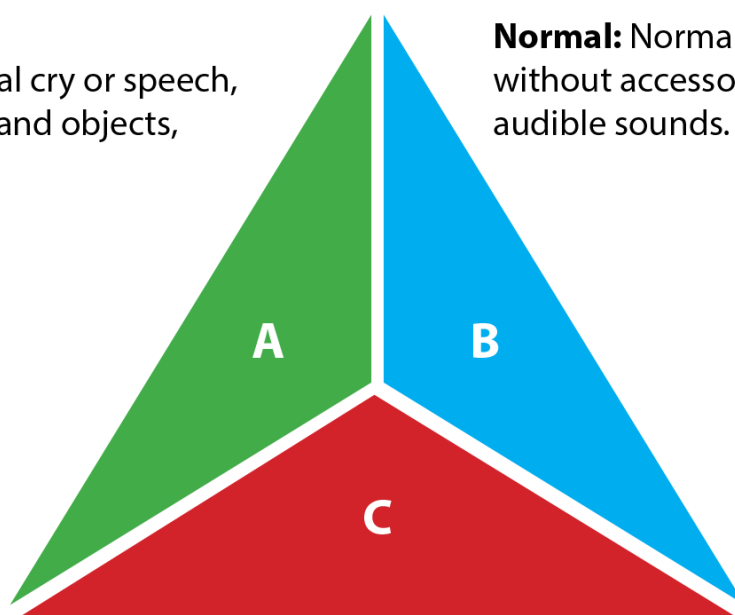
**Normal:** Active, normal cry or speech, interacts with people and objects, good muscle tone.

### Breathing

Respiratory rate, work of breathing

**Abnormal:** Tachypnoea, nasal flaring, indrawing, use of accessory muscles, grunting.

**Normal:** Normal regular breathing without accessory muscle use or audible sounds.



### Circulation

Heart rate, perfusion

**Abnormal:** Tachycardia, mottled skin, pale, cold, slow capillary refill time.

**Normal:** Normal heart rate, normal skin colour, warm, fast capillary refill time.

- The more abnormal the PAT is, the more severe the illness or injury is:
  - If the PAT is normal, the child is unlikely to have severe illness or injury.
  - If one segment of the PAT is abnormal, the child is showing signs of an important illness or injury.
  - If two segments of the PAT are abnormal, the child is seriously ill or injured and is likely to be status two.
  - If three segments of the PAT are abnormal, the child is severely ill or injured and is likely to be status one.
- Document the aspects of the PAT on the ePRF, but do not refer directly to the PAT as a title. This is because most hospital personnel are familiar with the individual aspects of the PAT, but not the actual title.

## Interaction and activity

- A child will usually have reduced interaction and activity if very unwell or badly injured.
- Signs of reduced activity include:
  - Lethargy.
  - Abnormal cry.
  - Failure to interact with people or objects.
  - Reduced tone or floppiness.

## The respiratory system

- Children rely heavily on the rate of breathing to compensate for respiratory difficulty. This is because they are unable to significantly increase the depth of respiration due to restriction of downward movement of the diaphragm against abdominal organs.
- Tachypnoea is an early sign of respiratory distress.
- Children have narrower airways with higher resistance than adults.
- Children have a higher resting respiratory rate than adults and higher oxygen consumption relative to size.
- In children, the diaphragm is the dominant respiratory muscle. They do not move their chest wall significantly during normal breathing and reliance on the diaphragm makes them more prone to respiratory fatigue.
- Children have lower functional residual capacity (FRC) than adults. This results in lower oxygen reserves and makes them more prone to hypoxia.

- Children have very compliant (elastic) ribs. This means that an increase in work of breathing will usually cause indrawing or retraction.
- Signs of respiratory distress include:
  - Tachypnoea.
  - Nasal flaring.
  - Grunting.
  - Weak cry.
  - Indrawing or retraction. Look for this in the supraclavicular, tracheal, intercostal and substernal sites.
  - Accessory muscle use.
  - Stridor.
  - Abnormal positioning for example sitting forward, the sniffing position, the tripod position or refusing to lie down.
  - Head bobbing.
- Hypoxia in children causes tachycardia, agitation, drowsiness and pallor. Cyanosis is a late sign.
- If hypoxia is very severe the heart rate will begin to fall. This is a very late sign of imminent cardiac arrest.

## The cardiovascular system

- Children have a higher blood volume and cardiac output per kilogram, a relatively fixed stroke volume and a higher resting heart rate than adults.
- Children have a significant capacity for vasoconstriction in the setting of falling cardiac output. This ability to vasoconstrict means that a fall in blood pressure is a very late sign of shock. However, the trend of blood pressure and pulse pressure over time are useful.
- The signs of shock in children are:
  - Tachycardia.
  - Tachypnoea.
  - Vasoconstriction with prolonged capillary refill time. This will also often produce mottled skin.
  - Reduced activity and interaction.
- Although children have a higher blood volume per kilogram, they have a lower total blood volume. This means that what may seem like a small amount of blood loss may represent a significant proportion of blood volume. For example, small children can become shocked from bleeding within their skull or from their scalp.



## Traumatic brain injury (TBI)

- Children have a large and heavy head relative to their body size and are more prone to TBI than adults.
- When unconscious, the upper airway tends to get obstructed by a relatively large, flaccid tongue, or kinked because of neck flexion induced by the prominent occiput.
- GCS scoring is more difficult in small children. Focus on the motor score as it is the most important component.

## Skeletal injury

- Children have more pliant and flexible bones than adults and are therefore subject to fewer fractures.
- Internal organ injuries in the absence of fractures of the overlying bones are more common than in adults. For example the rib cage is very compliant, so there may be internal injuries in the absence of rib fractures.

## Temperature control

- Children have a less mature thermoregulatory (temperature control) mechanism and a higher surface area to mass ratio compared to adults. This makes heat loss and hypothermia more common.
- Tympanic thermometers may be unreliable in small children if the probe will not adequately fit their external auditory meatus and an axillary measurement is preferred if a child is under six months of age.
- Temperature should be interpreted and managed in the context of the patient's age, illness and clinical picture, noting that infants may not be able to generate an elevated temperature in response to infection.

## Neonates

- There should be a very low threshold for transport to a medical facility by ambulance for neonates.
- Consideration must be given to the larger clinical picture (for example, anxious parents, poor feeding, dry nappies and change in behaviour) rather than relying on vital signs.
- Have a low threshold for seeking clinical advice.

## More Information

## Version history

|                                  |       |
|----------------------------------|-------|
| 28 October 2024                  | 1.0.5 |
| <b>Update:</b> Multiple changes. |       |

- For children, the doses of drugs, defibrillation energy and fluid volumes are based on weight (when this is known).
- If the weight is not known, it can be estimated from the child's age using formulae, noting that the formulae are a guide only and tend to underestimate true weight.
- Children may require a different sized SGA, ETT or length at lips than that predicted by the formulae.

## Estimated weight (kg)

|                  |                                      |
|------------------|--------------------------------------|
| Under 1 year old | 5                                    |
| 1 - 10 years     | $(\text{Age in years} + 4) \times 2$ |
| 11 - 14 years    | $\text{Age in years} \times 3$       |

## Cuffed endotracheal tube (ETT) size (mm)

|                   |                                      |
|-------------------|--------------------------------------|
| Newborn to 1 year | 3 - 4                                |
| 1 year and over   | $(\text{Age in years} \div 4) + 3.5$ |

## Endotracheal tube length at lips (cm)

|                 |                                     |
|-----------------|-------------------------------------|
| Newborn         | 6 + weight in kg                    |
| Under 1 year    | ETT size x 3                        |
| 1 year and over | $(\text{Age in years} \div 2) + 12$ |

## Defibrillation energy

|                        |        |
|------------------------|--------|
| Initial and subsequent | 4 J/kg |
|------------------------|--------|

## Paediatric drug doses

- The following pages contain tables of paediatric drug doses. They are based on rounding the child's weight up to 5, 10, 20, 30 or 40 kg. For example, a child known or estimated to weigh between 11 and 20 kg receives a dose based on 20 kg, and a child known or estimated to weigh between 31 and 40 kg receives a dose based on 40 kg.
- Children have a higher proportion of body water than adults and require higher doses of most drugs per kg, to achieve the same effect.
- Children weighing greater than 40 kg can usually be administered adult doses, but clinical judgement must be used if the dose may be potentially harmful, for example when administering paracetamol. If

there is any doubt, a paediatric dose should be administered.

- The tables do not incorporate all administration information and should be read in conjunction with the information within the '[medicines](#)' section of the comprehensive edition.
- The tables indicate the calculated dose of each drug for a given rounded weight, the concentration of drug that should be drawn up, and the volume that should be administered from that solution.
- Drug doses for rapid sequence intubation (RSI) are not contained within these tables, but are contained within the '[RSI](#)' guideline.
- Some of the doses within the tables differ slightly from the internationally recommended dose. This is to ensure a practical approach, and to help reduce administration errors. Where there are differences between the dose in the tables and the internationally recommended dose, these differences are not clinically significant.
- The usual recommended defibrillation energy for a child is 4 J/kg, but this may not be available for all defibrillators. If the recommended defibrillation energy level is not available, select the next highest level.

## Drug dilution and administration

### 1. Adrenaline 1:10,000 (0.1 mg/ml)

- Draw up 1 ml of adrenaline from a 1 mg/ml ampoule.
- Dilute to a total volume of 10 ml using 0.9% sodium chloride.
- Administer the volume from the tables as a IV bolus.

### 2. Adrenaline IV 1:1,000,000 (0.001 mg/ml)

- Use a 1 litre bag of 0.9% sodium chloride.
- Add 1 ml of adrenaline from a 1 mg/ml ampoule.
- Shake well and label.
- Administer the volume from the tables as a IV bolus.
- An IV infusion may be administered if aged 5 years or older.

### 3. Cefazolin IV 100 mg/ml

- Add approximately 4 ml of 0.9% sodium chloride to a 1 g ampoule of cefazolin powder and shake until dissolved.

- Draw up the contents of the ampoule.
- Dilute to a total volume of 10 ml using 0.9% sodium chloride.
- Discard unrequired drug and administer the volume from the tables as a slow IV push over 1-2 minutes, preferably into a running line.

#### 4. Ceftriaxone IV 200 mg/ml

- Add approximately 4ml of 0.9% sodium chloride to a 2 g ampoule of ceftriaxone powder and shake until dissolved.
- Draw up the contents of the ampoule.
- Dilute to a total volume of 10 ml using 0.9% sodium chloride.
- Discard unrequired drug and administer the volume from the tables as a slow IV push over 1-2 minutes, preferably into a running line.

#### 5. Ceftriaxone IM 400 mg/ml

- Add 4 ml of 0.9% sodium chloride to a 2 g ampoule of ceftriaxone powder and shake until dissolved. The total volume will be 5 ml.
- Draw up 2.5 ml in each of two syringes.
- Discard unrequired drug volume and administer one syringe into each lateral thigh. If this site is not suitable use each lateral upper arm.

#### 6. Fentanyl IV 10 mcg/ml (weight rounded up to 30 kg or more)

- Draw up 2 ml of fentanyl from a 100 mcg/2 ml ampoule.
- Dilute to a total volume of 10 ml using 0.9% sodium chloride.
- Administer the volume from the tables as an IV bolus.

#### 7. Fentanyl IV 1 mcg/ml (weight rounded up to 20 kg or less)

- Draw up 2 ml of fentanyl from a 100 mcg/2 ml ampoule.
- Place in a 100 ml bag of 5% glucose.
- Shake well and label.
- Administer the volume from the tables as an IV bolus.

## 8. Fentanyl and ketamine IV for post intubation sedation

- Draw up 200 mcg of fentanyl (two 100 mcg/2 ml ampoules) and 200 mg of ketamine (one 200 mg/2 ml ampoule), in the same syringe.
- Dilute to a total volume of 20 ml. This solution contains 10 mcg/ml of fentanyl and 10 mg/ml of ketamine.
- Administer the volume from the tables as an IV bolus.

## 9. Hydrocortisone IV 10 mg/ml

- Draw up 100 mg of hydrocortisone from a 100 mg/2 ml vial.
- Dilute to a total volume of 10 ml using 0.9% sodium chloride.
- Administer the volume from the tables IV over 1-2 minutes.

## 10. Ketamine IV for analgesia 1 mg/ml

- Draw up 1 ml of ketamine from a 200 mg/2 ml ampoule.
- Place into a 100 ml bag of 5% glucose. This solution contains 1 mg/ml.
- Draw up the volume from the tables and dilute this further to a total volume of 20 ml using 0.9% sodium chloride.
- Administer IV over approximately 15 minutes.

## 11. Ketamine IV for dissociation 10 mg/ml

- Draw up 1 ml of ketamine from a 200 mg/2 ml ampoule.
- Dilute to a total volume of 10 ml using 0.9% sodium chloride.
- Administer the volume from the tables as an IV bolus.

## 12. Levetiracetam

- Draw up the volume described in the tables from ampoule(s) containing 500 mg/5 ml.
- Administer IV over 5 minutes, preferably into a running line.

## 13. Magnesium IV 1 mmol/ml

- Draw up 5 ml of magnesium from a 10 mmol/5 ml ampoule.
- Dilute to a total volume of 10 ml using 0.9% sodium chloride.

- Discard unrequired drug and administer the volume from the tables IV over approximately 15 minutes, preferably into a running line.

## 14. Metaraminol IV 0.1 mg/ml

- Draw up 1 ml of metaraminol from a 10 mg/1 ml ampoule.
- Place into a 100 ml bag of 5% glucose. This solution contains 0.1 mg/ml.
- Shake well and label.
- Administer the volume from the tables as an IV bolus.

## 15. Midazolam IV 1 mg/ml

- Draw up 1 ml of midazolam from a 15 mg/3 ml ampoule.
- Dilute to a total volume of 5 ml using 0.9% sodium chloride.
- Discard unrequired drug and administer the volume from the tables as an IV bolus.

## 16. Morphine IV 1 mg/ml

- Draw up 1 ml of morphine from a 10 mg/1 ml ampoule.
- Dilute to a total volume of 10 ml using 0.9% sodium chloride.
- Administer the volume from the tables as an IV bolus.

## 17. Naloxone IV 0.1 mg/ml

- Draw up 1 ml of naloxone from a 0.4 mg/1 ml ampoule.
- Dilute to a total volume of 4 ml using 0.9% sodium chloride.
- Administer the volume from the tables as an IV bolus.

## 5 kg / less than one year

| Cardiac arrest        | Dose    | Volume             | Dilution number |
|-----------------------|---------|--------------------|-----------------|
| Adrenaline IV         | 0.05 mg | 0.5 ml (1:10,000)  | 1               |
| Amiodarone IV         | 25 mg   | 0.5 ml (undiluted) | -               |
| Manual defibrillation | 20-25   | Joules             | -               |

| LMA                                    | Size 1 (< 5 kg) | Cuff inflation 4 ml | -               |
|----------------------------------------|-----------------|---------------------|-----------------|
| ETT (cuffed)                           | Size 3          | 9 cm length at lips | -               |
| Other drugs                            |                 |                     | Dilution number |
| Adrenaline IV (not cardiac arrest)     | 0.001 mg        | 1 ml (1:1,000,000)  | 2               |
| Adrenaline IM                          | 0.05 mg         | 0.5 ml (1:10,000)   | 1               |
| Adrenaline IN (epistaxis)              | -               | -                   | -               |
| Cefazolin IV                           | 150 mg          | 1.5 ml (100 mg/ml)  | 3               |
| Ceftriaxone IV                         | 300 mg          | 1.5 ml (200 mg/ml)  | 4               |
| Ceftriaxone IM                         | 300 mg          | 0.75 ml (400 mg/ml) | 5               |
| Fentanyl IV (analgesia)                | 1-5 mcg         | 1-5 ml (1 mcg/ml)   | 7               |
| Fentanyl IM                            | 5 mcg           | 0.1 ml (undiluted)  | -               |
| Fentanyl IN                            | 5 mcg           | 0.1 ml (undiluted)  | -               |
| Fentanyl/ketamine IV (post intubation) | 5 mcg/5 mg      | 0.5 ml              | 8               |
| Glucagon IM                            | 0.5 mg          | 0.5 ml (undiluted)  | -               |
| 10% glucose IV                         | 2 ml/kg         | 10 ml               | -               |
| Hydrocortisone IV                      | 10 mg           | 1 ml (10 mg/ml)     | 9               |
| Hydrocortisone IM                      | 10 mg           | 0.2 ml (undiluted)  | -               |
| Ibuprofen liquid PO                    | 50 mg           | 2.5 ml (20 mg/ml)   | -               |
| Ibuprofen tablet PO                    | -               | -                   | -               |
| Ketamine IV (analgesia)                | -               | -                   | -               |
| Ketamine IV (dissociation)             | -               | -                   | -               |
| Ketamine IM/PO (analgesia)             | -               | -                   | -               |
| Ketamine IM (dissociation)             | -               | -                   | -               |
| Levetiracetam IV                       | 250 mg          | 2.5 ml (100 mg/ml)  | 12              |
| 1% lignocaine IO                       | 5 mg            | 0.5 ml (undiluted)  | -               |



|                                             |                |                      |    |
|---------------------------------------------|----------------|----------------------|----|
| 1% lignocaine SC                            | 15 mg (max)    | 1.5 ml (max)         | -  |
| Loratadine PO                               | -              | -                    | -  |
| Magnesium IV                                | -              | -                    | -  |
| Metaraminol IV                              | 0.05-0.1 mg    | 0.5-1 ml (0.1 mg/ml) | 14 |
| Midazolam IV (seizures)                     | 0.5 mg         | 0.5 ml (1 mg/ml)     | 15 |
| Midazolam IM (seizures)                     | 1 mg           | 0.2 ml (undiluted)   | -  |
| Morphine IV                                 | 0.25 mg        | 0.25 ml (1 mg/ml)    | 16 |
| Morphine IM/SC                              | 1 mg           | 0.1 ml (undiluted)   | -  |
| Naloxone IV                                 | 0.05 mg        | 0.5 ml (0.1 mg/ml)   | 17 |
| Naloxone IM                                 | 0.1 mg         | 0.25 ml (undiluted)  | -  |
| Ondansetron IV                              | -              | -                    | -  |
| Ondansetron IM                              | -              | -                    | -  |
| Paracetamol liquid PO                       | 75 mg          | 1.5 ml (50 mg/ml)    | -  |
| Paracetamol tablet PO                       | -              | -                    | -  |
| Prednisolone PO                             | 5 mg           | 1 ml (5 mg/ml)       | -  |
| Prednisone PO                               | -              | -                    | -  |
| Rocuronium IV                               | 5 mg           | 0.5 ml (undiluted)   | -  |
| 0.75% ropivacaine SC                        | 11.25 mg (max) | 1.5 ml (max)         | -  |
| 0.9% sodium chloride IV                     | 20 ml/kg       | 100 ml               | -  |
| 8.4% sodium bicarbonate                     | 10 mmol        | 10 ml (1 mmol/ml)    | -  |
| Tranexamic acid IN (epistaxis)              | -              | -                    | -  |
| Tranexamic acid IV (non-traumatic bleeding) | 100 mg         | 1 ml (undiluted)     | -  |
| Tranexamic acid IV (bleeding from trauma)   | 200 mg         | 2 ml (undiluted)     | -  |

## 10 kg / 1 year

| Cardiac arrest                         | Dose              | Volume               | Dilution number |
|----------------------------------------|-------------------|----------------------|-----------------|
| Adrenaline IV                          | 0.1 mg            | 1 ml (1:10,000)      | 1               |
| Amiodarone IV                          | 50 mg             | 1 ml (undiluted)     | -               |
| Manual defibrillation                  | 40-50             | Joules               | -               |
| LMA                                    | Size 2 (10-20 kg) | Cuff inflation 10 ml | -               |
| ETT (cuffed)                           | Size 3 or 4       | 12 cm length at lips | -               |
| Other drugs                            |                   |                      | Dilution number |
| Adrenaline IV (not cardiac arrest)     | 0.002 mg          | 2 ml (1:1,000,000)   | 2               |
| Adrenaline IM                          | 0.1 mg            | 0.1 ml (undiluted)   | -               |
| Adrenaline IN (epistaxis)              | -                 | -                    | -               |
| Cefazolin IV                           | 250 mg            | 2.5 ml (100 mg/ml)   | 3               |
| Ceftriaxone IV                         | 500 mg            | 2.5 ml (200 mg/ml)   | 4               |
| Ceftriaxone IM                         | 500 mg            | 1.25 ml (400 mg/ml)  | 5               |
| Fentanyl IV (analgesia)                | 2-10 mcg          | 2-10 ml (1 mcg/ml)   | 7               |
| Fentanyl IM                            | 10 mcg            | 0.2 ml (undiluted)   | -               |
| Fentanyl IN                            | 15 mcg            | 0.3 ml (undiluted)   | -               |
| Fentanyl/ketamine IV (post intubation) | 10 mcg/10 mg      | 1 ml                 | 8               |
| Glucagon IM                            | 0.5 mg            | 0.5 ml (undiluted)   | -               |
| 10% glucose IV                         | 2 ml/kg           | 20 ml                | -               |
| Hydrocortisone IV                      | 25 mg             | 2.5 ml (10 mg/ml)    | 9               |
| Hydrocortisone IM                      | 25 mg             | 0.5 ml (undiluted)   | -               |
| Ibuprofen liquid PO                    | 100 mg            | 5 ml (20 mg/ml)      | -               |
| Ibuprofen tablet PO                    | 100 mg            | ½ tablet             | -               |
| Ketamine IV (analgesia)                | 2 mg              | 2 ml (1 mg/ml)       | 10              |

|                            |               |                     |    |
|----------------------------|---------------|---------------------|----|
| Ketamine IV (dissociation) | 10 mg         | 1 ml (10 mg/ml)     | 11 |
| Ketamine IM/PO (analgesia) | 5 mg          | 0.05 ml (undiluted) | -  |
| Ketamine IM (dissociation) | 50 mg         | 0.5 ml (undiluted)  | -  |
| Levetiracetam IV           | 500 mg        | 5 ml (100 mg/ml)    | 12 |
| 1% lignocaine IO           | 10 mg         | 1 ml (undiluted)    | -  |
| 1% lignocaine SC           | 30 mg (max)   | 3 ml (max)          | -  |
| Loratadine PO              | 10 mg         | 1 tablet            | -  |
| Magnesium IV               | -             | -                   | -  |
| Metaraminol IV             | 0.1-0.2 mg    | 1-2 ml (0.1 mg/ml)  | 14 |
| Midazolam IV (seizures)    | 1 mg          | 1 ml (1 mg/ml)      | 15 |
| Midazolam IM (seizures)    | 2 mg          | 0.4 ml (undiluted)  | -  |
| Morphine IV                | 0.5 mg        | 0.5 ml (1 mg/ml)    | 16 |
| Morphine IM/SC             | 2 mg          | 0.2 ml (undiluted)  | -  |
| Naloxone IV                | 0.1 mg        | 1 ml (0.1 mg/ml)    | 17 |
| Naloxone IM                | 0.2 mg        | 0.5 ml (undiluted)  | -  |
| Ondansetron IV             | 2 mg          | 1 ml (undiluted)    | -  |
| Ondansetron IM             | 1 mg          | 0.5 ml (undiluted)  | -  |
| Paracetamol liquid PO      | 150 mg        | 3 ml (50 mg/ml)     | -  |
| Paracetamol tablet PO      | -             | -                   | -  |
| Prednisolone PO            | 10 mg         | 2 ml (5 mg/ml)      | -  |
| Prednisone PO              | 10 mg         | ½ tablet            | -  |
| Rocuronium IV              | 10 mg         | 1 ml (undiluted)    | -  |
| 0.75% ropivacaine SC       | 22.5 mg (max) | 3 ml (max)          | -  |
| 0.9% sodium chloride IV    | 20 ml/kg      | 200 ml              | -  |
| 8.4% sodium bicarbonate    | 20 mmol       | 20 ml (1 mmol/ml)   | -  |

|                                             |        |                  |   |
|---------------------------------------------|--------|------------------|---|
| Tranexamic acid IN (epistaxis)              | -      | -                | - |
| Tranexamic acid IV (non-traumatic bleeding) | 200 mg | 2 ml (undiluted) | - |
| Tranexamic acid IV (bleeding from trauma)   | 400 mg | 4 ml (undiluted) | - |

## 20 kg / 2-5 years

| Cardiac arrest                         | Dose              | Volume               | Dilution number |
|----------------------------------------|-------------------|----------------------|-----------------|
| Adrenaline IV                          | 0.2 mg            | 2 ml (1:10,000)      | 1               |
| Amiodarone IV                          | 100 mg            | 2 ml (undiluted)     | -               |
| Manual defibrillation                  | 80-100            | Joules               | -               |
| LMA                                    | Size 2 (10-20 kg) | Cuff inflation 10 ml | -               |
| ETT (cuffed)                           | Size 4 or 5       | 15 cm length at lips | -               |
| Other drugs                            |                   |                      | Dilution number |
| Adrenaline IV (not cardiac arrest)     | 0.004 mg          | 4 ml (1:1,000,000)   | 2               |
| Adrenaline IM                          | 0.2 mg            | 0.2 ml (undiluted)   | -               |
| Adrenaline IN (epistaxis)              | 0.1 mg            | 1 ml (1:10,000)      | 1               |
| Cefazolin IV                           | 500 mg            | 5 ml (100 mg/ml)     | 3               |
| Ceftriaxone IV                         | 1 g               | 5 ml (200 mg/ml)     | 4               |
| Ceftriaxone IM                         | 1 g               | 2.5 ml (400 mg/ml)   | 5               |
| Fentanyl IV (analgesia)                | 5-20 mcg          | 5-20 ml (1 mcg/ml)   | 7               |
| Fentanyl IM                            | 20 mcg            | 0.4 ml (undiluted)   | -               |
| Fentanyl IN                            | 30 mcg            | 0.6 ml (undiluted)   | -               |
| Fentanyl/ketamine IV (post intubation) | 20 mcg/20 mg      | 2 ml                 | 8               |
| Glucagon IM                            | 0.5-1 mg          | 0.5-1 ml (undiluted) | -               |
| 10% glucose IV                         | 2 ml/kg           | 40 ml                | -               |
| Hydrocortisone IV                      | 50 mg             | 5 ml (10 mg/ml)      | 9               |

|                            |             |                    |    |
|----------------------------|-------------|--------------------|----|
| Hydrocortisone IM          | 50 mg       | 1 ml (undiluted)   | -  |
| Ibuprofen liquid PO        | 150 mg      | 7.5 ml (20 mg/ml)  | -  |
| Ibuprofen tablet PO        | 100 mg      | ½ tablet           | -  |
| Ketamine IV (analgesia)    | 4 mg        | 4 ml (1 mg/ml)     | 10 |
| Ketamine IV (dissociation) | 20 mg       | 2 ml (10 mg/ml)    | 11 |
| Ketamine IM/PO (analgesia) | 10 mg       | 0.1 ml (undiluted) | -  |
| Ketamine IM (dissociation) | 100 mg      | 1 ml (undiluted)   | -  |
| Levetiracetam IV           | 1000 mg     | 10 ml (100 mg/ml)  | 12 |
| 1% lignocaine IO           | 20 mg       | 2 ml (undiluted)   | -  |
| 1% lignocaine SC           | 60 mg (max) | 6 ml (max)         | -  |
| Loratadine PO              | 10 mg       | 1 tablet           | -  |
| Magnesium IV               | 4 mmol      | 4 ml (1 mmol/ml)   | 13 |
| Metaraminol IV             | 0.2-0.4 mg  | 2-4 ml (0.1 mg/ml) | 14 |
| Midazolam IV (seizures)    | 2 mg        | 2 ml (1 mg/ml)     | 15 |
| Midazolam IM (seizures)    | 4 mg        | 0.8 ml (undiluted) | -  |
| Morphine IV                | 1 mg        | 1 ml (1 mg/ml)     | 16 |
| Morphine IM/SC             | 4 mg        | 0.4 ml (undiluted) | -  |
| Naloxone IV                | 0.1-0.2 mg  | 1-2 ml (0.1 mg/ml) | 17 |
| Naloxone IM                | 0.4 mg      | 1 ml (undiluted)   | -  |
| Ondansetron IV             | 4 mg        | 2 ml (undiluted)   | -  |
| Ondansetron IM             | 2 mg        | 1 ml (undiluted)   | -  |
| Paracetamol liquid PO      | 250 mg      | 5 ml (50 mg/ml)    | -  |
| Paracetamol tablet PO      | 250 mg      | ½ tablet           | -  |
| Prednisolone PO            | 20 mg       | 4 ml (5 mg/ml)     | -  |
| Prednisone PO              | 20 mg       | 1 tablet           | -  |

|                                             |             |                   |   |
|---------------------------------------------|-------------|-------------------|---|
| Rocuronium IV                               | 20 mg       | 2 ml (undiluted)  | - |
| 0.75% ropivacaine SC                        | 45 mg (max) | 6 ml (max)        | - |
| 0.9% sodium chloride IV                     | 20 ml/kg    | 400 ml            | - |
| 8.4% sodium bicarbonate                     | 40 mmol     | 40 ml (1 mmol/ml) |   |
| Tranexamic acid IN (epistaxis)              | 100 mg      | 1 ml (undiluted)  | - |
| Tranexamic acid IV (non-traumatic bleeding) | 400 mg      | 4 ml (undiluted)  | - |
| Tranexamic acid IV (bleeding from trauma)   | 800 mg      | 8 ml (undiluted)  | - |

## 30 kg / 6-10 years

| Cardiac arrest                     | Dose              | Volume               | Dilution number |
|------------------------------------|-------------------|----------------------|-----------------|
| Adrenaline IV                      | 0.3 mg            | 3 ml (1:10,000)      | 1               |
| Amiodarone IV                      | 150 mg            | 3 ml (undiluted)     | -               |
| Manual defibrillation              | 120-150           | Joules               | -               |
| LMA                                | Size 3 (30-50 kg) | Cuff inflation 20 ml | -               |
| ETT (cuffed)                       | Size 5 or 6       | 17 cm length at lips | -               |
| Other drugs                        |                   |                      | Dilution number |
| Adrenaline IV (not cardiac arrest) | 0.006 mg          | 6 ml (1:1,000,000)   | 2               |
| Adrenaline IM                      | 0.3 mg            | 0.3 ml (undiluted)   | -               |
| Adrenaline IN (epistaxis)          | 0.1 mg            | 1 ml (1:10,000)      | 1               |
| Cefazolin IV                       | 750 mg            | 7.5 ml (100 mg/ml)   | 3               |
| Ceftriaxone IV                     | 1.5 g             | 7.5 ml (200 mg/ml)   | 4               |
| Ceftriaxone IM                     | 1.5 g             | 3.75 ml (400 mg/ml)  | 5               |
| Fentanyl IV (analgesia)            | 10-30 mcg         | 1-3 ml (10 mcg/ml)   | 6               |
| Fentanyl IM                        | 30 mcg            | 0.6 ml (undiluted)   | -               |
| Fentanyl IN                        | 45 mcg            | 0.9 ml (undiluted)   | -               |

|                                        |              |                     |    |
|----------------------------------------|--------------|---------------------|----|
| Fentanyl/ketamine IV (post intubation) | 30 mcg/30 mg | 3 ml                | 8  |
| Glucagon IM                            | 1 mg         | 1 ml (undiluted)    | -  |
| 10% glucose IV                         | 2 ml/kg      | 60 ml               | -  |
| Hydrocortisone IV                      | 75 mg        | 7.5 ml (10 mg/ml)   | 9  |
| Hydrocortisone IM                      | 75 mg        | 1.5 ml (undiluted)  | -  |
| Ibuprofen liquid PO                    | 200 mg       | 10 ml (20 mg/ml)    | -  |
| Ibuprofen tablet PO                    | 200 mg       | 1 tablet            | -  |
| Ketamine IV (analgesia)                | 6 mg         | 6 ml (1 mg/ml)      | 10 |
| Ketamine IV (dissociation)             | 30 mg        | 3 ml (10 mg/ml)     | 11 |
| Ketamine IM/PO (analgesia)             | 15 mg        | 0.15 ml (undiluted) | -  |
| Ketamine IM (dissociation)             | 150 mg       | 1.5 ml (undiluted)  | -  |
| Levetiracetam IV                       | 1500 mg      | 15 ml (100 mg/ml)   | 12 |
| 1% lignocaine IO                       | 30 mg        | 3 ml (undiluted)    | -  |
| 1% lignocaine SC                       | 90 mg (max)  | 9 ml (max)          | -  |
| Loratadine PO                          | 10 mg        | 1 tablet            | -  |
| Magnesium IV                           | 6 mmol       | 6 ml (1 mmol/ml)    | 13 |
| Metaraminol IV                         | 0.3-0.6 mg   | 3-6 ml (0.1 mg/ml)  | 14 |
| Midazolam IV (seizures)                | 3 mg         | 3 ml (1 mg/ml)      | 15 |
| Midazolam IM (seizures)                | 6 mg         | 1.2 ml (undiluted)  | -  |
| Morphine IV                            | 1.5 mg       | 1.5 ml (1 mg/ml)    | 16 |
| Morphine IM/SC                         | 6 mg         | 0.6 ml (undiluted)  | -  |
| Naloxone IV                            | 0.1- 0.3 mg  | 1- 3 ml (0.1 mg/ml) | 17 |
| Naloxone IM                            | 0.6 mg       | 1.5 ml (undiluted)  | -  |
| Ondansetron IV                         | 6 mg         | 3 ml (undiluted)    | -  |
| Ondansetron IM                         | 3 mg         | 1.5 ml (undiluted)  | -  |

|                                             |               |                   |   |
|---------------------------------------------|---------------|-------------------|---|
| Paracetamol liquid PO                       | 375 mg        | 7.5 ml (50 mg/ml) | - |
| Paracetamol tablet PO                       | 250 mg        | ½ tablet          | - |
| Prednisolone PO                             | 30 mg         | 6 ml (5 mg/ml)    | - |
| Prednisone PO                               | 30 mg         | 1 ½ tablets       | - |
| Rocuronium IV                               | 30 mg         | 3 ml (undiluted)  | - |
| 0.75% ropivacaine SC                        | 67.5 mg (max) | 9 ml (max)        | - |
| 0.9% sodium chloride IV                     | 20 ml/kg      | 600 ml            | - |
| 8.4% sodium bicarbonate                     | 60 mmol       | 60 ml (1 mmol/ml) |   |
| Tranexamic acid IN (epistaxis)              | 100 mg        | 1 ml (undiluted)  | - |
| Tranexamic acid IV (non-traumatic bleeding) | 600 mg        | 6 ml (undiluted)  | - |
| Tranexamic acid IV (bleeding from trauma)   | 1200 mg       | 12 ml (undiluted) | - |

## 40 kg / 11-13 years

| Cardiac arrest                     | Dose              | Volume               | Dilution number |
|------------------------------------|-------------------|----------------------|-----------------|
| Adrenaline IV                      | 0.4 mg            | 4 ml (1:10,000)      | 1               |
| Amiodarone IV                      | 200 mg            | 4 ml (undiluted)     | -               |
| Manual defibrillation              | 160-200           | Joules               | -               |
| LMA                                | Size 3 (30-50 kg) | Cuff inflation 20 ml | -               |
| ETT (cuffed)                       | Size 6 or 7       | 19 cm length at lips | -               |
| Other drugs                        |                   |                      | Dilution number |
| Adrenaline IV (not cardiac arrest) | 0.008 mg          | 8 ml (1:1,000,000)   | 2               |
| Adrenaline IM                      | 0.4 mg            | 0.4 ml (undiluted)   | -               |
| Adrenaline IN (epistaxis)          | 0.2 mg            | 2 ml (1:10,000)      | 1               |
| Cefazolin IV                       | 1000 mg           | 10 ml (100 mg/ ml)   | 3               |
| Ceftriaxone IV                     | 2 g               | 10 ml (200 mg/ml)    | 4               |



|                                        |              |                    |    |
|----------------------------------------|--------------|--------------------|----|
| Ceftriaxone IM                         | 2 g          | 5 ml (400 mg/ml)   | 5  |
| Fentanyl IV (analgesia)                | 10-40 mcg    | 1-4 ml (10 mcg/ml) | 6  |
| Fentanyl IM                            | 40 mcg       | 0.8 ml (undiluted) | -  |
| Fentanyl IN                            | 60 mcg       | 1.2 ml (undiluted) | -  |
| Fentanyl/ketamine IV (post intubation) | 40 mcg/40 mg | 4 ml               | 8  |
| Glucagon IM                            | 1 mg         | 1 ml (undiluted)   | -  |
| 10% glucose IV                         | 2 ml/kg      | 80 ml              | -  |
| Hydrocortisone IV                      | 100 mg       | 10 ml (10 mg/ml)   | 9  |
| Hydrocortisone IM                      | 100 mg       | 2 ml (undiluted)   | -  |
| Ibuprofen liquid PO                    | 300 mg       | 15 ml (20 mg/ml)   | -  |
| Ibuprofen tablet PO                    | 300 mg       | 1 ½ tablets        | -  |
| Ketamine IV (analgesia)                | 8 mg         | 8 ml (1 mg/ml)     | 10 |
| Ketamine IV (dissociation)             | 40 mg        | 4 ml (10 mg/ml)    | 11 |
| Ketamine IM/PO (analgesia)             | 20 mg        | 0.2 ml (undiluted) | -  |
| Ketamine IM (dissociation)             | 200 mg       | 2 ml (undiluted)   | -  |
| Levetiracetam IV                       | 2000 mg      | 20 ml (100 mg/ml)  | 12 |
| 1% lignocaine IO                       | 40 mg        | 4 ml (undiluted)   | -  |
| 1% lignocaine SC                       | 120 mg (max) | 12 ml (max)        | -  |
| Loratadine PO                          | 20 mg        | 2 tablets          | -  |
| Magnesium IV                           | 8 mmol       | 8 ml (1 mmol/ml)   | 13 |
| Metaraminol IV                         | 0.4-0.8 mg   | 4-8 ml (0.1 mg/ml) | 14 |
| Midazolam IV (seizures)                | 4 mg         | 4 ml (1 mg/ml)     | 15 |
| Midazolam IM (seizures)                | 8 mg         | 1.6 ml (undiluted) | -  |
| Morphine IV                            | 2 mg         | 2 ml (1 mg/ml)     | 16 |
| Morphine IM/SC                         | 8 mg         | 0.8 ml (undiluted) | -  |

|                                             |             |                     |    |
|---------------------------------------------|-------------|---------------------|----|
| Naloxone IV                                 | 0.1-0.4 mg  | 1- 4 ml (0.1 mg/ml) | 17 |
| Naloxone IM                                 | 0.8 mg      | 2 ml (undiluted)    | -  |
| Ondansetron IV                              | 8 mg        | 4 ml (undiluted)    | -  |
| Ondansetron IM                              | 4 mg        | 2 ml (undiluted)    | -  |
| Paracetamol liquid PO                       | 500 mg      | 10 ml (50 mg/ml)    | -  |
| Paracetamol tablet PO                       | 500 mg      | 1 tablet            | -  |
| Prednisolone PO                             | 40 mg       | 8 ml (5 mg/ml)      | -  |
| Prednisone PO                               | 40 mg       | 2 tablets           | -  |
| Rocuronium IV                               | 40 mg       | 4 ml (undiluted)    | -  |
| 0.75% ropivacaine SC                        | 90 mg (max) | 12 ml (max)         | -  |
| 0.9% sodium chloride IV                     | 20 ml/kg    | 800 ml              | -  |
| 8.4% sodium bicarbonate                     | 80 mmol     | 80 ml (1 mmol/ml)   | -  |
| Tranexamic acid IN (epistaxis)              | 200 mg      | 2 ml (undiluted)    | -  |
| Tranexamic acid IV (non-traumatic bleeding) | 800 mg      | 8 ml (undiluted)    | -  |
| Tranexamic acid IV (bleeding from trauma)   | 1600 mg     | 16 ml (undiluted)   | -  |

## Paediatric vital signs

| Age        | Heart rate | Respiratory rate | Blood pressure (systolic) |
|------------|------------|------------------|---------------------------|
| <1         | 110-160    | 30-40            | 70-90                     |
| 1-2 years  | 100-150    | 25-35            | 80-95                     |
| 2-5 years  | 95-140     | 25-30            | 80-100                    |
| 5-12 years | 80-120     | 20-25            | 90-110                    |
| > 12 years | 60-100     | 15-20            | 100-120                   |

## Infant Glasgow Coma Scale

| Best eye opening                      |   |
|---------------------------------------|---|
| Spontaneously                         | 4 |
| To voice or touch                     | 3 |
| To pain or pressure                   | 2 |
| None                                  | 1 |
| Best verbal response                  |   |
| Smiles, babbles, coos                 | 5 |
| Cries normally                        | 4 |
| Cries only to pain or pressure        | 3 |
| Moans or grunts                       | 2 |
| None                                  | 1 |
| Best motor response                   |   |
| Normal spontaneous movement           | 6 |
| Localises or has purposeful movement  | 5 |
| Withdraws from pain or normal flexion | 4 |
| Abnormal (spastic) flexion            | 3 |
| Extension or rigid                    | 2 |
| None                                  | 1 |

## More Information

### Version history

**28 October 2024**

1.0.5

**Update:** Paediatric vital signs updates. Changes to loratadine, TXA, cefazolin and 8.4% sodium bicarbonate.**11 September 2023**

1.0.4

**Update:** Tables have been updated to include cefazolin and levetiracetam and removal of valproate.

22 November 2022

1.0.1

**Update:** Defibrillation energy, metaraminol dilution. Added morphine, removed blood volume from vital signs.

# Newborn Resuscitation

This guideline is for the resuscitation of babies immediately following birth.

## Initial actions

- Always clarify who is the team leader. If a lead maternity carer (LMC) is present, they are usually the team leader and are directing the treatment unless this responsibility is formally handed over to ambulance personnel.
- Place the baby skin to skin with the mother.
- Provide external stimulation while drying the baby.
- Maximise the external temperature and place a hat on the baby if possible.
- Continue to provide stimulation via rubbing/drying if breathing or activity is abnormal.
- Establish continual heart rate and SpO<sub>2</sub> monitoring if breathing or activity remain abnormal.
- Move to the appropriate section below at approximately one minute.

## If breathing is adequate and the heart rate is greater than or equal to 100/minute

- Do not provide ventilation.
- Clamp and cut the cord at 2-3 minutes.
- Move to the appropriate section if there is a sustained change in heart rate.
- Administer oxygen only if required to maintain a normal SpO<sub>2</sub>.

## If breathing is inadequate or the heart rate is 60-100/minute

- Clamp and cut the cord immediately if this is required to enable ventilation.
- Ventilate at a rate of 60/minute, using PEEP set to 5 cmH<sub>2</sub>O, initially without added oxygen.
- Continue to ventilate and add oxygen at 10 litres/minute if the heart rate fails to rise above 100/minute after two minutes.
- Move to the appropriate section if there is a sustained change in heart rate.

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## If the heart rate is less than 60/minute

- Clamp and cut the cord immediately if this is required to enable CPR.
- Start CPR at a ratio of 3:1.
- Focus on ventilation with PEEP set to 5 cmH<sub>2</sub>O and added oxygen at 10 litres/minute.
- Place an SGA.
- Consider placing an ETT.
- Gain IV access.
- Administer adrenaline IV every four minutes:
  - 0.01 mg (0.1 ml of 1:10,000) for gestation less than or equal to 26 weeks.
  - 0.025 mg (0.25 ml of 1:10,000) for gestation 27-37 weeks.
  - 0.05 mg (0.5 ml of 1:10,000) for gestation greater than or equal to 38 weeks.
- Move to the appropriate section if there is a sustained change in heart rate.

## In addition

- Continue to actively focus on keeping the baby warm, for example utilise heating pads if available.
- Monitor and record the SpO<sub>2</sub>. Normal SpO<sub>2</sub> post birth:
  - 60-70% at 1 minute.
  - 65-85% at 2 minutes.
  - 70-90% at 3 minutes.
  - 75-90% at 4 minutes.
  - 80-90% at 5 minutes.
  - 85-90% at 10 minutes.
  - 90-95% at more than 10 minutes.
- Titrate oxygen flow to an SpO<sub>2</sub> of 90-95% ten or more minutes after birth.
- Measure the blood glucose concentration if the baby's activity is not normal, but this is not a priority if resuscitation is required. Administer glucose if the blood glucose concentration is less than 2.5 mmol/litre:

# Newborn Resuscitation

- » Spread glucose gel on the gums, tongue and inside of the cheeks and repeat this every 5-10 minutes as required.
- » Administer 10 ml of 10% glucose IV if the blood glucose concentration does not rise 20 minutes after oral glucose.
- » Repeat the glucose measurement every ten minutes and administer further glucose as required until the glucose concentration is greater than or equal to 2.5 mmol/litre.

## Backup

- Backup from an ICP/CCP and/or LMC must be requested if ventilation is required at any stage.

## Referral and transport

- Transport to a hospital with neonatal facilities if the baby has required intervention at any stage, or has abnormal vital signs, whenever feasible and safe.
- The following hospitals have neonatal facilities:
  - » Whangārei Hospital.
  - » North Shore Hospital.
  - » Waitakere Hospital.
  - » Starship Children's Hospital.
  - » Middlemore Hospital.
  - » Waikato Hospital.
  - » Tauranga Hospital.
  - » Whakatane Hospital.
  - » Gisborne Hospital.
  - » Rotorua Hospital.
  - » Taranaki Base Hospital.
  - » Whanganui Hospital.
  - » Hawke's Bay Regional Hospital.
  - » Palmerston North Hospital.
  - » Wairarapa Hospital.
  - » Hutt Hospital.
  - » Wellington Regional Hospital.
  - » Nelson Hospital.
  - » Te Nikau Grey Hospital.

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- » Christchurch Hospital.
  - » Timaru Hospital.
  - » Dunedin Hospital.
  - » Southland Hospital.
- It is the responsibility of ambulance personnel to ensure all ambulance occupants are safely restrained during transport.

## Additional information

### General principles

- Assessment and interventions are based primarily on the baby's activity, breathing, heart rate and SpO<sub>2</sub>.
- Resuscitation of a newborn is focused primarily on supporting ventilation.
- A crying and/or active baby requires no specific intervention. Place skin to skin with the mother and dry the baby provided neither is requiring resuscitation. Place a hat on the baby if one is available, cover them both with a warm blanket and observe the baby's activity and breathing.
- It is normal for the peripheries of the baby to remain blue for several hours after birth.
- The heart rate is best monitored via ECG leads which may be placed on the back.
- Pulse oximetry should be monitored using the right hand whenever possible because a patent ductus arteriosus may produce falsely low measurements in the left hand and feet. Place the probe over several digits if the probe is too large for a single digit.
- Preventing heat loss is vital as hypothermia worsens outcomes:
  - » The torso and limbs of a premature baby requiring resuscitation should be immediately wrapped in plastic without being dried and placed in a hot environment as soon as possible.
  - » A term baby should be dried and resuscitated in a hot environment if possible.
  - » A hat should be placed on the baby if one is available.
  - » The interior of the ambulance should be made as hot as possible.
- Mother and baby must be safely restrained during transport.

### Oxygen administration, suctioning and ventilation

- Routine oxygen administration during neonatal resuscitation appears to make outcomes worse. This is why oxygen is reserved for deterioration despite initial ventilation.



# Newborn Resuscitation

- Suctioning the mouth and nose before the body is delivered is not required, even if meconium is present. If ventilation is required, this takes priority over suctioning meconium unless meconium is clearly occluding the airway.
- Ventilation via an SGA is preferred to ventilation via an ETT because intubation has a high failure rate, unless the person performing the intubation is very experienced. However, if ventilation is ineffective or resuscitation is ongoing, an ETT should be considered.

## Intravascular access, adrenaline and 0.9% sodium chloride

- Gaining IV access has a lower priority than providing good CPR/ventilation.
- If the baby is very small, IO access may be difficult to obtain because the paediatric IO needle may be too long and the softness of the bones may result in the IO needle being easily displaced.
- Administering adrenaline IV is not a priority, but may occur provided this does not compromise the focus on providing good ventilation and CPR.
- 0.9% sodium chloride IV does not have a significant role in resuscitation, but consider administering 10 ml/kg if there are signs that the baby has bled or the baby shows signs of shock.

## Clamping and cutting the umbilical cord

- Delaying clamping and cutting the cord until 2-3 minutes after birth appears to improve the outcome for the baby.
- There is no urgency to clamp and cut the cord provided neither the baby nor the mother are requiring resuscitation.
- Clamp and cut the cord at least 5 cm from the baby as this facilitates access to the cord vessels for later cannulation if required.

## Hypoglycaemia

- Hypoglycaemia is uncommon in neonates in the first few hours.
- The normal blood glucose concentration is lower in a neonate than in older children and a neonate is not hypoglycaemic unless the glucose concentration is less than 2.5 mmol/litre.
- Glucose concentration should be measured via heel prick.
- Spreading glucose gel on the mucous membranes of the mouth is usually an effective treatment.

# Newborn Resuscitation

## The role of hospital-based neonatal teams

- There is usually no role for requesting a neonatal team to attend a scene that is not a birthing unit.
- There may be a role for a neonatal team to attend a birthing unit, noting that this decision will be made by a neonatologist in consultation with the LMC and the team may take significant time to reach the scene.
- The decision to utilise a neonatal team is usually a trade-off between time to skilled intervention and the availability of dedicated equipment during transfer:
  - If the baby is requiring ventilation or CPR, it is usually faster to transport the baby to a neonatal team than to transport a neonatal team to the scene.
  - If the baby requires specific equipment for transfer, it is usually preferable to wait at the scene for a neonatal team with dedicated equipment.

## Starting and stopping resuscitation

- A resuscitation attempt should always be started unless an LMC directs ambulance personnel not to.
- Resuscitation should usually occur at the scene because the quality of resuscitation is compromised during transport. However, resuscitation en route to hospital may be appropriate if transport time is short and an ICP/CCP or LMC cannot reach the scene quickly.
- Resuscitation attempts should usually be stopped if the baby is in asystole for more than ten minutes.

## Response and/or transport by helicopter

- Response by helicopter may be indicated if newborn resuscitation is required and it will arrive at scene at least 15 minutes sooner than a road-based CCP.
- Transport by helicopter may be indicated if newborn resuscitation is required and it will save more than 15 minutes compared to road transport. For criteria to be met, the road transport time will usually need to be more than 75 minutes.

## More Information

### Version history

16 June 2026

1.1.0.1

**Update:** Assessment and treatment guidance updated.

28 October 2024

1.0.5

# Newborn Resuscitation

**Update:** Multiple changes.

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# General Principles in Pregnancy

- If a pregnant patient has a non-pregnancy-related condition, for example asthma or trauma, treat according to the appropriate guideline. In this situation:
  - Ambulance personnel are in charge of directing the treatment.
  - The patient must be transported to an ED and not an obstetric or birthing unit.
- If the patient has a pregnancy-related condition:
  - Always clarify and document who is in charge. If a lead maternity carer (LMC) is present, the LMC is in charge of directing the treatment unless this responsibility is formally handed over to ambulance personnel.
  - If an LMC is present, always utilise the pregnancy related transport checklist prior to commencing transport.
- If the patient is status one or status two, transport should usually be to an ED, preferably within a hospital with an obstetric unit. However, the patient may be transported direct to an obstetric unit, provided a direct phone call has been placed to clinical staff within the unit, and the patient has been accepted.
- When transporting to an obstetric or birthing unit, always contact the LMC if feasible, and always phone staff at the unit to provide patient and clinical details prior to arrival.
- Ambulance transfer (by both road and air) from primary birthing units is the responsibility of ambulance personnel. Personnel may ask if an LMC is available to travel in the ambulance with the patient, however if an LMC is not available, this does not preclude transport.

## Vaginal examination

- It is appropriate to view the patient's perineum with their consent, looking for signs of the baby, umbilical cord, or bleeding.
- It is inappropriate to place your fingers or hands inside the vagina except in the setting of severe PPH, shoulder dystocia, or breech presentation.
- Additional clinical judgement must be used when gaining consent in an emergency, for example if the patient has an altered level of consciousness. In this setting it is appropriate to initiate emergency interventions while explaining what is being done.

## Pregnancy and abdominal or pelvic injury

- A pregnant patient with possible abdominal or pelvic injury occurring after 20 weeks gestation must be given a clear recommendation to be transported to an ED by ambulance, preferably one within a hospital with obstetric facilities, even if the trauma is minor and the patient is asymptomatic.

# General Principles in Pregnancy

## Pre-labour rupture of membranes

- This is rupture of the membranes prior to the onset of labour.
- Contact the LMC if feasible.
- Exclude an obvious cord prolapse. If this is present, the patient requires immediate transport by ambulance.
- If the pregnancy is less than 37 weeks, clearly recommend the patient is assessed in a hospital with obstetric facilities, noting that transport by ambulance may not be required.
- If the pregnancy is greater than or equal to 37 weeks, clearly recommend the patient contacts their LMC. If the patient does not have an LMC seek telephone advice from clinical staff at a birthing or obstetric unit.

## Pre-term labour

- This is the onset of labour after 20 weeks and prior to 37 weeks of pregnancy.
- Contact the LMC if feasible.
- Provide a clear recommendation that transport to a hospital with obstetric facilities is required by ambulance.
- Prepare for the possibility of a rapid birth and be prepared to provide neonatal resuscitation if required.

## Supine hypotension

- After 20 weeks gestation, hypotension may occur in the supine position because the uterus can impede venous return through the inferior vena cava.
- To prevent supine hypotension always tilt the patient 30 degrees (or more) to their left by placing a rolled towel or pillow under their right hip. If this cannot be achieved, manually displace the uterus to the left if feasible.

## More Information

### Version history

9 June 2025

**New:** Guideline released.

1.0.5.3

# General Principles in Pregnancy

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This guideline is for patients with vaginal bleeding and confirmed or suspected pregnancy less than 20 weeks gestation. Refer to the '[antepartum haemorrhage guideline](#)' for patients with vaginal bleeding occurring after 20 weeks gestation and prior to birth.

- Administer 0.9% sodium chloride IV (using a fluid warming device if available) if the patient has signs of hypovolaemia:
  - 500 ml IV.
  - Repeat as required, minimising the total volume.
- If shock is severe:
  - Administer 1 g of tranexamic acid IV.
  - Provide active external warming.
  - Expedite transport.
  - Call for blood, provided there is an established protocol for this in the area, particularly if extrication will be delayed and/or transport will be prolonged.
- If shock is severe and unresponsive to treatment, seek clinical advice regarding the possibility of administering vasopressor support.

## Backup

- Backup from a Paramedic should be requested if the patient has signs of hypovolaemia.
- Backup from an ICP/CCP should be requested if the patient has severe shock.

## Referral and transport

- Provide a clear recommendation that the patient is transported to an ED by ambulance if:
  - Heavy vaginal bleeding is present, or
  - Hypovolaemia or significantly abnormal vital signs are present, or
  - Syncope has occurred, or
  - The patient is known to be more than 12 weeks pregnant, or
  - Abdominal or shoulder tip pain are present and intrauterine pregnancy has not been confirmed by imaging.
- Provide a clear recommendation that the patient is seen within 24 hours by their LMC or in primary care, (or in an ED if this is not possible) if:
  - Light vaginal bleeding is present, and

- » Intrauterine pregnancy has not been confirmed by imaging.
- Recommend the patient is seen by their LMC or in primary care non-urgently if:
  - » Light vaginal bleeding with a known intrauterine pregnancy, or
  - » It is likely that they have completed a miscarriage because products of conception have been expelled, and bleeding and pain are reducing.

## Additional information

### General principles

- Referral destination and urgency depends on a number of factors, most especially:
  - » The presence of heavy bleeding. For example, more than one pad is soaked per hour or bleeding is significantly more than a heavy menstrual period.
  - » Whether signs of hypovolaemia or abnormal vital signs are present.
  - » The presence or absence of significant abdominal or shoulder tip pain.
  - » The likely gestation. This can be estimated based on how many weeks have elapsed since the last menstrual period.
  - » Whether the patient has at any point had an ultrasound which confirms that this pregnancy is intrauterine.
  - » Whether miscarriage has likely been completed.
  - » Whether any risk factors for ectopic pregnancy are present.
- Vaginal bleeding occurs in roughly 1 in 2 pregnancies in the first 12 weeks, and miscarriage occurs in roughly 1 in 4 pregnancies. It may be associated with lower abdominal cramps similar to period cramps.
- To know that a pregnancy is intrauterine, the patient must have had an ultrasound confirming this.
- Once miscarriage is completed, the pain and bleeding will normally settle, and products of conception will normally be expelled. Follow up can be non-urgent in this setting.
- Pregnancy should be suspected in the following circumstances:
  - » The patient is aged between 10 and 50, is sexually active and has missed their last menstrual period.
  - » The patient has symptoms consistent with pregnancy, which may include breast tenderness, increased appetite, fatigue, frequent urination, and nausea and/or vomiting.

### Ectopic pregnancy

- Although ectopic pregnancy does not always present with vaginal bleeding, it is the most concerning cause for vaginal bleeding in early pregnancy.



- Ectopic pregnancy is usually associated with abdominal pain which is more generalised and/or severe than period cramps and may radiate to the shoulder tip (due to diaphragmatic irritation from haemoperitoneum).
- The risk of ectopic pregnancy is highest between 5-6 weeks gestation; however, it should also be considered in cases where the pregnancy has not been confirmed as intrauterine or the gestational age is uncertain.
- Risk factors for ectopic pregnancy include:
  - Previous ectopic pregnancy.
  - Previous pelvic or abdominal surgery.
  - Pelvic inflammatory disease.
  - Endometriosis.
  - Use of assisted reproductive technology, such as in-vitro fertilisation (IVF).

## Response and/or transport by helicopter

- Transport or response by helicopter may be indicated if the patient has clinically suspected or confirmed rupture of an ectopic pregnancy and:
  - The helicopter will arrive at scene at least 15 minutes sooner than a road-based CCP, or
  - Transport by helicopter will save more than 15 minutes compared to road transport. For criteria to be met, the road transport time will usually need to be more than 75 minutes.

## Administration of IV fluid

- Some patients may not become tachycardic despite significant hypovolaemia. This is because in the context of ectopic pregnancy, dilatation of the fallopian tube may cause vagal stimulation.
- 0.9% sodium chloride IV should be titrated to clinical signs of intravascular volume and perfusion.
- Large volumes of crystalloid may worsen bleeding by contributing to coagulopathy, particularly in the context of severe and uncontrolled bleeding:
  - This is why the total volume of 0.9% sodium chloride should be minimised.
  - Clinical judgment is required when determining the volume administered, but it should be uncommon to exceed 1 litre and it should be rare to exceed 2 litres.
  - Have a low threshold for seeking clinical advice if shock is severe and transport time to the most appropriate hospital is prolonged.
- Ambient temperature IV fluid can cause hypothermia which worsens bleeding by contributing to coagulopathy. IV fluid should be warmed using a dedicated warming device if one is available.

- Do not warm IV fluid in microwaves because this can result in unreliable fluid temperatures and/or may damage the bag.

## Tranexamic acid (TXA)

- The role for TXA in vaginal bleeding in early pregnancy is unclear.
- However, the balance of risk is in favour of TXA administration if shock is severe.

## Blood

- It is rare for blood to be required in the setting of vaginal bleeding in early pregnancy.
- Requests for blood to be delivered to the scene must not compromise rapid transport to the most appropriate hospital.
- Blood should only be requested if:
  - Shock is severe, and
  - There is an established protocol in the area for blood to be delivered to the scene, and
  - Blood can be administered 15 or more minutes before the patient can arrive in the most appropriate hospital.
- Blood may be available via staging at a small hospital/medical facility en route to the most appropriate hospital. In this setting personnel should seek clinical advice.
- ICPs/CCPs may continue blood administration initiated by a medical practitioner following their verbal instructions but should have a low threshold for seeking clinical advice.
- Requests for blood to be delivered to the scene outside of an established protocol must be made via the on-call doctor for the service. In this setting:
  - It will usually take approximately 15 minutes to arrange for blood to be prepared, plus transport time to the scene. This means that for such a request to be appropriate, the patient should usually have severe shock and a prolonged transport time.
  - Personnel cannot administer blood without specific instructions from the on-call doctor for the service.

## Cultural considerations after miscarriage

- After miscarriage, the tūpāpaku (deceased) should be handled with care. The whānau may have a kete (basket), or one may be provided. Unless the whānau permit otherwise, avoid using plastic bags.

➤ Karakia and waiata may be performed by the whānau – it is important to allow time for this.

## More Information

### Version history

|                                 |         |
|---------------------------------|---------|
| 9 June 2025                     | 1.0.5.3 |
| <b>New:</b> Guideline released. |         |

# Eclampsia

This guideline is for patients where eclampsia is suspected and the patient has had one or more seizures.

## If the patient is actively seizing

- Position the patient in the left lateral position or tilt the patient 30° to their left.
- Measure the blood glucose concentration and treat accordingly.
- Administer magnesium and midazolam concurrently if possible.
- Administer 20 mmol (5 g) of magnesium IV, over approximately 15 minutes.
- Administer midazolam IV:
  - 5 mg of midazolam IV.
  - This may be repeated once after five minutes.
- Administer midazolam IM if IV access has not already been obtained:
  - 10 mg of midazolam IM.
  - This may be repeated once after ten minutes.
- Commence transport as soon as possible.
- Seek urgent clinical advice if the seizure or hypertension is not rapidly controlled.
- Administer levetiracetam IV if the seizure continues or recurs after the above treatment:
  - 2 g of levetiracetam IV.
  - Repeat once if the seizure continues or recurs ten minutes after the first dose.
- Administer labetalol if the systolic BP is above 160 mmHg or the diastolic BP is above 110 mmHg, to maintain a systolic BP of 140-160 mmHg and a diastolic BP below 110 mmHg:
  - Administer a 20 mg bolus IV every five minutes, as required, while setting up an infusion.
  - Commence an IV infusion at 1 mg/kg/hour, up to a maximum of 150 mg/hour.
  - Adjust the rate as required by 20 mg/hour every 10 minutes, titrating to effect.
  - Administer additional boluses of 20 mg IV every five minutes as required.

## If the patient is no longer seizing

- Position the patient in the left lateral position or tilt the patient 30° to their left.
- Measure the blood glucose concentration and treat accordingly.

# Eclampsia

- Administer 20 mmol (5 g) of magnesium IV, over approximately 15 minutes.
- Administer labetalol if the systolic BP is above 160 mmHg or the diastolic BP is above 110 mmHg, to maintain a systolic BP of 140-160 mmHg and a diastolic BP below 110 mmHg:
  - Administer a 20 mg bolus IV every five minutes, as required, while setting up an infusion.
  - Commence an IV infusion at 1 mg/kg/hour, up to a maximum of 150 mg/hour.
  - Adjust the rate as required by 20 mg/hour every 10 minutes, titrating to effect.
  - Administer additional boluses of 20 mg IV every five minutes as required.

## Backup

- Backup from a CCP must be requested.

## Referral and transport

- The patient must receive a clear recommendation to be transported to an ED by ambulance.
- Transport the patient direct to a hospital with obstetric facilities whenever feasible and safe.

## Hospitals with obstetric facilities

### Northland and Auckland

- Whangarei Hospital.
- North Shore Hospital.
- Waitakere Hospital.
- Auckland City Hospital.
- Middlemore Hospital.

### Midland

- Waikato Hospital.
- Rotorua Hospital.
- Tauranga Hospital.
- Gisborne Hospital.

# Eclampsia

- Taranaki Base Hospital.

## Central/Lower North Island

- Hawke's Bay Hospital.
- Whanganui Hospital.
- Palmerston North Hospital.
- Wairarapa Hospital.
- Hutt Hospital.
- Wellington Hospital.

## South Island

- Nelson Hospital.
- Wairau Hospital.
- Te Nikau Grey Hospital.
- Christchurch Hospital.
- Timaru Hospital.
- Dunedin Hospital.
- Southland Hospital.

## Additional information

### General principles

- Eclampsia is the occurrence of one or more seizures in a patient with a clinical history of pre-eclampsia. This can occur any time after 20 weeks gestation and up to six weeks postpartum. Patients suspected or diagnosed with severe pre-eclampsia are considered high-risk for eclampsia.
- In the out of hospital setting, a pregnant patient who is between 20 weeks gestation and six weeks postpartum and has had one or more seizures should be presumed to have eclampsia, with treatment provided accordingly. This is because it is not possible to make a definitive diagnosis in the out of hospital setting and the balance of risk is in favour of administering magnesium.

# Eclampsia

## Pre-eclampsia

- Pre-eclampsia is a multisystem disorder that can occur during pregnancy after 20 weeks gestation and up to 6 weeks postpartum.
- Pre-eclampsia is variable in both clinical presentation and severity and can affect multiple organ systems. Clinical features can include:
  - Headache.
  - Visual changes.
  - Hyperreflexia.
  - Hypertension.
  - Oedema.
  - Jaundice.
- If pre-eclampsia is suspected, contact the LMC directly and arrange for an urgent assessment to occur.
- Magnesium and labetalol may have a role in the treatment of severe pre-eclampsia. Seek clinical advice if the patient has a systolic blood pressure greater than 160 mmHg, a diastolic blood pressure of greater than 110 mmHg, or has agitation.

## Response and/or transport by helicopter

- Response by helicopter may be indicated if the patient has eclampsia and it will arrive at scene at least 15 minutes sooner than a road-based CCP.
- Transport by helicopter may be indicated if the patient has eclampsia and it will save more than 15 minutes compared to road transport. For criteria to be met, the road transport time will usually need to be more than 75 minutes.

## More Information

### Version history

9 June 2025

1.0.5.3

**New:** Guideline released.

# Antepartum Haemorrhage

This guideline is for patients with any vaginal bleeding occurring after 20 weeks gestation and prior to birth.

- Tilt the patient 30° to their left to prevent supine hypotension.
- Commence transport as soon as possible.
- Administer 0.9% sodium chloride IV (using a fluid warming device if available) if the patient has signs of hypovolaemia:
  - 500 ml IV.
  - Repeat as required, minimising the total volume.
- If shock and/or bleeding is severe:
  - Administer 1 g of tranexamic acid IV.
  - Provide active external warming.
  - Call for blood, provided there is an established protocol for this in the area, particularly if extrication will be delayed and/or transport will be prolonged.
- If shock is severe and unresponsive to treatment, seek clinical advice regarding the possibility of administering vasopressor support.

## Backup

- Backup from a Paramedic should be requested if the patient has signs of hypovolaemia.
- Backup from an ICP/CCP should be requested if the patient has severe shock.
- Backup must not delay transport to an appropriate hospital.

## Referral and transport

- The patient must be transported to hospital by ambulance, preferably a hospital with obstetric facilities whenever this is feasible and safe.
- Provide early notification if signs of hypovolaemia are present.
- Notify the LMC en route to hospital whenever it is feasible to do so.
- The patient may be transported directly to an obstetric unit provided a direct phone call has been placed to clinical staff within the unit, and the patient has been accepted.



# Antepartum Haemorrhage

## Hospitals with obstetric facilities

### Northland and Auckland

- Whangarei Hospital.
- North Shore Hospital.
- Waitakere Hospital.
- Auckland City Hospital.
- Middlemore Hospital.

### Midland

- Waikato Hospital.
- Rotorua Hospital.
- Tauranga Hospital.
- Gisborne Hospital.
- Taranaki Base Hospital.

### Central/Lower North Island

- Hawke's Bay Hospital.
- Whanganui Hospital.
- Palmerston North Hospital.
- Wairarapa Hospital.
- Hutt Hospital.
- Wellington Hospital.

### South Island

- Nelson Hospital.
- Wairau Hospital.
- Te Nikau Grey Hospital.

# Antepartum Haemorrhage

- Christchurch Hospital.
- Timaru Hospital.
- Dunedin Hospital.
- Southland Hospital.

## Additional information

### General principles

- Any vaginal blood loss, even if only 'spotting', occurring after 20 weeks gestation must be considered antepartum haemorrhage (APH) until proven otherwise.
- APH may be associated with poor blood supply to the baby. This requires urgent investigation as immediate management may be required, and this is the reason for commencing transport as soon as possible.

### Administration of IV fluid

- During pregnancy the blood volume becomes expanded. Particularly after 28 weeks gestation, more than one litre of blood may be lost before the development of obvious signs of shock. Defining when shock is severe requires clinical judgement, but if a pregnant (or very recently pregnant) woman is showing clear signs of shock, then the shock is severe.
- 0.9% sodium chloride IV should be titrated to clinical signs of intravascular volume and perfusion.
- Large volumes of crystalloid may worsen bleeding by contributing to coagulopathy, particularly in the context of severe and uncontrolled bleeding:
  - This is why the total volume of 0.9% sodium chloride should be minimised.
  - Clinical judgment is required when determining the volume administered, but it should be uncommon to exceed 1 litre and it should be rare to exceed 2 litres.
  - Have a low threshold for seeking clinical advice if shock is severe and transport time to the most appropriate hospital is prolonged.
- Ambient temperature IV fluid can cause hypothermia which worsens bleeding by contributing to coagulopathy. IV fluid should be warmed using a dedicated warming device if one is available.
- Do not warm IV fluid in microwaves because this can result in unreliable fluid temperatures and/or may damage the bag.

# Antepartum Haemorrhage

## Tranexamic acid (TXA)

- The role for TXA in APH is unclear, however, the balance of risk is in favour of TXA administration if shock is severe.

## Blood

- It is rare for blood to be required in the setting of APH.
- Requests for blood to be delivered to the scene must not compromise rapid transport to the most appropriate hospital.
- Blood should only be requested if:
  - Shock is severe, and
  - There is an established protocol in the area for blood to be delivered to the scene, and
  - Blood can be administered 15 or more minutes before the patient can arrive in the most appropriate hospital.
- Blood may be available via staging at a small hospital/medical facility en route to the most appropriate hospital. In this setting personnel should seek clinical advice.
- ICPs/CCPs may continue blood administration initiated by a medical practitioner following their verbal instructions but should have a low threshold for seeking clinical advice.
- Requests for blood to be delivered to the scene outside of an established protocol must be made via the on-call doctor for the service. In this setting:
  - It will usually take approximately 15 minutes to arrange for blood to be prepared, plus transport time to the scene. This means that for such a request to be appropriate, the patient should usually have severe shock and a prolonged transport time.
  - Personnel cannot administer blood without specific instructions from the on-call doctor for the service.

## Response and/or transport by helicopter

- Transport or response by helicopter may be indicated if the patient has APH with signs of shock and:
  - The helicopter will arrive at scene at least 15 minutes sooner than a road-based CCP, or
  - Transport by helicopter will save more than 15 minutes compared to road transport. For criteria to be met, the road transport time will usually need to be more than 75 minutes.

# Antepartum Haemorrhage

## More Information

### Version history

|                                       |         |
|---------------------------------------|---------|
| 9 June 2025                           | 1.0.5.3 |
| <b>Update:</b> Guideline redeveloped. |         |

# Normal Birth

- Support the patient to adopt the position they prefer.
- Support the baby's head and shoulders as they appear, without applying traction.
- If the cord is wrapped around the neck:
  - If you can slip the cord over the head, then do so.
  - If you cannot slip the cord over the head, allow birth to continue.
- Support the baby's body as it appears, without applying traction.
- Document the time of birth.
- Following birth:
  - Place the baby skin to skin on the chest of the mother and initiate drying.
  - Regularly observe and document the baby's activity and breathing. If at any stage the baby's activity or breathing is abnormal, move to the ['newborn resuscitation'](#) guideline.
  - Ensure a warm environment and continue to keep the baby skin to skin with the mother, provided neither requires immediate treatment precluding a skin to skin position.
  - Place a hat on the baby if one is available, and cover the torso of the mother and baby with a blanket.
  - Ask the mother if they would like to encourage the baby to latch on to a nipple.
  - Administer 10 units of oxytocin IM into the mother's lateral thigh.
  - Clamp and cut the cord at least 5 cm from the baby, 5 or more minutes after birth, unless this is required earlier to facilitate urgent treatment of the mother or baby.
  - Allow the placenta to deliver spontaneously, without applying traction. Document the time of placental delivery. This usually occurs within 30 minutes but may take up to 60 minutes.
  - Place the placenta in a suitable container and keep it with the mother.

## Restraint during transport following birth

- The choice of restraint for the mother and baby requires clinical judgment that considers the need to maintain warmth, provide monitoring and/or treatment, and maintain safety during transport.
- The baby should be transported on a stretcher if active treatment and/or continuous monitoring of vital signs is required.
- If the baby is being transported on a stretcher, the baby should be restrained using a weight/size suitable restraint device.

# Normal Birth

- The baby may be transported in a baby car seat, provided the baby has never required any form of intervention or treatment, and has normal activity and breathing.
- The baby may be transported on the mother's chest, skin to skin, with a blanket over the torso of the mother and baby, with all stretcher seat belts over the blanket. This option may be best if maintaining warmth is the priority, for example the baby is premature or 'small for gestation' and is not requiring active treatment.
- The mother and baby should be transported in the same ambulance whenever this is feasible and safe.

## Cultural considerations

- Whenua (placenta) is sacred to Māori, and care should be taken in how it is handled. The whānau may have an ipu whenua (placental burial box) for it, and unless they permit otherwise, plastic bags should not be used.
- Karakia may be performed by the whānau after the birth of the child – it is important to allow time for this.
- The whānau/family may wish to use natural materials such as pounamu and flax to cut and tie the umbilical cord.

## More Information

### Version history

9 June 2025

1.0.5.3

**New:** Guideline released.

# Birth Emergency Action Lists

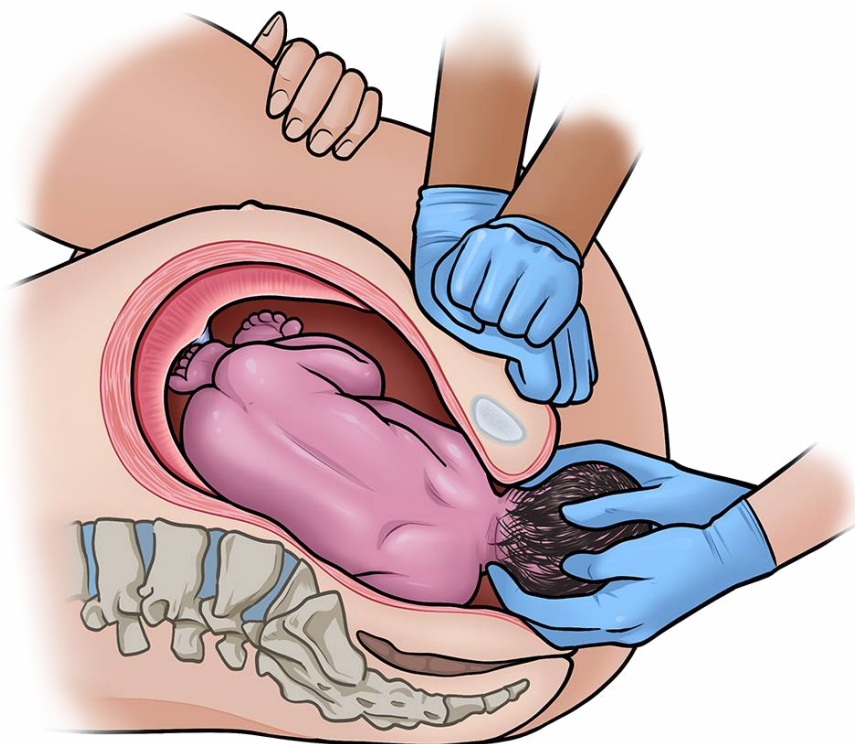
## Shoulder dystocia

This is when the baby's head is born, but birth does not occur with the next two contractions. The head may appear and then retract ('turtling'). The usual cause is that the baby's anterior shoulder is stuck behind the mother's pubic bone. It is potentially life-threatening for the baby.

Immediately take the following sequential actions. Move to the **'birth'** guideline if the baby is born.

- ✓ **Call for help and backup.**
  - » Call for help from an LMC if one is immediately available.
  - » Call for backup from an ICP/CCP.
- ✓ **Get the legs up.**
  - » Assist the mother to pull their knees as high as possible towards their armpits.
  - » Apply traction to the baby's head.
  - » If after the next contraction with the mother pushing the baby is not born, move to the next step.
- ✓ **Roll onto hands and knees.**
  - » Assist the mother to roll onto their hands and knees.
  - » Apply traction to the baby's head.
  - » If after the next contraction with the mother pushing the baby is not born, move to the next step.
- ✓ **Roll onto the back, get legs up and apply suprapubic pressure.**
  - » Assist the mother to roll onto their back.
  - » Assist the mother to pull their knees as high as possible towards their armpits.
  - » Have an assistant place their hands (using a CPR 'like' position) directly above the mother's pubic bone and push firmly and continuously, straight back. Whenever feasible do this from the side the baby is facing away from.
  - » Apply traction to the baby's head.
  - » If after the next contraction with the mother pushing the baby is not born, move to the next step.

# Birth Emergency Action Lists



- ✓ **Attempt to remove the posterior shoulder.**
  - » Keep suprapubic pressure in place.
  - » Place one hand into the vagina posteriorly.
  - » Feel for the baby's posterior shoulder and circle your first finger and thumb around the axilla.
  - » Apply firm traction on the axilla. The aim is to remove the shoulder. If successful, continue to provide axillary traction following the curve of the mother's sacrum.
  - » If after the next contraction with the mother pushing the baby is not born, move to the next step.
- ✓ **Commence urgent transport.**
  - » Commence urgent transport to the nearest birthing unit, obstetric unit, or ED, providing early notification of arrival.
  - » Position the mother tilted to the left with pillows/blankets under the pelvis and the stretcher head down (if possible) so that their head is below their pelvis.
  - » Seek clinical advice, putting the phone on speaker phone.
- ✓ **Prepare for newborn resuscitation.**

## Breech delivery

This is when the baby is coming out feet or buttocks first. It is potentially life-threatening for the baby.



# Birth Emergency Action Lists

## If the baby is not appearing at the vaginal opening

Immediately take the following sequential actions:

- ✓ **Call for help and backup.**
  - » Call for help from an LMC if one is immediately available.
  - » Call for backup from an ICP/CCP.
- ✓ **Instruct the patient not to push.**
- ✓ **Commence urgent transport.**
  - » Commence urgent transport to the nearest birthing or obstetric unit, providing early notification of arrival.
  - » Position the mother tilted to the left with pillows/blankets under the pelvis and the stretcher head down (if possible) so that their head is below their pelvis.
- ✓ **Prepare for newborn resuscitation.**
- ✓ **Allow birth to occur if this becomes imminent.**
  - » If birth becomes imminent, pull to the side of the road and move to the next action list.

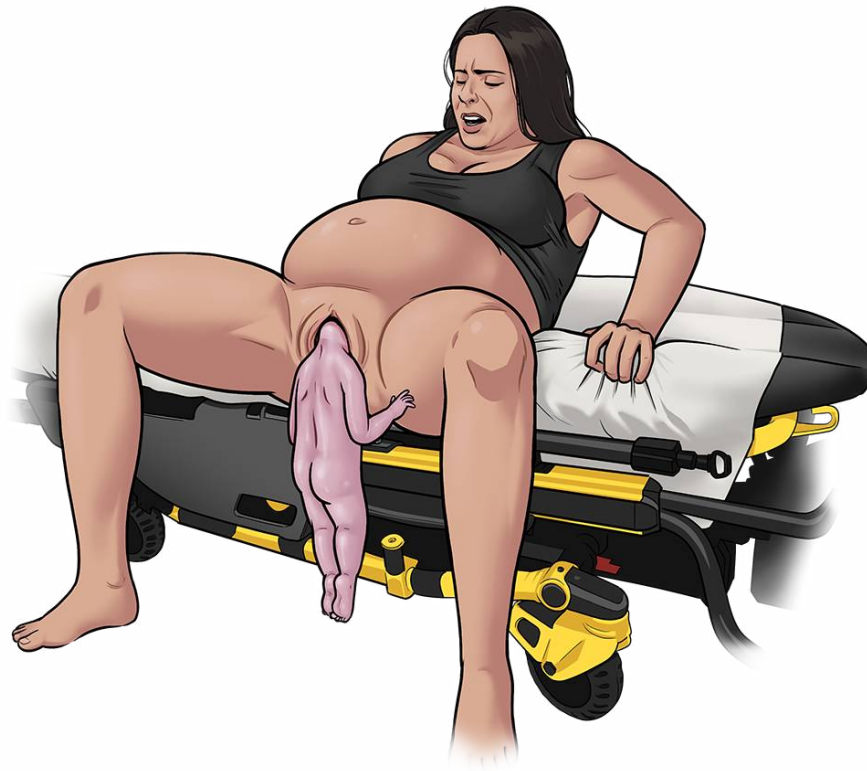
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## If the baby is appearing at the vaginal opening

Immediately take the following sequential actions:

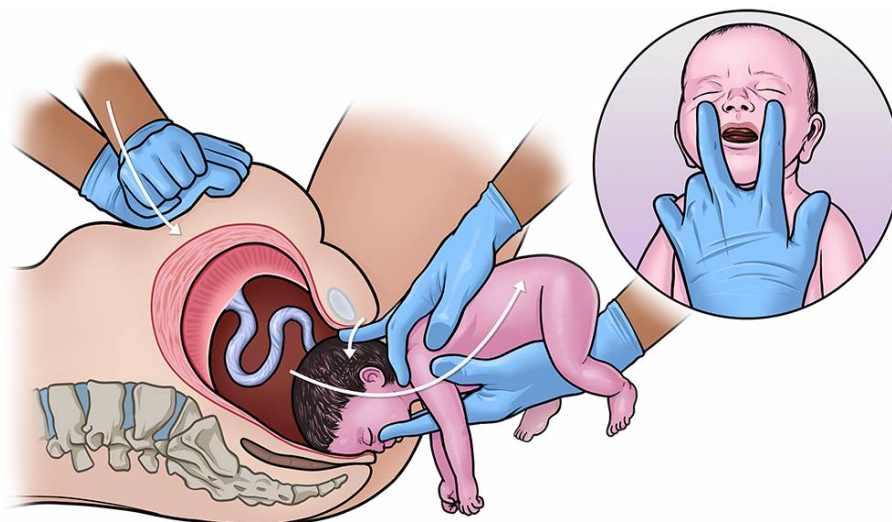
- ✓ **Call for help and backup if this has not already been done.**
  - » Call for help from an LMC if one is immediately available.
  - » Call for backup from an ICP/CCP.
  - » Seek clinical advice and put the phone on speaker phone.
- ✓ **Assist the patient to stand or sit.**
  - » Assist and support the patient to stand or sit on the edge of the stretcher/bed/chair.
  - » Remain 'hands off' the baby at this stage.
  - » If the baby's body is out but the head remains in the vagina, allow the baby to 'hang'.
  - » If the baby is not born within the next five minutes, or the baby has poor tone, move to the next step.

# Birth Emergency Action Lists



- ✓ **Intervene to assist birth.**
  - » Remove the legs one at a time, followed by the arms one at a time, if these are not already out.
  - » If the baby is not born within the next two minutes, or the baby has poor tone, move to the next step.
  
- ✓ **Deliver the head using head flexion and shoulder traction.**
  - » Place one arm under the baby with the limbs hanging on either side.
  - » Place the first and second fingers of this hand into the vagina and place them on either side of the cheek bones.
  - » Place your other arm over the baby and place your middle finger against the occiput with your other fingers over the shoulders.
  - » Have an assistant provide firm suprapubic pressure.
  - » Flex the baby's head forward and raise the baby by pulling on the shoulders, so that the face is delivered first.
  - » If the baby is not born within the next two minutes, move to the next step.

# Birth Emergency Action Lists



- ✓ **Commence urgent transport to the nearest birthing unit, obstetric unit, or ED, providing early notification of arrival.**
  - » Position the mother tilted to the left with pillows/blankets under the pelvis and the stretcher head down (if possible) so that their head is below their pelvis.
  - » Seek clinical advice, putting the phone on speaker phone.
- ✓ **Prepare for newborn resuscitation.**

## Prolapsed umbilical cord

This is when the umbilical cord appears in the vagina ahead of the baby. It is potentially life-threatening for the baby.

Immediately take the following sequential actions:

- ✓ **Call for help and backup.**
  - » Call for help from an LMC if one is immediately available.
  - » Call for backup from an ICP/CCP.
- ✓ **Instruct the patient not to push.**
- ✓ **Commence urgent transport.**
  - » Commence urgent transport to the nearest birthing or obstetric unit, providing early notification of arrival.
  - » Position the mother tilted to the left with pillows/blankets under the pelvis and the stretcher head down (if possible) so that their head is below their pelvis.
  - » Seek clinical advice, putting the phone on speaker phone.

# Birth Emergency Action Lists

- ✓ **Prepare for newborn resuscitation.**
- ✓ **Allow birth to occur if this becomes imminent.**
  - » If birth becomes imminent, pull to the side of the road and support the patient to adopt the position they prefer. Move to the ['birth'](#) guideline if the baby is born.

## Retained placenta

This is when the placenta has not been delivered within 60 minutes of birth.

Take the following sequential actions:

- ✓ **Call for help and backup.**
  - » Call for help from an LMC if one is immediately available.
  - » Call for backup from a Paramedic if not already on scene.
- ✓ **Gain IV access.**
- ✓ **Commence urgent transport.**
  - » Commence urgent transport to the nearest obstetric unit, providing early notification of arrival.
- ✓ **Monitor closely for signs of PPH.**
  - » Move to the ['postpartum haemorrhage'](#) guideline if appropriate.

# Postpartum Haemorrhage

Use this guideline for patients with vaginal bleeding estimated to be more than 500 ml, occurring within six weeks of birth.

- Examine the perineum externally and place pressure on any compressible site of bleeding.
- Seek help from an LMC if feasible, provided this does not delay transport.
- Massage the uterus firmly for approximately 10 minutes. Uterine massage is not a priority if there are insufficient people present to perform it and is not a priority during extrication and transport.
- Administer oxytocin:
  - 10 units IM if IV access cannot be easily gained, or
  - Place 10 units in one litre of 0.9% sodium chloride. Administer IV over approximately 60 minutes (approximately 4 drops per second). This rate of administration can be increased if shock is present.
- Administer 1 g of tranexamic acid IV.
- Commence transport as soon as possible.
- Administer additional 0.9% sodium chloride IV (using a fluid warming device if available) if the patient has signs of hypovolaemia:
  - 500 ml IV.
  - Repeat as required, minimising the total volume.
- If the placenta has not delivered and an LMC is not present, seek urgent clinical advice regarding controlled cord traction to help deliver the placenta.
- If shock and/or bleeding is severe:
  - Perform bimanual compression of the uterus. Consider dissociation with ketamine.
  - Provide active external warming.
  - Call for blood, provided there is an established protocol for this in the area, particularly if extrication will be delayed and/or transport will be prolonged.
- If shock is severe and unresponsive to treatment, or appears disproportionate to the severity of blood loss, seek clinical advice regarding the possibility of administering vasopressor support.

## Backup

- Have a low threshold for requesting backup from an ICP/CCP.
- Backup must not delay transport to an appropriate hospital.

# Postpartum Haemorrhage

## Referral and transport

- The patient must be transported to hospital by ambulance, preferably a hospital with obstetric facilities whenever this is feasible and safe.
- Provide early notification.
- Notify the LMC en route to hospital whenever feasible.
- The patient may be transported directly to an obstetric unit provided a direct phone call has been placed to clinical staff within the unit, and the patient has been accepted.

## Hospitals with obstetric facilities

### Northland and Auckland

- Whangarei Hospital.
- North Shore Hospital.
- Waitakere Hospital.
- Auckland City Hospital.
- Middlemore Hospital.

### Midland

- Waikato Hospital.
- Rotorua Hospital.
- Tauranga Hospital.
- Gisborne Hospital.
- Taranaki Base Hospital.

### Central/Lower North Island

- Hawke's Bay Hospital.
- Whanganui Hospital.
- Palmerston North Hospital.

# Postpartum Haemorrhage

- Wairarapa Hospital.
- Hutt Hospital.
- Wellington Hospital.

## South Island

- Nelson Hospital.
- Wairau Hospital.
- Te Nikau Grey Hospital.
- Christchurch Hospital.
- Timaru Hospital.
- Dunedin Hospital.
- Southland Hospital.

## Additional information

### General principles

- This guideline is focused primarily on the treatment of primary postpartum haemorrhage occurring within 24 hours of birth. However, the principles of treating secondary postpartum haemorrhage occurring between 24 hours and six weeks after birth, in the out-of-hospital setting, are very similar.
- Postpartum haemorrhage is most commonly due to uterine atony and/or retained products.
- It is difficult to estimate the volume of blood loss in PPH. Blood loss is usually spread over several locations (for example, in the bed or in the toilet), may be mixed with amniotic fluid and may be concealed (for example intra-uterine or intra-abdominal bleeding).
- The most important aspects of out-of-hospital treatment are to rapidly transport to an appropriate hospital, stop compressible bleeding, administer TXA and oxytocin, and provide most treatments en route whenever feasible.
- If oxytocin has already been administered after birth, an additional 10 units of oxytocin should be administered in the setting of PPH.

# Postpartum Haemorrhage

## Administration of IV fluid

- Consider gaining IV access in two sites, but do not delay commencing transport to achieve this.
- During pregnancy the blood volume becomes expanded. Particularly after 28 weeks gestation, more than one litre of blood may be lost before the development of obvious signs of shock. Defining when shock is severe requires clinical judgement, but if a pregnant (or very recently pregnant) woman is showing clear signs of shock, then the shock is severe.
- 0.9% sodium chloride IV should be titrated to clinical signs of intravascular volume and perfusion.
- Large volumes of crystalloid may worsen bleeding by contributing to coagulopathy, particularly in the context of severe and uncontrolled bleeding:
  - This is why the total volume of 0.9% sodium chloride should be minimised.
  - Clinical judgment is required when determining the volume administered, but it should be uncommon to exceed 1 litre and it should be rare to exceed 2 litres.
  - Have a low threshold for seeking clinical advice if shock is severe and transport time to the most appropriate hospital is prolonged.
- Ambient temperature IV fluid can cause hypothermia which worsens bleeding by contributing to coagulopathy. IV fluid should be warmed using a dedicated warming device if one is available.
- Do not warm IV fluid in microwaves because this can result in unreliable fluid temperatures and/or may damage the bag.

## Blood

- Requests for blood to be delivered to the scene must not compromise rapid transport to the most appropriate hospital.
- Blood should only be requested if:
  - Shock is severe, and
  - There is an established protocol in the area for blood to be delivered to the scene, and
  - Blood can be administered 15 or more minutes before the patient can arrive in the most appropriate hospital.
- Blood may be available via staging at a small hospital/medical facility en route to the most appropriate hospital. In this setting personnel must have a low threshold for seeking clinical advice.
- ICPs/CCPs may continue blood administration initiated by a medical practitioner following their verbal instructions but should have a low threshold for seeking clinical advice.



# Postpartum Haemorrhage

- Requests for blood to be delivered to the scene outside of an established protocol must be made via the on-call doctor for the service. In this setting:
  - It will usually take approximately 15 minutes to arrange for blood to be prepared, plus transport time to the scene. This means that for such a request to be appropriate, the patient should usually have severe shock and a prolonged transport time.
  - Personnel cannot administer blood without specific instructions from the on-call doctor for the service.

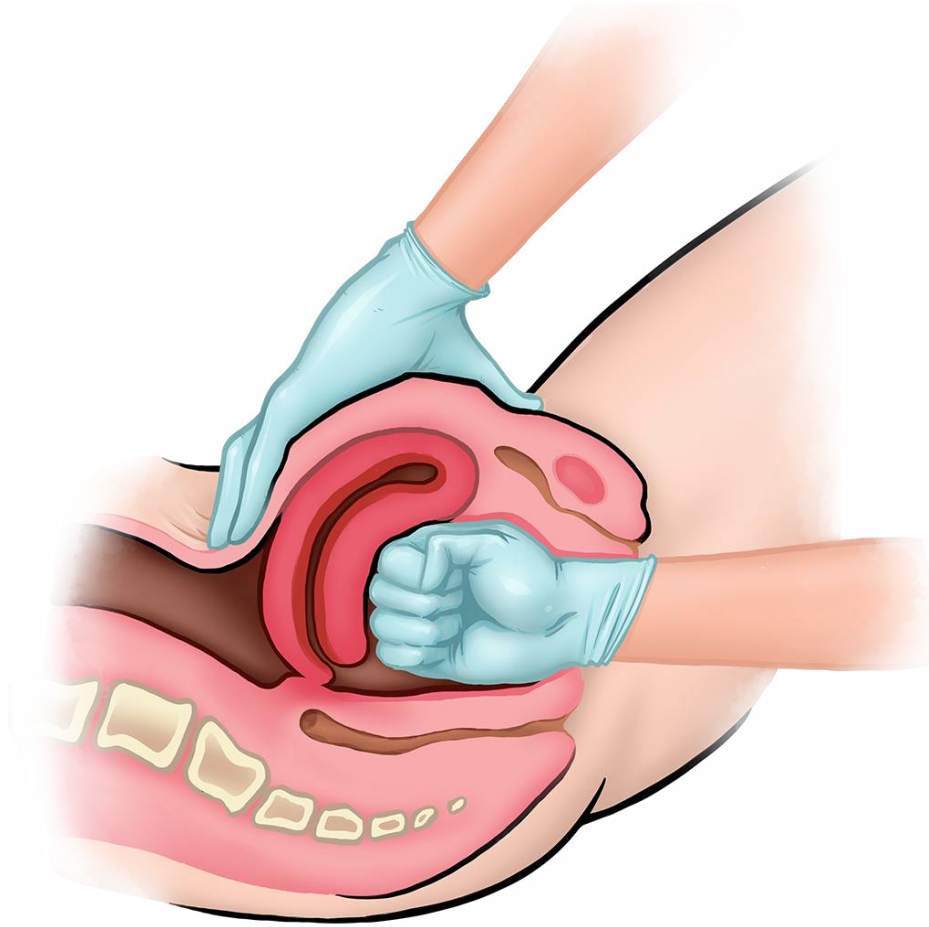
## Patient temperature

- It is important to keep the patient warm because hypothermia worsens bleeding by contributing to coagulopathy.
- Passive warming (for example with external blankets) is rarely effective, and the patient should be actively warmed.
- It is important to document the patient's temperature.

## Bimanual compression of the uterus

- This may be lifesaving and must not be delayed if shock and/or bleeding is severe.
- To perform bimanual compression of the uterus:
  - Explain what you are going to do and why.
  - Place one hand in the vagina as far as you can and form a fist.
  - Push upward with this hand toward the umbilicus.
  - Place your other hand on the abdomen, feel for the uterus and push both hands firmly toward each other.

# Postpartum Haemorrhage



**Bimanual compression of the uterus**

## Amniotic fluid embolism

- Amniotic fluid embolism is rare. It occurs when amniotic fluid enters the maternal circulation, causing a severe inflammatory response that has a high mortality rate.
- Signs include tachypnoea, hypoxia and shock.
- If shock is disproportionate to blood loss and unresponsive to treatment, amniotic fluid embolism may be present. IV adrenaline by infusion may have a role in treating amniotic fluid embolism and personnel should seek clinical advice if this is being considered.

## Response and/or transport by helicopter

- Transport or response by helicopter may be indicated if the patient has PPH with signs of shock and:
  - The helicopter will arrive at scene at least 15 minutes sooner than a road-based CCP, or
  - Transport by helicopter will save more than 15 minutes compared to road transport. For criteria to be met, the road transport time will usually need to be more than 75 minutes.

# Postpartum Haemorrhage

## More Information

## Version history

9 June 2025

1.0.5.3

**Update:** Guideline redeveloped.

This checklist is to be used prior to transport (including from birthing units) if an LMC is present.

- ✓ Confirm roles.
  - ✓ Confirm the treatment and monitoring plan for the baby. If the reason for transport is concern about the baby, the minimum monitoring required is pulse oximetry.
  - ✓ Confirm the treatment and monitoring plan for the mother. If the reason for transport is concern about the mother, the minimum monitoring required is pulse oximetry and NIBP.
  - ✓ Confirm the restraint requirement/plan for the baby.
  - ✓ Confirm heaters are on if appropriate.
  - ✓ Confirm if an ambulance crew member is required in the back during transport.
-

# Airway Backup Checklist

This checklist should be used to prepare a patient for RSI while awaiting CCP backup.

- ✓ Pre-oxygenate by mask.
  - ✓ Attach monitor, including ETCO<sub>2</sub>.
  - ✓ Gain IV access, preferably in two sites.
  - ✓ Prepare a running line of 0.9% sodium chloride.
  - ✓ Position the patient for optimal airway control.
  - ✓ Place an ETT holder under the patient's head.
  - ✓ Ensure suction is working and turn it off.
  - ✓ Prepare a manual ventilation bag with PEEP valve.
  - ✓ Maximise space and clear away unnecessary equipment.
  - ✓ Consider travelling toward backup.
-

# RSI (Rapid Sequence Intubation)

- RSI is indicated for a patient with a GCS less than or equal to 10, and
  - Clinically significant compromise of airway or ventilation, or
  - A clinical trajectory requiring RSI.
- Use the PODCASTS acronym to consider factors contributing to the balance of risk prior to proceeding with RSI.
  - P** Predicted difficulty of intubation.
  - O** Oxygenation and ventilation (severity of impairment).
  - D** Distance (time) to hospital.
  - C** Comorbidities that limit life expectancy.
  - A** Age, particularly those aged less than 5 years, or greater than 75 years.
  - S** Shock.
  - T** Trajectory of clinical course.
  - S** Special circumstances.

## RSI procedure

- Ensure adequate preparation and resuscitation before commencing RSI.
- Utilise the RSI checklist.
- Administer 2 mcg/kg of fentanyl IV (up to a maximum of 200 mcg) immediately prior to ketamine if the patient has:
  - Traumatic brain injury, or
  - Suspected intracranial pathology, or
  - Hypertension (SBP > 150 mmHg in an adult).
  - Withhold fentanyl if the above clinical indications are not present, or there are signs of shock.
- Administer 2 mg/kg of ketamine IV (up to a maximum of 200 mg) for most patients.
  - Consider reducing the dose to 1 mg/kg if:
    - There are signs of shock, or
    - Sedative medicines have already been administered or taken.

# RSI (Rapid Sequence Intubation)

- Administer neuromuscular blockade:
  - 2 mg/kg of rocuronium IV, or
  - 2 mg/kg of suxamethonium IV.
- Intubate and immediately confirm ETT position with waveform capnography over the first seven breaths.
- Utilise the post intubation checklist.
- Administer ongoing sedation, analgesia, neuromuscular blockade and vasopressor using the '[post intubation](#)' guideline.

## Adult dose summary table

|                            | 50 kg   | 60 kg   | 70 kg  | 80 kg  | 90 kg  | ≥ 100 kg |
|----------------------------|---------|---------|--------|--------|--------|----------|
| <b>Fentanyl (2 mcg/kg)</b> | 100 mcg | 120 mcg | 140mcg | 160mcg | 180mcg | 200mcg   |
| <b>Ketamine (1 mg/kg)</b>  | 50 mg   | 60 mg   | 70 mg  | 80 mg  | 90 mg  | 100 mg   |
| <b>Ketamine (2 mg/kg)</b>  | 100 mg  | 120 mg  | 140 mg | 160 mg | 180 mg | 200 mg   |
| <b>Rocuronium</b>          | 100 mg  | 100 mg  | 150 mg | 150 mg | 200 mg | 200 mg   |
| <b>Suxamethonium</b>       | 100 mg  | 100 mg  | 150 mg | 150 mg | 200 mg | 200 mg   |

## Paediatric dose summary table

|                            | 10 kg  | 20 kg  | 30 kg  | 40 kg  |
|----------------------------|--------|--------|--------|--------|
| <b>Fentanyl (2 mcg/kg)</b> | 20 mcg | 40 mcg | 60 mcg | 80 mcg |
| <b>Ketamine (1 mg/kg)</b>  | 10 mg  | 20 mg  | 30 mg  | 40 mg  |
| <b>Ketamine (2 mg/kg)</b>  | 20 mg  | 40 mg  | 60 mg  | 80 mg  |
| <b>Rocuronium</b>          | 20 mg  | 40 mg  | 60 mg  | 80 mg  |
| <b>Suxamethonium</b>       | 20 mg  | 40 mg  | 60 mg  | 80 mg  |

## Additional information

# RSI (Rapid Sequence Intubation)

## General principles

- The primary goal of performing RSI is to improve patient outcomes by securing the airway, treating hypoxia and controlling ventilation to prevent hypercarbia. This is particularly important in patients who are unconscious post cardiac arrest or have severe TBI, because secondary brain injury worsens long-term outcomes.
- Personnel calling for backup from a CCP must consider how long it will take for backup to arrive. To be appropriately utilised, such backup must arrive at least 15 minutes faster than the patient can be transported to an appropriate hospital.
- Including the clinical trajectory of the patient in the decision to perform RSI requires good clinical judgment. A clinical trajectory requiring RSI requires that the expectation is that the patient is likely to be admitted to an ICU ventilated, for example because the patient has severe TBI or is post cardiac arrest. A clinical trajectory that the patient may require intubation and ventilation for a procedure, such as surgery or a CT scan, is not usually considered an indication for RSI.
- Whenever feasible, an ICP or second CCP should be utilised as a dedicated assistant.
- If hypoxia is severe or the patient is at high risk of desaturation, consider oxygen via nasal prongs at 15 litres/minute during laryngoscopy.

## PODCASTS acronym

- The decision to perform RSI requires clinical judgement that considers all factors contributing to the balance of risk for each patient. Use of the PODCASTS mnemonic provides a structure to help ensure key factors are considered. Personnel must have a low threshold for seeking clinical advice if unsure.
- P** Predicted difficulty of intubation. If the intubation is predicted to be difficult, the balance of risk usually shifts toward the RSI occurring in hospital.
  - O** Oxygenation and ventilation (severity of impairment). If there is severe impairment of oxygenation or ventilation (particularly oxygenation) that cannot be managed with basic airway manoeuvres and/or adjuncts, the balance of risk usually shifts toward RSI occurring out of hospital.
  - D** Distance (time) to hospital. If the time to the most appropriate hospital, including extrication time, is prolonged, the balance of risks usually shifts toward RSI occurring out of hospital. There is no time 'cut off', but it should be unusual to perform RSI within 15 minutes of the most appropriate hospital.
  - C** Comorbidities that limit life expectancy. The presence of life limiting comorbidities usually shifts the balance of risk toward RSI occurring in hospital, because of the implications for ongoing treatment such as admission to an intensive care unit.
  - A** Age. For those aged less than 5 years or greater than 75 years, the balance of risk usually shifts



# RSI (Rapid Sequence Intubation)

toward RSI occurring in hospital.

- S** Shock. The presence of shock, particularly hypovolaemic shock, usually shifts the balance of risk toward RSI occurring in hospital. Positive pressure ventilation will further reduce venous return to the heart, and concurrent resuscitation of the patient's circulation, for example with blood, is usually required.
- T** Trajectory of clinical course. If the patient has an expected clinical trajectory of not improving rapidly, for example ROSC following prolonged cardiac arrest, the balance of risk usually shifts toward RSI occurring out of hospital. Conversely, a trajectory of expected improvement such as GHB poisoning usually shifts the balance of risk toward RSI occurring in hospital.
- S** Special circumstances. This includes all additional factors that alter the balance of risk. For example, scene factors such as availability of personnel to assist with the RSI, and specific patient factors such as morbid obesity and pregnancy.

## Choice and dosing of medicines

- Fentanyl should usually be administered immediately prior to ketamine when performing RSI for patients with traumatic brain injury, suspected intracranial pathology (for example haemorrhagic stroke) or hypertension (a SBP greater than 150 mmHg in an adult or a SBP significantly higher than normal in a child).
  - » This is because fentanyl may reduce the rise in BP and/or intracranial pressure, that may occur with laryngoscopy and intubation.
  - » 2 mcg/kg (to a maximum of 200 mcg) is an appropriate dose for most patients.
  - » However, fentanyl may contribute to hypotension. For this reason, fentanyl should be withheld if there are signs of shock.
- Ketamine is always administered during RSI.
  - » 2 mg/kg (to a maximum of 200 mg) is an appropriate dose for most patients.
  - » Although ketamine usually produces a small rise in BP, ketamine may cause a fall in BP if shock is present. For this reason, reduce the dose to 1 mg/kg if there are signs of shock.
  - » Consider reducing the dose to 1 mg/kg if sedative medicines have been administered, or taken. For example, consider reducing the dose to 1 mg/kg if:
    - » A sedative dose of ketamine has been administered as part of a delayed sequence intubation, or
    - » The patient has taken sedative medicines.
  - » Do not reduce the dose of ketamine in the setting of RSI for prolonged seizures. Even though midazolam will have been administered, an anticonvulsant dose of ketamine is required.
- Neuromuscular blockade is always administered during RSI.
  - » 2 mg/kg of rocuronium or suxamethonium (to a maximum of 200 mg) is an appropriate dose for all patients.

# RSI (Rapid Sequence Intubation)

- » The dose should usually be rounded to the nearest 50 mg in adults.

## Delayed sequence intubation

- Delayed sequence intubation is a term used to describe the administration of sedation prior to embarking on the RSI procedure.
- Delayed sequence intubation is indicated if the patient meets criteria for RSI, but is agitated and/or combative to a level that is preventing safe preparation prior to RSI:
  - » Administer sufficient ketamine (usually 1 mg/kg IV) to gain control of the agitation.
  - » Once control is gained, ensure adequate pre-oxygenation and complete the RSI checklist.
  - » Perform RSI and consider reducing the dose of ketamine, considering the dose of medicines already administered.

## Crash RSI

- Rarely, a patient may be deteriorating so rapidly that there is a high risk of death occurring prior to performing RSI, and a truncated (or 'crash') approach should be undertaken.
- Task one person to provide as much as pre-oxygenation as possible.
- Utilise the crash RSI checklist.
- Omit fentanyl and administer ketamine and neuromuscular blockade.

## Intubation without medicines

- For the majority of unconscious patients not in cardiac arrest, the risks of intubation without medicines outweigh the benefits.
- Intubation and ventilation without medicines must only occur if the patient has a GCS of 3 and ineffective breathing.

## Measurement of end tidal CO<sub>2</sub> (ETCO<sub>2</sub>)

- Intubation must not be attempted unless measurement of ETCO<sub>2</sub> via capnography is immediately available.
- Confirmation of correct placement of the ETT via capnography over the first seven breaths must occur as the first step following placement of an ETT. If the ETT is within the oesophagus and the stomach

# RSI (Rapid Sequence Intubation)

contains CO<sub>2</sub> (for example from drinks containing CO<sub>2</sub>, or following mouth to mouth ventilation), ETCO<sub>2</sub> may be detectable for the first 2-4 breaths and then fall rapidly below 7 mmHg.

- Continuous measurement of the ETCO<sub>2</sub> via capnography is compulsory for all patients intubated with an ETT, including those in cardiac arrest.
- The ETT must be removed if the ETCO<sub>2</sub> is below 7 mmHg (including during cardiac arrest), even if it is thought that the low ETCO<sub>2</sub> is due to technical error. This is because the risk of unrecognised oesophageal intubation is too high, even when the ETT is thought clinically to be within the trachea.

## More Information

### Version history

28 October 2024

1.0.5

**Update:** Multiple changes.

# RSI Checklist

- ✓ Roles assigned:
  - » Team leader.
  - » Airway.
  - » Airway assistant.
  - » Drugs and monitoring.
- ✓ Patient prepared:
  - » Check oxygen supply and pre-oxygenate.
  - » Consider nasal prongs.
  - » Position optimised.
  - » IV access patent. Running line attached.
  - » Consider marking surgical airway landmarks.
- ✓ Monitoring attached and visible:
  - » Baseline vital signs.
  - » Pulse oximetry, capnography, and cycling BP.
- ✓ Equipment checked and ready:
  - » PPE.
  - » Manual ventilation bag with PEEP valve set to 5 cmH<sub>2</sub>O.
  - » OPA, SGA, and surgical airway equipment.
  - » Laryngoscope.
  - » ETT and 10 ml cuff syringe.
  - » ETT holder in place.
  - » Suction.
  - » Bougie and stylet.
- ✓ Medicines drawn up, labelled and doses confirmed:
  - » Fentanyl (if indicated).
  - » Ketamine.
  - » Neuromuscular blocker.
  - » Post intubation sedation.
  - » Metaraminol.
- ✓ Team briefing:
  - » Airway plan, including failed intubation.

# RSI Checklist

- » Medication plan.
  - » Anticipated complications.
  - » Any concerns before proceeding?
-

# Post Intubation

Use this guideline if the patient has been intubated and ETT placement has been confirmed by capnography. If the patient has been intubated during cardiac arrest, this guideline should only be used if sustained ROSC occurs.

- Ventilate using the '[mechanical ventilation](#)' guideline.
- Administer fentanyl and ketamine:
  - Draw up 200 mcg of fentanyl in combination with 200 mg of ketamine. Dilute with 0.9% sodium chloride to a total of 20 ml. This solution contains 10 mcg/ml of fentanyl and 10 mg/ml of ketamine.
  - Commence an infusion at 0.2 ml/kg/hr using an infusion device, or
  - For an adult administer 5 ml of this solution IV (50 mcg of fentanyl and 50 mg of ketamine) every 15-30 minutes, titrating the frequency to the clinical signs of the patient's level of sedation.
  - For a child administer the volume described in the paediatric drug dose tables IV every 15-30 minutes, titrating the frequency to the clinical signs of the patient's level of sedation.
- Administer rocuronium as required:
  - 100 mg IV for an adult weighing greater than 80 kg.
  - 50 mg IV for an adult weighing 50-80 kg.
  - See the paediatric drug dose tables for a child.
- Administer 0.9% sodium chloride IV if there are signs of hypovolaemia or the patient is hypotensive:
  - 500 ml for an adult.
  - 10 ml/kg for a child.
  - Repeat as required.
- Administer metaraminol IV in addition to 0.9% sodium chloride, if the patient remains hypotensive.
 

**Metaraminol for an adult:**

  - Administer via an infusion device:
    - Administer an initial bolus of 0.5-1 mg metaraminol IV.
    - Start the infusion at 2 mg/hour and adjust the rate as required.
  - Alternatively, administer a bolus of 0.5-1 mg metaraminol IV every 5-10 minutes as required.

**Metaraminol for a child:**

  - Administer a bolus of metaraminol IV (see the paediatric drug dose tables) every 5-10 minutes as required.
- Consider administering adrenaline IV if the blood pressure is unresponsive to metaraminol or the patient is hypotensive and bradycardic.
 

**Adrenaline for an adult:**

# Post Intubation

- » Administer an adrenaline infusion via an infusion device. Start at a rate of 0.5 mg/hour and adjust the rate as required, or
- » Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
  - » Administer as an infusion. Start at 2 drops/second and adjust the rate as required, or
  - » Administer a bolus of 0.01 mg adrenaline (10 ml) IV every 1-2 minutes as required.

## Adrenaline for a child aged 5-14 years:

- » Administer an adrenaline infusion via an infusion device. Start at a rate of 0.25 mg/hour and adjust the rate as required, or
- » Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
  - » Administer as an infusion. Start at 1 drop/second and adjust the rate as required, or
  - » Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.

## Adrenaline for a child aged less than five years:

- » Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.
- » Do not administer adrenaline as an IV infusion.

## Additional information

### Sedation and neuromuscular blockade

- Provided the patient is not in cardiac arrest, all patients that have been intubated should receive sedation (preferably via an infusion device) and neuromuscular blockade.
- Consider administering sedation without neuromuscular blockade, provided this is safe, if it is considered likely that the patient will receive end of life care within the emergency department.
- When administering sedation:
  - » Use doses at the lower end of the described ranges if the patient is small, frail, or cardiovascularly unstable.
  - » Titrate the dose and frequency of administration to the clinical signs of the patient's level of sedation. Increase the dose and frequency of administration if the patient has hypertension and tachycardia or shows signs of tears. Decrease the dose and frequency of administration if the patient has hypotension.

### Ensuring a safe ETT cuff seal

- It is very easy to generate a high pressure within the ETT cuff and this may cause tracheal injury by impairing the blood supply to the tracheal mucosa. Tracheal injury may lead to tracheal stenosis and this may cause significant morbidity.

# Post Intubation

➤ As soon as feasible, the cuff should be checked to ensure that the cuff pressure is between 20-30 cmH<sub>2</sub>O.

## More Information

### Version history

|                                  |       |
|----------------------------------|-------|
| 28 October 2024                  | 1.0.5 |
| <b>Update:</b> Multiple changes. |       |



# Mechanical Ventilation

## Commencing mechanical ventilation action list

- Connect an oxygen source to the ventilator and check the oxygen supply.
- Connect and test the ventilator circuit/tubing.
- Select the initial ventilation settings as per the mechanical ventilation table (see below).
- Connect the circuit/filter/tubing to the patient and commence ventilation.
- Check ventilation, oxygenation, and continuous ETCO<sub>2</sub>.
- Ensure the tubing is secured and check for air leaks.
- Continue monitoring vital signs.
- Consider elevating the head of the stretcher if appropriate.
- Plan for ongoing sedation and neuromuscular blockade.
- Re-evaluate and adjust settings as required.

## Initial recommended settings table

|                                          | Normal lung                                                          | Obstructive lung<br>(e.g. Asthma, COPD) | Restrictive lung<br>(e.g. Aspiration)        |
|------------------------------------------|----------------------------------------------------------------------|-----------------------------------------|----------------------------------------------|
| <b>Ventilation mode</b>                  | SIMV/volume controlled                                               |                                         |                                              |
| <b>Tidal volume (Vt)</b>                 | 6-8 ml/kg (IBW)                                                      | 6-8 ml/kg (IBW)                         | 5-6 ml/kg (IBW)                              |
| <b>Ventilation frequency<br/>(adult)</b> | 12-15 /min                                                           | 6-8 /min                                | 18-25 /min                                   |
| <b>PEEP</b>                              | 5 cmH <sub>2</sub> O                                                 | 5 cmH <sub>2</sub> O                    | 10 cmH <sub>2</sub> O                        |
| <b>I:E ratio</b>                         | 1 : 2                                                                | 1 : 6                                   | 1 : 1                                        |
| <b>Relief airway pressure<br/>(pMax)</b> | 30-40 cmH <sub>2</sub> O                                             | 40-60 cmH <sub>2</sub> O                | 40-60 cmH <sub>2</sub> O                     |
| <b>FiO<sub>2</sub></b>                   | 1.0 (100% oxygen) and titrate level to an SpO <sub>2</sub> of 94-98% |                                         |                                              |
| <b>ETCO<sub>2</sub> target</b>           | 35-45 mmHg                                                           | Accept hypercapnia                      | 35-45 mmHg<br>May need to accept hypercapnia |

# Mechanical Ventilation

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Special considerations</b> | <p><b>Raised intracranial pressure:</b></p> <ul style="list-style-type: none"> <li>• Mild hyperventilation to <math>\text{ETCO}_2</math> of 30-35 mmHg.</li> <li>• Restrict PEEP to 5 <math>\text{cmH}_2\text{O}</math>.</li> <li>• Avoid hypoxaemia and hypotension.</li> </ul> <p><b>Severe TBI and other major trauma (combined chest and brain):</b></p> <ul style="list-style-type: none"> <li>• Preservation of perfusion/BP outweighs achieving <math>\text{ETCO}_2</math> targets.</li> </ul> <p><b>Bronchospasm:</b></p> <ul style="list-style-type: none"> <li>• Monitor for dynamic hyperinflation and tension pneumothorax.</li> <li>• Consider using manual ventilation.</li> </ul> <p><b>Severe acidosis:</b></p> <ul style="list-style-type: none"> <li>• These patients will require a higher minute volume to compensate for their acidosis.</li> <li>• Seek clinical advice.</li> </ul> <p><b>Paediatric ventilation:</b></p> <ul style="list-style-type: none"> <li>• See 'mechanical ventilation in children' in additional information.</li> <li>• Seek clinical advice if the child is aged less than 5 years.</li> </ul> |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Additional information

### General principles

- There are various types of mechanical ventilators on air and ground ambulances in New Zealand with substantial differences in features and capabilities.
- This guideline does not replace the training required to use a specific ventilator but provides a general overview, commonalities, and complications of mechanical ventilation.
- Mechanical ventilation is consistent, precise, and results in less lung injury even when compared to short-term manual ventilation.
- All intubated patients not in cardiac arrest should be ventilated via a mechanical ventilator unless there is a specific reason not to do so. Seek clinical advice if ventilating a child aged less than five years.

# Mechanical Ventilation

## Ventilation strategies

- There are three ventilation strategies outlined in the mechanical ventilation table:
  - **Normal lung:** For patients with normal lung physiology.
  - **Obstructive lung:** For patients with obstructive lung physiology, for example, acute bronchospasm, COPD, asthma or bronchiolitis.
  - **Restrictive lung:** For patients with restrictive or stiff lungs, for example, pneumonia, COVID, aspiration, ARDS, or obesity.

## Ventilation modes

- There are multiple ventilation modes with different names and abbreviations used to describe them. They can be categorised as:
  - **Volume controlled:** The delivered tidal volume is set and the peak airway pressure is dependent on the compliance of the lung, pleura and chest wall.
  - **Pressure controlled:** The airway pressure is set and the delivered tidal volume is dependent on the compliance of the lung, pleura and chest wall.
  - **Synchronised:** The patient can breathe spontaneously between mandatory ventilator-delivered breaths. This reduces ventilator/patient dyssynchrony during weaning of neuromuscular blocking medications.
  - **Non-synchronised (controlled or mandatory):** The patient is unable to take spontaneous breaths between ventilator-delivered breaths.
- A volume controlled mode is recommended for most patients.

## Tidal volume

- Tidal volume should usually be proportional to the patient's height rather than weight, as height correlates best with lung volume. For example, a tall patient should have a higher tidal volume than a short patient with the same weight. If using weight rather than height ideal body weight (IBW) should be used instead of total body weight.
- Some models of ventilator will automatically calculate IBW when inputting the height and sex of the patient.

## PEEP

- PEEP increases functional residual capacity, improves oxygenation and assists small and medium-sized airways to remain open during expiration.
- PEEP reduces venous return to the heart and should be used with caution in patients with signs of shock.

## Inspiratory expiratory ratio

# Mechanical Ventilation

- The proportion of time that each ventilation cycle is in the inspiratory phase compared with the expiratory phase is called the inspiratory expiratory ratio (I:E ratio).
- This may not be adjustable on all ventilators.

## Relief airway pressure

- This is often referred to as the maximum pressure or P max and is the airway pressure that the ventilator will not exceed.
- If the peak airway pressure reaches the relief airway pressure, the ventilator will stop delivering positive pressure and alarm.

## Fraction of inspired oxygen

- The fraction of inspired oxygen (FiO<sub>2</sub>) is the percentage of inspired oxygen expressed as a fraction. For example, 60% oxygen is an FiO<sub>2</sub> of 0.6.
- Provided pulse oximetry is reliable FiO<sub>2</sub> should be adjusted to maintain an SpO<sub>2</sub> of 94-98%. This reduces the risk of harm from high oxygen concentrations at a tissue level and conserves the oxygen supply.

## ETCO<sub>2</sub>

- ETCO<sub>2</sub> is inversely proportional to ventilation when cardiac output is near normal.
- The ETCO<sub>2</sub> will usually be above the target range immediately following intubation. Aim to reach the target over 5-10 minutes because rapid reductions in ETCO<sub>2</sub> may cause cerebral vasoconstriction.
- A patient with life-threatening bronchospasm will usually have a very high ETCO<sub>2</sub>. In this setting targeting normocapnia risks causing a reduced cardiac output from dynamic hyperinflation (or air trapping).
- In some patients (particularly those with clinically important aspiration) the ETCO<sub>2</sub> will remain high despite increasing ventilation. If the ETCO<sub>2</sub> remains high despite a ventilation rate of 20-30 breaths/minute, do not increase the ventilation any further and allow the ETCO<sub>2</sub> to remain high.

## Mechanical ventilation in children

- Seek clinical advice prior to commencing mechanical ventilation if the child is aged less than 5 years.
- Utilise the pre-set ventilator settings based on patient height, or set an initial ventilation rate (VR) of:
  - 2.5- 5 kg (newborn): VR 40/min
  - 5 kg (less than 1 year): VR 40/min
  - 10 kg (1 year): VR 25/min
  - 20 kg (2-5 years): VR 25/min

# Mechanical Ventilation

- » 30 kg (6-10 years): VR 20/min
- » 40 kg (11-13 years): VR 15/min
- Do not exceed a ventilation rate of > 40/min without seeking clinical advice.
- Use tidal volumes of 6-8 ml/kg.
- Dead space can impair ventilation. Use paediatric equipment if available.
- PEEP is usually required.
- Target an ETCO<sub>2</sub> of 35-45 mmHg and an SpO<sub>2</sub> of 94-98%.
- Regularly confirm tube position, bilateral chest rise and air entry, particularly after each patient movement.

## Troubleshooting

- If at any time troubleshooting does not quickly resolve the problem, disconnect the patient from the mechanical ventilator and commence manual ventilation.
- Use the action cards to troubleshoot the following problems:
  - » Hypoxia
  - » Hypotension
  - » High airway pressures
  - » Falling or absent ETCO<sub>2</sub>
  - » Rising ETCO<sub>2</sub>
- The **DOPES** acronym is another systematic approach to troubleshooting.
- D** **Displacement** of the ETT. Consider the possibility of inadvertent extubation, oesophageal placement or endobronchial intubation.
  - » Check the ETCO<sub>2</sub> level and waveform.
  - » Examine the chest for endobronchial intubation.
  - » Adjust the ETT length at the lips if required.
- O** **Obstruction** of the ETT or circuit/tubing. Consider the possibility the ETT is obstructed (for example by secretions or kinking) or the circuit is obstructed.

# Mechanical Ventilation

- » Check the filter is not obstructed with fluid or secretions.
- » Check the ETT is not kinked.
- » Check the patient is not biting the ETT.
- » Check the circuit/tubing is not kinked.
- » Check the ETT is not blocked by passing a suction catheter or bougie down the ETT.

## **P** **Pneumothorax.** Consider the possibility of pneumothorax.

- » Examine the chest.
- » Check for high peak inspiratory pressures.
- » Decompress the chest if tension pneumothorax is suspected, preferably by finger thoracostomy.
- » Consider other causes of abnormal lung function, for example pulmonary oedema, bronchospasm and aspiration.

## **E** **Equipment.** Consider the possibility of an equipment problem.

- » Check the cuff inflation.
- » Check the ventilator and the settings.
- » Check the circuit/tubing, ensuring all connections are secure.
- » Check the oxygen and power supply.

## **S** **Stacked breaths.** Consider the possibility of dynamic hyperinflation (air trapping). Disconnect the ventilator for 20-30 seconds to allow full expiration.

### More Information

#### Version history

28 October 2024

1.0.5

**Update:** Guideline redeveloped.

# Mental Health Conditions

## Introduction

- Mental health encompasses a wide range of conditions, and it is not the role of ambulance personnel to diagnose patients in this setting. It is however, important to have an understanding of the characteristics of mental health conditions that are commonly encountered in the out-of-hospital setting.
- Patients with particular mental health conditions are more commonly encountered in the out-of-hospital setting than others because:
  - There is a high incidence of the condition within the population (for example depression), or
  - The nature of the condition (for example schizophrenia) places the individual at increased risk of illness, injury or an acute deterioration in their mental health condition, requiring urgent assessment and/or treatment from a health professional.

## Assessment, treatment and referral

- The fundamental aspects of assessment and making decisions regarding treatment and/or referral for patients with a mental health condition are the same as for any other patient, with the addition of these important principles:
  - Assess the patient in a quiet and non-threatening environment whenever this is feasible.
  - Always assess and document the patient's mental status.
  - Assess and examine the patient carefully because physical illness (for example sepsis) or injury often co-exist with an acute deterioration in a mental health condition.
  - Involve family members and/or caregivers in assessment and decisions, whenever this is feasible.
  - Patients with a mental health condition are over-represented in mortality rates in several areas. A patient's mental health history and current mental status must be taken into consideration when making transport and/or referral decisions.
  - Seek advice from healthcare personnel that know the patient well, whenever this is feasible.
  - Further assessment and/or treatment from mental health personnel should occur in the community whenever this is feasible, provided it is clear that assessment and/or treatment in an ED is not required for a physical illness or injury.
  - Additional attention to safety is required if the patient has hallucinations, severe agitation or very abnormal behaviour.

## Depression

- Depression is the most commonly diagnosed mental health condition in New Zealand.
- Depression is characterised by signs and symptoms including, but not limited to:
  - Persistent negative mood.

# Mental Health Conditions

- » Decreased satisfaction or pleasure from engaging in activities.
  - » Insomnia or hypersomnia.
  - » Feelings of hopelessness or worthlessness.
  - » Inability to concentrate.
  - » Agitation.
  - » Suicidal ideation.
- Patients who have depression are at an increased risk of suicide, or attempted suicide.

## Anxiety disorders

- Anxiety disorders are common in New Zealand, with generalised anxiety disorder being the most frequently diagnosed.
- Each anxiety disorder has specific diagnostic criteria, but common features include:
  - » A symptom duration of at least six months, and
  - » A disproportionate fear or sympathetic nervous system response relative to the danger posed, and
  - » Maladaptive behaviour to avoid the source of anxiety.

## Bipolar disorder

- Bipolar disorder is a mood disorder, characterised by symptoms of depression that are punctuated by manic episodes.
- Manic episodes are commonly accompanied by signs and symptoms such as grandiosity, insomnia, flight of ideas, inattention, increased psychomotor activity, and excessive engagement in risky or unrestrained behaviours that are likely to have adverse outcomes.
- Patients with bipolar disorder are at increased risk of suicide, or attempting suicide.

## Personality disorders

- Personality disorders are common, with borderline personality disorder being the most frequently diagnosed.
- Personality disorders are usually characterised by unstable moods and emotions, abnormal behaviour and impaired social relationships.
- Patients with a personality disorder are at increased risk of non-suicidal self-harm, suicide, attempted suicide and threatened suicide.



# Mental Health Conditions

- Patients with a personality disorder may be intensely concerned with abandonment, and the perception of this impending loss or rejection can result in significant alterations to the patient's self-image, affect and behaviour.
- It is important to communicate with these patients in an optimistic manner, to encourage coping skills and to establish clear boundaries.

## Schizophrenia and schizoaffective disorder

- Schizophrenia is a psychotic disorder characterised by disturbances in the patient's perceptions, thoughts, behaviour and affect that last for at least six months. Signs and symptoms include, but are not limited to:
  - Hallucinations.
  - Delusions.
  - Disorganised thoughts and speech.
  - Abnormal behaviour.
  - Catatonia (a state of immobility and stupor).
  - Negative symptoms (for example, apathy and decreased emotional expressiveness).
- Schizoaffective disorder is characterised by schizophrenia with the addition of major mood abnormalities, for example depression or mania.
- These disorders are usually accompanied by major social and occupational dysfunction, and ambulance personnel are often requested at a point of acute deterioration.

# Assessing Mental Status

This guideline provides a guide to assessing mental status. The GCS is a very basic assessment of the level of consciousness and many patients with abnormal mental status will have a normal GCS. Abnormal mental status often only becomes obvious when a comprehensive assessment of mental status is undertaken. The BATOMI mnemonic provides a useful structure for an assessment of mental status.

## The BATOMI mnemonic

- B** Behaviour.
- A** Affect, appearance and mood.
- T** Thought and talk.
- O** Orientation.
- M** Memory.
- I** Intellect and insight.

### Behaviour

- Assess for signs of abnormal behaviour, for example:
  - Lack of eye contact.
  - Abnormal body language and/or posture.
  - Abnormal movement, for example restlessness, tremor or repetitive movement (such as lip smacking).

### Affect, appearance and mood

- Assess the patient's affect (observable behaviour that represents the expression of emotion). Examples of abnormal affect include:
  - **Blunted affect:** a significant reduction in the intensity of emotional expression.
  - **Flat affect:** the absence of signs of emotional expression.
  - **Inappropriate/incongruent affect:** the expression of emotion that does not match the content of what is being said or thought.
  - **Labile affect:** abnormal sudden rapid shifts in emotions or emotional expression.
- Assess the patient's appearance. For example, assess if they are dressed appropriately and if they maintain appropriate personal hygiene.

# Assessing Mental Status

- Assess for signs of abnormal mood.

## Thought and talk

- Assess the patient's thoughts and the expression of their thoughts. For example:
  - **Ideation.** In particular, note if the patient's ideas are abnormal and/or may be harmful, for example suicidal ideation.
  - **Confusion.** For example, assess whether the patient comprehends their current situation.
  - **Delusions (fixed and false beliefs).** For example, grandiose delusions, paranoid delusions or persecutory delusions.
- Assess the tone, flow, rate, volume and content of the patient's speech.
- Assess for the presence of hallucinations (false or distorted sensory perceptions).

## Orientation

- Assess the patient's orientation to person, place and time.

## Memory

- Assess the patient's short term memory by asking them to remember three random objects or words. After approximately one minute, ask the patient to recall the objects or words.
- Assess the patient's long-term memory by asking 2-3 questions that are easily verified. For example:
  - What is your address?
  - What political party is in power at the moment?

## Intellect and insight

- Assess the patient's intellect by noting their cognitive ability and comprehension of information.
- Assess the patient's insight by noting whether they can understand their situation and act appropriately.

## Additional information

### General principles

- Document the aspects of mental status on the ePRF, but do not refer directly to BATOMI as a title. This is because most hospital personnel are familiar with the individual aspects of BATOMI, but not the actual

# Assessing Mental Status

title.

## Delusions and hallucinations

- Delusions are defined as false beliefs that remain fixed despite what almost everyone else believes, and despite what usually constitutes incontrovertible evidence to the contrary. Take note of delusions as part of the mental status assessment.
- Hallucinations are defined as sensory perceptions that occur in the absence of external stimulation of the sensory organ. They are most common in patients with schizophrenia or other psychotic disorders.
  - » The majority of hallucinations experienced by patients are auditory (for example hearing voices), but hallucinations can also be visual, tactile (touch), olfactory (smell), gustatory (taste), and kinaesthetic (bodily or movement sense).
  - » Do not deny the patient's experience, but suggest your own perceptions. For example, "I can't see or hear what you are experiencing, but I believe that it's very real for you".
  - » Talk to the patient about what they are experiencing, reassure them that they are safe, and ask whether there is anything you can do to help.
  - » It may be difficult for the patient to concentrate on what you are saying because their hallucinations are distracting. Without being condescending, ensure you speak clearly and keep your sentences simple.

# Attempted Suicide

This guideline is for patients who have attempted suicide. See the [‘non-suicidal self-harm’](#) guideline if the suicide attempt is not considered to be genuine. Treat and/or transport in conjunction with the relevant guidelines if the suicide attempt is associated with poisoning and/or injury.

- Ensure scene safety and prevent imminent suicide, provided this is feasible and safe.
- Assess the threat to the patient's life from the suicide attempt.
- If the patient's condition is life-threatening:
  - Treatment and transport to an ED by ambulance should occur, provided this is feasible and safe.
  - The patient cannot decline this, even if they have capacity. See the [‘treatment and/or transport without informed consent’](#) guideline.
- If the patient's condition is not life-threatening:
  - Make a clear recommendation that a face to face mental health assessment is required in an ED.
  - The patient may decline this if they have capacity.
  - Make treatment and/or transport decisions that are in their best interest if the patient does not have capacity. See the [‘treatment and/or transport without informed consent’](#) guideline.
- If transport to an ED is not indicated or the patient is declining transport:
  - Refer the patient [via phone](#) to a mental health crisis team.
  - The patient cannot decline this referral, even if they have capacity.
  - Whenever feasible, remain on scene while the mental health crisis team speaks to the patient and/or their family/whānau.
  - Whenever feasible, create and implement a safety plan, preferably in conjunction with the mental health crisis team.
  - If a safety plan cannot be implemented, transport by ambulance for a mental health assessment should occur, provided this is feasible and safe.

## Referral and transport

- If the patient is well known to mental health services and clearly does not need transport to an ED for medical assessment or treatment, it is appropriate to attempt to contact the patient's usual mental healthcare provider for advice.
- Most patients requiring transport to a medical facility should be transported to an ED. If transport is occurring to a facility that is not an ED, clinical staff at the facility must be contacted by ambulance personnel, and the patient accepted.

# Attempted Suicide

- All patients requiring transport for a life-threatening condition must be transported to an ED by ambulance.
- Following discussion with mental health services personnel, it may be appropriate for the patient to be transported to an agreed place for mental health assessment to occur. This may not require transport by ambulance if safe and reliable private transport is available.

## Additional information

### General information

- In the absence of an immediately life-threatening problem, it is important to take the time required to attempt to establish rapport with the patient, and to attempt to gain their informed consent.
- Many people who attempt suicide do not want to die. Instead, they may feel a temporary sense of severe hopelessness and despair, which they are often able to recover from. It is important to let them know that ambulance personnel are going to help them.
- Section 41 of the Crimes Act 1961 gives legal authority for anyone to use such force as may be reasonably necessary to prevent suicide.
- Have a low threshold for seeking clinical advice if the situation is complex, difficult to resolve, or there is a significant risk of serious harm to the patient without treatment and/or transport.

### The threshold for treatment and/or transport without informed consent

- The threshold for treatment and transport without the patient's informed consent is deliberately high. To meet this threshold:
  - There must be a clear and genuine threat to the patient's life, or
  - The ongoing risk of the patient committing suicide is high and a safety plan cannot be implemented.
- A reasonable attempt to gain informed consent must occur whenever it is feasible to do so.
- The need for treatment and/or transport without informed consent must be balanced with the risks if the patient is physically resisting treatment and/or transport. See the [‘treatment and/or transport without informed consent’](#) guideline.

### Determining the threat to life from poisoning

- If the poisoning is clearly life-threatening on the initial clinical assessment, information from the Toxinz website is not routinely required.

# Attempted Suicide

- In other situations, determining that there is a genuine threat to life requires careful clinical judgement, taking into account:
  - The patient's vital signs, and
  - The estimated quantity and formulation taken, and
  - The estimated time since ingestion, and
  - The expected onset, peak and severity of effects, and
  - The compounding effects of multiple substances, and
  - The time 'window' for any expected treatment with antidotes or any investigations required, for example blood tests, and
  - Information from the Toxinz website ([www.toxinz.com](http://www.toxinz.com)).
- There is usually uncertainty regarding the accuracy of the described quantity of substances taken. Take into account all available information including empty containers, drug paraphernalia on scene, and access to poisons.

## Creating and implementing a safety plan

- The safety plan should include:
  - An appropriate adult being with the patient until a mental health assessment can occur, and
  - Recommending that access to medicines, poisons and weapons is restricted, whenever possible, and
  - Considering transport to a place of safety if required, and
  - Providing written advice on available helplines such as the 1737 free text/phone/internet service.
- Provide written worsening advice. If the suicide attempt involved poison, include specific worsening advice from the 'observation at home' subsection of the relevant Toxinz chapter/s.
- Always document the safety plan.

## More Information

### Version history

26 November 2025

1.0.7

**New:** Guideline released.

# Non-suicidal Self-harm

This guideline is for patients who intentionally harmed themselves, but are not intending to commit suicide.

- Assess the patient to rule out a suicide attempt.
  - If suicide has been attempted, see the '[attempted suicide](#)' guideline.
  - If the patient has thoughts of suicide, see the '[thoughts of suicide](#)' guideline.
- Assess the patient for injury and/or poisoning requiring transport to a medical facility.
- Ensure appropriate support and follow up if the patient is not transported to a medical facility.

## Additional information

### General principles

- Non-suicidal self-harm occurs when a patient deliberately injures their body, but is not intending to die. Examples include cutting or burning skin.
- It is important to differentiate between non-suicidal self-harm and attempting suicide. Although the methods used to engage in non-suicidal self-harm sometimes overlap with those of suicide attempts (for example, cutting the wrists), non-suicidal self-harm is distinct from suicide attempts in that these patients do not intend for their actions to be lethal.
- Patients often harm themselves repeatedly in one session, creating multiple injuries in the same area of the body. This is often a repeated behaviour and can result in extensive scarring.
- The motivations to engage in non-suicidal self-harm may include:
  - To distract themselves when they are experiencing pressing, unavoidable, and overwhelming negative thoughts and/or emotions.
  - To experience something physical when they are otherwise feeling dissociated or 'numb'.
  - To express themselves or communicate strong emotions that they cannot otherwise articulate.
  - As a form of self-punishment for perceived faults.
  - As a plea for help.

### Patient support and follow up

- Clinical judgement is required, but a patient who has engaged in non-suicidal self-harm does not routinely require referral for a mental health assessment.
- A patient who has engaged in non-suicidal self-harm requires support (for example from a competent adult) and follow up (for example in primary care).



# Non-suicidal Self-harm

- Have a low threshold for seeking clinical advice if uncertain.
  - Provide the patient and/or their support people with a telephone support number such as the 1737 free text/phone/internet mental health service.
-

# Thoughts of Suicide

This guideline is for patients aged greater than or equal to 12 years who have thoughts of suicide.

- Ensure scene safety and prevent imminent suicide, provided this is feasible and safe.
- If the patient is known to mental health services, attempt to contact their usual mental healthcare provider for advice.
- If the patient is not known to mental health services or their usual mental healthcare provider cannot be contacted, use the [‘suicide risk screening tool’](#) to assess their risk of committing suicide.
- If the patient is at low risk:
  - Recommend they are reviewed in primary care.
  - Recommend a safety plan is created and implemented.
  - The patient may decline these recommendations if they have capacity.
- If the patient is at medium risk:
  - Recommend the patient makes contact with the 1737 free text/phone/internet mental health service.
  - Recommend a safety plan is created and implemented.
  - The patient may decline these recommendations if they have capacity.
- If the patient is at high risk:
  - Refer the patient [via phone](#) to a mental health crisis team. The patient cannot decline this referral, even if they have capacity.
  - Whenever feasible, remain on scene while the mental health crisis team speaks to the patient and/or their family/whānau.
  - Whenever feasible, create and implement a safety plan, preferably in conjunction with the mental health crisis team.
  - Consider transport to a place of safety if a safety plan cannot be implemented.

## Referral and transport

- Most patients should not require transport to a medical facility.
- If transport is occurring to a medical facility that is not an ED, clinical staff at the medical facility must be contacted by ambulance personnel and the patient accepted.
- Following discussion with mental health services personnel, it may be appropriate to transport the patient to an agreed place for mental health assessment to occur. This may not require transport by ambulance if safe and reliable private transport is available.

# Thoughts of Suicide

- If the patient is being transported to ensure their safety, the preference is for transport to the home of family/whānau. Transport to ED should be a last resort.

## Suicide Risk Screening Tool

| Always ask questions 1 and 2                                                                                                                                                                                                                                                                            | Yes                          | No                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------|
| 1. In the past month, have you wished you were dead or wished you could go to sleep and not wake up?                                                                                                                                                                                                    |                              |                                                  |
| 2. In the past month, have you had any thoughts about killing yourself?                                                                                                                                                                                                                                 | Go to Q3                     | Go to Q6                                         |
| If YES to 2, ask questions 3, 4, 5 and 6<br>If NO to 2, skip to question 6                                                                                                                                                                                                                              | Yes                          | No                                               |
| 3. Have you been thinking about how you might do this?                                                                                                                                                                                                                                                  | Medium risk                  |                                                  |
| 4. Have you had these thoughts and had some intention of acting on them?                                                                                                                                                                                                                                | High risk                    |                                                  |
| 5. Have you started to work out or worked out the details of how to kill yourself? Did you intend to carry out this plan?                                                                                                                                                                               | High risk                    |                                                  |
| Always ask question 6                                                                                                                                                                                                                                                                                   | Yes                          | No                                               |
| 6. Have you ever done anything, started to do anything, or prepared to do anything to end your life?<br>Examples: Took pills, tried to hang yourself, collected pills, gave away valuables, wrote a will or suicide note, took out pills but didn't swallow any, went to the roof but didn't jump, etc. | Go to Q7                     | Low risk<br>(If no other high or medium risk)    |
| 7. If yes, was this within the past 3 months?                                                                                                                                                                                                                                                           | High risk<br>(Past 3 months) | Medium risk<br>(Lifetime and no other high risk) |

**Note:** If the answer to all questions is no then this guideline does not apply and no referral or specific advice is indicated.

# Thoughts of Suicide

## Additional information

### General information

- In the absence of an immediately life-threatening problem, it is important to take the time required to attempt to establish rapport with the patient, and to attempt to gain their informed consent.
- Many people who have thoughts of suicide do not want to die. Instead, they may feel a temporary sense of severe hopelessness and despair, which they are often able to recover from. It is important to let them know that ambulance personnel are going to help them.
- It is useful to briefly assess the situation before utilising the suicide risk screening tool, including:
  - The reason for the 111 call.
  - Any history of mental illness.
  - Any recent alcohol and/or recreational drug use.
  - What social support and/or stressors are present.
- Section 41 of the Crimes Act 1961 gives legal authority for anyone to use such force as may be reasonably necessary to prevent suicide.
- Have a low threshold for seeking clinical advice if the situation is complex, difficult to resolve, or there is a significant risk of serious harm to the patient without treatment and/or transport.

### Using the suicide risk screening tool

- The suicide risk screening tool is based on the Columbia Suicide Screening Tool.
- The tool may be used in all patients aged greater than or equal to 12 years.
- It is appropriate to utilise information from family/whānau, bystanders and the incident notes to contribute to a holistic understanding of the risks.
  - For example, if there is clear evidence they have held a knife to their neck and threatened to kill themselves, but they answer no to question 6, it can be assumed the answer is yes and the patient should be treated as high risk.
- Assume that the patient is at high risk if they refuse to engage with screening.

### The threshold for treatment and/or transport without informed consent

- Many people in the community have thoughts of committing suicide. Solely screening as high risk does not, by itself, meet the threshold for treatment and/or transport without informed consent.

# Thoughts of Suicide

- The threshold for treatment and/or transport without informed consent is deliberately high. To meet this threshold:
  - The risk of the patient committing suicide must be high, and
  - A safety plan must be unable to be implemented.
- The need for treatment and/or transport without informed consent must be balanced with the risks to personnel. See the [‘treatment and/or transport without informed consent’](#) guideline for more information.
- Whenever feasible, advice from mental healthcare personnel should be obtained prior to deciding that a patient requires treatment and/or transport without their informed consent.

## Creating and implementing a safety plan

- If the patient is at high risk of committing suicide the safety plan should include:
  - An appropriate adult being with them until a mental health assessment can occur, and
  - Recommending that access to medicines, poisons and weapons is restricted, whenever possible, and
  - Considering transport to a place of safety if required, and
  - Providing written advice on available helplines such as the 1737 free text/phone/internet service.
- If the patient is at medium or low risk of committing suicide, utilise the principles above noting that an appropriate adult should be available to be contacted for support but does not need to stay with the patient.
- Always document the safety plan and include it on an ACS form.

## More Information

### Version history

26 November 2025

1.0.7

**New:** Guideline released.

# Drowning

## If the patient is in cardiac arrest

- In addition to usual treatment, prioritise the ventilation aspect of CPR and use a CPR ratio of 15:2 unless an ETT is in place.
- Intubate with an ETT if ROSC is not achieved in the first few minutes.
- IV drugs have a very low priority.

## If the patient is not in cardiac arrest

- Assess for signs or symptoms of aspiration.
- Assess for signs of injury.
- Assess for signs or symptoms of hypothermia.

## Referral and transport

- Clearly recommend that the patient is transported to ED by ambulance if:
  - There was loss of consciousness associated with the drowning, or
  - There are signs or symptoms of aspiration, for example tachypnoea, crackles in the lungs on auscultation, persistent coughing, shortness of breath or an SpO<sub>2</sub> less than 94% on air, or
  - There are signs of injury requiring transport to ED, or
  - The patient has moderate to severe hypothermia.
- If the patient is not transported to an ED, clearly recommend they seek medical advice at an ED if respiratory symptoms develop within the next few days.

## Additional information

### General principles

- Drowning occurs when a patient has impaired lung function as a result of aspiration following immersion or submersion in liquid.
- Following drowning, the patient may or may not have a cardiac arrest and may or may not survive, but if the patient has abnormal lung function then the patient has drowned.
- The literature contains a number of imprecise and unhelpful terms. Do not use any of the following terms: near drowning, wet drowning, dry drowning, salt water drowning, fresh water drowning, silent drowning

# Drowning

or secondary drowning.

- Always consider the possibility that a medical event (for example, hypoglycaemia or a seizure) may have occurred prior to drowning.

## Cardiac arrest

- The key to surviving cardiac arrest secondary to drowning is rapid rescue and early CPR, with a focus on ventilation. The rapidity of the rescue is more important than the means by which rescue occurs, but the safety of rescuers must be maintained.
- A resuscitation attempt should be commenced unless there are very clear signs the patient is dead.
- There is no role for ventilation or CPR while the patient is still in the water.
- It is common for significant amounts of frothy fluid to be coming from the mouth and/or nose. Do not interrupt ventilation to remove or suction fluid as this is usually ineffective (fluid will continue to be produced) and interruptions to ventilation are harmful.
- Beware misdiagnosing severe bradycardia as asystole, especially in children.
- Adequate ventilation is often only achieved via an ETT because the lungs may contain significant volumes of liquid.
- Adequate ventilation may not be achieved via an LMA because of the high pressures required to achieve ventilation if the lungs contain liquid. If ventilation is not clearly achieved via an LMA, consider providing ventilation via a manual ventilation bag and mask using a two-person technique.
- Survivors tend to come from the group of patients that get ROSC within 10-15 minutes, usually with good CPR (with a focus on ventilation).
- In most patients it will be appropriate to continue resuscitation attempts for approximately 40 minutes, but it is appropriate to cease resuscitation attempts earlier than this if the rhythm has been asystole for more than ten minutes.
- Although there are case reports of patients surviving resuscitation attempts of several hours in duration, the patients were usually small children that had drowned in water containing ice and the overall mortality rate was very high. Even in winter, water temperatures in New Zealand are not cold enough to provide significant cerebral protection and therefore the duration of resuscitation should not be influenced by the temperature of the water the patient drowned in. Personnel should seek clinical advice if they are uncertain.

# Drowning

## General principles of treatment

- Liquid damages the endothelial lining of the lung and this may result in the development of significant amounts of pulmonary oedema.
  - Do not attempt to remove or suction fluid as this is usually ineffective (fluid will continue to be produced) and removing the oxygen mask increases the likelihood of hypoxia.
  - CPAP or PEEP is usually helpful.
  - The patient may be hypovolaemic (the pulmonary oedema fluid has come from the circulation) and may require 0.9% sodium chloride IV.
  - There is no role for treatment with GTN.
- Consider the possibility of cervical spine injury, but in the absence of trauma this is rare following drowning and is not an initial priority.
- The patient's stomach often contains significant amounts of water but there is usually no role for draining the stomach.
- There is no role for positioning the patient to try to achieve postural drainage of the lungs.
- Although there are significant electrolyte differences between salt water and fresh water, there are usually no clinically significant physiological differences between drowning in salt water and drowning in fresh water.
- Lung function may deteriorate several hours after aspiration, even if at the time lung function appears normal.

## Response and/or transport by helicopter

- Response by helicopter may be indicated if it will arrive at scene at least 15 minutes sooner than a road-based CCP and the patient has:
  - Prolonged altered level of consciousness, and/or
  - Severe respiratory compromise.
- Transport by helicopter may be indicated if it will save more than 30 minutes compared to road transport, and the patient has:
  - Prolonged altered level of consciousness, and/or
  - Severe respiratory compromise.
  - For criteria to be met, the road transport time will usually need to be more than 90 minutes.



# SCUBA Diving Emergencies

- Position the patient flat.
- Administer oxygen via a reservoir mask.
- Administer 0.9% sodium chloride IV if there are signs of hypovolaemia or poor perfusion:
  - 1 litre for an adult.
  - 20 ml/kg for a child.
  - Repeat once if required.

## Referral and transport

- Transport the patient to North Shore Hospital ED or Christchurch Hospital ED, provided the diagnosis of decompression sickness or arterial gas embolism is clear, and it is feasible and safe to do so. Seek clinical advice if there is uncertainty or transport will be prolonged.
- Do not transport directly to a recompression facility.
- Transport the patient to the nearest appropriate ED if the diagnosis is not clear.
- Discuss options to fly at the lowest safe altitude with the pilot. Aim for an altitude of no more than 300 metres above sea level provided this is feasible and safe.
- Transport the patient's dive computer with them if this is available.

## Additional information

### General principles

- The most common SCUBA diving emergency is decompression sickness. This occurs when gases (predominantly nitrogen) that are dissolved in body fluids expand to form bubbles.
- When a person has been SCUBA diving at depth, the increased ambient pressure results in an increase in the volume of gases dissolved in body fluids. As the diver ascends, these dissolved gases come out of body fluids and are exhaled via the lungs. If the volume of dissolved gases is high and/or the ascent is rapid, the gases leaving body fluids can form bubbles in the same way that occurs when a carbonated drink is opened.
- The patient should be positioned flat because this limits the number of bubbles that reach the brain.

# SCUBA Diving Emergencies

## Signs and symptoms of decompression sickness

- The patient may have any combination of the following:
  - Joint pain, particularly in large joints.
  - Headache.
  - Visual disturbance.
  - Itching skin or a feeling of 'insects crawling on the skin'.
  - Altered peripheral sensation or motor power.
  - Confusion.
  - Reduced level of consciousness.
  - Chest pain.
  - Shortness of breath.
- Although symptoms usually develop within one hour of SCUBA diving, they may be delayed until up to 24 hours later, and decompression sickness must be considered a possibility in any patient with unexplained symptoms occurring within 24 hours of SCUBA diving.

## Arterial gas embolism

- Arterial gas embolism occurs when gas bubbles directly enter arteries.
- The arterial bubbles usually come from a lung that has been damaged by expansion of gas during ascent.
- A patient with arterial gas embolism will most commonly develop sudden loss of consciousness or stroke-like symptoms, during or immediately after ascent.
- The treatment of a patient with arterial gas embolism is the same as the treatment of a patient with decompression sickness.

## Referral and transport

- Decompression sickness and arterial gas embolism may be made worse by transport at altitude. This is because the decreased ambient pressure causes gas bubbles to enlarge.
- The patient must not be transported direct to a recompression facility because assessment in an ED is required prior to referral to a recompression facility.
- Recompression facilities are only available at North Shore Hospital and Christchurch Hospital. This is why transport via North Shore Hospital ED or Christchurch Hospital ED is preferred, provided the diagnosis of decompression sickness or arterial gas embolism is clear.

# SCUBA Diving Emergencies

## Response and/or transport by helicopter

- Transport by helicopter may be indicated for patients with decompression sickness if it will save more than 30 minutes compared to road transport.
- For criteria to be met, the road transport time will usually need to be more than 90 minutes.

## Signs of spinal cord impairment

- Decompression sickness may cause signs and symptoms of spinal cord impairment.
- Treat the patient using this guideline and not the 'spinal cord injury' guideline if the patient has signs or symptoms of spinal cord impairment following SCUBA diving and has not had a traumatic injury.

# Hyperthermia

This guideline is for patients with hyperthermia following environmental exposure and/or exertion. If the patient is febrile due to another cause (for example sepsis or poisoning), see the appropriate guideline and utilise the cooling principles within this guideline if their temperature is greater than 40°C.

- Transfer the patient into a cool environment as soon as possible.
- Determine the severity of hyperthermia.

## Mild to moderate hyperthermia

- Cool the patient:
  - Remove excess clothing.
  - Apply cool water to the head and skin.
  - Ensure airflow over the patient, utilising a fan if available.
  - Provide a cool drink if feasible.
- Administer 0.9% sodium chloride IV if the patient has moderate hyperthermia and is not responding to initial treatment:
  - 1 litre for an adult.
  - 20 ml/kg for a child.
  - Repeat as required.
- Measure the blood glucose concentration and treat accordingly.
- Measure the temperature every 15-30 minutes until it is consistently below 38°C.

# Hyperthermia

## Severe hyperthermia

- Keep the patient supine and remove all excess clothing.
- Immediately commence rapid cooling at the scene, as a higher priority than commencing transport.
- Cooling options are dependent on what is available. In order of preference:
  - Immerse from the neck down in the coldest water available, adding ice if possible:
    - For example, in a bath or container.
    - Oscillate the water if feasible.
  - If immersion in cold water is not feasible, utilise the cold towel method:
    - Fill a container with the coldest water available, adding ice if possible.
    - Soak towels in the water.
    - Lay towels over the skin, with a focus on the torso and head.
    - Circulate towels every couple of minutes to ensure they remain as cold as possible.
  - If neither of the above methods are feasible, utilise a combination of the following:
    - Continuously wet the skin with the coldest water available, with a focus on the torso and head.
    - Apply ice packs, with a focus on the groins and armpits.
    - Fan continuously.
- Administer 0.9% sodium chloride IV:
  - 1-2 litres for an adult.
  - 20-40 ml/kg for a child.
  - Repeat as required.
- Measure the temperature every 15 minutes, or via continuous monitoring if available.
- Measure the blood glucose concentration and treat accordingly.
- Continue cooling until the patient's temperature is below 39°C or neurological symptoms resolve.
- Treat seizures as per the '[seizures](#)' guideline.

## Backup

- Backup from a Paramedic should be requested if the patient has moderate hyperthermia that is not responding to initial treatment.
- Backup from an ICP/CCP must be requested if the patient has severe hyperthermia.

# Hyperthermia

## Referral and transport

- Most patients with mild to moderate hyperthermia should be treated in the community without transport to an ED, provided:
  - The patient is cooled to below 38°C, and
  - The patient has normal vital signs and can mobilise normally following cooling, and
  - An underlying medical condition (for example, sepsis) contributing to hyperthermia is ruled out, and
  - A maximum of one dose of 0.9% sodium chloride has been administered.
- The patient must be transported to an ED by ambulance if severe hyperthermia is present.

## Additional information

### General principles

- Hyperthermia is defined as a core body temperature of greater than 38°C.
- Hyperthermia is caused by inadequate thermoregulation in response to excessive exposure to heat, excessive heat production or impaired heat loss, and usually occurs when exertion is combined with exposure to high ambient temperatures.
- When core body temperature exceeds 40°C proteins within the body may denature, resulting in severe cellular and organ dysfunction which may be life-threatening.
- Hyperthermia is distinct from fever which occurs when there is a physiological increase in the temperature 'set point'. However, when fever is very severe (for example greater than 40°C), the physiological effects are the same.
- The term 'heat exhaustion' is sometimes used to describe mild to moderate hyperthermia and the term 'heat stroke' is sometimes used to describe severe hyperthermia. However, use of these terms is discouraged because they are imprecise and have varying meaning.

### Determining the severity of hyperthermia table

This table describes the common clinical features associated with mild, moderate and severe hyperthermia. Not all clinical features need to be present.

# Hyperthermia

## Mild hyperthermia

- 38-39°C
- Tachycardia
- Tachypnoea

## Moderate hyperthermia

- 39-40°C
- Tachycardia
- Tachypnoea
- Lethargy
- Feeling faint
- Nausea and vomiting
- Muscle cramping
- Disorientation
- Headache

## Severe hyperthermia

- > 40°C
- Altered level of consciousness
- Confusion
- Agitation
- Dry and hot skin
- Signs of shock
- Dysrhythmias
- Seizures

## Factors that increase the risk of hyperthermia

- Extremes of age. The elderly and young children have impaired thermoregulatory mechanisms.
- Pre-existing medical conditions can increase the vulnerability to high ambient temperatures, including diabetes, hyperthyroidism, peripheral vascular disease, ischaemic heart disease and heart failure.
- Medications can cause increased heat production (for example thyroxine and cyclic antidepressants), decrease thirst (for example ACE inhibitors), decrease sweating (for example antihistamines,

# Hyperthermia

anticholinergics and beta blockers), or increase fluid loss (for example diuretics).

- Length and/or intensity of exposure. Hyperthermia is more likely to develop with an increase in both length and intensity of exposure to exertion and/or a hot environment.
- Humidity. When humidity is high, sweat will not evaporate as easily from the skin and heat loss is reduced.
- Strenuous activity combined with high ambient temperatures. This is a feature that is common amongst firefighters and endurance athletes.

## The pathophysiology of severe hyperthermia

- Dehydration from excessive sweating in combination with peripheral vasodilation can lead to hypovolaemia.
- Sodium and other electrolytes are lost through sweating which can lead to electrolyte abnormalities, particularly if sweat loss is severe and only water is consumed during prolonged exercise. The electrolyte abnormality of greatest concern is hyponatraemia which may be associated with seizures. Seek urgent clinical advice if hyponatremia is suspected as specific treatment (for example with sodium bicarbonate) may be required.
- Rhabdomyolysis (muscle breakdown) may occur.
  - Muscle cells are particularly prone to heat injury and as they die cellular proteins (including myoglobin), acid and potassium are released.
  - Release of acid into the circulation will interfere with normal cellular function, particularly in the heart.
  - Release of potassium from muscle cell death interferes with normal cardiac conduction and may cause dysrhythmias.
  - Release of myoglobin blocks kidney tubules and may cause renal failure.
- Neurological dysfunction including seizures may occur due to thermal brain injury.
- Dysrhythmias may occur. Electrolyte abnormalities (particularly hyperkalaemia) will interfere with normal cardiac conduction and may cause dysrhythmias.
- Coagulopathy may occur. The high temperature denatures proteins and this may disrupt normal coagulation mechanisms, leading to disseminated intravascular coagulation (DIC).

## Assessment and treatment

- Some tympanic thermometers may not be able to measure temperatures above 41°C. It is important to focus on the patient's clinical presentation rather than trying to obtain an accurate reading.



# Hyperthermia

- Ice or ice packs applied directly to skin may cause skin injury, however this is not a concern in the context of moderate to severe hyperthermia.
- Patients with severe hyperthermia should be positioned supine to maximise venous return and help maintain cardiac output.
- Hypoglycaemia may occur, particularly in the setting of severe hyperthermia.
- Agitation is common, particularly in the setting of severe hyperthermia. Have a low threshold for controlling agitation with midazolam or ketamine.

## More Information

### Version history

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9 June 2025

1.0.5.3

**Update:** Multiple changes.

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# Hypothermia

This guideline is for patients with hypothermia secondary to environmental exposure.

- Transfer the patient into a warm and dry environment as soon as possible and/or reduce environmental exposure.
- Remove wet clothing and ensure the patient is dry.
- If there is going to be a delay in getting the patient to a warm and dry environment, wrap them (still in their wet clothes) in a vapour barrier with external insulation and with a heat source (for example a heating pad/pack or chemical warming blanket) against the torso, if possible.
- Determine the severity of hypothermia.

## Mild hypothermia

- Actively warm the patient:
  - Place in a warm environment and dress in warm clothes.
  - Provide a drink containing carbohydrate, for example sugar, and food, provided the patient can safely swallow.
- Encourage gentle exercise provided this is safe.
- Measure the patient's temperature every 15-30 minutes until it is consistently above 35°C.

## Moderate to severe hypothermia

- Place in a warm environment.
- Keep the patient supine and minimise unnecessary movement.
- Continually monitor the cardiac rhythm.
- Prevent further cooling by wrapping the patient in a 'hypothermia wrap' with a heat source (for example a heating pad/pack) against the torso, if possible.
- Gain IV access but avoid IV fluid administration unless signs of severe hypovolaemic shock are clearly present. If IV fluid is administered, ensure this is warmed using a dedicated warming device if possible.
- Measure the patient's temperature every 15-30 minutes, or via continuous monitoring if available.

# Hypothermia

## Cardiac arrest

- Commence CPR if it is safe and feasible to do so, while the patient is further assessed:
  - Do not administer IV medicines.
  - **If the initial rhythm is VF or VT:** continue treatment and CPR, even if the rhythm deteriorates into asystole with defibrillation, providing a maximum of three shocks if the rhythm continues to be VF or VT.
  - **If the patient is in PEA or has severe bradycardia:** continue treatment but stop chest compressions and ventilate at a rate of six breaths/minute.
  - **If the initial rhythm is asystole:** stop treatment and stop CPR. Note that the patient may have severe bradycardia and a one minute period of observation of the cardiac rhythm is required before asystole can be confirmed. See below if the patient has been in an avalanche.
- If the patient has been in an avalanche, continue with treatment and CPR unless:
  - The chest is incompressible, or
  - There are visible injuries incompatible with life, or
  - The rhythm is asystole and the airway was completely occluded by snow/ice at the time of extrication.
- If the patient remains in cardiac arrest after ten minutes of resuscitation, seek clinical advice (if possible) on the appropriateness of transport to hospital with CPR en route.
  - Do not transport to hospital with CPR en route without clinical advice to do so, as the patient will need prolonged active warming and cardiovascular support such as cardiopulmonary bypass or extracorporeal membrane oxygenation (ECMO), and these will require significant advance warning and prior arrangements to be made with hospital specialists.
  - Focus on high performance CPR with a ventilation rate of 6 breaths/minute and utilise mechanical CPR if it is available.
  - Have the following additional information available if the patient has been in an avalanche:
    - The known or estimated time buried.
    - The core temperature if this is able to be measured.
    - Whether an air pocket was present in front of the face when extricated.
    - Whether the airway was completely occluded by snow/ice at the time of extrication.

## Backup

- Backup from a Paramedic should be requested if the patient has moderate hypothermia.
- Backup from an ICP/CCP should be requested if the patient has:

# Hypothermia

- » Severe hypothermia, or
- » Poor airway or poor breathing, or
- » Dysrhythmia.

## Referral and transport

- Most patients with mild hypothermia should be treated in the community without transport to an ED, provided:
  - » The patient is rewarmed to above 35°C, and
  - » The patient has normal vital signs and can mobilise normally following warming, and
  - » An underlying medical condition (for example sepsis) contributing to hypothermia is ruled out.
- The patient must be transported to an ED by ambulance if moderate to severe hypothermia is present. If there is a choice of hospital destination, transport a patient with severe hypothermia to a hospital with intensive care facilities, provided this is feasible and safe.

## Additional information

### General principles

- Hypothermia is defined as a core body temperature of less than 35°C. It is usually due to excessive exposure to a cold environment.
- Other conditions that alter a patient's ability to move (for example sepsis, stroke, hypoglycaemia, intoxication or injuries) may contribute to the development of hypothermia.
- Tympanic and digital thermometers are unreliable when the patient is hypothermic, and it is important to focus on the patient's clinical presentation when determining the severity of hypothermia, rather than relying on the measured temperature.
- Hypoglycaemia may occur in patients with hypothermia, due to depletion of blood glucose and glycogen stores following prolonged shivering.

### Determining the severity of hypothermia table

This table describes the common clinical features associated with mild, moderate and severe hypothermia. Not all clinical features need to be present.

# Hypothermia

## Mild hypothermia

- 32-35°C
- Shivering
- Increased muscle tone
- Tachycardia
- Hypertension
- Tachypnoea
- Poor coordination
- Lethargy
- Confusion

## Moderate hypothermia

- 28-32°C
- Absence of shivering
- Muscle rigidity
- Altered level of consciousness
- Bradycardia
- Hypotension
- Bradypnoea

## Severe hypothermia

- < 28°C
- Unconscious
- Severe bradycardia
- Severe bradypnoea
- Severe shock
- Unreactive pupils
- Cardiac arrest

## The physiological response to heat loss

- Exposure to cold causes reflex compensatory mechanisms in the body to:
  - Conserve heat through vasoconstriction (restricting blood flow to the peripheries) and piloerection (trapping warm air near the skin), and

# Hypothermia

- » Generate heat through increased cellular metabolism and shivering.
- As the body temperature falls, the body loses its ability to compensate for heat loss, causing a progressive slowing in cellular metabolism, enzyme activity, neural conduction and cardiac activity.
- As the body temperature continues to fall cellular metabolism also falls, further reducing heat production and resulting in a vicious cycle of falling temperature, falling metabolism and falling heat production.
- When core temperature drops below 32°C the tubular reabsorption ability within the kidneys is adversely affected, resulting in polyuria ('cold diuresis'). This is similar to the diuresis that occurs with severe hyperglycaemia and will contribute to hypovolaemia.

## Patients at increased risk of hypothermia

- Patients are at increased risk of hypothermia if they have an impaired ability to conserve and/or generate body heat, reduced physiological reserve, conditions that increase heat loss, or an inability to remove themselves from the cold environment.
- Patients at increased risk include those that are elderly, frail, intellectually impaired, children, immobile, intoxicated, malnourished or injured.
- Patients that develop hypothermia in New Zealand are predominantly from two groups:
  - » Those that are immobile and exposed to a cold environment in their home. For example, elderly patients with a fractured neck of femur who lie on a cold floor for a prolonged period.
  - » Those that are exposed to adverse weather during recreational or occupational activity. For example, in the bush or mountains.

## Warming

- Passive warming (for example the use of blankets) is only useful when the patient has a temperature over 35°C.
- Active warming is required when the patient has a temperature below 35°C.
- It is vital to remove wet clothing, dry the patient and place them in a warm environment, provided this is feasible and safe. If this is not feasible and safe, completely wrapping the patient in a 'hypothermia wrap' including a vapour barrier with external insulation, will trap moisture against the patient and help with warming, particularly if a heat source (for example a heating pad/pack or chemical warming blanket) is placed against the patient's torso, within the vapour barrier.
- Do not place a heat source directly against the skin as this may cause injury.
- Ideally three layers and a heat source are used to create a 'hypothermia wrap':

# Hypothermia

- » Place a heat source, for example a heating pack/pad against the patient's torso.
  - » Wrap the patient in a vapour barrier, for example using a commercially available rescue wrap/sack or multiple foil blankets. This reduces evaporative heat loss.
  - » Insulate around the vapour barrier, for example using a sleeping bag or blankets.
  - » Wrap the patient and the previously described layers in a further layer, for example a tarpaulin or sheet of plastic.
- Keep the interior of the ambulance as hot as possible. In particular, heat the interior of the ambulance before the patient is placed inside. This warms the physical environment, for example the stretcher, which acts as a 'heat sink' when cold.
  - Eating food, provided the patient can safely swallow, generates heat from the increase in metabolism that occurs as food is digested.

## Additional treatment considerations for moderate to severe hypothermia

- It is not feasible to provide sufficient heat to warm a patient with moderate to severe hypothermia in the out-of-hospital setting. The goal is to prevent further heat loss and transport the patient to a hospital with the facilities to provide appropriate and safe warming.
- Vasodilation may lead to a worsening of hypothermia as a result of cold blood returning to the patient's core. This is why it is important to focus on warming the torso (and not the extremities) and is why IV fluid is avoided unless the patient clearly has severe hypovolaemic shock. Even if IV fluid is warmed, it may make hypothermia worse by increasing blood supply to cold peripheries.
- Whenever feasible IV fluid should be warmed using a dedicated warming device. These may be available (for example via a helicopter service) and should be requested whenever feasible.
- Do not warm IV fluid in microwaves because this can result in unreliable fluid temperatures and/or may damage the plastic bag.
- Impaired cellular metabolism may impair the response to treatment. For example the patient may not respond to pacing, synchronised cardioversion or defibrillation.
- The risk of precipitating a dysrhythmia (particularly ventricular fibrillation) during patient movement is controversial. It is not clear from the literature that it is patient movement that causes the dysrhythmia, or if it is secondary to the change in temperature that usually coincides with the patient being moved. Moving the patient should not be unnecessarily delayed but avoid excessive and/or unnecessary movement, for example moving from a supine to a sitting position.
- Keep the patient flat to maximise venous return and help maintain cardiac output.

# Hypothermia

## Cardiac arrest

- Cardiac arrest secondary to hypothermia in New Zealand is uncommon and usually the circumstance is that a patient dies and their body cools after death. It is important to try to differentiate cardiac arrest secondary to hypothermia, from the circumstance where a patient has died and then cooled after death, particularly if the patient is elderly and/or frail.
- Because the metabolic rate drops significantly with severe hypothermia, it is possible for patients to survive prolonged cardiac arrest secondary to hypothermia.
- Survival following cardiac arrest secondary to severe hypothermia usually requires the patient to be transported to a hospital with the facilities to provide cardiopulmonary bypass or ECMO, while the patient is warmed.
- Resuscitate the patient in a warm environment if possible, because this will help prevent further heat loss. However, do not try to provide additional warming because this will not result in a significant increase in body temperature and will impair resuscitation.
- CPR is not recommended if the patient is in PEA or has severe bradycardia because:
  - The patient is likely to have cardiac output that cannot be detected clinically, and
  - CPR may precipitate ventricular fibrillation, and
  - CPR complicates extrication and transport.
- It is appropriate to provide intermittent manual chest compressions if required, for example while carrying the patient on a stretcher.
- Provision of chest compressions via a mechanical CPR device is preferred. If a mechanical CPR device is not available, and there are insufficient personnel available to provide continuous chest compressions, provide intermittent manual chest compressions (for example five minutes on and five minutes off).
- Provide a low ventilation rate of 6 breaths/minute because the metabolic rate will be very low and higher ventilation rates may compromise venous return to the heart.
- Attach and utilise capnography if an SGA or ETT is placed. A trace should be obtained if placement is correct, but the expected  $\text{ETCO}_2$  will be much lower than usual because of a low metabolic rate.

## More Information

### Version history

9 June 2025

1.0.5.3

**Update:** Multiple changes.



# Hypothermia

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# Anaphylaxis

- Administer adrenaline if the patient is showing systemic signs of anaphylaxis:
  - Administer 0.5 mg IM for an adult.
  - See the paediatric drug dose tables for the IM dose for a child.
- Administer 0.9% sodium chloride IV if the patient has signs of hypovolaemia:
  - 1 litre for an adult.
  - 20 ml/kg for a child.
  - Repeat as required.
- Repeat the adrenaline IM every ten minutes if the patient is not improving, or every five minutes if the patient is deteriorating.
- Administer adrenaline IV if the patient is deteriorating despite adrenaline IM:

## For an adult:

- Administer an adrenaline infusion via an infusion pump. Start at a rate of 0.5 mg/hour and adjust the rate as required, or
- Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
  - Administer as an infusion. Start at 2 drops/second and adjust the rate as required, or
  - Administer a bolus of 0.01 mg adrenaline (10 ml) IV every 1-2 minutes as required.

## For a child aged 5-14 years:

- Administer an adrenaline infusion via an infusion pump using service guidelines, or
- Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
  - Administer as an infusion. Start at 1 drop/second and adjust the rate as required, or
  - Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.

## For a child aged less than five years:

- Administer an adrenaline infusion via an infusion pump using service guidelines, or
- Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.

## In addition

- Personnel may consider additional treatments to manage specific symptoms, providing IM/IV adrenaline has been prioritised and systemic signs of anaphylaxis are improving.
- Consider 5 mg of nebulised adrenaline if the patient has airway oedema or prominent bronchospasm.

# Anaphylaxis

- Consider administration of loratadine PO if the patient has itch or rash, and systemic signs of anaphylaxis are improving:
  - 20 mg for a patient aged greater than or equal to 12 years.
  - 10 mg for a patient aged 1-11 years.

## Backup

- Backup from a Paramedic is usually sufficient as most patients improve 5-10 minutes after adrenaline IM and usually only require one dose.
- Backup from an ICP/CCP should be requested if the patient is peri-arrest or is not improving 5-10 minutes after one dose of adrenaline IM.

## Referral and transport

- All patients receiving treatment for anaphylaxis must be given a clear recommendation to be transported to a medical facility by ambulance.
- Transport should usually be to an ED, unless the patient:
  - Has previously had anaphylaxis, and
  - Has rapidly improved with a single dose of adrenaline IM, and
  - Is being taken to a primary care facility with the capacity to observe them for several hours.

## Additional information

### General principles

- Anaphylaxis is a rapid onset, multiple-organ, generalised hypersensitivity (allergic) syndrome. It is usually characterised by skin features of systemic inflammatory mediator release (urticaria, itch or flush, swollen lips and/or tongue) plus involvement of one or more of the following systems:
  - The respiratory system with any of the following: dyspnoea, chest or throat tightness, wheeze or stridor.
  - The cardiovascular system with any of the following: hypotension, poor perfusion, fainting, collapse or altered level of consciousness.
  - The gastrointestinal system with any of the following: severe nausea, vomiting, abdominal pain or diarrhoea.
- Exposure to an allergen results in the release of inflammatory mediators from mast cells and basophils which cause the signs and symptoms of anaphylaxis. While there are a number of mediators, histamine is the most widely recognised.

# Anaphylaxis

- Anaphylaxis can be triggered by almost anything, but most commonly it is caused by exposure to venom (particularly wasps and bees), food (particularly eggs, peanuts and shellfish) or medications.
- Patients with stings and only localised swelling, redness or pain do not have anaphylaxis.

## Recognising anaphylaxis

- To have anaphylaxis patients must have signs of systemic involvement. Skin features alone are insufficient.
- A very small proportion of patients do not have obvious skin features initially, particularly if the onset is sudden and severe.
- Consider the possibility of anaphylaxis in all patients with sudden onset of unexplained bronchospasm, shock or respiratory distress.

## Adrenaline administration

- The most important priority is the early administration of adrenaline.
- The risk of death is raised in patients whose need for adrenaline (or repeat adrenaline) is under-recognised.
- Have a low threshold for administering adrenaline if anaphylaxis is suspected, even if it is not immediately life-threatening.
- Have a low threshold for repeat adrenaline if the patient is not improving.
- A dose of 0.5 mg adrenaline IM is appropriate for the majority of adults. Paramedics and ICPs/CCPs may elect to reduce the dose, particularly if the patient is small, frail, or has ischaemic heart disease.
- When administering adrenaline IV, an IV infusion is preferred over IV boluses because this reduces the adverse effects of surges of adrenaline.

## Cardiac arrest secondary to anaphylaxis

- IV adrenaline has a high priority. See the ['cardiac arrest'](#) guideline.

## Angioedema

- Isolated oedema, particularly of the mouth and/or face in the absence of systemic signs of anaphylaxis is usually caused by angioedema and not anaphylaxis.

# Anaphylaxis

- Angioedema is a condition that results in intermittent, unpredictable and isolated swelling of the mouth and/or face. It often occurs in patients taking aspirin or an angiotensin converting enzyme (ACE) inhibitor and may occur following fibrinolytic therapy.
- Angioedema may respond to nebulised adrenaline.
- IM and IV adrenaline should not be administered because angioedema rarely responds to systemic adrenaline and the adverse effects of systemic adrenaline usually outweigh any possible benefit.

## Response and/or transport by helicopter

- Patients that have improved significantly following the administration of a single dose of adrenaline IM are usually appropriate for road transport and are not time dependent.
- Response by helicopter may be indicated if it will arrive at scene at least 15 minutes sooner than the road resource:
  - To administer adrenaline IM, or
  - The patient is known to require multiple doses of adrenaline, or
  - There is significant airway compromise.
- Transport by helicopter may be indicated if it will save more than 30 minutes compared to road transport and the patient is not improving with treatment and/or multiple doses of adrenaline are required. For criteria to be met, the road transport time will usually need to be more than 90 minutes.

## More Information

### Version history

28 October 2024

1.0.5

**Update:** Multiple changes.

# Minor Allergy

This guideline is for minor allergic reactions (including bites and stings) that are confined to skin involvement.

- Administer loratadine if itch is prominent:
  - 20 mg PO for a patient aged greater than or equal to 12 years.
  - 10 mg PO for a child aged 1-11 years.
- Administer prednisone or prednisolone in addition to loratadine, if the itch is associated with a rash:
  - 40 mg PO for an adult.
  - See the paediatric drug dose tables for a child.

## Referral and transport

- A patient may be administered loratadine, prednisone or prednisolone and be given a clear recommendation that immediate referral to a medical facility is not required, provided:
  - There are no signs of systemic involvement, and
  - There are no signs of spreading inflammation, and
  - There is no intraoral swelling of concern, and
  - No adrenaline (including self-administration) has been administered.

## More Information

### Version history

28 October 2024

1.0.5

**Update:** Loratadine dose updated.

# Non-traumatic Bleeding

This guideline is for patients with bleeding from non-traumatic causes that does not clearly fit into another guideline, for example gastrointestinal bleeding.

- Administer 0.9% sodium chloride IV (using a fluid warming device if available), if the patient has signs of hypovolaemia:
  - 500 ml IV for an adult.
  - 10 ml/kg IV for a child.
  - Repeat as required, minimising the total volume of IV fluid.

If shock is severe:

- Expedite transport.
- Provide active external warming whenever feasible.
- Call for blood provided there is an established protocol for this in the area, particularly if extrication will be delayed and/or transport will be prolonged.
- Administer TXA IV:
  - 1 g IV for an adult.
  - See the paediatric drug dose tables for a child.
- Consider seeking clinical advice and commencing a metaraminol infusion in an adult, if transport time is prolonged and the systolic BP is less than 80 mmHg systolic. Titrate the infusion rate to maintain a systolic BP of 80-90 mmHg.
- Notify receiving hospital personnel early.

## Backup

- Backup from an ICP/CCP should be requested if the patient has severe shock.
- Backup must not delay rapid transport to an appropriate hospital.

## Referral and transport

- Expedite transport if the patient has severe shock.
- The patient must be given a clear recommendation to be transported to an ED by ambulance if 0.9% sodium chloride is administered:
  - This should be to a hospital with acute surgical facilities, whenever this is feasible and safe.

# Non-traumatic Bleeding

» If transport time to a hospital with acute surgical facilities is prolonged, it may be appropriate to transport to a significantly closer hospital but have a low threshold for seeking clinical advice.

➤ The following hospitals have acute surgical facilities:

- » Whāngarei Hospital.
- » North Shore Hospital.
- » Auckland City Hospital.
- » Starship Children's Hospital.
- » Middlemore Hospital.
- » Waikato Hospital.
- » Tauranga Hospital.
- » Whakatane Hospital.
- » Gisborne Hospital.
- » Rotorua Hospital.
- » Taranaki Base Hospital.
- » Whanganui Hospital.
- » Hawke's Bay Regional Hospital.
- » Palmerston North Hospital.
- » Wairarapa Hospital.
- » Hutt Hospital.
- » Wellington Regional Hospital.
- » Wairau Hospital.
- » Nelson Hospital.
- » Te Nikau Grey Hospital.
- » Christchurch Hospital.
- » Timaru Hospital.
- » Dunedin Hospital.
- » Southland Hospital.

## Additional information

### Administration of 0.9% sodium chloride

➤ 0.9% sodium chloride IV should be titrated to clinical signs of intravascular volume and perfusion.



# Non-traumatic Bleeding

- Large volumes of crystalloid may worsen bleeding by contributing to coagulopathy, particularly in the context of severe and uncontrolled bleeding:
  - This is why the total volume of 0.9% sodium chloride should be minimised.
  - Clinical judgment is required when determining the volume administered, but it should be uncommon to exceed 1 litre in an adult or 20 ml/kg in a child, and it should be rare to exceed 2 litres in an adult or 40 ml/kg in a child.
  - Have a low threshold for seeking clinical advice if shock is severe and transport time to the most appropriate hospital is prolonged.
- Ambient temperature IV fluid can cause hypothermia which worsens bleeding by contributing to coagulopathy. IV fluid should be warmed using a dedicated warming device if one is available.
- Do not warm IV fluid in microwaves because this can result in unreliable fluid temperatures and/or may damage the bag.

## Tranexamic acid (TXA)

- Early administration of TXA has been shown to reduce bleeding and is a feature of most hospital massive haemorrhage protocols across New Zealand, regardless of whether the bleeding is caused by trauma or not.
- TXA should be administered when shock is severe, as this is a group of patients who have a higher risk of complications and/or mortality from their bleeding, regardless of the source. This is a lower priority than IV fluid administration, external warming, and rapid transport to hospital.

## Blood

- Requests for blood to be delivered to the scene must not compromise rapid transport to the most appropriate hospital.
- Blood should only be requested if:
  - Shock is severe, and
  - There is an established protocol in the area for blood to be delivered to the scene, and
  - Blood can be administered 15 or more minutes before the patient can arrive in the most appropriate hospital.
- Blood may be available via staging at a small hospital/medical facility en route to the most appropriate hospital. In this setting personnel must have a low threshold for seeking advice from the on-call doctor for the service.

# Non-traumatic Bleeding

- ICPs/CCPs may continue blood administration initiated by a medical practitioner following their verbal instructions but should have a low threshold for seeking clinical advice.
- Requests for blood to be delivered to the scene outside of an established protocol must be made via the on-call doctor for the service. In this setting:
  - It will usually take approximately 15 minutes to arrange for blood to be prepared, plus transport time to the scene. This means that for such a request to be appropriate, the patient should usually have severe shock and a prolonged transport time.
  - Personnel cannot administer blood without specific instructions from the on-call doctor for the service.

## Patient temperature

- It is important to keep the patient warm because hypothermia worsens bleeding by contributing to coagulopathy.
- Passive warming (for example with external blankets) is rarely effective if the patient has significant haemorrhage, and in this setting the patient should also be actively warmed.
- It is important to document the patient's temperature.

## Metaraminol

- Metaraminol is not a priority in patients with haemorrhage. However, as shock progresses vasodilation will often develop, and it is important to minimise the total volume of IV fluid administered.
- Metaraminol should only be administered to adult patients who continue to have severe hypotension following IV fluid administration.
- If a metaraminol infusion has been commenced, this should be accompanied by further fluid boluses if indicated.
- The aim of administering metaraminol is not to restore normal perfusion, but to maintain sufficient perfusion of vital organs without increasing blood loss.

## Response and/or transport by helicopter

- Response by helicopter may be indicated if the patient has severe shock and it will arrive at scene at least 15 minutes sooner than a road-based CCP.
- Transport by helicopter may be indicated if the patient has severe shock and it will save more than 15 minutes compared to road transport. For criteria to be met, the road transport time will usually need to

# Non-traumatic Bleeding

be more than 75 minutes.

## More Information

### Version history

28 October 2024

1.0.5

**New:** Guideline released.

# Bleeding following Tonsillectomy

This guideline is for patients with suspected or confirmed bleeding from the tonsil(s), occurring within three weeks of tonsillectomy.

- Administer tranexamic acid nebulised:
  - 500 mg nebulised for an adult or a child.
  - Repeat once after 30 minutes.
- Administer tranexamic acid IV:
  - 1 g IV for an adult.
  - See the paediatric drug dose tables for a child.
- Administer 0.9% sodium chloride IV, if the patient has signs of hypovolaemia:
  - 500 ml IV for an adult.
  - 10 ml/kg IV for a child.
  - Repeat as required, minimising the total volume.
- Have a low threshold for administering antiemetics, using the ['nausea and vomiting'](#) guideline.
- If shock is severe:
  - Provide active external warming, and
  - Consider calling for blood provided there is an established protocol for this in the area.
- RSI carries significant additional risks in this setting and personnel should usually seek clinical advice if this is being considered.
- Notify receiving hospital personnel early, particularly if hypovolaemia is present.

## Backup

- Backup from an ICP/CCP should be requested if the patient has severe shock, or has an altered level of consciousness.
- Backup must not delay transport to an appropriate hospital.

## Referral and transport

- Expedite transport, particularly when there are signs of hypovolaemia.
- Transport the patient direct to a hospital with ear nose and throat (ENT) surgical facilities, whenever it is feasible and safe to do so. The hospitals with ENT surgical facilities are:

# Bleeding following Tonsillectomy

- » Whangārei Hospital.
- » Auckland City Hospital.
- » Starship Children's Hospital.
- » Waikato Hospital.
- » Tauranga Hospital.
- » Gisborne Hospital.
- » Rotorua Hospital.
- » Taranaki Base Hospital.
- » Whanganui Hospital.
- » Hawke's Bay Regional Hospital.
- » Palmerston North Hospital.
- » Wellington Regional Hospital.
- » Wairau Hospital.
- » Nelson Hospital.
- » Te Nikau Grey Hospital.
- » Christchurch Hospital.
- » Dunedin Hospital.
- » Southland Hospital.

## Additional information

### General information

- It is common to underestimate the amount of blood loss.
- Morbidity and mortality are associated with airway obstruction and/or hypovolaemic shock.
- Bleeding most commonly occurs 5-10 days post-surgery and in this setting is often associated with infection of the tonsillar area.
- Bleeding often presents as vomiting of blood. Excessive swallowing and/or recurrent vomiting of blood is a common indicator of ongoing bleeding, particularly in young children.
- Parents of young children may report finding blood on a pillow or towel. Any report of bleeding should be assumed to be post tonsillectomy bleeding until proven otherwise.
- It is normal for the tonsillar area to look yellow/white and sloughy following tonsillectomy.

# Bleeding following Tonsillectomy

## Treatment and transport

- Although commonly associated with infection, there is no role for antibiotics in the out-of-hospital setting.
- Vomiting is common. Antiemetics may not always stop vomiting because vomiting is a normal physiological response to blood in the stomach.
- Nebulised adrenaline does not have a role in the treatment of post tonsillectomy bleeding.
- Although medical texts commonly describe applying direct pressure to the tonsillar bed using gauze on the end of a pair of forceps, this should not usually be attempted without seeking clinical advice, because it:
  - Requires a very cooperative patient, and
  - Requires an experienced clinician, and
  - Can rarely be achieved in the out of hospital setting, particularly during transport.
- Most patients will want to sit forward and intermittently spit or vomit into a container. Allow the patient to suction from the front of their mouth but do not insert suction catheters into the pharynx.
- Transport should occur to a hospital with ENT surgical facilities because surgery is often required. ENT is also commonly known as otorhinolaryngology (ORL).

## Response and/or transport by helicopter

- Transport by helicopter may be indicated for patients with bleeding following tonsillectomy if it will save more than 30 minutes compared with road transport.
- For criteria to be met, the road transport time will usually need to be more than 90 minutes.

## Administration of 0.9% sodium chloride

- 0.9% sodium chloride IV should be titrated to clinical signs of intravascular volume and perfusion.
- Large volumes of crystalloid may worsen bleeding by contributing to coagulopathy, particularly in the context of severe and uncontrolled bleeding:
  - This is why the total volume of 0.9% sodium chloride should be minimised.
  - Clinical judgment is required when determining the volume administered, but it should be uncommon to exceed one litre in an adult or 20 ml/kg in a child, and it should be rare to exceed two litres in an adult or 40 ml/kg in a child.
  - Have a low threshold for seeking clinical advice if shock is severe and transport time to the most appropriate hospital is prolonged.

# Bleeding following Tonsillectomy

- Ambient temperature IV fluid can cause hypothermia which worsens bleeding by contributing to coagulopathy. IV fluid should be warmed using a dedicated warming device if one is available.
- Do not warm IV fluid in microwaves because this can result in unreliable fluid temperatures and/or may damage the bag.

## Tranexamic acid (TXA)

- TXA reduces bleeding by reducing clot break down via fibrinolysis.
- TXA is indicated if there are any signs of bleeding, even if the patient does not have signs of hypovolaemia.
- The IV dose of TXA administered for post tonsillectomy bleeding is lower than the dose of TXA for bleeding following trauma.
- TXA IV should be administered in addition to nebulised TXA, provided IV access is able to be obtained. However, TXA IV is not a priority if access is difficult, for example in young children.

## Blood

- Requests for blood to be delivered to the scene must not compromise rapid transport to the most appropriate hospital.
- Blood should only be requested if:
  - Shock is severe, and
  - There is an established protocol in the area for blood to be delivered to the scene, and
  - Blood can be administered 15 or more minutes before the patient can arrive in the most appropriate hospital.
- Blood may be available via staging at a small hospital/medical facility en route to the most appropriate hospital. In this setting personnel must have a low threshold for seeking advice from the on-call doctor for the service.
- ICPs/CCPs may continue blood administration initiated by a medical practitioner following their verbal instructions but should have a low threshold for seeking clinical advice.
- Requests for blood to be delivered to the scene outside of an established protocol must be made via the on-call doctor for the service. In this setting:
  - It will usually take approximately 15 minutes to arrange for blood to be prepared, plus transport time to the scene. This means that for such a request to be appropriate, the patient should usually have severe shock and a prolonged transport time.

# Bleeding following Tonsillectomy

- » Personnel cannot administer blood without specific instructions from the on-call doctor for the service.

## Metaraminol

- Metaraminol administration does not usually have a role in the early phase of hypovolaemia treatment associated with bleeding following tonsillectomy.
- Metaraminol may have a role in the setting of severe shock and a prolonged transport time to hospital. ICPs/CCPs should usually seek clinical advice if metaraminol is being considered in this setting.

## More Information

### Version history

28 October 2024

1.0.5

**New:** Guideline released.



# Epistaxis

## Mild bleeding

- Firmly compress the fleshy part of the nose for 15 minutes.
- If the bleeding is not controlled administer adrenaline IN as below.

## Moderate or severe bleeding

- Instruct the patient to blow their nose to clear all blood clots.
- Administer adrenaline IN into each bleeding nostril using a mucosal atomising device, and firmly compress the fleshy part of the nose for 15 minutes:
  - Administer 0.2 mg of adrenaline (2 ml of 1:10,000) per bleeding nostril for a patient aged 12 years or over.
  - Administer 0.1 mg of adrenaline (1 ml of 1:10,000) per bleeding nostril for a child aged 5-11 years.
  - Seek clinical advice if the patient is a child aged less than five years.
  - Repeat once after 20 minutes if the bleeding does not stop or recurs and tranexamic acid is not available.
- Administer tranexamic acid IN into each bleeding nostril using a mucosal atomising device, and firmly compress the fleshy part of the nose for 15 minutes if the bleeding does not stop or recurs:
  - Administer 200 mg of tranexamic acid (2 ml undiluted) per bleeding nostril for a patient aged 12 years or over.
  - Administer 100 mg of tranexamic acid (1 ml undiluted) per bleeding nostril for a child aged 5-11 years.
  - Seek clinical advice if the patient is a child aged less than five years.

## Severe bleeding that remains uncontrolled

- Place a nasal tamponade device if one is available.
- Administer tranexamic acid IV:
  - Administer 1 g of tranexamic acid IV for an adult.
  - See the paediatric drug dose tables for a child.

## Backup

- Consider requesting backup from an ECP if placement of a nasal tamponade device is indicated.

# Epistaxis

- Call for backup from an ICP/CCP if there is severe bleeding that remains uncontrolled and an ECP is unavailable.
- Backup should not be requested for tranexamic acid IN alone.

## Referral and transport

- If the bleeding is mild and:
  - **Stops completely**, no follow up is required unless the patient is taking an anticoagulant or is hypertensive. If the patient is taking an anticoagulant or dual antiplatelet therapy, or their systolic blood pressure is greater than 180 mmHg and/or their diastolic blood pressure is greater than 110 mmHg, their treatment must be reviewed in primary care (preferably by their own GP) within 24 hours.
  - **Does not stop**, the patient should be seen by a doctor or ECP within a few hours. This could be in primary care unless the patient is taking an anticoagulant, in which case the patient should be seen in an ED.
- If the bleeding is moderate to severe and:
  - **Stops following intervention**, the patient must be seen by a doctor within a few hours. This is particularly important if the patient is taking an anticoagulant or dual antiplatelet therapy and transport to an ED by ambulance will usually be required.
  - **Does not stop**, the patient must be transported to ED by ambulance.
- If the patient is not transported by ambulance or immediately referred, provide advice to avoid hot showers, hot drinks or blowing their nose for as long as possible after the bleeding has been controlled.

## Additional information

- Dual antiplatelet therapy consists of more than one antiplatelet medicine such as aspirin and ticagrelor.
- Patients with hypertension are at increased risk of recurrent epistaxis, but hypertension should not be specifically treated by ambulance personnel without seeking clinical advice.

## More Information

### Version history

9 June 2025

1.0.5.3

**Update:** Multiple changes.

# Nausea and/or Vomiting

This guideline is for patients with clinically significant nausea and/or vomiting.

- Consider placing an alcohol swab under the patient's nose and advising them to inhale the vapour through their nose for several minutes. If effective, this may be repeated as required.
- Consider administering ondansetron:
  - Administer 8 mg of ondansetron IV once for an adult.
  - Administer 4 mg of ondansetron IM for an adult, if IV access cannot be obtained.
  - The IM dose may be repeated once after 20 minutes if required.
  - See the paediatric drug dose tables for a child.
- One IV dose of ondansetron may be administered in addition to one IM dose, if clinically significant nausea and/or vomiting persists, and IV access is subsequently obtained. For example, in this setting a maximum total dose of 12 mg may be administered to an adult.
- Consider administering 0.625 mg droperidol IV/IM to patients aged 12 years or older if:
  - Nausea and/or vomiting persists despite ondansetron, or
  - Motion sickness is a component of the patient's nausea and/or vomiting, or
  - The patient has cannabinoid hyperemesis syndrome.

## Backup

- Backup should not usually be requested for administration of droperidol unless the nausea and/or vomiting is severe and causing significant distress.

## Referral and transport

- A patient may be administered ondansetron and/or droperidol and be given a clear recommendation that immediate referral to a medical facility is not required, provided the patient:
  - Appears to have a minor clinical problem, and
  - Has no other significant symptoms, and
  - Has normal vital signs.

## Worsening advice

- All patients (and their family/whānau/carers) who are not immediately referred or transported to a medical facility, must be provided with the following worsening advice, preferably in writing:

# Nausea and/or Vomiting

- » Be seen in primary care or phone Healthline on 611 116 if their symptoms are not improving within 24 hours.
- » Remain in the care of an adult and do not drive a vehicle or operate machinery for 12 hours, if droperidol has been administered.
- » Call 111 for an ambulance if they develop:
  - » Severe abdominal pain.
  - » Vomiting of blood.
  - » Inability to mobilise independently.

## Additional information

- Prophylactic administration of an antiemetic is not routinely required. Consider administering an antiemetic if the nature of the patient's injuries, transport position and/or mode of transport (for example, helicopter) are such that vomiting would be particularly problematic.
- Prevention and treatment of motion sickness with ondansetron is usually ineffective. For this reason, droperidol is the preferred antiemetic in this setting.
- Antiemetics should not usually be administered for vomiting associated with an altered level of consciousness because they are rarely effective in this setting and can affect the neurological assessment of the patient.

## Cannabinoid hyperemesis syndrome

- Cannabinoid hyperemesis syndrome is episodes of nausea and vomiting due to high dose cannabis use that is chronic and/or frequent.
- Hot showers or bathing may reduce symptoms. Symptoms usually improve if cannabis use is tapered off and/or stopped.
- Cannabinoid hyperemesis often responds well to low dose droperidol.

## More Information

### Version history

9 June 2025

1.0.5.3

**Update:** Guideline redeveloped.

11 September 2023

1.0.4

**Update:** Droperidol added as second line antiemetic under certain circumstances.

# Nausea and/or Vomiting

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# Falls

Use this guideline to help determine which patients following a fall require referral to a medical facility and if so, to which type of facility. Use the '[transient loss of consciousness](#)' guideline if the patient is suspected to have fallen due to loss of consciousness.

- Assess the patient, including an assessment of the features contained within the flag table.
- In addition, perform a falls risk assessment if the patient is aged greater than or equal to 65 years, is living independently, and is not being transported to an ED by ambulance. Gain consent and refer the patient to a falls prevention service if any abnormalities are found. Alternatively, refer the patient to their GP if a falls referral pathway is not available.

## Red flags

### Should be seen in ED, usually via ambulance

- Clinically significant injury.
- Clinically significant pain.
- Abnormal vital signs.
- Signs of stroke.
- Headache with features of concern.
- New onset of visual disturbance.
- Unable to mobilise when normally able to do so.
- Unstable medical condition contributing to the fall.

## Orange flags

### Should be followed up in primary care

# Falls

- More than one fall in the last week.
- Postural hypotension.
- Recent change in medication.
- Minor injury requiring non-urgent treatment.
- New reduction in mobility but able to weight bear.

## Referral and transport

- If one or more red flags are present the patient must be given a clear recommendation to be assessed by a doctor within two hours and should usually be given a clear recommendation to be transported to ED, usually by ambulance.
- If the patient has one or more red flags and is being referred to primary care:
  - The patient's anticipated clinical needs must be able to be safely met in primary care, and
  - A clinician within primary care must be contacted directly by ambulance personnel who agrees to accept the patient, and
  - Safe transport (if required) must be available.
- If any orange flags are present (and no red flags) the patient should be given a clear recommendation to be followed up in primary care.
- If no red and no orange flags are present, the patient is usually suitable to be given a clear recommendation to remain in the community with self-care. Advise the patient to be seen in primary care or seek advice via Healthline on 0800611 116 if they have any concerns.

## Additional information

### General principles

- A patient that has fallen without a clear mechanical cause, for example a trip or slip, requires a thorough history and clinical examination to rule out a cause for collapse.
- A patient that has fallen always requires an assessment to rule out injury, even if they appear to be uninjured.
- Do not use the term 'lift assist' when referring to a patient that has fallen as this term should only be used when a crew is assisting another crew to lift a patient.

# Falls

- Falling is not a natural part of aging and all patients aged greater than or equal to 65 years who have fallen require an assessment that considers referral to a falls prevention service.
- Have a raised index for suspicion of injury if the patient:
  - Has fallen a significant height, for example greater than one metre or five stairs in an adult, or
  - Is taking an anticoagulant, dual antiplatelet therapy, or has a known bleeding disorder.
- Always consider the patient's comorbidities and social circumstances.
- Determining that a headache is associated with features of concern requires clinical judgement. Examples of features of concern include:
  - Features within the red flags of the headache guideline.
  - New onset of headache following the fall, especially if worsening, or associated with nausea and/or vomiting.
  - Taking an anticoagulant or dual antiplatelet therapy.
- Examples of unstable medical conditions contributing to the fall include diabetes with poor glucose control and poorly controlled Parkinson's disease. Clinical judgement is required when determining that a medical condition requires review in an ED or review in primary care.
- Postural hypotension is present if there is a fall of greater than 20 mmHg in the systolic BP or greater than 10 mmHg in the diastolic BP when standing.

## Falls risk assessment

- Falls are a common cause of injury and loss of independence in older patients.
- Ambulance personnel have an important role in referring older patients at risk of a fall to a falls referral pathway, as this reduces the risk of further falls and injury.
- Referral to a falls referral pathway is not required if the patient is living in an aged residential care facility because the facility personnel are required to manage this. Patients living in an independent unit (for example a villa) in the same complex as an aged residential care facility are designated as living independently.
- Ask the patient the following questions:
  - Have you slipped, tripped or fallen in the last year?
  - Do you need to use your hands to get out of a chair?
  - Are there any activities you've stopped doing because you are afraid of falling?
- Refer the patient to a falls referral pathway if:
  - The patient answered 'yes' to any of the questions, or



# Falls

» Personnel consider the patient is at risk of falling.

- Examine the environment for hazards which may contribute to the risk of falling. Examples include rugs, mats, cords and poor footwear. Eliminate these hazards with the patient's permission if feasible.

## Referring the patient to a falls prevention service

- Falls prevention services offer follow up and support, which may include:
  - » Access to strength and balance programmes.
  - » The provision of mobility aids.
  - » Vision testing.
  - » Medication review.
- To refer a patient to a falls prevention service via ePRF:
  - » HHStJ personnel: type 'referral@falls' in the Disposition section of the ePRF.
  - » WFA personnel: use the 'note to GP' function in ePRF.
- When making a referral, include as much information in the ePRF as possible to enable the falls prevention service to offer the patient the most appropriate support. For example:
  - » The patient's contact details (especially their phone number).
  - » The answers to the falls risk questions.
  - » Whether the patient uses a mobility aid.
  - » Whether the patient requires assistance with their activities of daily living, for example showering.
  - » Whether the patient can leave their home unassisted.
  - » Any obvious falls hazards in the patient's home.

## More Information

### Version history

9 June 2025

1.0.5.3

**Update:** Guideline redeveloped.

# Non-traumatic Lumbar Back Pain

This guideline is for adults with non-traumatic lumbar back pain. Use this guideline to help determine which patients require referral to a medical facility and if so, to which type of facility.

- Assess the patient, including an assessment of the features contained within the red flags.
- Administer oral analgesia as per the '[analgesia](#)' guideline.
- Withhold IV analgesia unless pain is very severe and transport to ED is planned.
- Consider administering one dose of methoxyflurane, if required, to enable the patient to be mobilised.
- Encourage the patient to mobilise and use non-medication analgesia options such as heat packs.

## Red flags

### Should be seen in ED, usually via ambulance

- Loss of bladder or bowel control.
- Temperature > 38°C.
- Rigors.
- Abnormal vital signs.
- Pain in the thoracic spine or chest.
- Abdominal pain or tenderness.
- Altered sensation in the saddle area.
- Altered sensation and/or power in both legs.
- Unable to mobilise.
- Signs or symptoms of generalised illness.
- Pain radiating down both legs.

## Backup

- Backup is rarely required, however, ECP backup should be considered to enable the provision of additional treatments.

# Non-traumatic Lumbar Back Pain

## Referral and transport

- If one or more red flags are present the patient must be given a clear recommendation to be assessed by a doctor within two hours and should usually be given a clear recommendation to be transported to ED, usually by ambulance.
- If the patient has one or more red flags and is being referred to primary care:
  - The patient's anticipated clinical needs must be able to be safely met in primary care, and
  - A clinician within primary care must be contacted directly by ambulance personnel who agrees to accept the patient, and
  - Safe transport (if required) must be available.
- If no red flags are present, the patient is usually suitable to be given a clear recommendation to remain in the community. In this setting one dose of methoxyflurane may be administered in addition to paracetamol, ibuprofen and/or tramadol.

## Worsening advice

- All patients (and their family/whānau/carers) who are not immediately referred or transported to a medical facility, must be provided with the following worsening advice, preferably in writing:
  - Provide them with a back pain information sheet if one is available.
  - Take regular pain relief and remain as mobile as possible.
  - Be seen in primary care or phone Healthline on 0800 611 116 if their symptoms are not improving within 48 hours.
  - Call 111 for an ambulance if they develop:
    - Loss of bowel or bladder control.
    - Fever.
    - Weakness in both legs.
    - Reduced sensation in their buttocks or both legs.
    - Inability to mobilise.

## Additional information

### General principles

- Non-traumatic back pain is usually precipitated by lifting or twisting, aggravated by movement and is often associated with muscle spasm.

# Non-traumatic Lumbar Back Pain

- A prolapsed disc may compress a nerve root causing altered sensation and/or motor power in one leg. Pain that radiates into one or both legs is usually a sign of sciatic nerve involvement. If the altered sensation and/or power are only in one leg, the patient does not need immediate transport to ED provided no red flags are present.
- Always examine the back and abdomen for signs of tenderness. This will help localise the pain and may help distinguish back pain from tenderness over the kidney, lower ribs or abdominal organs.
- Always observe the patient walking. A patient with severe lumbar back pain will usually find it difficult to move from a supine to a standing position and walking may be painful. However, provided the patient is able to walk, immediate transport to ED is not required if no red flags are present.

## Red flags

- Loss of bladder or bowel control, loss of sensation over the saddle area, loss of sensation and/or power in both legs and an inability to walk are all signs of compression of the spinal cord.
- The saddle area covers the perineum, buttocks and upper posterior thighs.
- Fever is a sign of possible epidural abscess.
- Always take a history with regard to abdominal pain and examine the abdomen for signs of tenderness. Abdominal pain radiating to the lumbar spine or flank may result from conditions such as pancreatitis, gastric or duodenal ulceration, cholecystitis, pyelonephritis or a leaking abdominal aortic aneurysm.

## More Information

### Version history

9 June 2025

1.0.5.3

**Update:** Guideline redeveloped.

# Hypovolaemia from Fluid Loss

This guideline is for hypovolaemia from fluid loss. For example, from gastroenteritis. See the '[hyperglycaemia](#)' guideline if the fluid loss is associated with a hyperglycaemic state.

- Administer 0.9% sodium chloride IV if the patient has signs of hypovolaemia:
  - 1 litre for an adult.
  - 20 ml/kg for a child.
  - Repeat as required.

## Backup

- Backup from a Paramedic should be requested if the patient has signs of hypovolaemia.
- Backup from an ICP/CCP should be requested if the patient has severe shock.

## Referral and transport

- The patient should usually be given a clear recommendation to be transported to an ED by ambulance if 0.9% sodium chloride is administered.
- The patient may receive a recommendation to remain in the community and/or be seen in primary care provided the following criteria are met in addition to the criteria within the generic non-transport checklist:
  - A maximum of one litre of 0.9% sodium chloride is administered for an adult, or 20 ml/kg for a child, and
  - The patient's underlying clinical condition is clearly minor and is improving.

## Additional information

- Fluid therapy should be titrated to signs of intravascular volume and perfusion.
- Rapid IV fluid administration is not required unless the patient has severe shock. In the absence of severe shock, IV fluid boluses should be administered over 30-60 minutes.
- Patients should be encouraged to drink fluids when appropriate.

## More Information

# Hypovolaemia from Fluid Loss

## Version history

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**28 October 2024**

1.0.5

**Update:** Referral and transport updated. New advice encouraging patients to drink fluids where appropriate.

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# Autonomic Dysreflexia

This guideline is for adult patients with chronic spinal cord impairment and autonomic dysreflexia. Seek clinical advice if the patient is a child.

- Look for a cause of stimulation and resolve if possible.
- Administer 20-30 mcg increments of fentanyl to a total of 100 mcg, or 2-3 mg increments of morphine to a total of 10 mg, even if the patient cannot feel pain.
- If the patient remains symptomatic with a systolic BP greater than 160 mmHg or greater than 40 mmHg above the patient's normal blood pressure, commence the following treatment to maintain a systolic BP of 140-160 mmHg and a diastolic BP below 110 mmHg:
  - Administer 0.4 mg of GTN SL every five minutes. Move to the next step if hypertension is not controlled within approximately 10 minutes of commencing GTN SL.
  - Continue to administer GTN SL and administer labetalol IV:
    - Administer a 20 mg bolus IV every five minutes, as required, while setting up an infusion.
    - Commence an IV infusion at 1 mg/kg/hour, up to a maximum of 150 mg/hour.
    - Adjust the rate as required by 20 mg/hour every 10 minutes, titrating to effect.
    - Administer additional boluses of 20 mg IV every five minutes as required.
- If labetalol is contraindicated, or the target systolic BP are not met within 30 minutes of commencing labetalol, seek clinical advice to discuss further options.

## Backup

- Backup from a CCP should be requested if hypertension is not controlled despite GTN SL.

## Referral and transport

- The patient should receive a clear recommendation that transport to a medical facility is not required if the cause has been identified and resolved, the systolic BP is controlled with an opiate and GTN SL alone, and the patient becomes asymptomatic.
- The patient must receive a recommendation to be transported to a medical facility by ambulance if a clear cause cannot be identified and resolved, GTN IV or labetalol IV is administered, or the patient remains symptomatic. The medical facility should usually be an ED, but may be a primary care facility:
  - Particularly if the patient is well known to staff at that facility, and
  - Staff at the facility are contacted prior to arrival and agree to accept the patient.

# Autonomic Dysreflexia

## Additional information

### General principles

- Autonomic dysreflexia (also known as autonomic hyperreflexia) is a term used to describe abnormal stimulation of the autonomic nervous system in patients with chronic spinal cord impairment. Autonomic dysreflexia usually occurs in association with significant stimulation below the level of the cord injury.
- It is not entirely clear how and why autonomic dysreflexia occurs. The most common theory is that:
  - Significant stimulation below the level of the spinal cord injury results in an increase in pain-like impulses within sensory nerves going to the spinal cord. The sensory nerves are intact and capable of transmitting pain impulses, even though the patient cannot feel them.
  - The pain-like impulses cannot be transmitted to the brain because the spinal cord is damaged. This results in stimulation of the sympathetic nervous system below the level of the spinal cord injury.
  - Stimulation of the sympathetic nervous system causes vasoconstriction below the level of the spinal cord injury and this causes hypertension.
  - Hypertension triggers reflex parasympathetic stimulation, causing bradycardia and vasodilation above the level of the spinal cord injury.

### Stimuli

- The most common forms of stimuli are:
  - Bladder distension, for example from a blocked urinary catheter. If the urinary catheter is blocked and cannot be unblocked, it needs to be replaced urgently.
  - Bowel distension, for example from severe constipation or bowel obstruction. If constipation is thought likely and a caregiver present, they may be able to administer an enema or disimpact the bowel. If a caregiver is not on scene or readily available, personnel may administer the patient's own enema.
- Other forms of stimuli include:
  - Acute injury, for example fractures or burns.
  - Labour. This is well recognised and most pregnant women with chronic spinal cord impairment will be electively admitted to hospital for caesarean section under anaesthesia.
  - Other sources of pain or infection, for example pressure areas.

### Signs and symptoms (not all need to be present)

- The patient will have chronic spinal cord impairment and:



# Autonomic Dysreflexia

- » Hypertension. This may be life-threatening. If hypertension is not present then the patient does not have autonomic dysreflexia.
- » Anxiety. This may be severe.
- » Headache. This occurs as a result of the hypertension.
- » Signs of vasoconstriction (such as mottling) below the level of the spinal cord injury. This may not always be clinically obvious.
- » Bradycardia. This occurs as a reflex cardiovascular response to hypertension.
- » Signs of vasodilation (such as flushing) above the level of the spinal cord injury. This may not always be clinically obvious.
- » Tachycardia. Rarely the patient can be tachycardic if there is significant stimulation of the sympathetic nervous system at a brain level.

## More Information

### Version history

11 September 2023

1.0.4

**Update:** Removal of GTN patches and metoprolol IV. Dosing for labetalol had changed.

# Blocked Urinary Catheter

Use this guideline for patients with blocked urinary catheters including suprapubic catheters, provided the patient has not had surgery on their renal tract or prostate in the last four weeks.

- Check that the:
  - Drainage bag is not full, and
  - Drainage bag is below pubic height, and
  - Tubing is not kinked or blocked, and
  - Tubing does not contain a one way valve that has been incorrectly inserted facing the wrong way.
- Examine the patient for signs of sepsis. Recommend transport to a medical facility without further intervention if clinically significant signs of sepsis are present.
- Flush the catheter using a clean technique if it cannot be replaced within the next few hours:
  - Draw up 50-100 ml of 0.9% sodium chloride or sterile water using a catheter tip syringe. Warm this under a tap if possible.
  - Detach the drainage tubing from the catheter and attach the syringe.
  - Flush the catheter firmly over 5-10 seconds. Some discomfort is expected but stop if there is severe pain.
  - Remove (if possible) the fluid using the syringe.
  - Re-attach the drainage tubing and ensure urine is flowing into the drainage bag.
- Arrange for the catheter to be replaced as soon as possible.

## Referral and transport

- If the problem was resolved by flushing the catheter, arrange for the catheter to be replaced as soon as possible, preferably within the community. Advise the patient to drink plenty of fluid to help maintain a good urine flow.
- If the problem was not resolved the patient should be given a clear recommendation to be transported to a medical facility to have the catheter replaced. Clinical judgement is required regarding the mode of transport and transport by private vehicle may be appropriate, provided this will not cause significant delay.

## Additional information

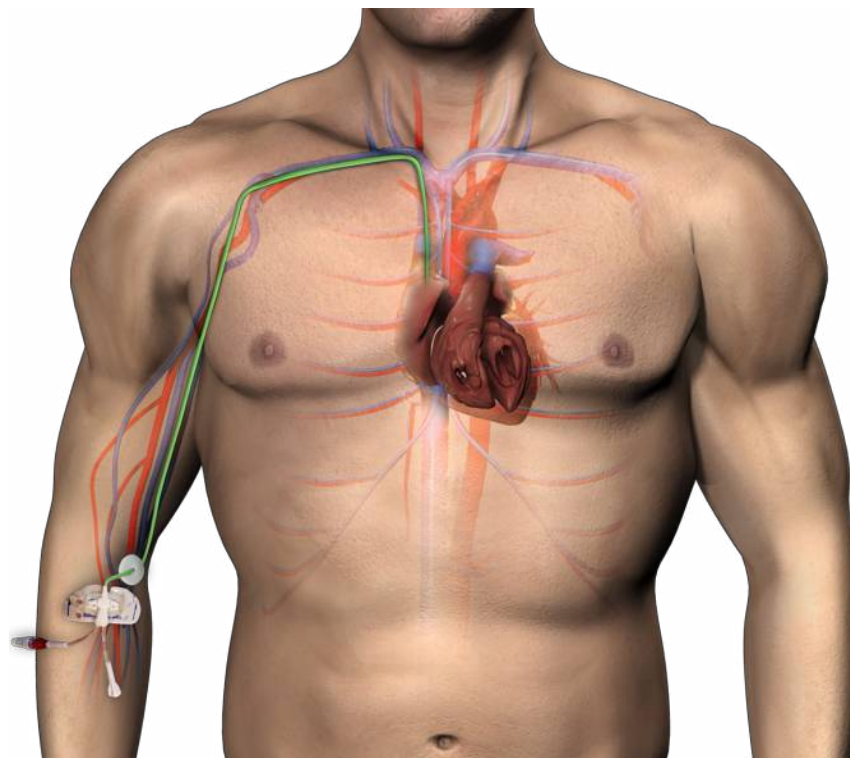
- The preferred approach is to have the catheter replaced, provided this is feasible. This may be available via an ECP or a local pathway such as a district nursing service.
- Even if the catheter is unblocked, the catheter should subsequently be replaced as soon as possible.

# Blocked Urinary Catheter

- Flushing a urinary catheter requires a clean, but not sterile technique. Perform hand hygiene before putting on gloves. Minimise contact between gloved hands and non-sterile surfaces. Do not contaminate the end of the catheter syringe with non-sterile surfaces prior to connecting it to the catheter. The end of the catheter and the drainage tubing do not require cleaning with an alcohol swab.
  - It is very common for sediment to be visible in the urine. This does not mean that a urinary tract infection is present.
  - It is very common for the urine to appear slightly pink and contain small blood clots following a catheter flush and this does not require specific treatment or advice. If there are large blood clots or there is red urine, the patient should be given a clear recommendation to be transported to a medical facility by ambulance.
  - If there are signs of clinically significant sepsis, flushing the catheter may precipitate severe sepsis and this is why transport must be recommended in this setting.
  - It is common for the catheter to be flushed, but for the fluid not to return when the syringe is aspirated. This is not of concern if urine subsequently flows into the drainage bag. If urine does not flow the catheter is still blocked.
-

This guideline describes when existing vascular access may be used by personnel at Paramedic, ICP and CCP level.

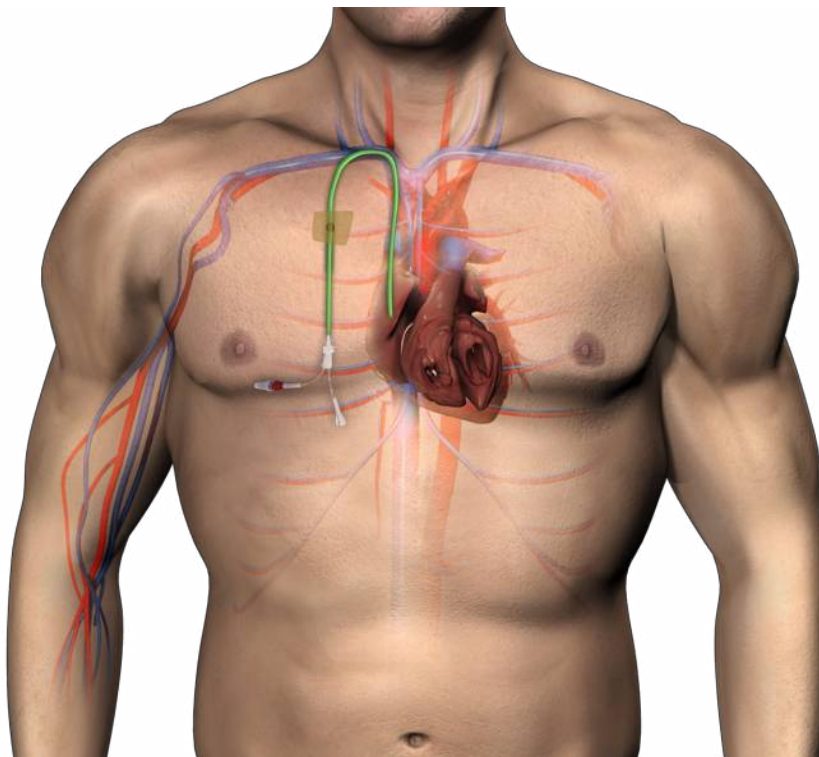
## Peripherally inserted central catheters (PICC)



### PICC line

- If a patient has an existing PICC line, IV access should be obtained in another limb whenever feasible. If this is not feasible, IV cannulae must not be placed in the same antecubital fossa as a PICC line because this risks damage to the line.
- PICC lines may be used for medicine or fluid administration if immediate IV treatment is required and alternative IV access cannot be obtained:
  - » Wear clean gloves.
  - » The luer plug may need to be changed if the existing one is not compatible with equipment carried by ambulance personnel.
  - » Clean the luer plug with an alcohol swab for fifteen seconds.
  - » Administer medicines or fluids as usual followed by a minimum of a 10 ml flush.
  - » Do not disconnect a pre-existing infusion (for example, IV nutrition) unless this is absolutely necessary.
  - » If a pre-existing infusion is disconnected do not reconnect it.

## Tunnelled central lines



### Tunnelled central line

- Tunnelled lines may only be used for medicine or fluid administration if a patient has an immediately life-threatening condition and alternative vascular access cannot be obtained.
- If the line is being used for dialysis:
  - It may contain a concentrated solution of anticoagulant to prevent blood clotting in the line. In this setting the line is usually labelled, but always assume that anticoagulant is present.
  - 5 ml of blood should be withdrawn and discarded to remove the anticoagulant before using the line.
  - If blood cannot be withdrawn, try another lumen.
  - If blood still cannot be withdrawn, consider using the lumen noting that it is possible the patient may get a bolus of anticoagulant. The anticoagulant dose will be small and is unlikely to be detrimental.
- The rest of the principles of using a tunnelled central line are the same as those for using a PICC line.

## Central lines



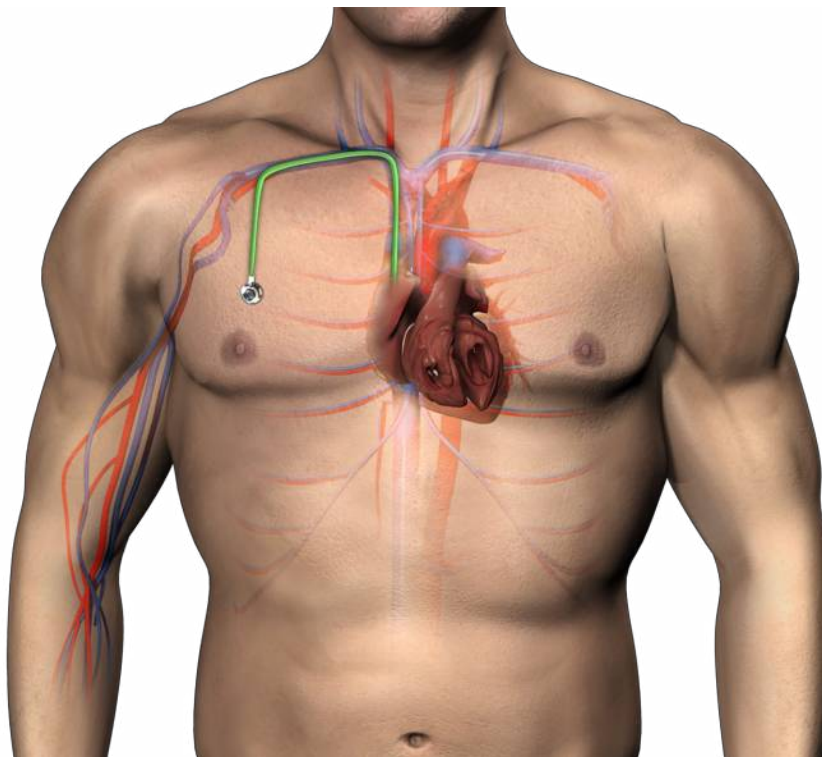
### Central line

- Central lines may only be used for medicine or fluid administration if a patient has an immediately life-threatening condition and alternative vascular access cannot be obtained.
- The principles of using a central line are the same as those for using a tunnelled line.

## Dialysis fistulae

- Do not gain IV access in the same limb as a fistula, unless IV access cannot be obtained elsewhere and medicine or fluid administration is required for an immediately life-threatening condition. In this setting, gain IV access as far from the fistula as possible, noting that if the IV access is proximal to the fistula there may be arterial flow within the vein.
- Do not measure blood pressure in the same limb as a fistula because the result will be altered.
- Dialysis fistulae may only be used for medicine or fluid administration if a patient has an immediately life-threatening condition and alternative vascular access cannot be obtained.

## Portacath lines



### Portacath line

- A portacath line may only be used for medicine or fluid administration if a patient has an immediately life-threatening condition and alternative vascular access cannot be obtained, noting that:
  - » The specifically designed needle must be available.
  - » The port must only be accessed by people trained to do so.
  - » An anticoagulant may be present.

## Introduction

- There is no age at which a patient can be automatically considered to be elderly. Each patient must be considered individually and their chronological age considered separately from their level of physical ability, comorbidities and physiological reserve (the combination of the latter three is often referred to as biological age or physiological age). However, patients aged greater than or equal to 75 years usually require a more considered and/or altered approach than younger patients and this is often reflected in the guidance within these CPGs.
- It is important to have an understanding of the differences in the elderly and to always take these into account.
- The elderly are more likely to have reduced physiological reserve, frailty, comorbidities, and to be taking multiple medicines (polypharmacy). The combination of these factors means that the clinical needs of the elderly are usually more complex as they age.

## Assessment

- Assessment may be challenging because of barriers to communication, the presence of comorbidities and polypharmacy.
- Communication may be more difficult because the patient may have reduced hearing and/or vision, or pre-existing conditions that reduce comprehension (such as receptive dysphasia or dementia). Strategies that can enhance communication include:
  - Speak slowly and clearly, but avoid being patronising.
  - Allow sufficient time for the patient to comprehend and answer questions.
  - Check that sensory aids are functional.
  - Reduce unnecessary background noise.
  - Place light sources behind the patient if possible.
- Always consider the influence of the patient's comorbidities:
  - Comorbidities may impair physiological reserve. For example, heart failure impairs the cardiovascular response to sepsis.
  - The signs and symptoms of a chronic disorder may mask the signs and symptoms of an acute disorder. For example, it may be more difficult to recognise respiratory infection in a patient with COPD and chronic production of sputum.
  - Treatment of a chronic disorder may mask the signs and symptoms of an acute disorder. For example, a patient taking a beta-blocker may not develop a tachycardia.
- Polypharmacy adds to the complexity. A detailed history is necessary to confirm compliance with medicines and the presence of self-medication, for example taking over the counter medicines or taking



medicines previously prescribed for another condition ('left over' medicines). Patients accessing multiple different health providers are at higher risk of polypharmacy than those who have coordinated care through one lead provider.

- The elderly are more likely to have an atypical presentation. For example, they are more likely to have an atypical presentation of myocardial ischaemia.
- The elderly are more likely to under-report symptoms.
- Always take a history from family members and caregivers in addition to taking a history from the patient, as they are the most likely to notice subtle changes, for example changes in mental status.

## Physiological changes of ageing

- Ageing results in physiological changes that occur in addition to the physiological changes associated with comorbidities. The physiological changes associated with ageing involve a general slowing of body processes and a reduction in physiological reserve, resulting in reduced ability to compensate during times of increased demand, for example during sepsis.
- The respiratory system changes include reduced respiratory muscle strength, increased thoracic cage stiffness, loss of alveolar structure (with less area for gas exchange) and a reduction in functional residual capacity, resulting in a lowering of SpO<sub>2</sub>.
- The cardiovascular system changes include reduced pacemaker cells, stiffening of arterial walls and reduced ability to vasoconstrict in response to shock, reduced baroreceptor sensitivity, reduced maximum heart rate, reduced stroke volume and a reduction in cardiac output.
- The nervous system changes include slowing of nerve conduction and reflexes. Brain size decreases allowing for more movement and stretching of bridging veins, with increased risk of cerebral bleeding (particularly subdural bleeding). The signs and symptoms of subdural bleeding may be subtle because of the increased intracranial free space.
- The renal system changes include reduced renal blood flow and function. This impairs excretion of waste products and medicines that are excreted renally.
- The hepatic system changes include a reduction in metabolic enzyme activity that prolongs the metabolism and clearance of most drugs.
- The musculoskeletal system changes include reduced muscle and bone mass, reduced flexibility and function, and loss of calcium in bones that increases the risk of fractures.
- The skin changes include skin becomes thinner and drier, has less elasticity and loses collagen. Sweat gland activity decreases and reduces the ability to thermoregulate in a hot environment.

- The metabolic changes include a reduction in body mass and the speed of physiological processes, resulting in a reduction in metabolic rate and heat production, increasing the risk of developing hypothermia in a cold environment.

## Decisions and recommendations regarding treatment and/or referral

- Decisions and recommendations regarding treatment and/or referral may be challenging because of the presence of reduced physiological reserve, comorbidities and polypharmacy.
- The elderly have a lowered percentage of body mass that is water and this reduces the volume of distribution of most medicines. As a result, the same dose of a medicine will usually have a greater effect (because of less dilution) than in a younger patient. Consider reducing the dose of medicines, particularly if the patient is frail, and whenever feasible administer IV medicines in low doses that are titrated to effect.
- There is some evidence that the elderly are less likely to receive pain relief than younger patients. Pain in the elderly may be masked by challenges with communication, reduced physiological response to pain, reduced cognition and a misconception that the elderly are stoic.
- Treatment decisions are often complicated by uncertainty regarding what ceiling (or maximum level) of treatment and/or intervention is appropriate, particularly if the patient is frail or has severe comorbidities. Always take into account all available information, including the patient's known wishes (for example in an advance directive) and information from health professionals that know the patient well. Seek clinical advice if uncertain.
- Have a lowered threshold for recommending the elderly are transported to a medical facility. However, there is good evidence that they are at increased risk of harm when admitted to hospital, particularly if they have dementia or are frail. These patients are preferably treated in the community by primary care personnel, particularly if they are in an aged residential care facility, whenever this is feasible and safe. In this setting direct communication with a doctor or nurse who accepts the referral should occur whenever possible.
- The elderly are at increased risk of falling, but falling is not a natural aspect of ageing. If the patient has fallen (or is at risk of falling) and does not require transport to a medical facility, ambulance personnel have an important role in ensuring the patient is referred to a falls prevention service. See the ['falls'](#) guideline.

## Dementia

- Dementia is a term for a broad category of diseases resulting in a long-term reduction in cognition, intellect and memory.
- The elderly are at increased risk of dementia, but dementia is not a natural aspect of ageing.

- The presence of dementia adds additional complexity when assessing the elderly and making decisions and/or recommendations regarding treatment and/or referral.
- Dementia is commonly associated with behavioural and psychiatric symptoms, including:
  - Physical aggression.
  - Loud vocalisation.
  - Agitation.
  - Wandering.
  - Anxiety.
  - Hallucinations.
  - Delusions.
- Acute worsening of the behavioural and psychiatric symptoms associated with dementia is a common cause for ambulance personnel to be called. The usual principles of assessment and making decisions and/or recommendations regarding treatment and/or referral apply, with an additional focus on:
  - Assessing for a treatable underlying cause, for example sepsis.
  - Assessing for an injury causing pain. Have a lowered threshold for administering pain relief, but avoid tramadol because drugs with anticholinergic activity may make confusion worse.
  - Assessing for a contribution from medicines with anticholinergic activity.
- Avoid sedation whenever it is feasible and safe to do so, because sedation may be associated with long-term worsening of the behavioural and psychiatric symptoms of dementia, and is associated with an increased risk of mortality. If sedation is required to ensure safety, the preferred medicines (in order) are olanzapine, droperidol and midazolam. Ketamine must be reserved for when safety is severely compromised and ketamine administration to patients with dementia should be rare.
- Patients with dementia often suffer from a significant worsening of their behavioural and psychiatric symptoms when admitted to hospital. Patients with dementia are preferably treated in the community by primary care personnel, particularly if they are in an aged residential care facility, whenever this is feasible and safe. In this setting direct communication with a doctor or nurse who accepts the referral should occur whenever possible.

# Obesity

## General principles

- Obesity increases the risk of associated comorbidities.
- Patients with obesity have an increased risk of mortality and morbidity for the same level of illness or injury, in comparison to non-obese patients.
- Have a lowered threshold for recommending the patient is assessed in a medical facility if the patient is obese.
- Obesity is associated with psychological problems including poor self-esteem and depression.

## Pathophysiological effects of obesity

- Total lung capacity and functional residual capacity (FRC) are reduced, causing the patient to desaturate faster than a non-obese patient.
- Work of breathing is raised due to the weight of the chest wall and abdominal contents. In particular, the work of breathing may be very high in the supine position, which should be avoided whenever possible.
- There is an increased risk of:
  - Obstructive sleep apnoea.
  - Chronic hypercarbia from obesity hypoventilation syndrome.
  - Hypertension.
  - Ischaemic heart disease.
  - Type two diabetes.
  - Osteoarthritis.
  - Acute infection, particularly soft tissue infection.

## Assessment

- Assessment is more difficult, but despite this obese patients require more careful assessment and greater attention to detail than non-obese patients.
- It is more difficult to auscultate heart and breath sounds.
- It is more difficult to examine the abdomen.
- 12 lead ECG acquisition is more difficult due to difficulty locating anatomical landmarks for lead placement. Flattening or inversion of the T wave may be due to obesity, but should not be attributed to obesity alone.

# Obesity

- Pulse oximetry may be unreliable due to excessive tissue thickness. Consider placing the probe on a little finger, little toe or an earlobe.
- Blood pressure measurement may be inaccurate due to the blood pressure cuff not being the correct size. Use the correct sized cuff if available or use the patient's forearm to measure the BP, noting that this technique may result in higher readings than those taken on the upper arm.

## Treatment

- Airway management is more difficult and a two-person bag/mask technique should be utilised if possible.
- Intubation is more difficult:
  - Routinely ramp the patient by elevating their upper body to align their ears with their sternal notch.
  - Use a video laryngoscope if one is available.
  - Routinely use a bougie and anterior laryngeal manipulation.
- Mechanical ventilation is more difficult:
  - Increase the peak inspiratory pressure if necessary.
  - Consider increasing the PEEP to 10 cmH<sub>2</sub>O.
  - Consider raising the upper body to aid ventilation.
- Finger thoracostomy is more difficult, due to the challenge of identifying landmarks and the thickness of the chest wall. Consider making a longer than usual incision and be assertive with dissection.
- IV cannulation is more difficult. Have a lowered threshold for gaining IO access.
- The patient is at higher risk of respiratory depression with opiates and midazolam.
- The patient is at higher risk of hypercarbia with oxygen administration.



Ramping

## Transport

- Ensure the patient's weight does not exceed the stretcher capacity.
- Call for a specific bariatric vehicle and/or stretcher if this is available.
- Call for extra personnel, including Fire personnel if required.
- Ambulance personnel may request lifting assistance from Fire personnel for obese patients requiring transport to a medical facility, when there is no other reasonable alternative. A patient status must be provided and personnel must not request assistance for patients being returned home or for pre-planned inter-facility transfers.
- Transport the patient sitting up whenever possible.
- Restrain the patient as per normal. If the patient cannot wear a seatbelt, consider other forms of restraint. The driver must also ensure that the nature of their driving is modified to keep the patient as safe as possible during transport.
- If the patient has a CPAP or BiPAP machine, ensure this is transported with the patient whenever feasible.

# Obesity

- Provide advanced warning of arrival at the medical facility.
  - If the patient is morbidly obese, it may be unsafe for them to be transported by air and whenever feasible this should be discussed with Air Desk personnel when making the request.
-

# Adrenal Insufficiency

This guideline is for patients at risk of adrenal crisis or suspected to have adrenal crisis. Follow the patient's management plan if they have one.

- Measure the blood glucose concentration and treat accordingly.
- Administer hydrocortisone:
  - 100 mg IV or IM for an adult.
  - See the paediatric drug dose tables for a child.
- Administer 0.9% sodium chloride IV if the patient has signs of hypovolaemia:
  - 1 litre for an adult.
  - 20 ml/kg for a child.
  - Repeat as required.
- In the absence of an obvious cause (for example, trauma), if the patient is not responding rapidly to the above treatment, consider the possibility of bacterial sepsis. If bacterial sepsis is suspected, refer to the ['bacterial sepsis'](#) guideline.
- Administer metaraminol IV if the systolic blood pressure is less than 100 mmHg in an adult, or significantly less than the normal predicted systolic blood pressure in a child.
- **Metaraminol for an adult:**
  - Administer via an infusion device:
    - Administer an initial bolus of 0.5-1 mg metaraminol IV.
    - Start the infusion at 2 mg/hour and adjust the rate as required.
  - Alternatively, administer a bolus of 0.5-1 mg metaraminol IV every 5-10 minutes as required.
- **Metaraminol for a child:**
  - Administer a bolus of metaraminol IV (see the paediatric drug dose tables) every 5-10 minutes as required.

## Backup

- Backup from an ICP/CCP should be requested if:
  - Hypotension persists despite hydrocortisone and 0.9% sodium chloride IV, or
  - Hyperkalaemia is suspected.



# Adrenal Insufficiency

## Referral and transport

- The patient must receive a clear recommendation to be transported to an ED by ambulance.

## Additional information

### General principles

- Adrenal insufficiency is an endocrine disorder caused by an inadequate production of adrenal hormones, including cortisol and aldosterone. These hormones are essential for a normal cardiovascular response to occur at times of physiological stress.
- Adrenal crisis is a life-threatening form of adrenal insufficiency, leading to shock and/or hypoglycaemia.
- Hydrocortisone is the primary treatment and is unlikely to cause harm if the diagnosis is unclear. There should be a low threshold for administration, with IM and IV routes being equally effective.
- Patients with known adrenal insufficiency are usually well educated about their illness and will likely have a written emergency treatment plan. They should be administered hydrocortisone for any illness or injury where their status is three or worse.
- Patients are at risk of adrenal insufficiency if they have been taking a steroid (for example prednisone or dexamethasone) for more than one month, and suddenly stop taking steroids, or suffer from physiological stress. They should be administered hydrocortisone for any illness or injury where their status is three or worse.
- The patient may have their own hydrocortisone for injection in the event of illness or injury. Personnel should follow any instructions (including verbal) regarding IM or IV administration of hydrocortisone.

### Adrenal crisis

- Adrenal crisis is a life-threatening condition that usually takes several hours to develop, although it can occur more quickly. Signs and symptoms can include:
  - Severe weakness.
  - Altered level of consciousness.
  - Syncope.
  - Shock.
  - Abdominal pain.
  - Back pain.
  - Nausea and vomiting.

# Adrenal Insufficiency

» Hypoglycaemia.

- Patients with adrenal crisis are at risk of hyperkalaemia. See the '[hyperkalaemia](#)' guideline for additional treatment if this is suspected.

## Adrenal insufficiency

- Patients with known adrenal insufficiency, or who are at risk of adrenal insufficiency, should be treated under this guideline if there is physiological stress that could lead to adrenal crisis.
- Examples of conditions causing physiological stress include:
  - » An illness such as infection, vomiting, or diarrhea (status three or worse).
  - » An injury (status three or worse).
  - » Labour.
- Adrenal insufficiency can be categorised into three types:
  - » Primary. This is caused by adrenal gland dysfunction and is most commonly caused by an autoimmune disease such as Addison's disease.
  - » Secondary. This is caused by pituitary gland dysfunction and is most commonly caused by a disease such as hypopituitarism.
  - » Tertiary. This is caused by hypothalamic dysfunction and is most commonly caused by the administration of long-term steroids.

## More Information

### Version history

13 February 2026

1.0.7.1

**Correction:** Paediatric doses of hydrocortisone in interactive flowchart.

9 June 2025

1.0.5.3

**New:** Guideline released.

# Abdominal Pain

Use this guideline to help determine which patients with abdominal pain require referral to a medical facility and if so, to which type of facility.

- Assess the patient, including an assessment of the features contained within the red flags.
- Acquire a 12 lead ECG if the abdominal pain is epigastric and the patient is aged over 35 years.

## Red flags

**Should be seen in ED, usually via ambulance**

# Abdominal Pain

- Severe pain.
- Abnormal vital signs.
- Pain radiating to the back.
- Loin or flank pain.
- Temperature  $\leq 35^{\circ}\text{C}$  or  $\geq 38^{\circ}\text{C}$ .
- Rigors.
- Female aged 10-50 years and menstrual period is overdue.
- Pregnant.
- Abdominal tenderness on palpation.
- Pain made worse by movement.
- Epigastric pain with suspicion of myocardial ischaemia.
- Persistent or recurrent vomiting.
- Frank haematuria.
- Known untreated AAA.
- Aged  $< 5$  years.
- Aged  $\geq 65$  years.
- Immunocompromised (for example on steroids or immunotherapy).
- Haematemesis

## Referral and transport

- If one or more red flags are present the patient must be given a clear recommendation to be assessed by a doctor within two hours and should usually be given a clear recommendation to be transported to ED, usually by ambulance.
- If the patient has one or more red flags and is being referred to primary care:
  - The patient's anticipated clinical needs must be able to be safely met in primary care, and

# Abdominal Pain

- » A clinician within primary care must be contacted directly by ambulance personnel who agrees to accept the patient, and
- » Safe transport (if required) must be available.
- If no red flags are present, the patient is usually suitable to be given a clear recommendation to remain in the community.

## Worsening advice

- All patients (and their family/whānau/carers) who are not immediately referred or transported to a medical facility, must be provided with the following worsening advice, preferably in writing:
  - » Be seen in primary care or phone Healthline on 0800611 116 if their symptoms are not improving within 24 hours.
  - » Call 111 for an ambulance if they develop:
    - » Severe worsening pain.
    - » Persistent vomiting.
    - » Fever.
    - » Inability to mobilise.

## Additional information

### Red flags

- Abdominal pain radiating to the spine or flank may result from conditions such as pancreatitis, gastric or duodenal ulceration, cholecystitis, pyelonephritis, or a leaking abdominal aortic aneurysm.
- An abdominal aortic aneurysm is usually asymptomatic prior to leaking. Although many references describe a pulsating mass, this may not be palpable. A leaking abdominal aortic aneurysm usually presents with abdominal pain that radiates to the back and signs of shock.
- Rigors indicate that bacteria may be present in the blood.
- A female aged 10-50 years whose menstrual period is overdue may have an ectopic pregnancy.
- Perforated bowel (for example from cancer, diverticular disease or ulceration) usually presents with non-specific abdominal pain for 1-2 days followed by signs of peritonitis (abdominal tenderness with pain made worse by movement).
- All patients with upper abdominal (epigastric) pain should have a 12 lead ECG acquired, noting that a normal ECG does not rule out myocardial ischaemia.

# Abdominal Pain

## More Information

### Version history

|                                       |         |
|---------------------------------------|---------|
| <b>9 June 2025</b>                    | 1.0.5.3 |
| <b>Update:</b> Guideline redeveloped. |         |
| <b>28 October 2024</b>                | 1.0.5   |
| <b>Update:</b> Red flags updated.     |         |

# Transient Loss of Consciousness

This guideline is for patients who have had a transient loss of consciousness.

- Assess the patient, including an assessment of the features contained within the flag tables.

## Red flags

**Should be seen in ED, usually via ambulance**

- Significantly abnormal vital signs.
- Clinically significant injury.
- Failure to recover to normal for the patient.
- Suspected cardiac cause, for example suspected dysrhythmia.
- Has a pacemaker or ICD.
- Abnormal 12 lead ECG with features of concern.
- New or unexplained shortness of breath.
- Moderate to severe headache.
- Pregnant.
- New or unexplained chest pain.
- Occurred during exertion.
- Occurred while supine.
- Aged less than 12 years.

## Orange flags

**Should be seen in primary care within 48 hours**

# Transient Loss of Consciousness

- Aged over 65 years.
- A history of heart disease.
- Postural hypotension.
- Family history of dysrhythmia or sudden cardiac death.
- Abnormal 12 lead ECG but no features of concern.

## Referral and transport

- If any red flags are present the patient must be given a clear recommendation to be assessed by a doctor within two hours and should usually be given a clear recommendation to be transported to ED, usually by ambulance. Note that additional judgement is required if the patient has previously been fully investigated for the same presentation, and in this setting follow up in primary care may be appropriate.
- If the patient has red flags and is being referred to primary care:
  - The patient's anticipated clinical needs must be able to be safely met in primary care, and
  - A clinician within primary care must be contacted directly by ambulance personnel who agrees to accept the patient, and
  - Safe transport (if required) must be available.
- If any orange flags are present (but no red flags) the patient should be given a clear recommendation to be followed up in primary care.
- If no red flags and no orange flags are present, the patient is usually suitable to be given a clear recommendation to remain in the community with self-care.

## Worsening advice

- All patients (and their family/whānau/carers) who are not immediately referred or transported to a medical facility, must be provided with the following worsening advice, preferably in writing:
  - Be seen in primary care or phone Healthline on 0800611 116 if their symptoms recur but they quickly recover to their normal self.
  - Call 111 for an ambulance if their symptoms recur and they do not quickly recover to their normal self.

## Additional information



# Transient Loss of Consciousness

## General principles

- Transient loss of consciousness is common and is usually due to benign syncope. However, it can be associated with clinically important disease, particularly heart disease.
- Benign syncope is due to a brief fall in cardiac output, usually associated with bradycardia, in the absence of important disease. The patient should then regain a normal level of consciousness within a few minutes of lying down. If the patient does not quickly regain a normal level of consciousness, consider an alternative provisional diagnosis.

## Assessment

- The history from a witness is important. A witness to the event who is no longer present should be contacted by phone if possible.
- What was the patient's posture before the event? Commonly there will be a prolonged period of standing prior to benign syncope.
- Were there any provoking factors? Commonly a procedure (for example an injection) or circumstance (for example the sight of blood) will provoke benign syncope.
- Were there any prodromal symptoms? Commonly the patient will complain of feeling sweaty and/or hot prior to benign syncope.
- What was the appearance and colour of the patient during the event? Most commonly the patient will be very pale as a result of low cardiac output.
- Were any abnormal movements observed during the event? It is possible for a patient to have some brief abnormal movements and/or twitching during benign syncope, but rhythmic jerking movements or more than brief abnormal movements suggest a seizure may have occurred.
- How long was the patient unconscious for? Most patients should rapidly recover to a normal level of consciousness within a few minutes. Failure to do so suggests a neurological cause for the event.
- Was the patient confused post the event? A very brief period of disorientation may occur, but confusion that persists beyond a few minutes suggests a neurological cause for the event.

## Physical assessment and vital signs

- Tongue biting and/or urinary incontinence suggest a seizure has occurred.
- Determining that an abnormal 12 lead ECG has features of concern requires clinical judgement. Examples of 12 lead ECG features of concern include:

# Transient Loss of Consciousness

- » ST segment elevation.
  - » ST segment depression.
  - » Left ventricular hypertrophy.
  - » Left or right bundle branch block.
  - » Second or third degree heart block.
  - » A persistent heart rate of less than 50/minute.
  - » A shortened (less than 350 ms) or prolonged (greater than 450 ms) QT interval.
  - » Any dysrhythmia.
  - » Ventricular pre-excitation. For example, Wolff-Parkinson-White syndrome.
  - » Paced rhythm.
  - » Additional clinical judgement is required if the features of concern are unchanged from a copy of a previous ECG (for example via shared care records), and the patient has previously been fully investigated. In this setting, follow up in primary care may be appropriate if no other red flags are present.
- Postural hypotension is present if there is a fall of greater than 20 mmHg in the systolic or greater than 10 mmHg in the diastolic BP, when standing.
-

# Medicines

## Introduction

- This guideline contains additional information on medicines, but does not incorporate all required knowledge.
- The information is limited to that which is particularly relevant. For example, only cautions and adverse effects relevant to medicine administration in the out-of-hospital setting are listed.

## Drawing up and administering medicines

- Medicines described for intravenous (IV) administration may be administered using the same dose via the intraosseous (IO) route.
- With the exception of adrenaline, medicines described for intramuscular (IM) administration may be administered using the same dose via the subcutaneous (SC) route.
- All medicines administered via the IV or IO route require an IV flush of 0.9% sodium chloride between medicines.
- Unless specifically described within these CPGs, medicines must not be combined within the same syringe.
- For each medicine, a standardised approach to drawing up, diluting and administering the medicine has been described. Taking a standardised approach reduces medication errors.
- The person with the medicine within their delegated scope of practice should usually be the person who draws up and administers the medicine, as this reduces drug errors. However, it is acceptable for other personnel to draw up and/or administer the medicine, provided the person with the medicine within their delegated scope of practice is responsible for all aspects of medicine administration. For example, an EMT, Paramedic, ICP or CCP may allow a First Responder to administer ondansetron IM as a learning exercise, but the EMT, Paramedic, ICP or CCP is responsible for ensuring the 'five rights' as below.
- The person responsible for the administration of the medicine is responsible for ensuring the 'five rights':
  - The right medicine is being administered.
  - The right dose is being administered.
  - The right patient is receiving the medicine. In particular, the contraindications and cautions have been considered.
  - The right route is being used.
  - The medicine is being administered at the right time. In particular, the dosing interval is correct.
- All personnel are responsible for ensuring good practice:
  - If a second clinical person is present they must be shown the ampoule and asked to name it.

# Medicines

- » The specified dilutions must be used.
- » The syringe must be labelled with name and concentration of the medicine unless the medicine is being drawn up and administered in one uninterrupted manoeuvre.
- » If a second clinical person is present they should be asked to check the calculation of a diluted solution.
- » If a medicine has a maximum dose and more than this has been drawn up, the excess dose must be discarded before administering the medicine.
- » The person administering the medicine should clearly say the medicine name, dose and route as it is administered.

## Administering medicines with caution

- The words 'use caution' or 'administer with caution' are used in a number of places within these CPGs.
- This indicates that personnel must consider the benefits and risks before administering the medicine, including the possibility of withholding the medicine, reducing the dose or extending the time between doses.

## Documentation

- All medicines administered must be documented, including:
  - » The time, dose and route of administration.
  - » The rate and total dose/volume administered if an infusion is used.

## More Information

### Version history

22 November 2022

1.0.1

**Update:** Documentation, drawing and administering medicines.

# Adenosine

## Mechanism of action

- Adenosine is an antidysrhythmic used for the treatment of paroxysmal supraventricular tachycardia (SVT).
- Adenosine is a nucleoside that depresses conduction through the AV node. This interrupts re-entry circuits within the heart and may restore sinus rhythm in patients with SVT.

## Authorised practice levels

- ICPs and CCPs.

## Indications

- ✓ Patients aged greater than or equal to 12 years with SVT and a ventricular rate greater than or equal to 150/minute, and
  - » The patient is not severely compromised, and
  - » The rhythm fails to revert following two Valsalva manoeuvres.

## Contraindications

- ✗ Known severe allergy.
- ✗ Known sick sinus syndrome without an internal pacemaker in place. Adenosine may cause severe bradycardia if the patient has sick sinus syndrome.
- ✗ Previous 2<sup>nd</sup> or 3<sup>rd</sup> degree heart block without an internal pacemaker in place. Adenosine may cause heart block if the patient has had previous heart block.
- ✗ Previous heart transplantation without an internal pacemaker in place. Following a heart transplant the heart is denervated and adenosine may cause severe bradycardia.

# Adenosine

## Cautions

- ⚠ Asthma. Adenosine may precipitate bronchospasm and should be withheld if the patient has had recurrent life-threatening attacks of bronchospasm, or is currently suffering an exacerbation of asthma.
- ⚠ COPD. Adenosine may precipitate bronchospasm and should be withheld if the patient has had recurrent life-threatening attacks of bronchospasm, or is currently suffering an exacerbation of COPD.
- ⚠ Wolff-Parkinson-White (WPW) syndrome if the rhythm is possibly fast atrial fibrillation. Adenosine is not contraindicated in a patient with known WPW syndrome provided the rhythm is clearly SVT. If the rhythm is possibly fast atrial fibrillation, adenosine should be withheld because of the risk of precipitating VF.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated. However, the balance of risk is in favour of administration if indicated.

## Dosage

- 6 mg.
- A second dose of 12 mg may be administered if the rhythm fails to revert.

## Administration

- The preferred site of injection is the antecubital fossa.
- Administer the dose as a rapid bolus followed immediately by a rapid flush of a minimum of 20 ml of 0.9% sodium chloride.

## Common adverse effects

- Bradycardia and/or sinus pause which may be up to 30 seconds.
- Shortness of breath and/or an urge to breathe deeply.
- Feeling light-headed.
- Nausea and flushing.
- Feeling of chest pressure and/or severe apprehension.

# Adenosine

## Usual onset of effect

- 5-10 seconds.

## Usual duration of effect

- 10-20 seconds.

## Pharmacokinetics

- Adenosine is rapidly taken up and metabolised within seconds by red blood cells and vascular endothelial cells.

## Usual preparation

- Ampoule containing 6 mg in 2 ml.

## Common interactions

- None.

## Additional information

- Adenosine usually causes a brief period of very low cardiac output and this often causes the patient to feel severe apprehension and/or an impending sense of doom. Warn the patient they may feel awful but reassure them this will pass very quickly.
- If the rhythm is possibly fast atrial fibrillation and an accessory pathway is present, adenosine may precipitate VF. If uncertain, treat the rhythm as atrial fibrillation and not SVT.
- Seek clinical advice if the patient is aged less than 12 years.

## More Information

### Version history

11 September 2023

1.0.4

**Update:** Indications updated. Addition to administration and additional information headings.

# Adenosine

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# Adrenaline

## Mechanism of action

- Adrenaline stimulates alpha and beta receptors, with the predominant effects occurring at alpha 1, beta 1 and beta 2 receptors.
- Alpha 1 stimulation causes smooth muscle contraction, vasoconstriction of blood vessels and stimulation of glycogenolysis and gluconeogenesis.
- Beta 1 stimulation causes an increase in inotropy (cardiac contractility), an increase in chronotropy (heart rate) and an increase in dromotropy (speed of electrical conduction within the heart).
- Beta 2 stimulation causes smooth muscle relaxation, skeletal muscle vasodilation, bronchodilation, and stabilisation of mast cell membranes, reducing histamine release.

## Authorised practice levels

- EMTs: nebulised, IM, IN and topical adrenaline.
- Paramedics: all of the above and adrenaline IV for cardiac arrest.
- ICPs and CCPs: all indications and all routes.

## Indications

- ✓ Cardiac arrest.
- ✓ Anaphylaxis.
- ✓ Severe asthma.
- ✓ Imminent respiratory arrest from COPD.
- ✓ Bradycardia with moderate or severe cardiovascular compromise.
- ✓ Blood pressure support if unresponsive to metaraminol.
- ✓ Septic shock, cardiogenic shock, and neurogenic shock, unresponsive to 0.9% sodium chloride IV and metaraminol IV.
- ✓ Stridor causing moderate or severe respiratory distress.
- ✓ IN for clinically significant epistaxis.
- ✓ Topical for clinically significant bleeding from a wound.

# Adrenaline

## Contraindications

- ✗ None.

## Cautions

- ⚠ Myocardial ischaemia. Adrenaline will increase myocardial oxygen consumption.
- ⚠ Tachydysrhythmias. Adrenaline will usually make tachydysrhythmias worse.

## Use in pregnancy or when breastfeeding

- Safe and should be administered when indicated.

## Dosage

- The dose of adrenaline is dependent on the indication and the route. See the individual guideline.

## Administration

- Topical: dilute each mg of adrenaline to a total of 10 ml using 0.9% sodium chloride. This solution is 1:10,000 and contains 0.1 mg/ml. Apply topically in addition to direct pressure.
- IN: dilute each mg of adrenaline to a total of 10 ml using 0.9% sodium chloride. This solution is 1:10,000 and contains 0.1 mg/ml. Administer the appropriate dose into each bleeding nostril using a mucosal atomising device, in addition to direct pressure.
- Nebulised: administer undiluted.
- IM: administer undiluted. The preferred IM site is the lateral thigh. If this site is not suitable use the lateral upper arm.
- Cardiac arrest:
  - Adults and children whose weight has been rounded to 50 kg or more: administer undiluted as an IV bolus.
  - Children whose weight has been rounded to 40 kg or less: dilute 1 mg of adrenaline to a total of 10 ml using 0.9% sodium chloride. This solution is 1:10,000 and contains 0.1 mg/ml. Draw up the dose from this solution and administer as an IV bolus.

# Adrenaline

- IV infusion: place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride. Shake well and label. This solution is 1:1,000,000 and contains 0.001 mg/ml.
  - For an adult: administer as an IV infusion starting at 2 drops/second. Adjust the rate to the patient's condition.
  - For a child aged 5-14 years: administer as an IV infusion starting at 1 drop/second. Adjust the rate to the patient's condition.
- When administering an IV infusion of adrenaline using 1 mg of adrenaline in a 1 litre bag of 0.9% sodium chloride:
  - 2 drops/second via a standard IV administration set will administer approximately 0.4 mg/hour of adrenaline.
  - Record an estimate of the total dose of adrenaline administered.
- For all other IV administration: place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride, Shake well and label. This solution is 1:1,000,000 and contains 0.001 mg/ml. Draw up the dose from this solution and administer as an IV bolus.

## Common adverse effects

- Tachycardia.
- Tachydysrhythmia.
- Myocardial ischaemia.
- Ventricular ectopy.
- Hypertension.
- Nausea and vomiting.
- Tremor, anxiety and sweating.
- Hyperglycaemia.

## Usual onset of effect

- IV: 5-10 seconds.
- IM: 2-5 minutes.
- Nebulised, IN and topical: on contact with the target site.

# Adrenaline

## Usual duration of effect

- The cardiovascular effects last 5-15 minutes.
- The mast cell membrane effects may last for several hours.

## Usual preparation

- Ampoule containing 1 mg in 1 ml.

## Pharmacokinetics

- Adrenaline is metabolised by the liver and taken up by sympathetic nerve endings.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- Increased doses may be required if the patient is taking a beta-blocker or a calcium channel blocker. This effect is particularly prominent in the setting of poisoning if a large dose of a beta-blocker and/or calcium channel blocker has been taken.

## More Information

### Version history

22 November 2022

1.0.1

**Update:** Indication to align with stridor and croup CPGs.

# Amiodarone

## Mechanism of action

- Amiodarone is an antidysrhythmic with a broad spectrum of activity.
- Amiodarone has predominantly class III activity. It prolongs the action potential duration, reduces automaticity and prolongs the refractory period of atrial, nodal and ventricular tissues.
- The electrophysiological effects result in a reduction in abnormal electrical activity (for example ectopy), a reduction in electrical conduction, a reduction in heart rate and a stabilisation of the SA and AV nodes.
- Amiodarone also causes a small increase in coronary blood flow (although this is not usually clinically significant) and a reduction in myocardial oxygen consumption by reducing inotropy (the force of cardiac contraction).

## Authorised practice levels

- Paramedics: cardiac arrest.
- ICPs and CCPs: all indications.

## Indications

- ✓ Cardiac arrest with VF or VT after three shocks.
- ✓ Adults with sustained VT in the absence of cardiac arrest.
- ✓ Adults with moderate cardiovascular compromise as a result of atrial fibrillation, where the ventricular rate is predominantly greater than 130/minute and metoprolol PO is contraindicated.
- ✓ Adults with moderate cardiovascular compromise as a result of atrial flutter, where the ventricular rate is predominantly greater than 130/minute.

## Contraindications

- ✗ Known severe allergy.
- ✗ Known severe allergy to iodine.
- ✗ VT secondary to cyclic antidepressant poisoning. In this setting amiodarone administration can be associated with severe worsening of shock, without resolution of the rhythm.
- ✗ If torsades de pointes is present or has occurred.

# Amiodarone

## Cautions

- ⚠ None if the patient is in cardiac arrest (with the exception of arrests involving tricyclic antidepressant poisoning or Torsades de Pointes, see contraindications).
- ⚠ Poor perfusion or signs of low cardiac output. Amiodarone reduces inotropy and may cause a significant fall in cardiac output, particularly when administered rapidly.
- ⚠ Hypotension. Amiodarone causes vasodilation and may worsen hypotension, particularly when administered rapidly.
- ⚠ Atrial fibrillation associated with sepsis. Amiodarone may cause a significant fall in cardiac output.
- ⚠ Known sick sinus syndrome without an internal pacemaker in place. Amiodarone slows the heart rate and severe bradycardia may occur following reversion of a tachydysrhythmia.
- ⚠ Previous 2<sup>nd</sup> or 3<sup>rd</sup> degree heart block without an internal pacemaker in place. Amiodarone slows the heart rate and severe bradycardia may occur following reversion of a tachydysrhythmia.
- ⚠ Pregnancy.

## Use in pregnancy or when breastfeeding

- May cause harm during pregnancy. Do not administer amiodarone unless there is a strong clinical indication.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- Cardiac arrest:
  - 300 mg for an adult.
  - If in VF or VT after six shocks, a second dose of 150 mg may be administered.
  - See the paediatric drug dose tables for a child.
- Tachydysrhythmia in an adult:
  - 300 mg IV over 30 minutes.
  - A further 150 mg IV over 30 minutes may be administered if the tachydysrhythmia persists.

# Amiodarone

## Administration

- Cardiac arrest: administer IV undiluted as a bolus.
- Tachydysrhythmia:
  - Place 300 mg of amiodarone in 100 ml of 5% glucose and label. 1 drop/second via a standard IV administration set will deliver 100 ml over approximately 30 minutes. Slow the rate of infusion if hypotension occurs.
  - The administration set will need to be flushed with 0.9% sodium chloride to ensure that all of the amiodarone has been administered.
  - An IV infusion over 30 minutes is the preferred method of administration. However, it is acceptable to dilute 300 mg of amiodarone to a total volume of 20-30 ml using 5% glucose or 0.9% sodium chloride. Administer this IV over 30 minutes and slow the rate of infusion if hypotension occurs.

## Common adverse effects

- Hypotension.
- Lightheadedness.
- Bradydysrhythmia.

## Usual onset of effect

- 5-10 minutes.

## Usual duration of effect

- 1-4 hours after a single dose.
- Amiodarone is taken up into tissues and slowly released. This may result in a prolonged half-life, particularly when more than one dose has been administered. This is why many references quote a half-life of 10-60 days, but the clinical duration of effect is much shorter than this.

## Usual preparation

- Ampoule containing 150 mg in 3 ml.

# Amiodarone

## Pharmacokinetics

- Amiodarone is metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- May potentiate the action of cyclic antidepressants in cyclic poisoning.
- May cause bradycardia following reversion of dysrhythmia if the patient is taking a beta-blocker and/or a centrally acting calcium channel blocker (for example diltiazem).

## Additional information

- If the indication is atrial fibrillation causing moderate cardiovascular compromise, the goal of treatment is to control the ventricular rate and not to revert the rhythm to sinus rhythm, although treatment with amiodarone may result in reversion of the rhythm to sinus rhythm. If a patient has been in atrial fibrillation for longer than a few days, there is a small risk that this may be associated with emboli leaving the left atrium. This is why amiodarone is reserved for patients with cardiovascular compromise that is clinically significant.
- If amiodarone is commenced, the full dose should be administered even if the rhythm reverts to sinus rhythm, unless severe hypotension or bradycardia occurs.
- Amiodarone is often described as relatively contraindicated in the presence of a prolonged QT interval, but this only applies to long-term administration.
- When diluting amiodarone with 0.9% sodium chloride the solution will go slightly cloudy. This does not adversely affect the amiodarone.
- Avoid shaking the ampoule because this will cause the solution to froth.

## More Information

### Version history

16 June 2026

1.1.0.1

**Update:** Indications, contraindications and dosage updated.



# Aspirin

## Mechanism of action

- Aspirin (acetylsalicylic acid) has antiplatelet, antipyretic, anti-inflammatory and analgesic effects. In the out-of-hospital setting aspirin is only administered for its antiplatelet activity.
- Aspirin inhibits the enzyme cyclooxygenase which results in a reduction in the formation of prostaglandins and thromboxane.

## Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.
- First Responders: refer to '[cardiac chest pain](#)' in the FR Field Guide.

## Indications

- ✓ Myocardial ischaemia.
- ✓ STEMI.

## Contraindications

- ✗ Known severe allergy.
- ✗ Third trimester of pregnancy.

## Cautions

- ⚠ Known bleeding disorder. Aspirin will increase the risk of bleeding, however the balance of risk is usually in favour of administering aspirin.
- ⚠ Clinically significant bleeding. Aspirin will increase the risk of bleeding and the nature and risk of the bleeding must be taken into account.
- ⚠ Known worsening of bronchospasm with NSAIDs. Some patients with asthma or COPD have known worsening of bronchospasm with NSAIDs (including aspirin) and a decision must be made based on the balance of risk. If there is a clear history of significant bronchospasm with NSAIDs, aspirin should be withheld.

# Aspirin

## Use in pregnancy or when breastfeeding

- May cause harm during pregnancy. Aspirin has been associated with premature delivery and premature closure of the ductus arteriosus, when administered in the third trimester of pregnancy.
- The likelihood of clinically important myocardial ischaemia occurring in a woman who is pregnant is so low that the balance of risk is usually in favour of aspirin being withheld.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- 300 mg.

## Administration

- Administer PO. Dispersible tablets may be chewed or dissolved in water.

## Common adverse effects

- Increased bleeding.
- Although indigestion, gastrointestinal ulceration and gastrointestinal bleeding are commonly listed as adverse effects, these are only associated with long-term administration.

## Usual onset of effect

- 30-60 minutes.

## Usual duration of effect

- 3-5 days for the antiplatelet activity. This is because platelets exposed to aspirin are impaired for the life of the platelet which is 7-10 days. Approximately 10% of platelets are replaced each day.

## Usual preparation

- 300 mg dispersible tablets.

## Pharmacokinetics

- Absorption occurs in the stomach and small intestine. The presence of food in the stomach will delay absorption, but this is not usually clinically significant.
- Aspirin is predominantly metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- Aspirin displaces warfarin from binding sites and increases the activity of warfarin. However, this effect is most prominent with chronic administration and aspirin is indicated if a patient taking warfarin has clinically significant myocardial ischaemia.

## Additional information

- 111 Call Handlers will give instructions for patients to take aspirin if they are suspected of having myocardial ischaemia. This means that some patients that do not have myocardial ischaemia will be advised to take aspirin. The risk for these patients is extremely low and there is potential benefit for those patients who are experiencing myocardial ischaemia.
- The contraindications to aspirin in the call handling protocols are different to those within these CPGs and these differences are not clinically significant.
- Many patients have enteric coated aspirin and not dispersible aspirin and they will be given instructions to chew and swallow their enteric coated aspirin.
- Enteric coated aspirin is not destroyed when it is chewed and will be absorbed. It is however, quite unpleasant to chew and swallow.
  - If the patient still has the tablets in their mouth, ask them to spit the tablets out and administer an additional 300 mg of dispersible aspirin.
  - If the patient has chewed and swallowed 300 mg of aspirin (including enteric coated aspirin), do not administer additional aspirin.
  - If the patient has swallowed (without chewing) 300 mg of enteric coated aspirin, administer an additional 300 mg of dispersible aspirin. This is because absorption of the enteric coated aspirin will be delayed.
  - If it is unclear what the patient has taken, administer an additional 300 mg of dispersible aspirin.
- Urea also impairs platelet function, but aspirin is not contraindicated in the setting of renal failure.

## More Information

### Version history

11 September 2023

1.0.4

**Update:** New indication to align with administtraion for STEMI.

# Atropine

- Atropine is an anticholinergic which is mostly used for the treatment of bradycardia.
- Atropine antagonises (blocks) muscarinic acetylcholine receptors, causing vagal inhibition resulting in:
  - An increase in heart rate.
  - Drying of salivary and bronchial secretions.
  - Bronchodilation.
  - Reduced gastrointestinal motility.

## Authorised practice levels

- ICPs and CCPs.

## Indications

- ✓ Adults with bradycardia, particularly if the rhythm is narrow complex.
- ✓ Organophosphate poisoning.

## Contraindications

- ✗ Known severe allergy.

## Cautions

- ⚠ Myocardial ischaemia. Atropine will increase myocardial oxygen consumption.

## Use in pregnancy or when breastfeeding

- Safe and should be administered when indicated.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- 0.6 mg for an adult. Repeat as required without a maximum dose, if the bradycardia is responsive to atropine.

# Atropine

- Repeated and escalating doses are likely to be required for organophosphate poisoning.

## Administration

- Administer undiluted as a rapid IV bolus. Slow administration may result in transient bradycardia.

## Common adverse effects

- Tachycardia.
- Confusion. Particularly in the elderly or those with intellectual impairment.
- Dry mouth.
- Blurred vision.

## Usual onset of effect

- 5-10 seconds.

## Usual duration of effect

- The cardiovascular effects last 15-60 minutes.
- The exocrine and smooth muscle effects last 4-6 hours.

## Usual preparation

- Ampoule containing 0.6 mg in 1 ml.

## Pharmacokinetics

- Atropine is predominantly metabolised in the liver, but some is excreted in urine.
- There are no significant effects from liver or kidney impairment on acute administration.

## Common interactions

# Atropine

- The action of atropine may be potentiated if the patient is taking other drugs with anticholinergic properties, such as phenothiazines, some antihistamines (such as promethazine, but not loratadine), tricyclic antidepressants and anti-parkinsonian medicines. These interactions are rarely clinically significant.

## More Information

### Version history

11 September 2023

1.0.4

**Update:** Indications updated to align with changes to bradycardia guideline.

# Calcium Chloride

## Mechanism of action

- Calcium is the active ingredient in calcium chloride.
- Calcium is a mineral that is essential for a number of normal body functions including cell membrane function, enzyme reactions, transmission of nerve impulses, cardiac electrophysiology, contraction of cardiac and skeletal muscle, and coagulation.
- Calcium raises the cardiac action potential threshold and protects cardiac cell membranes from the effects of hyperkalaemia, resulting in a reduction in dysrhythmias associated with hyperkalaemia.

## Authorised practice levels

- ICPs and CCPs.

## Indications

- ✓ Release syndrome following crush injury in adults.
- ✓ Suspected hyperkalaemia with severe ECG changes.
- ✓ Cardiac arrest secondary to suspected hyperkalaemia.

## Contraindications

- ✗ Known severe allergy.

## Cautions

- ⚠ None.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated, but calcium should be administered if indicated.

## Dosage

- Crush injury:



# Calcium Chloride

- » 6.8 mmol (1 g) of calcium chloride IV over one minute for an adult.
- » Repeat the dose if signs of hyperkalaemia persist or recur.
- » Seek clinical advice for a child.
- Hyperkalaemia with severe ECG changes:
  - » 6.8 mmol (1 g) of calcium chloride IV over one minute for an adult.
  - » Repeat the dose after 20 minutes if signs of severe ECG changes persist or recur.
  - » Seek clinical advice for a child.
- Cardiac arrest secondary to hyperkalaemia:
  - » 6.8 mmol (1 g) of calcium chloride IV as a bolus for an adult.
  - » Repeat the dose after 10 minutes if ROSC is not achieved.
  - » Seek clinical advice for a child.

## Administration

- Crush injury and hyperkalaemia with severe ECG changes:
  - » Administer IV over 1 minute into a large vein via a running line, as this reduces venous irritation.
- Cardiac arrest secondary to hyperkalaemia:
  - » Administer as a bolus via a running line.
- Do not mix with other medicines (and in particular do not mix with sodium bicarbonate) as precipitation will occur. If other medicines are being administered via the same vein, ensure a minimum flush of 50 ml of 0.9% sodium chloride between medicines.

## Common adverse effects

- Venous irritation including redness and pain at the site of injection.
- Tingling sensation.
- Rapid administration may cause dysrhythmias.

## Usual onset of effect

- 2-5 minutes.

# Calcium Chloride

## Usual duration of effect

- 1-4 hours.

## Usual preparation

- Ampoule containing 6.8 mmol (1 g) in 10 ml.

## Pharmacokinetics

- 80% is excreted in faeces and 20% is excreted in urine.
- There are no significant effects from liver or kidney impairment on acute administration.

## Common interactions

- None.

## Additional information

- Document the dose administered in mmol.
- Calcium gluconate is an alternative formulation of calcium, usually containing 2.2 mmol/10 ml. Administer two ampoules at a time if only calcium gluconate is available.

## More Information

### Version history

11 September 2023

1.0.4

**Update:** Indications updated. Information under dosage and administration headings updated.

# Cefazolin

## Mechanism of action

- Cefazolin is a first-generation cephalosporin antibiotic with activity against gram-negative and gram-positive bacteria. It inhibits production of the bacterial cell wall, causing bacteria to die.

## Authorised practice levels

- Paramedics, ICPs and CCPs.

## Indications

- ✓ Sepsis, where the source of infection appears to be the soft tissues or joint, and:
  - » The patient is aged greater than or equal to 12 years, and
  - » Has one or more clinical features indicating antibiotics, and
  - » Time to hospital is greater than 30 minutes.
- ✓ Compound fracture
- ✓ Large contaminated wound.
- ✓ Following chest decompression.
- ✓ Following amputation.
- ✓ Cellulitis. In this setting a single dose may be administered if the patient is not being directly referred (or transported) to a medical facility.

## Contraindications

- ✗ Known severe allergy to cephalosporins.

## Cautions

- ⚠ None.

## Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

# Cefazolin

## Dosage

- 1 g IV for an adult.
- See the paediatric drug dose tables for a child.

## Administration

- Add approximately 4 ml of 0.9% sodium chloride to a 1 g ampoule and shake until dissolved.
- Draw up the ampoule contents and dilute to a total of 10 ml.
- Administer IV over 1-2 minutes preferably into a running line.

## Common adverse effects

- None.

## Usual onset of effect

- 30-60 minutes

## Usual duration of effect

- 6-8 hours.

## Usual preparation

- Ampoule containing 1 g of cefazolin as a powder for reconstitution.

## Pharmacokinetics

- Cefazolin is predominately excreted in the urine.
- Clearance is prolonged if the patient has significant kidney impairment, but this does not alter the initial (loading) dose.

# Cefazolin

## Common interactions

- None.

## Additional information

- Rapid administration of cefazolin can cause pain and phlebitis, which is why it should be preferably administered into a running line.

## More Information

### Version history

|                                                               |       |
|---------------------------------------------------------------|-------|
| <b>28 October 2024</b><br><b>Update:</b> Indications updated. | 1.0.5 |
| <b>11 September 2023</b><br><b>New:</b> Guideline released.   | 1.0.4 |

# Ceftriaxone

## Mechanism of action

- Ceftriaxone is a cephalosporin antibiotic with broad activity against gram-negative and gram-positive bacteria. It inhibits production of the bacterial cell wall, causing bacteria to die.

## Authorised practice levels

- Paramedics, ICPs and CCPs.

## Indications

- ✓ Suspected meningococcal septicaemia.
- ✓ Sepsis, where cefazolin is not indicated, and:
  - » The patient is aged greater than or equal to 12 years, and
  - » One or more clinical features indicating antibiotics are present, and
  - » Time to hospital is greater than 30 minutes.

## Contraindications

- ✗ Anaphylaxis to cephalosporins.

## Cautions

- ⚠ None.

## Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

## Dosage

- 2 g IV for an adult.
- 2 g IM for an adult if IV access cannot be immediately obtained (meningococcal septicaemia only).
- See the paediatric drug dose tables for a child.

# Ceftriaxone

## Administration

- IV administration:
  - Add approximately 4 ml of 0.9% sodium chloride to a 2 g ampoule and shake until dissolved.
  - Draw up the ampoule and dilute to a total of 10 ml.
  - Discard unrequired volume before administration for a child.
  - Administer IV over 1-2 minutes preferably into a running line.
- IM administration:
  - Add 4 ml of 0.9% sodium chloride to a 2 g ampoule and shake until dissolved.
  - Draw up the ampoule using two syringes with approximately half in each. The total volume will be 5 ml.
  - Discard unrequired volume before administration for a child.
  - Administer one syringe into each lateral thigh. If this site is not suitable, use each lateral upper arm.

## Common adverse effects

- None.

## Usual onset of effect

- 30-60 minutes.

## Usual duration of effect

- 24 hours.

## Usual preparation

- Ampoule containing 2 g of ceftriaxone as a powder for reconstitution.

## Pharmacokinetics

- 50% is excreted in urine and 50% in bile.
- Neither renal impairment nor hepatic impairment alter the initial (loading) dose.

# Ceftriaxone

## Common interactions

- None.

## Additional information

- Some references describe dissolving ceftriaxone using 1% lignocaine for IM injection, to reduce the pain of injection. In the setting of suspected meningococcal septicaemia this reduction in pain is not significant and ceftriaxone should be dissolved using 0.9% sodium chloride.
- As meningococcal bacteria die they release endotoxins. The body's immune response to endotoxins can cause profound worsening of shock following antibiotic administration. Be prepared to treat this with 0.9% sodium chloride IV, and metaraminol or adrenaline IV. It is rare for significant amounts of endotoxins to be released from other bacteria.

## More Information

### Version history

11 September 2023

1.0.4

**Update:** Indications updated.



# Clopidogrel

## Mechanism of action

- Clopidogrel has antiplatelet activity.
- Clopidogrel antagonises (blocks) the binding of adenosine diphosphate (ADP) to platelets and impairs platelet function. Clopidogrel provides significantly more antiplatelet activity than aspirin.

## Authorised practice levels

- Paramedics, ICPs and CCPs.

## Indications

- ✓ STEMI in conjunction with fibrinolytic therapy.

## Contraindications

- ✗ Known severe allergy.

## Cautions

- ⚠ Clinically significant bleeding. Clopidogrel will increase bleeding.
- ⚠ At risk of bleeding. If there are any cautions present within the fibrinolytic therapy checklist, personnel must seek clinical advice.
- ⚠ Pregnancy.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy. The likelihood of STEMI occurring in a woman who is pregnant is so low that personnel must seek clinical advice.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- 300 mg if the patient is aged less than 75 years.

# Clopidogrel

- 75 mg if the patient is aged greater than or equal to 75 years.

## Administration

- Administer PO.

## Common adverse effects

- Increased bleeding.

## Usual onset of effect

- 30-60 minutes.

## Usual duration of effect

- 3-5 days. This is because platelets exposed to clopidogrel are impaired for the life of the platelet which is 7-10 days. Approximately 10% of platelets are replaced each day.

## Usual preparation

- 75 mg tablets.

## Pharmacokinetics

- Clopidogrel is a prodrug and must be metabolised to the active form in the liver.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- The risk of bleeding will be increased if the patient is taking an anticoagulant.

## More Information

# Clopidogrel

## Version history

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**11 September 2023**

1.0.4

**Update:** Cautions updated.

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# Droperidol

## Mechanism of action

- Droperidol blocks dopamine and alpha receptors centrally, resulting in sedation, reduced agitation and a state of mental detachment, and antiemetic action.

## Authorised practice levels

- Paramedics, ICPs and CCPs.

## Indications

- ✓ Patients aged greater than or equal to 12 years with acute behavioural disturbance causing a mild to moderate risk to safety, when olanzapine has not been administered or is ineffective.
- ✓ Severe pain, usually in conjunction with an opiate, associated with one of the following:
  - » Chronic use of opiates.
  - » Severe psychological distress.
  - » Nausea and/or vomiting.
- ✓ Management of nausea and agitation or pain that does not respond to an opiate during the last days of life.
- ✓ Nausea and/or vomiting:
  - » That persists despite ondansetron, or
  - » Where motion sickness is a component of the patient's nausea and/or vomiting, or
  - » The patient has cannabinoid hyperemesis syndrome.
- ✓ Vertigo.
- ✓ Primary headache that is severe.

## Contraindications

- ✗ Known severe allergy.
- ✗ Aged less than 12 years.
- ✗ Pregnancy (unless indicated for acute behavioural disturbance).

# Droperidol

## Cautions

- ⚠ Altered level of consciousness. This will increase and prolong the effects.
- ⚠ Parkinson's disease. There is a risk of worsening the movement disorder associated with Parkinson's disease.
- ⚠ Concurrent administration of ketamine or midazolam. This will increase and prolong the effects.
- ⚠ Aged greater than or equal to 75 years, particularly if frail. The effects of droperidol will be increased and prolonged in this cohort.
- ⚠ Signs of shock. Droperidol may make shock worse.

## Use in pregnancy or when breastfeeding

- The likelihood of a pregnant or breastfeeding patient requiring droperidol for acute behavioural disturbance is very low, however the balance of risk is such that it should be administered if indicated in this context.
- For all other indications, the balance of risk is not in favour of administration of droperidol and therefore it is contraindicated.

## Dosage

- For analgesia:
  - 0.625 mg IV/IM for an adult, once only.
- For acute behavioural disturbance:
  - 10 mg IV/IM for an adult.
  - Consider reducing the dose to 5 mg if a caution is present.
  - The dose may be repeated once after 20 minutes.
- For care during the last days of life:
  - 1.25 mg IV or 2.5 mg IM/SC.
  - The dose may be repeated as required.
- For nausea and/or vomiting:
  - 0.625 mg IV/IM for an adult, once only.
- For vertigo:

# Droperidol

- » 1.25mg IV/IM for an adult, once only.
- » Consider reducing the dose to 0.625 mg IV/IM if a caution is present.
- For primary headache that is severe:
  - » 1.25 mg IV/IM for an adult, once only.
  - » Consider reducing the dose to 0.625 mg IV/IM if a caution is present.

## Administration

- IV administration in acute behavioural disturbance:
  - » Draw up the required dose and dilute to a total of 10 ml using 0.9% sodium chloride.
  - » Administer IV over 1-2 minutes.
- IV administration for nausea and/or vomiting, an adjunct to analgesia, vertigo or primary headache that is severe:
  - » Draw up 2.5 mg and dilute to a total of 4 ml using 0.9% sodium chloride. This gives a 0.625 mg/ml solution.
  - » Administer the required dose as an IV bolus.
- IM administration: Administer undiluted. The preferred IM site is the lateral thigh. If this site is not suitable use the lateral upper arm.
- SC administration:
  - » Administer undiluted subcutaneously into the abdominal wall.
  - » There is no need to sterilise the skin at the site of injection unless the skin is visibly contaminated.

## Common adverse effects

- Hypotension. This may occur if a large IV dose is administered rapidly.

## Usual onset of effect

- IV/IM: 5-10 minutes.

## Usual duration of effect

- 4-6 hours.

# Droperidol

## Usual preparation

- Ampoule containing 2.5 mg in 1 ml.

## Pharmacokinetics

- Droperidol is predominantly metabolised in the liver with metabolites being excreted in the urine.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- Intoxication. Droperidol will have increased sedative effects if the patient is intoxicated with alcohol or has taken recreational drugs.
- Sedative drugs. Concurrent administration with other sedative drugs (such as olanzapine or midazolam) will result in an increased sedative effect.

## Additional information

- Droperidol has been reported to prolong the QT interval. This generally involved repeated and/or high doses and one or two doses are safe, even if the patient is known to have a prolonged QT interval.
- Droperidol may cause dyskinesia (abnormal, uncoordinated and involuntary movements) but this is unusual following one or two doses.

## More Information

### Version history

|                                                                                |         |
|--------------------------------------------------------------------------------|---------|
| <b>16 June 2026</b><br><b>Update:</b> Indications and dosage updated.          | 1.1.0.1 |
| <b>9 June 2025</b><br><b>Update:</b> Indications and contraindication updated. | 1.0.5.3 |
| <b>11 September 2023</b><br><b>Update:</b> Indications and dosage updated.     | 1.0.4   |
| <b>22 November 2022</b><br><b>Update:</b> Aligned with analgesia CPG.          | 1.0.1   |

# Droperidol

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# Enoxaparin

## Mechanism of action

- Enoxaparin is a low molecular weight heparin (LMWH) anticoagulant.
- Enoxaparin potentiates the activity of antithrombin III (a naturally occurring anticoagulant) causing inhibition of multiple coagulation factors, particularly factor Xa.

## Authorised practice levels

- Paramedics, ICPs and CCPs.

## Indications

- ✓ STEMI in conjunction with fibrinolytic therapy.

## Contraindications

- ✗ Known severe allergy.

## Cautions

- ⚠ Clinically significant bleeding. Enoxaparin will increase bleeding.
- ⚠ At risk of bleeding. If there are any cautions present within the fibrinolytic therapy checklist, personnel must seek clinical advice or discuss with the STEMI coordinator prior to administration.
- ⚠ Pregnancy.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated. The likelihood of STEMI occurring in a woman who is pregnant is so low that personnel must seek clinical advice prior to administration.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- IV: 30 mg, only if the patient is aged less than 75 years.

# Enoxaparin

- SC: Dosage is based on age and known (or estimated) weight.

| Weight   | Age less than 75 years  |                           | Age 75 years or older   |                           |
|----------|-------------------------|---------------------------|-------------------------|---------------------------|
|          | Enoxaparin<br>(dose SC) | Enoxaparin<br>(volume SC) | Enoxaparin<br>(dose SC) | Enoxaparin<br>(volume SC) |
| <60 kg   | 60 mg                   | 0.6 ml                    | 45 mg                   | 0.45 ml                   |
| 60-69 kg | 70 mg                   | 0.7 ml                    | 50 mg                   | 0.5 ml                    |
| 70-79 kg | 80 mg                   | 0.8 ml                    | 60 mg                   | 0.6 ml                    |
| 80-89 kg | 90 mg                   | 0.9 ml                    | 70 mg                   | 0.7 ml                    |
| ≥90 kg   | 100 mg                  | 1 ml                      | 75 mg                   | 0.75 ml                   |

## Administration

- IV:
  - Dilute 100 mg to a total of 10 ml using 0.9% sodium chloride. This gives a 10 mg/ml solution.
  - Discard 7 ml.
  - Administer 30 mg (3 ml) IV.
- SC:
  - Administer subcutaneously into the abdominal wall.
  - There is no need to sterilise the skin at the site of injection unless the skin is visibly contaminated.
  - Discard unwanted drug from the syringe before administration.
  - If an error is made in discarding unwanted drug volume and the dose remaining in the syringe is less than planned, administer the remaining dose.

## Common adverse effects

- Increased bleeding.

## Usual onset of effect

- 10-30 minutes.

# Enoxaparin

## Usual duration of effect

- 12-24 hours.

## Usual preparation

- Pre-filled syringe containing 100 mg in 1 ml.

## Pharmacokinetics

- Enoxaparin is predominately excreted in urine.
- Clearance is prolonged if the patient has significant renal impairment, but this does not alter the initial (loading) dose.

## Common interactions

- The risk of bleeding will be increased if the patient is taking an anticoagulant.

## More Information

### Version history

3 August 2026

1.1.0.6

**Update:** Added IV administration and dosing.

11 September 2023

1.0.4

**Update:** Cautions updated.

# Fentanyl

## Mechanism of action

- Fentanyl is an opiate analgesic. It is an opiate agonist (or stimulator) that binds to opiate receptors in the brain and spinal cord causing analgesia.

## Authorised practice levels

- Paramedics, ICPs and CCPs.

## Indications

- ✓ Moderate to severe pain.
- ✓ Cardiogenic pulmonary oedema with severe anxiety.
- ✓ Rapid sequence intubation.
- ✓ Sedation post intubation.
- ✓ Control of pain, agitation, or shortness of breath during the last days of life.
- ✓ Autonomic dysreflexia, even if the patient cannot feel pain.

## Contraindications

- ✗ Known severe allergy.

# Fentanyl

## Cautions

- ⚠ Altered level of consciousness. Fentanyl may further reduce the level of consciousness.
- ⚠ Aged less than one year. Children under the age of one year are at increased risk of respiratory depression following opiate administration.
- ⚠ Respiratory depression or at high risk of respiratory depression. For example, severe COPD, morbid obesity or on home BiPAP. Such patients may develop respiratory depression following opiate administration.
- ⚠ Labour. Opiates cross the placenta and may cause drowsiness and/or respiratory depression in the baby, particularly when administered within an hour or two of birth. Discuss administration with the lead maternity carer if possible. Following birth, close observation of the baby is required and personnel must be prepared to treat respiratory depression.
- ⚠ Concurrent administration of other opiates, ketamine or midazolam. This will increase and prolong the effects.
- ⚠ Aged greater than or equal to 75 years, particularly if frail. The effects of fentanyl will be increased and prolonged in this cohort.
- ⚠ Signs of shock. Fentanyl may make shock worse.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated in pregnancy, but fentanyl should be administered if indicated.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- IV bolus for analgesia:
  - 10-50 mcg every five minutes for an adult, as required. Use a dose at the lower end of the range if the patient is frail or has signs of shock.
  - See the paediatric drug dose tables for a child.
- IV infusion:
  - Start at 1 mcg/kg/hour, up to a maximum of 150 mcg/hour.
  - Administer additional boluses as required, every five minutes.

# Fentanyl

- » Adjust the infusion rate as required, every 15-20 minutes.
- IV for autonomic dysreflexia:
  - » 20-30 mcg increments to a total of 100 mcg.
- IN for analgesia:
  - » 100 mcg for an adult.
  - » See the paediatric drug dose tables for a child.
  - » Halve the dose if the patient is frail or has signs of shock.
  - » Repeat as required every 20 minutes.
- IM and SC for analgesia:
  - » 50-100 mcg IM/SC for an adult.
  - » See the paediatric drug dose tables for a child.
  - » Halve the dose if the patient is frail or has signs of shock.
  - » Repeat as required, every 20 minutes, up to a maximum of three doses.
- IV for RSI: see the '[RSI](#)' guideline.
- IV for sedation post intubation: see the '[post intubation](#)' guideline.

## Administration

- The preferred route for administration is IV.
- IV boluses for analgesia:
  - » Dilute 100 mcg to a total of 10 ml using 0.9% sodium chloride for an adult or a child whose weight has been rounded up to 30 kg or more. This final solution contains 10 mcg/ml.
  - » Dilute 100 mcg to a total of 100 ml using a 100 ml bag of 5% glucose for a child whose weight has been rounded up to 20 kg or less. This final solution contains 1 mcg/ml.
- IN administration:
  - » Draw up fentanyl undiluted, placing half of the total dose into two separate 1 ml syringes. When drawing up the first syringe, draw up an additional 0.1 ml of fentanyl over and above the planned volume and expel this slowly through the mucosal atomiser in order to fill the dead space. This does not need to be done with subsequent doses.
  - » Administer fentanyl IN by rapidly injecting one syringe (half of the total dose) into each nostril. Rapid injection is required to achieve a fine mist, which maximises absorption.

# Fentanyl

- IM administration: Administer undiluted. The preferred IM site is the lateral thigh. If this site is not suitable use the lateral upper arm.
- SC administration:
  - Administer undiluted subcutaneously into the abdominal wall.
  - There is no need to sterilise the skin at the site of injection unless the skin is visibly contaminated.
  - Pinch a fold of skin over the anterior abdominal wall between thumb and forefinger. Introduce the entire length of the needle using a dart technique and inject.

## Common adverse effects

- Respiratory depression.
- Bradycardia.
- Hypotension.
- Sedation.
- Nausea and vomiting.
- Itch.
- Euphoria.

## Usual onset of effect

- IV: 2-5 minutes. The maximal analgesic and respiratory depressant effects may not occur until 10-15 minutes and this may be longer in the elderly.
- IN: 5-10 minutes.
- IM/SC: 5-10 minutes.

## Usual duration of effect

- 30-60 minutes.
- The effect on respiration may last for several hours.

# Fentanyl

## Usual preparation

- Ampoule containing 100 mcg in 2 ml.

## Pharmacokinetics

- Fentanyl is more lipophilic (fat soluble) than morphine and this is why fentanyl is well absorbed through the nasal mucosa.
- Fentanyl may cause a small amount of histamine release. In combination with relief of pain this usually results in a small fall in blood pressure.
- Fentanyl is metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- The effects will be increased in the presence of other opiates and sedatives, for example, benzodiazepines or alcohol.

## Additional information

- A patient administered fentanyl must be given a clear recommendation to be transported to a medical facility by ambulance, unless the patient is being treated using the '[care during the last days of life](#)', '[patella dislocation](#)' or '[shoulder dislocation](#)' guidelines.
- Transport should usually be to an ED unless it is appropriate for the patient to be transported to a primary care facility.
- Fentanyl is usually the preferred opiate if:
  - The patient requires analgesia for a short period of time, or
  - The patient has renal impairment, or
  - The patient has signs of shock.

## More Information



# Fentanyl

## Version history

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**16 June 2026**

1.1.0.1

**Update:** Dosage updated.

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**22 November 2022**

1.0.1

**Update:** Contraindications, cautions and aligned with analgesia and end of life CPGs.

# Glucagon

## Mechanism of action

- Glucagon increases the blood glucose level by stimulating glycogenolysis (the breakdown of glycogen into glucose), predominantly within the liver.

## Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

## Indications

- ✓ Hypoglycaemia when the patient cannot safely swallow glucose/food and IV access cannot be obtained.

## Contraindications

- ✗ Known severe allergy.

## Cautions

- ⚠ None.

## Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

## Dosage

- 1 mg IM once for an adult or child aged greater than or equal to five years.
- 0.5 mg IM once for a child aged less than five years.

## Administration

- Dissolve the powder using the syringe within the kit and administer IM.
- The preferred site is the lateral thigh as this has the best absorption. If this site is not suitable use the lateral upper arm.

# Glucagon

## Common adverse effects

- None.

## Usual onset of effect

- 5-10 minutes.

## Usual duration of effect

- 15-60 minutes.

## Usual preparation

- Ampoule containing 1 mg as powder.

## Pharmacokinetics

- Glucagon is predominantly excreted unchanged into bile and urine.
- There are no significant effects from liver or kidney impairment on acute administration.

## Common interactions

- None.

## Additional information

- Glucagon relies on stored glycogen being available to exert its effect and if glycogen stores are not available, glucagon may be ineffective. Examples include if the patient:
  - Does not have diabetes.
  - Has sepsis.
  - Is a young child.
  - Has undergone strenuous exercise.
  - Has not eaten food for more than 12 hours.
  - Is suffering from adrenal insufficiency.

# Glucagon

- » Is suffering from chronic hypoglycaemia.
  - » Is suffering from alcohol-induced hypoglycaemia.
  - Following glucagon administration the patient's glycogen stores will be depleted. For this reason it is important the patient eats food as described in the ['hypoglycaemia'](#) guideline.
  - Glucagon also reduces the tone and motility of the smooth muscle in the gastrointestinal tract, but is not recommended in the setting of oesophageal obstruction.
  - Glucagon is sometimes suggested as part of the treatment for bradycardia caused by beta-blockers because it stimulates cardiac cells via a mechanism that is independent of the beta receptor. However, glucagon has almost no role in the out-of-hospital setting because it rarely provides a sustained heart rate rise in addition to adrenaline and requires much higher doses than that carried by ambulance personnel.
-

# Glucose Gel

## Mechanism of action

- Glucose gel provides a source of glucose that can be easily swallowed and is rapidly absorbed.

## Authorised practice levels

- All personnel. ATP is not required to administer glucose gel.
- First Responders: refer to '[diabetic emergencies](#)' in the FR Field Guide.

## Indications

- ✓ Hypoglycaemia in adults and children provided the patient is conscious enough to be able to swallow safely.
- ✓ Hypoglycaemia in neonates.

## Contraindications

- ✗ None.

## Cautions

- ⚠ None.

## Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

## Dosage

- 10-20 g for all ages.
- Administer one sachet and repeat every ten minutes if hypoglycaemia persists or recurs.

## Administration

- Administer PO.

# Glucose Gel

- Glucose gel may be spread on the gums, tongue and inside of the cheeks of a baby or small child.

## Common adverse effects

- None.

## Usual onset of effect

- 5-10 minutes.

## Usual duration of effect

- 30-60 minutes.

## Usual preparation

- There are multiple different brands.
- Most are a sachet containing 10-20 g glucose.

## Pharmacokinetics

- Glucose is absorbed in the stomach and small intestine.
- Glucose is rapidly metabolised by cells.

## Common interactions

- None.

## Additional information

- Document the approximate number of grams of oral glucose administered.

# GTN (Glyceryl Trinitrate) IV

## Mechanism of action

- GTN is a vasodilator. It acts on vascular smooth muscle to cause venous and arterial vasodilation, with the predominant effect being on veins.
- The mechanism of action is not clear, but it appears that GTN results in the formation of nitric oxide which is a vasodilator. GTN causes:
  - A reduction in venous return (preload) to the heart. This reduces ventricular filling and cardiac output which reduces myocardial oxygen demand.
  - Arterial dilation which reduces peripheral resistance (afterload). This reduces the force the left ventricle must overcome to eject blood into the arteries which reduces myocardial oxygen demand.
  - Dilation of the coronary arteries which may increase coronary blood supply, though this is not usually clinically significant.

## Authorised practice levels

- CCPs.

## Indications

- ✓ Cardiogenic pulmonary oedema and the patient is not rapidly improving with GTN SL and CPAP.

## Contraindications

- ✗ Known severe allergy.
- ✗ Systolic BP less than 110 mmHg.
- ✗ Heart rate less than 40/minute.
- ✗ Heart rate greater than 150/minute.
- ✗ Ventricular tachycardia.

# GTN (Glyceryl Trinitrate) IV

## Cautions

- ⚠ STEMI, particularly STEMI involving the right ventricle. GTN may cause a significant fall in cardiac output and if there are signs of low cardiac output GTN should be withheld.
- ⚠ The patient is frail.
- ⚠ Signs of shock. The patient's reduced cardiac output may fall further with GTN administration.
- ⚠ Dysrhythmia. This may cause a reduced cardiac output which may fall further with GTN administration.
- ⚠ Has taken a phosphodiesterase inhibitor in the last 24 hours. See common interactions.
- ⚠ Known aortic or mitral stenosis. With aortic or mitral stenosis, cardiac output may be reduced as a result of the narrowed valve and a fall in preload may cause further fall in cardiac output.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated. The likelihood of a pregnant or breastfeeding patient requiring GTN IV is very low, and personnel must seek clinical advice prior to administration.

## Dosage

- Start the infusion at 2 mg/hour. Adjust the rate as required by 0.5 mg/hour every 5-10 minutes, titrating to effect.

## Administration

- Draw up 50 mg of GTN from a 50 mg/10 ml ampoule and dilute to a total of 50 ml using 0.9% sodium chloride. This solution contains 1 mg/ml.
- Administer via an infusion device.

## Common adverse effects

- Hypotension.
- Flushing.
- Headache.



# GTN (Glyceryl Trinitrate) IV

- Tachycardia.
- Feeling light-headed.

## Usual onset of effect

- 1-2 minutes.

## Usual duration of effect

- 3-5 minutes following cessation of the infusion.

## Usual preparation

- Ampoule containing 50 mg/10 ml.

## Pharmacokinetics

- GTN is predominantly metabolised in the liver.

## Common interactions

- The effects may be increased if the patient is taking an antihypertensive medicine.
- Severe and/or prolonged hypotension may occur if a phosphodiesterase inhibitor has been taken within the last 24 hours:
  - Phosphodiesterase inhibitors can be used in the treatment of erectile dysfunction and pulmonary hypertension (particularly sildenafil).
  - These medicines are long-acting vasodilators and the administration of GTN may cause further vasodilation.

## Additional information

- Particular caution must be used in the presence of STEMI involving the right ventricle:
  - STEMI involving the right ventricle can result in a significant reduction in right ventricular contractility.
  - When the right ventricle is significantly impaired, it may provide little in the way of contribution to cardiac output and blood may be passively flowing down a pressure gradient between the inferior vena cava (IVC), the superior vena cava (SVC) and the left atrium.

# GTN (Glyceryl Trinitrate) IV

- » This may result in the preload (filling) of the left side of the heart being dependent on the venous pressures within the IVC and SVC.
- » GTN can result in a significant fall in venous pressure (and thus a fall in preload) which may cause a significant fall in cardiac output.

## More Information

### Version history

11 September 2023

1.0.4

**New:** Guideline released.

# GTN (Glyceryl Trinitrate) Spray

## Mechanism of action

- GTN is a vasodilator. It acts on vascular smooth muscle to cause venous and arterial vasodilation, with the predominant effect being on veins.
- The mechanism of action is not clear, but it appears that GTN results in the formation of nitric oxide which is a vasodilator. GTN causes:
  - A reduction in venous return (preload) to the heart. This reduces ventricular filling and cardiac output which reduces myocardial oxygen demand.
  - Arterial dilation which reduces peripheral resistance (afterload). This reduces the force the left ventricle must overcome to eject blood into the arteries which reduces myocardial oxygen demand.
  - Dilation of the coronary arteries which may increase coronary blood supply, though this is not usually clinically significant.

## Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

## Indications

- ✓ Myocardial ischaemia.
- ✓ Cardiogenic pulmonary oedema.
- ✓ Control of hypertension associated with autonomic dysreflexia.
- ✓ Control of hypertension (usually in conjunction with labetalol) prior to fibrinolytic treatment for STEMI.
- ✓ Control of hypertension (usually in conjunction with labetalol) during inter-hospital transfer for STEMI.
- ✓ STEMI

# GTN (Glyceryl Trinitrate) Spray

## Contraindications

- ✗ Known severe allergy.
- ✗ Systolic BP less than 110 mmHg.
- ✗ Heart rate less than 40/minute.
- ✗ Heart rate greater than 150/minute.
- ✗ Ventricular tachycardia.

## Cautions

- ⚠ STEMI, particularly STEMI involving the right ventricle. GTN may cause a significant fall in cardiac output and if there are signs of low cardiac output GTN should be withheld.
- ⚠ The patient is frail.
- ⚠ Signs of shock. The patient's reduced cardiac output may fall further with GTN administration.
- ⚠ Dysrhythmia. This may cause a reduced cardiac output which may fall further with GTN administration.
- ⚠ Has taken a phosphodiesterase inhibitor in the last 24 hours. See common interactions.
- ⚠ Known aortic or mitral stenosis. With aortic or mitral stenosis, cardiac output may be reduced as a result of the narrowed valve and a fall in preload may cause a further fall in cardiac output.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated. The likelihood of a pregnant or breastfeeding patient requiring GTN is very low, but GTN should be administered if indicated.

## Dosage

- Myocardial ischaemia: 0.4 mg every five minutes. Consider increasing the dosing interval to ten minutes if a caution is present.
- STEMI: 0.4 mg with caution. Withhold if signs of poor perfusion are present.
- Cardiogenic pulmonary oedema: 0.8 mg every five minutes. Consider increasing the dosing interval to ten minutes if a caution is present. Increase the dose and frequency if the patient is not improving.

# GTN (Glyceryl Trinitrate) Spray

- Control of hypertension: 0.4-0.8 mg every five minutes as required.

## Administration

- Spray under the tongue. If this cannot be achieved it is acceptable to spray into the mouth.
- If GTN is administered in the presence of a caution:
  - The patient should be lying flat.
  - IV access should have been obtained whenever possible.
  - The dosing interval should be increased to ten minutes.
  - Be ready to administer 0.9% sodium chloride IV if there is a significant fall in cardiac output or blood pressure.

## Common adverse effects

- Hypotension.
- Flushing.
- Headache.
- Tachycardia.
- Feeling light-headed.

## Usual onset of effect

- 1-2 minutes.

## Usual duration of effect

- 15-30 minutes.

## Usual preparation

- Metered dose bottle delivering 0.4 mg doses.

# GTN (Glyceryl Trinitrate) Spray

## Pharmacokinetics

- GTN is rapidly absorbed from the oral mucosa and reaches the vascular system without passing through the liver.
- GTN is predominantly metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- The effects may be increased if the patient is taking an antihypertensive medicine.
- Severe and/or prolonged hypotension may occur if a phosphodiesterase inhibitor has been taken within the last 24 hours:
  - Phosphodiesterase inhibitors can be used in the treatment of erectile dysfunction and pulmonary hypertension (particularly sildenafil).
  - These medicines are long-acting vasodilators and the administration of GTN may cause further vasodilation.

## Additional information

- GTN must be used with caution in the presence of STEMI because the risks may outweigh the benefits:
  - GTN may cause a significant fall in cardiac output.
  - GTN has a role in treating symptomatic myocardial ischaemia, but does not usually have a significant role in treating STEMI.
- Particular caution must be used in the presence of STEMI involving the right ventricle and personnel must have a low threshold for withholding GTN:
  - STEMI involving the right ventricle can result in a significant reduction in right ventricular contractility.
  - When the right ventricle is significantly impaired, it may provide little in the way of contribution to cardiac output and blood may be passively flowing down a pressure gradient between the inferior vena cava (IVC), the superior vena cava (SVC) and the left atrium.
  - This may result in the preload (filling) of the left side of the heart being dependent on the venous pressures within the IVC and SVC.
  - GTN can result in a significant fall in venous pressure (and thus a fall in preload) which may cause a significant fall in cardiac output.
- GTN and VT:

# GTN (Glyceryl Trinitrate) Spray

- » GTN is not administered for chest pain associated with VT because the risk of precipitating cardiac arrest outweighs the benefits.
- » During VT the atria and ventricles are contracting independently of each other (also called atrioventricular dissociation). This causes a reduction in ventricular preload (or filling) through loss of the 'atrial kick', which in combination with abnormal ventricular contraction may cause a substantial fall in cardiac output.
- » GTN reduces venous return to the heart and may further reduce cardiac output with the risk of precipitating severe hypotension or cardiac arrest.

## More Information

### Version history

|                                                              |         |
|--------------------------------------------------------------|---------|
| <b>16 June 2026</b><br><b>Update:</b> Dosage updated.        | 1.1.0.1 |
| <b>11 September 2023</b><br><b>Update:</b> Multiple changes. | 1.0.4   |
| <b>22 November 2022</b><br><b>Update:</b> Cautions.          | 1.0.1   |

# Hydrocortisone

## Mechanism of action

- Hydrocortisone is a corticosteroid with anti-inflammatory and immunosuppressant actions. It inhibits the production of inflammatory mediators, including prostaglandins and leukotrienes, resulting in a reduction in the inflammatory and immune response.

## Authorised practice levels

- Paramedics:
  - At risk of adrenal crisis or suspected to have adrenal crisis.
  - Severe or immediately life-threatening asthma.
  - Severe COPD or imminent respiratory arrest from COPD.
- ICPs and CCPs: All indications.

## Indications

- ✓ Adults with angioedema occurring during inter-hospital transfer for stroke clot retrieval.
- ✓ At risk of adrenal crisis or suspected to have adrenal crisis.
- ✓ Severe or immediately life-threatening asthma.
- ✓ Severe COPD or imminent respiratory arrest from COPD.

## Contraindications

- ✗ Known severe allergy.

## Cautions

- ⚠ None.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy. However, there is significant clinical experience with steroids, they appear to be safe and hydrocortisone should be administered if indicated.



# Hydrocortisone

- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- 100 mg for an adult.
- See the paediatric drug dose tables for a child.

## Administration

- Press down on the plastic cap to force the diluent into the vial.
- Mix the solution by turning the vial upside down several times. Do not shake the vial.
- Remove the central plastic cap and draw up the contents.
- For IV administration:
  - Dilute to approximately 10 ml using 0.9% sodium chloride and administer over 1-2 minutes.
- For IM administration:
  - Do not further dilute after reconstitution. The preferred IM site is the lateral thigh. If this site is not suitable use the lateral upper arm.

## Common adverse effects

- Fatigue.
- Sodium and water retention. This may worsen hypertension and heart failure, but is usually only of clinical significance with repeated and/or prolonged dosing.

## Usual onset of effect

- 10-20 minutes.

## Usual duration of effect

- 8-12 hours.

# Hydrocortisone

## Usual preparation

- Vial with two chambers, one containing 100 mg of hydrocortisone powder and the other containing 2 ml of sterile water for reconstitution.

## Pharmacokinetics

- Hydrocortisone is metabolised in the liver.
- Clearance is prolonged if the patient has significant liver impairment, but this does not alter the initial (loading) dose.

## Common interactions

- None.

## More Information

### Version history

|                                                                                                                  |         |
|------------------------------------------------------------------------------------------------------------------|---------|
| <b>16 June 2026</b><br><b>Update:</b> Indications updated.                                                       | 1.1.0.1 |
| <b>13 February 2026</b><br><b>Correction:</b> Removed advice of shaking vial until dissolved.                    | 1.0.7.1 |
| <b>9 June 2025</b><br><b>Update:</b> Clinical level, indications, dosage and administration information updated. | 1.0.5.3 |

# Ibuprofen

## Mechanism of action

- Ibuprofen is a non-steroidal anti-inflammatory drug (NSAID) that is predominantly used for treating pain.
- Ibuprofen inhibits the activity of the enzyme prostaglandin synthetase, reducing prostaglandin production and causing a reduction in inflammation, pain and fever.

## Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.
- First Responders: refer to '[pain relief](#)' in the FR Field Guide

## Indications

- ✓ Mild to moderate pain, usually in combination with paracetamol.
- ✓ Moderate to severe pain, usually in combination with other medicines.

## Contraindications

- ✗ Known severe allergy.
- ✗ Pregnancy.
- ✗ The presence of sepsis, dehydration, shock or clinically significant bleeding. Ibuprofen can worsen renal impairment and increase the risk of bleeding.
- ✗ Known worsening of bronchospasm with NSAIDs. Some patients with asthma or COPD have known worsening of bronchospasm with NSAIDs. If there is a clear history of significant bronchospasm with NSAIDs, ibuprofen must not be administered.

# Ibuprofen

## Cautions

- ⚠ The patient has taken ibuprofen within the last four hours. Ibuprofen is contained in many products such as cold and flu tablets/drinks, combination analgesics and migraine tablets. Additional ibuprofen may be administered provided it is clear that the total dose taken within a four hour period does not exceed the dose described within the CPGs. Withhold ibuprofen if there is any doubt.
- ⚠ Abdominal pain, particularly if the patient is very unwell or vomiting. Clinical judgement is required, but if the patient is very unwell or vomiting the possibility of significant intra-abdominal pathology exists and oral medicines should usually be withheld.
- ⚠ Age greater than or equal to 75 years, particularly if frail. In this setting, renal impairment is likely and ibuprofen may worsen renal impairment.

## Use in pregnancy or when breastfeeding

- May cause foetal harm during pregnancy and should not be administered. In particular, ibuprofen has been associated with premature delivery and premature closure of the ductus arteriosus when administered during the third trimester of pregnancy.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- 600 mg for an adult weighing greater than 80 kg.
- 400 mg for an adult weighing 80 kg or less.
- See the paediatric drug dose tables for a child.

## Administration

- Administer PO.
- Children unable to swallow tablets may be administered ibuprofen syrup, or tablets that have been crushed and placed in a soft food such as jam or honey.

## Common adverse effects

- Renal impairment.

# Ibuprofen

- Increased bleeding.
- Although indigestion, gastrointestinal ulceration and gastrointestinal bleeding are commonly listed as adverse effects, these are only associated with chronic administration.

## Usual onset of effect

- 30-60 minutes.

## Usual duration of effect

- 4-6 hours.

## Usual preparation

- 200 mg tablets.
- Syrup containing 20 mg/ml.

## Pharmacokinetics

- Ibuprofen is absorbed in the stomach and small intestine. The presence of food in the stomach will delay absorption, but this is not usually clinically significant.
- Ibuprofen is metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- Warfarin. Ibuprofen displaces warfarin from binding sites and increases the activity of warfarin.

## Additional information

- Ibuprofen is not indicated for pain associated with myocardial ischaemia.
- All personnel (including those without ATP) may give ibuprofen to a patient for self-administration, provided the package instructions are followed.

# Ibuprofen

- Ibuprofen is useful in the context of soft tissue pain, musculoskeletal pain and renal colic, particularly if an opiate is not going to be administered.
- Ibuprofen may be administered in addition to other medicines for mild to moderate pain, particularly those not requiring transport or opiate analgesia.
- A patient may be administered ibuprofen for pain associated with a minor condition and be given a recommendation that immediate referral to a medical facility is not required.

## More Information

### Version history

22 November 2022

1.0.1

**Update:** Indications, contraindications and cautions.

# Ipratropium

## Mechanism of action

- Ipratropium is a bronchodilator.
- Ipratropium is an anticholinergic agent with predominantly antimuscarinic activity. It antagonises (blocks) acetylcholine receptors, causing vagal inhibition resulting in bronchodilation.

## Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

## Indications

- ✓ Bronchospasm secondary to asthma or COPD.
- ✓ Prominent bronchospasm secondary to airway burns, smoke inhalation or chest infection.

## Contraindications

- ✗ Known severe allergy.

## Cautions

- ⚠ None.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy, but ipratropium should be administered if indicated.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- 0.5 mg for adults and children once only.

# Ipratropium

## Administration

- Administer nebulised undiluted (in combination with 5 mg of salbutamol) using:
  - Air as the driving gas for COPD.
  - Oxygen as the driving gas for asthma.

## Common adverse effects

- Tachycardia.
- Dry mouth.
- Blurred vision. This usually only occurs with repeated doses.

## Usual onset of effect

- 2-5 minutes.

## Usual duration of effect

- 6 hours.

## Usual preparation

- Ampoule containing 0.5 mg in 2 ml.

## Pharmacokinetics

- Only a small amount of the nebulised dose is absorbed, with most of the dose being nebulised to the atmosphere. The inhaled ipratropium is absorbed through the lungs and some is swallowed.
- Excretion is predominantly via the urine.
- Clearance is prolonged if the patient has significant kidney impairment, but this does not alter the initial (loading) dose.



# Ipratropium

## Common interactions

- None.

## Additional information

- Ipratropium does not have a significant role in the treatment of bronchospasm as a result of smoke, toxic gas inhalation or chest infection. However, it may be administered if bronchospasm is prominent.
- Ipratropium has been reported to cause worsening of glaucoma, but only with frequent doses of nebulised ipratropium (with the nebulised drug contacting the eyes) in the setting of poorly controlled glaucoma. Glaucoma is not a caution when administering a single dose.

## More Information

### Version history

22 November 2022

1.0.1

**Update:** Aligned with asthma and COPD CPGs.

# Ketamine

## Mechanism of action

- Ketamine has complex actions but is predominantly an N-methyl-d-aspartate (NMDA) receptor antagonist (blocker), resulting in inhibition of excitatory neurotransmitters in the brain.
- Low doses cause analgesia, larger doses cause amnesia and dissociation, and high doses cause anaesthesia.

## Authorised practice levels

- Paramedics: analgesia.
- ICPs and CCPs: all indications.

## Indications

- ✓ Severe pain that has not been adequately controlled with an opiate.
- ✓ Inducing dissociation.
- ✓ Acute behavioural disturbance causing a severe to immediately life-threatening risk to safety.
- ✓ Rapid sequence intubation (RSI).
- ✓ Significant movement during CPR that is interfering with resuscitation.
- ✓ Asthma with severe agitation that is impairing the ability to safely provide treatment or transport.

## Contraindications

- ✗ Known severe allergy.

# Ketamine

## Cautions

- ⚠ Altered level of consciousness. Ketamine may reduce the level of consciousness.
- ⚠ Signs of shock. Ketamine may make shock worse.
- ⚠ Concurrent administration of opiates or midazolam. This will increase and prolong the effects.
- ⚠ Aged greater than or equal to 75 years, particularly if frail. The effects of ketamine will be increased and prolonged in this cohort.
- ⚠ Aged less than five years.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy, but ketamine should be administered if indicated.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- For analgesia:
  - IV:
    - 0.2 mg/kg IV up to a maximum of 20 mg for an adult, administered over approximately 15 minutes. Repeat as required.
    - Consider commencing an infusion using service guidelines.
    - See the paediatric drug dose tables for a child and administer IV over approximately 15 minutes. Repeat as required.
  - IM or PO:
    - 0.5 mg/kg IM or PO for an adult, up to a maximum of 50 mg, if IV access cannot be obtained. This may be repeated once after 20 minutes.
    - See the paediatric drug dose tables for a child. This may be repeated once after 20 minutes.
- For dissociation:
  - 1 mg/kg IV (up to a maximum of 150 mg) for an adult. This may be repeated once after five minutes.
  - 5 mg/kg IM (up to a maximum of 400 mg) for an adult, once only.
  - See the paediatric drug dose tables for a child.
  - Halve all doses if a caution is present.

# Ketamine

- For acute behavioural disturbance:
  - 1 mg/kg IV (up to a maximum of 150 mg) every five minutes as required, or
  - 5 mg/kg IM, rounded to the nearest 100 mg (up to a maximum of 400 mg).
  - The IM dose may be repeated once after 20 minutes.
- For asthma with severe agitation: 0.5 mg/kg IV, up to a maximum of 50 mg. This may be repeated every 10 minutes as required.
- For RSI: see the '[RSI](#)' guideline.
- For significant movement during CPR that is interfering with resuscitation administer 1 mg/kg (up to a maximum of 150 mg) of ketamine. This may be repeated once if required.

## Administration

- The preferred route of administration is IV.
- Administer analgesia doses IV over approximately 15 minutes as this appears to reduce adverse side effects.
- For analgesia: Dilute 100 mg (1 ml) of ketamine in a 100 ml bag of 5% glucose. Shake well and label:
  - For adults: draw up the dose and administer 2 mg (2 ml) every 1-2 minutes.
  - For children: draw up the dose, dilute the volume further to a total of 20 ml using 0.9% sodium chloride and administer 2 ml every 1-2 minutes.
- Administer dissociation doses by diluting 100 mg (1 ml) to a total of 10 ml using 0.9% sodium chloride and administering the dose as an IV bolus.
- The IM route is preferred over the PO route as IM absorption is more reliable. Administer IM ketamine undiluted. The preferred site is the lateral thigh as this has the best absorption. If this site is not suitable use the lateral upper arm.
- The PO route should be reserved for the very unusual circumstance in which IM injection is contraindicated or not feasible. Administer ketamine PO undiluted in a liquid, for example paracetamol syrup or water.

## Common adverse effects

- Transient hypertension.
- Tachycardia.

# Ketamine

- Apnoea.
- Nausea and vomiting.
- Sedation.
- Hallucinations.

## Usual onset of effect

- IV: 1-2 minutes.
- IM: 5-10 minutes.
- PO: 10-20 minutes.

## Usual duration of effect

- 10-60 minutes.

## Usual preparation

- Ampoule containing 200 mg in 2 ml.

## Pharmacokinetics

- Ketamine is predominantly metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- The effects will be increased in the presence of other analgesic medicines or sedatives, for example opiates, benzodiazepines or alcohol.

## Additional information

- Warn the patient they may feel strange following ketamine administration.

# Ketamine

- Prior to administering ketamine for pain, sufficient opiates should be administered so that further doses are not providing additional analgesia. This will usually require 150-200 mcg of fentanyl or 15-20 mg of morphine for an adult.
- Do not treat hallucinations routinely with midazolam because the combination of midazolam and ketamine is commonly associated with a reduced level of consciousness, particularly if an opiate has also been administered. Most hallucinations will settle with a combination of explanation and time. However, midazolam in low doses may be administered IV if the hallucinations are severe provided the patient is obeying commands and physiologically stable.
- Round the patient's weight up to the nearest 10 kg when calculating doses.

## More Information

### Version history

|                                                                                             |         |
|---------------------------------------------------------------------------------------------|---------|
| <b>16 June 2026</b>                                                                         | 1.1.0.1 |
| <b>Update:</b> Contraindications, cautions and dosage updated.                              |         |
| <b>11 September 2023</b>                                                                    | 1.0.4   |
| <b>Update:</b> Multiple changes.                                                            |         |
| <b>22 November 2022</b>                                                                     | 1.0.1   |
| <b>Update:</b> Indications, cautions, infusion instructions and aligned with analgesia CPG. |         |

# Labetalol

## Mechanism of action

- Labetalol is a beta-blocker and an alpha-blocker. It antagonises (blocks) beta-1 receptors in the heart, causing a decrease in heart rate, cardiac output and blood pressure and it antagonises (blocks) alpha-1 receptors peripherally, causing vasodilation and a decrease in blood pressure.

## Authorised practice levels

- ICPs and CCPs.
- Paramedics may only administer labetalol after seeking clinical advice.

## Indications

- ✓ Control of hypertension prior to fibrinolytic therapy for STEMI.
- ✓ Control of hypertension during inter-hospital transfer for STEMI.
- ✓ Control of hypertension during inter-hospital transfer for stroke clot retrieval.
- ✓ Control of hypertension associated with autonomic dysreflexia.
- ✓ Control of hypertension associated with eclampsia.
- ✓ Control of hypertension during inter-hospital transfer for acute aortic dissection.

## Contraindications

- ✗ Known severe allergy.
- ✗ Bradycardia. Labetalol will further reduce the heart rate.
- ✗ Hypotension. Labetalol will further reduce the blood pressure.

# Labetalol

## Cautions

- ⚠ 1<sup>st</sup> degree heart block. Labetalol may cause bradycardia.
- ⚠ Known sick sinus syndrome without an internal pacemaker in place. Labetalol may cause bradycardia.
- ⚠ Previous 2<sup>nd</sup> or 3<sup>rd</sup> degree heart block without an internal pacemaker in place. Labetalol may cause worsening of heart block and bradycardia.
- ⚠ Asthma or COPD. Labetalol may cause bronchospasm and should usually be withheld if the patient regularly takes bronchodilators.
- ⚠ Heart failure. Labetalol will reduce cardiac output and may make heart failure worse.

## Use in pregnancy or when breastfeeding

- Safe and should be administered when indicated.

## Dosage

- Administer a 20 mg bolus IV every five minutes, as required, while setting up an infusion.
- Commence an IV infusion at 1 mg/kg/hour, up to a maximum of 150 mg/hour.
- Adjust the rate as required by 20 mg/hour every 10 minutes, titrating to effect.
- Administer additional boluses of 20 mg IV every five minutes as required.

The table below provides an overview of the blood pressure threshold for commencing treatment and the BP maintenance goal.

| Indication                              | BP to commence treatment              | BP maintenance goal                         |
|-----------------------------------------|---------------------------------------|---------------------------------------------|
| Prior to fibrinolytic therapy for STEMI | SBP > 180 mmHg, or<br>DBP > 110 mmHg. | SBP 140-160 mmHg,<br>and<br>DBP < 110 mmHg. |
| IHT for STEMI                           | SBP > 180 mmHg, or<br>DBP > 110 mmHg. | SBP 140-160 mmHg,<br>and<br>DBP < 110 mmHg. |



# Labetalol

|                                 |                                                                    |                                             |
|---------------------------------|--------------------------------------------------------------------|---------------------------------------------|
| IHT for SCR                     | SBP > 180 mmHg, or<br>DBP > 110 mmHg.                              | SBP 140-160 mmHg,<br>and<br>DBP < 110 mmHg. |
| Autonomic dysreflexia           | SBP > 160 mmHg, or<br>SBP > 40 mmHg above the patient's<br>normal. | SBP 140-160 mmHg,<br>and<br>DBP < 110 mmHg. |
| Eclampsia                       | SBP > 160 mmHg, or<br>DBP > 110 mmHg.                              | SBP 140-160 mmHg,<br>and<br>DBP < 110 mmHg. |
| IHT for acute aortic dissection | SBP > 120 mmHg.                                                    | SBP 100-120 mmHg.                           |

## Administration

- Draw up 100 mg undiluted and administer 20 mg as a bolus, before commencing an infusion via an infusion device.

## Common adverse effects

- Hypotension.
- Bradycardia.
- Bronchospasm.

## Usual onset of effect

- 2-3 minutes.

## Usual duration of effect

- 1-2 hours.

## Usual preparation

- Ampoule containing 100 mg in 20 ml.

# Labetalol

## Pharmacokinetics

- Labetalol is metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- The blood pressure effect will be potentiated by other medicines that lower blood pressure. For example, GTN, antihypertensive medicines and amiodarone.
- The heart rate effects will be potentiated by other medicines that lower heart rate. For example, amiodarone and centrally acting calcium channel blockers such as diltiazem.

## More Information

### Version history

|                                                           |         |
|-----------------------------------------------------------|---------|
| <b>16 June 2026</b>                                       | 1.1.0.1 |
| <b>Update:</b> Indications and dosage updated.            |         |
| <b>9 June 2025</b>                                        | 1.0.5.3 |
| <b>Update:</b> Indications and dosage updated.            |         |
| <b>11 September 2023</b>                                  | 1.0.4   |
| <b>Update:</b> Dosage and additional information updated. |         |

# Levetiracetam

## Mechanism of action

- Levetiracetam is an anticonvulsant. It is thought to inhibit seizure activity by blocking some calcium channels and also binding synaptic proteins to modulate neurotransmitter release.

## Authorised practice levels

- Paramedics, ICPs and CCPs.

## Indications

- ✓ Seizure that continues or recurs after two doses of parenteral midazolam.
- ✓ Seizure activity associated with severe TBI, even if seizure activity has ceased following midazolam.

## Contraindications

- ✗ Known severe allergy.

## Cautions

- ⚠ None.

## Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

## Dosage

- 2 g IV for an adult, this may be repeated once if the seizure continues or recurs ten minutes after the first dose.
- See the paediatric drug dose tables for a child. This may be only administered once.

## Administration

- Draw up the required dose.

# Levetiracetam

- Administer this IV over 5 minutes preferably into a running line.
- Do not administer IM.

## Common adverse effects

- None.

## Usual onset of effect

- 10 minutes.

## Usual duration of effect

- 6-8 hours.

## Usual preparation

- Ampoule containing 500 mg in 5 ml.

## Pharmacokinetics

- Levetiracetam is excreted by the kidneys.
- There are no significant effects from renal impairment on the initial (loading) dose.

## Common interactions

- None.

## More Information

### Version history

9 June 2025

1.0.5.3

**Update:** Indications updated.

28 October 2024

1.0.5

**Update:** Indications updated.

# Levetiracetam

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**11 September 2023**

1.0.4

**New:** Guideline released.

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# Lignocaine 1%

## Mechanism of action

- Lignocaine is a local anaesthetic.
- Lignocaine blocks the initiation and transmission of nerve impulses by blocking the movement of sodium ions across the nerve cell membrane.

## Authorised practice levels

- Paramedics, ICPs and CCPs.

## Indications

- ✓ Subcutaneous injection for prophylaxis of pain associated with IV cannulation.
- ✓ Subcutaneous injection for digital ring blocks for analgesia.
- ✓ Intraosseous injection for significant bone pain associated with intraosseous administration.

## Contraindications

- ✓ Known severe allergy.
- ✓ Local infection in the area of injection.

## Cautions

- ⚠ Taking an anticoagulant (ring blocks).

## Use in pregnancy or when breastfeeding

- Safe. May be administered if indicated.

## Dosage

- Subcutaneous:
  - The maximum subcutaneous dose for an adult is 200 mg (20 ml of 1% lignocaine).
  - See the paediatric drug dose tables for the maximum dose for a child.

# Lignocaine 1%

» The subcutaneous dose may be repeated once after 30 minutes.

- Intraosseous:
  - » 50 mg (5 ml of 1% lignocaine) for an adult.
  - » See the paediatric drug dose tables for a child.
  - » The intraosseous dose may be repeated once after 15 minutes.

## Administration

- Subcutaneous for IV insertion: administer into the subcutaneous tissue at the site of cannulation. Raise a bleb and wait approximately one minute before insertion.
- Digital ring blocks: administer approximately 1-2 ml of 1% lignocaine into the tissue on either side of the web space of the digit.
- Intraosseous: administer slowly over 1-2 minutes and wait one further minute before infusing fluid. This is intended to limit the amount of lignocaine flushed into the circulation.

## Common adverse effects

- Stinging at the time of injection.

## Usual onset of effect

- 1-2 minutes for IV cannulation.
- 5-10 minutes for digital ring blocks.

## Usual duration of effect

- 30-60 minutes.

## Usual preparation

- Ampoule containing 50 mg in 5 ml.

# Lignocaine 1%

## Pharmacokinetics

- Lignocaine is metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- None.

## Additional information

- Do not apply lignocaine topically to the eye because the solution contains a preservative that may cause harm.
- Warming lignocaine, for example in your pocket or hand, may reduce the stinging associated with subcutaneous injection.
- Overdose of lignocaine when administered subcutaneously is very rare, but can occur if doses exceed 3 mg/kg or more than 1 mg/kg is inadvertently administered IV. If this occurs the following may develop:
  - Tingling around the mouth.
  - Seizures.
  - Dysrhythmias, particularly bradycardias.
  - Hypotension.
  - Cardiac arrest.

## More Information

### Version history

|                                            |         |
|--------------------------------------------|---------|
| <b>16 June 2026</b>                        | 1.1.0.1 |
| <b>Update:</b> Indications updated.        |         |
| <b>22 November 2022</b>                    | 1.0.1   |
| <b>Update:</b> Aligned with analgesia CPG. |         |



# Loratadine

## Mechanism of action

- Loratadine is a non-sedating antihistamine.
- Loratadine antagonises (blocks) peripheral histamine receptors, blocking the action of histamine and reducing itching and redness.

## Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

## Indications

- ✓ Minor allergic reactions confined to skin involvement.
- ✓ Itch or rash associated with anaphylaxis, provided all systemic signs of anaphylaxis are improving.

## Contraindications

- ✗ Known severe allergy.
- ✗ Age less than one year.

## Cautions

- ⚠ Pregnancy.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy. Because minor allergic reactions rarely require specific treatment, the balance of risk is such that loratadine should usually be withheld.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- 20 mg for an adult or child aged greater than or equal to 12 years.

# Loratadine

- 10 mg for a child aged 1-11 years.

## Administration

- Administer PO.
- Always ask a parent (or guardian) if a young child can swallow tablets. Loratadine may be crushed and placed in a soft food such as jam or honey.

## Common adverse effects

- None.

## Usual onset of effect

- 30-60 minutes.

## Usual duration of effect

- 12-24 hours.

## Usual preparation

- 10 mg tablets.

## Pharmacokinetics

- Loratadine is predominantly metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- None.
- Increases in plasma concentrations of loratadine have been reported after concurrent administration of ketoconazole, erythromycin and roxithromycin. In a small number of patients this was associated with a prolonged QT interval, but this only occurred with long-term administration.

# Loratadine

## Additional information

- All personnel (including those without ATP) may give loratadine to a patient for self-administration, provided the package instructions are followed.

## More Information

### Version history

28 October 2024

1.0.5

**Update:** Dosage updated.

# Magnesium Sulphate

## Mechanism of action

- Magnesium is the active ingredient in magnesium sulphate.
- Magnesium reduces bronchial smooth muscle contraction resulting in bronchodilation.
- Magnesium is thought to have multiple mechanisms when treating eclampsia, including vasodilation, blood-brain barrier protection, reduction of cerebral oedema, and central anticonvulsant effects.
- Magnesium has been shown to stabilise the cardiac membrane during torsades de pointes.

## Authorised practice levels

- ICPs and CCPs.

## Indications

- ✓ Bronchospasm secondary to severe or immediately life-threatening asthma.
- ✓ Bronchospasm secondary to severe COPD or imminent respiratory arrest from COPD.
- ✓ If torsades de pointes is present or has occurred.
- ✓ Suspected eclampsia.

## Contraindications

- ✗ Known severe allergy.

## Cautions

- ⚠ Hypotension. Magnesium is a vasodilator and may make hypotension worse.

## Use in pregnancy or when breastfeeding

- Safe and should be administered when indicated.

# Magnesium Sulphate

## Dosage

- Asthma:
  - 10 mmol (2.5 g) IV for an adult.
  - See the paediatric drug dose tables for a child.
  - A second dose may be administered if transport time is longer than 30 minutes and the patient is not improving.
- COPD:
  - 10 mmol (2.5 g) IV for an adult.
  - A second dose may be administered if transport time is longer than 30 minutes and the patient is not improving.
- Torsades de pointes is present or has occurred:
  - 10 mmol (2.5 g) IV for an adult.
  - Repeat once if episodes of torsades de pointes continue.
- Suspected eclampsia:
  - 20 mmol (5 g) IV.

## Administration

- Cardiac arrest if torsades de pointes is present or has occurred:
  - Dilute to a total of 10 ml using 0.9% sodium chloride and administer over 1-2 minutes.
- All other indications:
  - Administer IV over approximately 15 minutes.
  - Draw up the dose and dilute to a total of 10 ml using 0.9% sodium chloride (if required) and administer 1 ml every 1-2 minutes, preferably into a running line.
  - Alternatively, in adults or children weighing over 40 kg only, add to a 100 ml bag of 5% glucose, shake well, label and infuse over approximately 15 minutes:
    - 2-3 drops per second via a standard IV administration set will deliver 100 ml over approximately 15 minutes.
    - The administration set will need to be flushed with 0.9% sodium chloride to ensure all the magnesium has been administered.

# Magnesium Sulphate

## Common adverse effects

- Flushing. Particularly if administered rapidly.
- Hypotension. Particularly if administered rapidly.
- Muscle weakness. This is usually only seen with doses exceeding 20 mmol.

## Usual onset of effect

- 5-10 minutes.

## Usual duration of effect

- 30-60 minutes.

## Pharmacokinetics

- Magnesium is primarily excreted in the urine.
- There are no significant effects from renal impairment on the initial (loading) dose.

## Usual preparation

- 5 ml ampoule containing 10 mmol (2.5 g) of magnesium.

## Common interactions

- None.

## Additional information

- Do not administer magnesium into an IV line that has an adrenaline infusion running through it concurrently, because precipitation may occur.
- Rarely, magnesium may have a role in the treatment of severe pre-eclampsia. In this setting seek clinical advice.
- Document the dose administered in mmol.

# Magnesium Sulphate

## More Information

### Version history

9 June 2025

1.0.5.3

**Update:** Dosage updated.

11 September 2023

1.0.4

**Update:** Indications have been added. Dosage, administration and additional information has been updated.

22 November 2022

1.0.1

**Update:** Indication for COPD.

# Metaraminol

## Mechanism of action

- Metaraminol is an alpha receptor agonist. It stimulates alpha-1 receptors in peripheral blood vessels, causing vasoconstriction and a rise in blood pressure.
- Some references describe a small degree of stimulation of beta receptors, but this is usually clinically insignificant.

## Authorised practice levels

- ICPs and CCPs.

## Indications

- ✓ Hypotension in the setting of septic shock, post cardiac arrest, cardiogenic shock, adrenal crisis, severe traumatic brain injury, neurogenic shock, rapid sequence intubation and post intubation.

## Contraindications

- ✗ Known severe allergy.

## Cautions

- ⚠ Bradycardia. Metaraminol may make bradycardia worse as reduced endogenous release of adrenaline may occur as blood pressure is raised. Consider administering adrenaline if the patient is hypotensive and bradycardic.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated. The likelihood of a pregnant or breastfeeding patient requiring metaraminol is very low, but it should be administered if indicated.

## Dosage

- Titrate the dose and frequency to the patient's blood pressure:
  - Administer an infusion via an infusion pump, or
  - Administer 0.5-1 mg IV every 5-10 minutes as required for an adult, or



# Metaraminol

- » See the paediatric drug dose tables for bolus doses for a child, and administer IV every 5-10 minutes as required.
- » If an infusion is administered in an adult:
  - » Administer an initial bolus of 0.5-1 mg IV.
  - » Start the infusion at 2 mg/hour and adjust the rate as required.
  - » The patient is likely to require approximately 2-3 mg/hour.

## Administration

- » IV bolus administration in an adult:
  - » Dilute 10 mg of metaraminol to a total of 10 ml using 0.9% sodium chloride. This final solution contains 1 mg/ml.
  - » Administer the required dose, preferably via a running line.
- » IV bolus administration in a child:
  - » Draw up 1 ml of metaraminol from a 10 mg/1 ml ampoule.
  - » Place into a 100 ml bag of 5% glucose.
  - » Shake well and label. This final solution contains 0.1 mg/ml.
  - » See the paediatric drug dose tables for dosing/volume and draw up doses from the bag of 5% glucose.
- » IV infusion:
  - » Follow service guidelines for dilution and infusion pump set up.
  - » Adjust the infusion rate as required.

## Common adverse effects

- » Hypertension. This usually only occurs when boluses greater than 1 mg are administered in an adult.
- » Bradycardia. See cautions.

## Usual onset of effect

- » 1-2 minutes.

# Metaraminol

## Usual duration of effect

- 10-15 minutes.

## Usual preparation

- Ampoule containing 10 mg in 1 ml.
- Pre-filled syringe containing 10 mg in 10 ml.

## Pharmacokinetics

- Metaraminol is predominantly metabolised in the liver with metabolites being excreted in the urine.
- Liver impairment does not alter the dose which is titrated to effect.

## Common interactions

- None.

## Additional information

- Some references describe the IM administration of metaraminol but this has no role in the out-of-hospital setting.

## More Information

### Version history

9 June 2025

1.0.5.3

**Update:** Indications updated.

22 November 2022

1.0.1

**Update:** Usual preparation and administration.

# Methoxyflurane

## Mechanism of action

- Methoxyflurane is an inhalational analgesic.
- The mechanism of action is not clear.

## Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

## Indications

- ✓ Acute moderate to severe pain if:
  - An opiate cannot be administered because of a contraindication or caution, or
  - Opiate administration will be significantly delayed, or
  - There will be severe worsening of pain during a procedure or extrication.

## Contraindications

- ✗ Known severe allergy.
- ✗ Personal or family history of malignant hyperthermia.
- ✗ Known renal impairment.
- ✗ 15 ml methoxyflurane (5 doses) administered within the last week. Frequent administration increases the risk of renal impairment.

## Cautions

- ⚠ Age greater than or equal to 75 years, particularly if frail.
- ⚠ Pre-eclampsia. In this setting, renal impairment is likely and methoxyflurane may worsen renal impairment.
- ⚠ Administration within a confined space.
- ⚠ Acute exacerbation of chronic pain.

# Methoxyflurane

## Use in pregnancy or when breastfeeding

- Safety has not been formally demonstrated in pregnancy, but methoxyflurane may be administered. Methoxyflurane has been extensively used during labour in Australia and New Zealand for many years without adverse effects.
- Methoxyflurane may cause temporary drowsiness in the baby and administration should be discussed with the lead maternity carer if there are known signs of foetal distress.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- Maximum of 6 ml (two doses) for a patient aged greater than or equal to 12 years.
- Maximum of 3 ml (one dose) for a child aged less than 12 years.

## Administration

- Whenever possible, have the patient self-administer methoxyflurane.
- Administer 3 ml (one dose) at a time and always use the charcoal filter. Using a charcoal filter maximises the amount of exhaled methoxyflurane that is absorbed, limiting exposure to personnel.
- Instruct the patient to breathe out through the inhaler.
- Do not administer supplementary oxygen via the inhaler as this significantly increases the amount of methoxyflurane lost through evaporation.
- Place the inhaler in a closed plastic bag if the methoxyflurane has not been fully used. It may subsequently be reused by the same patient.
- An ambulance is not considered a confined space. Maximising ventilation (for example, having ventilation fans on) reduces occupational exposure. Consider not administering methoxyflurane in an ambulance if the patient cannot cooperate with breathing out through the inhaler.

## Common adverse effects

- Sedation.
- Feeling light-headed.

# Methoxyflurane

## Usual onset of effect

- 1-2 minutes.

## Usual duration of effect

- 2-5 minutes after stopping administration.

## Usual preparation

- 3 ml bottle accompanying a plastic inhaler.

## Pharmacokinetics

- Approximately 20% is exhaled. The remainder is metabolised in the liver.
- One of the metabolites is fluoride ions. High concentrations of fluoride ions have been associated with renal impairment and this is the reason for known renal impairment being a contraindication and for having a maximum dose.

## Common interactions

- The effects will be increased in the presence of other pain relieving medicines or sedatives, for example opiates, benzodiazepines or alcohol.

## Additional information

- Malignant hyperthermia (MH) is a rare, inherited disorder of muscle metabolism affecting approximately 20 families in New Zealand, many of whom are in the Manawatu area. When exposed to some anaesthetic agents (including methoxyflurane), the patient may develop a life-threatening hypermetabolic state with severe hyperthermia. Patients with MH (or a family history of MH) usually know about it.
- Renal failure with dialysis is not a contraindication or a caution to methoxyflurane administration because once a patient is receiving dialysis further renal impairment is of no clinical consequence.
- Kidney stones and/or renal colic are not a contraindication or a caution to methoxyflurane administration because these are rarely associated with renal impairment.
- Methoxyflurane must not be left with a patient who is not being transported to a medical facility by ambulance.

# Methoxyflurane

## More Information

### Version history

|                                                                                   |         |
|-----------------------------------------------------------------------------------|---------|
| <b>16 June 2026</b>                                                               | 1.1.0.1 |
| <b>Update:</b> Indications updated.                                               |         |
| <b>11 September 2023</b>                                                          | 1.0.4   |
| <b>Update:</b> A new caution has been added (acute exacerbation of chronic pain). |         |
| <b>22 November 2022</b>                                                           | 1.0.1   |
| <b>Update:</b> Contraindications, cautions and aligned with analgesia CPG.        |         |

# Metoprolol Tartrate

## Mechanism of action

- Metoprolol is an immediate release beta blocker. It antagonises (blocks) beta-1 receptors in the heart, causing a decrease in heart rate, cardiac output and blood pressure.

## Authorised practice levels

- Paramedics, ICPs and CCPs.

## Indications

- Adults with moderate cardiovascular compromise as a result of atrial fibrillation, where the ventricular rate is predominantly greater than 130/minute.

## Contraindications

- Known severe allergy.
- Hypotension. Metoprolol will further reduce the blood pressure.

## Cautions

- First degree heart block. Metoprolol may cause bradycardia.
- Known sick sinus syndrome without an internal pacemaker in place. Metoprolol may cause bradycardia.
- Previous second or third degree heart block without an internal pacemaker in place. Metoprolol may cause worsening of heart block and bradycardia.
- Asthma or COPD. Metoprolol may cause bronchospasm and should usually be withheld if the patient regularly takes bronchodilators.
- Heart failure. Metoprolol will reduce cardiac output and may make heart failure worse.
- Pregnancy.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy. The likelihood of administration being required in a woman who is pregnant is so low that personnel must seek clinical advice prior to administration.

# Metoprolol Tartrate

- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- 25 mg.

## Administration

- Administer PO.

## Common adverse effects

- None.

## Usual onset of effect

- 10-20 minutes

## Usual duration of effect

- 4-6 hours

## Usual preparation

- 50 mg tablets.

## Pharmacokinetics

- Metoprolol is absorbed within the gastrointestinal tract and metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- The blood pressure effect will be potentiated by other medicines that lower blood pressure. For example, GTN, antihypertensive medicines and amiodarone.



# Metoprolol Tartrate

- The heart rate effects will be potentiated by other medicines that lower the heart rate. For example, amiodarone and centrally acting calcium channel blockers such as diltiazem.

## Additional information

- Patients with chronic or paroxysmal atrial fibrillation may be prescribed a beta blocker. Personnel should still administer metoprolol as indicated, even if the patient has taken their own beta blocker.

## More Information

### Version history

|                                                |         |
|------------------------------------------------|---------|
| <b>16 June 2026</b>                            | 1.1.0.1 |
| <b>Update:</b> Indications and dosage updated. |         |
| <b>11 September 2023</b>                       | 1.0.4   |
| <b>New:</b> Guideline released.                |         |

# Midazolam

## Mechanism of action

- Midazolam is a benzodiazepine.
- Midazolam enhances the activity of gamma-aminobutyric acid (GABA) at GABA receptors within the central nervous system, resulting in anticonvulsant activity, sedation, amnesia, anxiolysis and muscle relaxation.

## Authorised practice levels

- Paramedics:
  - IM for acute behavioural disturbance, seizures and care during the last days of life.
  - IV for seizures and care during the last days of life.
- ICPs and CCPs: all routes and all indications.

## Indications

- ✓ Generalised seizures that continue for more than five minutes, or seizures are recurrent.
- ✓ Acute behavioural disturbance causing a mild to moderate risk to safety and droperidol is unavailable or ineffective.
- ✓ Pain associated with severe muscle spasm or severe anxiety if adequate analgesia is not being achieved with an opiate and ketamine is not appropriate.
- ✓ Sedation, for example, for joint relocation.
- ✓ Severe anxiety associated with COPD.
- ✓ Control of anxiety or shortness of breath that does not respond to an opiate during the last days of life.

## Contraindications

- ✗ Known severe allergy.
- ✗ Aged less than 12 years (analgesia only).

# Midazolam

## Cautions

- ⚠ Altered level of consciousness. Midazolam will further reduce the level of consciousness.
- ⚠ Respiratory depression or at high risk of respiratory depression. For example, severe COPD, morbid obesity or on home BiPAP. Such patients may develop respiratory depression following midazolam administration.
- ⚠ Signs of shock. Midazolam may worsen shock.
- ⚠ Concurrent administration of opiates or ketamine. This will increase and prolong the effects.
- ⚠ Aged greater than or equal to 75 years, particularly if frail. The effects of midazolam will be increased and prolonged in this cohort.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy, but midazolam should be administered if indicated.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- The dose of midazolam is dependent on the indication and the route. See the individual guidelines.
- If a dose range is described, use a dose at the lower end of the range if a caution is present.
- If midazolam is being administered as an adjunct to analgesia or sedation, the patient must be rousable at all times.

## Administration

- IV administration:
  - Dilute 2 ml from a 15 mg/3 ml ampoule to a total of 10 ml using 0.9% sodium chloride. This solution contains 1 mg/ml.
  - Administer IV as a bolus.
- IM administration:
  - Draw up the dose from a 15 mg/3 ml ampoule. Do not dilute.

# Midazolam

- » The preferred site is the lateral thigh as this has the best absorption. If this site is not suitable use the lateral upper arm.

## Common adverse effects

- Sedation.
- Respiratory depression.
- Hypotension.
- Amnesia.

## Usual onset of effect

- IV: 2-3 minutes.
- IM: 3-5 minutes.

## Usual duration of effect

- 30-60 minutes. The sedative effect may be longer, particularly in the elderly.

## Usual preparation

- Ampoule containing 15 mg in 3 ml.

## Pharmacokinetics

- Midazolam is predominantly metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- The effects will be increased and prolonged in the presence of other sedatives or pain relieving medicines (for example, other benzodiazepines, opiates, ketamine and alcohol).

# Midazolam

## More Information

### Version history

|                                                                                             |         |
|---------------------------------------------------------------------------------------------|---------|
| <b>9 June 2025</b>                                                                          | 1.0.5.3 |
| <b>Update:</b> Indications updated.                                                         |         |
| <b>11 September 2023</b>                                                                    | 1.0.4   |
| <b>Update:</b> Indications have been updated to align with changes to EAS CPG 5.1 and 5.8.  |         |
| <b>22 November 2022</b>                                                                     | 1.0.1   |
| <b>Update:</b> Contraindications, cautions and aligned with analgesia and end of life CPGs. |         |

# Morphine

## Mechanism of action

- Morphine is an opiate analgesic. It is an opiate agonist that binds to opiate receptors in the brain and spinal cord causing analgesia.

## Authorised practice levels

- Paramedics, ICPs and CCPs.

## Indications

- ✓ Moderate to severe pain.
- ✓ Cardiogenic pulmonary oedema with severe anxiety.
- ✓ Control of pain, agitation, or shortness of breath during the last days of life.
- ✓ Autonomic dysreflexia, even if patient cannot feel pain.

## Contraindications

- ✗ Known severe allergy.
- ✗ Renal failure.

# Naloxone

## Mechanism of action

- Naloxone is an opiate receptor antagonist (blocker). By blocking opiate receptors, naloxone reverses the effects of opiates, particularly respiratory depression and sedation.

## Authorised practice levels

- Paramedics, ICPs and CCPs.

## Indications

- ✓ Opiate poisoning is suspected and the patient has a significantly impaired level of consciousness or significantly impaired breathing.
- ✓ Excess adverse effects from administration of opiates.

## Contraindications

- ✗ Known severe allergy.

## Cautions

- ⚠ Chronic opiate use. If the patient is taking an opiate chronically, there is a risk of adverse physiological effects associated with rapid opiate withdrawal.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated in pregnancy, but naloxone should be administered if indicated.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- 0.1-0.4 mg IV every five minutes as required for an adult.
- 0.8 mg IM as required for an adult. This may be repeated every 10 minutes.
- See the paediatric drug dose tables for a child.

# Naloxone

## Administration

- The preferred route for administration is IV.
- IV: dilute 0.4 mg to a total of 4 ml. This final solution contains 0.1 mg/ml.
- Administer the minimum dose required to produce improvement. Rapid reversal of opiates may be associated with seizures, hypertension, pulmonary oedema or severe agitation, particularly if the patient takes opiates regularly.
- IM: administer undiluted. The preferred site is the lateral thigh as this has the best absorption. If this site is not suitable use the lateral upper arm.

## Common adverse effects

- Sweating.
- Tachycardia.
- Hypertension.

## Usual onset of effect

- IV: 1-2 minutes.
- IM: 5-10 minutes.

## Usual duration of effect

- 30-60 minutes.
- The duration of action of naloxone may be shorter than the duration of action of the opiate that has been administered/taken and naloxone may need to be repeated.

## Usual preparation

- Ampoule containing 0.4 mg in 1 ml.



# Naloxone

## Pharmacokinetics

- Naloxone is predominantly metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- None.

## Additional information

- There is no role for naloxone in the treatment of cardiac arrest associated with opiate poisoning. In this setting cardiac arrest is secondary to respiratory arrest and once cardiac arrest has occurred naloxone has no useful effect. The best treatment is CPR that includes a focus on ventilation. If ROSC occurs, naloxone should still not be administered because it may be associated with seizures, hypertension, pulmonary oedema or severe agitation.
- There is no evidence to support the commonly held view that adequate oxygenation prior to naloxone administration reduces the severity of agitation following naloxone administration. However, treatment of severe hypoxia takes precedence over the administration of naloxone.
- Acute opiate withdrawal following naloxone administration can lead to patient agitation and/or aggression, particularly when there has also been a period of hypoxia. Ensure a planned team approach to maintain safety of both the patient and personnel.

# Olanzapine

## Mechanism of action

- Olanzapine is an atypical antipsychotic.
- Olanzapine has actions at multiple receptors within the brain causing a reduction in agitation, sedation, anxiolysis and stabilisation of mood.

## Authorised practice levels

- Paramedics, ICPs and CCPs.

## Indications

- ✓ Patients aged greater than or equal to 12 years with acute behavioural disturbance causing a mild to moderate risk to safety, when the patient will take an oral medicine.

## Contraindications

- ✗ Known severe allergy.
- ✗ Poisoning with an antipsychotic, for example quetiapine, risperidone or olanzapine.

## Cautions

- ⚠ Pregnancy.
- ⚠ Intoxication. This will increase and prolong the effects.
- ⚠ Aged greater than or equal to 75 years, particularly if frail. The effects of olanzapine will be increased and prolonged in this cohort.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy and the balance of risk is such that olanzapine should usually be withheld.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

# Olanzapine

## Dosage

- 10 mg PO.
- Consider reducing the dose to 5 mg PO if a caution is present.
- The dose may be repeated once after 20 minutes.

## Administration

- Administer PO. The tablet is dispersible and will dissolve in the mouth. A sip of liquid will aid absorption but is not routinely required.
- The tablet may be dissolved in liquid. For example, water, tea or coffee.

## Common adverse effects

- Sedation.

## Usual onset of effect

- 10-20 minutes.

## Usual duration of effect

- 12-24 hours.

## Usual preparation

- 5 mg dispersible tablets (wafers).

## Pharmacokinetics

- Olanzapine is predominantly metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

# Olanzapine

## Common interactions

- Intoxication. Olanzapine will have increased and prolonged effects if the patient is intoxicated with alcohol or has taken recreational drugs.
- Sedative drugs. Concurrent administration with other sedative drugs (such as midazolam or droperidol) will result in an increased and prolonged effect.

## More Information

### Version history

|                                                                 |       |
|-----------------------------------------------------------------|-------|
| <b>11 September 2023</b><br><b>Update:</b> Indications updated. | 1.0.4 |
| <b>22 November 2022</b><br><b>Update:</b> Indications.          | 1.0.1 |

# Ondansetron

## Mechanism of action

- Ondansetron is an antiemetic.
- Ondansetron antagonises (blocks) serotonin receptors centrally in the brain and peripherally in the gastrointestinal tract, resulting in a reduction in nausea and vomiting.

## Authorised practice levels

- EMTs: IM.
- Paramedics, ICPs and CCPs: IV and IM.

## Indications

- ✓ Clinically significant nausea and/or vomiting.

## Contraindications

- ✗ Known severe allergy.
- ✗ Age less than one year.
- ✗ Suspected serotonin syndrome.

## Cautions

- ⚠ None.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy, but ondansetron may be administered if nausea and/or vomiting is severe.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

# Ondansetron

## Dosage

- 8 mg IV once for an adult. See the paediatric drug dose tables for a child.
- 4 mg IM for an adult, if IV access cannot be obtained. See the paediatric drug dose tables for a child.
- The IM dose may be repeated once after 20 minutes if required.
- One IV dose may be administered in addition to one IM dose, if clinically significant nausea and/or vomiting persists and IV access is subsequently obtained. For example, in this setting a maximum total dose of 12 mg may be administered to an adult.

## Administration

- The preferred route of administration is IV.
- Administer IV undiluted.
- Administer IM undiluted. The preferred site is the lateral thigh as this has the best absorption. If this site is not suitable use the lateral upper arm.

## Common adverse effects

- Headache.
- Flushing.

## Usual onset of effect

- IV: 2-5 minutes.
- IM: 5-10 minutes.

## Usual duration of effect

- 4-8 hours.

## Usual preparation

- Ampoule containing 4 mg in 2 ml.

# Ondansetron

- Ampoule containing 8 mg in 4 ml.

## Pharmacokinetics

- Ondansetron is predominantly metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- Ondansetron has been reported to prolong the QT interval, particularly if high doses are administered in conjunction with other medicines that also prolong the QT interval, for example erythromycin. However, one or two doses in this setting is safe.

## Additional information

- Prophylactic administration of ondansetron is not routinely required. Consider administering ondansetron if the nature of the patient's injuries and transport position are such that vomiting would be particularly problematic.
- Ondansetron should not be administered for vomiting associated with an altered level of consciousness because it is rarely effective in this setting.
- Ondansetron has been reported to further prolong the QT interval in patients known to have a prolonged QT syndrome. This generally involved repeated and/or high doses and one or two doses is safe, even if the patient is known to have a prolonged QT syndrome.
- Nausea and/or vomiting in the context of serotonin syndrome must not be treated with ondansetron as this may worsen serotonin syndrome.

## More Information

### Version history

11 September 2023

1.0.4

**Update:** Additional information updated to include serotonin syndrome.

# Oxytocin

## Mechanism of action

- Oxytocin is a synthetic version of the naturally occurring hormone oxytocin which is normally released from the pituitary gland.
- Oxytocin stimulates oxytocin receptors on the uterus, causing increased uterine contraction and reducing blood loss from the uterus.

## Authorised practice levels

- Paramedics, ICPs and CCPs.

## Indications

- ✓ Following normal birth.
- ✓ Postpartum haemorrhage.

## Contraindications

- ✗ Known severe allergy.

## Cautions

- ⚠ None.

## Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

## Dosage

- For following normal birth: 10 units IM.
- For postpartum haemorrhage:
  - 10 units IV (preferred), or
  - 10 units IM.



# Oxytocin

- If oxytocin IM has already been administered as part of routine treatment following normal birth, an additional dose of oxytocin should be administered (in the other thigh if administering IM) if postpartum haemorrhage develops. This may require meeting another vehicle.

## Administration

- IV administration:
  - Place 10 units into a 1 litre bag of 0.9% sodium chloride. This solution contains 0.01 units/ml.
  - Shake well and label.
  - Administer IV over approximately 60 minutes (approx. 4 drops per second).
  - The rate of administration may be increased if shock is present.
- IM administration:
  - Administer IM undiluted.
  - The preferred site is the lateral thigh as this has the best absorption. If this site is not suitable use the lateral upper arm.
  - If multiple babies are present, administration must occur after delivery of the last baby.

## Common adverse effects

- Abdominal cramping.
- Tachycardia.
- Flushing.

## Usual onset of effect

- 5-10 minutes.

## Usual duration of effect

- 30-60 minutes.

## Usual preparation

- Ampoule containing 10 units in 1 ml.

# Oxytocin

## Pharmacokinetics

- Oxytocin is metabolised in the liver and kidneys.
- There are no significant effects from liver or kidney impairment on acute administration.

## Common interactions

- None.

## Additional information

- Oxytocin that is not within kits must be refrigerated:
  - Without refrigeration it loses approximately 5-10% of activity per month.
  - It may be safely stored in kits provided it is labelled with the discard date which is eight weeks after removal from the fridge.
  - It may be transported for up to 12 hours between stores/stations in a container that keeps it cool, without affecting the expiry date, provided it is placed into a fridge immediately on arrival.
- Routine administration of oxytocin following normal birth is controversial, but appears to reduce the incidence of postpartum haemorrhage.
- Oxytocin has been reported to cause prolongation of the QT interval if the patient is taking other medicines that also prolong the QT interval. However, this is usually associated with prolonged IV infusions and is not a clinically significant consideration when administering one or two IM doses.

## More Information

### Version history

9 June 2025

1.0.5.3

**Update:** Dosage and administration updated.

# Paracetamol

## Mechanism of action

- Paracetamol inhibits the production of prostaglandins resulting in a reduction in pain and fever.

## Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.
- First Responders: refer to '[pain relief](#)' in the FR Field Guide

## Indications

- ✓ Mild or moderate pain.
- ✓ Moderate to severe pain, in addition to other medicines.

## Contraindications

- ✗ Known severe allergy.

## Cautions

- ⚠ The patient has taken paracetamol within the last four hours. Paracetamol is contained in many products such as cold and flu tablets/drinks, cough mixtures, combination pain relievers and migraine tablets. Additional paracetamol may be administered provided it is clear that the total dose taken within a four hour period does not exceed the dose described within the CPGs. Withhold paracetamol if there is any doubt.
- ⚠ Abdominal pain, particularly if the patient is very unwell or vomiting. Clinical judgement is required, but if the patient is very unwell or vomiting, the possibility of significant intra-abdominal pathology exists and oral medicines should usually be withheld.
- ⚠ Known severe liver disease. Liver disease must be severely impaired before paracetamol clearance is altered, but the balance of risk is such that paracetamol should usually be withheld in this setting.

## Use in pregnancy or when breastfeeding

- Safe and may be administered if indicated.

# Paracetamol

## Dosage

- 1.5 g PO for an adult weighing greater than 80 kg.
- 1 g PO for an adult weighing 80 kg or less.
- See the paediatric drug dose tables for a child.
- **First Responders:** refer to '[pain relief](#)' in the FR Field Guide and administer the dose listed there.

## Administration

- Administer PO.
- Children unable to swallow tablets may be administered paracetamol syrup, or tablets that have been crushed and placed in a soft food such as jam or honey.

## Common adverse effects

- None.

## Usual onset of effect

- 30-60 minutes.

## Usual duration of effect

- 4-6 hours.

## Usual preparation

- 500 mg tablets.
- Syrup containing 50 mg/ml.

## Pharmacokinetics

- Paracetamol is metabolised in the liver.

# Paracetamol

- If liver impairment is severe, paracetamol clearance will be significantly delayed, but this does not affect the initial (loading) dose.

## Common interactions

- None.

## Additional information

- Paracetamol should be administered in addition to other medicines for severe pain, particularly if the transport time is long. This is not a priority but will reduce the need for subsequent analgesia and improve the quality of pain relief.
- The described doses of paracetamol are higher than those usually recommended in many references. However, the described doses are safe provided they are not administered repeatedly.
- Paracetamol is not indicated for the treatment of fever because fever may confer some benefit if the patient has an infection. However, paracetamol may be administered if the patient has a temperature greater than 39°C and the fever is causing discomfort.
- Paracetamol is not indicated for pain associated with myocardial ischaemia.
- All personnel (including those without ATP) may give paracetamol to a patient for self-administration, provided the package instructions are followed.
- A patient may be administered paracetamol for pain associated with a minor condition and be given a recommendation that immediate referral to a medical facility is not required.

## More Information

### Version history

22 November 2022

1.0.1

**Update:** Aligned with analgesia CPG.

# Prednisone and Prednisolone

## Mechanism of action

- Prednisone is a prodrug that is metabolised to prednisolone in the liver.
- Prednisolone is a corticosteroid with anti-inflammatory and immunosuppressant actions. It inhibits the production of inflammatory mediators, including prostaglandins and leukotrienes, resulting in a reduction in the inflammatory and immune response.

## Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

## Indications

- ✓ Bronchospasm associated with asthma or COPD.
- ✓ Croup.
- ✓ Minor allergy associated with rash.

## Contraindications

- ✗ Known severe allergy.

## Cautions

- ⚠ Age less than five years with asthma. Steroids do not usually have a role in children aged less than five years because they do not generally alter the course of their asthma exacerbation. However, a steroid is indicated if the child has a clear history of asthma and has previously received oral steroids.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy. However, there is significant clinical experience with steroids and they appear to be safe. A steroid should be administered if there is bronchospasm, but should be withheld if the clinical problem is minor, for example rash or itch.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

# Prednisone and Prednisolone

## Dosage

- 40 mg for an adult.
- See the paediatric drug dose tables for a child.
- If the patient is already taking prednisone:
  - Administer an additional full dose if the patient is taking a dose that is lower than that described in these CPGs. If the patient is not transported by ambulance to a medical facility, advise the patient to discontinue their usual prednisone, take the prednisone supplied by personnel and have their treatment reviewed in primary care (preferably by their own GP) within two days.
  - Do not administer an additional dose if the patient is taking a dose equal to or higher than that described in these CPGs. If the patient is not transported by ambulance to a medical facility, advise the patient to continue taking their usual prednisone and have their treatment reviewed by a doctor (preferably their own GP) within two days.

## Administration

- Administer PO.
- Prednisone tablets are very bitter. Do not crush prednisone tablets because this may cause vomiting.
- The tablets may be divided. The tablet does not always break evenly, and this is not of clinical importance. The larger piece of the divided tablet should be chosen for administration.
- Patients aged greater than or equal to ten years should usually be administered prednisone tablets but may be administered the same dose of prednisolone syrup.
- Patients aged less than ten years should usually be administered prednisolone syrup but may be administered the same dose of prednisone tablets.
- Always ask a parent (or guardian) of a young child if the child can swallow tablets.

## Common adverse effects

- Fatigue.
- Sodium and water retention. This may worsen hypertension and heart failure, but is usually only of clinical significance with prolonged dosing.
- Gastrointestinal reflux.

# Prednisone and Prednisolone

## Usual onset of effect

- 30-60 minutes.

## Usual duration of effect

- 24 hours.

## Usual preparation

- 5 mg prednisone tablets.
- 20 mg prednisone tablets.
- Prednisolone syrup containing 5 mg/ml.

## Pharmacokinetics

- Prednisone and prednisolone are predominantly metabolised by the liver.
- Prednisone is a prodrug that is metabolised to prednisolone in the liver. A dose of prednisone provides slightly less steroid equivalence than the same dose of prednisolone, but this is not clinically significant.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- None.

## Providing the patient with prednisone

- If the patient is aged greater than or equal to 12 years and is not transported, additional prednisone should be provided to them by ambulance personnel, unless the patient already has an asthma/COPD action plan and their own supply of steroid.
- Provide the patient with an additional four days (four x 40 mg) of prednisone. Advise the patient and any carers, including in writing, that the patient:
  - Takes 40 mg (two 20 mg tablets) at approximately the same time each day, for the next four days until finished, in addition to their other usual medicines.



# Prednisone and Prednisolone

» Should be reviewed in primary care for a review of their treatment within two days.

## More Information

### Version history

|                                             |         |
|---------------------------------------------|---------|
| <b>16 June 2026</b>                         | 1.1.0.1 |
| <b>Update:</b> Treatment guidance updated.  |         |
| <b>28 October 2024</b>                      | 1.0.5   |
| <b>Update:</b> Indications updated.         |         |
| <b>22 November 2022</b>                     | 1.0.1   |
| <b>Update:</b> Prednisone pack information. |         |

# Rocuronium

## Mechanism of action

- Rocuronium is a neuromuscular blocker. It antagonises (blocks) nicotinic acetylcholine receptors at the neuromuscular junction (motor nerve end plate) of skeletal muscle. This results in the inability of skeletal muscles to contract.

## Authorised practice levels

- Only CCPs may administer rocuronium for the purposes of rapid sequence intubation (RSI).
- All ICPs may administer rocuronium post intubation, provided the endotracheal tube position has been confirmed with capnography.

## Indications

- ✓ Neuromuscular blockade for RSI.
- ✓ Neuromuscular blockade following endotracheal intubation:
  - » Rocuronium is routinely administered following RSI if suxamethonium is used as the neuromuscular blocker.
  - » Rocuronium is administered if the patient shows clinically significant signs of moving following intubation without RSI.
- ✓ Patient movement during cardiac arrest that is interfering with resuscitation, despite ketamine administration, provided the patient has been intubated and the endotracheal tube position has been confirmed with capnography.

## Contraindications

- ✗ Known severe allergy.
- ✗ Endotracheal tube position has not been confirmed by capnography (this does not apply if the indication is for RSI).

# Rocuronium

## Cautions

- ⚠ Predicted difficult intubation (this only applies if the indication is for RSI).
- ⚠ If the indication is for neuromuscular blockade following endotracheal intubation:
  - » Chronic muscle weakness. Examples include myasthenia gravis, motor neuron disease and muscular dystrophy. Rocuronium may cause prolonged muscle weakness and should be withheld if possible. If rocuronium is administered the dose should be halved.
  - » An adult with a very poor prognosis. Examples include severe comorbidities requiring long-term care, unwitnessed cardiac arrest and cardiac arrest with a first rhythm of asystole. Rocuronium should be withheld if possible because the patient is unlikely to benefit from admission to an intensive care unit and may be extubated in the ED.

## Use in pregnancy or when breastfeeding

- Safe and should be administered when indicated.

## Dosage

- For RSI: see the '[RSI](#)' guideline.
- For neuromuscular blockade following endotracheal intubation:
  - » 100 mg IV for an adult weighing greater than 80 kg.
  - » 50 mg IV for an adult weighing less than or equal to 80 kg.
  - » See the paediatric drug dose tables for a child.
  - » Repeat as required.

## Administration

- Administer IV as a bolus.

## Common adverse effects

- None.

# Rocuronium

## Usual onset of effect

- 30-60 seconds. This is predominantly affected by cardiac output and will be prolonged if cardiac output is low.

## Usual duration of effect

- 30-60 minutes.

## Usual preparation

- Ampoule containing 50 mg in 5 ml.

## Pharmacokinetics

- Rocuronium is metabolised in the liver and excreted in urine.
- Significant hepatic or renal impairment will delay clearance and prolong the duration of effect.

## Common interactions

- None.

## Additional information

- Rocuronium that is not within kits must be refrigerated:
  - Without refrigeration it loses approximately 5-10% of activity per month.
  - It may be safely stored in kits provided it is labelled with the discard date which is eight weeks after removal from the fridge.
  - It may be transported for up to 12 hours between stores/stations in a container that keeps it cool, without affecting the expiry date, provided it is placed into a fridge immediately on arrival.
- Rocuronium is a non-depolarising neuromuscular blocker. This means that the acetylcholine receptor at the neuromuscular junction is not stimulated prior to being blocked and no fasciculations will occur. This is in contrast to suxamethonium which is a depolarising neuromuscular blocker. Suxamethonium binds to the acetylcholine receptor at the neuromuscular junction and stimulates the receptor before blocking it, resulting in brief muscle contractions which are seen as fasciculations.

# Rocuronium

- During cardiac arrest the blood flow generated by CPR may be insufficient to deliver adequate levels of rocuronium to skeletal muscle and neuromuscular blockade may not occur. If sustained ROSC is achieved, neuromuscular blockade will occur and the patient will require an adequate level of sedation.
- Adequate sedation and analgesia must always be administered to a patient if rocuronium is administered. See the [‘post intubation’](#) guideline.

## More Information

### Version history

22 November 2022

1.0.1

**Update:** Discard date following removal from the fridge.

# Ropivacaine 0.75%

## Mechanism of action

- Ropivacaine is a local anaesthetic. It blocks the initiation and transmission of nerve impulses by blocking the movement of sodium ions across the nerve cell membrane.

## Authorised practice levels

- ICPs and CCPs.

## Indications

- ✓ Severe pain associated with clinically obvious fractured neck of femur or fractured proximal shaft of femur, where pain has not been adequately controlled with an opiate and transport time (including extrication time) to hospital is greater than 30 minutes.

## Contraindications

- ✗ Known severe allergy.
- ✗ Local infection at the site of injection.
- ✗ Additionally, for fascia iliaca block:
  - » Previous surgery in the groin.
  - » Aged less than 12 years.

## Cautions

- ⚠ Taking an anticoagulant.

## Use in pregnancy or when breastfeeding

- Safe and may be administered if indicated.

## Dosage

- 40 ml of 0.375% (20 ml of 0.75% diluted to 40 ml) ropivacaine for an adult weighing greater than 60 kg.
- 30 ml of 0.375% (15 ml of 0.75% diluted to 30 ml) ropivacaine for an adult weighing 60 kg or less.

# Ropivacaine 0.75%

- Administer the fascia iliaca block once only.

## Administration

- Administer the dose described into the fascia iliaca space.

## Common adverse effects

- Stinging at the time of injection.

## Usual onset of effect

- 5-10 minutes.

## Usual duration of effect

- 1-2 hours.

## Usual preparation

- Ampoule containing 150 mg in 20 ml (0.75%).

## Pharmacokinetics

- Ropivacaine is metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- None.

## Additional information

- A fascia iliaca block may be performed if the patient is taking antiplatelet medicines such as aspirin, clopidogrel or ticagrelor.

# Ropivacaine 0.75%

- A fascia iliaca block may be performed if the patient is taking an anticoagulant but have a low threshold for stopping the procedure if it is difficult, and do not attempt multiple passes with the needle.
- Overdose of ropivacaine is very rare but can occur if doses exceed 150 mg in an adult or it is inadvertently administered IV. If this occurs the following may develop:
  - Tingling around the mouth.
  - Seizures.
  - Dysrhythmias, particularly bradycardias.
  - Hypotension.
  - Cardiac arrest.

## More Information

### Version history

22 November 2022

1.0.1

**Update:** Aligned with analgesia CPG.



# Salbutamol

## Mechanism of action

- Salbutamol is a bronchodilator. It is an agonist (stimulator) of beta-2 receptors.

## Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

## Indications

- ✓ Bronchospasm secondary to asthma or COPD.
- ✓ Prominent bronchospasm secondary to airway burns, smoke inhalation or chest infection.
- ✓ Release syndrome following crush injury.
- ✓ Known or suspected hyperkalaemia with ECG changes.

## Contraindications

- ✗ Known severe allergy.

## Cautions

- ⚠ None.

## Use in pregnancy or when breastfeeding

- Safe and may be administered if indicated.

## Dosage

- MDI: One puff into the MDI and spacer, followed by six breaths. Repeat this six times.
- Nebulised for bronchospasm: 5 mg for adults and children, repeated as required. The initial dose is combined with 0.5 mg ipratropium, but subsequent doses are not.
- Nebulised for release syndrome following crush injury or known or suspected hyperkalaemia: 5 mg for adults and children, repeat as required.

# Salbutamol

## Administration

- MDI: Administer one puff of salbutamol using an MDI and spacer, followed by six breaths. Repeat this six times.
- Nebulised: Administer nebulised undiluted using:
  - Air as the driving gas for COPD.
  - Oxygen as the driving gas for all other indications.

## Common adverse effects

- Tremor.
- Tachycardia.

## Usual onset of effect

- 2-5 minutes.

## Usual duration of effect

- 1-2 hours.

## Usual preparation

- Metered dose inhaler delivering 100 mcg doses per puff.
- Ampoule containing 5 mg in 2.5 ml.
- Ampoule containing 2.5 mg in 2.5 ml.

## Pharmacokinetics

- Only a small amount of the nebulised dose is absorbed, with most of the dose being nebulised to the atmosphere. The inhaled salbutamol is absorbed through the lungs and some is swallowed.
- Salbutamol is metabolised in the liver and excreted in urine.
- There are no significant effects from liver or kidney impairment on acute administration.

# Salbutamol

## Common interactions

- Salbutamol will be less effective in the presence of a beta-blocker, with the reduction in effect being most pronounced with a non-selective beta-blocker such as propranolol.

## Additional information

- Salbutamol does not have a significant role in the treatment of bronchospasm as a result of smoke, toxic gas inhalation or chest infection. However, it may be administered if bronchospasm is prominent.

## More Information

### Version history

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**11 September 2023**

1.0.4

**Update:** Indications, dosage and administration information updated.

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**22 November 2022**

1.0.1

**Update:** Aligned with asthma and COPD CPGs.

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# Sodium Bicarbonate 8.4%

## Mechanism of action

- 8.4% sodium bicarbonate is a systemic alkalinising agent. It increases plasma bicarbonate, buffers hydrogen ions and raises the blood pH.
- Following crush injury 8.4% sodium bicarbonate has a role because:
  - Sodium ions help protect cardiac cell membranes from the effects of hyperkalaemia.
  - A rise in pH results in potassium moving into the cells, which lowers the potassium concentration in blood.
  - A rise in urinary pH reduces myoglobin deposition in the kidneys.

## Authorised practice levels

- ICPs and CCPs.

## Indications

- ✓ Release syndrome following crush injury in an adult.
- ✓ Known or suspected hyperkalaemia with severe ECG changes.
- ✓ Cardiac arrest secondary to hyperkalaemia.
- ✓ Suspected cyclic antidepressant poisoning with QRS prolongation.
- ✓ Severe TBI and intubated with clinical signs of raised intracranial pressure.

## Contraindications

- ✗ Known severe allergy.

## Cautions

- ⚠ IV access via a small vein. 8.4% sodium bicarbonate is hyperosmolar and will cause venous irritation if administered via a small vein.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated, but 8.4% sodium bicarbonate should be administered if indicated.

# Sodium Bicarbonate 8.4%

## Dosage

- Cyclic antidepressant poisoning:
  - 100 ml IV for an adult.
  - 2 ml/kg for a child.
- Severe TBI, intubated and clinical signs of raised intracranial pressure:
  - 100 ml IV for an adult.
  - 2 ml/kg for a child.
- Hyperkalaemia with severe ECG changes:
  - 100 ml IV for an adult.
  - Repeat the dose after 20 minutes if severe ECG changes persist or recur.
  - Seek clinical advice for a child.
- Cardiac arrest secondary to hyperkalaemia:
  - 100 ml IV for an adult.
  - Repeat the dose if ROSC is not achieved within 10 minutes of the first dose.
  - Seek clinical advice for a child.
- Release syndrome following crush injury:
  - 100 ml IV for an adult.
  - Administer further doses if signs of hyperkalaemia (including dysrhythmia) occur.
  - Seek clinical advice for a child.

## Administration

- Cardiac arrest secondary to hyperkalaemia: Administer IV as a bolus, via a running line.
- All other indications: Administer IV over approximately one minute via a running line.
- Do not mix with other medicines as precipitation will occur. If other medicines are being administered via the same vein, ensure a minimum IV flush of 50 ml of 0.9% sodium chloride between medicines.

## Common adverse effects

- None.

# Sodium Bicarbonate 8.4%

## Usual onset of effect

- 1-2 minutes.

## Usual duration of effect

- 1-2 hours.

## Usual preparation

- Ampoule containing 1 mmol/ml of sodium bicarbonate.
- Ampoules may be 10 ml, 50 ml or 100 ml.

## Pharmacokinetics

- 8.4% sodium bicarbonate dissociates into sodium and bicarbonate ions which are excreted in urine.
- There are no significant effects from renal impairment on administration.

## Common interactions

- None.

## Additional information

- Document the dose administered as mmol.

## More Information

### Version history

28 October 2024

1.0.5

**Update:** Indications and dosage updated.

11 September 2023

1.0.4

**Update:** New indications have been added. Dosage and administration information updated.

# Suxamethonium

## Mechanism of action

- Suxamethonium is a neuromuscular blocker. It antagonises (blocks) nicotinic acetylcholine receptors at the neuromuscular junction (motor nerve end plate) of skeletal muscle. This results in the inability of skeletal muscles to contract.

## Authorised practice levels

- CCPs.

## Indications

- ✓ Neuromuscular blockade for rapid sequence intubation (RSI).

## Contraindications

- ✗ Known severe allergy.
- ✗ Known personal or family history of malignant hyperthermia (MH). In susceptible patients suxamethonium will trigger life-threatening MH.
- ✗ Pre-existing paraplegia or quadriplegia. Long-term muscle weakness causes proliferation of acetylcholine receptors on skeletal muscle and suxamethonium may cause life-threatening hyperkalaemia.
- ✗ Muscle disorder with long-term weakness. Examples include muscular dystrophy and motor neurone disease. Long-term muscle weakness causes proliferation of acetylcholine receptors on skeletal muscle and suxamethonium may cause life-threatening hyperkalaemia.
- ✗ Hyperkalaemia is strongly suspected. Suxamethonium will cause a brief rise in the potassium concentration within blood of 0.5-1 mmol/litre. In the presence of hyperkalemia this rise may be significant.

## Cautions

- ⚠ Predicted difficult intubation.

## Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

# Suxamethonium

## Dosage

- See the '[RSI](#)' guideline.
- The onset of adequate neuromuscular blockade may be delayed if cardiac output is very low. Consider increasing the dose of suxamethonium to approximately 3 mg/kg, up to a maximum of 200 mg.

## Administration

- Administer undiluted for an adult or a large child.
- Dilute 100 mg to a total volume of 10 ml for a small child. This solution contains 10 mg/ml.
- Administer IV as a bolus, preferably into a running IV line.

## Common adverse effects

- Bradycardia. This is due to stimulation of acetylcholine receptors in the parasympathetic nervous system and is usually only clinically significant in young children.

## Usual onset of effect

- 30-60 seconds. This is predominantly affected by cardiac output and will be prolonged if cardiac output is low.

## Usual duration of effect

- 4-8 minutes.

## Usual preparation

- Ampoule containing 100 mg in 2 ml.

## Pharmacokinetics

- Suxamethonium is metabolised by the enzyme pseudocholinesterase. In most patients suxamethonium is cleared within 4-8 minutes.



# Suxamethonium

- The neuromuscular blocking effects of suxamethonium can be increased to 4-12 hours if the patient has pseudocholinesterase deficiency.
- Pseudocholinesterase deficiency is rare.

## Common interactions

- None.

## Additional information

- Suxamethonium that is not within kits must be refrigerated:
  - Without refrigeration it loses approximately 5-10% of activity per month.
  - It may be safely stored in kits provided it is labelled with the discard date which is 8 weeks after removal from the fridge.
  - It may be transported for up to 12 hours between stores/stations in a container that keeps it cool, without affecting the expiry date, provided it is placed into a fridge immediately on arrival.
- Suxamethonium is a depolarising neuromuscular blocker. This means that the acetylcholine receptor at the neuromuscular junction is stimulated prior to being blocked and muscle fasciculations occur prior to the onset of neuromuscular blockade.
- Malignant hyperthermia (MH) is a rare, inherited disorder of muscle metabolism affecting approximately 20 families in New Zealand, many of whom are in the Manawatu area. When exposed to some anaesthetic agents (including suxamethonium) the patient may develop a life-threatening hypermetabolic state with severe hyperthermia. Patients with MH (or a family history of MH) usually know about it.
- Burn injury is often quoted as a contraindication to the administration of suxamethonium because life-threatening hyperkalaemia may occur. However, this is not a contraindication immediately following burn injury or once the patient is well enough to be discharged from hospital.

## More Information

### Version history

22 November 2022

1.0.1

**Update:** Discard date following removal from the fridge.

# Tenecteplase

## Mechanism of action

- Tenecteplase is a fibrinolytic that accelerates the breakdown of blood clots. It converts the plasma protein plasminogen into the active enzyme plasmin, which breaks down fibrin within blood clots.

## Authorised practice levels

- Paramedics, ICPs and CCPs.

## Indications

- ✓ STEMI when following the fibrinolytic pathway.

## Contraindications

- ✗ Known severe allergy.

# Tenecteplase

## Cautions

- ⚠ More than ten minutes of CPR.
- ⚠ Non-compressible vascular puncture within the last 24 hours.
- ⚠ Internal bleeding within the last six weeks.
- ⚠ Lumbar puncture or epidural insertion within the last six weeks.
- ⚠ TIA within the last three months.
- ⚠ Known bleeding disorder.
- ⚠ Taking an anticoagulant.
- ⚠ Systolic BP greater than 180 mmHg or diastolic BP greater than 110 mmHg.
- ⚠ Known to be pregnant or less than two weeks postpartum.
- ⚠ Time of onset of symptoms was greater than 12 hours ago.
- ⚠ Requires assistance with their activities of daily living.
- ⚠ Severe comorbidities that significantly limit their life expectancy.
- ⚠ Suspected aortic dissection.
- ⚠ Major surgery, major trauma or severe brain injury within the last six weeks.
- ⚠ Intracranial surgery within the last six months.
- ⚠ Ischaemic stroke within the last six months.
- ⚠ Previous intracerebral haemorrhage.
- ⚠ Known cerebral aneurysm, arteriovenous malformation or tumour.

## Use in pregnancy or when breastfeeding

- Administration during pregnancy or within two weeks of birth carries a significant risk of bleeding. The likelihood of STEMI occurring in a woman who is pregnant or within two weeks of birth is very low and personnel must seek clinical advice prior to administration.

# Tenecteplase

- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- Dosage is based on age and known (or estimated) weight.

| Weight   | Age less than 75 years    |                             | Age 75 years or older     |                             |
|----------|---------------------------|-----------------------------|---------------------------|-----------------------------|
|          | Tenecteplase<br>(dose IV) | Tenecteplase<br>(volume IV) | Tenecteplase<br>(dose IV) | Tenecteplase<br>(volume IV) |
| <60 kg   | 30 mg                     | 6 ml                        | 15 mg                     | 3 ml                        |
| 60-69 kg | 35 mg                     | 7 ml                        | 17.5 mg                   | 3.5 ml                      |
| 70-79 kg | 40 mg                     | 8 ml                        | 20 mg                     | 4 ml                        |
| 80-89 kg | 45 mg                     | 9 ml                        | 22.5 mg                   | 4.5 ml                      |
| ≥90 kg   | 50 mg                     | 10 ml                       | 25 mg                     | 5 ml                        |

## Administration

- Dissolve the powder using the syringe within the kit.
- Carefully discard unwanted drug from the syringe, preferably into the ampoule before administration, ensuring the correct dose remains in the syringe.
- If an error is made in discarding unwanted drug and the correct dose cannot be drawn up, administer the remaining drug, document this well, and notify the receiving clinician.
- Administer undiluted as an IV bolus, preferably into a running IV line.

## Common adverse effects

- Bleeding. Tenecteplase commonly causes superficial bleeding, including epistaxis, bruising and bleeding from IV sites.
- Dysrhythmia. It is common for dysrhythmia to occur if the coronary artery reperfuses. Most commonly the rhythm is accelerated idioventricular rhythm (AIVR) which does not require specific treatment. Other dysrhythmias should be treated using the appropriate guideline.

# Tenecteplase

## Usual onset of effect

- 5-10 minutes.

## Usual duration of effect

- 2-6 hours.

## Pharmacokinetics

- Tenecteplase is metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

## Usual preparation

- Glass ampoule containing 50 mg of tenecteplase, in powder form with a pre-filled syringe containing 10 ml of sterile water.

## Common interactions

- None.

## More Information

### Version history

11 September 2023

1.0.4

**Update:** Indication wording has been changes. Cautions and contraindications have been updated.

# Ticagrelor

## Mechanism of action

- Ticagrelor has antiplatelet activity. It produces reversible inhibition of the ADP receptor on platelets, decreasing platelet aggregation.

## Authorised practice levels

- Paramedics, ICPs and CCPs.

## Indications

- STEMI, in conjunction with the primary PCI pathway, when requested by the accepting clinician.

## Contraindications

- Known severe allergy.

## Cautions

- Clinically significant bleeding. Ticagrelor will increase bleeding.
- At risk of bleeding.
- Pregnancy.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy. The likelihood of STEMI occurring in a woman who is pregnant is so low that personnel must seek clinical advice prior to administration.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- 180 mg for an adult.

# Ticagrelor

## Administration

- Administer PO.

## Common adverse effects

- Increased bleeding.

## Usual onset of effect

- 30-60 minutes.

## Usual duration of effect

- 24 hours.

## Usual preparation

- 90 mg tablet.

## Pharmacokinetics

- Ticagrelor is metabolised in the liver and excreted predominantly in the bile.
- There are no significant effects from liver impairment on acute administration.

## Common interactions

- The risk of bleeding will be increased if the patient is taking an anticoagulant.

## More Information

### Version history

11 September 2023

**New:** Guideline released.

1.0.4

# Tramadol

## Mechanism of action

- Tramadol is an analgesic. It has multiple actions within the central nervous system, including opiate receptor stimulation and inhibition of the re-uptake of noradrenaline and serotonin.

## Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

## Indications

- ✓ Moderate to severe pain, particularly if an opiate and/or ketamine is not going to be administered.

## Contraindications

- ✗ Known severe allergy.
- ✗ Aged less than 12 years.

## Cautions

- ⚠ Tramadol has been taken within the last four hours. Additional tramadol may be administered provided it is clear that the total dose taken within a four hour period does not exceed the dose described within the CPGs. Withhold tramadol if there is any doubt.
- ⚠ Abdominal pain, particularly if the patient is very unwell or vomiting. Clinical judgement is required, but if the patient is very unwell or vomiting, the possibility of significant intra-abdominal pathology exists and oral medicines should usually be withheld.
- ⚠ Aged greater than or equal to 75 years, particularly if there is a previous history of dementia or confusion. Tramadol has anticholinergic activity and this may cause confusion, particularly in the elderly.
- ⚠ Confusion. Tramadol has anticholinergic activity and this may worsen confusion.
- ⚠ Pregnancy.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated in pregnancy and tramadol should usually be withheld.



# Tramadol

- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- 50 mg PO.

## Administration

- Administer PO.

## Common adverse effects

- Nausea and/or vomiting.
- Feeling light-headed or unusual.
- Sedation.
- Dry mouth.

## Usual onset of effect

- 30-60 minutes.

## Usual duration of effect

- 4-8 hours.

## Usual preparation

- 50 mg capsules.

## Pharmacokinetics

- Tramadol is metabolised in the liver and excreted by the kidneys.
- There are no significant effects from liver or kidney impairment on the initial (loading) dose.

# Tramadol

## Common interactions

- Tramadol has been reported to cause serotonin syndrome in patients taking other medicines or recreational drugs that also raise serotonin levels within the brain. Examples include selective serotonin re-uptake inhibitors (SSRIs), tricyclic antidepressants and Ecstasy. However, this usually only occurs when doses of tramadol higher than 50 mg are taken regularly.

## Additional information

- Tramadol is not indicated for pain associated with myocardial ischaemia.
- Do not routinely administer tramadol if an opiate or ketamine is administered because tramadol does not usually provide significant additional pain relief and may worsen side effects. However, tramadol may be administered if the patient has taken tramadol before and found it effective.
- Some patients experience nausea and/or feel unusual with tramadol and may refuse to have it again. These are side effects, not an allergy and are most likely to occur with doses higher than 50 mg administered IV.
- Tramadol has been reported to lower the seizure threshold in patients with epilepsy. However, this usually only occurs with doses higher than 50 mg taken regularly.
- A patient may be administered tramadol (in combination with paracetamol and/or ibuprofen) for pain associated with a minor condition and be given a recommendation that immediate referral to a medical facility is not required provided:
  - Adequate pain control is achieved, and
  - The patient is advised to be seen in primary care for a review of their condition.

## More Information

### Version history

22 November 2022

1.0.1

**Update:** Aligned with analgesia CPG and caution added.

# Tranexamic Acid

## Mechanism of action

- Tranexamic acid is an antifibrinolytic medicine. It blocks the conversion of plasminogen to plasmin, reducing fibrinolysis (breakdown of blood clots) and bleeding.

## Authorised practice levels

- Paramedics, ICPs and CCPs.

## Indications

- ✓ Postpartum haemorrhage.
- ✓ Clinically significant bleeding or signs of hypovolaemia following trauma.
- ✓ Non-traumatic bleeding and shock is severe.
- ✓ Cardiac arrest secondary to trauma.
- ✓ Bleeding following tonsillectomy.
- ✓ Crush injury.
- ✓ Moderate to severe epistaxis, not controlled following one dose of adrenaline IN.
- ✓ Antepartum haemorrhage if shock and/or bleeding is severe.

## Contraindications

- ✗ Known severe allergy.
- ✗ Trauma when tranexamic acid will be administered more than three hours after the time of injury. This appears to increase mortality rates, the cause of which is not clear.

## Cautions

- ⚠ None.

# Tranexamic Acid

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated, but the balance of risk is such that it should be administered if indicated.

## Dosage

- For bleeding following tonsillectomy:
  - 500 mg nebulised for an adult or child. Repeat once after 30 minutes.
  - 1 g IV for an adult. See the paediatric drug dose tables for a child.
- For bleeding following trauma:
  - 2 g IV for an adult.
  - See the paediatric drug dose tables for a child.
- Non-traumatic bleeding:
  - 1 g IV for an adult.
  - See the paediatric drug dose tables for a child.
- For moderate to severe epistaxis:
  - Administer 200 mg (2 ml undiluted) per bleeding nostril for a patient aged 12 years or over.
  - Administer 100 mg (1 ml undiluted) per bleeding nostril for a child aged 5-11 years.
  - Seek clinical advice if the patient is a child aged less than five years.

## Administration

- IV: administer over 1-2 minutes.
- Nebulised: administer undiluted.
- IN: administer undiluted.

## Common adverse effects

- None.

# Tranexamic Acid

## Usual onset of effect

- 30-60 minutes.

## Usual duration of effect

- 8 hours.

## Usual preparation

- 10 ml ampoule containing 1 g.
- 5 ml ampoule containing 500 mg.

## Pharmacokinetics

- Tranexamic acid is predominantly excreted in the urine.
- Renal impairment does not alter the initial (loading) dose.

## Common interactions

- None.

## Additional information

- Some references describe the topical administration of tranexamic acid for control of superficial bleeding. This usually has no role in the out-of-hospital setting and personnel should seek clinical advice if it is being considered.

## More Information

### Version history

9 June 2025

1.0.5.3

**Update:** Indications and dosage updated.

28 October 2024

1.0.5

**Update:** Indications, dosage and route updated.

# Tranexamic Acid

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11 September 2023

1.0.4

**Update:** Indications have been updated. Reference to PATCH trial removed.

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